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JEE Main and Advanced

15 MOCK TEST

COMBINED

PHYSICS, CHEMISTRY AND MATHEMATICS

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JEE Main and Advanced

15 MOCK TEST

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PHYSICS, CHEMISTRY AND MATHEMATICS

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IIT-JEE JEE MAIN AND ADVANCED
15 MOCK TEST
COMBINED
PHYSICS, CHEMISTRY AND MATHEMATICS

by Team Prabhat

Published by **PRABHAT PRAKASHAN**

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Test Booklet Code

A

Mock Test "JEE-Main"

Do not open this Test Booklet until you are asked to do so.
Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions:

1. Immediately fill in the particulars on this page of the Test Booklet with *Blue/Black Ball Point Pen*. *Use of pencil is strictly prohibited.*
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75** questions. The maximum marks are **300.**
5. There are ***three*** parts in the question paper A, B, C consisting of, **Physics, Chemistry** and **Mathematics** having 25 questions in each part of equal weightage.
6. Each question is allotted **4 (four)** marks for each correct response. For MCQs - 4 Marks will be awarded for every correct answer and 1 Mark will be deducted for every incorrect answer.
7. For answer with numeric value - 4 Marks will be awarded for every correct answer and 0 Mark will be deducted for every incorrect answer.
8. Use ***Blue/Black Ball Point Pen only*** for writing particulars/ marking responses on **Side-1** and **Side-2** of the Answer Sheet. ***Use of pencil is strictly prohibited***.
9. No candidates is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc., except the Admit Card inside the examination hall/room.
10. Rough work is to be done on the space provided for this purpose in the Test Booklet only. This space is given at the bottom of each page and at the end of the booklet.
11. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room / Hall. ***However, the candidates are allowed to take away this Test Booklet with them.***
12. The CODE for this Booklet is **A**. Make sure that the CODE printed on **Side-2** of the Answer Sheet and also tally the serial number of the Test Booklet and Answer Sheet are the same as that on this booklet. In case of discrepancy, the candidate should immediately report the matter to the invigilator for replacement of both the Test Booklet and the Answer Sheet.
13. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capital letters): _____

Roll Number : in figures ☐☐☐☐☐☐☐☐

in words _____

Examination Centre Number: ☐☐☐☐☐☐☐☐

Name of Examination Centre (in Capital letters):_____

Candidate's Signature:_____ Invigilator's Signature:_____

Sample

***Read the following instructions carefully*:**

1. The candidates should fill in the required particulars on the Test Booklet and Answer Sheet (***Side-1***) with ***Blue/Black Ball Point Pen.***
2. For writing/marking particulars on ***Side-2*** of the Answer Sheet, use ***Blue/Black Ball Point Pen only.***
3. The candidates should not write their Roll Numbers anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
4. Out of the four options given for each question, only one option is the correct answer.
5. For each ***incorrect response, one-fourth (¼)*** of the total marks allotted to the question would be deducted for MCQs and ***No deduction*** for numeric questions from the total score. ***No deduction*** from the total score, however, will be made ***if no response*** is indicated for an item in the Answer Sheet.
6. Handle the Test Booklet and Answer Sheet with care, *as under no circumstances (except for discrepancy in Test Booklet Code and Answer Sheet Code), another set will be provided.*
7. The candidates are not allowed to do any rough work or writing work on the Answer Sheet. All calculations/writing work are to be done in the space provided for this purpose in the Test Booklet itself, marked 'Space for Rough Work'. This space is given at the bottom of each page and at the end of the booklet.
8. On completion of the test, the candidates must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
9. Each candidate must show on demand his/her Admit Card to the Invigilator.
10. No candidate, without special permission of the Superintendent or Invigilator, should leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign the Attendance Sheet again. Cases where a candidate has not signed the Attendance Sheet a second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case. **The candidates are also required to put their left hand THUMB impression in the space provided in the Attendance Sheet.**
12. Use of Electronic/Manual Calculator and any Electronic Item like mobile phone, pager etc. is prohibited
13. The candidates are governed by all Rules and Regulations of the JAB/Board with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of the JAB/Board.
14. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
15. **Candidates are not allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, electronic device or any other material except the Admit Card inside the examination hall/room.**

Sample

JEE-MAIN: PHYSICS MOCK TEST-1

SECTION 1 (Multiple Choice Question)

1. If a particle moves from point P (2,3,5) to point Q (3,4,5). Its displacement vector be:

a. $\hat{i}+\hat{j}+10\hat{k}$ **b.** $\hat{i}+\hat{j}+5\hat{k}$

c. $\hat{i}+\hat{j}$ **d.** $2\hat{i}+4\hat{j}+6\hat{k}$

2. Long horizontal rod has a bead which can slide along its length and is initially placed at a distance L from one end A of the rod. The rod is set in angular motion about A with a constant angular acceleration α. If the coefficient of friction between the rod and bead is μ, and gravity is neglected, then the time after which the bead starts slipping is:

a. $\sqrt{\frac{\mu}{\alpha}}$ **b.** $\frac{\mu}{\sqrt{\alpha}}$

c. $\frac{1}{\sqrt{\mu\alpha}}$ **d.** infinitesimal

3. A wind-powered generator converts wind energy into electric energy. Assume that the generator converts a fixed fraction of the wind energy intercepted by its blades into electrical energy. For wind speed v, the electrical power output will be proportional to

a. v **b.** v^2

c. v^3 **d.** v^4

4. A spherical hollow is made in a lead sphere of radius R, such that its surface touches the outside surface of lead sphere and passes through the center. What is the shift in the center of mass of lead sphere due to the hollowing?

a. $\frac{R}{7}$ **b.** $\frac{R}{14}$

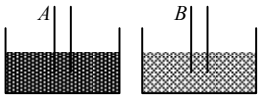
c. $\frac{R}{2}$ **d.** R

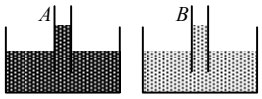
5. The bob of a simple pendulum executes simple harmonic motion in water with a period t, while the period of oscillation of the bob is t_0 in air. Neglect frictional force of water and given that the density of the bob is $(4/3)\times1000$ kg/m^3. What relationship is true between t and t_0?

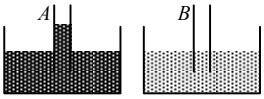
a. $t=t_0$ **b.** $t=t_0/2$

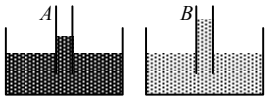
c. $t=2t_0$ **d.** $t=4t_0$

6. A capillary tube A is dipped in water. Another identical tube B is dipped in a soap-water solution. Which of the following shows the relative nature of the liquid columns in the two tubes?

a. 

b. 

c. 

d. 

7. Two rods of different materials having coefficients of linear expansion α_1, α_2 and Young's moduli Y_1, Y_2 respectively are fixed between two rigid massive walls. The rods are heated such that they undergo the same increase in temperature. There is no bending of the rods. If $\alpha_1:\alpha_2=2:3$, the thermal stresses developed in the two rods are equal provided $Y_1:Y_2$ is equal to:

a. 2 : 3 **b.** 1 : 1

c. 3 : 2 **d.** 4 : 9

8. Three liquids of equal volumes are thoroughly mixed. If their specific heats are s_1, s_2, s_3 and their temperature $\theta_1, \theta_2, \theta_3$ and their densities d_1, d_2, d_3 respectively then the final temperature of the mixture is:

a. $\frac{s_1\theta_1+s_2\theta_2+s_3\theta_3}{d_1s_1+d_2s_2+d_3s_3}$ **b.** $\frac{d_1s_1\theta_1+d_2s_2\theta_2+d_3s_3\theta_3}{d_1s_1+d_2s_2+d_3s_3}$

c. $\frac{d_1s_1\theta_1+d_2s_2\theta_2+d_3s_3\theta_3}{d_1\theta_1+d_2\theta_2+d_3\theta_3}$ **d.** $\frac{d_1\theta_1+d_2\theta_2+d_3\theta_3}{s_1\theta_1+s_2\theta_2+s_3\theta_3}$

9. Two monatomic ideal gases 1 and 2 of molecular masses m_1 and m_2 respectively are enclosed in separate containers kept at the same temperature. The ratio of the speed of sound in gas 1 to the gas 2 is given by:

a. $\sqrt{\frac{m_1}{m_2}}$ **b.** $\sqrt{\frac{m_2}{m_1}}$

c. $\frac{m_1}{m_2}$ **d.** $\frac{m_2}{m_1}$

10. A gas is compressed adiabatically till its temperature is doubled. The ratio of its find volume to initial volume will be:

a. $\frac{1}{2}$ **b.** more than $\frac{1}{2}$

c. less than $\frac{1}{2}$ **d.** between 1and 2

11. A black body is at a temperature of 2880 K. The energy of radiation emitted by this body with wavelength between 499 nm and 500 nm is U_1, between 999 nm and 1000 nm is U_2 and between 1499 nm and 1500 nm is U_3. The Wien constant, $b = 2.88 \times 10^6$ nm-K. Then,

a. $U_1 = 0$ **b.** $U_3 = 0$

c. $U_1 > U_2$ **d.** $U_2 > U_1$

12. In moving from *A* to *B* along an electric field line, the electric field does 6.4×10^{-19}J of work on an electron. If ϕ_1, ϕ_2 are equipotential surfaces, then the potential difference $(V_C - V_A)$ is

a. – 4V **b.** 4V

c. Zero **d.** 64 V

13. Force acting on a charged particle kept between the plates of a charged condenser is *F*. If one of the plates of the condenser is removed, the force acting on the same particle becomes:

a. 0 **b.** $\frac{F}{2}$

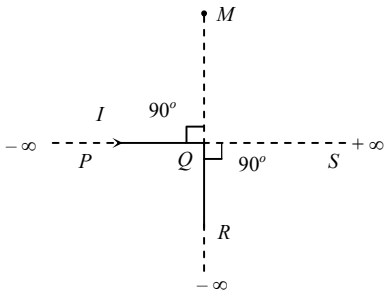
c. F **d.** $2F$

14. An electric cable of copper has just one wire of radius 9 mm. Its resistance is 50 Ω. This single copper wire of cable is replaced by 6 different well insulated copper wires, each of radius 3 mm. The total resistance of the cable will now be equal to:

a. 7.5 Ω **b.** 45 Ω

c. 90 Ω **d.** 270 Ω

15. An infinitely long conductor *PQR* is bent to form a right angle as shown. A current *I* flows through *PQR*. The magnetic field due to this current at the point *M* is H_1. Now another infinitely long straight conductor *QS* is connected at *Q* so that the current is *I*/2 in *QR* as well as in *QS,* The current in *PQ* remaining unchanged. The magnetic field at *M* is now H_2. .The ratio H_1 / H_2 is given by



a. $\frac{1}{2}$ **b.** 1

c. $\frac{2}{3}$ **d.** 2

16. Double slit interference experiment is carried out with monochromatic light and interference fringes are observed. If now monochromatic light is replaced by white light, what change is expected in interference pattern?

a. no change

b. pattern disappears

c. white and dark fringes are observed throughout the pattern

d. a few coloured fringes are observed on either side of central white fringe

17. The electric field of an electromagnetic wave in free space is given by $E = 10\cos(10^7 t + kx)\hat{j}$ V/m, Where, *t* and *x* are in seconds and metres respectively. It can be inferred that

(a) the wavelength λ is 188.4 m

(b) the wave number *k* is 0.33 rad/m

(c) the wave amplitude is 10 V/m

(d) the wave is propagating along + *x* direction

Which one of the following pairs of statements is correct?

a. (c) and (d) **b.** (a) and (b)

c. (b) and (c) **d.** (a) and (c)

18. Light of wavelength λ strikes a photosensitive surface and electrons are ejected with kinetic energy *E*. If the kinetic energy is to be increased to 2*E*, the wavelength must be changed to λ' where:

a. $\lambda' = \frac{\lambda}{2}$ **b.** $\lambda' = 2\lambda$

c. $\frac{\lambda}{2} < \lambda' < \lambda$ **d.** $\lambda' = \lambda$

19. The ionisation potential of hydrogen atom is 13.6 V. The energy required to remove an electron in the *n* = 2 state of the hydrogen atom is

a. 27.2 eV **b.** 13.6 eV

c. 6.8 eV **d.** 3.4 eV

20. Hydrogen bomb is based on which of the following phenomenon?

a. Nuclear fission

b. Nuclear fusion

c. Radioactive decay

d. None of these

SECTION 2 (Numeric Value Question)

21. A cube has a side of length 1.2×10^{-2} m. Calculate its volume:

a. $1.7\times10^{-6}\,m^3$ **b.** $1.73\times10^{-6}\,m^3$

c. $1.70\times10^{-6}\,m^3$ **d.** $1.732\times10^{-6}\,m^3$

22. A body of mass 2 kg is released at the top of a smooth inclined plane having inclination $30°$. It takes 3 seconds to reach the bottom. If the angle of inclination is doubled keeping same height (*i.e.,* made $60°$), what will be the time taken?

a. 3 sec **b.** $\sqrt{3}$ sec

c. $3\sqrt{3}$ sec **d.** 1.5 sec

23. There are two bodies of masses 100 kg and 10000 kg separated by a distance 1m. At what distance from the smaller body, the intensity of gravitational field will be zero:

a. $\frac{1}{9}$ m **b.** $\frac{1}{10}$ m

c. $\frac{1}{11}$ m **d.** $\frac{10}{11}$ m

24. Due to a small magnet intensity at a distance x in the end on position is 9 gauss. What will be the intensity at a distance $\frac{x}{2}$ on broad side on position?

a. 9 gauss **b.** 4 gauss

c. 36 gauss **d.** 4.5 gauss

25. When both source and observe approach each other with a velocity equal to the half the velocity of sound the change in frequency of sound as detected by the listener is:

a. 0 **b.** 25 %

c. 50% **d.** 150%

Space for rough work

JEE ADVANCE PAPER-I

Time 3 Hours. **Max. Marks 180 (60 for Physics)**

Read the Instructions Carefully

Question Paper Format and Marking Scheme:

1. The question paper has three parts: Physics, Chemistry and Mathematics. Each part has three sections.
2. **Section 1** contains 6 multiple choice questions with one or more than one correct option.
   Marking Scheme: +4 if only (all) the correct option(s) is (are) chosen, +3 if all the four options are correct but only three options are chosen, +2 if three or more options are correct but only two options are chosen, both of the options must be correct, +1 if two or more options are correct buy only one option is chosen and it must be correct, -2 (in all other cases).
3. **Section 2** contains 8 Numerical value answer type questions. The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).
   Marking Scheme: +3 if only correct numerical value is given.
4. **Section 3** contains 2 Paragraph based questions (2 paragraphs, each having 2 MCQs with one correct answer only)
   Marking Scheme: +3 if only the correct option is selected, -1 in all other cases.

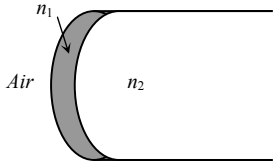
Note:

Possible response	Section 1	Section 2	Section 3
Not attempted	0	0	0
Partial Correct	+1 for each correct option selected	0	0
Correct	+4	+3	3
incorrect	-2	0	-1

SECTION 1 (Maximum Marks: 24)

MCQs with one or more than one correct answer

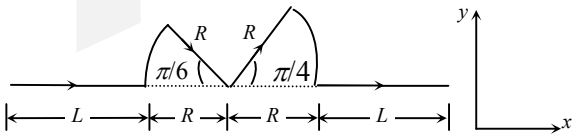
1. A transparent thin film of uniform thickness and refractive index $n_1 = 1.4$ is coated on the convex spherical surface of radius R at one end of a long solid glass cylinder of refractive index $n_2 = 1.5,$ as shown in the figure. Rays of light parallel to the axis of the cylinder traversing through the film from air to glass get focused at distance f_1 from the film, while rays of light traversing from glass to air get focused at distance f_2 from the film. Then



a. $|f_1| = 3R$ **b.** $|f_1| = 2.8R$

c. $|f_2| = 2R$ **d.** $|f_2| = 1.4R$

2. A conductor (shown in the figure) carrying constant current I is kept in the x-y plane in a uniform magnetic field $\vec{B}$. If F is the magnitude of the total magnetic force acting on the conductor, then the correct statement(s) is(are)



a. If $\vec{B}$ is along $\hat{z}, F \propto (L + R)$

b. If $\vec{B}$ is along $\hat{x}, F = 0$

c. If $\vec{B}$ is along $\hat{y}, F \propto (L + R)$

d. If $\vec{B}$ is along $\hat{z}, F = 0$

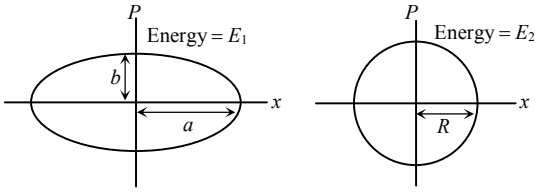
3. Planck's constant h, speed of light c and gravitational constant G are used to form a unit of length L and a unit of mass M. Then the correct option(s) is(are)

a. $M \propto \sqrt{c}$ **b.** $M \propto \sqrt{G}$

c. $L \propto \sqrt{h}$ **d.** $L \propto \sqrt{G}$

4. Two independent harmonic oscillators of equal mass are oscillating about the origin with angular frequencies ω_1 and ω_2 and have total energies E_1 and $E_2,$ respectively. The variations of their momenta p with positions x are

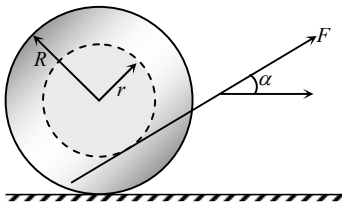
shown in the figures. If $\frac{a}{b}=n^2$ and $\frac{a}{R}=n$, then the correct equation(s) is(are)



a. $E_1\omega_1 = E_2\omega_2$ **b.** $\frac{\omega_2}{\omega_1}=n^2$

c. $\omega_1\omega_2 = n^2$ **d.** $\frac{E_1}{\omega_1}=\frac{E_2}{\omega_2}$

5. Inner and outer radii of a spool are r and R respectively. A thread is wound over its inner surface and spool is placed over a rough horizontal surface. Thread is pulled by a force F as shown in figure. In case of pure rolling, which of the following statements are false?



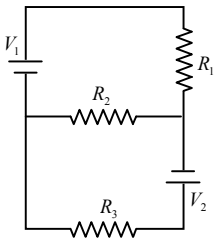
a. Thread unwinds, spool rotates anticlockwise and friction acts leftwards

b. Thread winds, spool rotates clockwise and friction acts leftwards

c. Thread winds, spool moves to the right and friction acts rightwards

d. Thread winds, spool moves to the right and friction does not come into existence

6. Two ideal batteries of *emf* V_1 and V_2 and three resistances R_1, R_2 and R_3 are connected as shown in the figure. The current in resistance R_2 would be zero if



a. $V_1 = V_2$ and $R_1 = R_2 = R_3$

b. $V_1 = V_2$ and $R_1 = 2R_2 = R_3$

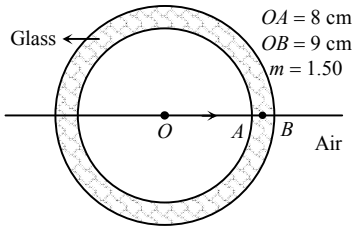
c. $V_1 = 2V_2$ and $2R_1 = 2R_2 = R_3$

d. $2V_1 = V_2$ and $2R_1 = R_2 = R_3$

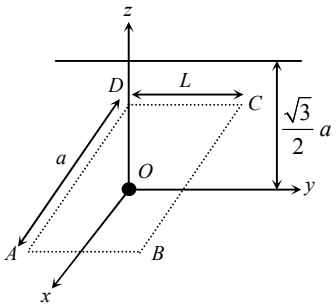
SECTION 2 (Maximum Marks: 24)

Numerical value answer type questions

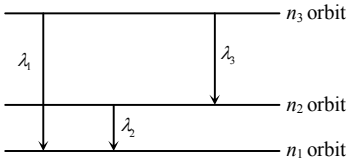
7. A small filament is at the centre of a hollow glass sphere of inner and outer radii 8 cm and 9 cm respectively. The refractive index of glass is 1.50. Calculate the position of the image of the filament when viewed from outside the sphere.



8. An infinitely long uniform line charge distribution of charge per unit length λ lies parallel to the y-axis in the y-z plane at $z=\frac{\sqrt{3}}{2}a$ (see figure). If the magnitude of the flux of the electric field through the rectangular surface *ABCD* lying in the *x*-*y* plane with its center at the origin is $\frac{\lambda L}{n\varepsilon_0}$ (ε_0 = permittivity of free space), then the value of n is

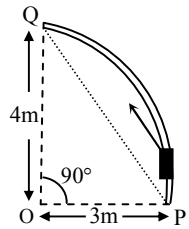


9. For an atom of an ion having single electron, the wavelength observed $\lambda_1 = 2$ are units and $\lambda_3 = 3$ units figure. The value of missing wavelength λ_2 is

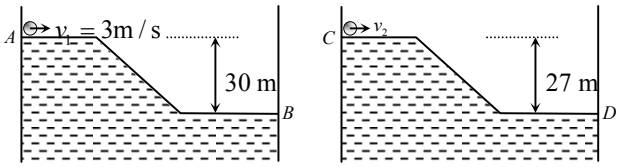


10. Consider an elliptically shaped rail PQ in the vertical plane with $OP = 3$ m and $OQ = 4$ m. A block of mass 1 kg is pulled along the rail from P to Q with a force of 18 N, which is always parallel to line PQ (see the given figure). Assuming no frictional losses, the kinetic energy

of the block when it's reaches Q in $(n \times 10)$ joules. The value of n is (take acceleration due to gravity $10\ ms^{-2}$)



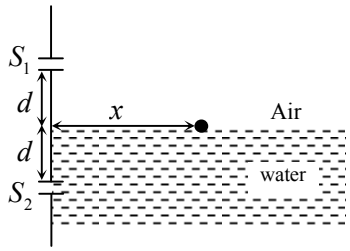
11. Two identical uniform discs roll without slipping on two different surfaces AB and CD (see figure) starting at A and C with linear speeds v_1 and v_2, respectively, and always remain in contact with the surfaces. If they reach B and D with the same linear speed and $v_1 = 3\,m/s$, then v_2 in m/s is $(g = 10\ m/s^2)$



12. Two spherical stars A and B emit blackbody radiation. The radius of A is 400 times that of B and A emits 10^4 times the power emitted from B. The ratio (λ_A / λ_B) of their wavelengths λ_A and λ_B at which the peaks occur in their respective radiation curves is

13. The half life of a freshly prepared radioactive sample is 2 h. If the sample emits radiation of intensity, which is 16 times the permissible safe level, then the minimum time taken after which it would be possible to work safely with source is

14. A Young's double slit interference arrangement with slits S_1 and S_2 is immersed in water (refractive index = 4/3) as shown in the figure. The positions of maxima on the surface of water are given by $x^2 = p^2 m^2 \lambda^2 - d^2$, where λ is the wavelength of light in air (refractive index = 1), $2d$ is the separation between the slits and m is an integer. The value of p is

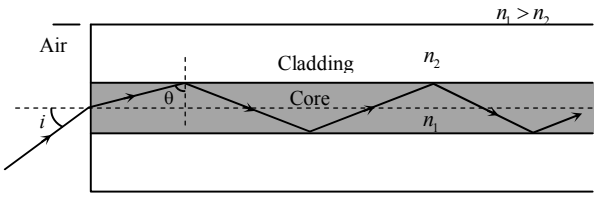


SECTION 3 (Maximum Marks: 12)

2 Paragraph based questions (2 paragraphs, each having 2 MCQs with one correct answer only)

Paragraph for Question No. 15 to 16

Light guidance in an optical fiber can be understood by considering a structure comprising of thin solid glass cylinder of refractive index n_1 surrounded by a medium of lower refractive index n_2. The light guidance in the structure takes place due to successive total internal reflections at the interface of the media n_1 and n_2 as shown in the figure. All rays with the angle of incidence i less than a particular value i_m are confined in the medium of refractive index n_1. The numerical aperture (NA) of the structure is defined as sin i_m.



15. For two structures namely S_1 with $n_1 = \sqrt{45}/4$ and $n_2 = 3/2$, and S_2 with $n_1 = 8/5$ and $n_2 = 7/5$ and taking the refractive index of water to be 4/3 and that of air to be 1, the correct option(s) is(are)

a. NA of S_1 immersed in water is the same as that of S_2 immersed in a liquid of refractive index $\frac{16}{3\sqrt{15}}$

b. NA of S_1 immersed in liquid of refractive index $\frac{6}{\sqrt{15}}$ is the same as that of S_2 immersed in water

c. NA of S_1 placed in air is the same as that of S_2 immersed in liquid of refractive index $\frac{4}{\sqrt{15}}$.

d. NA of S_1 placed in air is the same as that of S_2 placed in water

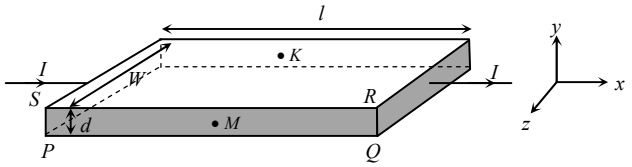
16. If two structures of same cross-sectional area, but different numerical apertures NA_1 and $NA_2\,(NA_2 < NA_1)$ are joined longitudinally, the numerical aperture of the combined structure is

a. $\frac{NA_1NA_2}{NA_1+NA_2}$ **b.** NA_1+NA_2

c. NA_1 **d.** NA_2

Paragraph for Question No. 17 to 18

In a thin rectangular metallic strip a constant current I flows along the positive x-direction, as shown in the figure. The length, width and thickness of the strip are ℓ, w and d, respectively. A uniform magnetic field $\vec{B}$ is applied on the strip along the positive y-direction. Due to this, the charge carriers experience a net deflection along the z-direction. This results in accumulation of charge carriers on the surface $PQRS$ and appearance of equal and opposite charges on the face opposite to $PQRS$. A potential difference along the z-direction is thus developed. Charge accumulation continues until the magnetic force is balanced by the electric force. The current is assumed to be uniformly distributed on the cross-section of the strip and carried by electrons.



17. Consider two different metallic strips (1 and 2) of the same material. Their lengths are the same, widths are w_1 and w_2 and thicknesses are d_1 and d_2, respectively. Two points K and M are symmetrically located on the opposite faces parallel to the x-y plane (see figure). V_1 and V_2 are the potential differences between K and M in strips 1 and 2, respectively. Then, for a given current I flowing through them in a given magnetic field strength B, the correct statement(s) is(are)

a. If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = 2V_1$

b. If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = V_1$

c. If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = 2V_1$

d. If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = V_1$

18. Consider two different metallic strips (1 and 2) of same dimensions (lengths ℓ, width w and thickness d) with carrier densities n_1 and n_2, respectively. Strip 1 is placed in magnetic field B_1 and strip 2 is placed in magnetic field B_2, both along positive y-directions. Then V_1 and V_2 are the potential differences developed between K and M in strips 1 and 2, respectively. Assuming that the current I is the same for both the strips, the correct option(s) is(are)

a. If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = 2V_1$

b. If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = V_1$

c. If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = 0.5V_1$

d. If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = V_1$

Space for rough work

JEE ADVANCE PAPER-II

Time 3 Hours. **Max. Marks 180 (60 for Physics)**

Read the Instructions Carefully

Question Paper Format and Marking Scheme:

1. The question paper has three parts: Physics, Chemistry and Mathematics. Each part has three sections.
2. **Section 1** contains 6 MCQs with one or more than one correct answer
   Marking Scheme: +4 if only (all) the correct option(s) is (are) chosen, +3 if all the four options are correct but only three options are chosen, +2 if three or more options are correct but only two options are chosen, both of the options must be correct, +1 if two or more options are correct buy only one option is chosen and it must be correct, -2 (in all other cases)
3. **Section 2** contains 8 Numerical value answer type questions.
   Marking Scheme: +3 if only correct numerical value is given
4. **Section 3** contains 4 Matching type questions with 4 options.
   Marking Scheme: +3 if only the correct option is selected, -1 in all other cases

Note:

Possible response	Section 1	Section 2	Section 3
Not attempted	0	0	0
Partial Correct	+1 for each correct option selected	0	0
Correct	+3	+4	+3
Incorrect	-1	-2	0

SECTION 1 (Maximum Marks: 24)

MCQs with one or more than one correct answer

1. A fission reaction is given by ${}^{236}_{92}\mathrm{U} \longrightarrow {}^{140}_{54}\mathrm{Xe} + {}^{94}_{38}\mathrm{Sr} + x + y$, where x and y are two particles. Considering ${}^{236}_{92}\mathrm{U}$ to at rest, the kinetic energies of the products are denoted by K_{Xe}, K_{Sr}, $K_{\mathrm{x}}(2\mathrm{MeV})$ and $K_{\mathrm{y}}(2\mathrm{MeV})$, respectively. Let the binding energies per nucleon of ${}^{236}_{92}\mathrm{U}$, ${}^{140}_{54}\mathrm{Xe}$ and ${}^{94}_{38}\mathrm{Sr}$ be 7.5 MeV, 8.5 MeV and 8.5 MeV respectively. Considering different conservation laws, the correct option(s) is(are)

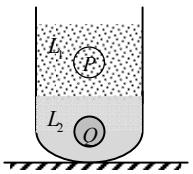
a. $x = n,\ y = n,\ K_{\mathrm{Sr}} = 129\ \mathrm{MeV},\ K_{\mathrm{Xe}} = 86\ \mathrm{MeV}$

b. $x = p,\ y = e^{-},\ K_{\mathrm{Sr}} = 129\ \mathrm{MeV},\ K_{\mathrm{Xe}} = 86\ \mathrm{MeV}$

c. $x = p,\ y = n,\ K_{\mathrm{Sr}} = 129\ \mathrm{MeV},\ K_{\mathrm{Xe}} = 86\ \mathrm{MeV}$

d. $x = n,\ y = n,\ K_{\mathrm{Sr}} = 86\ \mathrm{MeV},\ K_{\mathrm{Xe}} = 129\ \mathrm{MeV}$

2. Two spheres P and Q of equal radii have densities ρ_1 and ρ_2, respectively. The spheres are connected by a mass less string and placed in liquids L_1 and L_2 of densities σ_1 and σ_2 and viscosities η_1 and η_2, respectively. They float in equilibrium with the sphere P in L_1 and sphere Q in L_2 and the string being taut (see figure). If sphere P alone in L_2 has terminal velocity $\vec{V}_P$ and Q alone in L_1 has terminal velocity $\vec{V}_Q$, then



a. $\dfrac{|\vec{V}_P|}{|\vec{V}_Q|} = \dfrac{\eta_1}{\eta_2}$ **b.** $\dfrac{|\vec{V}_P|}{|\vec{V}_Q|} = \dfrac{\eta_2}{\eta_1}$

c. $\vec{V}_P \cdot \vec{V}_Q > 0$ **d.** $\vec{V}_P \cdot \vec{V}_Q < 0$

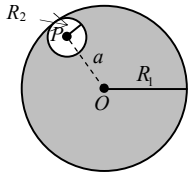
3. In terms of potential difference V, electric current I, permittivity ε_0, permeability μ_0 and speed of light c, the dimensionally correct equation(s) is(are)

a. $\mu_0 I^2 = \varepsilon_0 V^2$ **b.** $\varepsilon_0 I = \mu_0 V$

c. $I = \varepsilon_0 c V$ **d.** $\mu_0 c I = \varepsilon_0 V$

4. Consider a uniform spherical charge distribution of radius R_1 centred at the origin O. In this distribution, a spherical

cavity of radius R_2, centred at P with distance $OP = a = R_1 - R_2$ (see figure) is made. If the electric field inside the cavity at position $\vec{r}$ is $\vec{E}(\vec{r})$, then the correct statement(s) is(are)



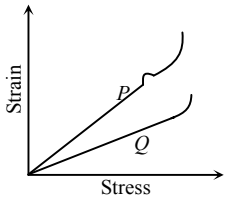
a. $\vec{E}$ is uniform, its magnitude is independent of R_2 but its direction depends on $\vec{r}$

b. $\vec{E}$ is uniform, its magnitude depends on R_2 and its direction depends on $\vec{r}$

c. $\vec{E}$ is uniform, its magnitude is independent of a but its direction depends on $\vec{a}$

d. $\vec{E}$ is uniform and both its magnitude and direction depend on $\vec{a}$

5. In plotting stress versus strain curves for two materials P and Q, a student by mistake puts strain on the y-axis and stress on the x-axis as shown in the figure. Then the correct statement(s) is(are)



a. P has more tensile strength than Q

b. P is more ductile than Q

c. P is more brittle than Q

d. The Young's modulus of P is more than that of Q

6. A spherical body of radius R consists of a fluid of constant density and is in equilibrium under its own gravity. If $P(r)$ is the pressure at $r(r < R)$, then the correct option(s) is(are)

a. $P(r = 0) = 0$

b. $\dfrac{P(r = 3R/4)}{P(r = 2R/3)} = \dfrac{63}{80}$

c. $\dfrac{P(r = 3R/5)}{P(r = 2R/5)} = \dfrac{16}{21}$

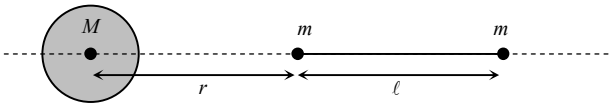
d. $\dfrac{P(r = R/2)}{P(r = R/3)} = \dfrac{20}{27}$

SECTION 2 (Maximum Marks: 24)

Numerical value answer type questions

7. An electron in an excited state of Li^{2+} ion has angular momentum $3h/2\pi$. The de Broglie wavelength of the electron in this state is $p\pi a_0$ (where a_0 is the Bohr radius). The value of p is

8. A large spherical mass M is fixed at one position and two identical point masses m are kept on a line passing through the centre of M (see figure). The point masses are connected by a rigid mass less rod of length ℓ and this assembly is free to move along the line connecting them. All three masses interact only through their mutual gravitational interaction. When the point mass nearer to M is at a distance $r = 3\ell$ from M, the tension in the rod is zero for $m = k\left(\dfrac{M}{288}\right)$. The value of k is



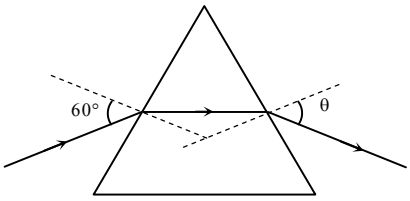
9. The energy of a system as a function of time t is given as $E(t) = A^2 \exp(-\alpha t)$, where $\alpha = 0.2\ s^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%, the percentage error in the value of $E(t)$ at $t = 5\ s$ is

10. The densities of two solid spheres A and B of the same radii R vary with radial distance r as $\rho_A(r) = k\left(\dfrac{r}{R}\right)$ and $\rho_B(r) = k\left(\dfrac{r}{R}\right)^5$, respectively, where k is a constant. The moments of inertia of the individual spheres about axes passing through their centres are I_A and I_B, respectively. If $\dfrac{I_B}{I_A} = \dfrac{n}{10}$, the value of n is

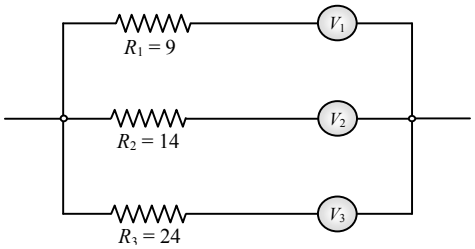
11. Four harmonic waves of equal frequencies and equal intensities I_0 have phase angles 0, $\dfrac{\pi}{3}, \dfrac{2\pi}{3}$ and π. When they are superposed, the intensity of the resulting wave is nI_0. The value of n is

12. For a radioactive material, its activity A and rate of change of its activity R are defined as $A = -\dfrac{dN}{dt}$ and $R = -\dfrac{dA}{dt}$, where $N(t)$ is the number of nuclei at time t. Two radioactive sources P (mean life τ) and Q (mean life 2τ) have the same activity at $t = 0$. Their rates of change of activities at $t = 2\tau$ are R_P and R_Q, respectively. If $\dfrac{R_P}{R_Q} = \dfrac{n}{e}$, then the value of n is

13. A monochromatic beam of light is incident at $60°$ on one face of an equilateral prism of refractive index n and emerges from the opposite face making an angle $\theta(n)$ with the normal (see the figure). For $n = \sqrt{3}$ the value of θ is $60°$ and $\frac{d\theta}{dn} = m$. The value of m is



14. The figure shows a portion of an electric circuit. Resistors are known and are indicated on the diagram and the voltmeters are identical. If the voltmeters V_1 and V_2 read 7.5 V and 5.0 V respectively, find the reading of the voltmeter V_3.



SECTION 3 (Maximum Marks: 12)

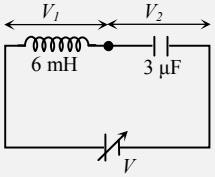
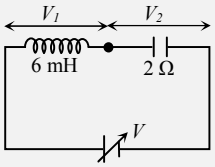
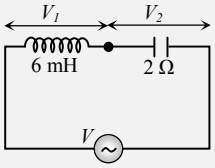
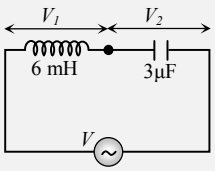
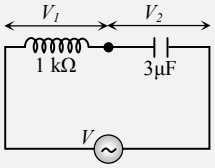
Matching type questions with 4 options

15. Match the statement of Column with those in Column II:

Column I	Column II
(A) In any Bohr orbit of the hydrogen atom, the ratio of kinetic energy to potential energy of the electron is	**1.** $-\frac{1}{2}$
(B) The ratio of the kinetic energy to the total energy of an electron in a Bohr orbit is	**2.** 2
(C) In the lowest energy level of hydrogen atom, the electron has the angular momentum	**3.** $\frac{h}{2\pi}$
(D) Ratio of the wavelengths of first line of Lyman series and first line of Balmer series is	**4.** 5 : 27
	5. 0

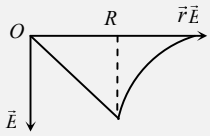
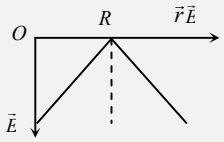
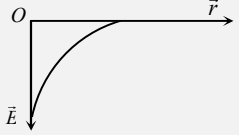
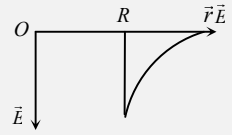
a. A→ 1; B→ 2, 4; C→3; D→ 4
b. A→3; B→1, 2; C→1; D→ 2
c. A→4; B→3, 4; C→2; D→ 1
d. A→2; B→2, 3; C→4; D→ 3

16. You are given many resistances, capacitors and inductors. These are connected to a variable DC voltage source (the first two circuits) or an AC voltage source of 50 Hz frequency (the next three circuits) in different ways as shown in Column II. When a current I (steady state for DC or r.m.s. for AC) flows through the circuit, the corresponding voltage V_1 and V_2. (indicated in circuits) are related as shown in Column I.

Column I	Column II
(A) $I \neq 0, V_1$ is proportional to I	**1.** 
(B) $I \neq 0, V_2 > V_1$	**2.** 
(C) $V_1 = 0, V_2 = V$	**3.** 
(D) $I \neq 0, V_2$ is proportional to I	**4.** 
	5. 

a. A→3, 4; B→2, 3, 4; C→1, 2; D→2, 3, 4
b. A→1, 2; B→1, 2, 3; C→3, 4; D→1, 3 ,4
c. A→2, 4; B→2, 3, 4; C→2, 3; D→1, 2, 3
d. A→1, 3; B→1, 3, 4; C→1, 3; D→2, 3, 4

17. The gravitational field intensity $\vec{E}$ of earth at any point is defined as the gravitational force per unit mass at that point. If varies from place to place. The variation is shown in column II with form of position $\vec{r}$ *v s* the $\vec{E}$ in the form of graphs. The variation of $\vec{r}$ is given in column I. Choose the correct form of graphs for the corresponding variations of $\vec{r}$.

Column I	Column II
(A) Position $\vec{r}$ of body measured from surface earth upward	**1.** 
(B) Position $\vec{r}$ of body measured from surface of earth along diameter to opposite point on surface of earth	**2.** 
(C) Position $\vec{r}$ measured from center of earth to any point	**3.** 
(D) Position $\vec{r}$ measured from center of hollow sphere to any point	**4.** 

a. A→1; B→2; C→3; D→4
b. A→4; B→3; C→2; D→1
c. A→4; B→1; C→3; D→2
d. A→3; B→2; C→1; D→4

18. A person in a lift is holding a water jar, which has a small hole at the lower end of its side. When the lift is at rest, the water jet coming out of the hole hits the floor of the lift at a distance *d* of 1.2 m from the person. In the following, state of the lift's motion is given in Column I and the distance where the water jet hits the floor of the lift is given in Column II. Match the statements from Column I with those in Column II and select the correct using the code given below the lists.

Column I	Column II
(A) Lift is accelerating vertically up.	**1.** $d = 1.2$ m
(B) Lift is accelerating vertically down with an acceleration less than the gravitational acceleration.	**2.** $d > 1.2$ m
(C) Lift is moving vertically up with constant speed.	**3.** $d < 1.2$ m
(D) Lift is falling freely.	**4.** No water leaks out of the jar

a. A→1; B→1; C→1; D→4
b. A→4; B→3; C→2; D→1
c. A→4; B→1; C→3; D→2
d. A→3; B→2; C→1; D→4

Space for rough work

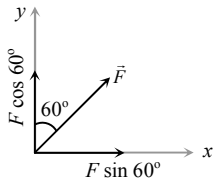
ANSWER & SOLUTIONS

JEE-Main

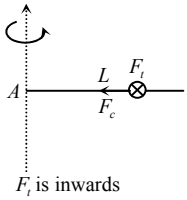
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
c	a	c	b	c	b	c	b	b	b
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
d	b	b	a	c	d	d	c	d	b
21.	**22.**	**23.**	**24.**	**25.**					
a	b	c	c	c					

1. **(c)** Displacement vector $\vec{r} = \Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k}$

$= (3-2)\hat{i} + (4-3)\hat{j} + (5-5)\hat{k} = \hat{i} + \hat{j}$



2. **(a)** Tangential force (F_t) of the bead will be given by the normal reaction (N), while centripetal force (F_c) is provided by friction (F_r). The bead starts sliding when the centripetal force is just equal to the limiting friction. Therefore, $F_t = ma = m\alpha\, L = N$



$\therefore$ Limiting value of friction $(f_r)_{\max} = \mu N = \mu m\alpha L \ldots (i)$

Angular velocity at time t is $\omega = \alpha t$

$\therefore$ Centripetal force at time t will be

$F_c = mL\ \omega^2 = mL\ \alpha^2\ t^2 \quad \ldots (ii)$

Equation (i) and (ii), we get $t = \sqrt{\frac{\mu}{\alpha}}$

For $t > \sqrt{\frac{\mu}{\alpha}}, F_c > (f_r)_{\max}$ *i.e.*, the bead starts sliding.

In the figure F_t is perpendicular to the paper inwards.

3. **(c)** Power $= \vec{F} \cdot \vec{v} = Fv$

$$F = v\left(\frac{dm}{dt}\right) = v\left\{\frac{d\ (\rho \times \text{volume})}{dt}\right\}$$

$$= \rho v\left\{\frac{d(\text{volume})}{dt}\right\} = \rho v(Av) = \rho A v^2$$

$\therefore$ Power $P = \rho A v^3$ or $P \propto v^3$

4. **(b)** Let ρ be the density of lead. Then $M = \frac{4}{3}\pi R^3 \rho$ = mass of total sphere

$$m_1 = \frac{4}{3}\pi\left(\frac{R}{2}\right)^3 \rho = \text{ mass of removed part } = \left(\frac{M}{8}\right)$$

$$m_2 = M - \frac{M}{8} = \frac{7M}{8} = \text{mass of remaining sphere}$$

Choosing the centre of big sphere as the origin,

$$X_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$0 = \frac{(M/8)\times(R/2) + (7M/8)\times x_2}{M}$$

Solving, we get $x_2 = \frac{-R}{14}$

5. **(c)** In air $g_{eff} = g$

In water, $g_{eff} = g - \frac{\text{Buoyant force}}{\text{Mass}} = g - \frac{d_w}{d_b}g = \frac{g}{4}$

Now $t = 2\pi\sqrt{\frac{t}{g_{eff}}}$, Hence $2t_0 = t$

6. **(b)** Soap solution has lower surface tension as compared to pure water so h is less for soap solution.

7. **(c)** $F_1 = F_2$

$\Rightarrow \quad Y_1 A\,\alpha_1 \Delta\theta = Y_2 A\,\alpha_2 \Delta\theta \Rightarrow \frac{Y_1}{Y_2} = \frac{r_2}{r_1} = \frac{3}{2}$

8. **(b)** Let 0°C be the reference temperature for zero heat, then initial heat energy = final heat energy

$m_1 s_1\theta_1 + m_2 s_2\theta_2 + m_3 s_3\theta_3 = (m_1 s_1 + m_2 s_2 + m_3 s_3)$

$\Rightarrow \quad \theta = \frac{m_1 s_1\theta_1 + m_2 s_2\theta_2 + m_3 s_3\theta_3}{m_1 s_1 + m_2 s_2 + m_3 s_3}$

$$= \frac{Vd_1 s_1\theta_1 + Vd_2 s_2\theta_2 + Vd_3 s_3\theta_3}{Vd_1 s_1 + Vd_2 s_2 + Vd_3 s_3} = \frac{d_1 s_1\theta_1 + d_2 s_2\theta_2 + d_3 s_3\theta_3}{d_1 s_1 + d_2 s_2 + d_3 s_3}$$

9. **(b)** Speed of sound in a gas is given by $v = \sqrt{\frac{\gamma RT}{M}}, v \propto \frac{1}{\sqrt{M}}$

$\therefore \quad \frac{v_1}{v_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{m_2}{m_1}}$

Here, $\gamma = \frac{C_p}{C_V} = \frac{5}{3}$ for both the gases $\left(\gamma_{\text{monoatomic}} = \frac{5}{3}\right)$

10. **(b)** $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \Rightarrow \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \frac{T_1}{T_2}$

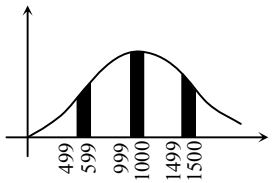
or $\quad \frac{V_1}{V_2} = \left(\frac{T_1}{T_2}\right)^{\gamma-1} = \left(\frac{T_1}{2T_2}\right)^{1-\gamma} = \frac{1}{(2)^{1-\gamma}} > \frac{1}{2}$

11. **(d)** Wien's displacement law is $\lambda_m T = b$ (b = Wien's constant)

$\therefore \quad \lambda_m = \frac{b}{T} = \frac{2.88\times10^6\ \text{nm-K}}{2880\text{K}}$

$\therefore$ $\lambda = 1000$ nm

Energy distribution with wavelength will be as follows:



From the fig. it is clear that $\frac{U_1}{U_2}$ (In fact U_2 is maximum)

12. **(b)** Work done by the field $W = q(-dV) = -e(V_A - V_B)$

$= e(V_B - V_A) = e(V_C - V_A)$ $(\because V_B = V_C)$

$\Rightarrow$ $(V_C - V_A) = \frac{W}{e} = \frac{6.4\times10^{-19}}{1.6\times10^{-19}} = 4\text{V}$

13. **(b)** In the electric field between plates of parallel plate capacitor $E = \frac{\sigma}{\varepsilon_0}$

$\therefore$ $F = qE = \frac{q\sigma}{\varepsilon_0}$ . . . (i)

When one plate is removed, the electric field becomes

$E' = \frac{\sigma}{2\varepsilon_0}$

$\therefore$ $F' = qE' = \frac{\sigma}{2\varepsilon_0}$ . . . (ii)

$\frac{F'}{F} = \frac{1}{2}$ $\Rightarrow$ $F' = \frac{F}{2}$

14. **(a)** Resistance of 9 mm cable $= 5\Omega$

As $R \propto \frac{1}{A}$ or $R \propto \frac{1}{r^2}$; resistance of 3 mm

Cable $9\times5 = 45\,\Omega$

In second case 6 wires are connected in parallel, so total resistance of cables $= \frac{45}{6} = 7\cdot5\Omega$

15. **(c)** Magnetic field at any point lying on the current carrying straight conductor is zero. Here

$H_1 =$ Magnetic field at M due to current in PQ.

$H_2 =$ Magnetic field at M due to QR + magnetic field at M due to QS + magnetic field at M due to PQ

$= 0 + \frac{H_1}{2} + H_1 = \frac{3}{2}H_1 \Rightarrow \frac{H_1}{H_2} = \frac{2}{3}$

16. **(d)** If monochromatic light is replaced by white light, a few coloured fringes are seen on either side of a central white fringe.

17. **(d)** Electric field of an electromagnetic wave in free space is given by $\vec{E} = 10\cos(10^7 t + kx)\hat{j}$ which is acting along y-direction. As E is varying with x and t, hence propagation of electromagnetic wave takes place along $-x$ axis. Thus statement (d) is wrong. Comparing the relation, $E = 10\cos(10^7 t + kx)$ with standard equation of electromagnetic wave

$E = E_0\cos\frac{2\pi}{\lambda}(\upsilon t + x) = E_0\cos\left(\frac{2\pi\upsilon}{\lambda}t + \frac{2\pi}{\lambda}x\right)$

We have, $E_0 = 10V/m.$ Thus statement (c) is correct.

$\frac{2\pi\upsilon}{\lambda} = 10^7$ or $\frac{2\pi\times(3\times10^8)}{\lambda} = 10^7$

or $\lambda = 60\pi = 60\times\frac{22}{7} \approx 188.4\,\text{m}$

Thus, statement (a) is correct.

$\Rightarrow$ $k = \frac{2\pi}{\lambda} = \frac{2\pi}{60\pi} = \frac{1}{30} = 0.033$ rad / m

Thus, statement (b) is wrong.

18. **(c)** $E = \frac{hc}{\lambda} = -W$ . . . (i)

$2E = \frac{hc}{\lambda'} = -W$ . . . (ii)

Dividing (ii) by (i) $2 = \frac{\frac{hc}{\lambda'} - W}{\frac{hc}{\lambda} - W} = \frac{2hc}{\lambda} - 2W = \frac{hc}{\lambda'} - W$

$\Rightarrow$ $\frac{hc}{\lambda'} = \frac{2hc}{\lambda} - W$ $\Rightarrow$ $\frac{1}{\lambda'} = \frac{2}{\lambda} - \frac{W}{hc}$

$\Rightarrow$ $\lambda' = \frac{1}{\frac{2}{\lambda} - \frac{W}{hc}}$ $\Rightarrow$ $\lambda' > \frac{\lambda}{2}$ but less than λ

19. **(d)** Energy required to remove electron in the n = 2 state $= +\frac{13.6}{(2)^2} = +3.4$ eV

20. **(b)** Hydrogen bomb is based on nuclear fusion.

21. **(a)** $V = l^3 = (1.2\times10^{-2}\,\text{m})^3 = 1.728\times10^{-6}\,\text{m}^3$

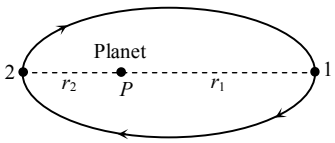
$\because$ Length (l) has two significant figures, the volume (V) will also have two significant figures. Therefore, the correct answer is $V = lV = 1.7\times10^{-6}\,\text{m}^3$

22. **(b)** $a = g\sin\theta, l = \frac{h}{\sin\theta}$

$\Rightarrow$ $\frac{h}{\sin\theta} = \frac{1}{2}g\sin\theta t^2 \Rightarrow t \propto \frac{1}{\sin\theta}$

$\Rightarrow$ $\frac{t_2}{t_1} = \frac{\sin\theta_1}{\sin\theta_2} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$ $/\, t_2 = \frac{t_1}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$ s

23. **(c)** $\frac{GM_1}{x^2}=\frac{GM_2}{(1-x)^2}$



$\Rightarrow \quad \frac{100}{x^2}=\frac{10,000}{(1-x)^2}$

or $\quad \frac{1}{100}=\frac{x^2}{(1-x)^2}\Rightarrow\frac{1}{10}=\frac{x^2}{1-x} \Rightarrow x=\frac{1}{11}$ m

24. **(c)** In C.G.S. $B_{axial}=9=\frac{2M}{x^3}$ ... (*i*)

$$B_{\text{equaterial}}=\frac{M}{\left(\frac{x}{2}\right)^3}=\frac{8M}{x^3} \quad \ldots (ii)$$

From equation (*i*) and (*ii*) $B_{\text{equaterial}}=36$ gauss.

25. **(c)** $n'=n+\frac{v_r}{r}n=n+\frac{v/2}{v}n=\frac{3}{2}n$

% change in frequency

$$=\frac{n'-n}{n}\times100\%=\left(\frac{n'}{n}-1\right)\times100\%$$

$$=\left(\frac{3}{2}-1\right)\times100\%=50\%$$

JEE Advance Paper -I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
c	a,b,c	a,c,d	b,d	a,c,d	a,b,d	9	6	6	5
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**		
7	2	8	5	a,c	d	a,d	a,c		

1. **(c)** For air to glass $\frac{1.5}{f_1}=\frac{1.4-1}{R}+\frac{1.5-1.4}{R} \quad \therefore f_1=3R$

For glass to air. $\frac{1}{f_2}=\frac{1.4-1.5}{-R}+\frac{1-1.4}{-R} \quad \therefore f_2=2R$

2. **(a, b, c)** $\vec{F}=2I(L+R)[\hat{i}\times\vec{B}]$ $\xrightarrow{\qquad\qquad}$ $2(L+R)$

3. **(a, c, d)** $h\equiv[ML^2T^{-1}], c\equiv[LT^{-1}], G\equiv[M^{-1}L^3T^{-2}]$

$\Rightarrow \quad M\propto\sqrt{\frac{hc}{G}}, L\propto\sqrt{\frac{hG}{c^3}}$

4. **(b, d)** For first oscillator $E_1=\frac{1}{2}m\omega_1^2a^2$

and $p=mv=m\omega_1a=b$

$\Rightarrow \quad \frac{a}{b}=\frac{1}{m\omega_1}$ ...(*i*)

For second oscillator

$E_2=\frac{1}{2}m\omega_2^2R^2$, and $m\omega_2=1$ ...(*ii*)

$\Rightarrow \quad \left(\frac{a}{b}\right)=\frac{\omega_2}{\omega_1}=n^2$

$\Rightarrow \quad \frac{E_1}{\omega_1^2a^2}=\frac{E_2}{\omega_2^2a^2}\Rightarrow\frac{E_1}{\omega_1}=\frac{E_2}{\omega_2}$

5. **(a, c, d)** Since, the spool rolls over the horizontal surface, therefore, instantaneous axis of rotation passes through the point of contact of spool with the horizontal surface. About the instantaneous axis of rotation, moment produced by F is clockwise. Therefore, the spool rotates clockwise. In that case acceleration will be rightward and thread will wind. If rotational motion of spool is considered about its own axis then resultant moment on it must be clockwise. But moment produced by the force F is anticlockwise and its magnitude is equal to $F.r$,

Hence, moment produced by the friction (about its own axis) must be clockwise and its magnitude must be greater than $F.r$. It is possible only when friction acts leftwards. Therefore, option (b) is correct.

6. **(a, b, d)** $V_1=\frac{R_1(V_1+V_2)}{R_1+R_3}\Rightarrow V_1R_3=V_2R_1$

$\Rightarrow \quad V_2=\frac{R_3(V_1+V_2)}{R_1+R_3}\Rightarrow V_2R_1=V_2R_3$

7. **(9)** For refraction at the first surface, $u=-8$ cm,

$R_1=-8$ cm, $\mu_1=1$, $\mu_2=1.5$

$\Rightarrow \quad \frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R_1}$

$\Rightarrow \quad \frac{1.5}{v'}+\frac{1}{8}=\frac{0.5}{-8}\Rightarrow v'=-8$ cm

It means due to the first surface the image is formed at the centre.

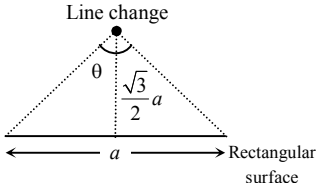
For the second surface $u=-9$ cm, $\mu_1=1.5$, $R_2=-9$ cm

$\Rightarrow \quad \frac{m_2}{v}-\frac{m_1}{u}=\frac{m_2-m_1}{R_2}$

$\Rightarrow \quad \frac{1}{v}+\frac{1.9}{5}=\frac{1-1.5}{-9}\Rightarrow v=-9$ cm

Thus, the final image is formed at the centre of the sphere.

8. **(6)** From the figure θ = 60°



So, No. of rectangular surfaces used to form a $m=\frac{360°}{60°}=6$

9. **(6)** As is clear from figure. $E_1 = E_2 + E_3$

$$\frac{hc}{\lambda_1} = \frac{hc}{\lambda_2} + \frac{hc}{\lambda_3}$$

$$\therefore \quad \frac{1}{\lambda_2} = \frac{1}{\lambda_1} - \frac{1}{\lambda_3} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} = 6 \text{ units}$$

10. **(5)** Using work energy theorem $W_{mg} + W_F = \Delta KE$

$$-mgh + Fd = \Delta KE \quad -1 \times 10 \times 4 + 18(5) = \Delta KE$$

$$\Rightarrow \quad \Delta KE = 50 \quad \therefore \quad n = 5$$

11. **(7)** Kinetic energy of a pure rolling disc having velocity of centre of mas $v = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{mR^2}{2}\right)\frac{v^2}{R^2} = \frac{3}{4}mv^2$

So, $\frac{3}{4}m(3)^2 + mg(30) = \frac{3}{4}m(v_2)^2 + mg(27)$

$$\therefore \quad v_2 = 7 \text{ m/s}$$

12. **(2)** $\left(\frac{dQ}{dt}\right)_A = 10^4\left(\frac{dQ}{dt}\right)_B \quad (400R)^2 T_A^4 = 10^4 (R^2 T_B^4)$

So, $2T_A = T_B$ and $\frac{\lambda_A}{\lambda_B} = \frac{T_B}{T_A} = 2$

13. **(8)** Here, $T = 2\text{h}, t = ?$

To work safely with the sample, its activity must be reduced to $\frac{1}{16}$.

From $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n = \frac{1}{16} = \left(\frac{1}{2}\right)^4 \quad \therefore \quad n = 4$

$$\Rightarrow \quad t = n\,T = 2 \times 4 = 8\text{h}$$

14. **(3)** For maxima, $\frac{4}{3}\sqrt{d^2 + x^2} - \sqrt{d^2 + x^2} = m\lambda$, m is an integer

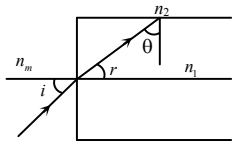
So, $x^2 = 9m^2\lambda^2 - d^2 \Rightarrow p = 3$

15. **(a, c)** $\theta \geq c$

$$\Rightarrow \quad 90° - r \geq c$$

$$\Rightarrow \quad \sin(90° - r) \geq c$$

$$\Rightarrow \quad \cos r \geq \sin c$$



Using $\frac{\sin i}{\sin r} = \frac{n_i}{n_m}$ and $\sin = \frac{n_2}{n_3}$

We get, $\sin^2 i_m = \frac{n_1^2 - n_2^2}{n_m^2}$

Putting values, we get, correct options as *a* & c

16. **(d)** For total internal reflection to take place in both structures, the numerical aperture should be the least one for the combined structure and hence, correct option is d.

17. **(a, d)** $I_1 = I_2$

$$\Rightarrow \quad neA_1v_1 = neA_2v_2$$

$$\Rightarrow \quad d_1w_1v_1 = d_2w_2v_2$$

Now, potential difference developed across MK, $V = Bvw$

$$\Rightarrow \quad \frac{V_1}{V_2} = \frac{v_1w_1}{v_2w_2} = \frac{d_2}{d_1} \text{ and hence correct choice is a \& d.}$$

18. **(a, c)** As $I_1 = I_2$

$$\Rightarrow \quad n_1w_1d_1v_1 = n_2w_2d_2v_2$$

Now, $\frac{V_2}{V_1} = \frac{B_2v_2w_2}{B_2v_1w_1} = \left(\frac{B_2w_2}{B_1w_1}\right)\left(\frac{n_1w_1d_1}{n_2w_2d_2}\right) = \frac{B_2n_1}{B_1n_2}$

JEE Advance Paper -II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	a,d	a,c	d	a,b	b,c	2	7	4	6
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**		
3	2	2	3	a	a	d	a		

1. **(a)** Q value of reaction

$= (140 + 94) \times 8.5 - 236 \times 7.5 = 219$ MeV

So, Total kinetic energy of Xe and Sr $= 219 - 2 - 2 = 215$ MeV

So, by conservation of momentum, energy, mass and charge, only option a. is correct.

2. **(a, d)** From the given conditions, $\rho_1 < \sigma_1 < \sigma_2 < \rho_2$

From equilibrium, $\sigma_1 + < \sigma_2 = \rho_1 + \rho_2$

$$V_p = \frac{2}{9}\left(\frac{\rho_1 - \sigma_2}{\eta_2}\right)g \text{ and } V_Q = \frac{2}{9}\left(\frac{\rho_2 - \sigma_1}{\eta_1}\right)g$$

So, $\frac{|\vec{V}_P|}{|\vec{V}_Q|} = \frac{\eta_1}{\eta_1}$ and $\vec{V}_P.\vec{V}_Q < 0$

3. **(a, c)** $BI\ell c \equiv VI$

$$\Rightarrow \quad \mu_0 I^2 c \equiv VI$$

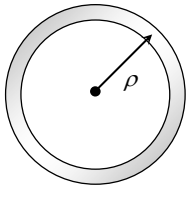
$$\Rightarrow \quad \mu_0 Ic = V \qquad \Rightarrow \mu_0^2 I^2 c^2 = V^2$$

$$\Rightarrow \quad \mu_0 I^2 = \varepsilon_0 V^2 \qquad \Rightarrow \varepsilon_0 cV = I$$

4. **(d)** $\vec{E} = \frac{\rho}{3\varepsilon_0}\overrightarrow{C_1C_2}$

$C_1 \Rightarrow$ Centre of sphere and

$C_2 \Rightarrow$ centre of cavity.



5. **(a, b)** $Y = \frac{\text{stress}}{\text{strain}}$

$$\Rightarrow \quad \frac{1}{Y} = \frac{\text{strain}}{\text{stress}}$$

$\Rightarrow \quad \frac{1}{Y_P} > \frac{1}{Y_Q} \quad \Rightarrow Y_P < Y_Q$

6. **(b, c)** $P(r) = K\left(1 - \frac{r^2}{R^2}\right)$

7. **(2)** $mvr = \frac{nh}{2\pi} = \frac{3h}{2\pi}$

de-Broglie Wavelength $\lambda = \frac{h}{mv}$

$= \frac{2\pi r}{3} = \frac{2\pi}{3}\frac{a_0(3)^2}{z_{Li}} = 2\pi a_0$

8. **(7)** For m closer to M; $\frac{GMm}{9\ell^2} - \frac{Gm^2}{\ell^2} = ma$ . . .(*i*)

and for the other m: $\frac{Gm^2}{\ell^2} + \frac{GMm}{16\ell^2} = ma$ . . .(*ii*)

From both the equations, $k = 7$

9. **(4)** $E(T) = A^2 e^{-at}$

$\Rightarrow \quad dE = -\alpha A^2 e^{-at} dt + 2AdAe^{-at}$

Putting the values for maximum error,

$\Rightarrow \quad \frac{dE}{E} = \frac{4}{100} \Rightarrow$ % error = 3.

10. **(6)** $I = \int \frac{2}{3}\rho 4\pi r^2 r^2 dr$

$\Rightarrow \quad I_A \propto \int (r)(r^2)(r^2)\, dr$

$\Rightarrow \quad I_B \propto \int (r^5)(r^2)(r^2)\, dr$

$\therefore \quad \frac{I_B}{I_A} = \frac{6}{10}$

11. **(3)** First and fourth wave interfere destructively. So from the interference of 2nd and 3rd wave only,

$\Rightarrow \quad I_{net} = I_0 + I_0 + 2\sqrt{I_0}\sqrt{I_0}\cos\left(\frac{2\pi}{3} - \frac{\pi}{3}\right) = 3I_0 \Rightarrow n = 3$

12. **(2)** $\lambda_P = \frac{1}{\tau}; \lambda_Q = \frac{1}{2\tau} \Rightarrow \frac{R_P}{R_Q} = \frac{(A_0\lambda_P)e^{-\lambda_P t}}{A_0\lambda_Q e^{-\lambda_Q t}}$

At $t = 2\tau$; $\frac{R_P}{R_Q} = \frac{2}{e}$

13. **(2)** Snell's Law on 1st surface: $\frac{\sqrt{3}}{2} = n\sin r_1$

$\sin r_1 = \frac{\sqrt{3}}{2n}$ . . .(*i*)

$\Rightarrow \quad \cos r_1 = \sqrt{1 - \frac{3}{4n^2}} = \sqrt{\frac{4n^2 - 3}{2n}}$

$r_1 + r_2 = 60°$ . . .(*ii*)

Snell's Law on 2nd surface: $n\sin r_2 = \sin\theta$

Using equation (*i*) and (*ii*) $n\sin(60° - r_1) = \sin\theta$

$\Rightarrow \quad n\left[\frac{\sqrt{3}}{2}\cos r_1 - \frac{1}{2}\sin r_1\right] = \sin\theta$

$\Rightarrow \quad \frac{d}{dn}\left[\frac{\sqrt{3}}{4}\left(\sqrt{4n^2 - 3} - 1\right)\right] = \cos\theta\frac{d\theta}{dn}$

For $\theta = 60°$ and $n = \sqrt{3} \Rightarrow \frac{d\theta}{dn} = 2$

14. **(3)** $V_3 = \frac{V_1V_2(R_2 - R_1)}{V_1(R_3 - R_1) - V_2(R_3 - R_2)} = 3.0\,\text{V}$

15. **(a)** A→ 1; B→ 2, 4; C→ 3; D→ 4

16. **(a)** A→ 3, 4; B→ 2, 3, 4; C→ 1, 2; D→ 2, 3, 4

17. **(d)** Gravitational field intensity at any point outside due to earth is given by $\vec{E} = \vec{g} = \frac{GM}{(R+h)^2} = \frac{GM}{r^2}$

where h is height from surface or $\vec{E} \propto \frac{1}{r^2}$ but – *ve* in sign.

Graph is parabolic.

Inside earth g decreases are r increases form surface of earth.

$\vec{E} = g\left(1 + \frac{x}{R}\right) = g - \frac{g}{R}x \qquad -E \propto \frac{1}{r^2}$

or $E \propto x$ is straight line decreasing to 0 at centre of earth and then increases in magnitude but – ve as directed towards the centre always.

At any point outside $E \propto \frac{1}{r^2}$ (parabolic graph)

Inside earth $E \propto r$ (linear graph)

In hollow sphere inside $E = 0$ graph along r-axis; outside as if whole mass lies at centre $E \propto \frac{1}{r^2}$ decreasing parabolic graph after sudden rise at surface of earth.

18. **(a)** A→1; B→1; C→1; D→4

In A, B, C no horizontal velocity is imparted to falling water, so d remains same.

In D, since its free fall, $a_{eff} = 0$

$\therefore$ Liquid won't fall with respect to lift.

□ □ □

JEE-MAIN: PHYSICS MOCK TEST-2

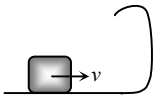
SECTION 1 (Multiple Choice Question)

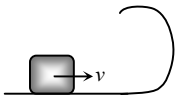
1. In the relation $p = \frac{\alpha}{\beta} e^{-\frac{\alpha Z}{k\theta}}$ p is pressure, Z is distance, k is Boltzmann constant and θ is the temperature. The dimensional formula of α will be:

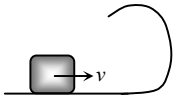
a. $[MLT^{-2}]$ **b.** $[ML^2T]$

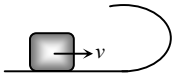
c. $[ML^0T^{-1}]$ **d.** $[M^0L^2T^{-1}]$

2. A small block is shot into each of the four tracks as shown below. Each of the tracks rises to the same height. The speed with which the block enters the track is the same in all cases. At the highest point of the track, the normal reaction is maximum in

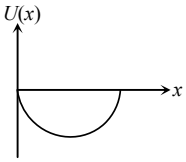
a. 

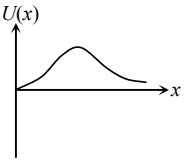
b. 

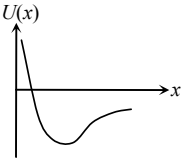
c. 

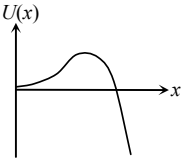
d. 

3. A particle, which is constrained to move along x-axis, is subjected to a force in the same direction which varies with the distance x of the particle from the origin as $F(x) = -kx + ax^3$. Here, k and a are positive constants. For $x \geq 0$, the functional form of the potential energy $U(x)$ of the particle is

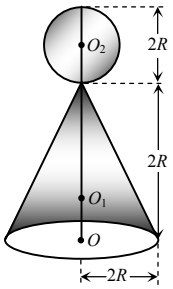
a. 

b. 

c. 

d. 

4. A carpenter has constructed a toy as shown in the adjoining figure. If the density of the material of the sphere is 12 times that of cone, the position of the center of mass of the toy is given by.



a. at a distance of $2R$ from O

b. at a distance of $3R$ from O

c. at a distance of $4R$ from O

d. at a distance of $5R$ from O

5. The total energy of a particle executing simple harmonic motion is:

a. $\propto x$ **b.** $\propto x^2$

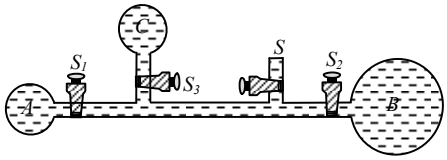
c. independent x **d.** $\propto x^{1/2}$

6. Two satellites A and B go around a planet P in circular orbits having radii $4R$ and R respectively. If the speed of satellite A is $3v$, the speed of satellite B will be:

a. $12v$ **b.** $6v$

c. $\frac{4}{3}v$ **d.** $\frac{3}{2}v$

7. The following diagram shows three soap bubbles A, B and C prepared by blowing the capillary tube fitted with stop cocks, S_1, S_2 and S_3. With stop cock S closed and stop cocks S_1, S_2 and S_3 opened.



a. B will start collapsing with volumes of A and C increasing

b. C will start collapsing with volumes of A and B increasing

c. C and A both will start collapsing with the volume of B increasing

d. Volumes of A, B and C will becomes equal at equilibrium

8. A steel wire of length 20 cm and uniform cross-section 1 mm^2 is tied rigidly at both the ends. The temperature of the wire is altered from 40°C to 20°C. What is the magnitude of force developed in the wire? (Coefficient of linear expansion for steel, $\alpha = 1.1 \times 10^{-5}$ /°C and Y for steel is 2.0×10^{11} N/m^2).

a. 2.2×10^6 N **b.** 16 N

c. 8 N **d.** 44 N

9. The intensity level of two waves of same frequency in a given medium are 20 dB and 60 dB. Then the ratio of their amplitudes is:

a. 1 : 4 **b.** 1 : 16

c. $1 : 10^4$ **d.** 1 : 100

10. The specific heat of a substance varies as $(2t^2 + t) \times 10^{-3}$ cal/g °C. What is the amount of heat required to raise the temperature of 100 g of substance through 20°C to 40°C.

a. 37.9 kilocal **b.** 3.79 kilocal
c. 379 kilocal **d.** 82 cal

11. A real gas behaves like an ideal gas if its
a. pressure and temperature are both high
b. pressure and temperature are both low
c. pressure is high and temperature is low
d. pressure is low and temperature is high

12. The plates of a parallel plate capacitor of capacity of 50μF are charged by a battery to a potential of 100 volt. The battery remains connected the plates are separated from each other so that the distance between them is doubled. How much is the energy spent by battery in doing so?
a. 25×10^{-2} J **b.** -12.5×10^{-2} J
c. -25×10^{-2} J **d.** 12.5×10^{-2} J

13. The magnetic moment produced in a substance of 1gm is 6×10^{-7} ampere-metre^2. If its density is 5 gm/cm^3, then the intensity of magnetisation in A/m will be
a. 8.3×10^{6} **b.** 3.0
c. 1.2×10^{-7} **d.** 3×10^{-6}

14. An ionized gas contains both positive and negative ions. If it is subjected simultaneously to an electric field along the $+x$ direction and a magnetic field along the $+z$ direction, then
a. Positive ions deflect towards $+y$ direction and negative ions towards $-y$ direction
b. All ions deflect towards $+y$ direction
c. All ions deflect towards $-y$ direction
d. Positive ions deflect towards $-y$ direction and negative ions towards $+y$ direction

15. A hundred turns of insulated copper wire are wrapped around an iron cylinder of area 1×10^{-3} m^2 are connected to a resistor. The total resistance in the circuit is 10 ohms. If the longitudinal magnetic induction in the iron changes from 1 Wb m^{-2}, in one direction to 1 Wb m^{-2} in the opposite direction, how much charge flows through the circuit?
a. 2×10^{-2} C **b.** 2×10^{-3} C
c. 2×10^{-4} C **d.** 2×10^{-5} C

16. A resistor R, an inductor L and a capacitor C are connected in series to an oscillator of frequency n. If the resonant frequency is n_r, then the current lags behind voltage, when
a. $n = 0$ **b.** $n < n_r$
c. $n = n_r$ **d.** $n > n_r$

17. The intensity ratio at a point of observation due to two coherent waves is 100 : 1. The ratio between their amplitudes is:
a. 1 : 1 **b.** 1 : 10
c. 1 : 100 **d.** 10 : 1

18. An electromagnetic wave propagating along north has its electric field vector upwards. Its magnetic field vector points towards
a. north **b.** east
c. west **d.** downwards

19. The ionisation energy of 10 times ionised sodium atom is
a. 13.6 eV **b.** 13.6×11 eV
c. $\frac{13.6}{11}$ eV **d.** $13.6 \times (11)^2$ eV

20. The example of nuclear fusion is
a. Formation of barium and krypton from uranium
b. Formation of helium from hydrogen
c. Formation of plutonium 235 from uranium 235
d. Formation of water from hydrogen and oxygen

SECTION 2 (Numeric Value Question)

21. A force of 5 N acts on a particle along a direction making an angle of 60° with vertical. Its vertical component be:
a. 10 N **b.** 3 N
c. 4 N **d.** 2.5 N

22. A neutral water molecule (H_2O) in it's vapor state has an electric dipole moment of magnitude 6.4×10^{-30} C–m. How far apart are the molecules centres of positive and negative charge?
a. 4 m **b.** 4 mm
c. 4 μm **d.** 4 pm

23. A cell of e.m.f. 1.5 V having a finite internal resistance is connected to a load resistance of 2Ω. For maximum power transfer, the internal resistance of the cell in ohms should be:
a. 4 **b.** 0.5
c. 2 **d.** None of these

24. Two plane mirrors are at 45° to each other. If an object is placed between them, then the number of images will be
a. 5 **b.** 9
c. 7 **d.** 8

25. A photon of energy 8 eV is incident on a metal surface of Threshold frequency 1.6×10^{15}Hz. The kinetic energy of the photoelectrons emitted (in eV) (Take $h = 6 \times 10^{-34}$ J – s)
a. 1.6 **b.** 6
c. 2 **d.** 1.2

JEE ADVANCE PAPER-I

SECTION 1 (Maximum Marks: 24)

MCQs with one or more than one correct answer

1. A piece of wire is bent in the shape of a parabola $y = kx^2$ (y-axis vertical) with a bead of mass m on it. The bead can slide on the wire without friction. It stays at the lowest point of the parabola when the wire is at rest. The wire is now accelerated parallel to the x-axis with a constant acceleration a. The distance of the new equilibrium position of the bead, where the bead can stays at rest with respect to the wire, from the y-axis is

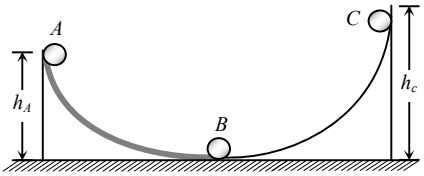
a. $\frac{a}{gk}$ **b.** $\frac{a}{2gk}$ **c.** $\frac{2a}{gk}$ **d.** $\frac{a}{4gk}$

2. A block of mass m is placed on a surface with a vertical cross-section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is:

a. $\frac{1}{3}$m **b.** $\frac{1}{2}$m

c. $\frac{1}{6}$m **d.** $\frac{2}{3}$m

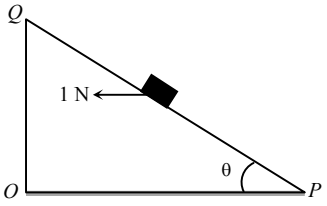
3. A ball moves over a fixed track as shown in the figure. From A to B the ball rolls without slipping. Surface BC is frictionless. K_A, K_B and K_C are kinetic energies of the ball at A, B and C, respectively. Then



a. $h_A > h_C$; $K_B > K_C$ **b.** $h_A > h_C$; $K_C > K_A$

c. $h_A = h_C$; $K_B = K_A$ **d.** $h_A < h_C$; $K_B > K_C$

4. A small block of mass of 0.1 kg lies on a fixed inclined plane PQ which makes an angle θ with the horizontal. A horizontal force of 1 N acts on the block through its centre of mass as shown in the figure. The block remains stationary if (take g = 10 m/s^2).



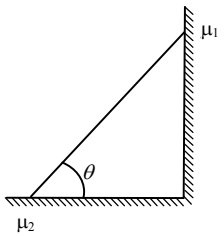
a. $\theta = 45°$

b. $\theta > 45°$ and a frictional force acts on the block towards P.

c. $\theta < 45°$ and a frictional force acts on the block towards

d. a frictional force acts on the block towards

5. In the figure, a ladder of mass m is shown leaning against a wall. It is in static equilibrium making an angle θ with the horizontal floor. The coefficient of friction between the wall and the ladder is μ_1 and that between the floor and the ladder is μ_2. The normal reaction of the wall on the ladder is N_1 and that of the floor is N_2. If the ladder is about to slip, then



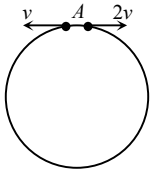
a. $\mu_1 = 0, \mu_2 \neq 0$ and $N_2 \tan\theta = \frac{mg}{2}$

b. $\mu_1 \neq 0, \mu_2 = 0$ and $N_1 \tan\theta = \frac{mg}{2}$

c. $\mu_1 \neq 0, \mu_2 \neq 0$ and $N_1 = \frac{mg}{1+\mu_1\mu_2}$

d. $\mu_1 = 0, \mu_2 \neq 0$ and $N_1 \tan\theta = \frac{mg}{2}$

6. Two small particles of equal masses start moving in opposite directions from a point A in a horizontal circular orbit. Their tangential velocities are v and $2v$ respectively, as shown in the figure. Between collisions, the particles move with constant speeds. After making how many elastic collisions, other than that at A, these two particles will again reach the point A?



a. 4 **b.** 3

c. 2 **d.** 1

SECTION 2 (Maximum Marks: 24)

Numerical value answer type questions

7. The densities of two solid spheres *A* and *B* of the same radii *R* vary with radial distance *r* as $\rho_A(r) = k\left(\frac{r}{R}\right)$ and $\rho_B(r) = k\left(\frac{r}{R}\right)^5$, respectively, where *k* is a constant. The moments of inertia of the individual spheres about axes passing through their centres are I_A and I_B, respectively. If $\frac{I_B}{I_A} = \frac{n}{10}$, the value of *n* is

8. Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}g$, where g is the gravitational acceleration on the surface of the earth. The average mass density of the planet is $\frac{1}{3}$ times that of the earth. If the escape speed on the surface of the earth is taken to be 11 kms^{-1}, the escape speed on the surface of the planet in kms^{-1} will be

9. A bullet is fired vertically upwards with velocity *v* from the surface of a spherical planet. When it reaches its maximum height, its acceleration due to the planet's gravity is $1/4^{th}$ of its value at the surface of the planet. If the escape velocity from the planet is $v_{esc} = v\sqrt{N}$, then the value of *N* is (ignore energy loss due to atmosphere)

10. The ends *Q* and *R* of two thin wires, *PQ* and *RS*, are soldered (joined) together. Initially each of the wires has a length of 1m at 10°C. Now the end *P* is maintained at $10°C$, while the end *S* is heated and maintained at $400\ °C$. The system is thermally insulated from its surroundings. If the thermal conductivity of wire *PQ* is twice that of the wire *RS* and the coefficient of linear thermal expansion of *PQ* is $1.2\times10^{-5}K^{-1}$, the change in length of the wire *PQ* is:

11. Two soap bubbles *A* and *B* are kept in a closed chamber where the air is maintained at pressure 8 N/m^2. The radii of bubbles *A* and *B* are 2 cm and 4 cm, respectively. Surface tension of the soap-water used to make bubbles is 0.04 N/m. Find the ratio $\frac{n_B}{n_A}$, where n_A and n_B are the number of moles of air in bubbles *A* and *B*, respectively. [Neglect the effect of gravity.]

12. A cylindrical vessel of height 500 mm has an orifice (small hole) at its bottom. The orifice is initially closed and water is filled in it up to height H. Now the top is completely sealed with a cap and the orifice at the bottom is opened. Some water comes out from the orifice and the water level in the vessel becomes steady with height of water column being 200 mm. Find the fall in height (in mm) of water level due to opening of the orifice. [Take atmospheric pressure $=1.0\times10^5\ N/m^2$, density of water = 1000 kg/m^3 and g = 10 m/s^2. Neglect any effect of surface tension.]

13. Consider two solid spheres *P* and *Q* each of density 8 gm cm^{-3} and diameters 1 cm and 0.5 cm, respectively. Sphere *P* is dropped into a liquid of density 0.8 gm cm^{-3} and viscosity $\eta = 3$ poiseulles. Sphere *Q* is dropped into a liquid of density 1.6 gm cm^{-3} and viscosity $\eta = 2$ poiseulles. The ratio of the terminal velocities of *P* and *Q* is:

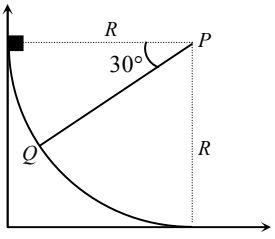
14. A 0.1 kg mass is suspended from a wire of negligible mass. The length of the wire is 1m and its cross-sectional area is $4.9\times10^{-7}\ m^2$. If the mass is pulled a little in the vertically downward direction and released, it performs simple harmonic motion of angular frequency 140 rad s^{-1}. If the Young's modulus of the material of the wire is $n\times10^9\ Nm^{-1}$, the value of *n* is

SECTION 3 (Maximum Marks: 12)

Paragraph based questions (2 paragraphs, each having 2 MCQs with one correct answer only)

Paragraph for Question No. 15 to 16

A small block of mass 1 kg is released from rest at the top of a rough track. The track is circular arc of radius 40 m. The block slides along the track without toppling and a frictional force acts on it in the direction opposite to the instantaneous velocity. The work done in overcoming the friction upto the point *Q*, as shown in the figure, below, is 150 J. (Take the acceleration due to gravity, $g = 10\ m/s^{-2}$).



15. The speed of the block when it reaches the point Q is

a. $5\ \text{ms}^{-1}$ **b.** $10\ \text{ms}^{-1}$

c. $10\sqrt{3}\ \text{ms}^{-1}$ **d.** $20\ \text{ms}^{-1}$

16. The magnitude of the normal reaction that acts on the block at the point Q is

a. 7.5 N

b. 8.6 N

c. 11.5 N

d. 22.5 N

Paragraph for Question No. 17 to 18

Two waves $y_1 = A\cos(0.5\,\pi x - 100\pi t)$ $y_2 = A\cos$ $(0.4\,\pi x - 92\pi t)$ are travelling in a pipe placed along x-axis.

17. Find the number of times intensity is maximum in time interval of 1 sec

a. 4 **b.** 6

c. 8 **d.** 10

18. Find wave velocity of louder sound

a. 100 m/s **b.** 192 m/s

c. 200 m/s **d.** 96 m/s

Space for rough work

JEE ADVANCE PAPER-II

SECTION 1 (Maximum Marks: 24)

MCQs with one or more than one correct answer

1. A person trying to lose weight by burning fat lifts a mass of 10 kg upto a height of 1 m 1000 times. Assume that the potential energy lost each time he lowers the mass is dissipated. How much fat will he use up considering the work done only when the weight is lifted up? Fat supplies $3.8\times10^{7}\,\text{J}$ of energy per kg which is converted to mechanical energy with a 20% efficiency rate (Take $g = 9.8\ \text{ms}^{-2}$):

a. $2.45\times10^{-3}\,\text{kg}$ **b.** $6.45\times10^{-3}\,\text{kg}$

c. $9.89\times10^{-3}\,\text{kg}$ **d.** $12.89\times10^{-3}\,\text{kg}$

2. If the resultant of all the external forces acting on a system of particles is zero, then from an inertial frame, one can surely say that

a. linear momentum of the system does not change in time

b. kinetic energy of the system does not change in time

c. angular momentum of the system does not change in time

d. potential energy of the system does not change in time

3. A point mass of 1 kg collides elastically with a stationary point mass of 5 kg. After their collision, the 1 kg mass reverses its direction and moves with a speed of 2 ms^{-1}. Which of the following statement(s) is (are) correct for the system of these two masses?

a. Total momentum of the system is 3 kg ms^{-1}

b. Momentum of 5 kg mass after collision is 4 kg ms^{-1}

c. Kinetic energy of the centre of mass is 0.75 J

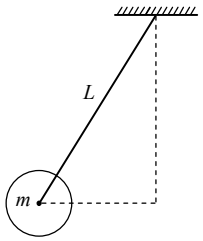
d. Total kinetic energy of the system is 4 J

4. A solid sphere of mass *M*, radius *R* and having moment of inertia about an axis passing through the centre of mass as *I*, is recast into a disc of thickness *t*, whose moment of inertia about an axis passing through its edge and perpendicular to its plane remains *I*. Then, radius of the disc will be

a. $\frac{2R}{\sqrt{15}}$ **b.** $R\sqrt{\frac{2}{15}}$

c. $\frac{4R}{\sqrt{15}}$ **d.** $\frac{R}{4}$

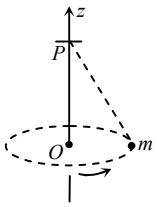
5. A ball of mass (*m*) 0.5 kg is attached to the end of a string having length (*L*) 0.5 m. The ball is rotated on a horizontal circular path about vertical axis. The maximum tension that the string can bear is 324 N. The maximum possible value of angular velocity of ball (in rad/s) is



a. 9 **b.** 18

c. 27 **d.** 36

6. A small mass *m* is attached to a massless string whose other end is fixed at *P* as shown in the figure. The mass is undergoing circular motion in the *x*-*y* plane with centre at *O* and constant angular speed ω. If the angular momentum of the system, calculated about *O* and *P* are denoted by $\vec{L}_O$ and $\vec{L}_P$ respectively, then.



a. $\vec{L}_O$ and $\vec{L}_P$ do not vary with time

b. $\vec{L}_O$ varies with time while $\vec{L}_P$ remains constant.

c. $\vec{L}_O$ remains constant while $\vec{L}_P$ varies with time

d. $\vec{L}_O$ and $\vec{L}_P$ both vary with time.

SECTION 2 (Maximum Marks: 24)

Numerical value answer type questions

7. A body sends waves 500 mm long through medium *A* and 0.25 m long in medium *B*. If velocity of waves in medium *A* is 16 m/s, what is the velocity (in m/s) of waves in medium *B*?

8. A light pointer fixed to one prong of a tuning fork touches gently a smoked vertical plate. The fork is set vibrating and the plate is allowed to fall freely. Two complete oscillations are traced when the plate falls through 40 cm. What is the frequency (in Hz) of the tuning fork?

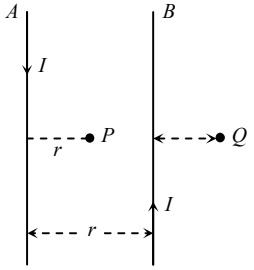
9. A water tank is 20 m deep. If the water barometer reads 10 m at that place, then what is the pressure at the bottom of the tank in atmosphere?

10. There is a soap bubble of radius 2.4×10^{-4} m in air cylinder which is originally at the pressure of $10^5\,\text{N/m}^2$. The air in the cylinder is now compressed isothermally until the radius of the bubble is halved. The pressure of air in the cylinder now becomes $n\times10^5\,\text{N/m}^2$. The surface tension of soap film is $0.08\,\text{Nm}^{-1}$. Find the integer value of n.

11. A length of wire carries a steady current I. It is bent first to form a circular plane coil of one turn. The same length is now best more sharply to give double loop of smaller radius. If the same current I is passed, the ratio of the magnitude of magnetic field at the centre with its first value is.

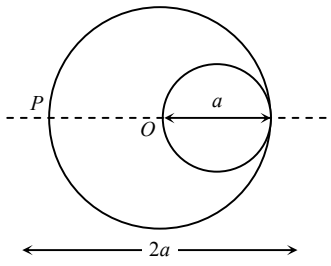
12. A current 1 amp is flowing in the sides of an equilateral triangle of side $4{\cdot}5\times10^{-2}$ m. the magnetic field at the centroid of the triangle in the unit of $(10^{-5}\,\text{T})$ is.

13. There are two infinite long parallel straight current carrying wires, A and B separated by a distance r (Fig.)



The current in each wire is I. The ratio of magnitude of magnetic field at points P and Q when points P and Q lie in the plane of wires is

14. A length of wire carries a steady current I. It is bent first to form a circular plane coil of one turn. The same length is now bent more sharply to give double loop of smaller radius. If the same current I is passed, the ratio of the magnitude of magnetic field at the centre with its first value is



SECTION 3 (Maximum Marks: 12)

Matching type questions with 4 options

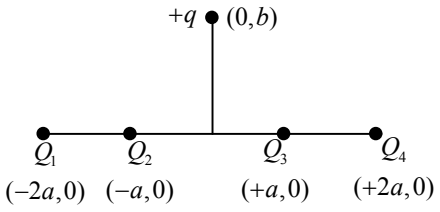
15. Work is defined as dot product of force and displacement $W=\int dW=\int \vec{F}\cdot\overrightarrow{dS}$. It is a scalar quantity. The total work done will depend on the displacement and the force, which may be constant or variable.

Thus in different situations of variable force applied in column I, the final expression for work done can be expressed as in column II.

Column I	Column II
(A) Force constant in magnitude acts at constant angle θ with direction of motion	**1.** $\frac{1}{2}kx^2\cos\theta$
(B) Force constant in magnitude acts at angle θ which varies as $\theta = kx$	**2.** $\frac{F}{K}\sin\theta$
(C) Force varies with distance x as $F = kx$ but angle θ is constant	**3.** $Fx\sin\theta$
(D) Force is constant in magnitude but changes in direction with changing angle and always acts along radius of circular path	**4.** zero

a. A→1, B→2, C→3, D→4 **b.** A→3, B→2, C→1, D→4
c. A→4, B→3, C→2, D→1 **d.** A→2, B→1, C→4, D→3

16. Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the x-axis at $x=-2a,-a,+a$ and $+2a$ respectively. A positive charge q is placed on the positive y-axis at a distance $b>0$. Four options of the signs of these charges are given in column I. the direction of the forces on the charge q is given in column II. Match column I with column II and select the correct answer using the code given below the lists.



Column I	Column II
(A) Q_1, Q_2, Q_3, Q_4 all positive	**1.** $+x$
(B) Q_1, Q_2 positive; Q_3, Q_4 negative	**2.** $-x$
(C) Q_1, Q_4 positive; Q_2, Q_3 negative	**3.** $+y$
(D) Q_1, Q_3 positive; Q_2, Q_4 negative	**4.** $-y$

a. A→3, B→1, C→4, D→2
b. A→4, B→2, C→3, D→1
c. A→3, B→1, C→2, D→4
d. A→4, B→2, C→1, D→3

17. The vibration of body can be under various types of forces. The vibration are classified mentioned in column I under the conditions mentioned in column II. Match the type of vibrations in column I with conditions in column II.

Column I	Column II
(A) Free vibrations	**1.** A body vibrating in viscous medium
(B) Forced vibrations	**2.** A body vibrating under its natural restoring force
(C) Resonant vibrations	**3.** A body vibrating under the influence of another vibrating body
(D) Damped vibrations	**4.** A body vibrating with its natural frequency under the influence of another vibrating body of same frequency

a. A→1, B→2, C→3, D→4
b. A→4, B→3, C→2, D→1
c. A→2, B→3, C→4, D→1
d. A→3, B→4, C→1, D→2

18. To determine specific heat of different substances we use different types of calorimeters. Can you match the type of colorimeter used named in column I to the substance whose specific heat is determined by corresponding calorimeter in column II.

Column I	Column II
(A) Regnault's calorimeter	**1.** To determine specific heat of solids at very low temperatures
(B) Joly's differential calorimeter	**2.** To determine specific heat of solids or gases at constant pressures
(C) Calender and Barmer's calorimeter	**3.** Used to measure specific heat at constant volume
(D) Nernst's vacuum calorimeter	**4.** Specific heat of liquids and gases at constant pressure

a. A→1, B→2, C→3, D→4
b. A→2, B→3, C→4, D→1
c. A→3, B→4, C→1, D→2
d. A→4, B→3, C→2, D→1

Space for rough work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	a	d	c	c	b	c	d	d	a
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
d	a	b	c	a	d	d	b	d	c
21.	**22.**	**23.**	**24.**	**25.**					
d	d	c	c	c					

1. **(a)** $[\mathrm{V}]=\dfrac{[\mathrm{GM}]}{[r]}=\dfrac{[\mathrm{M^{-1}L^3T^{-2}M}]}{[\mathrm{L}]}=[\mathrm{L^2T^{-2}}]\Rightarrow[\alpha]=\left[\dfrac{k\theta}{\mathrm{Z}}\right]$

Further $[p]=\left[\dfrac{\alpha}{\beta}\right]$

$\therefore\quad [\beta]=\left[\dfrac{\alpha}{p}\right]=\left[\dfrac{k\theta}{Zp}\right]$

Dimensions of $k\ \theta$ are that to energy.

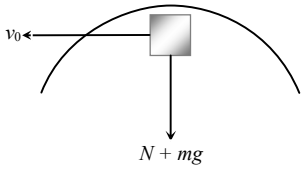
Hence, $[\alpha]=\dfrac{[\mathrm{K}\theta]}{[z]}=\dfrac{[\mathrm{ML^2T^{-2}}]}{[\mathrm{L}]}=[\mathrm{MLT^{-2}}]$

2. **(a)** Since, the block rises to the same heights in all the four cases, from conservation of energy, speed of the block at highest point will be same in all four cases. Say it is v_0

Equation of motion will be $N+mg=\dfrac{mv_0^2}{R}$

or $N=\dfrac{mv_0^2}{R}-mg$

R (the radius of curvature) in first case is minimum. Therefore, normal reaction N will be maximum in first case.



Note: In the question it should be mentioned that all the four tracks are frictionless. Otherwise, v_0 will be different in different tracks.

3. **(d)** $F=-\dfrac{dU}{dx}$

$\therefore\quad dU=-F.\,dx\quad$ or $U(x)=-\int_0^x(-kx+ax^3)\,dx$

$U(x)=\dfrac{kx^2}{2}-\dfrac{ax^4}{4}$

$U(x)=0$ at $x=0$ and $x=\sqrt{\dfrac{2k}{a}}$

$U(x)=$ Negative for $x>\sqrt{\dfrac{2k}{a}}$

From the given function we can see that

$F=0$ at $x=0$ *i.e.,* slope of $U-x$ graph is zero at $x=0$.

4. **(c)** If the density of cone be ρ, then its mass will be

$m_1=\dfrac{1}{3}\pi(2R)^2(4R)\rho=\dfrac{16}{3}\pi R^3\rho$

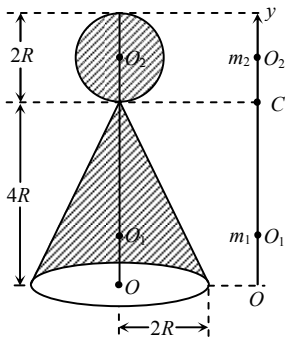
And its centre of mass will be at a height $\dfrac{h}{4}=\dfrac{4R}{4}=R$

From O on the line of symmetry, i.e., $y_1=R$

Similarly the mass of the sphere

$m_2=\dfrac{4}{3}\pi R^3(12\rho)=16\pi R^3\rho=3m_1$

And its centre of mass will be at its centre O_2, *i.e.,* $y_2=4R+R=5R$ (from O).



Now treating the sphere and cone as point masses with their masses concentrated at their centres of mass respectively and taking the line of symmetry as y-axis with origin at O, the centre of mass of the toy is given by

$Y_{CM}=\dfrac{m_1y_1+m_2y_2}{m_1+m_2}=\dfrac{m_1\times R+3m_1\times 5R}{m_1+3m_1}=4R$

i.e., centre of mass of the toy is at a distance $4R$ from O on the line of symmetry, *i.e.,* at the apex of the cone.

5. **(c)** The total energy of a particle in simple harmonic motion is constant.

6. **(b)** Orbital speed of a satellite $v_0=\sqrt{\dfrac{GM}{r}}$

$\Rightarrow\quad \dfrac{v_B}{v_A}=\sqrt{\dfrac{r_A}{r_B}}=\sqrt{\dfrac{4R}{R}}=2$

$\Rightarrow\quad v_B=2v_A=2\times(3v)=6v$

7. **(c)** Excess pressure inside soap bubble is inversely proportional to the radius of bubble, *i.e.,* $\Delta P\propto 1/r'$. This means that bubbles A and C posses greater pressure inside it than B. So the air will move from A and C, towards B.

8. **(d)** $F = Y A\alpha\Delta\theta$

$= \left(2.0\times10^{11}\right)\times\left(1\times10^{-6}\right)\times\left(1.1\times10^{-5}\right)(40-20) = 44$ N

9. **(d)** $L_2 - L_1 = 10\log_{10}\dfrac{I_2}{I_1}$

$\Rightarrow \quad 60-20 = 10\log_{10}\dfrac{I_2}{I_1}$

$\dfrac{I_2}{I_1} = 10^4 \quad \Rightarrow \quad \dfrac{a_2^2}{a_1^2} = 10^4$

$\Rightarrow \quad \dfrac{a_2}{a_1} = 100 \quad \Rightarrow a_1 : a_2 = 1 : 100$

10. **(a)** $dQ = mc\,dt$

$Q = \int_{t_1}^{t_2} mc\,dt$

$= \int_{t_1=20}^{t_2=40} 100\times\left(2t^2+t\right)\times10^{-3}\,dt \;= 10^{-1}\left[\dfrac{2t^3}{3}+\dfrac{t^2}{2}\right]_{20}^{40}$

$= 10^{-1}\left[\dfrac{2}{3}\left[(40)^3-(20)^3\right]+\dfrac{1}{2}(40)-(20)^2\right]$

$= 3.79\times10^3$ cal $= 3.79$ kilocal

11. **(d)** A real gas behaves like an ideal gas at low pressure and high temperature.

12. **(a)** When separation between the plates is doubled the capacitance becomes one half *i.e.,* $C' = 25\,\mu$F

Energy spent by battery $= qV$

$= (C'V)V = C'V^2$

$= 25\times10^{-6}\times(100)^2 = 25\times10^{-2}$ J

13. **(b)** $I = \dfrac{M}{V} = \dfrac{M}{\text{mass/density}}$,

Given mass $= 1gm = 10^{-3}$ kg

And density $= 5gm/cm^3 = \dfrac{5\times10^{-3}kg}{(10^{-2})^3 m^3} = 5\times10^3 kg/m^3$

Hence $I = \dfrac{6\times10^{-7}\times5\times10^3}{10^{-3}} = 3$

14. **(c)** As the electric field is switched on, positive ion will start to move along positive *x*-direction and negative ion along negative *x*-direction. Current associated with motion of both types of ions is along positive *x*-direction. According to Fleming's left hand rule force on both types of ions will be along negative *y*-direction.

15. **(a)** $dQ = \dfrac{d\phi}{R} = \dfrac{nA\,dB}{R} = \dfrac{100\times1\times10^{-3}\times2}{10} = 2\times10^{-2}$C

16. **(d)** The current will lag behind the voltage when reactance of inductance is more than the reactance of condenser.

Thus, $\omega L > \dfrac{1}{\omega C}$ or $\omega > \dfrac{1}{\sqrt{LC}}$

or $\quad n > \dfrac{1}{2\pi\sqrt{LC}}$ or $n > n_r$

where, n_r = resonant frequency.

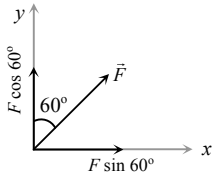
17. **(d)** $I \propto a^2 \Rightarrow \dfrac{a_1}{a_2} = \sqrt{\dfrac{I_1}{I_2}} = \sqrt{\dfrac{100}{1}} = \dfrac{10}{1}$

18. **(b)** The direction of propagation of electromagnetic wave is given by the direction of $(\vec{E}\times\vec{B})$. Here, the *em* wave is propagating along north. The electric field vector is acting upwards, so the magnetic field vector will point towards east.

19. **(d)** $(E_{ion})_{Na} = Z^2(E_{ion})_H = (11)^2 13.6$ eV

20. **(c)** Fast neutrons can escape from the reaction. So as to proceed the chain reaction. Slow neutrons are best.

21. **(d)** The component of force in vertical direction



$= F\cos\theta = F\cos60° = 5\times\dfrac{1}{2} = 2.5$ N

22. **(d)** There are 10 electrons and 10 protons in a neutral water molecule.

So it's dipole moment is $p = q(2l) = 10e(2l)$

Hence length of the dipole *i.e,.* distance between centres of positive and negative charges is

$2l = \dfrac{p}{10e} = \dfrac{6.4\times10^{-20}}{10\times1.6\times10^{-19}} = 4\times10^{-12}$ m $=4$ pm

23. **(c)** For maximum power $R_{ext} = R_{\text{int}} = 2\Omega$

24. **(c)** $n = \dfrac{360°}{45°} - 1 = 7$

25. **(c)** $W = hv_0 = 6\times10^{-34}\times1\cdot6\times10^{15}$ joule

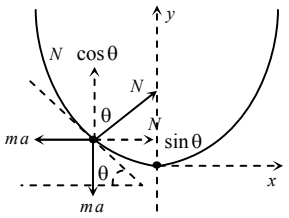
$= \dfrac{6\times10^{-34}\times1\cdot6\times10^{15}}{1\cdot6\times10^{-19}}$eV $= 6$ eV

$\therefore \quad E_k = hv - W = 8$ eV -6 eV $= 2$eV

JEE Advance Paper -I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	c	a,b,c	a,c	c,d	c	6	3	2	7
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**		
6	6	3	4	b	a	a	c		

1. **(b)** $\tan\theta = \dfrac{a}{g}$

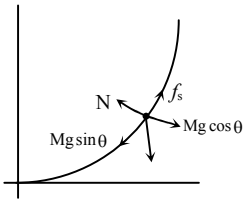


$\tan\theta = \dfrac{dy}{dx} = 2kx$

$\Rightarrow \quad x = \dfrac{a}{2gk}$

2. **(c)** $mg\sin\theta = \mu_s mg\cos\theta$ [when partied is just balanced]

$\Rightarrow \quad \tan\theta = u_s \Rightarrow \tan\theta = \dfrac{dy}{dx} = \left(\dfrac{x^2}{2}\right)$



$\therefore \quad \dfrac{x^2}{2} = 0.5 \Rightarrow x = 1 \Rightarrow y = \dfrac{1}{6}$

3. **(a, b, c)** $E_A = mgh_A + K_A$

$\Rightarrow \quad E_B = K_B$

$\Rightarrow \quad E_C = mgh_C + K_C$

Using conservation of energy $E_A = E_B = E_C$

$\Rightarrow \quad K_B > K_C$

$\Rightarrow \quad K_B > K_A$

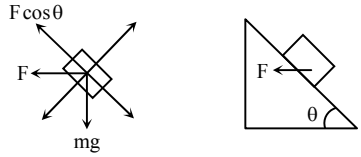
$\Rightarrow \quad Mg(h_A - h_C) + (K_A - K_C) = 0 \Rightarrow h_A - h_C = \dfrac{K_C - K_A}{Mg}$

4. **(a, c)** $f = 0 \quad (\theta = 45°)$

So, for block to be at rest $F\cos\theta = mg\sin\theta$

LHS $F\cos\theta = 1\times\cos 45° = \dfrac{1}{\sqrt{2}}$

RHS $mg\sin\theta = 0.1\times 10\times\dfrac{1}{\sqrt{2}} = \dfrac{1}{\sqrt{2}}$



{LHS = RHS)

If $\theta > 45°;\ mg\sin\theta > F\cos\theta$

$\therefore$ Friction acts towards Q.

If $\theta < 45° \ F\cos\theta > mg\sin\theta$

$\therefore$ Friction acts towards *P*.

5. **(c, d)** Condition of translational equilibrium $N_1 = \mu_2 N_2$

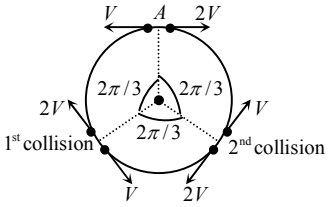
$\Rightarrow \quad N_2 = \mu_1 N_1 = Mg$, Solving $N_2 = \dfrac{mg}{1+\mu_1\mu_2}$

$\Rightarrow \quad N_1 = \dfrac{\mu_2 mg}{1+\mu_1\mu_2}$ Applying torque equation about corner (left) point on the floor

$\Rightarrow \quad mg\dfrac{\ell}{2}\cos\theta = N_1\ell\sin\theta + \mu_1 N_1\ell\cos\theta$

Solving $\tan\theta = \dfrac{1-\mu_1\mu_2}{2\mu_2}$

6. **(c)** Velocity will exchange after each collision



7. **(6)** $I = \int \dfrac{2}{3}\rho 4\pi r^2 r^2 dr$

$\Rightarrow \quad I_A \propto \int (r)(r^2)(r^2)\,dr$

$\Rightarrow \quad I_B \propto \int (r^5)(r^2)(r^2)\,dr$

$\therefore \quad \dfrac{I_B}{I_A} = \dfrac{6}{10}$

8. **(3)** $\dfrac{g'}{g} = \dfrac{\sqrt{6}}{11}; \dfrac{\rho'}{\rho} = \dfrac{2}{3}$

Hence, $\dfrac{R'}{R} = \dfrac{3\sqrt{6}}{22}$

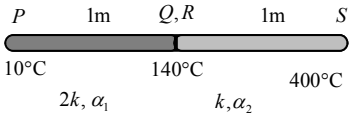
$\Rightarrow \quad \dfrac{V'_{esc}}{V_{sec}} \propto \sqrt{\dfrac{R'^2\rho'}{R^2\rho}} = \dfrac{3}{11} \Rightarrow v'_{esc} = 3\text{km/s}.$

9. **(2)** At height *R* from the surface of planet acceleration due to planet's gravity is $\dfrac{1}{4}th$ in comparison to the value at the surface

So, $-\dfrac{GMm}{R} + \dfrac{1}{2}mv^2 = -\dfrac{GMm}{R+R}$ and $-\dfrac{GMn}{R} + \dfrac{1}{2}mv_{esc}^2 = 0$

$\therefore \quad v_{esc} = v\sqrt{2}$

10. (1)



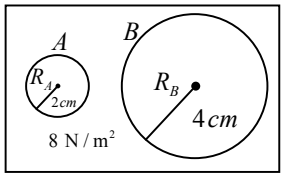
$d\ell = dx\alpha_1(\theta - 10)$ $\quad \Delta\ell = \int d\ell$

$\because \quad \left(\frac{\theta - 10}{x}\right) = \frac{130}{1} \Rightarrow \quad \theta = 10 + 130x$

$\Delta\ell = \int_0^1 (130x)\alpha_1 dx \quad \Delta\ell = 130\alpha_1 \frac{x^2}{2}$

$\Delta\ell = 130 \times 1.2 \times 10^{-5} \times \frac{1}{2} = 0.78 \text{ mm} \simeq 1$

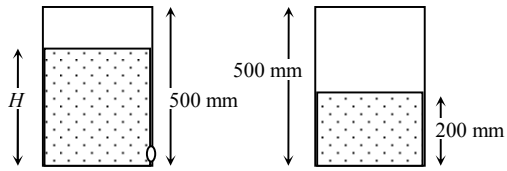
11. (6) $P_A = P_0 + \frac{4T}{R_A} = 16 \text{N/m}^2$



$\Rightarrow \quad P_B = P_0 + \frac{4T}{R_B} = 12 \text{N/m}^2 \Rightarrow \quad \frac{n_B}{n_A} = \frac{P_B}{P_A}\left(\frac{R_B}{R_A}\right)^3 = 6$

12. (6) $P = P_0 - \rho gh = 98 \times 10^3 \text{ N/m}^2$

$\Rightarrow \quad P_0 V_0 = PV$



$\Rightarrow \quad 10^5[A(500 - H)] = 98 \times 10^3[A(500 - 200)]$

$\Rightarrow \quad H = 206 \text{ mm}$

Level fall = 206 – 200 = 6 mm

13. (3) $6\pi\eta r v + \rho_L V_g = \rho_0 V g$

$\frac{V_P}{V_Q} = \frac{(\rho_P V_P - \rho_L V_P)g}{6\pi\eta_P r_P} \times \frac{6\pi\eta_Q r_Q}{(\rho_Q V_Q - \rho_L V_Q)}$

$= \frac{r_P^3(8 - 0.8)}{\eta_P . r_P (8 - 1.6)} \times \frac{r_Q . \eta_Q}{r_Q^3}$

$= \left(\frac{r_P}{r_Q}\right)^2 \times \left(\frac{\eta_Q}{\eta_P}\right) \times \left(\frac{7.2}{6.4}\right) = 4 \times \frac{7.2}{6.4} \times \frac{2}{3} = 3$

14. (4) $\omega = \sqrt{\frac{YA}{mL}}$

15. (b) Using work energy theorem

$mg\ R\sin 30° + W_f = \frac{1}{2}mv^2$

$\Rightarrow \quad 200 - 150 = \frac{v^2}{2}$

$\Rightarrow \quad v = 10 \text{ m/s}$

16. (a) $N - mg\cos 60° = \frac{mv^2}{R}$

$\Rightarrow \quad N = 5 + \frac{5}{2} = 7.5 \text{ N}$

17. (a) $|f_1 - f_2| = 4 s^{-1}$

18. (c) $v_1 = v_2 = 200 \text{ m/s}$

JEE Advance Paper -II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	a	a,c	a	d	c	8	7	3	8
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**		
4	4	8	4	b	a	c	a		

1. (d) Let m mass of fat is used.

$m(3.8 \times 10^7)\frac{1}{5} = 10(9.8)(1)(1000)$

$m = \frac{9.8 \times 5}{3.8 \times 10^3} = 12.89 \times 10^{-3} \text{ kg}$

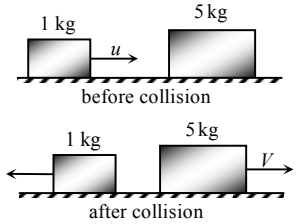
2. (a) Linear momentum remains constant if net external force on the system of particles is zero.

3. (a, c) By conservation of linear momentum

$u = 5v - 2 \quad \ldots(i)$

By Newton's experimental law of collision

$u = v + 2 \quad \ldots(ii)$



Using (i) and (ii) we have $v = 1 \text{m/s}$ and $u = 3 \text{ m/s}$

Kinetic energy of the centre of mass

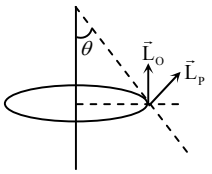
$= \frac{1}{2} m_{\text{system}} v^2_{\text{cm}} = 0.75 \text{J}$

4. (a) $\frac{2}{5}MR^2 = \frac{3}{2}Mr^2 \Rightarrow r = \frac{2R}{\sqrt{15}}$

5. (d) $m\omega^2_{\max} r = T_{\max}$

$\Rightarrow \quad \omega_{\max} = \sqrt{\frac{T_{\max}}{mr}} = \sqrt{\frac{324}{0.5 \times 0.5}} = \sqrt{1296} = 36 \text{ rad/s}$

6. **(c)** $\vec{L}_O = (m\omega r^2)\hat{k}$



Direction of $\vec{L}_P$ varies with time as its direction will be perpendicular to string i.e. changing with time.

$\vec{L}_O$ = Constant , $\vec{L}_P$ = Variable

7. **(8)** Here, $\lambda_1 = 500\,\text{mm} = 0.5\text{m}$

$\lambda_2 = 0.25\text{ m}; \upsilon_1 = 16\text{ m/s}, \upsilon_2 = ?$

As frequency of body is fixed, say *n*, therefore,

$\upsilon_1 = \lambda_1 n; \upsilon_2 = \lambda_2 n$

$\Rightarrow \quad \frac{\upsilon_2}{\upsilon_1} = \frac{\lambda_2}{\lambda_1} = \frac{0.25}{0.5} = \frac{1}{2}$

$\Rightarrow \quad \upsilon_2 = \frac{\upsilon_1}{\upsilon_2} = \frac{16}{2} = 8\text{m/s}$

8. **(7)** Here, $m = 2, h = 40\,\text{cm}, n = ?$

Time taken by the plate to fall down,

$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times 40}{980}} = \frac{2}{7}\text{sec}$

Frequency of fork, $n = \frac{m}{t} = \frac{2}{2/7} = 7\text{Hz}.$

9. **(3)** The pressur at the bottom, $P = P_0 + h\rho g$

Where, p_0 = atmospheric pressure, therefore,

$P = 1 + \frac{h\rho g}{P_0}\text{ atmosphere}$

$= 1 + \frac{20\rho g}{10\rho g} = 3\,\text{atmosphere}.$

10. **(8)** Here, $p = 10^5\,\text{N/m}^2, R = 2.4\times 10^{-4}\text{m}; S = 0.08\,\text{N/m}$

Initial pressure inside the soap bubble in air cylinder,

$P_1 = P + \frac{4S}{R}.$

Let V_1 be the initial volume of the soap bubble;

$V_1 = \frac{4}{3}\pi R^3$

After compression, volume of the soap bubble;

$V_2 = \frac{4}{3}\pi\left(\frac{R}{2}\right)^3 = \frac{V_1}{8}$

If is the air pressure after compression in the cylinder, then pressure inside the bubble is

$P_2 = P' + \frac{4S}{(R/2)} = \left(P' + \frac{8S}{R}\right)$

Using Boyle's law, we have, $P_1 V_1 = P_2 V_2$

or $\left(P + \frac{4S}{R}\right)V_1 = \left(P' + \frac{8S}{R}\right)\frac{V_1}{8}$

or $P + \frac{4S}{R} = \frac{1}{9}\left(P' + \frac{8S}{R}\right)$

or $P' = 8P + \frac{24S}{R} = 8\times 10^5 + \frac{24 + 0.08}{2.4\times 10^{-4}} = 8.08\times 10^5 = n\times 10^5$

$\therefore \quad n = 0.08 \approx 8.$

11. **(4)** When wires is taken in the form of one turn circular coil, then length, $l = 2\pi r$ or $r = \frac{l}{2\pi}, n = 1$ Magnetic field induction at the centre of circular coil due to current *I* is

$B = \frac{\mu_0}{4\pi}\frac{2\pi nI}{r} = \frac{\mu_0}{4\pi}\frac{2\pi\times 1\times I}{(l/2\pi)} = \frac{\mu_0 \pi I}{l}$

When wire is taken in the form of double loop, then

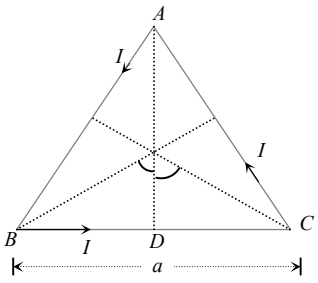
$l = 2\times 2\pi r_1$ or $r_1 = \frac{I}{4\pi}$ and $n = 2$

$\therefore$ Magnetic field induction at the centre of the circular coil.

$B_1 = \frac{\mu_0}{4\pi}\frac{2\pi\times n\times I}{r_1} = \frac{\mu_0}{4\pi}\frac{2\pi\times 2\times I}{(l/4\pi)} = 4\times\mu_0\frac{\pi I}{l}$

$\therefore \quad \frac{B_1}{B} = 4$

12. **(4)** Refer Fig.



The magnetic field induction at the centriod *O* due to current I through one side *BC* of the triangle will be

$B_1 = \frac{\mu_0}{4\pi}\frac{I}{r}(\sin\theta_1 + \sin\theta_2)$

It will be acting perpendicular to the plane of triangle upwards. Total magnetic field induction at *O* due to current through all the three sides of the triangle will be

$B = 3B_1 = \frac{3\mu_0}{4\pi}\frac{I}{\pi}[\sin\theta_1 + \sin\theta_2]$

Here, $I = 1A, \theta_1 = 60° = \theta_2$ and $r = OD = \frac{BD}{\tan 60°} = \frac{a/2}{\sqrt{3}}$

$$= \frac{a}{\sqrt[2]{3}} = \frac{4 \cdot 5 \times 10^{-2}}{2\sqrt{3}} m.$$

$\therefore$ $B = 3 \times 10^{-7} \times \frac{1}{(4 \cdot 5 \times 10^{-2} / 2\sqrt{3})} \times [\sin 60° + \sin 60°]$

On solving, $B = 4 \times 10^{-5}$ T.

13. **(8)** Magnetic field at P due to currents in two wires will be acting perpendicular to the plane of wires, upwards and is given by.

$$B_P = \frac{\mu_0}{4\pi} \frac{2I}{(r/2)} + \frac{\mu_0}{4\pi} \frac{2I}{(r/2)} = \frac{2\mu_0 I}{\pi r}$$

Magnetic field at Q due to current in A is perpendicular to the plane of wire upwards and due to current in B is perpendicular to the plane of wire downwards and is given by $B_Q = \frac{\mu_0 2I}{4\pi 2r} + \frac{\mu_0 2I}{4\pi r} = \frac{\mu_0 I}{4\pi r}$

$\therefore$ $\frac{B_P}{B_Q} = \frac{(2\mu_0 I / \pi r)}{(\mu_0 I / 4\pi r)} = 8.$

14. **(4)** When wire is taken in the form of one turn circular coil, then length $l = 2\pi r$

or $r = \frac{l}{2\pi}$, $n = 1$

Magnetic field induction at the centre of circular coil due to current I is

$$B = \frac{\mu_0}{4\pi} \frac{2\pi n I}{r} = \frac{\mu_0}{4\pi} \frac{2\pi \times 1 \times I}{(l/2\pi)} = \frac{\mu_0 \pi I}{l}$$

When wire is taken in the form of double loop, then

$l = 2 \times 2\pi r_1$ or $r_1 = \frac{l}{4\pi}$ and $n = 2$

$\therefore$ Magnetic field induction at the centre of the circular coil,

$$B_1 = \frac{\mu_0}{4\pi} \frac{2\pi \times n \times I}{r_1} = \frac{\mu_0}{4\pi} \frac{2\pi \times 2 \times I}{(l/4\pi)} = 4 \times \mu_0 \frac{\pi I}{l}$$

$\Rightarrow$ $\frac{B_1}{B} = 4.$

15. **(b)** A→3, B→2, C→1, D→4

$W = \vec{F} \cdot \vec{S} = Fx \cos\theta$

$\therefore$ Displacements $\vec{S} = \vec{x}$ while angle θ between $\vec{F}$ and $\vec{x}$ is constant

$\theta = kx$, hence $\frac{d\theta}{dx} = K$ or $dx = \frac{d\theta}{k}$

$\therefore$ $\int dW = \int \vec{F} \cdot \vec{dS}$

$= \int \vec{F} \cdot \vec{dx} = \int F \cdot \cos\theta \frac{d\theta}{K} = \frac{F}{K} \sin\theta$

$W = \int dW = \int \vec{F} \cdot \vec{dx} = \int \vec{Kx} \cdot \vec{dx}$

$= K \int x\, dx \cos\theta = \frac{1}{2} Kx^2 \cos\theta$

$\vec{F} \perp \vec{x}$ at every point

$\therefore$ $dW = \vec{F} \cdot \vec{dx} = 0$

$\because \theta = 90° = Fdx \cos 90° = 0$

16. **(a)** A→3; B→1; C→4; D→2

17. **(c)** A→2, B→3, C→4, D→1

When a body is disturbed from its mean position and left to vibrate under restoring force, e.g., A tunning force struck with rubber pad, vibrations are called free or natural vibrations. When a body is mode to vibrate with another vibrating body placed nearby, e.g., stem of a vibrating tunning fork when pressed on top of sonometer makes the sonometer wire to vibrate having forced vibrations. When a body is forced to vibrate by another vibrating body the vibrations are forced on other body. But if frequency of forced vibrations is equal to the natural frequency of forced body then vibrations of forced body are called resonant vibrations. When a body is vibrating under the action of viscous force, gradually the amplitude of vibration decreases because energy of vibrating body is dissipated as work done against viscous force in the form of heat etc. and body gradually stops vibrating. Such vibratious are called damped vibrations.

18. **(a)** A→1, B→2, C→3, D→4

The particular names of calorimeters are after the names of scientists who designed them for specific heat measurements in different conditions.

□ □ □

JEE-MAIN: PHYSICS MOCK TEST-3

SECTION 1 (Multiple Choice Question)

1. If $\vec{A} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ the direction of cosines of the vector $\vec{A}$ are:

a. $\frac{2}{\sqrt{45}}, \frac{4}{\sqrt{45}}$ and $\frac{-5}{\sqrt{45}}$ **b.** $\frac{1}{\sqrt{45}}, \frac{2}{\sqrt{45}}$ and $\frac{3}{\sqrt{45}}$

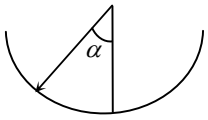
c. $\frac{4}{\sqrt{45}}, 0$ and $\frac{4}{\sqrt{45}}$ **d.** $\frac{3}{\sqrt{45}}, \frac{2}{\sqrt{45}}$ and $\frac{5}{\sqrt{45}}$

2. A particle is projected vertically upwards and it is at a height h after 2 seconds and again after 10 seconds. The height h is:

a. 196 m **b.** 98 m

c. 9.8 m **d.** 19.8 m

3. An insect crawls up a hemispherical surface very slowly (see the figure). The coefficient of friction between the surface and the insect is 1/3. If the line joining the centre of the hemispherical surface to the insect makes an angle α is given



a. $\cot\alpha = 3$ **b.** $\tan\alpha = 3$

c. $\sec\alpha = 3$ **d.** $\text{cosec}\alpha = 3$

4. An ideal spring with spring constant k is hung from the ceiling and a block of mass M is attached to its lower end. The mass is released with the spring initially unscratched. Then the maximum extension in the spring is

a. $\frac{4Mg}{k}$ **b.** $\frac{2Mg}{k}$

c. $\frac{Mg}{k}$ **d.** $\frac{Mg}{2k}$

5. A particle executes simple harmonic motion between $x = -A$ and $x = +A$. The time taken for it to go from 0 to $A/2$ is T_1 and to go from $A/2$ to A is T_2. Then:

a. $T_1 < T_2$ **b.** $T_1 > T_2$

c. $T_1 = T_2$ **d.** $T_1 = 2T_2$

6. A large number of liquid drops each of radius r coalesce to form a single drop of radius R. The energy released in the process is converted into the kinetic energy of the big drop so formed. The speed of the big is (Given surface tension of liquid is T, density of liquid is ρ)

a. $\sqrt{\frac{T}{\rho}\left(\frac{1}{r} - \frac{1}{R}\right)}$ **b.** $\sqrt{\frac{2T}{\rho}\left(\frac{1}{r} - \frac{1}{R}\right)}$

c. $\sqrt{\frac{4T}{\rho}\left(\frac{1}{r} - \frac{1}{R}\right)}$ **d.** $\sqrt{\frac{6T}{\rho}\left(\frac{1}{r} - \frac{1}{R}\right)}$

7. A solid sphere of radius R made of material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid when a mass m is placed on the piston to compress the liquid, the fractional change in the radius of the sphere, $\frac{\delta R}{R}$ is:

a. $\frac{mg}{AK}$ **b.** $\frac{mg}{3AK}$

c. $\frac{mg}{A}$ **d.** $\frac{mg}{3AK}$

8. The wavelength of light observed on the earth, from a moving star is found to decreases by 0.05%. Relative to the earth, the star is:

a. moving away with a velocity of 1.5×10^5 m/s

b. coming closer with a velocity of 1.5×10^5 m/s

c. moving away with a velocity of 1.5×10^4 m/s

d. coming closer with a velocity of 1.5×10^4 m/s

9. An ideal gas expands isothermally from a volume V_1 to V_2 and then compressed to original volume V_1 adiabatically. Initial pressure is p_1 and final pressure is p_3. The total work done is W. Then,

a. $p_3 > p_1, W > 0$ **b.** $p_3 < p_1, W < 0$

c. $p_3 > p_1, W < 0$ **d.** $p_3 = p_1, W = 0$

10. At NTP one mole of diatomic gas is compressed adiabatically to half of its volume ($\gamma = 1.40$). The work done on the gas will be $(2)^{0.4} = 1.40$:

a. 1280 J **b.** 1610 J

c. 1792 J **d.** 2025 J

11. Two identical conducting rods are first connected independently to two vessels, one containing water at 100°C and the other containing ice at 0ºC. In the second case, the rods are joined end to end connected to the same vessels. Let q_1 and q_2 gram per second be the rate of melting of ice in the two cases respectively. The ratio q_1/q_2 is

a. $\frac{1}{2}$ **b.** $\frac{2}{1}$

c. $\frac{4}{1}$ **d.** $\frac{1}{4}$

12. The energy density (energy per unit volume) in an electric field caused by a point charge falls off with the distance from the point charge as:

a. $1/r$ **b.** $1/r^2$

c. $1/r^3$ **d.** $1/r^4$

13. The needle of a deflection galvanometer shows a deflection of 60° due to a short bar magnet at a certain distance in $\tan A$ position. If the distance is doubled, the deflection is

a. $\sin^{-1}\left(\frac{\sqrt{3}}{8}\right)$ **b.** $\cos^{-1}\left(\frac{\sqrt{3}}{8}\right)$

c. $\tan^{-1}\left(\frac{\sqrt{3}}{8}\right)$ **d.** $\cot^{-1}\left(\frac{\sqrt{3}}{8}\right)$

14 If power factor is $\frac{1}{2}$ in a series RL circuit $R = 100\Omega$. ac mains is used then L is

a. $\frac{3}{\pi}$ henry **b.** π henry

c. $\frac{\pi}{\sqrt{2}}$ henry **d.** None of these

15. The amplitude ratio of two superposing waves is 2 : 1. The ratio of maximum and minimum intensities is:

a. 1 : 1 **b.** 2 : 1

c. 4 : 1 **d.** 9 : 1

16. The electric and the magnetic field, associated with an e.m. wave, propagating along the + z-axis, can be represented by

a. $\left[\vec{E} = E_0\hat{i}, \vec{B} = B_0\hat{j}\right]$ **b.** $\left[\vec{E} = E_0\hat{k}, \vec{B} = B_0\hat{i}\right]$

c. $\left[\vec{E} = E_0\hat{j}, \vec{B} = B_0\hat{i}\right]$ **d.** $\left[\vec{E} = E_0\hat{j}, \vec{B} = B_0\hat{k}\right]$

17. If the wavelength of the first line of the Balmer series of hydrogen is 6561 Å, the wavelength of the second line of the series should be

a. 13122 Å **b.** 3280 Å

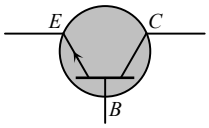
c. 4860 Å **d.** 2187 Å

18. In the nuclear reaction ${}_6C^{11} \rightarrow {}_5B^{11} + \beta^+ + X$, what does X stand for

a. An electron **b.** A proton

c. A neutron **d.** A neutrino

19. The symbol given in figure represents



a. NPN transistor

b. PNP transistor

c. Forward biased PN junction diode

d. Reverse biased NP junction diode

SECTION 2 (Numeric Value Question)

20. For television broadcasting, the frequency employed is normally

a. 30–300 MHz **b.** 30–300 GHz

c. 30–300 kHz **d.** 30–300 Hz

21. The potential difference V and current i flowing through an appliance in an ac circuit are given by $V = 5\cos\omega t$ volt, $i = 5\sin\omega t$ amp and power dissipated in the appliance is

a. 0 W **b.** 10 W

c. 5 W **d.** 2.5 W

22. A wire has a mass $(03. \pm 0.003)$ g, radius (0.5 ± 0.005) mm and length (6 ± 0.06) cm. The maximum percentage error in the measurement of its density is:

a. 1 **b.** 2

c. 3 **d.** 4

23. An aircraft with a wing-span of 40 m flies with a speed of 1080 km h^{-1} in the eastward direction at a constant altitude in the northern hemisphere, where the vertical component of earth's magnetic field is 8.3×10^{-4} T. Then the e.m.f. that develops between the tips of the wings is:

a. 0.5 V **b.** 0.35 V

c. 1 V **d.** 2.1 V

24. If an observer is walking away from the plane mirror with 3 m/sec. Then the velocity of the image with respect to observer will be

a. 6 m/sec **b.** –6 m/sec

c. 12 m/sec **d.** 3 m/sec

25. The eccentricity of earth's orbit is 0.0167. The ratio of its maximum speed in its orbit to its minimum speed is:

a. 2.507 **b.** 1.033

c. 8.324 **d.** 1.000

Space for Rough Work

JEE ADVANCE PAPER-I

SECTION 1 (Maximum Marks: 24)

MCQs with one or more than one correct answer

1. In a large building, there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 heater of 1 kW. The voltage of the electric mains is 220 V. The minimum capacity of the main fuse of the building will be:

a. 12 A **b.** 14 A

c. 8 A **d.** 10 A

2. When 5V potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is $2.5\times10^{-4}\,\text{ms}^{-1}$. If the electron density in the wire is $8\times10^{28}\,\text{m}^{-3}$, the resistivity of the material is close to:

a. $1.6\times10^{-8}\,\Omega\text{m}$ **b.** $1.6\times10^{-7}\,\Omega\text{m}$

c. $1.6\times10^{-6}\,\Omega\text{m}$ **d.** $1.6\times10^{-5}\,\Omega\text{m}$

3. A galvanometer having a coil resistance of 100 Ω gives a full scale deflection, when a current of 1 mA is passed through it. The value of the resistance, which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10 A, is:

a. $0.01\,\Omega$ **b.** $2\,\Omega$

c. $0.1\,\Omega$ **d.** $3\,\Omega$

4. The temperature dependence of resistances of Cu and undoped Si in the temperature range 300–400 K, is best described by:

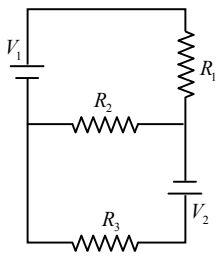
a. Linear increase for Cu, linear increase for Si

b. Linear increase for Cu, exponential increase for Si

c. Linear increase for Cu, exponential decrease for Si

d. Linear decrease for Cu, linear decrease for Si

5. Two ideal batteries of emf V_1 and V_2 and three resistances R_1, R_2 and are R_3 connected as shown in the figure. The current in resistance R_2 would be zero if



a. $V_1 = V_2$ and $R_1 = R_2 = R_3$

b. $V_1 = V_2$ and $R_1 = 2R_2 = R_3$

c. $V_1 = 2V_2$ and $2R_1 = 2R_2 = R_3$

d. $2V_1 = V_2$ and $2R_1 = R_2 = R_3$

6. Heater of an electric kettle is made of a wire of length L and diameter d. It takes 4 minutes to raise the temperature of 0.5 kg water by 40 K. This heater is replaced by a new heater having two wires of the same material, each of length L and diameter $2d$. The way these wires are connected is given in the options. How much time in minutes will it take to raise the temperature of the same amount of water by 40 K?

a. 4 if wires are in parallel

b. 2 if wires are in series

c. 1 if wires are in series

d. 0.5 if wires are in parallel

SECTION 2 (Maximum Marks: 24)

Numerical value answer type questions

7. A current 1 amp is flowing in the sides of an equilateral triangle of side $4.5\times10^{-2}\,\text{m}$. The magnetic field at the centroid of the triangle in the units of $\left(10^{-5}\,\text{T}\right)$ is

8. If the maximum values of signal and carrier waves are 4 V and 5V respectively, the percentage of amplitude modulation is a × 10%. What is the value of a ?

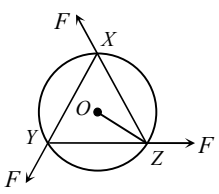
9. A signal wave of frequency 4.5 kHz is modulated with a carrier wave of frequency 3.45 MHz. The bandwidth of FM wave is kHz is

10. What is the maximum usuable frequency (in MHz) for E-layer of atmosphere having critical frequency 4 MHz, when the angle of incidence is 60°?

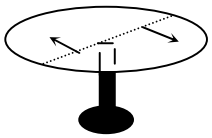
11. A T.V. Tower has a height 100 m. In order to triple its coverage range, the height of tower to be increased is $a\times10^2\,\text{m}$. What is the integer value of a?

12. A microwave telephone link operating at the central frequency of 10 GHz has been established. If 2% of this is available for microwave communication channel and each telephone is allotted a bandwidth of 8 kHz, the number of telephone channels which can be simultaneously granted is 2.5×10^a. What is the integer value of a ?

13. A uniform circular disc of mass $1.5\,\text{kg}$ and radius is initially at rest on a horizontal frictionless surface. Three forces of equal magnitude $F = 0.5\ N$ are applied simultaneously along the three sides of an equilateral triangle XYZ with its vertices on the perimeter of the disc (see figure). One second after applying the forces, the angular speed of the disc in rad s^{-1} is



14. A horizontal circular platform of radius 0.5 m and mass 0.45 kg is free to rotate about its axis. Two mass less spring toyguns, each carrying a steel ball of mass 0.05 kg are attached to the platform at a distance 0.25 m from the centre on its either sides along its diameter (see figure). Each gun simultaneously fires the balls horizontally and perpendicular to the diameter in opposite directions. After leaving the platform, the balls have horizontal speed of $9\,\text{ms}^{-1}$ with respect to the ground. The rotational speed of the platform in rad s^{-1} after the balls leave the platform is

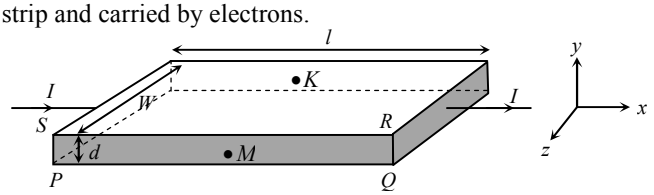


SECTION 3 (Maximum Marks: 12**)**

Paragraph based questions (2 paragraphs, each having 2 MCQs with one correct answer only)

Paragraph for Question No. 15 to 16

In a thin rectangular metallic strip a constant current I flows along the positive x-direction, as shown in the figure. The length, width and thickness of the strip are l, w and d, respectively. A uniform magnetic field $\vec{B}$ is applied on the strip along the positive y-direction. Due to this, the charge carriers experience a net deflection along the z-direction. This results in accumulation of charge carriers on the surface $PQRS$ and appearance of equal and opposite charges on the face opposite to $PQRS$. A potential difference along the z-direction is thus developed. Charge accumulation continues until the magnetic force is balanced by the electric force. The current is assumed to be uniformly distributed on the cross-section of the strip and carried by electrons.



15. Consider two different metallic strips (1 and 2) of the same material. Their lengths are the same, widths are w_1 and w_2 and thicknesses are d_1 and d_2, respectively. Two points K and M are symmetrically located on the opposite faces parallel to the x-y plane (see figure). V_1 and V_2 are the potential differences between K and M in strips 1 and 2, respectively. Then, for a given current I flowing through them in a given magnetic field strength B, the correct statement(s) is(are)

a. If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = 2V_1$

b. If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = V_1$

c. If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = 2V_1$

d. If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = V_1$

16. Consider two different metallic strips (1 and 2) of same dimensions (lengths ℓ, width w and thickness d) with carrier densities n_1 and n_2, respectively. Strip 1 is placed in magnetic field B_1 and strip 2 is placed in magnetic field B_2, both along positive y-directions. Then V_1 and V_2 are the potential differences developed between K and M in strips 1 and 2, respectively. Assuming that the current I is the same for both the strips, the correct option(s) is(are)

a. If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = 2V_1$

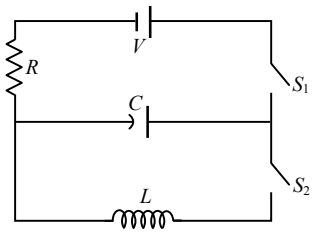
b. If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = V_1$

c. If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = 0.5V_1$

d. If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = V_1$

Paragraph for Question No. 17 to 18

The capacitor of capacitance C can be charged (with the help of a resistance R) by a voltage source V, by closing switch S_1 while keeping switch S_2 open. The capacitor can be connected in series with an inductor 'L' by closing switch S_2 and opening S_1



17. Initially, the capacitor was uncharged. Now, switch S_1 is closed and S_2 is kept open. If time constant of this circuit is τ, then

a. after time interval τ, charge on the capacitor is $CV/2$

b. after time interval 2τ, charge on the capacitor is $CV(1-e^{-2})$

c. the work done by the voltage source will be half of the heat dissipated when the capacitor is fully charged

d. after time interval 2τ, charge on the capacitor is $CV(1-e^{-1})$

18. After the capacitor gets fully charged, S_1 is opened and S_2 is closed so that the inductor is connected in series with the capacitor. Then

a. at $t = 0$, energy stored in the circuit is purely in the form of magnetic energy

b. at any time $t > 0$, current in the circuit is in the same direction

c. at $t > 0$, there is no exchange of energy between the inductor and capacitor

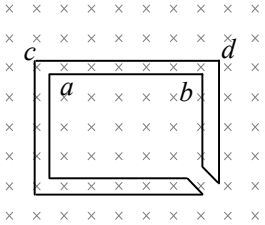
d. at any time $t > 0$, instantaneous current in the circuit may $V\sqrt{\dfrac{C}{L}}$

JEE ADVANCE PAPER-II

SECTION 1 (Maximum Marks: 24)

MCQs with one or more than one correct answer

1. The figure shows certain wire segments joined together to form a coplanar loop. The loop is placed in a perpendicular magnetic field in the direction going into the plane of the figure. The magnitude of the field increases with time. I_1 and I_2 are the currents in the segments *ab* and *cd*. Then,



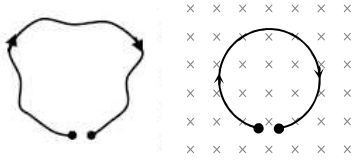
a. $I_1 > I_2$

b. $I_1 < I_2$

c. I_1 is in the direction ba and I_2 is in the direction cd

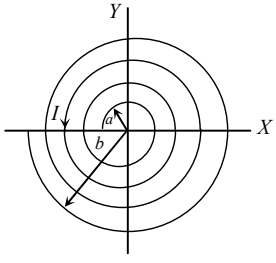
d. I_1 is in the direction ab and I_2 is in the direction dc

2. A thin flexible wire of length L is connected to two adjacent fixed points and carries a current I in the clockwise direction, as shown in the figure. When the system is put in a uniform magnetic field of strength B going into the plane of the paper, the wire takes the shape of a circle. The tension in the wire is



a. IBL **b.** $\frac{IBL}{\pi}$ **c.** $\frac{IBL}{2\pi}$ **d.** $\frac{IBL}{4\pi}$

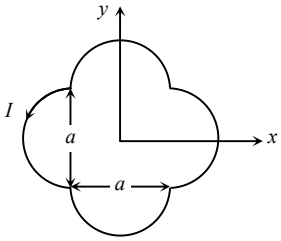
3. Along insulated copper wire is closely wound as a spiral of '*N*' turns. The spiral has inner radius '*a*' and outer radius '*b*'. The spiral lies in the X-Y plane and a steady current '*I*' flows through the wire. The Z component of the magnetic field at the center of the spiral is



a. $\frac{\mu_0 NI}{2(b-a)}\ln\left(\frac{b}{a}\right)$ **b.** $\frac{\mu_0 NI}{2(b-a)}\ln\left(\frac{b+a}{b-a}\right)$

c. $\frac{\mu_0 NI}{2b}\ln\left(\frac{b}{a}\right)$ **d.** $\frac{\mu_0 NI}{2b}\ln\left(\frac{b+a}{b-a}\right)$

4. A loop carrying current I lies in the x-y plane as shown in the figure. The unit vector $\hat{k}$ is coming out of the plane of the paper. The magnetic moment of the current loop is

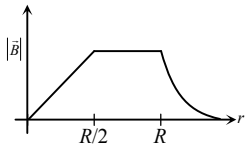


a. $a^2 I\hat{k}$ **b.** $\left(\frac{\pi}{2}+1\right)a^2 I\hat{k}$

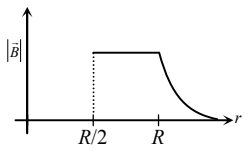
c. $-\left(\frac{\pi}{2}+1\right)a^2 I\hat{k}$ **d.** $(2\pi+1)a^2 I\hat{k}$

5. An infinitely long hollow conducting cylinder with inner radius $R/2$ and outer radius R carries a uniform current density along its length. The magnitude of the magnetic field, $|\vec{B}|$ as a function of the radial distance r from the axis is best represented by.

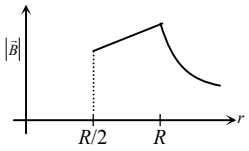
a.



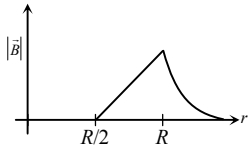
b.



c.



d.



6. A circular loop of radius 0.3 cm lies parallel to a much bigger circular loop of radius 20 cm. The centre of the small loop is on the axis of the bigger loop. The distance between their centres is 15 cm. If a current of 2.0 A flows through the smaller loop, then the flux linked with bigger loop is

a. 9.1×10^{-11} weber **b.** 6×10^{-11} weber

c. 3.3×10^{-11} weber **d.** 6.6×10^{-9} weber

SECTION 2 (Maximum Marks: 24)

Numerical value answer type questions

7. The isotope $^{12}_{5}B$ having a mass 12.014 *u* undergoes β - decay to $^{12}_{6}C$. $^{12}_{6}C$ has an excited state of the nucleus ($^{12}_{6}C^{*}$) at 4.041 MeV above its grounds state. If $^{12}_{5}B$ decays to $^{12}_{6}C^{*}$, the maximum kinetic energy of the β - particle in units of MeV is:

($1u = 931.5 MeV/c^2$, where c is the speed of light in vacuum)

8. A binary star consists of two stars *A* (mass 2.2Ms) and *B* (mass 11Ms), where Ms is the mass of the sun. They are separated by distance *d* and are rotating about their centre of mass, which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star *B* about the centre of mass is

9. Two homogeneous spheres *A* and *B* of masses *m* and 2*m* having radii 2*a* and *a* respectively are placed in contact. The ratio of distance of c.m. from first sphere to the distance of c.m. from second sphere is:

10. A non-uniform thin rod of length *L* is placed along *X*-axis so that one of its ends is at the origin. The linear mass density of rod is $\lambda = \lambda_0 x \cdot$ The centre of mass of rod divides the length of the rod in the ratio:

11. A sphere of mass 5 kg and diameter 2 m rotates about a tangent. What is its moment of inertia?

12. A uniform rod of length 1 m and mass 0.5 kg rotates at angular speed of 6 rad/sec about one of its ends. What is the KE of the rod?

13. A particle performing uniform circular motion has angular momentum *L*. When its angular velocity is doubled and KE is also doubled, the new angular momentum becomes *x* times. What is *x*?

14. The fundamental frequency of a closed organ pipe is equal to first overtone frequency of an open organ pipe. If the length of the open pipe is 36 cm, what is the length (in cm) of the closed pipe?

SECTION 3 (Maximum Marks: 12)

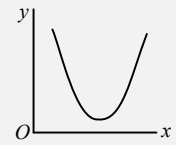
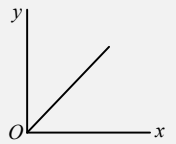
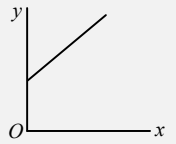
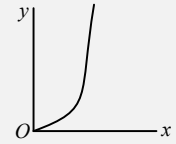
Matching type questions with 4 options

15. Match Column I with Column II and select the correct answer using the codes given below the Columns:

Column I	Column II
(A) Boltzmann Constant	**1.** $[ML^2T^{-1}]$
(B) Coefficient of viscosity	**2.** $[ML^{-1}T^{-1}]$
(C) Plank Constant	**3.** $[MLT^{-3}K^{-1}]$
(D) Thermal conductivity	**4.** $[ML^2T^{-2}K^{-1}]$

a. A-4; B-1, 4; C- 4; D-2
b. A-1; B- 1,4; C-4; D-4
c. A-4; B-1,4; C-4; D-2
d. A-1; B- 1,4; C-2; D-4

16. Column I give a list of possible set of parameters measured in some experiments. The variations of the parameters in the form of graphs are shown in Column II. Match the set of parameters given in Column I with the graph given in Column II.

Column I	Column II
(A) Potential energy of a simple pendulum (*y* axis) as a function of displacement (*x* axis)	**1.** 
(B) Displacement(*y* axis) as a function of time (*x* axis) for a one dimensional motion at zero or constant acceleration when the body is moving along the positive *x*-direction	**2.** 
(C) Range of a projectile (*y*-axis) as a function of its velocity (*x*-axis) when projected at a fixed angle	**3.** 
(D) The square of the time period (*y*-axis) of a simple pendulum as a function of its length (*x* -axis)	**5.** 

a. A-4; B- 1,4; C- 4; D- 2
b. A-1; B- 1,4; C- 4; D- 4
c. A-2; B- 1,4; C- 4; D- 2
d. A-1; B- 1,4; C- 2; D- 4

17. Motion is defined as rate of change of position. In the motions described in column I, match the facts about that motion in column II for the changes in position vector.

Column I	Column II
(A) Magnitude only	**1.** A coin dropped from roof of house
(B) Direction only	**2.** A coin thrown at any angle with horizontal
(C) Both in magnitude and direction	**3.** A coin held in your hand
(D) Remains invariant	**4.** A coin in rotated in circular path with variable speed

a. A-1; B-2; C-3; D-4
b. A-4; B-2; C-3; D-1
c. A-1; B-4; C-2; D-3
d. A-3; B-2; C-4; D-1

20. Match the statement of Column with those in Column II:

Column I	Column II
(A) In any Bohr orbit of the hydrogen atom, the ratio of kinetic energy to potential energy of the electron is	**1.** $-\frac{1}{2}$
(B) The ratio of the kinetic energy to the total energy of an electron in a Bohr orbit is	**2.** 2
(C) In the lowest energy level of hydrogen atom, the electron has the angular momentum	**3.** $\frac{h}{2\pi}$
(D) Ratio of the wavelengths of first line of Lyman series and first line of Balmer series is	**4.** 5 : 27

a. A-1; B-3,4; C-4; D-2,3
b. A-1; B-3,4; C-4; D-3
c. A-2; B-3,4; C-1; D-3
d. A-1; B-2,4; C-3; D-4

Space for Rough Work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	b	a	b	b	d	b	b	c	c
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	d	c	a	d	a	c	d	a	a
21.	**22.**	**23.**	**24.**	**25.**					
a	d	c	a	b					

1. **(a)** $\vec{A} = 2\hat{i} + 4\hat{j} - 5\hat{k}$

$\therefore \quad |\vec{A}| = \sqrt{(2)^2 + (4)^2 + (-5)^2} = \sqrt{45}$

$\therefore \quad \cos\alpha = \frac{2}{\sqrt{45}},\ \cos\beta = \frac{4}{\sqrt{45}},\ \cos\gamma = \frac{-5}{\sqrt{45}}$

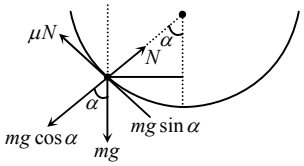
2. **(b)** $h = ut - \frac{1}{2}gt^2$

$\Rightarrow \quad h = u \times 2 - \frac{1}{2}g \times (10)^2 = u \times 10 - \frac{1}{2}g \times (10)^2$

$\Rightarrow \quad 2u - 2g = 10u - 50g \Rightarrow u = 6g$

$\therefore \quad h = 2u - 2g = 2 \times 6g - 2g = 10g = 98\ m$

3. **(a)** Equilibrium of insect gives

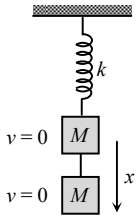


$N = mg\cos\alpha \quad \ldots(i)$

$\mu N = mg\ \sin\alpha \quad \ldots(ii)$

From equation (i) and (ii) we get $\cot\alpha = \frac{1}{\mu} = 3$

4. **(b)** Let x be the maximum extension of the spring. From conservation of mechanical energy decrease in gravitational potential energy = increase in elastic potential energy



$\therefore \quad Mg\ x = \frac{1}{2}kx^2$ or $x = \frac{2Mg}{k}$

5. **(b)** For block P, friction will provide the necessary restoring force.

$\therefore \quad f_{max} = m\omega^2 A$ with $\omega^2 = \frac{k}{m+m} = \frac{k}{2m}$

Hence, $f_{max} = m\left(\frac{k}{2m}\right)A = \frac{kA}{2}$

6. **(d)** Work done, $W = T\Delta A$

or $W = T[(4\pi R^2)_n - 4\pi R^2] = T.4\pi[nr^2 - R^2]$

Where $n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$ and $W = \frac{1}{2}m\upsilon^2$

$\therefore \quad \frac{1}{2}m\upsilon^2 = T.4\pi(nr^2 - R^2)$

or $\frac{1}{2} \times \frac{4}{3}\pi R^3 \rho\upsilon^2 = T.4\pi(nr^2 - R^2)$

or $\upsilon = \sqrt{\frac{6T}{\rho}\left(\frac{nr^2}{R^3} - \frac{R^2}{R^3}\right)} = \sqrt{\frac{6T}{\rho}\left(\frac{nr^2}{nr^3} - \frac{1}{R}\right)} = \sqrt{\frac{6T}{\rho}\left(\frac{1}{r} - \frac{1}{R}\right)}$

7. **(b)** $\frac{\Delta P}{\Delta V/V} = \frac{mg/A}{\Delta V/V}$ $\frac{\Delta V}{V} = \frac{mg}{AK} \quad \ldots(i)$

But for a sphere $V = \frac{4}{3}\pi R^2$

Differentiating $\delta V = \frac{4}{3}\pi 3R^2(\delta R)$

$\Rightarrow \quad \frac{\delta V}{V} = \frac{\frac{4}{3}\pi(3R^2)\delta R}{\frac{4}{3}\pi R^3} = \frac{3\delta R}{R}$

$\therefore \quad$ From (i) $\frac{3\delta V}{R} = \frac{mg}{AK} \Rightarrow \frac{\delta R}{R} = \frac{mg}{3AK}$

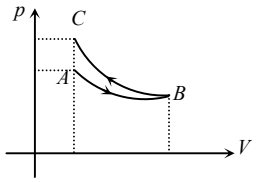
8. **(b)** $\Delta\lambda = \frac{0.05}{100}\lambda$, As $\Delta\lambda = \frac{v}{c}\lambda$

$\Rightarrow \quad \frac{0.05}{100}\lambda = \frac{v}{c}\lambda$ or $v = 5 \times 10^{-4}c$

or $v = 5 \times 10^{-4} \times 3 \times 10^8 = 1.5 \times 10^5$ m/s

As λ decreases, the star is approaching the observer.

9. **(c)** Slope of adiabatic process at a given state (p, V, T) is more than the slope of isothermal process. The corresponding p-V graph for the two processes is as shown in figure. In the figure, AB is isothermal and BC is adiabatic.



W_{AB} = positive (as volume is increasing) and

W_{BC} = negative (as volume is decreasing) plus,

$|W_{BC}| > |W_{AB}|$, as area under p-V graph gives the work done.

Hence, $W_{AB} + W_{BC} = W < 0$

From the graph itself, it is clear that $p_3 > p_1$.

Hence, the correct option is

Note: At point B, slope of adiabatic (process BC) is greater than the slope of isothermal (process AB).

10. **(c)** $P_1 V_1^{\gamma} = P_2 V_2^{\gamma} \Rightarrow P_2 = P_1\left(\frac{V_1}{V_2}\right)^{\gamma}$

$$\text{work} = \frac{R}{1-\gamma}\left(P_2 V_2 - P_1 V_1\right) = \frac{1}{1-\gamma}\left[P_1\left(\frac{V_1}{V_2}\right)^{\gamma-1} - 1\right]$$

$$= \frac{1}{1-1.40}\times 10^5 \times 22.4\times 10^{-3}\times\left[(2)^{0.40} - 1\right]$$

$$= \frac{1}{0.40}\times 22.4\times 10^2\left[1.32 - 1\right] = 1792\ \text{J}$$

11. **(c)** $\frac{dQ}{dt} = L\left(\frac{dm}{dt}\right)$ or $\frac{\text{Temperature difference}}{\text{Thermal resistance}} = L\left(\frac{dm}{dt}\right)$

or $\frac{dm}{dt} \propto \frac{1}{\text{Thermal resistance}} \Rightarrow q \propto \frac{1}{R}$

In the first case rods are in parallel and thermal resistance is $\frac{R}{2}$ while in second case rods are in series and thermal resistance is $2R$.

$\Rightarrow \quad \frac{q_1}{q_2} = \frac{2R}{R/2} = \frac{4}{1}$

12. **(d)** Energy $u \propto E^2$ and $u \propto \frac{1}{r^2}$ for a point charge; so $u \propto \frac{1}{r^4}$

13. **(c)** For short bar magnet in tan A-position

$$\frac{\mu_0}{4\pi}\frac{2M}{d^3} = H\tan\theta \qquad \ldots (i)$$

When distance is doubled, then new deflection θ' is given by

$$\frac{\mu_0}{4\pi}\frac{2M}{(2d)^3} = H\tan\theta' \qquad \ldots (ii)$$

$\therefore \quad \frac{\tan\theta'}{\tan\theta} = \frac{1}{8}$

$\Rightarrow \quad \tan\theta' = \frac{\tan\theta}{8} = \frac{\tan 60^\circ}{8} = \frac{\sqrt{3}}{8} \Rightarrow \theta' = \tan^{-1}\left(\frac{\sqrt{3}}{8}\right)$

14. **(a)** $\cos\phi = \frac{1}{2} \Rightarrow \phi = 60^\circ$

$$\tan 60^\circ = \frac{\omega L}{R} \Rightarrow L = \frac{\sqrt{3}}{\pi}\text{H}$$

15. **(d)** $\frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \left(\frac{2+1}{2-1}\right)^2 = \frac{9}{1}$

16. **(a)** In the given question, the electromagnetic wave is propagating along $+z$ axis. In e.m. wave, the electric field $(\vec{E})$ and magnetic field $(\vec{B})$ are perpendicular to each other and also perpendicular to the direction of propagation of wave,

So, $\vec{E} = E_0\,\hat{i}$ and $\vec{B} = B_0\,\hat{j}$.

17. **(c)** The wavelength of spectral line in Balmer series is given by $\frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{n^2}\right]$

For first line of Balmer series, $n = 3$

$\Rightarrow \quad \frac{1}{\lambda_1} = R\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = \frac{5R}{36}$; For second line $n = 4$.

$\Rightarrow \quad \frac{1}{\lambda_2} = R\left[\frac{1}{2^2} - \frac{1}{4^2}\right] = \frac{3R}{16}$

$\therefore \quad \frac{\lambda_2}{\lambda_1} = \frac{20}{27} \Rightarrow \lambda_1 = \frac{20}{27}\times 6561 = 4860\,\text{Å}$

18. **(d)** ${}_6C^{11} \rightarrow {}_5B^{11} + \beta^+ + \gamma$ because $\beta^+ = {}_1e^0$

19. **(a)** The base is always thin

20. **(a)** VHF (Very High Frequency) band having frequency range 30 MHz to 300 MHz is typically used for TV and radar transmission.

21. **(a)** $V = 5\cos\omega t = 5\sin\left(\omega t + \frac{\pi}{2}\right) \Rightarrow I = 2\sin\omega t$

$\therefore \quad \phi = \frac{\pi}{2} \Rightarrow P = E_{rms} I_{rsm}\cos\frac{\pi}{2} = 0$ W

22. **(d)** Density $\rho = \frac{m}{\pi r^2 L}$

$\therefore \quad \frac{\Delta\rho}{\rho}\times 100 = \left(\frac{\Delta m}{m} + 2\frac{\Delta r}{r} + \frac{\Delta L}{L}\right)\times 100$

After substituting the values, we get the maximum percentage error in density = 4%

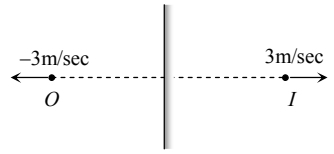
23. **(c)** $L = 40$ m, $v = 1080$ km,

$h^{-1} = 300$ m sec^{-1} and $B = 8.3\times 10^{-4}$ T

$\Rightarrow \quad e = Blv = 8.3\times 10^{-4}\times 40\times 300 = 1$ V

24. **(a)** Relative velocity of image *w.r.t.* object

$= 3 - (-3) = 6$ m/sec



25. **(b)** Angular momentum of satellite about center of earth remains constant, i.e., mvr = constant

$\Rightarrow \quad v \propto \frac{1}{2} \quad \Rightarrow \quad \frac{v_2}{v_1} = \frac{r_1}{r_2}$

Speed is maximum at position 2,

$\therefore \quad \frac{v_{\max}}{v_{\min}} = \frac{r_1}{r_2} = \frac{(1+\varepsilon)a}{(1-\varepsilon)a} = \frac{1+\varepsilon}{1-\varepsilon} = \frac{1+0.0167}{1-0.0167} = 1.033$

JEE Advance Paper -I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	d	a	c	a,b,d	b,d	4	8	9	8
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**		
8	4	2	4	a,d	a,c	b	d		

1. **(a)** $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$\Rightarrow \quad P = \frac{V^2}{R_{eq}} = \frac{V^2}{R_1} + \frac{V^2}{R_2} + ...$

$= 15\times 40 + 5\times 100 + 80\times 5 + 1\times 1000 = 2500\,\text{W}$

$\Rightarrow \quad P = VI$

$\Rightarrow \quad P = 2500 = 220\,I \Rightarrow \; I = \frac{250}{22} = \frac{125}{11}$

2. **(d)** $\upsilon_d = \frac{i}{Ane} \qquad \therefore \; \frac{i}{A} = \upsilon_d \cdot ne$

$\Rightarrow \quad i = \frac{V}{R} = \frac{V}{\rho\frac{\ell}{A}} \qquad \therefore \; \rho = \frac{V}{\frac{i}{A}\ell}$

$\Rightarrow \quad \rho = \frac{V}{\upsilon_d \cdot ne\,\ell}$

$\Rightarrow \quad \rho = \frac{5}{2.5\times 10^{-4}\times 8\times 10^{28}\times 1.6\times 10^{-19}\times 0.1}$

$= \frac{2}{8\times 1.6\times 10^{-4}} = \frac{1}{6.4}\times 10^{-4} = 1.6\times 10^{-5}\,\Omega\,\text{m}$

3. **(a)** $S = \frac{i_g G}{1 - i_g}$ Here $i_g = 10^{-3}\,\text{A}$

$\Rightarrow \quad G = 10^2\,\Omega,\; I = 10A \Rightarrow S = 10^{-2}\,\Omega$

4. **(c)** For conductor (Cu) resistance increases linearly and for semiconductor resistance decreases Exponentially in given temperature range.

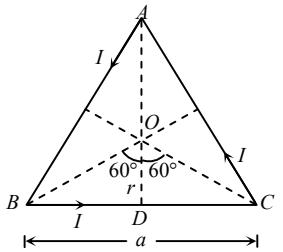
5. **(a, b, d)** $V_1 = \frac{R_1(V_1 + V_2)}{R_1 + R_3} \qquad \Rightarrow V_1 R_3 = V_2 R_1$

$\Rightarrow \quad V_2 = \frac{R_3(V_1 + V_2)}{R_1 + R_3} \qquad \Rightarrow V_2 R_1 = V_2 R_3$

6. **(b, d)** $H = \frac{V^2}{R}4 = \frac{V^2}{R/2}t_1 \frac{V^2}{R/8}t_2$

$\Rightarrow \quad t_1 = 2\,\text{min}. \; \Rightarrow \; t_2 = 0.5\,\text{min}.$

7. **(4)** Refer Fig., The magnetic field induction at the centroid O due to current I through one side BC of the triangle will be $B_1 = \frac{\mu_0}{4\pi}\frac{I}{r}.\,(\sin\theta_1 + \sin\theta_2)$



It will be acting perpendicular to the plane of triangle upwards. Total magnetic field induction at O due to current through all the three sides of the triangle will be

$B = 3B_1 = \frac{3\mu_0}{4\pi}\frac{I}{r}[\sin\theta_1 + \sin\theta_2]$

Here, $I = 1$ A, $\theta_1 = 60° = \theta_2$

And $r = OD = \frac{BD}{\tan 60°} = \frac{a/2}{\sqrt{3}} = \frac{a}{2\sqrt{3}} = \frac{4.5\times 10^{-2}}{2\sqrt{3}}\,\text{m}$

$\therefore \quad B = 3\times 10^{-7}\times\frac{1}{\left(4.5\times 10^{-2}/2\sqrt{3}\right)}\times\left[\sin 60° + \sin 60°\right]$

On solving, $B = 4 \times 10$–5 T.

8. **(8)** Here, $A_m = 4$ V ; $A_c = 5$ V

Percentage of modulation,

$\mu = \frac{A_m}{A_c}\times 100 = \frac{4}{5}\times 100 = 80\% = a\times 10\%. \Rightarrow a = 8$

9. **(9)** Here, $v_s = 4.5\,\text{kHz}; v_c = 3.45\,\text{MHz} = 3450\,\text{kHz}$

Bandwidth $= 2v_s = 2\times 4.5 = 9.0\,\text{kHz}.$

10. **(8)** Here, $v_c = 4\,\text{MHz}, i = 90°$

Maximum usuable frequency $= v_c \sec i$

$= 4\times\sec 60° = 4\times 2 = 8\,\text{MHz}$

11. **(8)** $d' = \sqrt{2h'R} = 3d = 3\sqrt{2hR}$

or $h' = 9h = 9\times 100 = 900\,m$

Increase in height of tower $= 900 - 100 = 800 = a\times 10^2$

$\Rightarrow \quad a = 8$

12. **(4)** Microwave frequency used in telephone link = 10 GHz

$= 10 \times 10^9$ Hz $= 10^{10}$ Hz

Frequency available for microwave communication

$= 2\%$ of 10 GHz $= \frac{2}{100}\times 10^{10} = 2\times 10^8\,\text{Hz}$

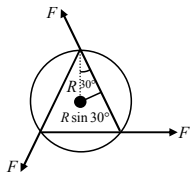
Bandwidth of each telephone channel = 8 kHz

$= 8\times 10^3$ Hz

Number of microwave telephone channels

$= \frac{2\times 10^8}{8\times 10^3} = 2.5\times 10^4 = 2.5\times 10^a \;\; \therefore \; a = 4$

13. **(2)** $\tau = I\alpha$



$\Rightarrow \quad 3FR\sin 30° = I\alpha$

$\Rightarrow \quad I = \dfrac{MR^2}{2}$

$\Rightarrow \quad \alpha = 2$

$\Rightarrow \quad \omega = \omega_0 + \alpha t \Rightarrow \omega = 2\ rad/s$

14. **(4)** Since net torque about centre of rotation is zero, so we can apply conservation of angular momentum of the system about centre of disc $L_i = L_f$

$0 = I\omega + 2mv\,(r/2);$ comparing magnitude

$\therefore \quad \left(\dfrac{0.45\times 0.5\times 0.5}{2}\right)\omega = 0.05\times 9\times\dfrac{0.5}{2}\times 2$

$\therefore \quad \omega = 4$

15. **(a, d)** $I_1 = I_2$

$\Rightarrow \quad neA_1v_1 = neA_2v_2 \Rightarrow d_1w_1v_1 = d_2w_2v_2$

Now, potential difference developed across MK

$V = Bvw$

$\Rightarrow \quad \dfrac{V_1}{V_2} = \dfrac{v_1w_1}{v_2w_2} = \dfrac{d_2}{d_1}$ and hence, correct choice is a & d.

16. **(a, c)** As $I_1 = I_2\ n_1w_1d_1v_1 = n_2w_2d_2v_2$

Now, $\dfrac{V_2}{V_1} = \dfrac{B_2v_2w_2}{B_2v_1w_1} = \left(\dfrac{B_2w_2}{B_1w_1}\right)\left(\dfrac{n_1w_1d_1}{n_2w_2d_2}\right) = \dfrac{B_2n_1}{B_1n_2}$

$\therefore$ Correct options are a & c.

17. **(b)** $Q = Q_0(1 - e^{-t/\tau})$

$Q = CV(1 - e^{-t/\tau})$ after time interval 2τ.

18. **(d)** $q = Q_0\cos\omega t$

$\Rightarrow \quad i = -\dfrac{dq}{dt} = Q_0\omega\sin\omega t$

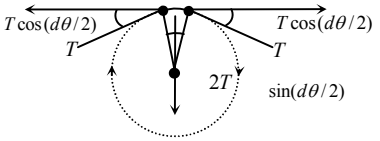
$\Rightarrow \quad i_{kax} = C\omega V = V\sqrt{\dfrac{C}{L}}$

JEE Advance Paper -II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	c	a	b	d	a	9	6	2	2/3
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**		
7	3	1	9	c	a	c	d		

1. **(d)** According to Lenz's Law, current will be in anticlockwise sense as magnetic field is increasing into the plane of paper.

2. **(c)** $2T\sin\dfrac{d\theta}{2} = BiRd\theta$ (for θ small)

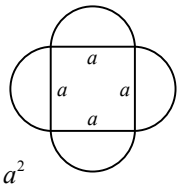


$\Rightarrow \quad Td\theta = BiRd\theta$

$\Rightarrow \quad T = BiR = \dfrac{BiL}{2\pi}$

3. **(a)** $\displaystyle\int_a^b \frac{\mu_0 IN}{2r(b-a)}dr = \frac{\mu_0 IN}{2(b-a)}\int_a^b \frac{dr}{r} = \frac{\mu_0 IN}{2(b-a)}\ln\left(\frac{b}{a}\right)$

4. **(b)** $M = IA$



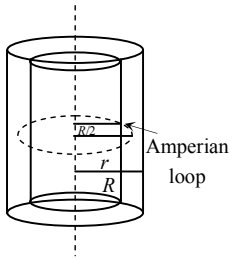
$\Rightarrow \quad A = 4\times\dfrac{\pi\left(\dfrac{a}{2}\right)^2}{2} + a^2$

$= 2\times\dfrac{\pi a^2}{4} + a^2 = \pi\left(\dfrac{a^2}{2}\right) + a^2\left(\dfrac{\pi}{2} + 1\right)a^2$

$\therefore \quad M = \left(\dfrac{\pi}{2} + 1\right)a^2 I\hat{k}$

5. **(d)** Let current be *I*.

Current density $J = \dfrac{I}{\pi\left(R^2 - \dfrac{R^2}{4}\right)}$



$\Rightarrow \quad J = \dfrac{I}{\pi\dfrac{3R^2}{4}} = \dfrac{4I}{3\pi R^2}$

Field for $\left(r \le \dfrac{R}{2}\right)B = 0$ For $\dfrac{R}{2} < r \le R$

Applying Ampere's Law $B.2\pi r$

$= \mu_0\dfrac{4I}{3\pi R^2}.\pi\left(r^2 - \dfrac{R^2}{4}\right) = \dfrac{4\mu_0 I}{3R^2}\left(r^2 - \dfrac{R^2}{4}\right)$

$B = \dfrac{2\mu_0 I}{3\pi R^2}\left(r - \dfrac{R^2}{4r}\right) \quad \Rightarrow$ At $r = \dfrac{R}{2}, B = 0$

$\therefore$ At $r = \dfrac{R}{2}$ filed is continuous.

From the above expression as *r* increases *B* increases.

For $r \ge R\ \ B.2\pi r = \mu_0 I$

$\Rightarrow \quad B = \dfrac{\mu_0 I}{2\pi r} \qquad \therefore B \propto \dfrac{1}{r}$

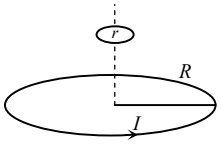
$\Rightarrow$ At $r = R, \qquad B = \dfrac{\mu_0 I}{2\pi R}$

From inside expression at $r = R$

$\Rightarrow \quad B = \dfrac{2\mu_0 I}{3\pi R^2}.\left(R - \dfrac{R^2}{4R}\right) = \dfrac{\mu_0 I}{2\pi R}$

This proves the continuity in the graph at $r = R$. From the above only correct option is (d).

6. **(a)** First we find the mutual inductance of the assembly.



Let current I flow through the larger loop.

The strength of induction at the smaller loop

$$= \frac{\mu_0 I R^2}{2(R^2+d^2)^{3/d}}$$

$\therefore$ Flux through smaller loop $= \dfrac{\mu_0 \pi I R^2 r^2}{2(R^2+d^2)^{3/2}}$

$\therefore$ Mutual inductance $= \dfrac{\mu_0 R^2 r^2}{2(R^2+d^2)^{3/2}}$

$\therefore$ Flux linked through coil $= \dfrac{\mu_0 \pi R^2 r^2}{2(R^2+d^2)^{3/2}} .i$

$$= \frac{4\pi \times 10^{-7} \times \pi \times 10^{-2} \times 9 \times 10^{-6} \times 2}{2(4\times 10^{-2} + 2.25 \times 10^{-2})^{3/2}}$$

$$= \frac{16}{2} \times \frac{10 \times 18}{15.625} \times (10^{+3-15}) = 9.1 \times 10^{-11} \text{ weber}$$

7. **(9)** ${}^{12}_{5}B \longrightarrow {}^{12}_{6}C + \beta^- + \gamma^-$

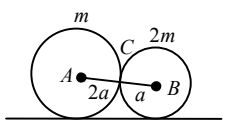
$$Q = (m_5 B^{12} - m_6^{12} C^*)c^2 = [m_5 B^{12} - (m_6 C^{12} + \Delta m)]C^2$$

$$= (m_5 B^{12} - m_6 C^{12})C^2 - \Delta m C^2 = 0.014 \times 931 - 4.041 = 9$$

8. **(6)** $\dfrac{L_{\text{total}}}{L_B} = \dfrac{m_1 r_1^2}{m_2 r_2^2} + 1$

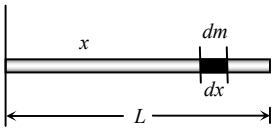
9. **(2)** Assuming that masses of the two spheres are concentrated at their centres *A* and *B* and taking $x = 0$ at *A*, Fig., we get

$$x_{cm} = \frac{m\times 0 + 2m \times 3a}{(m+2m)} = 2a = AC$$



$$CB = a \quad \therefore \quad \frac{AC}{CB} = \frac{2a}{a} = 2 .$$

10. **(2/3)** As shown in Fig., mass of small element of length *dx* is $dm = \lambda\, dx = \lambda_0\, x\, dx$



$$\Rightarrow \quad x_{cm} = \frac{\int_0^L x\,dm}{\int dm} = \frac{\int_0^L x(\lambda_0 x\,dx)}{\int_0^L \lambda_0 x\,dx}$$

$$x_{cm} = \frac{[x^3/3]_0^L}{[x^2/2]_0^L} = \frac{L^3/3}{L^2/2} = \frac{2L}{3}$$

If $x_1 = \dfrac{2L}{3}$, $x_2 = L - \dfrac{2L}{3} = \dfrac{L}{3} \Rightarrow \dfrac{x_1}{x_2} = \dfrac{2L/3}{L/3} = \dfrac{1}{2} .$

11. **(7)** Here, $m = 5$ kg; $r = \dfrac{2}{2} = 1m$; $I = ?$

Moment of inertia of the sphere about a tangent to the sphere, $I = \dfrac{7}{5}mr^2 = \dfrac{7}{5} \times 5 \times 1^2 = 7 \text{kgm}^2 .$

12. **(3)** Here, $l = 1$ m, $m = 0.5$ kg, $\omega = 6$ rad/sec

$$KE = \frac{1}{2} I\omega^2 = \frac{1}{2}\left(\frac{ml^2}{3}\right)\omega^2 = \frac{1}{2} \times \frac{0\cdot 5 \times 1^2}{3} \times 6^2 = 3 \text{ J} .$$

13. **(1)** Here, $L_1 = I_1\omega_1$, When $\omega_2 = 2\omega_1$,

and $KE = \dfrac{1}{2} I_2 \omega_2^2 = 2\left(\dfrac{1}{2} I_1 \omega_1^2\right)$

$$\frac{1}{2} I_2 (2\omega_1)^2 = 2 \times \frac{1}{2} I_1 \omega_1^2; \quad I_2 = \frac{1}{2} I_1$$

$\therefore$ $L_2 = I_2\omega_2 = \dfrac{1}{2} I_1 \times 2\omega_1 = I_1\omega_1 = L_1 \Rightarrow x = \dfrac{L_2}{L_1} = 1 .$

14. **(9)** Let l_1 be the length of closed pipe $l_2 = 36\text{cm}$= length of open pipe

Fundamental frequency of closed pipe $n_1 = \dfrac{\upsilon}{4l_1}$

Frequency of first overtone of open pipe,

$$n_2 = 2 \times \frac{\upsilon}{2l_1} = \frac{\upsilon}{l_2} = \frac{\upsilon}{36} \quad \text{As } n_1 = n_2$$

$\therefore$ $\dfrac{\upsilon}{4l_1} = \dfrac{\upsilon}{36}; 4l_1 = 36 \quad \Rightarrow l_1 = \dfrac{36}{4} = 9\text{cm}.$

15. **(c)** A-4; B-1, 4; C- 4; D-2

A. $KE = \dfrac{3}{2}KT \Rightarrow [ML^2T^{-2}] = K'[K] \Rightarrow K' = [ML^2T^{-2}K^{-1}]$

B. $F = 6\pi\eta r v \Rightarrow [MLT^{-2}] = \eta[L][LT^{-1}] \Rightarrow \eta[ML^{-1}T^{-1}]$

C. $E = hf \Rightarrow [ML^2T^{-2}] = \dfrac{h}{[T]} \Rightarrow h = [ML^2T^{-1}]$

D. $\dfrac{dQ}{dt} = \dfrac{K'A(\Delta T)}{\Delta x} \Rightarrow \dfrac{[ML^2T^{-2}]}{[T]} = \dfrac{k[L^2][K']}{[L]}$

$\Rightarrow$ $K' = [MLT^{-3}K^{-1}]$

16. **(a)** A-4; B- 1,4; C- 4; D- 2

17. **(c) c.** A-1; B-4; C-2; D-3

When coin is dropped if falls is straight line where direction is same but magnitude changes with change in position. When a coin is rotated in circular path direction goes on changing but radius remains constant. In projectile motion both direction and magnitude of *r* changes. *A* coin held in your hand has constant position vector and hence it is invariant.

18. **(d)** A-1, B-2,4, C-3, D-4

□ □ □

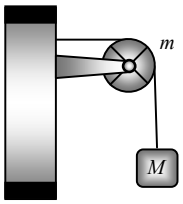
JEE-MAIN: PHYSICS MOCK TEST-4

SECTION 1 (Multiple Choice Question)

1. Which of the following sets have different dimensions?
a. Pressure, Young's modulus, Stress
b. Emf, Potential difference, Electric potential
c. Heat, Work done, Energy
d. Dipole moment, Electric flux, Electric field

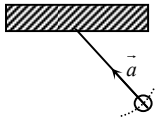
2. The 100 coplanar forces each equal to 10 N act on a body. Each force makes angle $\frac{\pi}{50}$ with the preceding force. What is the resultant of the forces?
a. 1000 N **b.** 500 N
c. 250 N **d.** Zero

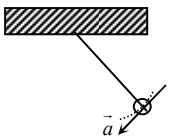
3 A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by

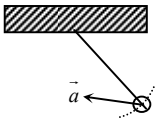


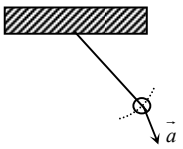
a. $\sqrt{2}Mg$ **b.** $\sqrt{2}mg$
c. $\sqrt{(M+m)^2+m^2}g$ **d.** $(\sqrt{(M+m)^2+M^2})g$

4. A simple pendulum is oscillating without damping. When the displacement of the bob is less than maximum, its acceleration vector $\vec{a}$ is correctly shown in

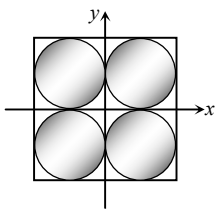
a.


b.


c.


d.


5. Four holes of radius R are cut form of a thin square plate of side $4R$ and mass M. The moment of inertia of the remaining portion about z-axis is:



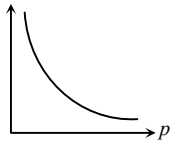
a. $\frac{\pi}{12}MR^2$ **b.** $\left(\frac{4}{3}-\frac{\pi}{4}\right)MR^2$
c. $\left(\frac{8}{3}-\frac{10\pi}{16}\right)MR^2$ **d.** $\left(\frac{4}{3}-\frac{\pi}{6}\right)MR^2$

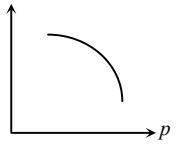
6. The radius of earth is 6400 km and g = 10 m/s^2. In order that a body of 5 kg weighs zero at the equator, the angular speed of earth in radian/sec is:
a. $\frac{1}{80}$ **b.** $\frac{1}{400}$
c. $\frac{1}{800}$ **d.** $\frac{1}{1600}$

7. The lower end of a capillary tube of radius r is placed vertically in water. Then with the rise of water in the capillary, heat evolved is:
a. $+\frac{\pi^2r^2h^2}{J}dg$ **b.** $+\frac{\pi^2r^2h^2dg}{2J}$
c. $-\frac{\pi r^2h^2dg}{2J}$ **d.** $-\frac{\pi r^2h^2dg}{J}$

8. A wire of length L and cross-sectional area A is made of a material of Young's modulus Y. If the wire is stretched by the amount x, the work done is:
a. $\frac{YAx^2}{2L}$ **b.** $\frac{YAx^2}{L}$
c. $\frac{YAx}{2L}$ **d.** Yax^2L

9. Which of the following graphs correctly represent the variation of $\beta = -\frac{dV/dp}{V}$ with p for an ideal gas at constant temperature?

a.


b.


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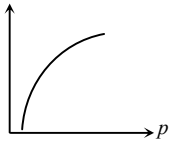

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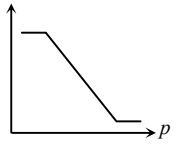


10. Two rods of different materials and identical cross-sectional area are joined face to face at one end and their free ends are fixed to the rigid walls. If the temperature of the surroundings is increased by 30°C, the magnitude of the displacement of the joint of rods is: (length of the

rods $l_1 = l_2 = 1$ unit, ratio of their Young's modulii, $Y_1 / Y_2 = 2$; coefficients of linear expansion are α_1 and α_2)

a. $5(\alpha_2 - a_1)$ **b.** $10(\alpha_1 - a_2)$

c. $10(\alpha_2 - 2a_1)$ **d.** $5(2\alpha_1 - a_2)$

11 A steel ball of mass $m_1 = 1\,\text{kg}$ moving with velocity 50 ms^{-1} collides with another stationary steel ball of mass $m_2 = 0.200\,\text{kg}$. During the collision, their internal energies change equally. If T_1 and T_2 are rise in temperature of balls m_1 and m_2 respectively, specific heat of steel = 0.105 and mechanical equivalent of heat J = 4.2 J/cal, then:

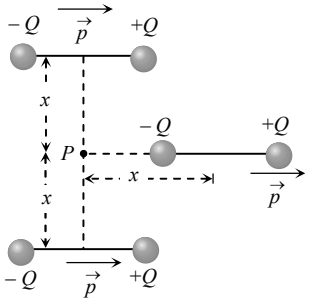
a. $T_1 = 7.0\times10^3\ ^\circ\text{C}, T_2 = 1.4\times10^3\ ^\circ\text{C},$

b. $T_1 = 1.4\times10^3\ ^\circ\text{C}, T_2 = 7.0\times10^3\ ^\circ\text{C},$

c. $T_1 = 1.4\ ^\circ\text{C}, T_2 = 7.0\ ^\circ\text{C},$

d. $T_1 = 7.0\ ^\circ\text{C}, T_2 = 1.4\ ^\circ\text{C},$

12. Three identical dipoles are arranged as shown below. What will be the net electric field at P $\left(k = \dfrac{1}{4\pi\varepsilon_0}\right)$?



a. $\dfrac{k.p}{x^3}$ **b.** $\dfrac{2kp}{x^3}$

c. Zero **d.** $\dfrac{\sqrt{2}\,kp}{x^3}$

13. The area of the plates of a parallel plate capacitor is A and the distance between the plates is 10 mm. There are two dielectric sheets in it, one of dielectric constant 10 and thickness 6 mm and the other of dielectric constant 5 and thickness 4 mm. The capacity of the capacitor is:

a. $\dfrac{12}{35}\varepsilon_0 \text{A}$ **b.** $\dfrac{2}{3}\varepsilon_0 \text{A}$

c. $\dfrac{5000}{7}\varepsilon_0 \text{A}$ **d.** $1500\,\varepsilon_0 \text{A}$

14. If a magnet is suspended at an angle 30° to the magnetic meridian, it makes an angle of 45° with the horizontal. The real dip is

a. $\tan^{-1}(\sqrt{3}/2)$ **b.** $\tan^{-1}(\sqrt{3})$

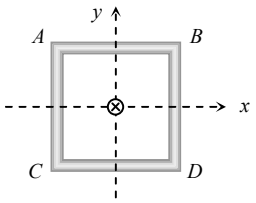
c. $\tan^{-1}(\sqrt{3/2})$ **d.** $\tan^{-1}(2/\sqrt{3})$

15. Two straight long conductors *AOB* and *COD* are perpendicular to each other and carry currents i_1 and i_2. The magnitude of the magnetic induction at a point *P* at a distance *a* from the point *O* in a direction perpendicular to the plane *ACBD* is

a. $\dfrac{\mu_0}{2\pi a}(i_1 + i_2)$ **b.** $\dfrac{\mu_0}{2\pi a}(i_1 - i_2)$

c. $\dfrac{\mu_0}{2\pi a}(I_1^2 + I_2^2)^{1/2}$ **d.** $\dfrac{\mu_0}{2\pi a}\dfrac{I_1 I_2}{(I_1 + I_2)}$

16. A circular coil and a bar magnet placed nearby are made to move in the same direction. The coil covers a distance of 1 m in 0.5 sec and the magnet a distance of 2 m in 1 sec. The induced e.m.f. produced in the coil:



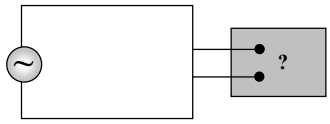
a. Zero

b. 1 V

c. 0.5 V

d. Cannot be determined from the given information

17. Following figure shows an ac generator connected to a "box" through a pair of terminals. The box contains possible *R*, *L*, *C* or their combination, whose elements and arrangements are not known to us. Measurements outside the box reveals that $e = 75\sin(\sin\omega t)$ volt, i = 1.5 sin (ωt + 45°) amp, then, the wrong statement is



a. There must be a capacitor in the box

b. There must be an inductor in the box

c. There must be a resistance in the box

d. The power factor is 0.707

18. The focal length of a concave mirror is *f* and the distance from the object to the principle focus is *x*. The ratio of the size of the image to the size of the object is

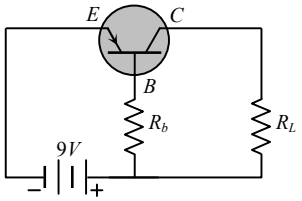
a. $\dfrac{f+x}{f}$ **b.** $\dfrac{f}{x}$

c. $\sqrt{\dfrac{f}{x}}$ **d.** $\dfrac{f^2}{x^2}$

19. The energy of incident photon is 12.375 eV while the energy of scattered photon is 9.375 eV. Then the kinetic energy of recoil electron is

a. 3 eV **b.** less than 3 eV
c. more than 3 eV **d.** 21.75 eV

20. In a transistor circuit shown here the base current is 35 μA. The value of the resistor R_b is



a. 123.5 kΩ **b.** 257 k Ω
c. 380.05 k Ω **d.** None of these

SECTION 2 (Numeric Value Question)

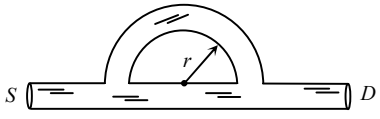
21. A wooden block of mass 10 *g* is dropped from the top of a cliff 100 m high. Simultaneously a bullet of mass 10 *g* is fired from the foot of the cliff upwards with a velocity of 100 m/s. The bullet and the wooden block will meet, each other, after a time:

a. 10 s **b.** 0.5 s
c. 1 s **d.** 7 s

22. The displacement of a particle varies according to the relation $x = 4(\cos \pi t + \sin \pi t)$. The amplitude of the particle is:

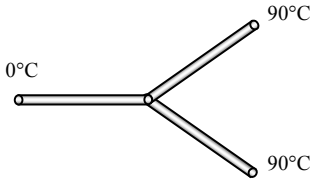
a. 8 **b.** – 4
c. 4 **d.** $4\sqrt{5}$

23. A sound wave of wavelength 32 cm enters the tube at *S* as shown in the figure. Then the smallest radius *r* so that a minimum of sound is heard at detector *D* is.



a. 7cm **b.** 14 cm
c.21cm **d.** 28 cm

24. Three rods made of the same material and having the same cross-section have been joined as shown in the figure. Each rod is of the same length. The left and right ends are kept at 0°C and 90ºC respectively. The temperature of junction of the three rods will be



a. 45°C **b.** 60°C
c. 30° C **d.** 20° C

25. Six resistance of 6 ohm each are connected to form a hexagon. The resistance between any two adjacent terminals is:

a. 6 Ω **b.** 36 Ω
c. 5 Ω **d.** 1 ohm

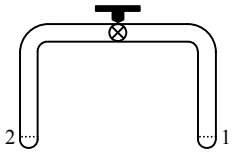
Space for Rough Work

JEE ADVANCE PAPER-I

SECTION 1 (Maximum Marks: 24**)**

MCQs with one or more than one correct answer

1. A glass tube of uniform internal radius r has a valve separating the two identical ends. Initially, the valve is in a tightly closed position. End 1 has a hemispherical soap bubble of radius r. End 2 has sub-hemispherical soap bubble as shown in figure. Just after opening the valve,



a. air from end 1 flows towards end 2. No change in the volume of the soap bubbles

b. air from end 1 flows towards end 2. Volume of the soap bubble at end 1 decreases

c. no changes occurs

d. air from end 2 flows towards end 1. volume of the soap bubble at end 1 increases

2. A thin uniform cylinder shell, closed at both ends is partially filled with water. It is floating vertically in water in half-submerged state. If ρ_c is the relative density of the material of the shell with respect to water, then the correct statement is that the shell is.

a. more than half-filled if ρ_c is less than 0.5

b. more than half-filled if ρ_c is less than 1.0

c. half-filled if ρ_c is more than 0.5

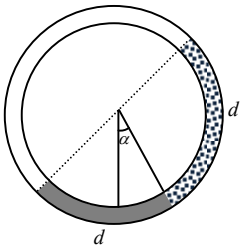
d. less than half- filled if ρ_c is less than 0.5

3. An open glass tube is immersed in mercury in such a way that a length of 8 cm extends above the mercury level. The open end of the tube is then closed and sealed and the tube is raised vertically up by additional 46 cm. What will be length of the air column above mercury in the tube now? (Atmospheric pressure = 76 cm of Hg)

a. 38 cm **b.** 6 cm

c. 16 cm **d.** 22 cm

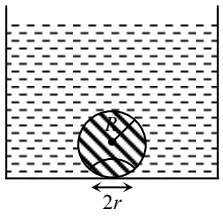
4. There is a circular tube in a vertical plane. Two liquids which do not mix and of densities d_1 and d_2 are filled in the tube. Each liquid subtends 90° angle at centre. Radius joining their interface makes an angle α with vertical. Ratio $\frac{d_1}{d_2}$ is



a. $\frac{1+\tan\alpha}{1-\tan\alpha}$ **b.** $\frac{1+\sin\alpha}{1-\cos\alpha}$

c. $\frac{1+\sin\alpha}{1-\sin\alpha}$ **d.** $\frac{1+\cos\alpha}{1-\cos\alpha}$

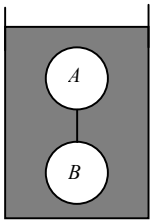
5. On heating water, bubbles being formed at the bottom of the vessel detaches and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If $r << R$, and the surface tension of water is T, value of r just before bubbles detaches is: (density of water is ρ_w)



a. $R^2\sqrt{\frac{\rho_w g}{T}}$ **b.** $R^2\sqrt{\frac{3\rho_w g}{T}}$

c. $R^2\sqrt{\frac{2\rho_w g}{3T}}$ **d.** $R^2\sqrt{\frac{\rho_w g}{6T}}$

6. Two solid spheres A and B of equal volumes but of different densities d_A and d_B are connected by a string. They are fully immersed in a fluid of density d_F. They get arranged into an equilibrium state as shown in the figure with a tension in the string. The arrangement is possible only if



a. $d_A < d_F$ **b.** $d_B > d_F$

c. $d_A > d_F$ **d.** $d_A + d_B = 2d_F$

SECTION 2 (Maximum Marks: 24)

Numerical value answer type questions

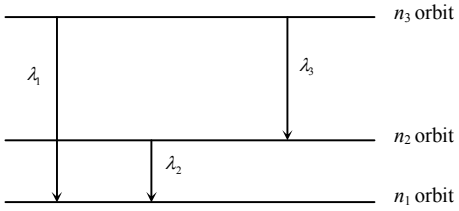
7. A 25 watt bulb, which is producing monochromatic light of wavelength 6600 Å is used to illuminated a metal surface. If the surface has 3% efficiency for photoelectric effect, then the photoelectric current produced in deci ampere is (use $h = 6.6 \times 10^{-34}$ Js):

8. de-Broglie wavelength associated with an electron accelerated through a potential difference 4 V is λ_1. When accelerating voltage is decreased by 3 V, its de-Broglie wavelength is λ_2. The ratio $\frac{\lambda_1}{\lambda_2}$ is

9. What is the energy in eV that should be added to an electron of energy 2 eV to reduce its de-Broglie wavelength from 1 nm to 0.5 nm?

10. The angular momentum of electron in 2nd orbit of hydrogen atom is L. The angular momentum of electron in 4th orbit of hydrogen atom is $n\,L$ where $n = ?$

11. For an atom of an ion having single electron, the wavelength observed $\lambda_1 = 2$ are units and $\lambda_3 = 3$ units figure. The value of missing wavelength λ_2 is



12. How many different wavelength may be observed in the spectrum from a hydrogen sample if the atoms are excited to 3rd excited state?

13. Light waves from two coherent sources having intensities I and $2I$ cross each other at a point with a phase diff. of 60º. What is the resultant intensity at the point?

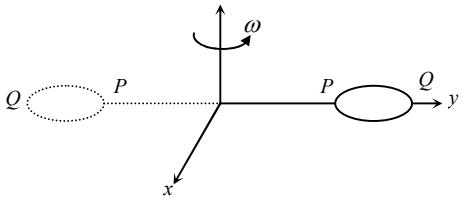
14. A *pn*-junction diode can withstand currents upto 10 mA. When it is forward biased, the potential drop across it is 1.0 V. Assuming that this potential drop is independent of the current, find the maximum voltage of the battery used to forward bias the diode when a resistance of is connected in series with the diode.

SECTION 3 (Maximum Marks: 12)

Paragraph based questions (2 paragraphs, each having 2 MCQs with one correct answer only)

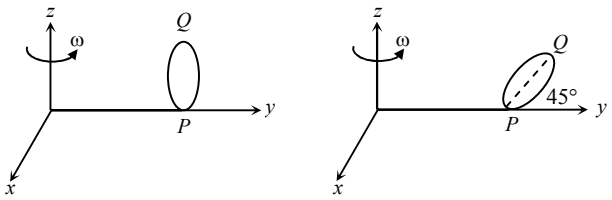
Paragraph for Question No. 15 to 16

The general motion of a rigid body can be considered to be a combination of (i) a motion of its centre of mass about an axis, and (ii) its motion about an instantaneous axis passing through the centre of mass. These axes need not be stationary. Consider, for example, a thin uniform disc welded (rigidly fixed) horizontally at its rim to a massless stick, as shown in the figure.



When the disc-stick system is rotated about the origin on a horizontal frictionless plane with angular speed ω, the motion at any instant can be taken as a combination of (i) a rotation of the centre of mass of the disc about the z-axis, and (ii) a rotating of the disc through an instantaneous vertical axis passing through its centre of mass (as is seen from the changed orientation of points P and Q). Both these motions have the same angular speed ω in this case.

Now consider two similar systems as shown in the figure: Case (a) the disc with its face vertical and parallel to x-z plane; Case (b) the disc with its face making an angle of 45° with x-y plane and its horizontal diameter parallel to x-axis. In both the cases, the disc is welded at point P, and the systems are rotated with constant angular speed ω about the z-axis.



15. Which of the following statements regarding the angular speed about the instantaneous axis (passing through the centre of mass) is correct?

a. It is $\sqrt{2}\omega$ for both the cases

b. It is ω for case *a.*; and $\frac{\omega}{\sqrt{2}}$ for case *b.*

c. It is ω for case *a.*; and ω for case *b.*

d. It is ω for both the cases

16. Which of the following statements about the instantaneous axis (passing through the centre of mass) is correct?

a. It is vertical for both the cases *a*. and *b*.

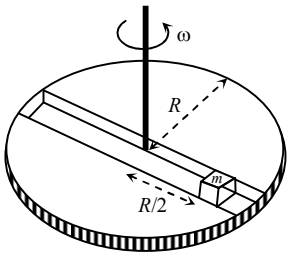
b. It is vertical for case *a*.; and is at 45°to the x-z plane and lies in the plane of the disc for case *b*.

c. It is horizontal for case *a*.; and is at 45° to the x-z plane and is normal to the plane of the disc for case *b*.

d. It is vertical for case *a*.; and is at 45° to the x-z plane and is normal to the plane of the disc for case *b*.

Paragraph for Question No. 17 to 18

A frame of reference that is accelerated with respect to an inertial frame of reference is called a non-inertial frame of reference. A coordinate system fixed on a circular disc rotating about a fixed axis with a constant angular velocity ω is an example of a non-inertial frame of reference.



The relationship between the force $\vec{F}_{\text{rot}}$ experienced by a particle of mass *m* moving on the rotating disc and the force $\vec{F}_{\text{in}}$ experienced by the particle in an inertial frame of reference is, $\vec{F}_{\text{rot}} = \vec{F}_{\text{in}} + 2m(\vec{v}_{\text{rot}} \times \vec{\omega}) + m(\vec{\omega} \times \vec{r}) \times \vec{\omega}$,

Where $\vec{v}_{\text{rot}}$ is the velocity of the particle in the rotating frame of reference and $\vec{r}$ is the position vector of the particle with respect to the centre of the disc.

Now consider a smooth slot along a diameter of a disc of radius *R* rotating counter-clockwise with a constant angular speed ω about its vertical axis through its center. We assign a coordinate system with the origin at the center of the disc, the *x*-axis along the slot, the *y*-axis perpendicular to the slot and the *z*-axis along the rotation axis $(\vec{\omega} = \omega\hat{k})$. A small block of mass *m* is gently placed in the slot at $\vec{r} = (R/2)\hat{i}$ at $t = 0$ and is constrained to move only along the slot.

17. The distance *r* of the block at time *t* is:

a. $\frac{R}{2}\cos 2\omega t$ **b.** $\frac{R}{4}(e^{2\omega t} + e^{-2\omega t})$

c. $\frac{R}{2}\cos \omega t$ **d.** $\frac{R}{4}(e^{\omega t} + e^{-\omega t})$

18. The net reaction of the disc on the block is :

a. $m\omega^2 R \sin \omega t\hat{j} - mg\hat{k}$

b. $-m\omega^2 R \cos \omega t\hat{j} - mgk\hat{6}$

c. $\frac{1}{2}m\omega^2 R(e^{\omega t} - e^{-\omega t})\hat{j} + mg\hat{k}$

d. $\frac{1}{2}m\omega^2 R(e^{2\omega t} - e^{-2\omega t})\hat{j} + mg\hat{k}$

Space for Rough Work

JEE ADVANCE PAPER-II

SECTION 1 (Maximum Marks: 24)

MCQs with one or more than one correct answer

1. One end of a horizontal thick copper wire of length $2L$ and radius $2R$ is welded to an end of another horizontal thin copper wire of length L and radius R. When the arrangement is stretched by applying forces at two ends, the ratio of the elongation in the thin wire to that in the thick wire is

a. 0.25 **b.** 0.50

c. 2.00 **d.** 4.00

2. The pressure that has to be applied to the ends of a steel wire of length 10 cm to keep its length constant when its temperature is raised by 100°C is: (For steel Young's modulus 2×10^{11} Nm^{-2} and coefficient of thermal expansion is 1.1×10^{-5} K^{-1})

a. 2.2×10^{7} Pa **b.** 2.2×10^{6} Pa

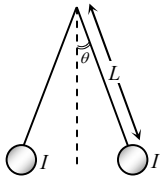
c. 2.2×10^{8} Pa **d.** 2.2×10^{9} Pa

3. A pendulum made of a uniform wire of cross-sectional area A has time period T. When an additional mass M is added to its bob, the time period changes to T_M. If the Young's modulus of the material of the wire is Y then $1/Y$ is equal to: (g = gravitational acceleration)

a. $\left[\left(\frac{T_M}{T}\right)^2-1\right]\frac{A}{Mg}$ **b.** $\left[\left(\frac{T_M}{T}\right)^2-1\right]\frac{Mg}{A}$

c. $\left[1-\left(\frac{T_M}{T}\right)^2\right]\frac{A}{Mg}$ **d.** $\left[1-\left(\frac{T}{T_M}\right)^2\right]\frac{A}{Mg}$

4. Function $x = A\sin^2\omega t + B\cos^2\omega t + C\sin\omega t\cos\omega t$ represents SHM.



a. For any value of A, B and C (except $C = 0$)

b. If $A = -B$; $C = 2B$, amplitude $= |B\sqrt{2}|$

c. If $A = B$; $C = 0$

d. If $A = B$; $C = 2B$ amplitude $= |B|$

5. Two coaxial solenoids of different radii carry current I in the same direction. Let $\vec{F}_1$ be the magnetic force on the inner solenoid due to the outer one and $\vec{F}_2$ be the magnetic force on the outer solenoid due to the inner one. Then:

a. $\vec{F}_1 = \vec{F}_2 = 0$

b. $\vec{F}_1$ is radially inwards and $\vec{F}_2$ is radially outwards

c. $\vec{F}_1$ is radially inwards and $\vec{F}_2 = 0$

d. $\vec{F}_1$ is radially outwards and $\vec{F}_2 = 0$

6. Consider a spherical shell of radius R at temperature T. The black body radiation inside it can be considered as an ideal gas of photons with internal energy per unit volume $u = \frac{U}{V} \propto T^4$ and pressure $P = \frac{1}{3}\left(\frac{U}{V}\right)$. If the shell now undergoes an adiabatic expansion the relation between T and R is:

a. $T \propto e^{-R}$ **b.** $T \propto e^{-3R}$

c. $T \propto \frac{1}{R}$ **d.** $T \propto \frac{1}{R^3}$

SECTION 2 (Maximum Marks: 24)

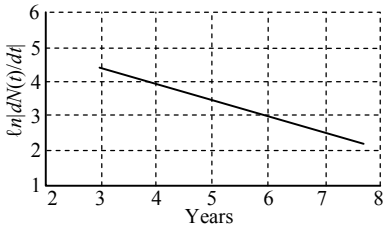
Numerical value answer type questions

7. An electron in an excited state of Li^{2+} ion has angular momentum $3h/2\pi$. The de-Broglie wavelength of the electron in this state is $p\pi a_0$ (where a_0 is the Bohr radius). The value of p is

8. Consider a hydrogen atom with its electron in the n^{th} orbital. An electromagnetic radiation of wavelength 90 nm is used to ionize the atom. If the kinetic energy of the ejected electron is 10.4 eV, then the value of n is (hc = 1242 eV nm)

9. A hydrogen atom in its ground state is irradiated by light of wavelength 970 Å. Taking $hc/e = 1.237\times10^{-6}$ eV m and the ground state energy of hydrogen atom as –13.6 eV, the number of lines present in the emission spectrum is:

10. To determine the half life of a radioactive element, a student plots a graph of $\ell_n\left|\frac{dN(t)}{dt}\right|$ versus t.



Here $dN(t)/dt$ is the rate of radioactive decay at time t. If the number of radioactive nuclei of this element decreases by a factor of p after 4.16 years, the value of p is

11. The activity of a freshly prepared radioactive sample is 10^{10} disintegrations per second, whose mean life is 10^9 s. The mass of an atom of this radioisotope is 10^{-25} kg. kg. The mass (in mg) of the radioactive sample is

12. A freshly prepared sample of a radioisotope of half-life 1386s has activity 10^3 disintegrations per second. Given that ln 2 = 0.693, the fraction of the initial number of nuclei (expressed in nearest integer percentage) that will decay in the first 80s after preparation of the sample is

13. A nuclear power plant supplying electrical power to a village uses a radioactive material of half life T years as the fuel. The amount of fuel at the beginning is such that the total power requirement of the village is 12.5 % of the electrical power available from the plant at that time. If the plant is able to meet the total power needs of the village for a maximum period of nT years, then the value of n is

14. For a radioactive material, its activity A and rate of change of its activity R are defined as $A = -\frac{dN}{dt}$ and $R = -\frac{dA}{dt}$, where $N(t)$ is the number of nuclei at time t. Two radioactive sources P (mean life τ) and Q (mean life 2τ) have the same activity at $t = 0$. Their rates of change of activities at $t = 2\tau$ are R_P and R_Q, respectively. If $\frac{R_P}{R_Q} = \frac{n}{e}$, then the value of n is

SECTION 3 (Maximum Marks: 12**)**

Matching type questions with 4 options

15. There is a definite relation between velocity, mass and acceleration of body to know about the work done by the forces applied on body. Column I has some statements showing some relations which are related with some statements in Column II. Match the correct options.

Column I	Column II
(A) If kinetic energy is K the momentum P is 0	**1.** zero
(B) If momentum p is zero the kinetic energy is 0	**2.** $\sqrt{2mk}$
(C) If different masses have same kinetic energy the ratio of their momenta is 0	**3.** $\frac{m_2}{m_1}$
(D) If different masses have same momentum, the ratio of their kinetic energy is	**4.** $\sqrt{\frac{m_1}{m_2}}$

a. A-1, B-2, C-3, D-4 **b.** A-2, B-1, C-4, D-3
c. A-4, B-3, C-2, D-1 **d.** A-3, B-2, C-1, D-4

16. Waves are classified on the basis of frequency range. Column I mentions the type of waves while column II gives the corresponding frequency range. Can you match the proper options?

Column I	Column II
(A) Infrasonic	**1.** Objects having velocity more than velocity of sound in air at 0°C
(B) Audible	**2.** Frequency < 20 Hz
(C) Ultrasonics	**3.** Frequency $> 20{,}000$ Hz
(D) Supersonics	**4.** Frequency lies between 20 Hz to 20, 000 Hz

a. A-4, B-3, C-2, D-1 **b.** A-2, B-4, C-3, D-1
c. A-4, B-4, C-1, D-2 **d.** A-3, B-2, C-1, D-4

17. Various electromagnetic waves are given in column I and various frequency ranges in column II

Column I	Column II
(A) Radiowaves	**1.** 1×10^{16} to 3×10^{21} Hz
(B) γ-rays	**2.** 1×10^{9} to 3×10^{11} Hz
(C) Microwaves	**3.** 3×10^{18} to 5×10^{22} Hz
(D) X-rays	**4.** 5×10^{5} to 10^{9} Hz

a. A-1, B-2, C-3, D-4 **b.** A-2, B-3, C-4, D-1
c. A-3, B-4, C-1, D-2 **d.** A-4, B-3, C-2, D-1

18. In each of the following questions, match column I and column II, and select the correct match out of four given choices.

Column I	Column II
(A) Transducer	**1.** Range of frequencies over which communication system works
(B) Attenuation	**2.** A device that has input in electrical form or provides output in electrical form
(C) Range	**3.** Loss of strength of a signal during propagation.
(D) Bandwidth	**4.** Argest distance between transmitter and receiver.

a. A-,2 B-4, C-1, D-3 **b.** A-4, B-3, C-1, D-2
c. A-1, B-2, C-3, D-4 **d.** A-2, B-3, C-4, D-1

ANSWERS & SOLUTIONS

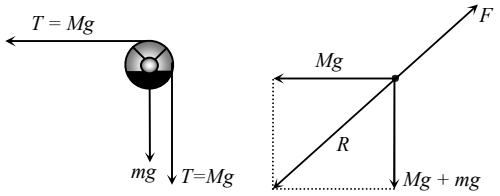
JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	d	d	c	c	c	b	a	a	c
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	c	c	a	c	a	b	b	a	b
21.	**22.**	**23.**	**24.**	**25.**					
c	d	b	b	c					

1. **(d)** Dipole moment = (charge) × (distance)
Electric flux = (electric field) × (area)

2. **(d)** Total angle $= 100 \times \frac{\pi}{50} = 2\pi$
So, all the force will pass through one point and all forces will be balanced. *i.e.* their resultant will be zero.

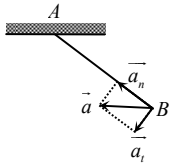
3. **(d)** Free body diagram of pulley is shown in figure.



Pulley is in equilibrium under four forces. Three forces as shown in figure and the fourth, which is equal and opposite to the resultant of these three forces, is the force applied by the clamp on the pulley (*say F*).
Resultant *R* of these three forces is $R = (\sqrt{(M+m)^2 + M^2})g$
Therefore, the force *F* is equal and opposite to *R* as shown in figure.

$\therefore \quad F = (\sqrt{(M+m)^2 + M^2})g.$

4. **(c)** Net acceleration $\vec{a}$ of the bob in position



B has two components.
(a) $\vec{a}_n$ = radial acceleration (towards *BA*)
(b) $\vec{a}_t$ = tangential acceleration (perpendicular to *BA*)
Therefore, direction of $\vec{a}$ is correctly shown in option (c).

5. **(c)** *M* = Mass of the square plate before cutting the holes
Mass of one hole $m = \left(\frac{M}{16R^2}\right)\pi R^2 = \frac{\pi M}{16}$

$\therefore$ Moment of inertia of the remaining portion,
$I = I_{square} - 4I_{hole}$
$= \frac{M}{12}\left(16R^2 + 16R^2\right) - 4\left[\frac{mR^2}{2} + m\left(2R^2\right)\right]$
$= \frac{8}{3}MR^2 - 10mR^2 = \left(\frac{8}{3} - \frac{10\pi}{16}\right)MR^2$

6. **(c)** $g' = g - R\omega^2$ (at equator)
Given $g' = 0 \Rightarrow g = R\omega^2$

$\Rightarrow \quad \omega = \sqrt{\frac{g}{R}} = \sqrt{\frac{10}{6400 \times 10^3}} = \frac{1}{800}$ rad/s

7. **(b)** When the tube is placed vertically in water, water rises through height *h* given by $h = \frac{2T\cos\theta}{rdg}$
Upward force $= 2\pi r \times T\cos\theta$
Work done by this force in raising water column through height *h* is given by $\Delta W = (2\pi rT\cos\theta)h$
$= (2\pi rh\cos\theta)T = (2\pi rh\cos\theta)\left(\frac{rhdg}{2\cos\theta}\right) = \pi r^2h^2dg$
However, the increase in potential energy ΔE_p of the raised water column $= mg\frac{h}{2}$, where *m* is the mass of the raised column of water.
$\therefore \quad m = \pi r^2hd$
So, $\Delta E_p = (\pi r^2hd)\left(\frac{gh}{2}\right) = \frac{\pi r^2h^2dg}{2}$
Further, $\Delta W - \Delta E_p = \frac{\pi r^2h^2dg}{2}$
The part $(\Delta W - \Delta E_p)$ is used in doing work against viscous forces and frictional forces between water and glass surface and appears as heat.
So, Heat released $\frac{\Delta W - \Delta E_p}{J} = \frac{\pi r^2h^2dg}{2J}$

8. **(a)** Energy stored, $U = \frac{1}{2} \times$ (force)×(extension) $= \frac{1}{2}Fx$
As $Y = \frac{FL}{Ax} \Rightarrow F = \frac{YA_x}{L}$
$\therefore \quad U = \frac{1}{2}\left(\frac{YA_x}{L}\right)x = \frac{1}{2}\frac{YAx^2}{L}$

9. **(a)** $\beta = -\frac{dV/dp}{V}$ = compressibility of gas
$= \frac{1}{\text{Bulk modulus of elasticity}}$ and $\beta = \frac{1}{p}$ under isothermal

conditions. Thus, β *versus p* graph will be a rectangular hyperbola.

10. **(c)** Applying, $Q=\dfrac{KA(\theta_1-\theta_2)}{l}$

temperature difference across the rods are 20°C and 10°C respectively.

$\therefore$ Displacement of the joint $=\alpha_2 t_2-\alpha_1 t_1$

$=10\alpha_2-20\alpha_1=10(\alpha_2-2\alpha_1)$

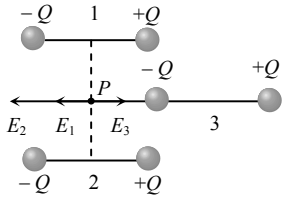
11. **(c)** $\dfrac{1}{2}\cdot\dfrac{1}{2}m_1v^2=\mathrm{J}\,m_1 s\,\Delta\theta_1$

$\Rightarrow\quad \Delta\theta_1=\dfrac{v^2}{4\mathrm{Js}}=\dfrac{(50)^2}{4\times4.2\times10^3\times0.105}=1.4\ ^\circ\mathrm{C}$

$m_1 s\,\Delta\theta_1=m_2 s\,\Delta\theta_2$

$\Rightarrow\quad \Delta\theta_2=\dfrac{m_1}{m_2}\Delta\theta_1=\dfrac{1}{0.2}\times1.4=7^\circ\mathrm{C}$

12. **(c)** Point P lies at equatorial positions of dipole 1 and 2 and axial position of dipole 3. Hence field at P



Due to dipole 1 $\quad E_1=\dfrac{k.p}{x^3}$ (towards left)

Due to dipole 2 $\quad E_2=\dfrac{k.p}{x^2}$ (towards left)

Due to dipole 3 $\quad E_3=\dfrac{k.(2p)}{x^3}$ (towards right)

So, Net field at P will be zero.

13. **(c)** The two capacitor C_1 and C_2 are connected in series

$C_1=\dfrac{K_1\varepsilon_0 A}{d_1},\ C_2=\dfrac{K_2\varepsilon_0 A}{d_2}$

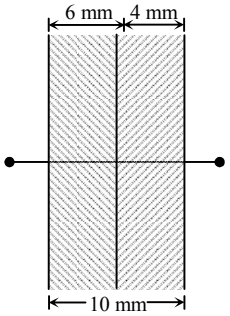
$\dfrac{1}{C}=\dfrac{1}{C_1}+\dfrac{1}{C_2}=\dfrac{d_1}{K_1\varepsilon_0 A},\dfrac{d_2}{K_2\varepsilon_0 A}$

$=\dfrac{1}{\varepsilon_0 A}\left[\dfrac{d_1}{K_1}+\dfrac{d_2}{K_2}\right]$

$=\dfrac{1}{\varepsilon_0 A}\left[\dfrac{6\times10^{-3}}{10}+\dfrac{4\times10^{-3}}{5}\right]$

$=\dfrac{1}{\varepsilon_0 A}\dfrac{14\times10^{-3}}{10}$

This gives $C=\dfrac{5000}{7}\varepsilon_0 A$



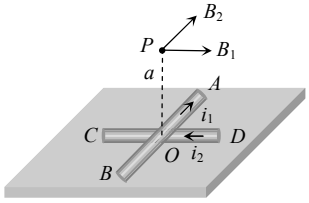
14. **(a)** Let the real dip be ϕ, then $\tan\phi=\dfrac{B_V}{B_H}$

For apparent dip,

$\tan\phi'=\dfrac{B_V}{B_H\cos\beta}=\dfrac{B_V}{B_H\cos30^\circ}=\dfrac{2B_V}{\sqrt{3}B_H}$

or $\tan45^\circ=\dfrac{2}{\sqrt{3}}.\tan\phi$ or $\phi=\tan^{-1}\left(\dfrac{\sqrt{3}}{2}\right)$

15. **(c)** At P: $B_{net}=\sqrt{B_1^2+B_2^2}$

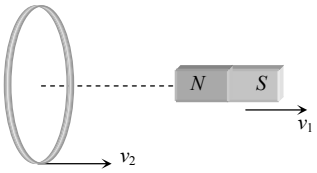


$=\sqrt{\left(\dfrac{\mu_0}{4\pi}\dfrac{2i_1}{a}\right)^2+\left(\dfrac{\mu_0}{4\pi}\dfrac{2i_2}{a}\right)^2}=\dfrac{\mu_0}{2\pi a}(i_1^2+i_2^2)^{1/2}$

16. **(a)** Speed of the magnet $v_1=\dfrac{2}{1}=2$ m/s

Speed of the coil $v_2=\dfrac{1}{0.5}=2\mathrm{m/s}$

Relative speed between coil and magnet is zero, so there is no induced emf in the coil.



17. **(b)** Since voltage is lagging behind the current, so there must be no inductor in the box.

18. **(b)** $\dfrac{I}{O}=\dfrac{f}{f-u}$; where $u=f+x\quad\therefore\dfrac{I}{O}=-\dfrac{f}{x}$

19. **(a)** Kinetic energy of recoil electron, $E_k=hv-hv'$

$=12.375-9.375=3\mathrm{eV}$

20. **(b)** $V_b=i_bR_b\Rightarrow R_b=\dfrac{9}{35\times10^{-6}}=257\ \mathrm{k\Omega}$

21. **(c)** $s_1+s_2=100$

$\Rightarrow\quad \dfrac{1}{2}gt^2+ut-\dfrac{1}{2}gt^2=100\ m$

$\Rightarrow\quad ut=100\ m\Rightarrow t=\dfrac{100}{u}=\dfrac{100}{100}=1\mathrm{s}$

22. **(d)** $x = 4(\cos \pi t + \sin \pi t)$

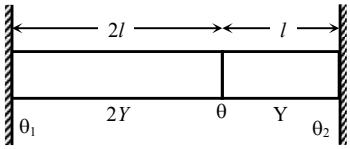
$$4\sqrt{2}\left[\frac{1}{\sqrt{2}}\cos \pi t + \frac{1}{\sqrt{2}}\sin \pi t\right]$$

$$= 4\sqrt{2}\left[\sin\frac{\pi}{2}\cos \pi t + \cos\frac{\pi}{2}\sin \pi t\right] = 4\sqrt{2}\sin\left(\pi t + \frac{\pi}{4}\right)$$

Standard equation of displacement is $x = a\sin(\omega t + \phi)$

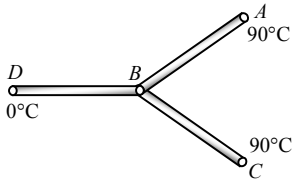
Comparing the given equation with standard equation $a = 4\sqrt{2}$

23. **(b)** Path difference $(\pi r - 2r) = \frac{\lambda}{2} = \frac{32}{2} = 16$



$\Rightarrow \quad r = \frac{16}{\pi - 2} = 14\text{cm}$

24. **(b)** Let θ be the temperature of the junction (say B). Thermal resistance of all the three rods is equal. Rate of heat flow through AB + Rate of heat flow through CB = Rate of heat flow through BD



$\therefore \quad \frac{90° - \theta}{R} + \frac{90° - \theta}{R} = \frac{\theta - 0}{R}$, Here, R =Thermal resistance

$\therefore \quad 3\theta = 180°, \text{ or } \theta = 60°\text{C}$

25. **(c)** Resistance $(6 \times 5) = 30\,\Omega$ and $6\,\Omega$ are in parallel; so equivalent resistance $R = \frac{30 \times 6}{30 + 6} = 5\,\Omega$

JEE Advance Paper -I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	a	a	a	c	a,b,d	4	2	6	2
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**		
6	6	4	2	d	a	d	c		

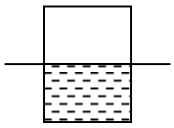
1. **(b)** $P_1 =$ pressure just inside the bubble at the end

$$2 = P_0 + \frac{4T}{R}$$

2. **(a)** If water is half filled $M_{\text{shell}}g = \rho(\Delta V)_{\text{displaced}}g$

$\Rightarrow \quad (\Delta V) = \left(\frac{V_{\text{shell}}}{2}\right)$

$\Rightarrow \quad M_{\text{shell}} = \frac{\rho_{\text{liwuid}}V_{\text{shell}}}{2} \Rightarrow \left(\frac{M_{\text{shell}}}{V_{\text{shell}}}\right) = \frac{\rho_{\text{liquid}}}{2}$



$\Rightarrow \quad \sigma_{\text{shell}} = \frac{\rho_{\text{liquid}}}{2} \qquad \Rightarrow R.d = \frac{1}{2}.$

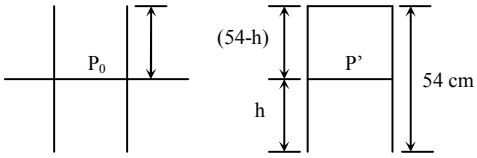
If $R.d < \frac{1}{2}$, Then the cylinder should be more than half filled so that it is half submerged and floating.

3. **(a)** $P'(54 - h) = P_0 8$ (Boyel's Law)

$\Rightarrow \quad P' = \frac{P_0 8}{54 - h} \qquad \Rightarrow P' + \rho gh = P_0$

$\Rightarrow \quad \frac{8P_0}{54 - h} + \rho gh = P_0$

$\Rightarrow \quad \rho gh = \left(\frac{46 - h}{54 - h}\right)P_0 \ (P_0 = \rho g(76))$



$\Rightarrow \quad h = \left(\frac{46 - h}{54 - h}\right)(76) \qquad \Rightarrow 54h - h^2 = 76 \times 46 - 76h$

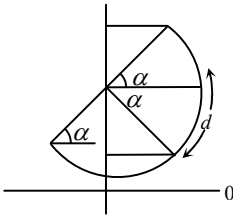
$\Rightarrow \quad h^2 - 130h + 76 \times 46 = 0 \qquad \Rightarrow (h - 38)(h - 92) = 0$

$\Rightarrow \quad h = 38, 92 \qquad \Rightarrow h = 38\text{cm}$

$\downarrow$

Not possible

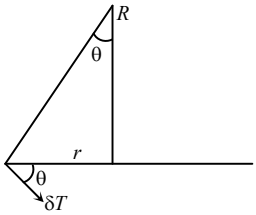
4. **(a)** Pressure at 0 level is same from both sides.



$$d_1(1 - \sin\alpha) = d_1(1 - \cos\alpha) + d_2[\cos\alpha + \sin\alpha]$$

$\Rightarrow \quad \frac{d_1}{d_2} = \frac{\sin\alpha + \cos\alpha}{\cos\alpha - \sin\alpha}$

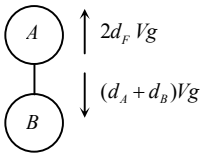
5. **(c)** $R^2\sqrt{\frac{2\rho_w g}{3T}}$, The downward force on the bubble due to surface tension $= 2\pi r.T\sin\theta = \frac{2\pi Tr^2}{R}$



The upward buoyant force exceeds, the surface tension force then the bubble detaches.

$$\therefore \quad \frac{4}{3}\pi R^3 \rho_w g = \frac{2\pi T r^2}{R} \Rightarrow r = R^2\sqrt{\frac{2\rho_w g}{3T}}$$

6. **(a, b, d)**



7. **(4)** No. of photons emitted per second by 25 watt source will be

$$n = \frac{E}{hc/\lambda} = \frac{E\lambda}{hc} = \frac{25\times(6600\times10^{-10})}{(6.6\times10^{-34})\times(3\times10^{8})} = \frac{25}{3\times10^{-19}}$$

Current, I = 3% of number of photoelectrons emitted per sec × charge of electron

$$= \frac{3}{100}ne\frac{3}{100}\times\frac{25}{3\times10^{-19}}\times1.6\times10^{-19} = 0.4\,\text{A} \quad = 4 \text{ deciampere}$$

8. **(2)** de-Broglie wavelength, $\lambda = \frac{4}{\sqrt{2mq\text{V}}}$ or $\lambda \propto \frac{1}{\sqrt{\text{V}}}$

$$\therefore \quad \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{\text{V}_1}{\text{V}_2}} = \sqrt{\frac{4\text{V}}{4\text{V}-3\text{V}}} = 2$$

9. **(6)** As $\lambda = \frac{h}{\sqrt{2mE_K}}$ or $E_K = \frac{h^2}{2m\lambda^2}$ or $E_K \propto \frac{1}{\lambda^2}$

$$\therefore \quad \frac{E_{K_1}}{E_{K_2}} = \left[\frac{\lambda_2}{\lambda_1}\right]^2 = \left[\frac{0.5}{1}\right]^2 = \frac{1}{4}$$

or $E_{K_1} = 4E_{K_1} = 4\times2eV = 8$ eV

Increase in energy $= E_{K_2} - E_{K_1} = 8-2 = 6$eV

10. **(2)** $L = 2\left(\frac{h}{2\pi}\right)$ and $L' = 4\left(\frac{h}{2\pi}\right)$

$\therefore \quad L' = 2\,L = n\,L,$ where $n = 2$

11. **(6)** As is clear from figure. $E_1 = E_2 + E_3$

$$\frac{hc}{\lambda_1} = \frac{hc}{\lambda_2} + \frac{hc}{\lambda_3}$$

$$\therefore \quad \frac{1}{\lambda_2} = \frac{1}{\lambda_1} - \frac{1}{\lambda_3} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} = 6 \text{ units}$$

12. **(6)** 3rd excited state means n = 4. No. of different wavelength observed $N = \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$

13. **(4)** Here, $I_1 = I$

$I_2 = 2I, \phi = 60°$

Amplitude R of resultant wave is

$$R = \sqrt{a^2 + b^2 + 2ab\cos\phi}$$

$$R^2 = a^2 + b^2 + 2ab\cos\phi$$

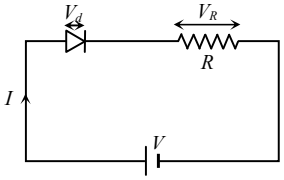
As intensity $\propto$ (amplitude)^2

$\therefore$ Resultant intensity, $I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$$= I + 2I + 2\sqrt{I\times2I}\times\cos60°$$

$$= 3I + I\sqrt{2} = I(3+\sqrt{2}) = 4.414\,I$$

14. **(2)** See figure



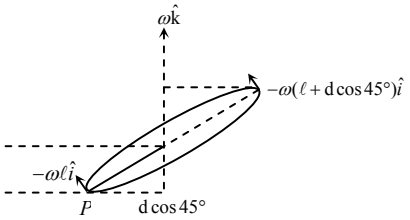
Voltage drop across diode is V_d= 1.0V

Voltage drop across R is $V_R = RI$

$$= 100\times10\times10^{-3} = 1.0\text{V}$$

$$V_{\max} = V_d + V_R = 1.0 + 1.0 = 2.0\text{V}$$

15. **(d)** It is ω for both case.



As angular velocity of a rigid body about any point should be same. $\vec{v} = \vec{v}_{PC} + \vec{v}_{CO}$

$$\Rightarrow \quad \vec{v}_{PO} = \vec{r}_{PC} + \vec{\omega}' + \vec{v}_{CO}$$

$$-\omega L\hat{i} = -\frac{d}{2\sqrt{2}}(\hat{k}+\hat{j})\times\vec{\omega} + -\left(\omega L + \frac{d}{2}.45°\right)\hat{i}$$

$$\Rightarrow \quad \vec{r}_p = \ell\hat{j}$$

$$\Rightarrow \quad \vec{r}_c = \left(\ell + \left(\frac{d}{2}\cos45°\right)\right)\hat{j} + \left(\frac{d}{2}\sin45°\right)\hat{k}$$

$$\Rightarrow \quad \vec{r}_{PC} = -\left(\frac{d}{2}\cos45°\right)\hat{j} - \left(\frac{d}{2}\sin45°\right)\hat{k}$$

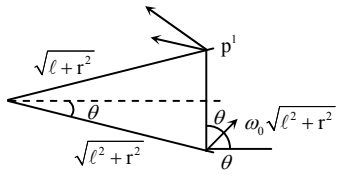
$$\Rightarrow \quad \vec{v}_{CO} = -\left(\ell + \frac{d}{2}\cos45°\right)\omega\hat{i}$$

$$\Rightarrow \quad \frac{\omega d}{2\sqrt{2}}\hat{i} = -\frac{d}{2\sqrt{2}}(\hat{i}+j)\times\vec{\omega}' \Rightarrow \vec{\omega}' = \omega\hat{k}$$

16. **(a)** $V = \omega\sqrt{\ell^2 + r^2}$ TOP View

$$W = \left(\frac{2V\sin\theta}{2r}\right) = \frac{\omega\sqrt{\ell^2+r^2}\,r}{r\sqrt{\ell^2+r^2}} = \omega$$

Here there is a v_{rel} of P' and Q' (not for P, Q) $(v_{rel})_{P'Q'}$ will be in horizontal plane



$\Rightarrow$ ω must be along vertical. Same situation will be there for P as mentioned ω should be same for both cases.

17. **(d)** $mr\omega^2 = ma$

$\Rightarrow a = r\omega^2 \qquad \Rightarrow \dfrac{vdv}{dr} = r\omega^2$

$\Rightarrow \int_0^v vdv = \omega^2 \int_{R/2}^r rdr \qquad \Rightarrow v = \omega\sqrt{r^2 - \dfrac{R^2}{4}}$

$$\int_{R/2}^{v} \frac{dr}{\sqrt{r^2 - \frac{R^2}{4}}} = \int_0^t \omega\, dt \qquad \ldots(i)$$

Assume: $r = \dfrac{R}{2}\sec\theta$

$\Rightarrow dr = \dfrac{R}{2}\sec\theta\tan\theta d\theta$

$$\Rightarrow \int \frac{\frac{R}{2}\sec\theta\tan\theta d\theta}{\sqrt{\frac{R^2}{4}\tan^2\theta}} = \int_0^t \omega dt$$

$$\Rightarrow \omega t = \ell n\left[\frac{2r}{R} + \frac{\sqrt{4r^2 - R^2}}{R}\right]$$

$$\Rightarrow r = \frac{R}{4}[e^{\omega t} + e^{-\omega t}]$$

18. **(c)** $\vec{F}_{rot} = \vec{F}_{in} + 2m(v_{rot}\hat{i}) \times \omega\hat{k} + m(\omega\hat{k} \times r\hat{i}) \times \omega\hat{k}$

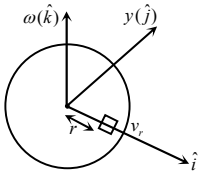
$mr\omega^2\hat{i} = \vec{F}_{in} + 2mv_{rot}\omega(-\hat{j}) + m\omega^2 r\hat{i}$

$\vec{F}_{in} = 2mv_r\omega\hat{j} \qquad \ldots(i)$

$r = \dfrac{R}{4}\left[e^{\omega t} + e^{-\omega t}\right]$

$\dfrac{dr}{dt} = v_r = \dfrac{R}{4}\left[\omega e^{\omega t} - \omega e^{-\omega t}\right]$

$\vec{F}_{in} = 2m\dfrac{R\omega}{4}\left[e^{\omega t} - e^{-\omega t}\right]\omega\hat{j}$



Also reaction is due to disc surface then

$$\vec{F}_{reaction} = \frac{mR\omega^2}{2}\left[e^{\omega t} - e^{-\omega t}\right]\hat{j} + mg\hat{k}$$

JEE Advance Paper -II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
c	c	a	a,b,c	a	a,b	2	2	6	8
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**		
1	4	3	2	b	b	d	d		

1. **(c)**

$\Rightarrow k_1 = \dfrac{\pi 4R^2 x}{2L}, k_2 = \dfrac{\pi R^2 y}{L}$

$\Rightarrow F = k_1 x = k_2 y \Rightarrow \dfrac{y}{x} = \dfrac{k_1}{k_2} = 2$

2. **(c)** $\dfrac{F}{A} = Y\dfrac{\Delta\ell}{\ell} \Rightarrow \dfrac{\Delta\ell}{\ell} = \alpha\Delta T$

$\Rightarrow P = \dfrac{F}{A} = Y\alpha\Delta T$

$= 2\times10^{11}\times1.1\times10^{-5}\times100 = 2.2\times10^8$ Pa

3. **(a)** $T = 2\pi\sqrt{\dfrac{\ell_1}{g}}, \quad T_M = 2\pi\sqrt{\dfrac{\ell_2}{g}}$

$T^2 = 4\pi^2\cdot\dfrac{\ell_1}{g}, \quad T_M^2 = 2\pi^2\dfrac{\ell_2}{g}$

$\Rightarrow \dfrac{T_M^2}{T^2} = \dfrac{\ell_2}{\ell_1}x$

$\Rightarrow \dfrac{T_M^2}{T^2} - 1 = \dfrac{\ell_2}{\ell_1} - 1 = \dfrac{\ell_2 - \ell_1}{\ell_1}$

$\Rightarrow \gamma = \dfrac{Mg}{A}\cdot\dfrac{\ell_1}{(\ell_2 - \ell_1)}$

$\Rightarrow \dfrac{1}{\gamma} = \dfrac{A}{Mg}\cdot\dfrac{(\ell_2 - \ell_1)}{\ell_1} = \dfrac{A}{Mg}\left[\left(\dfrac{T_M}{T}\right)^2 - 1\right]$

4. **(a, b, c)** $x = \dfrac{A}{2}(1 - \cos 2\omega t) + \dfrac{B}{2}(1 + \cos 2\omega t) + \dfrac{C}{2}\sin 2\omega t$

For $A = 0, B = 0$ $\qquad x = \dfrac{C}{2}\sin 2\omega t$

For $A = -B$ and $C = 2B$ $\qquad x = B\cos 2\omega t + B\sin 2\omega t$

Amplitude $= |B\sqrt{2}|$

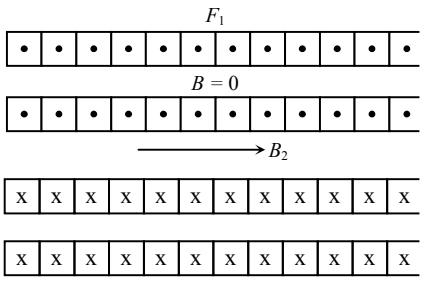
For $A = B; C = 0$ $\qquad x = A$

Hence this is not correct option

For $A = B, C = 2B$ $\qquad x = B + B\sin 2\omega t$

It is also represents SHM.

5. **(a)** Solenoid consists of circular current loops which are placed in uniform field, so magnetic fore on both solenoids $= 0$.



$\Rightarrow \quad \vec{F}_1 = \vec{F}_2 = 0$

6. **(c)** $P = \frac{1}{3}\left(\frac{U}{V}\right) \propto T^4$

For an ideal gas $PV = nRT$

$\therefore \quad P = \frac{nR'T}{V}$ (R' -molar gas constant)

$\therefore \quad T \propto e^{-R} \quad \Rightarrow \quad \frac{nR'}{V} \propto T^3$

$\therefore \quad \frac{1}{V} \propto T^3 \quad \Rightarrow \quad \frac{1}{\frac{4\pi}{3}R^3} \propto T^3 \quad \therefore \; T^3 \propto \frac{1}{R^3} \quad \Rightarrow \quad T \propto \frac{1}{R}$

7. **(2)** $mvr = \frac{nh}{2\pi} = \frac{3h}{2\pi}$ de-Broglie wavelength

$\lambda = \frac{h}{mv} = \frac{2\pi r}{3} = \frac{2\pi}{3}\frac{a_0(3)^2}{z_{Li}} = 2\pi a_0$

8. **(2)** $E_{photon} = E_{ionize\ atom} + E_{kinetic\ energy}$

$\Rightarrow \quad \frac{1242}{90} = \frac{13.6}{n^2} + 10.4$. From this, $n = 2$

9. **(6)** [Here we are assuming that in original paper atom $\Rightarrow$ atoms] $13.6\left[\frac{1}{1^2} - \frac{1}{n^2}\right] = \frac{12370}{970}$

$\Rightarrow \quad n^2 = 16 \Rightarrow n = 4$

No. of lines $= {}^4C_2 = \frac{4 \times 3}{2} = 6$

10. **(8)** $N = N_0 e^{-\lambda t}$

$\Rightarrow \quad \ell n \,|\, dN/dt \,| \, \ell n(N_0\lambda) - \lambda t$

From graph, $\lambda = \frac{1}{2}$ per year

$\Rightarrow \quad t_{1/2} = \frac{0.693}{1/2} = 1.386$ year

$\Rightarrow \quad 4.16$ yrs $= 3t_{1/2} \; \therefore \; p = 8$

11. **(1)** $A = \lambda N \Rightarrow N = \frac{A}{\lambda} = A\tau$

So $M = mN = mA\tau = 10^{-25} \times 10^{10} \times 10^9 = 10^{-6}\,\text{kg} = 1\,\text{mg}$

12. **(4)** $f = (1 - e^{-\lambda t}) = 1 - e^{-\lambda t} \approx 1 - (1 - \lambda t) = \lambda t$

$\Rightarrow \quad f = 0.04$. Hence % decay $\approx 4\%$

13. **(3)** $\frac{A}{A_0} = \left(\frac{1}{2}\right)^{\frac{t}{T}}$

Where, A_0 is the initial activity of the radioactive material and A is the activity at t.

So, $\frac{12.5}{100} = \left(\frac{1}{2}\right)^{\frac{t}{T}} \quad \therefore \; t = 3T.$

14. **(2)** $\lambda_P = \frac{1}{\tau}; \lambda_Q = \frac{1}{2\tau}$

$\Rightarrow \quad \frac{R_P}{R_Q} = \frac{(A_0\lambda_P)e^{-\lambda_P t}}{A_0\lambda_Q e^{-\lambda_Q t}}$, At $t = 2\tau; \frac{R_P}{R_Q} = \frac{2}{e}$

15. **(b)** A-2, B-1, C-4, D-3

Kinetic energy $K = \frac{1}{2}mv^2$ or, $K = \frac{p^2}{2m}$ or $p = \sqrt{2mK}$

If $K = 0, \; p = 0$ as above

For different mass m_1 and m_2 if $E_1 = E_2$

$\frac{p_1^2}{2m_1} = \frac{p_2^2}{2m_2}$ or $\frac{p_1}{p_2} = \sqrt{\frac{m_1}{m_2}}$

If $p_1 = p_2$ for different masses. $\frac{E_1}{E_2} = \frac{p_1^2}{2m_1} \times \frac{2m_2}{p_2} = \frac{m_2}{m_1}$

16. **(b)** A-2, B-4, C-3, D-1

Waves having frequency less than minimum audible frequency are infrasonics.

Waves of frequency between 20Hz to 20,000Hz are called audible waves. Waves having frequency > 20,000Hz are ultrasonic waves. Supersonic objects are those which have velocity more than velocity of sound.

17. **(d)** A-4, B-3, C-2, D-1

The frequency ranges of various waves are as under :

Radiowaves; 5×10^5 to 10^9 Hz;

γ-rays; 3×10^{18} to 5×10^{22} Hz

Microwaves; 1×10^9 to 3×10^{11} Hz;

X-rays; 1×10^{16} to 3×10^{21} Hz.

18. **(d)** A-2, B-3, C-4, D-1

A transducer is a device that has input in electrical form or provides output in electrical form. Attenuation is loss of strength of a signal during propagation. Range is largest distance between transmitter and receiver. Bandwidth is range of frequencies over which communication system works.

□ □ □

JEE-MAIN: PHYSICS MOCK TEST-5

SECTION 1 (Multiple Choice Question)

1. Frequency is the function of density (ρ), length (a) and surface tension (T). Then its value is

a. $k\rho^{1/2}a^{3/2}/\sqrt{T}$ **b.** $k\rho^{3/2}a^{3/2}/\sqrt{T}$

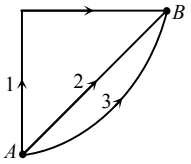
c. $k\rho^{1/2}a^{3/2}/T^{3/4}$ **d.** $k\rho^{1/2}a^{1/2}/T^{3/2}$

2. With respect to a rectangular cartesian coordinate system, three vectors are expressed as: $\vec{a}=4\hat{i}-\hat{j},\vec{b}=-3\hat{i}+2\hat{j}$ and $\vec{c}=-\hat{k}$ where $\hat{i},\hat{j},\hat{k}$ are unit vectors, along the *X*, *Y* and *Z*-axis respectively. The unit vectors $\hat{r}$ along the direction of sum of these vector is:

a. $\hat{r}=\frac{1}{\sqrt{3}}(\hat{i}+\hat{j}-\hat{k})$ **b.** $\hat{r}=\frac{1}{\sqrt{2}}(\hat{i}+\hat{j}-\hat{k})$

c. $\hat{r}=\frac{1}{3}(\hat{i}-\hat{j}+\hat{k})$ **d.** $\hat{r}=\frac{1}{\sqrt{2}}(\hat{i}+\hat{j}+\hat{k})$

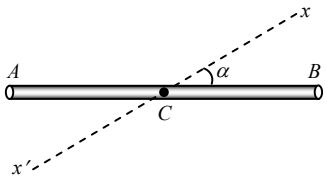
3. If W_1,W_2 and W_3 represent the work done in moving a particle from *A* to *B* along three different paths 1, 2 and 3 respectively (as shown) in the gravitational field of a point mass *m*. Find the correct relation between W_1,W_2 and W_3.



a. $W_1 > W_2 > W_3$ **b.** $W_1 = W_2 = W_3$

c. $W_1 < W_2 < W_3$ **d.** $W_2 > W_1 > W_3$

4. The moment of inertia of a uniform rod of length $2l$ and mass *m* about an axis xx' passing through its center and inclined at an angle α is:



a. $\frac{ml^2}{3}\sin^2\alpha$ **b.** $\frac{ml^2}{12}\sin^2\alpha$

c. $\frac{ml^2}{6}\cos^2\alpha$ **d.** $\frac{ml^2}{2}\cos^2\alpha$

5. A mass *M* is suspended from a spring of negligible mass. The spring is pulled a little and then released, so that the mass executes SHM of time period *T*. If the mass is increased by *m*, the time period becomes $5T/3$. The ratio of m/M is

a. $\frac{5}{3}$ **b.** $\frac{3}{5}$ **c.** $\frac{16}{9}$ **d.** $\frac{28}{9}$

6. The gravitational potential difference between the surface of a planet and a point 10 m above is 4.0 J/kg. The gravitational field in this region, assumed uniform is

a. 0.025 N/kg **b.** 0.40 N/kg

c. 40 N/kg **d.** 4.0 N/kg

7. 1000 drops of water all of same size join together to form a single drop and the energy released raises the temperature of the drop. Given the *T* is the surface tension of water. *r* the radius of each small drop, ρ the density of liquid, *J* the mechanical equivalent of heat. What is the rise in temperature?

a. T/Jr **b.** $10T/Jr$

c. $100T/Jr$ **d.** None of these

8. A simple pendulum is made by attaching a 1 kg bob to 5 m long copper wire. Its period is *T*. now if 1 kg bob is replaced by 10 kg bob, the period of oscillations:

a. remains *T*

b. becomes greater than *T*

c. becomes less than *T*

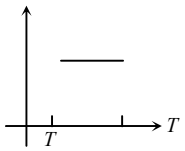
d. any of above answer depends on locality

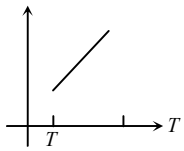
9. On which of the given scales of temperature, the temperature is inverse negative?

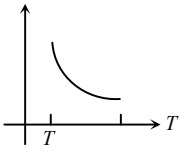
a. Celsius **b.** Fahrenheit

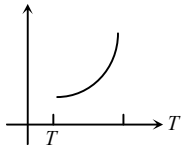
c. Reaumur **d.** Kelvin

10. An ideal gas is initially at temperature *T* and volume *V*. Its volume is increased by ΔV due to an increase in temperature ΔT, pressure remaining constant. The quantity $\delta=\Delta V/V\Delta T$ varies with temperature is:

a.


b.


c.


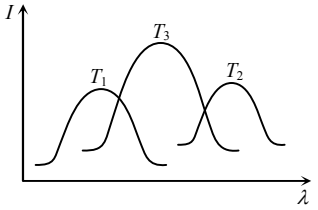
d.


11. A lead ball moving with velocity V strikes a wall and stops. If 50T of its energy is converted into heat, then what will be the increases in temperature (specific heat of lead is s)?

a. $\frac{2V^2}{Js}$ **b.** $\frac{V^2}{4Js}$

c. $\frac{V^2 s}{J}$ **d.** $\frac{V^2 s}{2J}$

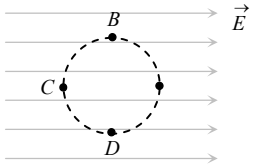
12. The plots of intensity *versus* wavelength for three black bodies at temperatures T_1, T_2 and T_3 respectively are as shown. Their temperatures are such that



a. $T_1 > T_2 > T_3$ **b.** $T_1 > T_3 > T_2$

c. $T_2 > T_3 > T_1$ **d.** $T_3 > T_2 > T_1$

13. The electric field in a region surrounding the origin is uniform and along the x-axis. A small circle is drawn with the centre at the origin cutting the axes at points A, B, C, D having co-ordinates $(a, 0)$, $(0, a)$, $(-a, 0)$, $(0, -a)$; respectively as shown in figure then potential in minimum at the point



a. A **b.** B

c. C **d.** D

14. A spherical capacitor consists of two concentric spherical conductors of inner one of radius R_1 maintained at potential V_1 and the outer one of radius R_2 at potential V_2. The potential centre $(R_2 > x > R_1)$ is:

a. $\frac{V_1 - V_2}{R_2 - R_1}(x - R_1)$ **b.** $\frac{V_1 R_1 (R_2 - x) + V_2 R_2 (x - R_1)}{(R_2 - R_1)x}$

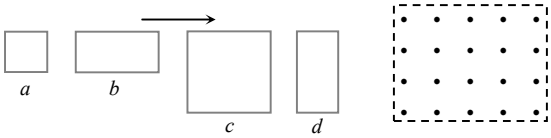
c. $V_1 + \frac{V_2 x}{R_1 - R_1}$ **d.** $\frac{V_1 + V_2}{R_2 + R_1}x$

15. Two identical short bar magnets, each having magnetic moment M, are placed a distance of $2d$ apart with axes perpendicular to each other in a horizontal plane. The magnetic induction at a point midway between them is

a. $\frac{\mu_0}{4\pi}(\sqrt{2})\frac{M}{d^3}$ **b.** $\frac{\mu_0}{4\pi}(\sqrt{3})\frac{M}{d^3}$

c. $\left(\frac{2\mu_0}{\pi}\right)\frac{M}{d^3}$ **d.** $\frac{\mu_0}{4\pi}(\sqrt{5})\frac{M}{d^3}$

16. The figure shows four wire loops, with edge lengths of either L or $2L$. All four loops will move through a region of uniform magnetic field $\vec{B}$ (directed out of the page) at the same constant velocity. Rank the four loops according to the maximum magnitude of the e.m.f. induced as they move through the field, greatest first:



a. $(e_c = e_d) < (e_a = e_b)$ **b.** $(e_c = e_d) > (e_a = e_b)$

c. $e_c > e_d > e_b > e_a$ **d.** $e_c < e_d < e_b < e_a$

17. A resistance is connected to an AC source. The average power dissipation in the resistance is

a. 50% of the peak power

b. 25% of the peak power

c. 41.4% of the peak power

d. equal to peak power

18. When a point source of monochromatic light is at a distance of 0.2 m from a photoelectric cell, the cut-off voltage and the saturation current are 0.6 V and 18 mA respectively. If the same source is placed 0.6 m away from the photoelectric cell, then

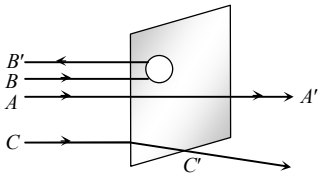
a. the stopping potential will be 0.2 V

b. the stopping potential will be 0.6 V

c. the saturation current will be 6 mA

d. the saturation current will be 18 mA

19. A beam of fast moving alpha particles were directed towards a thin film of gold. The parts A', B' and C' of the transmitted and reflected beams corresponding to the incident parts A, B and C of the beam, are shown in the adjoining diagram. The number of alpha particles in



a. B' will be minimum and in C' maximum

b. A' will be maximum and in B' minimum

c. A' will be minimum and in B' maximum

d. C' will be minimum and in B' maximum

20. Energy released in the fission of a single $_{92}U^{235}$ nucleus is 200 MeV. The fission rate of a $_{92}U^{235}$ fuelled reactor operating at a power level of 5W is

a. $1.56 \times 10^{+10}\, s^{-1}$ **b.** $1.56 \times 10^{+11}\, s^{-1}$

c. $1.56 \times 10^{+16}\, s^{-1}$ **d.** $1.56 \times 10^{+17}\, s^{-1}$

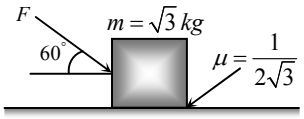
SECTION 2 (Numeric Value Question)

21. A train of 150 m in length is going towards north direction at a speed of 10 m/s. A parrot flies at a speed of 5 m/s towards south direction parallel to the railway track. Then the time taken by the parrot to cross the train is equal to:

a. 12 sec **b.** 8 sec

c. 10 sec **d.** 15 sec

22. What is the maximum value of the force F such that the block shown in the arrangement, does not move?



a. 20 N **b.** 10 N

c. 12 N **d.** 15 N

23. When an astronaut in a rocket reaches near the moon, he sends a radiowave of frequency 5000 MHz towards the moon. He finds that the reflected wave from the moon has a frequency 86 kHz more than the real frequency. The velocity of the rocket with respect to the moon is:

a. 1.29 km/sec **b.** 2.58 km/sec

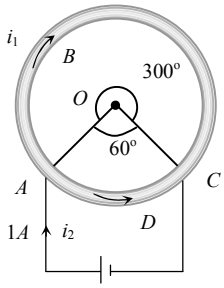
c. 3.87 km/sec **d.** 5.16 km/sec

24. A battery of 10 cells each of e.m.f. E = 1.5 V, and internal resistance 0.5Ω has 1 cell wrongly connected. It is being charged by 220V power supply with an external resistance of $47\,\Omega$ in series. The potential difference across the battery is:

a. 32 V **b.** 8 V

c. 180 V **d.** 188 V

25. A cell is connected between the points A and C of a circular conductor $ABCD$ of centre O with angle $AOC = 60°$. If B_1 and B_2 are the magnitudes of the magnetic fields at O due to the currents in ABC and ADC respectively, the ratio $\frac{B_1}{B_2}$ is



a. 0.2 **b.** 6

c. 1 **d.** 5

Space for Rough Work

JEE ADVANCE PAPER-I

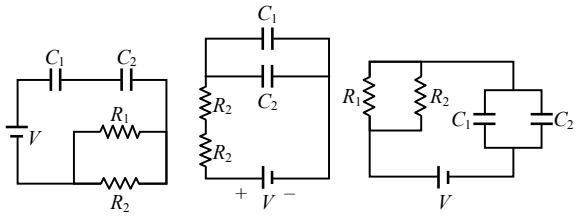
SECTION 1 (Maximum Marks: 24)

MCQs with one or more than one correct answer

1. Given,

$R_1 = 1\,\Omega$ $C_1 = 2\ \mu F$

$R_2 = 2\,\Omega$ $C_2 = 4\ \mu F$



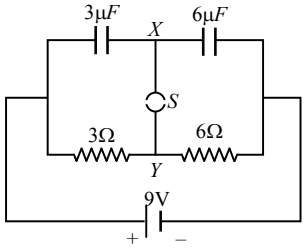
(I) (II) (III)

The time constants (in μS) for the circuits I, II, III are respectively

a. 18, 8/9, 4 **b.** 18, 4, 8/9

c. 4, 8/9, 18 **d.** 8/9, 18, 4

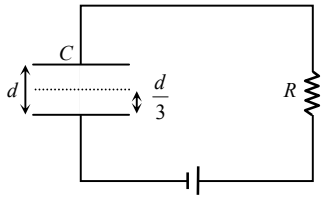
2. A circuit is connected as shown in the figure with the switch S open. When the switch is closed the total amount of charge that flows from Y to X is



a. 0 **b.** 54 μC

c. 27 μC **d.** 81 μC

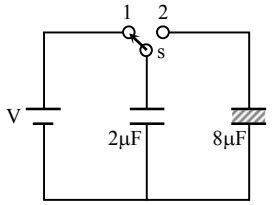
3. A parallel plate capacitor C with plates of unit area and separation d is filled with a liquid of dielectric constant $K = 2$. The level of liquid is $\frac{d}{3}$ initially. Suppose the liquid level decreases at a constant speed V, the time constant as a function of time t is



a. $\frac{6\varepsilon_0 R}{5d + 3Vt}$ **b.** $\frac{(15d + 9Vt)\varepsilon_0 R}{2d^2 - 3dVt - 9V^2t^2}$

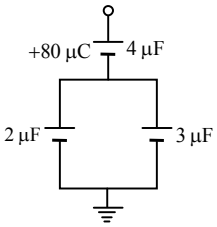
c. $\frac{6\varepsilon_0 R}{5d - 3Vt}$ **d.** $\frac{(15d - 9Vt)\varepsilon_0 R}{2d^2 + 3dVt - 9V^2t^2}$

4. A $2\mu F$ capacitor is charged as shown in figure. The percentage of its stored energy dissipated after the switch S is turned to position 2 is



a. 0% **b.** 20% **c.** 75% **d.** 80%

5. In the given circuit, a charge of +80 μF is given to the upper plate of the 4μF capacitor. Then in the steady state, the charge on the upper plate of the 3μF capacitor is.



a. +32 μC **b.** +40 μC

c. +48 μC **d.** +80 μC

6. Two capacitors C_1 and C_2 are charged to 120 V and 200 V respectively. It is found that by connecting them together the potential on each one can be made zero. Then

a. $5C_1 = 3C_2$ **b.** $3C_1 = 5C_2$

c. $3C_1 + 5C_2 = 0$ **d.** $9C_1 = 4C_2$

SECTION 2 (Maximum Marks: 24)

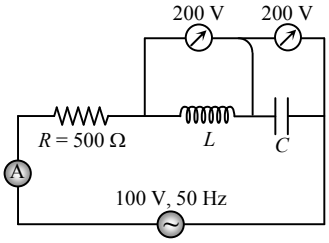
Numerical value answer type questions

7. The magnetic flux of ϕ (in weber) in a closed circuit of resistance 3 ohm varies with time t (in second) according to the equation $\phi = 2t^2 - 10t + 3$. What will be the magnetic of induced current at t = 0.25 s ?

8. An a.c. generator gives an output voltage of E = 170 sin 56·52t. What is the frequency of alternating voltage produced?

9. A condenser of capacitance 0.144 pH is used in a transmitter to transmit at wavelength λ. If inductance of $\frac{1}{\pi^2}$ mH is used for resonance, what is the value of λ?

10. In the series RLC circuit as shown in Fig., what would be the ammeter reading?



11. An a.c. source of frequency 1000 Hz is connected to a coil of $\frac{200}{\pi}$ mH and negligible resistance. If effective current through the coil is 7.5 mA, what is the voltage across the coil?

12. Two alternating currents are given by $I_1 = I_0 \sin(\omega t - \phi)$ and $I_2 = I_0 \cos(\omega t + \phi)$. What is the ratio of virtual values of the two currents?

13. A silver sphere of radius 1 cm and work function 4.7 eV is suspended from an insulating thread in free-space. It is under continuous illumination of 20 nm wavelength light. As photoelectrons are emitted, the sphere gets charged and acquires a potential. The maximum number of photo-electrons emitted from the sphere is $A \times 10^Z$ (where $1 < A < 10$). The value of Z is

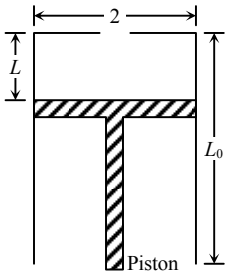
14. When a piece of metal is illuminated by monochromatic light of wavelength λ then the stopping potential for photoelectric current is $3V_0$. When the same surface is illustrated by light of wavelength 2λ, then stopping potential becomes the V_0 value of Threshold wavelength for photoelectric current is $a\lambda$, where the value of a is

SECTION 3 (Maximum Marks: 12)

Paragraph based questions (2 paragraphs, each having 2 MCQs with one correct answer only)

Paragraph for Question No. 15 to 16

A fixed thermally conducting cylinder has a radius R and height L_0. The cylinder is open at its bottom and has a small hole at its top. A piston of mass M is held at a distance L from the top surface, as shown in the figure. The atmospheric pressure is P_0.



15. The piston is now pulled out slowly and held at a distance $2L$ from the top. The pressure in the cylinder between its top and the piston will then be

a. P_0 **b.** $\frac{P_0}{2}$

c. $\frac{P_0}{2} + \frac{Mg}{\pi R^2}$ **d.** $\frac{P_0}{2} - \frac{Mg}{\pi R^2}$

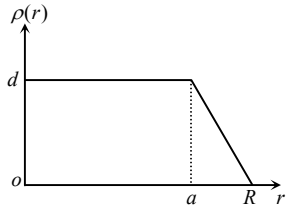
16. While the piston is at a distance $2L$ from the top, the hole at the top is sealed. The piston is then released, to a position where it can stay in equilibrium. In this condition, the distance of the piston from the top is

a. $\left(\frac{2P_0\pi R^2}{\pi R^2 P_0 + Mg}\right)(2L)$ **b.** $\left(\frac{P_0\pi R^2 - Mg}{\pi R^2 P_0}\right)(2L)$

c. $\left(\frac{P_0\pi R^2 + Mg}{\pi R^2 P_0}\right)(2L)$ **d.** $\left(\frac{P_0\pi R^2}{\pi R^2 P_0 - Mg}\right)(2L)$

Paragraph for Question No. 17 to 18

The nuclear charge (Ze) is non-uniformly distributed within a nucleus of radius R. The charge density $\rho(r)$ [charge per unit volume] is dependent only on the radial distance r from the centre of the nucleus as shown in figure. The electric field is only along the radial direction.



17. The electric field at $r = R$ is

a. independent of a

b. directly proportional to a

c. directly proportional to a^2

d. inversely proportional to a

18. For $a = 0$, the value of d (maximum value of ρ as shown in the figure) is

a. $\frac{3Ze}{4\pi R^3}$ **b.** $\frac{3Ze}{\pi R^3}$

c. $\frac{4Ze}{3\pi R^3}$ **d.** $\frac{Ze}{3\pi R^3}$

Space for Rough Work

JEE ADVANCE PAPER-II

SECTION 1 (Maximum Marks: 24)

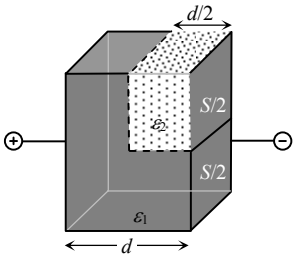
MCQs with one or more than one correct answer

1. A parallel plate capacitor has a dielectric slab of dielectric constant K between its plates that covers 1/3 of the area of its plates, as shown in the figure. The total capacitance of the capacitor is C while that of the portion with dielectric in between is C_1. When the capacitor is charged, the plate area covered by the dielectric gets charge Q_1 and the rest of the area gets charge Q_2. The electric field in the dielectric is E_1 and that in the other portion is E_2. Choose the correct option/options, ignoring edge effects.

a. $\frac{E_1}{E_2}=1$ **b.** $\frac{E_1}{E_2}=\frac{1}{K}$

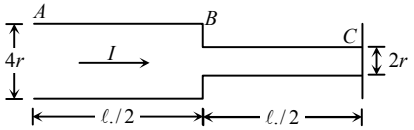
c. $\frac{Q_1}{Q_2}=\frac{3}{K}$ **d.** $\frac{C}{C_1}=\frac{2+K}{K}$

2. A parallel plate capacitor having plates of area S and plate separation d, has capacitance C_1 in air. When two dielectrics of different relative permittivities ($\varepsilon_1 = 2$ and $\varepsilon_2 = 4$) are introduced between the two plates as shown in the figure, the capacitance becomes C_2. The ratio $\frac{C_2}{C_1}$ is



a. 6/5 **b.** 5/3 **c.** 7/5 **d.** 7/3

3. Consider a cylindrical element as shown in the figure. Current flowing the through element is I and resistivity of material of the cylinder is ρ. Choose the correct option out the following



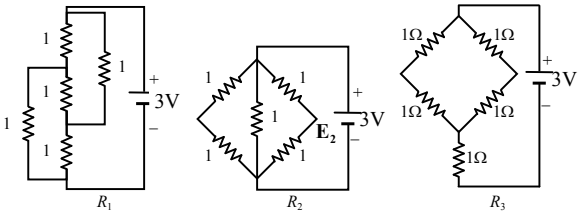
a. Power loss in first half is four times the power loss in second half

b. Voltage drop in first half is twice of voltage drop in second half

c. Current density in both halves are equal

d. Electric field in both halves is equal

4. Figure shows three resistor configurations R_1, R_2, and R_3 connected to 3V battery. If the power dissipated by the configuration R_1, R_2, and R_3 is P_1, P_2, and P_3, respectively, then



a. $P_1 > P_2 > P_3$ **b.** $P_1 > P_3 > P_2$

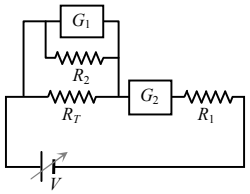
c. $P_2 > P_1 > P_3$ **d.** $P_3 > P_2 > P_1$

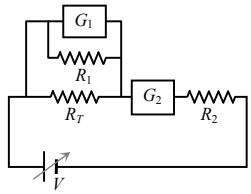
5. Incandescent bulbs are designed by keeping in mind that the resistance of their filament increases with the increase in temperature. If at room temperature, 100 W, 60 W and 40 W bulbs have filament resistances $R100$, $R60$ and $R40$, respectively, the relation between these resistances is

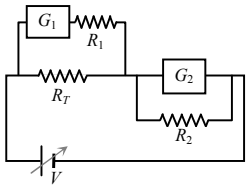
a. $\frac{1}{R_{100}}=\frac{1}{R_{40}}+\frac{1}{R_{60}}$ **b.** $R_{100}=R_{40}+R_{60}$

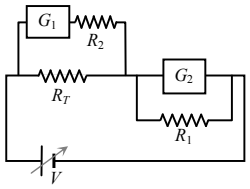
c. $R_{100} > R_{60} > R_{40}$ **d.** $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

6. To verify Ohm's law, a student is provided with a test resistor R_T, a high resistance R_1, a small resistance R_2, two identical galvanometers G_1 and G_2, and a variable voltage source V. The correct circuit to carry out the experiment is

a.


b.


c.


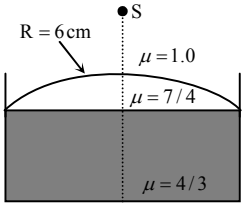
d.


SECTION 2 (Maximum Marks: 24)

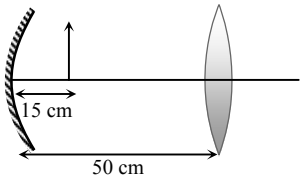
Numerical value answer type questions

7. Image of an object approaching a convex mirror of radius of curvature 20 m along its optical axis is observed to move from $\frac{25}{3}$ m to $\frac{50}{7}$ m in 30 seconds. What is the speed of the object in km per hour?

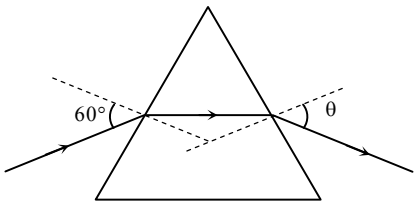
8. Water (with refractive index = 4/3) in a tank is 18 cm deep. Oil of refractive index $\frac{7}{4}$ lies on water making a convex surface of radius of curvature '$R = 6$ cm' as shown. Consider oil to act as a thin lens. An object 'S' is placed 24 cm above water surface. The location of its image is at 'x' cm above the bottom of the tank. Then 'x' is



9. Consider a concave mirror and a convex lens (refractive index = 1.5) of focal length 10 cm each, separated by a distance of 50 cm in air (refractive index = 1) as shown in the figure. An object is placed at a distance of 15 cm from the mirror. Its erect image formed by this combination has magnification M_1. When the set-up is kept in a medium of refractive index 7/6, the magnification becomes M_2. The magnitude $\left|\frac{M_2}{M_1}\right|$ is



10. A monochromatic beam of light is incident at 60° on one face of an equilateral prism of refractive index n and emerges from the opposite face making an angle θ (n) with the normal (see the figure). For $n = \sqrt{3}$ the value of θ is 60° and $\frac{d\theta}{dn} = m$. The value of m is



11. An α-particle and a proton are accelerated from rest by a potential difference of 100 V. After this, their de Broglie wavelengths are λ_α and λ_p respectively. The ratio $\frac{\lambda_p}{\lambda_\alpha}$, to the nearest integer, is

12. A silver sphere of radius 1 cm and work function 4.7 eV is suspended from an insulating thread in free space. It is under continuous illumination of 200 nm wavelength light. As photoelectrons are emitted, the sphere gets charged and acquires a potential. The maximum number of photoelectrons emitted from the sphere is $A \times 10^z$ (where $1 < A < 10$). The value of 'Z' is

13. A proton is fired from very far away towards a nucleus with charge $Q = 120$ e, where e is the electronic charge. It makes a closest approach of 10 fm to the nucleus. The de Broglie wavelength (in units of fm) of the proton at its start is: (take the proton mass, $m_p (5/3) \times 10^{-27}$ kg; $h/e = 4.2 \times 10^{-15}\ J.s/C; \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ m/F; 1\ fm = 10^{-15}\ m$)

14. The work functions of silver and sodium are 4.6 eV and 2.3 eV, respectively. The ratio of the slope of the stopping potential versus frequency plot for silver to that of sodium is

SECTION 3 (Maximum Marks: 12**)**

Matching type questions with 4 options

15. Some laws / processes are given in Column I. Match these with the physical phenomena given in Column II.

Column I	Column II
(A) Intensity of light received by lens	**1.** Radius of aperture (R)
(B) Angular magnification	**2.** Dispersion of lens
(C) Length of telescope	**3.** Focal length f_a, f_e
(D) Sharpness of image	**4.** Spherical aberration

a. A-1; B-3; C-3; D-1,2,3
b. A-1,2,3; B-4; C-2,3, D-1
c. A-1; B-3; C-1,2,3; D-2
d. A-2,3; B-1; C-4; D-3

16. Study of properties of matter helps the student to know some phenomenon having important applications of the principles learnt by him in his daily life. Can you match some of the phenomenon you observe with the principles you have studied.

Column I	Column II
(A) A cricketer spins his ball while bowling to change the direction and momentum of ball is	**1.** Surface tension

according to principle of	
(B) A painter askes you for primers coating on walls before panting to remove sewage of water through bricks in walls	**2.** Gravitational pull
(C) Formation of stars in universe	**3.** Bernoulli's pull
(D) Ships are asked to spread oil on sea surface during his tides	**4.** Capillarity

a. A-1; B-2; C-3; D-4 **b.** A-4; B-3; C-2; D-1
c. A-3; B-2; C-1; D-4 **d.** A-3; B-4; C-2; D-1

17. Match the statement of Column I with those in Column II:

Column I	Column II
(A) Magnetic field intensity is defined as	**1.** The spin motion of electron
(B) Which of the following, the most suitable material for making permanent magnet is	**2.** Steel
(C) In the case of bar magnet, lines of magnetic induction	**3.** Run continuously through the bar and outside
(D) A sensitive magnetic instrument can be shielded very effectively from outside magnetic fields by placing it inside a box of	**4.** Soft iron of high permeability

a. A-4; B-3; C-2; D-1 **b.** A-2; B-4; C-3; D-1
c. A-1; B-2; C-3; D-4 **d.** A-4; B-1; C-3; D-2

18. Column I defines the type of modulus or coefficient of elasticity. Column II gives the type of corresponding modulus. Match the definition with proper type of elasticity.

Column I	Column II
(A) Ratio of longitudinal or tensile stress to longitudinal strain	**1.** Modulus of Rigidly
(B) Ratio of normal or hydrostatic stress to volumetric strain	**2.** Poisson's ratio
(C) Ratio of lateral strain to longitudinal strain	**3.** Bulk modulus
(D) Ratio of tangential stress to shear strain	**4.** Young's modulus

a. A-1; B-2; C-3; D-4 **b.** A-4; B-3; C-2; D-1
c. A-3; B-2; C-1; D-4 **d.** A-2; B-1; C-4; D-3

Space for Rough Work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	a	b	a	c	b	d	b	d	c
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b	b	a	b	d	b	a	b	b	b
21.	**22.**	**23.**	**24.**	**25.**					
c	a	d	a	c					

1. **(d)** Velocity gradient $= \frac{v}{x} = \frac{[LT^{-1}]}{[L]} = [T^{-1}]$

Potential gradient $= \frac{V}{x} = \frac{[ML^2T^{-3}A^{-1}]}{[L]} = [MLT^{-3}A^{-1}]$

Energy gradient $= \frac{E}{x} = \frac{[ML^2T^2]}{[L]} = [MLT^{-2}]$

Pressure gradient $= \frac{P}{x} = \frac{[ML^{-1}T^{-2}]}{[L]} = [ML^{-2}T^{-2}]$

2. **(a)** $\vec{r} = \vec{a} + \vec{b} + \vec{c} = 4\hat{i} - \hat{j} - 3\hat{i} + 2\hat{j} - \hat{k} = \hat{i} + \hat{j} - \hat{k}$

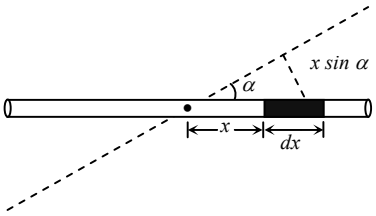
$\hat{r} = \frac{\vec{r}}{|r|} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{1^2 + 1^2 + (-1)^2}} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$

3. **(b)** Gravitational field is a conservative force field. In a conservative force field work done is path independent

$\therefore \quad W_1 = W_2 = W_3$

4. **(a)** The desired moment of inertia is $I = \int_{x=-l}^{x=+l} dI$

$= \int_{-l}^{+l} \left(\frac{m}{2l} dx\right)(x \sin\alpha)^2 = \frac{ml^2}{3}\sin^2\alpha$



5. **(c)** $m_1 = M, T_1 = T$

$\Rightarrow \quad m_2 = M + m, T_2 = \frac{5T}{3}$

$\Rightarrow \quad \frac{T_1}{T_2} = \frac{2\pi\sqrt{m_1/K}}{2\pi\sqrt{m_2/K}} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{M}{M+m}}$

or, $\frac{M+m}{M} = \left(\frac{T_2}{T_1}\right)^2 = \left(\frac{5T/3}{T}\right) = \frac{25}{9}$

$\Rightarrow \quad \frac{m}{M} = \frac{16}{9}$

6. **(b)** $f_g = \frac{\Delta V_g}{\Delta r} = \frac{4}{10} = 0.4$ N/kg

7. **(d)** $\frac{4}{3}\pi R^3 = n \times \frac{4}{3}\pi r^3$

$R = (n)^{1/3} r = 10r$

Decrease in surface area $\Delta A = n(4\pi r^2) - 4\pi R^2$

$= 1000(4\pi r^2) - 4\pi(10r)^2$

$= 4\pi r^2[1000 - 100] = 900 \times 4\pi r^2$

Energy released $= \Delta A \times T = \frac{900 \times 4\pi r^2 \mathrm{T}}{J}$ cal

Let θ be the rise in temperature;' then

$1000 \times \frac{4}{3}\pi r^3 \times 1 \times d\theta = \frac{900 \times 4\pi r^2 \mathrm{T}}{J}$

$\therefore \quad d\theta = \frac{2.7\mathrm{T}}{Jr}$

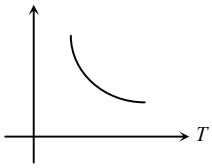
8. **(b)** For simple pendulum $T = 2\pi\sqrt{\frac{l}{g}}$ when weight suspended on elastic wire is increased, it length increases, so time period increases.

9. **(d)** Zero Kelvin $= -237\ ^\circ\mathrm{C}$ (absolute temperature). As no matter can attain this temperature, hence temperature can never be negative on Kelvin scale.

10. **(c)** For an ideal gas : $pV = nRT$

For p = constant $p\Delta V = nT\Delta T$

$\therefore \quad \frac{\Delta V}{\Delta T} = \frac{nR}{p} = \frac{nR}{\frac{nRT}{V}} = \frac{V}{T}$



$\therefore \quad \frac{\Delta V}{V\Delta T} = \frac{1}{T}$ or $\delta = \frac{1}{T}$

Therefore, δ is inversely proportional to temperature T. *i.e.*, when T increases, δ decreases and *vice-versa*. Hence, $\delta - T$ graph will be a rectangular hyperbola as shown in the above figure.

11. **(b)** $\frac{1}{2} \cdot \frac{1}{2} mv^2 = J\ ms\ \Delta\theta$

$\Rightarrow \quad \Delta\theta = \frac{v^2}{4\ Js}$

12. **(b)** Wien's displacement law for a perfectly black body is $\lambda_m T =$ constant = Wien's constant b

Here, λ_m is the minimum wavelength corresponding to maximum intensity I.

or $\lambda_m \propto \frac{1}{T}$

From the figure $(\lambda_m)_1 < (\lambda_m)_3 < (\lambda_m)_2$

Therefore, $T_1 > T_3 > T_2$

13. **(a)** In the direction of electric field, potential decreases.

14. **(b)** Let q_1 and q_2 be charges on inner and outer spherical conductors.

Then $V_1 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{R_1} + \frac{q_2}{R_2}\right)$ . . .(*i*)

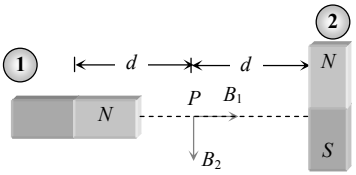
Also $V_2 = \frac{1}{4\pi\varepsilon_0}\frac{q_1 + q_2}{R_2}$ . . .(*ii*)

Then potential at point *P*, $V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{x} + \frac{q_2}{R_2}\right)$ . . .(*iii*)

Solving (*i*) and (*ii*), q_1 and q_2 may be found. If these are substituted in (*iii*), we get

$$V = \frac{V_1 R_1 (R_2 - x) + V_2 R_2 (x - R_1)}{(R_2 - R_1)x}$$

15. **(d)** At point *P* net magnetic field $B_{net} = \sqrt{B_1^2 + B_2^2}$



where $B_1 = \frac{\mu_0}{4\pi}.\frac{2M}{d^3}$ and $B_2 = \frac{\mu_0}{4\pi}.\frac{M}{d^3}$

$\Rightarrow$ $B_{net} = \frac{\mu_0}{4\pi}.\frac{\sqrt{5}M}{d^3}$

16. **(b)** Emf induces across the length of the wire which cuts the magnetic field.

(Length of *c* = Length *d*) > (Length of *a* = *b*).

So, $(e_c = e_d) > (e_a = e_b)$

17. **(a)** Peak power, $P_0 = V_0 I_0$

Average power dissipation, $P_{av} = \frac{V_0 I_0}{2}\cos\phi$

For a resistive circuit $\phi = 0$

$\Rightarrow$ $\cos\phi = 1$

$\Rightarrow$ $P_{av} = \frac{V_0 I_0}{2} = \frac{P_0}{2} = 50\%$ of P_0

18. **(b)** By changing distance the intensity change but frequency remains same; so stopping potential remains same.

19. **(b)** Because atom is hollow and whole mass of atom is concentrated in a small centre called nucleus.

20. **(b)** $\frac{\text{Energy}}{\text{Fission}} = 200 \text{ MeV} = 200 \times 10^6 \times 1.6 \times 10^{-19}$ J

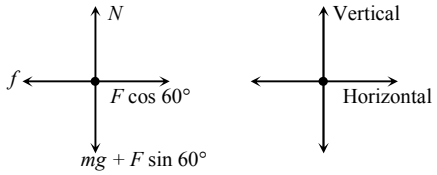
Fission rate $= \frac{5}{200 \text{ MeV}} = 1.56 \times 10^{11}$ fission/sec.

21. **(c)** Relative velocity of parrot and train

$= 10 + 5 = 15$ m/s

$\therefore$ Time $= \frac{s}{v} = \frac{150}{15} = 10$ s

22. **(a)** Free body diagram (FBD) of the block (shown by a dot) is shown in figure. For vertical equilibrium of the block



$N = mg + F\sin 60° = \sqrt{3g} + \sqrt{3}\frac{F}{2}$

For no motion, force of friction or $f \geq F\cos 60°$

$\Rightarrow$ $\mu N \geq F\cos 60°$ or $\frac{1}{2\sqrt{3}}\left(\sqrt{3g} + \frac{\sqrt{3}F}{2}\right) \geq \frac{F}{2}$

or $g \geq \frac{F}{2}$

or $F \leq 2g$ or 20N

Therefore, maximum value of *F* is 20 N.

23. **(d)** Here change in frequency takes place in two steps

$\therefore$ $\Delta v = \frac{2v}{c}v$

$\Rightarrow$ $86 \times 10^3 = 2\frac{v}{c} \times 5000 \times 10^6$

$\Rightarrow$ $v = \frac{86 \times 10^3}{10^{10}}$

$c = 86 \times 10^{-7} \times 3 \times 10^8$ m/s

$= 2.58 \times 10^3$ m/s

24. **(a)** Current in circuit

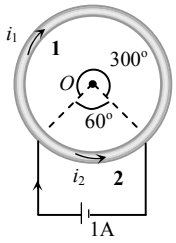
$$i = \frac{220 - (9-1)\times 1\cdot 5}{47 + 0\cdot 5 \times 10} = \frac{220 - 12}{52} = 4 \text{ A}$$

$\therefore$ Potential difference across battery which is being charged

$V = E + ir$

$= (8 \times 1\cdot 5) + 4 \times (0\cdot 5 \times 10) = 32$ V

25. **(c)** $B = \frac{\mu_0}{4\pi}\frac{\theta i}{r} \Rightarrow B \propto \theta\, i$ (but $\frac{i_1}{i_2} = \frac{l_2}{l_1} = \frac{\theta_2}{\theta_1}$)



$\Rightarrow \quad \frac{B_1}{B_2} = \frac{\theta_1}{\theta_2}.\frac{i_1}{i_2}$

So, $\frac{B_1}{B_2} = \frac{\theta_1}{\theta_2} \times \frac{\theta_2}{\theta_1} \Rightarrow B_1 = B_2$

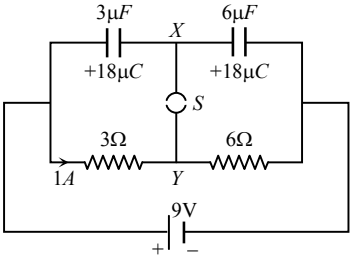
JEE Advance Paper-I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	c	a	d	c	b	3	9	7	2
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**		
3	1	7	4	a	d	a	b		

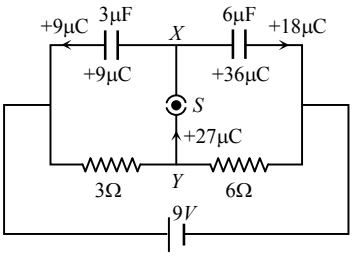
1. **(d)** $\tau_1 = 8/9\ \mu S$

$\Rightarrow \quad \tau_2 = 18\ \mu S \Rightarrow \tau_3 = 4\ \mu S$

2. **(c)** $27\ \mu C$



Initial charge distribution (when switch S is open)



Final charge distribution (when switch S is closed)

3. **(a)** $C_{\text{equivalent}} = \dfrac{\frac{2\varepsilon_0}{\frac{d}{3}-vt}.\frac{\varepsilon_0}{\frac{2d}{3}+vt}}{\frac{2\varepsilon_0}{\frac{d}{3}-vt}+\frac{\varepsilon_0}{\frac{2d}{3}+vt}}$

$\therefore \quad \tau = C_{\text{equivalent}} R$

4. **(d)** $U_i = \frac{1}{2}\times(2)V^2$

$\Rightarrow \quad \Delta U_{loss} = \frac{1}{2}\times\frac{(2)\times(8)}{10}\times V^2$

$\Rightarrow \quad \eta_{lost} = \frac{\Delta U_{loss}}{U_i} = \frac{4}{5} = 80\%$

5. **(c)** Charge on capacitor of $3\ \mu F$

$\Rightarrow \quad Q = 80\frac{3}{2+3} = 48\mu C$

6. **(b)** Potential $= 0$ on connecting them together *i.e.,* $Q = 0$

i.e. $C_1V_1 = C_2V_2$ [capacitance is positive but they are connected with opposite polarity]

$\Rightarrow \quad 120\, C_1 = 200\, C_2$

$\Rightarrow \quad 3C_1 = 5C_2$

7. **(3)** Here, $R = 3\ \Omega$

$\Rightarrow \quad \phi = 2t^2 - 10t + 3$

$I = ?t = 0.25\,g$

$\Rightarrow \quad e = \frac{d\phi}{dt} = -\frac{d}{dt}(2t^2 - 10t + 3)$

$\Rightarrow e = -4\,t + 10$

At $t = 0.25\ s$, $e = -4\times0.25 + 10 = 9$ volt

$\Rightarrow \quad I = \frac{e}{R} = \frac{9}{3} = 3$ A

8. **(9)** Here, $E = 170\sin 56.52t$

Compare with the standard form of equation of alternating emf

$E = E_0 \sin\omega t$

$\Rightarrow \quad \omega = 56.52$

$\Rightarrow \quad 2\pi v = 56.52$

$\Rightarrow \quad v = \frac{56.52}{2\pi} = \frac{56.52}{2\times3.14} = 9$ Hz.

9. **(7)** Here, $C = 0.144\,\text{pF} = 0.144\times10^{-12}\,\text{F}$

$\Rightarrow \quad L = \frac{1}{\pi^2}\,\text{mH} = \frac{10^{-3}}{\pi^2}\,\text{H}$

$\Rightarrow \quad \upsilon = \frac{1}{2\pi\sqrt{LC}}$

$\Rightarrow \quad \lambda = \frac{v}{\upsilon} = \upsilon.2\pi\sqrt{LC}$

$= 3\times10^8 \times 2\pi\sqrt{\frac{10^{-3}}{\pi^2}\times0.144\times10^{-12}}$

$\Rightarrow \quad \lambda = 3\times10^8\times2\pi\times\frac{1.2\times10^{-8}}{\pi}$

$\lambda = 7.2$ m

The correct answer is 7 m.

10. **(2)** As $V_L = V_C = 200\,\text{V}$

$\therefore$ The circuit is series resonance circuit $Z = R$

$\Rightarrow \quad I_v = \dfrac{E_v}{Z} = \dfrac{E_v}{R} = \dfrac{100}{50} = 2\text{A}$

11. **(3)** Here, $v = 1000$ Hz,

$L = \dfrac{200}{\pi}\text{mH} = \dfrac{0.2}{\pi}\text{H}$

$I_v = 7.5\,\text{mA} = 7.5\times10^{-3}\,\text{A}$

$E_v = ?$

As resistance of coil is negligible,

$\therefore \quad E_v = I_v\left(X_L\right) = I_v\left(2\pi vL\right)$

$= 7.5\times10^{-3}\left(2\pi\times1000\times\dfrac{0.2}{\pi}\right) = 3\text{V}$

12. **(1)** $\dfrac{I_{v_1}}{I_{v_2}} = \dfrac{I_0/\sqrt{2}}{I_0/\sqrt{2}} = 1$

Phase difference has no impact on virtual value of current.

13. **(7)** Here, $r = 1\text{ cm} = 10^{-2}\text{ m}; \phi_0 = 4.7$ eV, $\lambda = 200\,\text{nm}$.

Let V_s be the stopping potential. According to Einstein's photoelectric equation (using hc = 1240 eV nm), we have

$\text{eV}_s\dfrac{hc}{\lambda} - \phi_0 = \dfrac{1240\text{eV}}{200\text{nm}} - 4.7\text{eV}$

$= 6.2\text{eV} - 4.7\text{eV} = 1.5\text{eV} = 1.5\text{eV}$

$\therefore \quad V_s = 1.5$

The sphere will stop emitting photoelectrons, when the potential on its surface becomes 1.5 V. Let n be the no. of photoelectrons emitted from sphere.

The charge on sphere, $q = ne, V_s = 1.5\,\text{V}$

$\therefore \quad n = \dfrac{V_s\times r\times4\pi\varepsilon_0}{e} = \dfrac{(1.5)\times(10^{-2})}{1.6\times10^{-19}}\times\dfrac{1}{(9\times10^9)} = 1.05\times10^7$

As per questions, $A\times10^Z = 1.05\times10^7$ or $Z = 7$

14. **(4)** $3\text{eV}_0 = \dfrac{hc}{\lambda} - \phi_0$ and $\text{eV}_0 = \dfrac{hc}{2\lambda} - \phi_0$

or $2\text{eV}_0 = \dfrac{hc}{\lambda}\left[1 - \dfrac{1}{2}\right] = \dfrac{hf}{2\lambda}$ or $\dfrac{hc}{\text{eV}_0} = 4\lambda$

If λ_0 is the stopping wavelength, then $\text{eV}_0 = hc/\lambda_0$

or $\quad \lambda_0 = \dfrac{hc}{eV_0} = 4\lambda$

15. **(a)** Po

16. **(d)** $Mg + P(\pi R^2) = P_0\pi R^2$

$\Rightarrow \quad P_0(2L\pi R^2) = P(x\pi R^2)$

$(P_1V_1 = P_2V_2$ for isothermal process)

$\Rightarrow \quad x = \left(\dfrac{P_0\pi R^2}{\pi R^2P_0 - Mg}\right)(2L)$

17. **(a)** independent of a

18. **(b)** $q = \int_0^R \dfrac{d}{2}(R-x)4\pi x^2dx = Ze$

$\Rightarrow \quad d = \dfrac{3Ze}{\pi R^3}$

JEE Advance Paper-II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a,d	d	b	c	d	c	3	2	7	2
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**		
3	7	7	1	a	d	c	b		

1. **(a, d)** As E = V/d

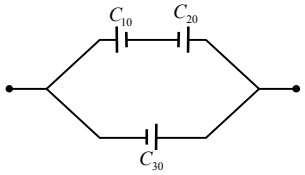
$E_1/E_2 = 1$ (both parts have common potential difference)

Assume C_0 be the capacitance without dielectric for whole capacitor.

$\Rightarrow \quad k\dfrac{C_0}{3} + \dfrac{2C_0}{3} = C$

$\Rightarrow \quad \dfrac{C}{C_1} + \dfrac{2+k}{k} \Rightarrow \dfrac{Q_1}{Q_2} + \dfrac{k}{2}.$

2. **(d)** $C_{10} = \dfrac{4\varepsilon_0\dfrac{S}{2}}{d/2} = \dfrac{4\varepsilon_0S}{d}$



$\Rightarrow \quad C_{20} = \dfrac{2\varepsilon_0S}{d}, C_{30}\dfrac{\varepsilon_0S}{d}$

$\Rightarrow \quad \dfrac{1}{C_{10}'} = \dfrac{1}{C_{10}} + \dfrac{1}{C_{10}} = \dfrac{d}{2\varepsilon_0S}\left[1 + \dfrac{1}{2}\right]$

$\Rightarrow \quad C_{10}' = \dfrac{4\varepsilon_0S}{3d}$

$\Rightarrow \quad C_2 = C_{30} + C_{10}' = \dfrac{7\varepsilon_0S}{3d}$

$\Rightarrow \quad \dfrac{C_2}{C_1} = \dfrac{7}{3}$

3. **(b)** $\frac{R_1}{R_2}=\frac{A_1}{A_2}=\frac{4}{1}$

$\Rightarrow \quad \frac{P_1}{P_2}=\frac{I^2R_1}{I^2R_2}=\frac{4}{1}$

$\Rightarrow \quad \frac{V_1}{V_2}=\frac{IR_1}{IR_2}=\frac{4}{1}\Rightarrow\frac{J_1}{J_2}=\frac{1}{4}$

4. **(c)** $P=\frac{V^2}{R}$

$\Rightarrow \quad R_1=1\,\Omega,\ R_2=1/2\,\Omega,\ R_3=2\,\Omega$

$\therefore \quad P_2>P_1>P_3$

5. **(d)** Power $\propto 1/R$

6. **(c)** G_1 is acting as voltmeter and G_2 is acting as ammeter.

7. **(3)** For $v_1=\frac{50}{7}m,\ u_1=-25\text{m}$

$\Rightarrow \ v_2=\frac{25}{3}m,\ u_2=-50\text{m}$

Speed of object $=\frac{25}{30}\times\frac{18}{5}=3\,\text{kmph}.$

8. **(2)** $\frac{7}{4V_1}-\frac{1}{-24}=\frac{\frac{7}{4}-1}{6}$

$\Rightarrow \ V_1=21\,\text{cm}$

$\Rightarrow \ \frac{4/3}{V_2}-\frac{7/4}{21}=0$

$\Rightarrow \ V_2=16\,\text{cm}$

$\Rightarrow \ x=18-16=2\ \text{cm}$

9. **(7)** Image by mirror is formed at 30 cm from mirror at its right and finally by the combination it is formed at 20 cm on right of the lens. So in air medium, magnification by lens is unity. In second medium, $\mu=\frac{7}{6}$, focal length of the lens is given by,

$$\frac{\frac{1}{10}}{\frac{1}{f}}=\frac{(1.5-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)}{\left(\frac{1.5}{7/6}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)}$$

$\Rightarrow \quad f=\frac{35}{2}\text{cm}$

So, in second medium, final image is formed at 140 cm to the right of the lens. Second medium does not change the magnification by mirror.

So $\left|\frac{M_2}{M_1}\right|=\left|\frac{M_{m_2}M_{\ell_2}}{M_{m_1}M_{\ell_1}}\right|=7$

10. **(2)** Snell's Law on 1st surface: $\frac{\sqrt{3}}{2}=n\sin r_1$

$$\sin r_1=\frac{\sqrt{3}}{2n} \qquad \ldots(i)$$

$\Rightarrow \quad \cos r_1=\sqrt{1-\frac{3}{4n^2}}=\sqrt{\frac{4n^2-3}{2n}}$

$$r_1+r_2=60^\circ \qquad \ldots(ii)$$

Snell's Law on 2nd surface: $n\sin r_2=\sin\theta$

Using equation (i) and (ii) $n\sin(60^\circ-r_1)=\sin\theta$

$$n\left[\frac{\sqrt{3}}{2}\cos r_1-\frac{1}{2}\sin r_1\right]=\sin\theta$$

$$\frac{d}{dn}\left[\frac{\sqrt{3}}{4}\left(\sqrt{4n^2-3}-1\right)\right]=\cos\theta\frac{d\theta}{dn}$$

For $\theta=60^\circ$ and $n=\sqrt{3}$

$\Rightarrow \quad \frac{d\theta}{dn}=2$

11. **(3)** $\frac{1}{2}mv^2=qV$

$\Rightarrow \quad \lambda=\frac{h}{mv}$

$\Rightarrow \quad \lambda=\sqrt{8}\simeq 3.$

12. **(7)** Stopping potential $=\frac{hc}{\lambda}-\text{W}$

$=6.2\ \text{eV}-4.7\ \text{eV}=1.5\ \text{eV}$

$\Rightarrow \quad V=\frac{Kq}{r}=1.5$

$\Rightarrow \quad n=\frac{1.5\times10^{-2}}{9\times10^9\times1.6\times10^{-19}}=1.05\times10^7$

$\Rightarrow \quad Z=7$

13. **(7)** $\frac{1}{2}mv_\infty^2=\frac{1}{4\pi\varepsilon_0}\frac{(120)e^2}{(10\,fm)}$ [By Conservation of energy]

Assuming the nucleus to be considerably massive, we can disregard its motion.

$\therefore$ Let momentum of proton be $p=mv_\infty$

$\therefore \quad \frac{p^2}{2m_p}=\frac{1}{4\pi\varepsilon_0}\frac{(120)e^2}{(10\times10^{-15})}$

$\therefore \quad p=9\times10^9\times2\times\frac{5}{3}\times10^{-27}\times\frac{120\times e^2}{10^{-14}}$

$\Rightarrow \quad p = 30 \times 120 \times 10^{9-27+14} \times e^2$

$\therefore \quad p = \sqrt{3600 \times 10^{-4} \times e^2}$

$\therefore \quad p = 60 \times 10^{-2} \times e$

$\therefore \quad \lambda = \dfrac{h}{p} = \dfrac{h}{e \times 60 \times 10^5} = \dfrac{42 \times 10^{-15}}{60 \times 10^{-2}} = \dfrac{42}{6} \times 10^{-15}\,\text{m}$

$\therefore \quad \lambda = 7\text{fm}$

14. **(1)** Slope of graph is h/e = constant $\Rightarrow$ 1

15. **(a)** A→1, B→3, C→3, D→1,2,3
A, C and D in case of concave mirror convex lens image can be real, virtual, diminished magnified or of same size. B is case of convex mirror image is always virtual (for real object)

16. **(d)** A-3; B-4; C-2; D-1
According to Bernoulli's theorem the spinning of ball along with horizontal throw will change the velocity of streamlines of air above and below the ball which will change the pressure on ball above and below it which cause change in its direction of motion. This is called magnus effect.
Primers contain materials which make obtuse angle of contact due to their surface tension and do not allow capillary rise of water in pours of bricks.
Interstellar dust particles attract each other by gravitational pull to build a star.
Oil has smaller surface tension than water and is lighter than water so it spreads move over the surface of water to decrease height of tides.

17. **(c)** A-1; B-2; C-3; D-4

18. **(b)** A-4; B-3; C-2; D-1

$$Y = \frac{F}{A} \times \frac{l}{\Delta l} = \frac{\text{Longitudinal stress}}{\text{Longitudinal strain}} = \text{Young's modulus}$$

$$K = \frac{F}{A} \times \frac{V}{\Delta V} = \frac{\text{Normal stress}}{\text{Volumetric strain}} = \text{Bulk modulus}$$

= Volume elasticity

$$\text{Poisson's ratio} = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$$

$$\Rightarrow \quad \sigma = \frac{\frac{\Delta D}{D}}{\frac{\Delta l}{l}} = \frac{\Delta D}{D} \times \frac{l}{\Delta l}$$

□ □ □

Test Booklet code

A

Mock Test 1 "JEE-Main"

Do not open this Test Booklet until you are asked to do so.
Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions:

1. Immediately fill in the particulars on this page of the Test Booklet with *Blue/Black Ball Point Pen*. *Use of pencil is strictly prohibited.*
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **360.**
5. There are ***three*** parts in the question paper A, B, C consisting of, **Physics, Chemistry** and **Mathematics** having 30 questions in each part of equal weightage. Each question is allotted **4 (four)** marks for each correct response.
6. *Candidates will be awarded marks as stated above in instruction No. 5 for correct response of each question. 1/4 (one fourth) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made if no response is indicated for an item in the answer sheet.*
7. There is only one correct response for each question. Filling up more than one response in each question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instruction 6 above.
8. Use ***Blue/Black Ball Point Pen only*** for writing particulars/ marking responses on **Side-1** and **Side-2** of the Answer Sheet. ***Use of pencil is strictly prohibited****.*
9. No candidates is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc., except the Admit Card inside the examination hall/room.
10. Rough work is to be done on the space provided for this purpose in the Test Booklet only. This space is given at the bottom of each page and at the end of the booklet.
11. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room / Hall. ***However, the candidates are allowed to take away this Test Booklet with them.***
12. The CODE for this Booklet is **A**. Make sure that the CODE printed on **Side-2** of the Answer Sheet and also tally the serial number of the Test Booklet and Answer Sheet are the same as that on this booklet. In case of discrepancy, the candidate should immediately report the matter to the invigilator for replacement of both the Test Booklet and the Answer Sheet.
13. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capital letters): _____

Roll Number : in figures ☐☐☐☐☐☐☐☐

in words _____

Examination Centre Number: ☐☐☐☐☐☐☐☐

Name of Examination Centre (in Capital letters):_____

Candidate's Signature:_____ Invigilator's Signature:_____

Sample

***Read the following instructions carefully*:**

1. The candidates should fill in the required particulars on the Test Booklet and Answer Sheet (***Side-1***) with ***Blue/Black Ball Point Pen.***
2. For writing/marking particulars on ***Side-2*** of the Answer Sheet, use ***Blue/Black Ball Point Pen only.***
3. The candidates should not write their Roll Numbers anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
4. Out of the four options given for each question, only one option is the correct answer.
5. For each ***incorrect response, one-fourth (¼)*** of the total marks allotted to the question would be deducted from the total score. ***No deduction*** from the total score, however, will be made ***if no response*** is indicated for an item in the Answer Sheet.
6. Handle the Test Booklet and Answer Sheet with care, *as under no circumstances (except for discrepancy in Test Booklet Code and Answer Sheet Code), another set will be provided.*
7. The candidates are not allowed to do any rough work or writing work on the Answer Sheet. All calculations/writing work are to be done in the space provided for this purpose in the Test Booklet itself, marked 'Space for Rough Work'. This space is given at the bottom of each page and at the end of the booklet.
8. On completion of the test, the candidates must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
9. Each candidate must show on demand his/her Admit Card to the Invigilator.
10. No candidate, without special permission of the Superintendent or Invigilator, should leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign the Attendance Sheet again. Cases where a candidate has not signed the Attendance Sheet a second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case. **The candidates are also required to put their left hand THUMB impression in the space provided in the Attendance Sheet.**
12. Use of Electronic/Manual Calculator and any Electronic Item like mobile phone, pager etc. is prohibited
13. The candidates are governed by all Rules and Regulations of the JAB/Board with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of the JAB/Board.
14. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
15. **Candidates are not allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, electronic device or any other material except the Admit Card inside the examination hall/room.**

Sample

JEE-MAIN: CHEMISTRY MOCK TEST-1

1. In a galvanic cell, the salt bridge
a. does not participate chemically in the cell reaction.
b. stops the diffusion of ions from one electrode to another.
c. is necessary for the occurrence of the cell reaction
d. ensures mixing of the two electrolytic solutions

2. Sodium metal crystallizes in a body centred cubic lattice with a unit cell edge of 4.29 Å. The radius of sodium atom is approximately
a. 1.86 Å **b.** 3.22 Å
c. 5.72 Å **d.** 0.93 Å

3. Which of the following is the energy of a possible excited state of hydrogen?
a. $+13.6\,eV$ **b.** $-6.8\,eV$
c. $-3.4\,eV$ **d.** $+6.8\,eV$

4. The number of atoms in 100 g of an fcc crystal with density $=10.0\,g\,cm^{-3}$ and cell edge equal to 200 pm is equal to
a. 5×10^{24} **b.** 5×10^{25} **c.** 6×10^{23} **d.** 2×10^{25}

5. The following reaction is performed at 298 K.

$2NO(g) + O_2(g) \qquad 2NO_2(g)$

The standard free energy of formation of $NO(g)$ is 86.6 kJ/mol at 298 K. What is the standard free energy of formation of $NO_2(g)$ at 298 K? $(K_p = 1.6\times10^{12})$

a. $R(298)\ n(1.6\times10^{12}) - 86600$

b. $86600 + R(298)\ n(1.6\times10^{12})$

c. $86600 - \dfrac{n(1.6\times10^{12})}{R(298)}$

d. $0.5[2\times86600 - R(298)\ n(1.6\times10^{12})]$

6. The vapour pressure of acetone at $20°C$ is 185 torr. When 1.2 g of a non-volatile substance was dissolved in 100 g of acetone at $20°C$, its vapour pressure was 183 torr. The molar mass ($g\ mol^{-1}$) of the substance is:
a. 32 **b.** 64
c. 128 **d.** 488

7. The standard Gibbs energy change at 300 K for the reaction $2A \qquad B+C$ is 2494.2 J. At a given time, the composition of the reaction mixture is $[A]=\frac{1}{2}, [B]=2$ and $[C]=\frac{1}{2}$. The reaction proceeds in the:
$[R = 8.314\ J/K/mol, e = 2718]$
a. forward direction because $Q > K_c$
b. reverse direction because $Q > K_c$
c. forward direction because $Q < K_c$
d. reverse direction because $Q < K_c$

8. Two Faraday of electricity is passed through a solution of $CuSO_4$. The mass of copper deposited at the cathode is:
a. 0 g **b.** 63.5 g
c. 2 g **d.** 127 g

9. What is the half life period of a radioactive substance if 87.5% of any given amount of the substance disintegrates in 40 minutes?
a. 160 min **b.** 10 min
c. 20 min **d.** 13 min 20 sec

10. 3 g of activated charcoal was added to 50 mL of acetic acid solution (0.06 N) in a flask. After an hour it was filtered and the strength of the filtrate was found to be 0.042 N. The amount of acetic acid adsorbed (per gram of charcoal) is :
a. 18 mg **b.** 36 mg
c. 42 mg **d.** 54 mg

11. The ionic radii (in Å) of N^{3-}, O^{2-} and F^- are respectively:
a. 1.36, 1.40 and 1.71 **b.** 1.36, 1.71 and 1.40
c. 1.71, 1.40 and 1.36 **d.** 1.71, 1.36 and 1.40

12. In the context of the Hall-Heroult process for the extraction of A, which of the following statements is false?
a. CO and CO_2 are produced in this process
b. A_2O_3 is mixed with CaF_2 which lowers the melting point of the mixture and brings conductivity
c. A^{3+} is reduced at the cathode to form A
d. Na_3AlF_6 serves as the electrolyte

13. From the following statements regarding H_2O_2, choose the incorrect statement
a. It can act only as an oxidising agent
b. It decomposes on exposure to light
c. It has to be stored in plastic or wax lined glass bottles in dark
d. It has to be kept away from dust

14. Which one of the following alkaline earth metal sulphates has its hydration enthalpy greater than its lattice enthalpy?
a. $CaSO_4$ **b.** $BeSO_4$
c. $BaSO_4$ **d.** $SrSO_4$

15. Which among the following is the most reactive?

a. Cl_2 **b.** Br_2 **c.** I_2 **d.** ICl

16. Which one has the highest boiling point?

a. He **b.** Ne **c.** Kr **d.** Xe

17. The number of geometric isomers that can exist for square planar $[Pt(Cl)(py)(NH_3)(NH_2OH)]^+$ is (py = pyridine):

a. 2 **b.** 3 **c.** 4 **d.** 6

18. The color of $KMnO_4$ is due to

a. $M \rightarrow L$ charge transfer transition

b. $d - d$ transition

c. $L \rightarrow M$ charge transfer transition

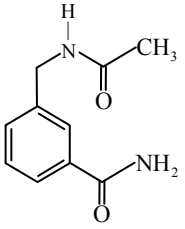
d. $\sigma - \sigma^*$ transition

19. In Carius method of estimation of halogens, 250 mg of an organic compound gave 141 mg of AgBr. The percentage of bromine in the compound is: (Atomic mass Ag = 108, Br = 80)

a. 24 **b.** 36 **c.** 48 **d.** 60

20. In the reaction shown below, the major product(s) formed is/are

a.



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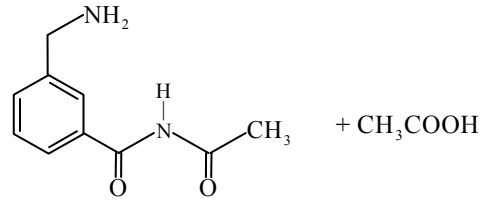
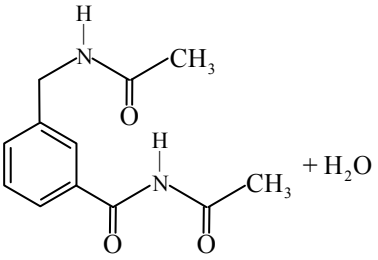
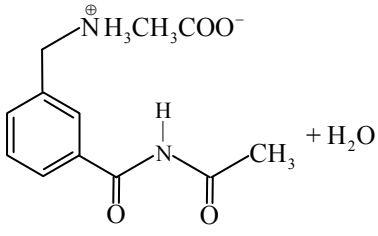


c.

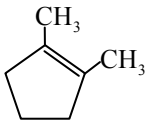


d.

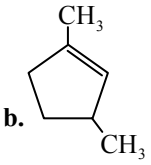


21. Which compound would give 5-keto-2-methyl hexanal upon ozonolysis?

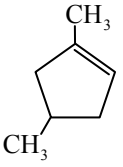
a.



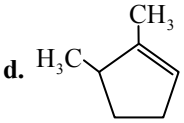
b.



c.



d.



22. The synthesis of alkyl fluorides is best accomplished by :

a. Free radical fluorination

b. Sandmeyer's reaction

c. Finkelstein reaction

d. Swarts reaction

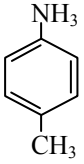
23. In the following sequence of reactions:

Toluene $\xrightarrow{KMnO_2}$ A $\xrightarrow{SOCl_2}$ B $\xrightarrow[BaSO_4]{H_2/Pd}$ C, the product C is:

a. C_6H_5COOH **b.** $C_6H_5CH_3$

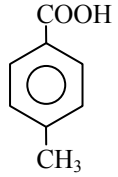
c. $C_6H_5CH_2OH$ **d.** C_6H_5CHO

24. In the reaction

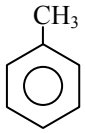


$\xrightarrow[0-5°C]{NaNO_2/HCl}$ D $\xrightarrow[\Delta]{CuCN/KCN}$ $E + N_2$ The product E is

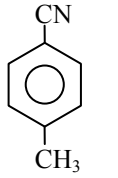
a.



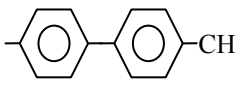
b.



c.



d.



25. Which polymer is used in the manufacture of paints and lacquers?

a. Bakelite **b.** Glyptal

c. Polypropene **d.** Poly vinyl chloride

26. Which of the vitamins given below is water soluble?

a. Vitamin C **b.** Vitamin D

c. Vitamin E **d.** Vitamin K

27. Which of the following compounds is not colored yellow?

a. $Zn_2[Fe(CN)_6]$ **b.** $K_3[Co(NO_2)_6]$

c. $(NH_4)_3\ [As(Mo_3O_{10})_4]$ **d.** $BaCrO_4$

28. Match the statement of Column with those in Column II:

Column I	Column II
(A) $C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$	**1.** Pseudo first order
(B) $CH_3COOC_2H_5 \xrightarrow[H^+ \text{ or } OH^-]{HOH} CH_3COOH + C_2H_5OH$	**2.** Zero order
(C) $H_2 + Cl_2 \xrightarrow{hv} 2HCl$	**3.** Second order
(D) $CH_3Cl + OH^- \longrightarrow CH_3OH + Cl^-$	**4.** First order

a. A→1; B→3, 4; C→2; D→3

b. A→2; B→4; C→3; D→1

c. A→1; B→3, 2; C→2; D→4

d. A→4; B→1; C→3; D→2

Space for rough work

Assertion and Reason

Note: Read the Assertion (A) and Reason (R**)** carefully to mark the correct option out of the options given below:

a. If both assertion and reason are true and the reason is the correct explanation of the assertion.

b. If both assertion and reason are true but reason is not the correct explanation of the assertion.

c. If assertion is true but reason is false.

d. If the assertion and reason both are false.

e. If assertion is false but reason is true

29. **Assertion:** Enthalpy and entropy of any elementary substance in the standard state are taken as zero.
Reason: At zero degree absolute, the constituent particles become completely motionless.

30. **Assertion:** Molecularity has no meaning for a complex reaction.
Reason: The overall molecularity of a complex reaction is equal to the molecularity of the slowest step.

JEE ADVANCE PAPER-I

Time 3 Hours. **Max. Marks 264 (88 for Chemistry)**

Read The Instructions Carefully

Question Paper Format and Marking Scheme:

1. The question paper has three parts: Physics, Chemistry and Mathematics. Each part has three sections.
2. **Section 1** contains 8 questions. The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).
   Marking Scheme: +4 for correct answer and 0 in all other cases.
3. **Section 2** contains 10 multiple choice questions with one or more than one correct option.
   Marking Scheme: +4 for correct answer, 0 if not attempted and –2 in all other cases.
4. **Section 3** contains 2 "match the following" type questions and you will have to match entries in Column I with the entries in Column II.
   Marking Scheme: for each entry in Column I, +2 for correct answer, 0 if not attempted and – 1 in all other cases.

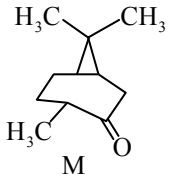
NOTE: *It's the mock test as per previous year's papers but sometimes IIT changes the test paper pattern and marking scheme too.*

SECTION 1 (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- Marking scheme:
  +4 If the bubble corresponding to the answer is darkened.
  0 In all other cases.

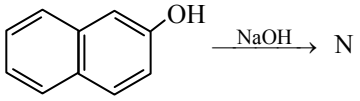
1. If the freezing point of a 0.01 molal aqueous solution of a cobalt (III) chloride-ammonia complex (which behaves as a strong electrolyte) is $-0.0558°C$, the number of chloride(s) in the coordination sphere of the complex is [K_f of water = 1.86 K kg mol^{-1}]

2. The total number of stereoisomers that can exist for M is



M

3. The number of resonance structures for N is



4. The total number of lone pairs of electrons in N_2O_3 is

5. For the octahedral complexes of Fe^{3+} in SCN^- (thiocyanato-S) and in CN^- ligand environments, the difference between the spin-only magnetic moments in Bohr magnetons (When approximated to the nearest integer) is [Atomic number of Fe = 26]

6. Among the triatomic molecules/ions, $BeCl_2, N_3^-, N_2O, NO_2^+, O_3, SCl_2, ICl_2^-, I_3^-$ and XeF_2, the total number of linear molecule(s)/ion(s) where the hybridization of the central atom does not have contribution from the d-orbital(s) is
[Atomic number: S = 16, Cl = 17, I = 53 and Xe = 54]

7. Not considering the electronic spin, the degeneracy of the second excited state (n = 3) of H atom is 9, while the degeneracy of the second excited state of H^- is

8. All the energy released from the reaction $X \rightarrow Y$. $\Delta_r G^0 = -193$ kJ mol^{-1} is used for oxidizing M^+ as $M^+ \rightarrow M^{3+} + 2e^-, E^0 = -0.25V$. Under standard conditions, the number of moles of M^+ oxidized when one mole of X is converted to Y is [F = 96500 C mol^{-1}]

SECTION 2 (Maximum Marks: 40)

- This section contains **TEN** questions.
- Each question has **FOUR** options (a), (b), (c) and (d). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Marking scheme:
  - +4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened.
  - 0 If none of the bubbles is darkened
  - –2 In all other cases

9. For the reaction: $I^- + ClO_3^- + H_2SO_4 \longrightarrow HSO_4^- + I_2$

The correct statement(s) in the balanced equation is/are

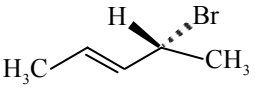
a. Stoichiometric coefficient of HSO_4^- is 6

b. Iodide is oxidised

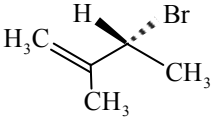
c. Sulphur is reduced

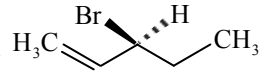
d. H_2O is one of the products.

10. Compound(s) that on hydrogenation produce(s) optically inactive compound(s) is (are)

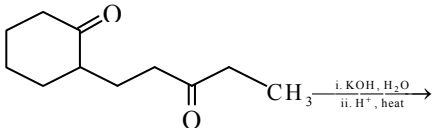
a. 

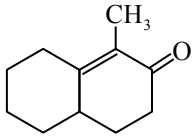
b.

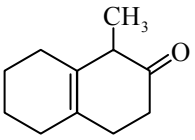
c. 

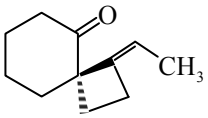
d. 

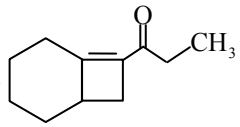
11. The major product of the following reaction is



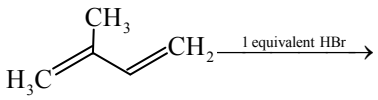
a. 

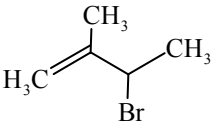
b. 

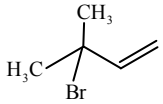
c. 

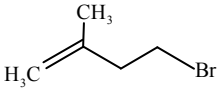
d. 

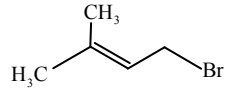
12. In the following reaction, the major product is



a. 

b. 

c. 

d. 

13. The correct statement(s) for orthoboric acid is/are

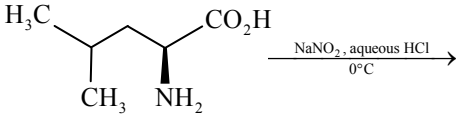
a. It behaves as a weak acid in water due to self ionisation

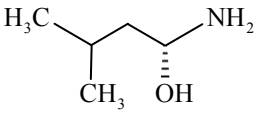
b. Acidity of its aqueous solution increases upon addition of ethylene glycol

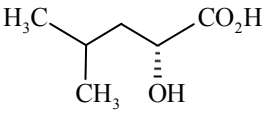
c. It has a three dimensional structure due to hydrogen bonding.

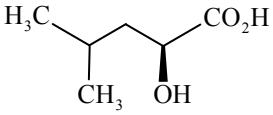
d. It is a weak electrolyte in water

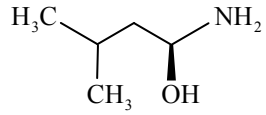
14. The major product of the reaction is



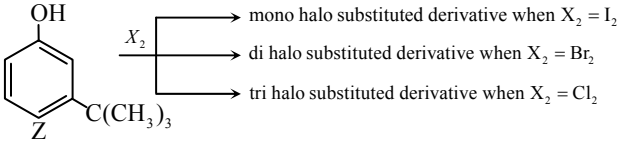
a. 

b. 

c. 

d. 

15. The reactivity of compound Z with different halogens under appropriate conditions is given below:



The observed pattern of electrophilic substitution can be explained by

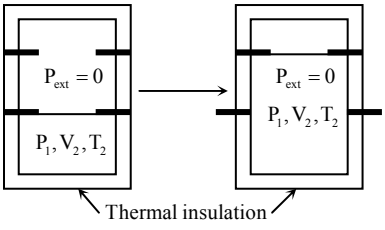
a. the steric effect of the halogen

b. the steric effect of the tert-butyl group

c. the electronic effect of the phenolic group

d. the electronic effect of the tert-butyl group

16. An ideal gas in thermally insulated vessel at internal pressure $= P_1$, volume $= V_1$ and absolute temperature $= T_1$ expands irreversibly against zero external presssure, as shown in the diagram. The final internal pressure, volume and absolute temperature of the gas are P_2, V_2 and T_2, respectively. For this expansion.



a. $q = 0$ **b.** $T_2 = T_1$

c. $P_2V_2 = P_1V_1$ **d.** $P_2V_2^{\gamma} = P_1V_1^{\gamma}$

17. Fe^{3+} is reduced to Fe^{2+} by using

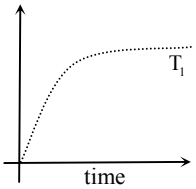
a. H_2O_2 in presence of NaOH

b. Na_2O_2 in water

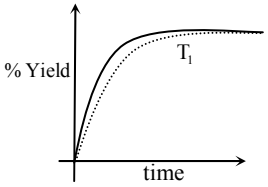
c. H_2O_2 in presence of H_2SO_4

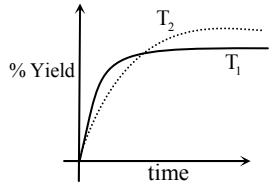
d. Na_2O_2 in presence of H_2SO_4

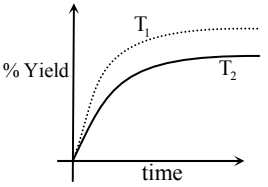
18. The % yield of ammonia as a function of time in the reaction $N_2(g) + 3H_2(g) \quad 2NH_3(g), \Delta H < 0$ at (P, T_1) is given below:

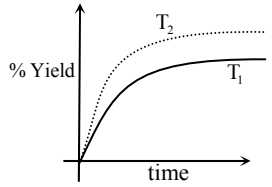


If this reaction is conducted at (P, T_2), with $T_2 > T_1$, the % yield of ammonia as a function of time is represented by

a.


b.


c.


d.


SECTION 3 (Maximum Marks: 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column I** and **Column II**
- **Column I** has **four** entries (A), (B), (C) and (D)
- **Column II** has **five** entries (1), (2), (3), (4) and (5)
- Match the entries in **Column I** with the entries in **Column II**
- One or more entries in **Column I** may match with one or more entries in **Column II**
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below:

  (A) (1) (2) (3) (4) (5)

  (B) (1) (2) (3) (4) (5)

  (C) (1) (2) (3) (4) (5)

  (D) (1) (2) (3) (4) (5)

- For each entry in **Column I,** darken the bubbles of all the matching entries. For example, if entry (A) in **Column I,** matches with entries (2), (3) and (4), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- Marking scheme:

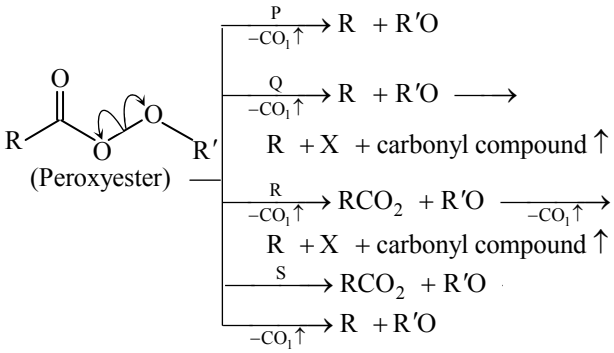
  For each entry in **Column I**

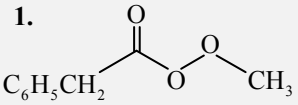
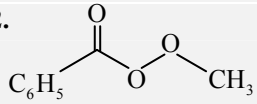
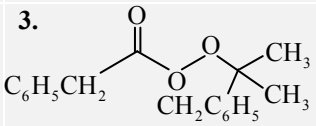
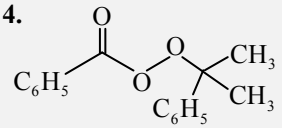
  +2 If only the bubble(s) corresponding to all the correct match(es) is(are) darkened

  0 If none of the bubbles is darkened

  –1 In all other cases

19. Different possible thermal decomposition pathways for peroxyesters are shown below. Match each pathway from Column -I with an appropriate structure from Column –II is:



Column I	Column II
(A) Pathway A	**1.** 
(B) Pathway B	**2.** 
(C) Pathway C	**3.** 
(D) Pathway D	**4.** 

a. A→1; B→3; C→4; D→2

b. A→2; B→4; C→4; D→1

c. A→1; B→1; C→2; D→4

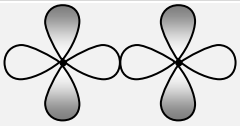
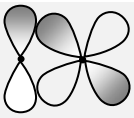
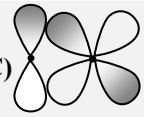
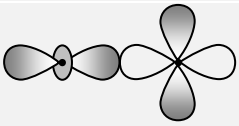
d. A→3.; B→2; C→1; D→3

20. Match the orbital overlap figures shown in Column -I with the description given in Column -II and select the correct

answer using the code given below the lists.

(en $H_2NCH_2CH_2NH_2$; atomic numbers :

Ti = 22, Cr = 24, Co = 27; Pt = 78)

Column I	Column II
(A) 	**1.** p-d π antibonding
(B) 	**2.** d-d σ bonding
(C) 	**3.** p-d π bonding
(D) 	**4.** d-d σ antibonding

a. A→2; B→3; C→1; D→4

b. A→2; B→4; C→4; D→1

c. A→1; B→1; C→2; D→4

d. A→3.; B→2; C→1; D→3

Space for rough work

JEE ADVANCE PAPER-II

Time 3 Hours. **Max. Marks 240 (80 for Chemistry)**

Read The Instructions Carefully

Question Paper Format and Marking Scheme:

1. The question paper has three parts: Physics, Chemistry and Mathematics. Each part has three sections.
2. Section 1 contains 8 questions. The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).
   Marking Scheme: +4 for correct answer and 0 in all other cases.
3. Section 2 contains 8 multiple choice questions with one or more than one correct option.
   Marking Scheme: +4 for correct answer, 0 if not attempted and –2 in all other cases.
4. Section 3 contains 2 "paragraph" type questions. Each paragraph describes an experiment, a situation or a problem. Two multiple choice questions will be asked based on this paragraph. One or more than one option can be correct.
   Marking Scheme: +4 for correct answer, 0 if not attempted and – 2 in all other cases.

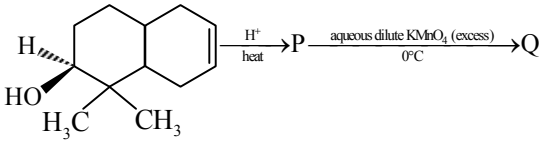
NOTE: *It's the mock test as per previous year's papers but sometimes IIT changes the test paper pattern and marking scheme too.*

SECTION 1 (Maximum Marks: 32)

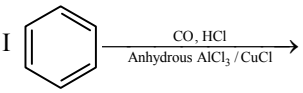
- This section contains **EIGHT** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- Marking scheme:
  - +4 If the bubble corresponding to the answer is darkened.
  - 0 In all other cases.

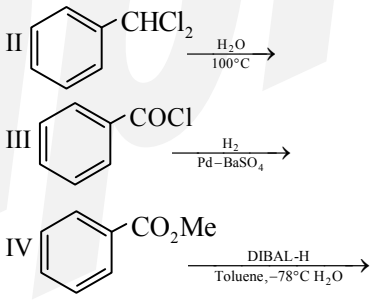
1. In dilute aqueous H_2SO_4, the complex diaquodioxalato-ferrate (II) is oxidized by MnO_4^-. For this reaction, the ratio of the rate of change of $[H^+]$ to the rate of change of $[MnO_4^-]$ is

2. The number of hydroxyl group(s) in Q is

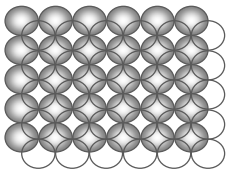


3. Among the following, the number of reaction(s) that produce(s) benzaldehyde is





4. The crystal of a solid is square packing of identical spheres in each layer and spheres of one layer are placed just above the voids made by spheres in previous layer. The packing efficiency of such crystal (in %) is $[\pi = 3.15, \sqrt{2} = 1.4]$



5. Among the complex ions, $[Co(NH_2\text{-}CH_2\text{-}CH_2\text{-}NH_2)_2Cl_2]^+$, $[CrCl_2(C_2O_4)_2]^{3-}$, $[Fe(H_2O)_4(OH)_2]^+$, $[Fe(NH_3)_2](CN)_4]^-$, $[Co(NH_2-CH_2-CH_2-NH_2)_2(NH_3)Cl]^{2+}$ and $[Co(NH_3)_4(H_2O)Cl]^{2+}$, the number of complex ion(s) that show(s) cis-trans isomerism is

6. Three moles of B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is

7. The molar conductivity of a solution of a weak acid HX (0.01 M) is 10 times smaller than the molar conductivity of a solution of a weak acid HY (0.10 M). If $\lambda^0_{X^-} \approx \lambda^0_{Y^-}$, the difference in their pK_a values, pK_a (HX) pK_a (HY), is (consider degree of ionisation of both acids to be << 1)

8. A closed vessel with rigid walls contains 1 mol of $^{238}_{92}U$ and 1 mol of air at 298 K. Considering complete decay of $^{238}_{92}U$ to $^{206}_{82}Pb$, the ratio of the final pressure to the initial pressure of the system at 298 K is

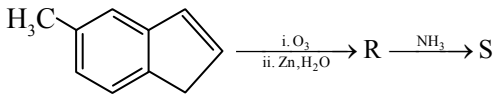
SECTION 2 (Maximum Marks: 32)

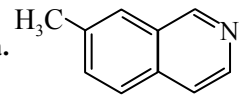
- This section contains **EIGHT** questions.
- Each question has **FOUR** options (a), (b), (c) and (d). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Marking scheme:
  +4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened.
  0 If none of the bubbles is darkened
  –2 In all other cases

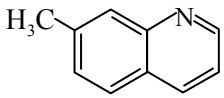
9. Which of the following compounds will exhibit geometrical isomerism?

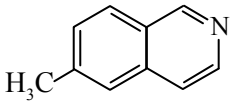
a. 1-Phenyl-2-butene
b. 3-Phenyl-1-butene
c. 2-Phenyl-1-butene
d. 1,1-Diphenyl-2-propane

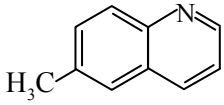
10. In the following reactions, the product S is



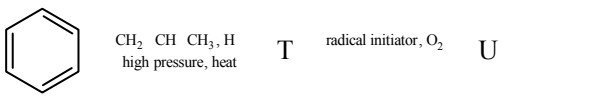
a. 

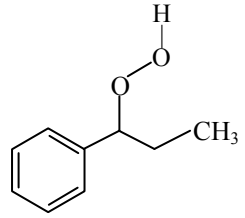
b. 

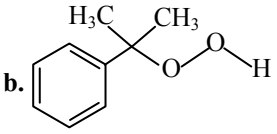
c. 

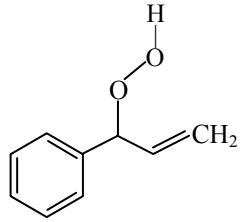
d. 

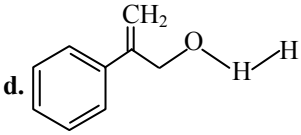
11. The major product U in the following reactions is



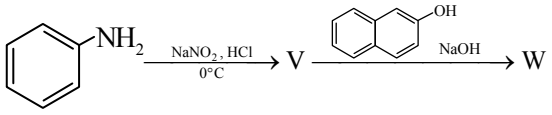
a. 

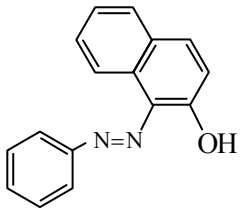
b. 

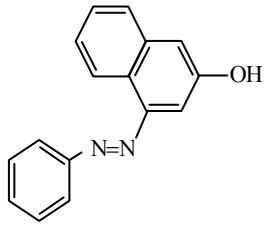
c. 

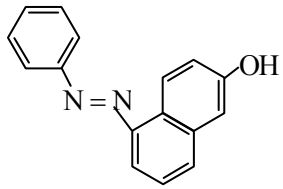
d. 

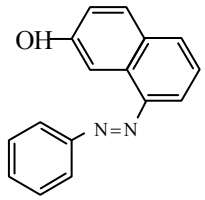
12. In the following reactions, the major product W is



a. 

b. 

c. 

d. 

13. The correct statement(s) regarding, (i) HClO, (ii) $HClO_2$, (iii) $HClO_3$ and (iv) $HClO_4$, is (are)

a. The number of Cl = O bonds in (ii) and (iii) together is two
b. The number of lone pairs of electrons on Cl in (ii) and (iii) together is three
c. The hybridisation of Cl in (iv) is sp^3
d. Amongst (i) to (iv), the strongest acid is (i)

14. The pair(s) of ions where BOTH the ions are precipitated upon passing H_2S gas in presence of dilute HCl, is(are)

a. Ba^{2+}, Zn^{2+} **b.** Bi^{3+}, Fe^{3+}

c. Cu^{2+}, Pb^{2+} **d.** Hg^{2+}, Bi^{3+}

15. Under hydrolytic conditions, the compounds used for preparation of linear polymer and for chain termination, respectively, are

a. CH_3SiCl_3 and $Si(CH_3)_4$

b. $(CH_3)_2SiCl_2$ and $(CH_3)_3SiCl$

c. $(CH_3)_2SiCl_2$ and $(CH_3)_3SiCl_3$

d. $SiCl_4$ and $(CH_3)_3SiCl$

16. When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2. The TRUE statement(s) regarding this adsorption is(are)

a. O_2 is physisorbed

b. heat is released

c. occupancy of π^*_{2p} of O_2 is increased

d. bond length of O_2 is increased

SECTION 3 (Maximum Marks: 16)

- This section contains **TWO** questions.
- Based on each paragraph, there will be **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Marking scheme:
- +4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened.
- 0 If none of the bubbles is darkened
- –2 In all other cases

Paragraph for Question Nos. 17 to 18

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

17. Energy of the state S_1 in units of the hydrogen atom ground state energy is :

a. 0.75 **b.** 1.50

c. 2.25 **d.** 4.50

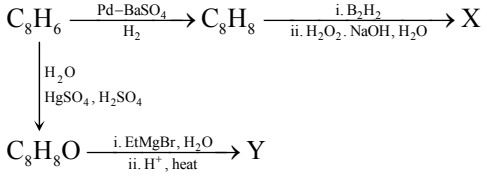
18. The orbital angular momentum quantum number of the state S_2 is :

a. 0 **b.** 1

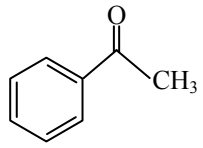
c. 2 **d.** 3

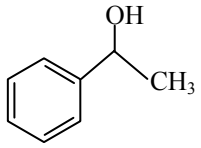
Paragraph for Question No. 19 to 20

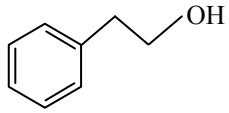
In the following reactions

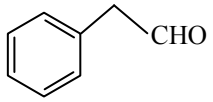


19. Compound X is

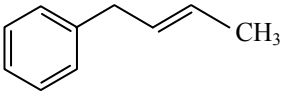
a. 

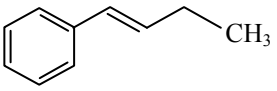
b. 

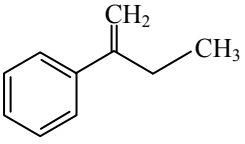
c. 

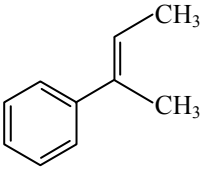
d. 

20. The major compound Y is

a. 

b. 

c. 

d. 

Space for rough work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	a	c	a	d	b	b	b	d	a
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	d	a	b	d	d	b	c	a	a
21.	**22.**	**23.**	**24.**	**25.**	**26.**	**27.**	**28.**	**29.**	**30.**
b	d	d	c	b	a	a	a	c	b

1. (a)

2. (a) For BCC unit 6 cell, $\sqrt{3}a = 4r$

$$r = \frac{\sqrt{3}}{4}a = \frac{\sqrt{3}}{4} \times 4.29 \text{ Å} = 1.85 \text{ Å}$$

3. (c) Energy in 1st excited state $= -3.4$ eV

4. (a) $\rho = \dfrac{Z_{eff} \times Mw}{a^3 \times 10^{-30} \times N_A}$ [For fcc, $Z_{eff} = 4$ unit cell]

$$\therefore \quad Mw = \frac{\rho \times a^3 \times 10^{-30} \times N_A}{Z_{eff}}$$

$$= \frac{(10.0 \text{ g cm}^{-3} \times (200 \text{ pm})^3 \times 10^{-30} \text{ cm}^3 \times 6 \times 10^{23} \text{ atoms})}{4}$$

Thus, 12 g mol^{-1} contains $= N_A$ atoms $= 6 \times 10^{23}$ atoms

$$\therefore \quad 100 \text{ g contains } = \frac{6 \times 10^{23}}{12} \times 100 = 5 \times 10^{24} \text{ atoms}$$

5. (d) $\Delta G_{Rx^n} = 2\Delta G_{NO_2} - 2\Delta G_{NO}$

$$\Delta G_{NO_2} = \Delta G_{NO} + \tfrac{1}{2}\Delta G_{Rx}$$

$$= \Delta G_{NO} + \tfrac{1}{2}(-RT \ln_e K_p)$$

$$= 0.5[2 \times 86600 - R(298)\ell n\,(1.6 \times 10^{12})]$$

6. (b) $\dfrac{P^0 - P}{P^0} = X_{solute} = \dfrac{n_1}{n_1 + n_2}$

$$\frac{185 - 183}{185} = \frac{2}{185} = \frac{1.2/M}{\frac{1.2}{M} + \frac{100}{58}}$$

$\Rightarrow \quad M = 64$

7. (b) $\Delta G = -RT \ln_e K_c$

$$2494.2 = 8.314 \times 300 \ln_e K_c \Rightarrow K_c = e^{-1}$$

$$K_c = e^{-1} = \frac{1}{2.718} = 0.36$$

$$Q = \frac{(B)(C)}{[A]^2} = \frac{2 \times \frac{1}{2}}{[1/2]^2} = 4$$

$Q > K_c$, i.e. backward reaction.

8. (b) $Cu^{+2} + 2e^- \longrightarrow Cu(s)$

$2 \text{ mol} \quad 1 \text{ mol} = 63.5$

9. (d) $t = 40$ min

$$k = \frac{2.303}{t}\log\frac{a}{a-x}$$

$$k = \frac{2.303}{40}\log\frac{a}{a - 0.875\,a}$$

$$= \frac{2.303}{40}\log\frac{a}{0.125\,a}$$

$$= \frac{2.303}{40}\log 8 = 0.051 \text{ min}$$

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.051} = 13.58 \text{ min}$$

10. (a) Amount of acetic acid adsorbed

$$= \frac{(0.06 - 0.042) \times 50 \times 10^{-3} \times 60}{3} = 16 \times 10^{-3} = 18 \text{ mg}$$

11. (c) Ionic Radii order: $N^{3-} > O^{2-} > F^-$

12. (d)

13. (a) It can act as an oxidising as well as reducing agent.

14. (b) $BaSO_4$ is least soluble, $BeSO_4$ is most soluble.

15. (d) The interhalogen compounds are generally more reactive than halogens (except F_2).

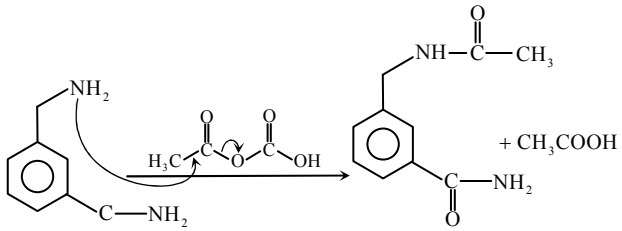
16. (d) Xe has the highest boiling point.

17. (b) No. of Geometrical isomers of $[Pt(Cl)(py)(NH_3)(NH_2OH)]^+ = 3$

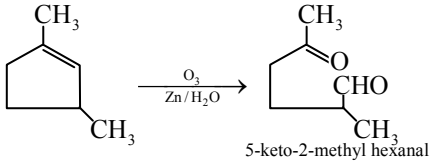
18. (c)

19. (a) % of Bromine $= \dfrac{80}{188} \times \dfrac{141 \text{ mg}}{250 \text{ mg}} \times 100 = 24\%$

20. (a) Only amines undergo acetylation and not acid amides.



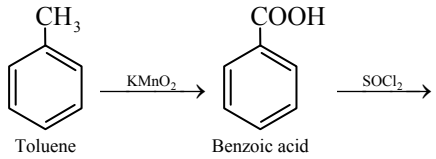
21. (b)



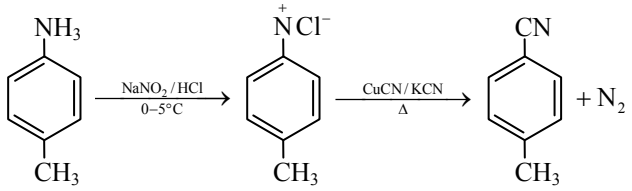
5-keto-2-methyl hexanal

22. **(d)** $R-I+AgF \longrightarrow R-F+AgI$ (Swarts Reaction)

23. **(d)**



24. **(c)**



25. **(b)** Glyptal is used in the manufacture of paints and lacquers.

26. **(a)** Vitamin 'B' and 'C' are water soluble.

27. **(a)** $Zn_2[Fe(CN)_6]$ is bluish white ppt.

28. **(a)** A→1; B→3, 4; C→2; D→3

(A) Inversion of cane sugar is pseudo first order reaction.

(B) Hydrolysis of ester in the presence of acid is first order while in the presence of base is second order reaction.

(C) Photochemical reactions are of zero order.

(D) SN^2 reactions are of second order.

29. **(c)** Enthalpy is zero but entropy is not zero. Vibrational motion exists even at absolute zero.

30. **(b)** Molecularity of a reaction can be defined only for an elementary reaction because complex reaction does not take place in one single step and it is almost impossible for all the total molecules of the reactants to be in a state of encounter simultaneously.

JEE Advance Paper -I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
1	2	9	8	4	4	3	4	a,b,d	b,d
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
a	d	b,d	c	a,b,c	a,b,c	a,b	b	a	a

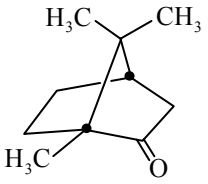
1. **(1)** $\Delta T_f = iK_f m$

$0.0558 = i \times 1.86 \times 0.01 \Rightarrow i = 3$

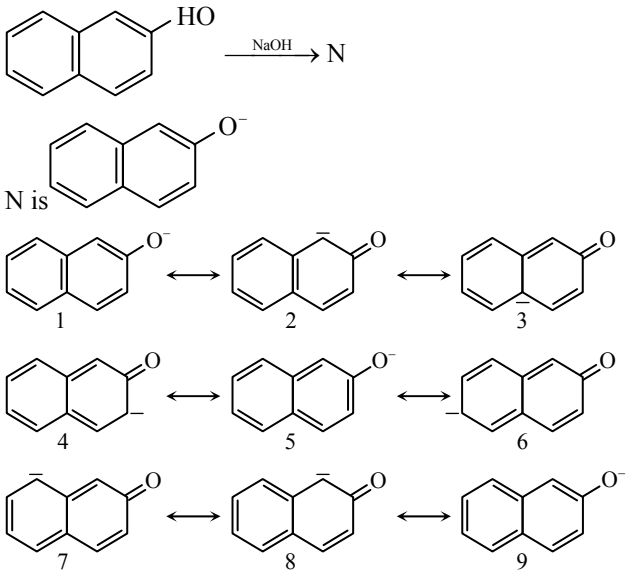
$\therefore$ Complex is $[Co(NH_3)_5Cl]Cl_2$.

2. **(2)** Bridging does not allow the other 2 variants to exist.

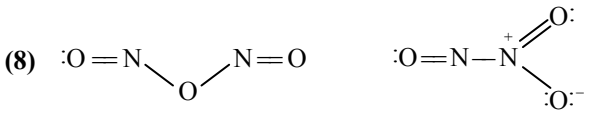
Total no. of stereoisomers of M = 2



3. **(9)**



4. **(8)**

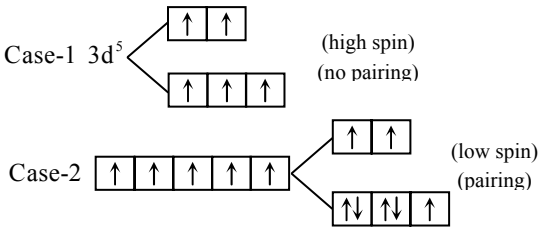


Total no. of lone pairs = 8

5. **(4)** $[Fe(SCN)_6]^{3-}$ and $[Fe(CN)_6]^{3-}$

In both the cases the electronic configuration of Fe^{3+} will be $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5$.

Since SCN is a weak field ligand and $\overline{C}N$ is a strong field ligand, the pairing will occur only in case of $[Fe(CN)_6]^3$.



Case – 1 $\mu = \sqrt{n(n+2)} = \sqrt{5(5+2)} = \sqrt{35} = 5.91$ BM

Case – 2 $\mu = \sqrt{n(n+2)} = \sqrt{1(1+2)} = \sqrt{3} = 1.73$ BM

Difference in spin only magnetic moment

$= 5.91 - 1.73 = 4.18 \approx 4$

6. **(4)** $BeCl_2, N_3^-, N_2O, NO_2^+, O_3, SCl_2, ICl_2^-, I_3^-, XeF_2$

$BeC_2 \longrightarrow sp \longrightarrow$ linear

$N_3^- \longrightarrow sp \longrightarrow$ linear

$N_2O \longrightarrow sp \longrightarrow$ linear

$\overset{\oplus}{N}O_2 \longrightarrow sp \longrightarrow$ linear

$O_3 \longrightarrow sp^2 \longrightarrow$ linear

$SCl_2 \longrightarrow sp^3 \longrightarrow$ linear

$I_3^- \longrightarrow sp^3d \longrightarrow$ linear

$ICl_2^- \longrightarrow sp^3d \longrightarrow$ linear

$XeF_2 \longrightarrow sp^3d \longrightarrow$ linear

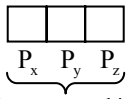
So, among the following only four (4) has linear shape and no d-orbital is involved in hybridization.

7. **(3)** Single electron species don't follow the $(n+\ )$ rule but multi electron species do.

Ground state of $H^- = 1s^2$

First excited state of $H^- = 1s^1, 2s^1$

Second excited state of $H^- = 1s^1, 2s^0, 2p^1$



(3 degenerate orbitals)

8. **(4)** $X \longrightarrow Y \ \Delta_r G^0 = -193 KJ/mol$

$M^+ \longrightarrow M^{3+} + 2e^- \ E^0 = -0.25V$

ΔG^0 for the this reaction is

$\Delta G^0 = -nFE^0 = -2 \times (-0.25) \times 96500 = 48250 \ J/mol$

48.25 KJ/mol

So the number of moles of M^+ oxidized using

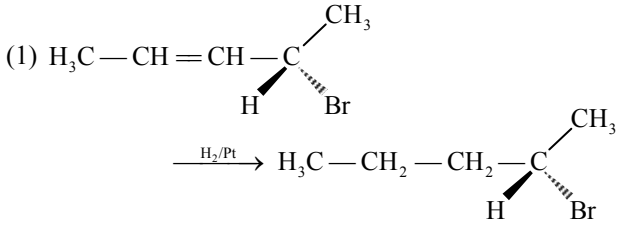
$X \longrightarrow Y$ will be

$= \dfrac{193}{48.25} = 4$ moles

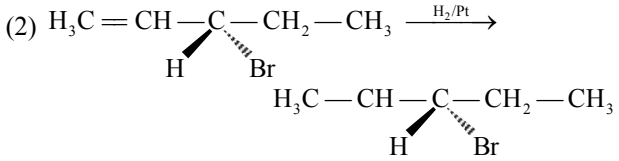
9. **(a, b, d)** $6I^- + ClO_3^- + 6H_2SO_4 \longrightarrow$

$Cl^- + 6HSO_4^- + 3I_2 + 3H_2O$

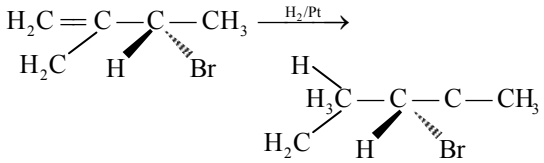
10. **(b, d)**

(1) 

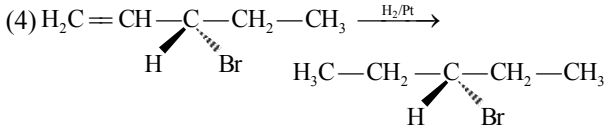
Optically active

(2) 

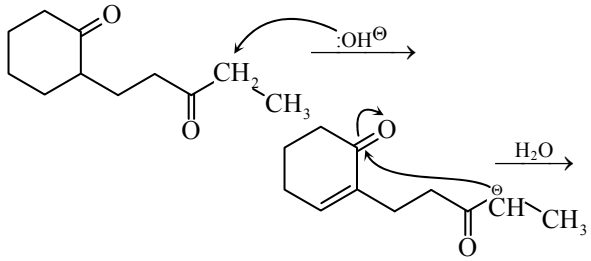
Optically inactive

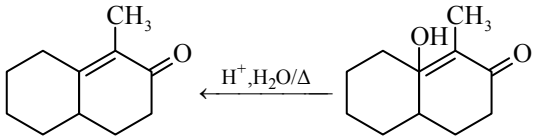
(3) 

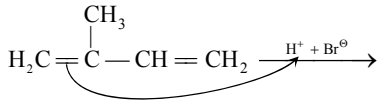
Optically inactive

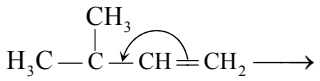
(4) 

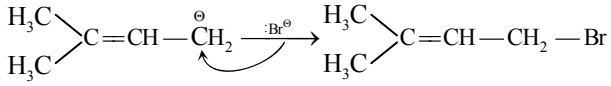
Optically inactive

11. **(a)** 



12. **(d)** 

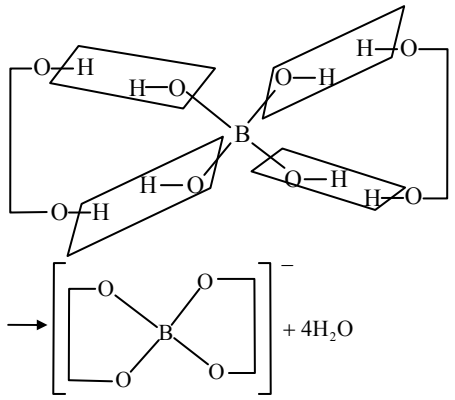




13. **(b, d)** **(a)** H_3BO_3 is a weak monobasic Lewis acid.

$H_3BO_3 + H-OH \qquad B(OH)_4^- + H^+ \qquad \ldots(i)$

(b) Equilibrium (i) is shifted in forward direction by the addition of syn-diols like ethylene glycol which forms a stable complex with $B(OH)_4^-$.

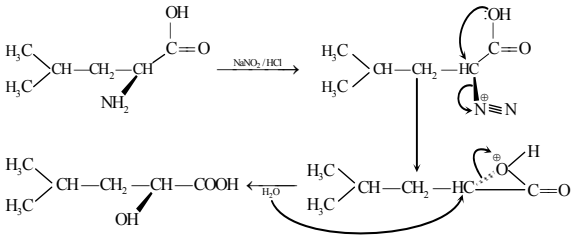


(stable complex)

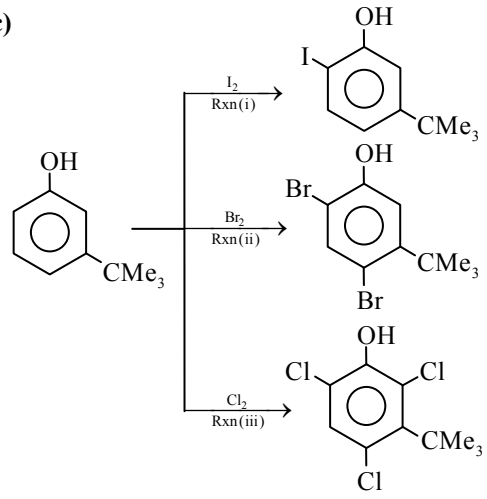
(c) It has a planar sheet like structure due to hydrogen bonding.

(d) H_3BO_3 is a weak electrolyte in water.

14. (c)



15. (a, b, c)



16. (a, b, c) Since container is thermally insulated so, q = 0, and it is a case of free expansion therefore W = 0 and E 0. So, T_1 T_2. Also, P_1V_1 P_2V_2.

17. (a, b) $2Fe^{3}$ H_2O_2 $2OH$ $2Fe^{2}$ $2H_2O$ O_2

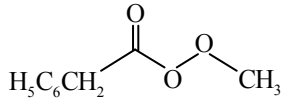
$Na_2O_2 + H_2O \longrightarrow H_2O_2 + NaOH$

18. (b) $N_2(g) + 2H_2(g)$

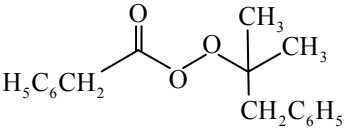
$N_2(g) + 2H_2(g)$ $2NH_3(g); \Delta H < 0$

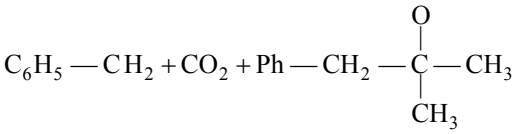
Increasing the temperature lowers equilibrium yield of ammonia. However, at higher temperature the initial rate of forward reaction would be greater than at lower temperature that is why the percentage yield of NH_3 too would be more initially.

19. (a) A→1; B→3; C→4; D→2

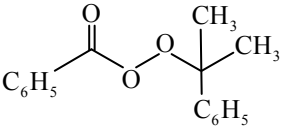
(A) 

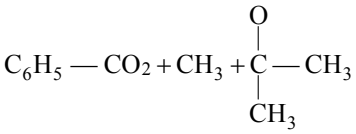
$C_6H_5 - CH_2 + CO_2 + CH_2O$

(B) 

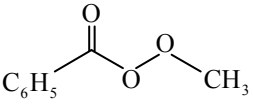


$\longrightarrow Ph - CH_2 + CH_3 - CO - CH_3$

(C) 

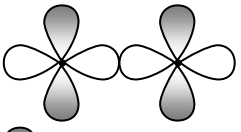


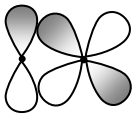
$\longrightarrow Ph + CH_3 - CO - Ph + CH_3$

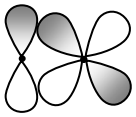
(D) 

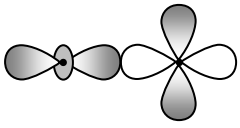
$C_6H_5 - CO_2 + CH_3O \longrightarrow C_6H_5 + CO_2$

20. (a) A→2; B→3; C→1; D→4

(A)  → d-d σ bonding

(B)  → p-d π bonding

(C)  → p-d π antibonding

(D)  → d-d σ antibonding

JEE Advance Paper-II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
8	4	4	75	5	6	3	9	a	a
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b	a	b,c	c,d	b	b,c,d	a	b	c	d

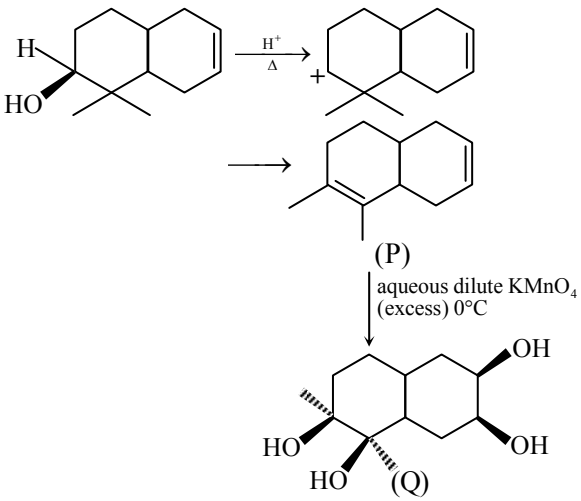
1. **(8)** $[Fe(C_2O_4)(H_2O)]^{2-} + MnO_4^{2-} + 8H^+ \longrightarrow$

$Mn^{2+} + Fe^{3+} + 4CO_2 + 6H_2O$

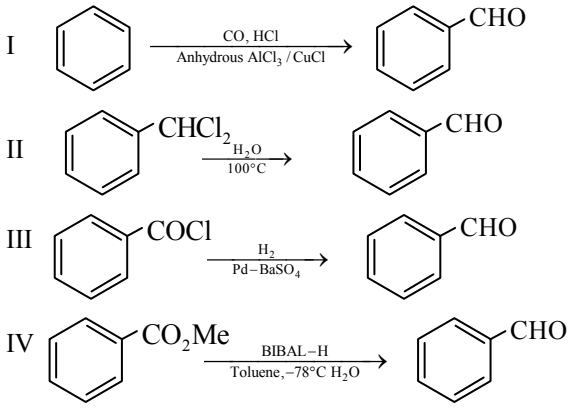
So, the ratio of rate of change of

$[H^+]$ to that of rate of change of $[MnO_4]$ is 8.

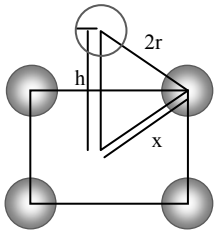
2. **(4)**



3. **(4)**



4. **(75)**



$Z = 1[\frac{1}{8}$th is shared in first layer and

$\frac{1}{2}$ is shared in second layer

$\therefore \quad Z = \frac{1}{8} \times 4 + 0.5 = 1$

$(2r^2) = h^2 + x^2 \Rightarrow h = \sqrt{2}r$

Area $= 4r^2$ [$a^2 = (2r)^2 \Rightarrow a = 2r$]

Vol $= 4r^2 \times \sqrt{2}r = 4\sqrt{2}r^3$

$$D = \frac{1 \times \frac{4}{3}\pi r^3}{4\sqrt{2}r^3} = \frac{\pi}{3\sqrt{2}} = 0.75 = 75\%$$

5. **(5)** $[Co(en)_2Cl_2]^+ \longrightarrow$ will show cis-trans isomerism

$[CrCl_2(C_2O_4)_2]^{3-} \longrightarrow$ will show cis-trans isomerism

$[Fe(H_2O)_4(OH)_2]^+ \longrightarrow$ will show cis-trans isomerism

$[Fe(CN)_4(NH_3)_2]^- \longrightarrow$ will show cis-trans isomerism

$[Co(en)_2(NH_3)Cl]^{2+} \longrightarrow$ will show cis-trans isomerism

$[Co(NH_3)_4(H_2O)Cl]^{2+} \longrightarrow$ will not show cis-trans isomerism (although it will show geometrical isomerism)

6. **(6)** $B_2H_6 + 6MeOH \longrightarrow 2B(OMe)_3 + 6H_2$

1 mole of B_2H_6 reacts with 6 moles of MeOH to give 2 moles of $B(OMe)_3$

3 moles of B_2H_6 will react with 18 moles of MeOH to give 6 moles of $B(OMe)_3$.

7. **(3)** $HX \qquad H^+ + X^- \qquad Ka = \frac{[H^+][X^-]}{[HX]}$

$HY \qquad H^+ + Y^- \qquad Ka = \frac{[H^+][Y^-]}{[HY]}$

Λ_m for $HX = \Lambda_{m_1} \qquad \Lambda_m$ for $HY = \Lambda_{m_2}$

$\Lambda_{m_1} = \frac{1}{10}\Lambda_{m_2} \qquad Ka = C\alpha^2$

$$Ka_1 = C_1 \times \left(\frac{\Lambda_{m_1}}{\Lambda^0_{m_1}}\right)^2$$

$$Ka_2 = C_2 \times \left(\frac{\Lambda_{m_2}}{\Lambda^0_{m_2}}\right)^2$$

$$\frac{Ka_1}{Ka_2} = \frac{C_1}{C_2} \times \left(\frac{\Lambda m_1}{\Lambda m_2}\right)^2$$

$$= \frac{0.01}{0.1} \times \left(\frac{1}{10}\right)^2 = 0.001$$

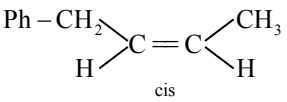
$pKa_1 - pKa_2 = 3$

8. **(9)** In conversion of $^{238}_{92}U$ to $^{206}_{82}Pb$, 8α-particles and 6β particles are ejected.

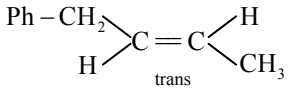
The number of gaseous moles initially $= 1$ mol

The number of gaseous moles finally $= 1 + 8$ mol; (1 mol from air and 8 mol of $_2He^4$) So the ratio $= 9/1 = 9$

9. **(a)**

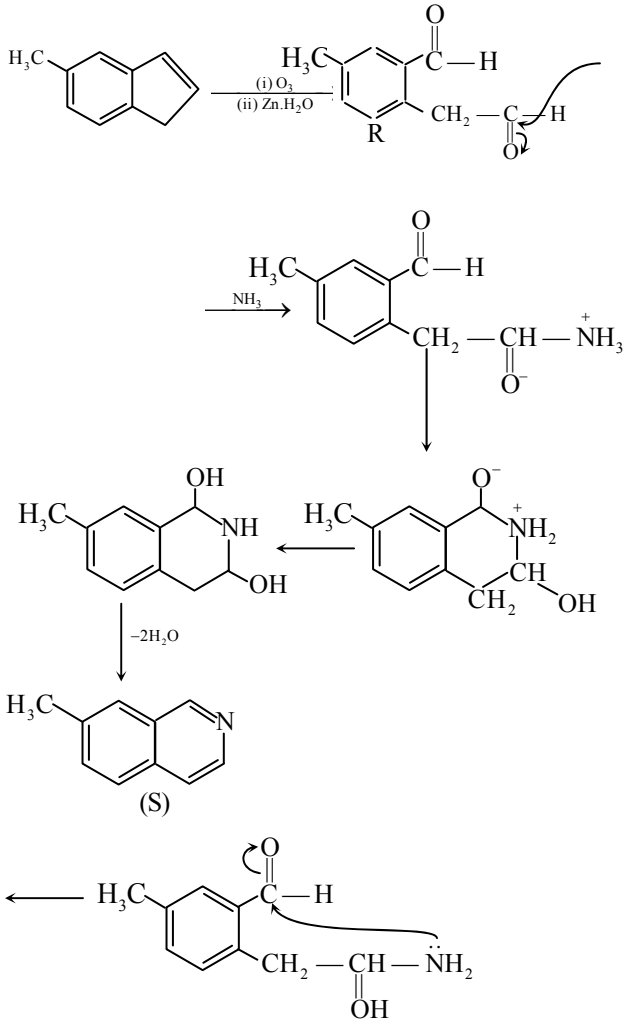


cis



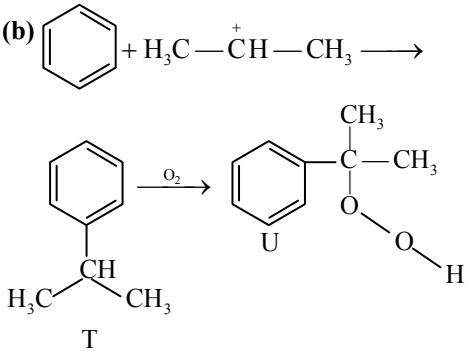
trans

10. **(a)**



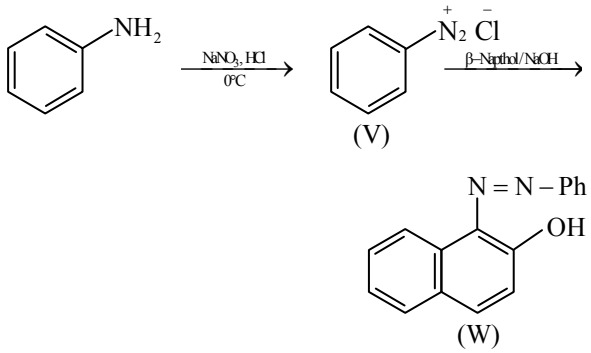
(S)

11. **(b)**



T

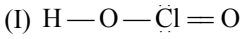
12. **(a)**

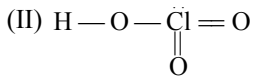


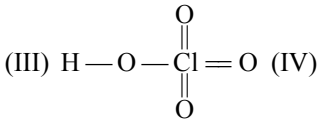
(V)

(W)

13. **(b, c)** H — O — Cl

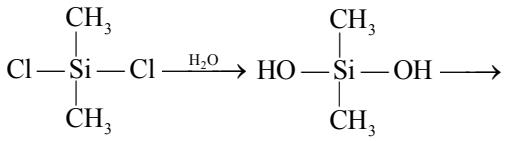
(I) 

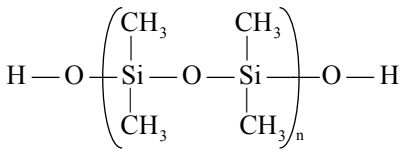
(II) 

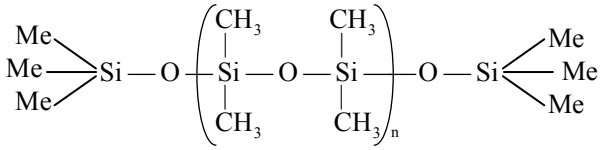
(III)  (IV)

14. **(c, d)** Cu^{2+}, Pb^{2+}, Hg^{2+}, Bi^{3+} give ppt. with H_2S in presence of dilute HCl.

15. **(b)**







16. **(b, c, d)** * Adsorption of O_2 on metal surface is exothermic.

* During electron transfer from metal to O_2 electron occupies π^*_{2p} orbital of O_2.

* Due to electron transfer to O_2 the bond order of O_2 decreases hence bond length increases.

For Question Nos. 17 to 18

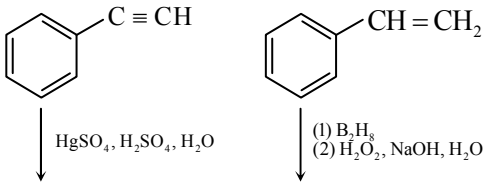
17. **(c)** $E_{S_1} = \dfrac{E_{1H} \times 3^2}{2^2} = \dfrac{9}{4} = 2.25 \times E_{1H}$.

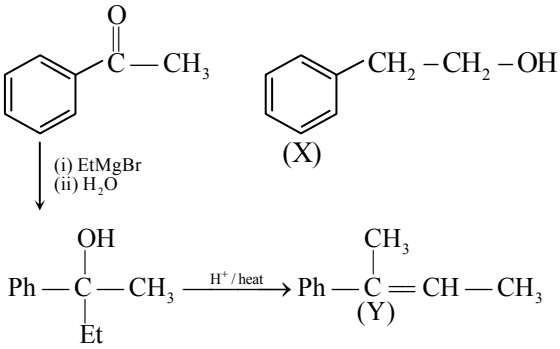
18. **(b)** S_2 is 3p $\therefore l = 1$

For Question Nos. 19 to 20

$C_8H_6 \longrightarrow$ = double bond equivalent

$= 8 + 1 - \frac{6}{2} = 6$





19. **(c)**

20. **(d)**

□ □ □

JEE-MAIN: CHEMISTRY MOCK TEST-2

1. 2 g of oxygen contain same number of atoms as contained by
a. 0.5 g hydrogen **b.** 4.0 g sulphur
c. 7.0 g nitrogen **d.** 2.3 g sodium

2. The axial angles in triclinic crystal system are
a. $\alpha = \beta = \gamma = 90°$ **b.** $\alpha = \gamma = 90°,\ \beta \neq 90°$
c. $\alpha \neq \beta \neq \gamma \neq 90°$ **d.** $\alpha = \beta = \gamma \neq 90°$

3. A sample of drinking water was found to be severely contaminated with chloroform $CHCl_3$, supposed to be a carcinogen. The level of contamination was 15 ppm (by mass). Determine the molality of chloroform in the water sample.
a. $2.12\times10^{-4}\,\text{mol kg}^{-1}$ **b.** $1.26\times10^{-4}\,\text{mol kg}^{-1}$
c. $0.12\times10^{-4}\,\text{mol kg}^{-1}$ **d.** $5.36\times10^{-4}\,\text{mol kg}^{-1}$

4. The temperature at which hydrogen molecules will have the same root mean square velocity as oxygen molecules have at 27°C is
a. 25°C **b.** 7.93°C
c. –248°C **d.** 127°C

5. An electron makes a transition from orbit n = 4 to the orbit n = 2 of a hydrogen atom. The wave number of the emitted radiations (R = Rydberg's constant) will be
a. $\frac{16}{3R}$ **b.** $\frac{2R}{16}$ **c.** $\frac{3R}{16}$ **d.** $\frac{4R}{16}$

6. Select the compound in which chlorine is assigned the oxidation number +5
a. $HClO_4$ **b.** $HClO_2$
c. $HClO_3$ **d.** HCl

7. The molar conductivity is maximum for the solution of concentration
a. 0.001 M **b.** 0.005 M
c. 0.002 M **d.** 0.004 M

8. The number and type of bonds between two carbon atoms in calcium carbide are
a. One sigma, one pi **b.** One sigma, two pi
c.Two sigma, one pi **d.** Two singma, two pi

9. The standard state gibbs free energy change for the given isomerization reaction cis-2-pentene trans-2-pentene is $-3.67\,\text{kJ/mol}$ at 400K. If more trans-2-pentene is added to the reaction vessel, then
a. More cis-2-pentene is formed
b. Equilibrium is shifted in the forward direction
c. Equilibrium remains unaffected
d. Additional trans-2-pentene is formed

10. The dissociation constant of CH_3COOH is 1.8×10^{-5}. The hydrolysis constant for 0.1 M sodium acetate is
a. 5.56×10^{4} **b.** 5.56×10^{-10}
c. 1.8×10^{-5} **d.** 1.8×10^{9}

11. The half-life of a first order reaction having rate constant $K = 1.7\times10^{-5}\,s^{-1}$ is
a. 12.1 h **b.** 9.7 h
c. 11.3 h **d.** 1.8 h

12. Which characteristic is true in respect of colloidal particle
a. They always have two phases
b. They are only in liquid state
c. They can't be electrolysed
d. They are only hydrophilic

13. The heat of transition (ΔH_t) of graphite into diamond would be, where
$C(\text{graphite}) + O_2(g) \longrightarrow CO_2(g);\ \Delta H = x\,\text{kJ}$
$C(\text{diamond}) + C_2(g) \longrightarrow CO_2(g)\ ;\ \Delta H = y\,\text{kJ}$
a. $(x+y)\,\text{kJ mol}^{-1}$ **b.** $(x-y)\,\text{kJ mol}^{-1}$
c. $(y-x)\,\text{kJ mol}^{-1}$ **d.** None of these

14. All the nuclei from the initial element to the final element constitute a series which is called
a. g-series **b.** b-series
c. b-g series **d.** Disintegration series

15. The following compound will undergo electrophilic substitution more readily than benzene
a. Nitrobenzene **b.** Benzoic acid
c. Benzaldehyde **d.** Phenol

16. Cis and trans 2-butene are
a. Conformational isomers
b. Optical isomers
c. Position isomers
d. Geometrical isomers

17. Electrolysis of a concentrated solution of sodium fumarate gives
a. Ethylene **b.** Ethane
c. Acetylene **d.** Vinyl alcohol

18. Identify *X* and *Y* in the following sequence $C_2H_5Br \xrightarrow{X}$ product $\xrightarrow{Y} C_3H_7\ NH_2$

a. $X = KCN, Y = LiAlH_4$

b. $X = KCN, Y = H_3O^+$

c. $X = CH_3Cl, Y = AlCl_3 / HCl$

d. $X = CH_3NH_2, Y = HNO_2$

19. Consider the following alcohols

(i) 1-Phenyl-1-propanol

(ii) 3-Phenyl-1-propanol

(iii) 1-Phenyl-2-propanol

The correct sequence of increasing order of reactivity of these alcohol in their reaction with HBr is

a. (i), (ii), (iii) **b.** (ii), (i), (iii)

c. (i), (iii), (ii) **d.** (ii), (iii), (i)

20. Reaction between diethyl cadmium and acetyl chloride leads to the formation of

a. dimethyl ketone **b.** ethylmethyl ketone

c. diethyl ketone **d.** acetaldehyde

21. Which of the following is basic?

a. CH_3CH_2OH **b.** H_2O_2

c. $HOCH_2CH_2OH$ **d.** CH_3COOH

22. $CH_3CN \xrightarrow{Na, C_2H_5OH} X$. The compound X is:

a. $CH_3CH_2NO_2$ **b.** CH_3CH_2COOH

c. $C_6H_5N(CH_3)_2$ **d.** $C_6H_5CONH_2$

23. Which of the following is a synthetic polymer

a. Rubber **b.** Perspex

c. Protein **d.** Cellulose

24. The correct statement in respect of protein haemoglobin is that it

a. Acts as an oxygen carrier in the blood

b. Forms antibodies and offers resistance to diseases

c. Functions as a catalyst for biological reactions

d. Maintains blood sugar level

25. Which one of the following is known as broad spectrum antibiotics

a. Streptomycine **b.** Ampicillin

c. Chloramphenicol **d.** Penicillin G

26. The element or elements whose position is anomalous in the periodic table is

a. Halogens **b.** Fe, Co and Ni

c. Inert gases **d.** Hydrogen

27. Flux added in the extraction of iron is

a. Silica **b.** Felspar

c. Limestone **d.** Flint

28. The correct order of the increasing ionic character is

a. $BeCl_2 < MgCl_2 < CaCl_2 < BaCl_2$

b. $BeCl_2 < MgCl_2 < BaCl_2 < CaCl_2$

c. $BeCl_2 < BaCl_2 < MgCl_2 < CaCl_2$

d. $BaCl_2 < CaCl_2 < MgCl_2 < BeCl_2$

29. Nesseler's reagent is

a. K_2HgI_4 **b.** $K_2HgI_4 + KOH$

c. $K_2HgI_2 + KOH$ **d.** $K_2HgI_4 + Hg$

30. The primary valence of the metal ion in the co-ordination compound $K_2\left[Ni(CN)_4\right]$ is

a. Four **b.** Zero **c.** Two **d.** Six

Space for rough work

JEE ADVANCE PAPER-I

SECTION 1 Contains 8 Questions

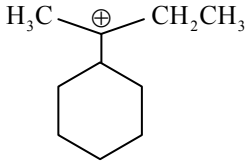
The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. At 400 K, the root mean square (rms) speed of a gas X (molecular weight = 40) is equal to the most probable speed of gas Y at 60 K. The molecular weight of the gas Y is

2. Based on VSEPR theory, the number of 90 degree F−Br−F angles in BrF_5 is

3. 20% surface sites have adsorbed N_2. On heating N_2 gas evolved from sites and were collected at 0.001 atm and 298 K in a container of volume is 2.46 cm^3. Density of surface sites is $6.023\times10^{14}/cm^2$ and surface area is 1000 cm^2, find out the no. of surface sites occupied per molecule of N_2.

4. The total number of contributing structure showing hyperconjugation (involving C–H bonds) for the following carbocation is

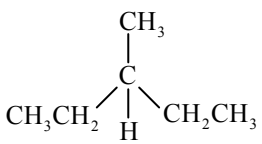


5. The volume (in mL) of 0.1 M $AgNO_3$ required for complete precipitation of chloride ions present in 30 mL of 0.01 M solution of $[Cr(H_2O)_5Cl]Cl_2$, as silver chloride is close to

6. Among PbS, CuS, HgS, MnS, Ag_2S, NiS, CoS, Bi_2S_3 and SnS_2, the total number of black coloured sulphides is

7. 29.2% (w/w) HCl stock solution has a density of 1.25 g mL^{-1}. The molecular weight of HCl is 36.5 g mol^{-1}. The volume (mL) of stock solution required to prepare a 200 mL solution of 0.4 M HCl is

8. The maximum number of isomers (including stereoisomers) that are possible on monochlorination of the following compound is



SECTION 2 Contains 10 Multiple Choice Questions

With one or more than one correct option

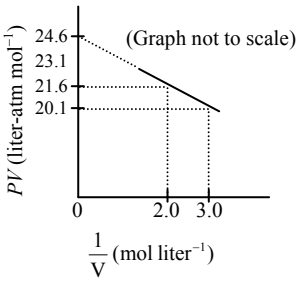
9. Sodium metal crystallises in a body centred cubic lattice with a unit cell edge of 4.29 Å. The radius of sodium atom is approximately

a. 1.86 Å **b.** 3.22 Å **c.** 5.72 Å **d.** 0.93 Å

10. 18 g glucose $(C_6H_{12}P_6)$ is added to 178.2 g water. The vapor pressure of water (in torr) for this aqueous solution is :

a. 7.6 **b.** 76.0 **c.** 752.4 **d.** 759.0

11. For one mole of a van der Waals gas when b = 0 and T = 300 K, the PV vs. 1/V plot is shown below. The value of the van der Waals constant a (atm. $liter^2$ mol^2) is.



a. 1.0 **b.** 4.5 **c.** 1.5 **d.** 3.0

12. A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0 mL to 375 mL at a constant temperature of 37.0°C. As it does so, it absorbs 208 J of heat. The values of q and w for the process will be $(R = 8.314\ J/mol\ K)$ $(\ln 7.5 = 2.01)$

a. q = +208 J, w = −208 J **b.** q = −208 J, w = −208 J

c. q = −208 J, w = +208 J **d.** q = +208 J, w = +208 J

13. The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be

a. −2.55 eV **b.** −5.1 eV

c. −10.2 eV **d.** +2.55 eV

14. In a galvanic cell, the salt bridge

a. does not participate chemically in the cell reaction.

b. stops the diffusion of ions from one electrode to another.

c. is necessary for the occurrence of the cell reaction

d. ensures mixing of the two electrolytic solutions

15. Stability of the species of Li_2, Li_2^- and Li_2^+ increase in the order of

a. $Li_2 < Li_2^+ < Li_2^-$ **b.** $Li_2^- < Li_2^+ < Li_2$

c. $Li_2 < Li_2^- < Li_2^+$ **d.** $Li_2^- < Li_2 < Li_2^+$

16. How many litres of water must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2?

a. 0.1 L **b.** 0.9 L **c.** 2.0 L **d.** 9.0 L

17. $NiCl_2\{P(C_2H_5)_2(C_6H_5)\}_2$ exhibits temperature dependent magnetic behavior (paramagnetic / diamagnetic). The coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively.

a. tetrahedral and tetrahedral

b. square planar and square planar

c. tetrahedral and square planar

d. square planar and tetrahedral

18. Four successive members of the first row transition elements are listed below with atomic numbers. Which one of them is expected to have the highest $E^0_{M^{3+}/M^{2+}}$ value?

a. $Cr(Z = 24)$ **b.** $Mn(Z = 25)$

c. $Fe(Z = 26)$ **d.** $Co(Z = 27)$

SECTION 3 Contains 2 Match The Following Type Questions

You will have to match entries in Column I with the entries in Column II.

19. Match gases under specified conditions listed in Column-I with their properties/laws in Column-II.

Column I	Column II
(A) Hydrogen gas ($P = 200$ atm, $T = 273K$)	**1.** Compressibility factor $\neq 1$
(B) Hydrogen gas ($P \to 0$, $T = 273K$)	**2.** Attractive forces are dominant
(C) CO_2 ($P = 1$ atm, $T = 273K$)	**3.** $P = nRT$
(D) Real gas with very large molar volume	**4.** $P(V - nb) = nRT$

a. A $\to$ 2; B $\to$ 3; C $\to$ 2; D $\to$ 1, 4

b. A $\to$ 1, 3; B $\to$ 1; C $\to$ 2; D $\to$ 4

c. A $\to$ 3; B $\to$ 4; C $\to$ 3, 4; D $\to$ 1, 2

d. A $\to$ 1, 2; B $\to$ 3; C $\to$ 1, 2; D $\to$ 1, 4

20. The standard reduction potential data at 25ºC is given below:

$E°(Fe^{3+}, Fe^{2+}) = +0.77$ V; $E°(Fe^{2+}, Fe) = -0.44$V

$E°(Cu^{2+}, Cu) = +0.34$V; $E°(Cu^{+}, Cu) = +0.52$V

$E°[O_2(g) + 4H^+ + 4e^- \longrightarrow 2H_2O] = +1.23$V;

$E°[O_2(g) + 2H_2O + 4e^- \longrightarrow 4OH^-] = +0.40$V

$E°(Cr^{3+}, Cr) = -0.74$V; $E°(Cr^{2+}, Cr) = -0.91$ V

Match $E°$ of the redox pair in Column I with the values given in Column II.

Column I	Column II
(A) $E°(Fe^{3+}, Fe)$	**1.** -0.18 V
(B) $E°(4H_2O \quad 4^+ + 4OH^-)$	**2.** -0.4 V
(C) $E°(Cu^{2+} + Cu \longrightarrow 2Cu^+)$	**3.** -0.04 V
(D) $E°(Cr^{3+}, Cr^{2+})$	**4.** -0.83 V

a. A $\to$ 2; B $\to$ 3; C $\to$ 1; D $\to$ 4

b. A $\to$ 1; B $\to$ 2; C $\to$ 3; D $\to$ 4

c. A $\to$ 3; B $\to$ 4 ; C $\to$ 1; D $\to$ 2

d. A $\to$ 2; B $\to$ 3; C $\to$ 1; D $\to$ 4

Space for rough work

JEE ADVANCE PAPER-II

SECTION 1 Contains 8 Questions.

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. The maximum number of electrons that can have principal quantum number, n = 3, and spin quantum number, $m_s = -\frac{1}{2}$, is

2. The dissociation constant of a substituted benzoic acid at 25ºC is 1.0×10^{-4}. The pH of a 0.01M solution of its sodium salt is

3. The total number of cyclic structural as well as stereo isomers possible for a compound with the molecular formula C_5H_{10} is

4. To an evacuated vessel with movable piston under external pressure of 1 atm, 0.1 mol of He and 1.0 mol. of an unknown compound (vapour pressure 0.68 atm. At 0ºC) are introduced. Considering the ideal gas behaviour, the total volume (in litre) of the gases at 0ºC is close to

5. The coordination number of Al in the crystalline state of $AlCl_3$ is

6. Among the following, the number of compounds than can react with PCl_5 to give $POCl_3$ is

$O_2, CO_2, SO_2, H_2O, H_2SO_4, P_4O_{10}$

7. If the freezing point of a 0.01 molal aqueous solution of a cobalt (III) chloride-ammonia complex (which behaves as a strong electrolyte) is –0.0558ºC, the number of chloride(s) in the coordination sphere of the complex is [K_f of water = 1.86 K kg mol^{-1}]

8. A student performs a titration with different burettes and finds titre values of 25.2 mL, 25.25 mL and 25.0 mL. The number of significant figures in the average titer value is

SECTION 2 Contains 8 Multiple Choice Questions

With one or more than one correct option

9. The correct statement(s) regarding defects in solids is(are)

a. Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion

b. Frenkel defect is a dislocation defect

c. Trapping of an electron in the lattice leads to the formation of F-centre

d. Schottky defects have no effect on the physical properties of solids

10. Mixture(s) showing positive deviation from Raoult's law at 35ºC is(are)

a. carbon, tetrahedral + methanol

b. carbon disulphide + acetone

c. benzene + toluene

d. phenol + aniline

11. For a gaseous state, if most probable speed is denoted by C*, average speed by $\overline{C}$ and mean square speed by C, then for a large number of molecules the ratio of these speeds are

a. $C^* : \overline{C} : C = 1.225 : 1.128 : 1$

b. $C^* : \overline{C} : C = 1.128 : 1.1225 : 1$

c. $C^* : \overline{C} : C = 1 : 1.228 : 1.225$

d. $C^* : \overline{C} : C = 1 : 1.225 : 1.128$

12. The species having bond order different from that in CO is

a. NO^- **b.** NO^+ **c.** CN^- **d.** N_2

13. Which of the following is the wrong statement?

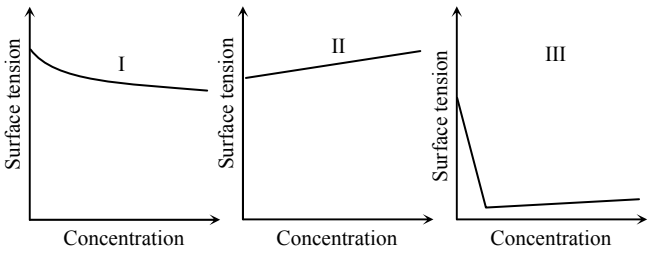
a. ONCF and ONO^- are not isoelectronic

b. O_3 molecules is bent

c. Ozone is violet–black in solid state

d. Ozone is diamagnetic gas

14. The qualitative sketches I, II and III given below show the variation of surface tension with molar concentration of three different aqueous solutions of KCl, CH_3OH and $CH_3(CH_2)_{11}OSO_3^- Na^+$ at room temperature. The correct assignment of the sketches is



a. I : KCl II : CH_3OH III : $CH_3(CH_2)_{11}OSO_3^- Na^+$

b. I : $CH_3(CH_2)_{11}OSO_3^- Na^+$ II : CH_3OH III : KCl

c. I : KCl II : $CH_3(CH_2)_{11}OSO_3^- Na^+$ III : CH_3OH

d. I : CH_3OH II : KCl III: $CH_3(CH_2)_{11}OSO_3^- Na^+$

15. Using the data provided, calculate the multiple bond energy (kJ mol^{-1}) of C≡C bond in C_2H_2. That energy is (take the bond energy of a C—H bond as 350 kJ mol^{-1})

$2C(s) + H_2(g) \longrightarrow C_2H_2(g);\ \Delta H = 225\,kJ\,mol^{-1}$

$2C(s) \longrightarrow 2C(g)$

$\Delta H = 1410\,kJ\,mol^{-1}$

$H_2(g) \longrightarrow 2H(g)$

$\Delta H = 330\,kJ\,mol^{-1}$

a. 1165 **b.** 837
c. 865 **d.** 815

16. A solution of (–)-1-chloro-1-phenylethane in toluene racemises slowly in the presence of a small amount of $SbCl_5$, due to the formation of

a. carbanion **b.** carbene
c. carbocation **d.** free radical

SECTION 3 Contains 2 Paragraph Type Questions

Each paragraph describes an experiment, a situation or a problem. Two multiple choice questions will be asked based on this paragraph. One or more than one option can be correct.

Paragraph for Question No. 17 to 18

The electrochemical cell shown below is a concentration cell. $M|M^2 +$ (saturated solution of sparingly soluble salt, $MX_2) \,||\, M^{2+}(0.001\,mol\,dm^{-3})\,|M$. The emf of the cell depends on the difference in concentration of M^{2+} ions at the two electrodes. The emf of the cell at 298 K is 0.059 V.

17. The value of $\Delta G\ (kJ\ mol^{-1})$ for the given cell is (take 1 F = 96500 C mol^{-1})

a. –5.7 **b.** 5.7
c. 11.4 **d.** –11.4

18. The solubility product $(K_{sp}; mol^3\ dm^{-9})$ of MX_2 at 298 K based on the information available for the given concentration cell is (take 2.303 × R × 298 / F = 0.059 V)

a. 1×10^{-15} **b.** 4×10^{-15}
c. 1×10^{-12} **d.** 4×10^{-12}

Paragraph for Question Nos. 19 to 20

Chemical reactions involve interaction of atoms and molecules. A large number of atoms/molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic/molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical/electrochemical reaction, which requires a clear understanding of the mole concept.

A 4.0 molar aqueous solution of NaCl is prepared and 500mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (atomic mass: Na = 23, Hg = 200, 1 Faraday = 96500 coulombs)

19. The total number of moles of chlorine gas evolved is

a. 0.5 **b.** 1.0
c. 2.0 **d.** 3.0

20. If the cathode is a Hg electrode, the maximum weight (g) of amalgam formed from this solution is

a. 200 **b.** 225
c. 400 **d.** 446

Space for rough work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	c	b	c	c	c	a	b	a	b
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	a	b	d	d	d	c	a	d	b
21.	**22.**	**23.**	**24.**	**25.**	**26.**	**27.**	**28.**	**29.**	**30.**
a	a	b	a	c	d	c	a	b	c

1. **(b)** Number of atoms in 2 g of oxygen $= \frac{2\times 6\times 10^{23}}{16}$

(1) $6\times 10^{23}\times 0.5 = 3\times 10^{23}$

(2) $\frac{6\times 10^{23}\times 4}{32} = 7.5\times 10^{22}$

(3) $\frac{6\times 10^{23}\times 7}{28} = 1.5\times 10^{23}$

(4) $\frac{6\times 10^{23}\times 2.3}{23} = 6\times 10^{22}$

$\therefore$ The correct answer is (b).

2. **(c)** The axial angles in triclinic crystal system are different and none is perpendicular to any of the others i.e., $\alpha \neq \beta \neq \gamma \neq 90°$.

3. **(b)** Level of contamination = 15 ppm

= 15 parts in 10^6 parts Mass % of $CHCl_3$

$= \frac{15}{10^6}\times 100 = 1.5\times 10^{-3}$

Thus, the water sample contains

1.5×10^{-3}% (by mass) of $CHCl_3$

Amount of $CHCl_3 = \frac{15\text{g}}{119.5\,\text{g mol}^{-1}} = 0.126\text{mol}$

Mass of water (solvent) = $10^6\,\text{g} = 10^3\,\text{kg}$

So, molality of $CHCl_3$ in solution

$= \frac{0.126\,\text{mol}}{10^3\,\text{kg}} = 1.26\times 10^{-4}\,\text{mol kg}^{-1}$

4. **(c)** r.m.s. $(H_2) = \sqrt{\frac{3RT}{2}}$

r.m.s. $(O_2) = \sqrt{\frac{3R\times 400}{32}}$

$(T = 273+127 = 400\text{ K})$

$\sqrt{\frac{3RT}{2}} = \sqrt{\frac{3R\times 300}{32}}$

$\Rightarrow \quad \frac{T}{2} = \frac{400}{32}$

$\therefore \quad T = \frac{400\times 2}{32} = 25\text{ K}$ or $T = 25-273 = -248°\text{C}$

5. **(c)** Wave number $\frac{1}{\lambda} = R\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right] = R\left[\frac{1}{4}-\frac{1}{16}\right] = \frac{3R}{16}$

6. **(c)** $H\overset{*}{Cl}O_3$

$\Rightarrow \quad 1+x-2\times 3 = 0\ ; \quad x = 6-1 = +5$

7. **(a)** Since molar conductance $\propto \frac{1}{\text{Molarity}}$

8. **(b)** 1σ and 2π

9. **(a)** Equilibrium shifts backward by Le-chatelier's principle.

10. **(b)** $K_b = \frac{K_w}{K_a} = \frac{1\times 10^{-14}}{1.8\times 10^{-5}} = 5.56\times 10^{-10}$

11. **(c)** $K = 1.7\times 10^{-5}\,\text{s}^{-1}$

$t_{1/2} = \frac{0.693}{K} = \frac{0.693}{1.7}\times 10^5 = 11.32\text{h}$

12. **(a)** Dispersion medium and dispersed phase are phases of colloid.

13. **(b)** Graphite $\longrightarrow$ diamond $\Delta H_t = (x-y)\,\text{kJ mol}^{-1}$.

14. **(d)** Definition of disintegration series.

15. **(d)** Phenol will undergo electrophilic substitution more readily than benzene.

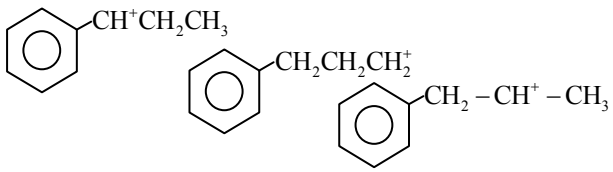
16. **(d)** Cis and trans 2-butene are geometrical isomers.

17. **(c)** $\begin{matrix}\text{CHCOONa}\\ \|\\ \text{CHCOONa}\end{matrix} + 2H_2O \xrightarrow{\text{Electrolysis}} \underset{\text{Acetylene}}{\begin{matrix}\text{CH}\\ \|\\ \text{CH}\end{matrix}} + 2CO_2 + 2KOH + H_2$

18. **(a)** $C_2H_5Br \xrightarrow{KCN(X)} C_2H_5CN \xrightarrow{LiAlH_4(Y)} C_2H_5CH_2NH_2 (C_3H_7NH_2)$

$X = KCN, Y = LiAlH_4$

19. **(d)** HBr reacts with alcohols through the formation of carbocation. The stable is the carbocation formed, more is the reactivity of alcohol with HBr. The carbocation formed are



(1) (2) (3)

20. **(b)** $Cd\begin{matrix}CH_2CH_3\\CH_2CH_3\end{matrix} + 2CH_3COCl \longrightarrow 2CH_3COCH_2CH_3 + CaCl_2$

21. **(a)** Ethanol is the weakest acid among these, hence it is most basic.

22. **(a)** $CH_3CN + 4H \xrightarrow{Na,\ C_2H_5OH} CH_3CH_2NH_2(X)$

23. **(b)** Perspex is a synthesised polymer.

24. **(a)** Four Fe^{2+} ions of each haemoglobin can bind with 4 molecules of O_2 and it is carried as oxyhaemoglobin.

$Hb_4 + 4O_2 \longrightarrow Hb_4O_8$

25. **(c)** Chloramphenicol is broad spectrum antibiotic used in the treatment of typhoid, dysentry, acute fever.

26. **(d)** It shows similarities with both alkali metals as well as halogens.

27. **(c)** Impurities of SiO_2 is present in iron ore so basic flux $CaCO_3$ is added.

$\underset{\text{Flux}}{CaO} + \underset{\text{Impurity}}{SiO_2} \longrightarrow \underset{\text{Slag}}{CaSiO_3}$

28. **(a)** $BeCl_2 < MgCl_2 < CaCl_2 < BaCl_2$

As we go down the group I.E. decreases. Hence ionic character increases.

29. **(b)** $2KI + \underset{\text{Nessler's reagent}}{HgI_2} \longrightarrow K_2HgI_4 + KOH$

30. **(c)** Primary valencies are also known as oxidation state.

$K_2[Ni(CN)_4],\ 2 + x - 4 = 0 \Rightarrow x = +2$

JEE Advance Paper -I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
4	0	2	6	6	6	8	8	a	c
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	a	b	a	b	d	c	d	d	c

1. **(4)** $V_{rms_{(Xgas)(400K)}} = V_{mp_{(Ygas)(60K)}}$

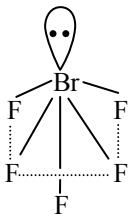
$M.W.(X\,gas) = 40; M.W.(Y\,gas) = x$

$\sqrt{\frac{3RT_1}{M_1}} = \sqrt{\frac{2RT_1}{M_2}}$

$\sqrt{\frac{400\times 3}{40}} = \sqrt{\frac{2\times 60}{x}}$

$\Rightarrow \quad 30 = \frac{120}{x}, \quad x = 4$

2. **(0)**



All four planar bonds (F−Br−F) will reduce from 90° to 84.8° after $p-bp$ repulsion.

So among the following only four (4) has linear shape and no d-orbital is involved in hybridization.

3. **(2)** $P_{N_2} = 0.001$ atm, $T = 298$ K, $V = 2.46\ cm^2$

By ideal gas, $PV = nRT$

$n_{N_2} = \frac{PV}{RT} = \frac{0.001\times 2.46\times 10^{-3}}{0.0821\times 298} = 1.0\times 10^{-7}$

Now molecules of

$N_2 = 6.023\times 10^{23}\times 1\times 10^{-7} = 6.023\times 10^{16}$

Now total surface sites available

$= 6.023\times 10^{14}\times 1000 = 6.023\times 10^{17}$

$\therefore$ Surface site used to adsorb

$N_2 = \frac{20}{100}\times 6.023\times 10^{17} = 12.04\times 10^{16}$

$\therefore$ Sites occupied per molecule of

$N_2 = \frac{12.04\times 10^{16}}{6.02\times 10^{16}} = 2$

4. **(6)** $6\times H-$ atoms are there

5. **(6)** Number of ionisable Cl^- in $[Cr(H_2O)_5Cl]Cl_2$ is 2

$\therefore$ Millimoles of

$Cl^- = 30\times 0.01\times 2 = 0.6$

$\therefore$ Millimoles of Ag^+ required $= 0.6$

$\therefore \quad 0.6 = 0.1\,V$

$\Rightarrow \quad V = 6$ ml

6. **(6)** Black coloured sulphides

$\{PbS, CuS, HgS, Ag_2S, NiS, CoS\}$

* Bi_2S_3 in its crystalline form is dark brown but Bi_2S_3 precipitate obtained is black in colour.

7. **(8)** Molarity

$= \frac{n}{V} = \frac{W}{M_w\times v} = \frac{W\times d_{sol}}{M_w\times W_{sol}} = \frac{W}{W_{sol}}\times\frac{100\times d^{soln}}{M_w\times 100}\times 1000$

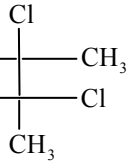
$= \frac{\frac{W}{W}\%\times d\times 10}{M_w}$

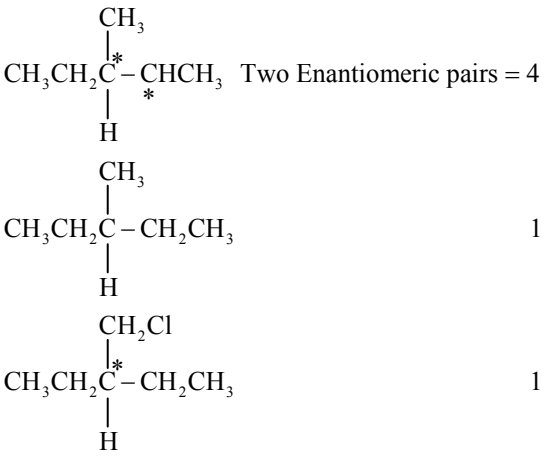
$= \frac{29.2 \times 1.25 \times 10}{36.5} = 10\,M \quad M_1V_1 = M_2V_2$

$\Rightarrow \quad 10 \times V_1 = 0.4 \times 200$

$\Rightarrow \quad V = \frac{0.4 \times 200}{10} = 8ml$

8. **(8)**





Two Enantiomeric pairs = 4

1

1

Total = 2 + 4 + 1 + 1 = 8

9. **(a)** For BCC 6 unit cell, $\sqrt{3}a = 4r$

$r = \frac{\sqrt{3}}{4}a = \frac{\sqrt{3}}{4} \times 4.29\ Å = 1.85\ Å$

10. **(c)** Moles of glucose = $\frac{18}{180} = 0.1$

Moles of water = $\frac{178.2}{18} = 9.9$

$\Rightarrow \quad n_{Total} = 10$

$\Rightarrow \quad \frac{\Delta P}{P°} = \frac{0.1}{10}$

$\Rightarrow \quad \Delta P = 0.01P°$

$= 0.01 \times 760 = 7.6$ torr

$P_S = 760 - 7.6 = 752.4$ torr

11. **(c)** $b = 0, T = 300\,K, n = 1$

$\left(P + \frac{an^2}{V^2}\right)(V - nb) = RT$

$\left(P + \frac{a}{V^2}\right)(V) = RT$

$\Rightarrow \quad PV + \frac{a}{V} = RT$

$PV = -a \times \left(\frac{1}{V}\right) + RT$

$y = mx + C \ \Rightarrow$ Slope = – a

Slope $\frac{y_2 - y_1}{x_2 - x_1} = \frac{20.1 - 21.6}{3 - 2} = 1.5$

12. **(a)** q = +208 J, (as it absorb heat)

$w_{Rew} = -2.303\,nRT \log_{10}\left(\frac{v_2}{v_1}\right)$

$= -2.303 \times (0.04 \times 8.314 \times (310) \log_{10}\left(\frac{375}{50}\right) = -208\ J$

13. **(b)** E.A = Ionisation potential

$\therefore$ EA of $Na^+ = -5.1\,eV$

14. **(a)**

15. **(b)** B.O. of $Li_2^+ = 0.5$, B.O. of $Li_2^- = 0.5$

Hence stability order $= Li_2^- < Li_2^+ < Li_2$

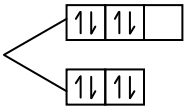
16. **(d)** $0.1 \times 1 = (1 + v) \times 0.01 \ \Rightarrow \ 1 + v = \frac{0.1}{0.01}$

$\Rightarrow \quad 1 + v = 10 \ \Rightarrow \ v = 10 - 1 = 9L$

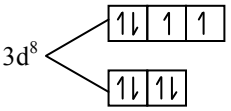
17. **(c)** $[NiCl_2\,P(C_2H_5)_2(C_6H_5)_2]^{+2}$

$Ni^{+2}(28) : [Ar]4s°3d^8$

In strong ligand $3d^8$



In weak ligand $3d^8$



Hybridisation is dsp^2 (diamagnetic)

→ Square planar

Hybridisation is sp^3 (paramagnetic)

→ tetrahedral

18. **(d)** Factual

19. **(d)** A → 1, 2 ; B → 3 ; C → 1, 2 ; D → 1, 4

(A) $Z = \frac{PV_m}{RT}$ at high pressure and low temperature.

Equation $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ reduces to

$P(V - nb) = nRT.$

(B) For hydrogen gas value of Z = 1 at P = 0 and it increases continuously on increasing pressure.

(C) CO_2 molecules have larger attractive forces, under normal conditions.

(D) $Z = \frac{PV_m}{RT}$, at very large molar volume $Z \neq 1$.

20. **(c)** $A \rightarrow 3; B \rightarrow 4; C \rightarrow 1; D \rightarrow 2$

(A) $\Delta G^{o}_{Fe^{3+}/Fe} = \Delta G^{o}_{Fe^{3+}/Fe^{2+}} + \Delta G^{o}_{Fe^{2+}/Fe}$

$\Rightarrow \quad -3\times FE^{o}_{(Fe^{+3}/Fe)} = -1\times FE^{o}_{(Fe^{+3}/Fe^{+2})} + (-2\times FE^{o}_{Fe^{+2}/Fe})$

$\Rightarrow \quad E^{o}_{Fe^{+3}/Fe} = -0.04\ V$

(B) $O_2(g) + 2H_2O + 4e^- \longrightarrow 4\bar{O}H \quad E° = 0.40\ V \ldots(i)$

$2H_2O \longrightarrow O_2(g) + 4H^+ + 4e^- \quad E° = -1.23\ V \quad \ldots(ii)$

So $4H_2O \qquad 4H^+ + 4\bar{O}H \qquad \ldots(iii)$

E° For III[rd] reduction $= 0.40 - 1.23 = -0.83\ V$.

(C) $\Delta G^{o}_{(Cu^{+2}/Cu)} = -1\times FE^{o}_{Cu^{+2}/Cu^{+}} + (-1\times F\times E^{o}_{Cu^{+}/Cu})$

$-2\times FE^{o}_{Cu^{+2}/Cu} = -1\times FE^{o}_{Cu^{+2}/Cu^{+}} + (-1\times F\times E^{o}_{Cu^{+}/Cu})$

$\Rightarrow \quad E^{o}_{Cu^{+2}/Cu} = -0.18\ V.$

(D) $\Delta G^{o}_{Cr^{+3}/Cr^{+2}} = \Delta G^{o}_{Cr^{+3}/Cr} + \Delta G^{o}_{Cr/Cr^{+2}}$

$-1\times F\times E^{o}_{Cr^{+3}/Cr^{+2}} = -3\times F\times E^{o}_{Cr^{+3}/Cr} + (-2\times F\times E^{o}_{Cr/Cr^{+2}})$

$\Rightarrow \quad E^{o}_{Cr^{+3}/Cr^{+2}} = -0.4\ V.$

JEE Advance Paper -II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
9	8	7	7	6	5	1	3	b, c	a, b
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	a	a	d	d	c	d	b	b	d

1. **(9)** Number of orbital for $n = 3$ is $= n^2 = 9$

Number of electrons for $n = 3$ and $m_s = -\frac{1}{2} = 9$

2. **(8)** $K_a(C_6H_5COOH) = 1 \times 10^{-4}$

pH of $0.01M\ C_6H_5COONa$

$\underset{0.01(1-h)}{C_6H_5COO^-} + H_2 \qquad \underset{0.01h}{C_6H_5COOH} + \underset{0.01h}{OH^{-1}}$

$K_h = \frac{K_w}{K_a} = \frac{0.01h^2}{1-h}$

$\frac{10^{-14}}{10^{-4}} = \frac{10^{-2}h^2}{1-h} \qquad (1-h \quad 1)$

$[OH^-] = 0.01h = 0.01\times10^{-4} = 10^{-6}$

$[H^+] = 10^{-8} \quad pH = 8$

3. **(7)** Cyclic C_5H_{10}

For 3rd structure 2 cis-trans and 1 optical isomer are possible. Total 7 isomers.

4. **(7)** Let unknown is X.

$p_{He} = p_{total} - P_x = (1 - 0.68)\ atm = 0.32\ atm$

Now $p_{He} = n_{He}\frac{RT}{V}$

$\therefore \quad v = \frac{RT}{p_{He}} = \frac{0.10\times0.082\times273}{0.32} = 7$

5. **(6)** Coordination number of Al is 6. It exists in ccp lattice with 6 coordinate layer structure.

6. **(5)**

7. **(1)** $\Delta T_f = iK_f m$

$0.0558 = i\times1.86\times0.01$

$i = 3$

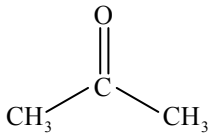
$\therefore \quad$ Complex is $[Co(NH_3)_5Cl]Cl_2$

8. **(3)**

9. **(b, c)**

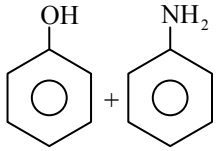
10. **(a, b)**

(a) $CCl_4 + CH_3OH \Rightarrow$ Positive deviation from Raoult's law

(b) $CS_2 +$ 

$\Rightarrow$ Positive deviation from Raoult's law

(c) $C_6H_6 + C_7H_8 \Rightarrow$ Ideal solution

(d) 

$\Rightarrow$ Negative deviation from Raoult's law.

11. (c) $C^* : \bar{C} : C = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{3RT}{M}}$

$\sqrt{2} : \sqrt{\frac{8}{3.14}} = \sqrt{3}$

1 : 1.128 : 1.225

12. **(a)** NO^- (16 electron system)

Bond order =2.

NO^+, CN^- and N_2 are isoelectronic with CO therefore all have same bond order (=3).

13. **(a)** ONCF and ONO^- are isoelectronic in nature.

14. **(d)** Impurities affect surface tension appreciably. It is observed that impurities which tend to concentrate on surface of liquids, compared to its bulk lower the surface tension.

Substances like detergents, soaps $[CH_3(CH_2)_{11}SO_3\ Na\ ]$ decreases the surface tension sharply.

Those like alcohol (e.g., $-CH_3OH$, C_2H_5OH) lower the surface tension slightly. This can also be related to the fact that CH_3OH has smaller dielectric constant. Dielectric constant is directly proportional to surface tension. So, on adding CH_3OH in water, overall dielectric constant decreases and surface tension decreases.

Inorganic impurities present in bulk of a liquid such as KCl tend to increase the surface tension of water.

15. **(d)** $2C(s) + H_2(g) \longrightarrow C_2H_2(g)(H-C\equiv C-H)$

$\Rightarrow$ $B{\cdot}E(H_2) + \Delta B{\cdot}E(H_2) + \Delta H_{sub}(C) - B{\cdot}E(C-H)\times 2 + B{\cdot}E(C\equiv C)\ \Delta H_{rxn}$

$\Rightarrow$ $330 + 1410 - [350\times 2 + x] = 225 \Rightarrow x = 815$

16. **(c)** Racemises slowly due to formation of intermediate carbocation.

For Question Nos. 17 to 18

17. **(d)** $\Delta G = -nFE_{cell}$

$= -2\times 96500\times 0.059$

$= -11.387\ \text{kJ mol}^{-1}$

$= -11.4\ \text{kJ}$

18. **(b)** $E = \dfrac{-0.0591}{2}\log_{10}\dfrac{C}{10^{-3}}$

$0.059 = \dfrac{-0.0591}{2}\log_{10}\left(\dfrac{C}{10^{-3}}\right)$

$10^{-2} = \dfrac{C}{10^{-3}}\ \ C = [M^{2+}] = 10^{-5}\ \text{M}.$

$K_{sp} = [M^{+2}][X^-]^2 = 4s^3 = 4(10^{-5})^3 = 4\times 10^{-15}$

For Question Nos. 19 to 20

19. **(b)** $NaCl \longrightarrow Na^+ + Cl^-$

At anode: $2Cl^- \longrightarrow Cl_2$

Moles of $Cl^- = 2$ in 500 ml.

Therefore 1 mole of Cl_2 evolves.

20. **(d)** Na—Hg (amalgam) formed = 2 moles at cathode.

□ □ □

JEE-MAIN: CHEMISTRY MOCK TEST-3

1. The volume of 1.0 g of hydrogen in litres at N.T.P. is
a. 2.24 **b.** 22.4 **c.** 1.12 **d.** 11.2

2. Which set of characteristics of ZnS crystal is correct?
a. Coordination number (4: 4) : ccp; Zn^{2+} ion in the alternate tetrahedral voids
b. Coordination number (6: 6); hcp; Zn^{2+} ion in all tetrahedral voids
c. Coordination number (6: 4) : hcp; Zn^{2+} ion in all octahedral voids
d. Coordination number (4: 4); ccp; Zn^{2+} ion in all tetrahedral voids

3. $PtCl_4.6H_2O$ can exist as a hydrated complex. 1 molal aqueous solution has depression in freezing point of 3.72º Assume 100% ionisation and $K_f(H_2O) = 1.86° \text{ mol}^{-1}$ kg, then complex is
a. $[Pt(H_2O)_6]Cl_4$ **b.** $[Pt(H_2O)_4Cl_2]Cl_2 2H_2O$
c. $[Pt(H_2O)_3Cl_3], Cl.3H_2O$ **d.** $[Pt(H_2O_4)Cl_4]4H_2O$

4. A weather balloon filled with hydrogen gas at 1 atm and 27ºC has volume equal to 12000 litres. On ascending, it reaches a place where temperature is –23ºC and pressure 0.5 atm. The volume of the balloon is
a. 24000 L **b.** 12000 L
c. 10000 L **d.** 20000 L

5. In Bohr model of the hydrogen atom, the lowest orbit corresponds to
a. Infinite energy **b.** The maximum energy
c. The minimum energy **d.** Zero energy

6. When $KMnO_4$ is reduced with oxalic acid in acidic solution, the oxidation number of Mn changes from
a. 7 to 4 **b.** 6 to 4
c. 7 to 2 **d.** 4 to 2

7. The unit of molar conductivity is
a. $\Omega^{-1}cm^{-2}\,mol^{-1}$ **b.** $\Omega\,cm^{-2}\,mol^{-1}$
c. $\Omega^{-1}cm^{2}\,mol^{-1}$ **d.** $\Omega\,cm^{2}\,mol^{-1}$

8. In a double bond connecting two atoms, there is a sharing of
a. 2 electrons **b.** 1 electron
c. 4 electrons **d.** All electrons

9. In a reversible reaction, the catalyst
a. Increases the activation energy of the backward reaction
b. Increases the activation energy of the forward reaction
c. Decreases the activation energy of both, forward and backward reaction
d. Decreases the activation energy of forward reaction

10. Ammonium cyanide is salt of $NH_4OH(K_b = 1.8 \times 10^{-5})$ and HCN ($K_b = 4.0 \times 10^{-10}$). The hydrolysis constant of 0.1 M NH_4CN at 25ºC is
a. 1.4 **b.** 7.2×10^{-15}
c. 7.2×10^{-1} **d.** 1.4×10^{-6}

11. For the reaction A + B → C it is found that doubling the concentration of A increases the rate by 4 times, and doubling the concentration of B doubles the reaction rate. What is the overall order of the reaction?
a. 4 **b.** 3/2
c. 3 **d.** 1

12. The coagulation power of an electrolyte for arsenious sulphide decreases in the order
a. Na^+, Al^{+3}, Ba^{+2} **b.** $PO_4^{-3}, SO_4^{-2}, Cl^-$
c. Al^{+3}, Ba^{+2}, Na^+ **d.** None of these

13. Correct relationship between heat of fusion (ΔH_{fus}), heat of vaporisation (ΔH_{vap}) and heat of sublimation (ΔH_{sub}) is
a. $\Delta H_{fus} = \Delta H_{vap} + \Delta H_{sub}$
b. $\Delta H_{vap} = \Delta H_{fus} + \Delta H_{sub}$
c. $\Delta H_{sub} = \Delta H_{vap} + \Delta H_{fus}$
d. $\Delta H_{sub} = \Delta H_{vap} - \Delta H_{fus}$

14. The number of neutrons in the parent nucleus which gives N^{14} on β- emission is
a. 7 **b.** 14 **c.** 6 **d.** 8

15. Which represents nucleophilic aromatic substitution reaction?
a. Reaction of benzene with Cl_2 in sunlight
b. Benzyl bromide hydrolysis
c. Reaction of NaOH with dinitrofluorobenzene
d. Sulphonation of benzene

16. Which one of the following is the chiral molecule?
a. CH_3Cl **b.** CH_2Cl_2
c. $CHBr_3$ **d.** CHClBrI

17. n-Propyl chloride and benzene react in the presence of anhydrous $AlCl_3$ to form
a. ethyl benzene **b.** methyl benzene
c. n-propyl benzene **d.** iso-propyl benzene

18. 1-chlorobutane reacts with alcoholic KOH to form
a. 1-butene **b.** 2-butane
c. 1-butanol **d.** 2-butanol

19. An alcohol having molecular formula $C_5H_{11}OH$ on dehydration gives an alkene, which on oxidation yield a mixture of ketone and an acid. The alcohol is
a. $CH_3CH_2CH(OH)CH_2CH_3$ **b.** $CH_3\underset{\underset{OH}{|}}{C}HCH_2CH_2CH_3$
c. $(CH_3)_2CHCH(OH)CH_3$ **d.** $(CH_3)_2CCH_2OH$

20. m-chlorobenzaldehyde on reaction with conc. KOH at room temperature gives
a. potassium m-chlorobenzate and m-chlorobenzyl alcohol
b. m-hydroxy benzaldehyde and m-chlorobenzyle alcohol
c. m-chlorobenzyl alcohol and m-hydroxy benzyle alcohol
d. potassium m-chlorobenzoate and m-hydroxy bezaldehyde

21. Acetamide is treated separately with the following reagents. Which one of these would give methyl amine?
a. PCl_5 **b.** $NaOH + Br_2$
c. Sodalime **d.** Hot conc. H_2SO_4

22. On treating aniline with nitrous acid and HCl at 0–5ºC gives
a. An alcohol
b. Diazonium salt
c. Nitro aniline
d. Aniline hydrogen chloride

23. 'Rayon' is
a. Natural silk **b.** Artificial silk
c. Natural plastic or rubber **d.** Synthetic plastic

24. The waxes are long chain compounds of fatty acids, which belong to the class of
a. Esters **b.** Ethers
c. Alcohols **d.** Acetic acid

25. Which of the following is a local anaesthetic?
a. Diazepam **b.** Procaine
c. Mescaline **d.** None of these

26. The electronic configuration of the element which is just above the element with atomic number 43 in the same periodic group is
a. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$
b. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^5$
c. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^1$
d. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1\ 4p^6$

27. Complex is formed in the extraction of
a. Na **b.** Cu **c.** Ag **d.** Fe

28. $MgCl_2.6H_2O$ when heated gives
a. Magnesium oxychloride **b.** Magnesium dichloride
c. Magnesium oxide **d.** Magnesium chloride

29. Acidified potassium dichromate on reacting with a sulphite is reduced to
a. CrO_2Cl_2 **b.** CrO_4^{2-}
c. Cr^{3+} **d.** Cr^{2+}

30. In the extraction of which of the following, complex ion forms?
a. Cu **b.** Ag **c.** Fe **d.** Na

Space for rough work

JEE ADVANCE PAPER-I

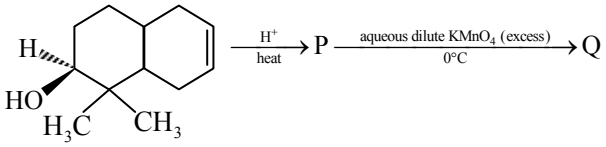
SECTION 1 Contains 8 Questions

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

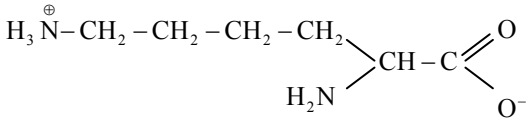
1. The molar conductivity of a solution of a weak acid HX (0.01 M) is 10 times smaller than the molar conductivity of a solution of a weak acid HY (0.10 M). If $\lambda^0_{X^-} \approx \lambda^0_{Y^-}$, the difference in their pK_a values, pK_a (HX) pK_a (HY), is (consider degree of ionization of both acids to be << 1)

2. The total number of α and β particles emitted in the nuclear reaction $^{238}_{92}U \rightarrow ^{214}_{82}Pb$ is

3. The number of hydroxyl group(s) in Q is



4. The total number of basic groups in the following form of lysine is



5. In 1 L saturated solution of AgCl [Ksp(AgCl)=1.6×10^{-10}], 0.1 mol of CuCl [$K_{sp}(CuCl) = 1.0\times10^{-6}$] is added. The resultant concentration of Ag^+ in the solution is 1.6×10^{-x}. The value of "x" is

6. $EDTA^{4-}$ is ethylenediaminetetraacetate ion. The total number of N – Co – O bond angles in $[Co(EDTA)^{1-}]$ complex ion is

7. Reaction of Br_2 with Na_2CO_3 in aqueous solution gives sodium bromide and sodium bromated with evolution of CO_2 gas. The number of sodium bromide molecules involved in the balanced chemical equation is

8. A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt.75), alanine and phenylanine. Glycine contributes 47.0% to the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is

SECTION 2 Contains 10 Multiple Choice Questions

With one or more than one correct option

9. If the unit cell of a mineral has cubic close packed (ccp) array of oxygen atoms with m fraction of octahedral holes occupied by aluminium ions and n fraction of tetrahedral holes occupied by magnesium ions, m and n, respectively, are

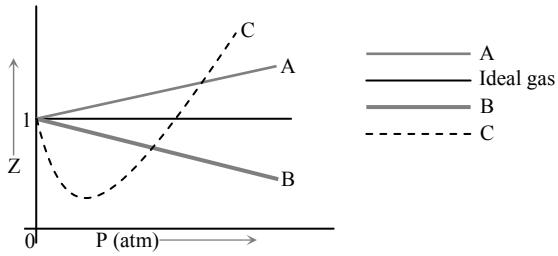
a. $\frac{1}{2}, \frac{1}{8}$ **b.** $1, \frac{1}{4}$

c. $\frac{1}{2}, \frac{1}{2}$ **d.** $\frac{1}{4}, \frac{1}{8}$

10. A monatomic ideal gas undergoes a process in which the ratio of P to V at any instant is constant and equals to 1. What is the molar heat capacity of the gas?

a. $\frac{4R}{2}$ **b.** $\frac{3R}{2}$ **c.** $\frac{5R}{2}$ **d.** 0

11. The given graph represents the variation of Z (compressibility factor $= \frac{PV}{nRT}$) versus P, for three real gases A, B and C. Identify the only incorrect statement.



a. For the gas A, a = 0 and its dependence on P is linear at all pressure

b. For the gas B, b = 0 and its dependence on P is linear at all pressure

c. For the gas C, which is typical real gas for which neither a nor $b = 0$. By knowing the minima and the point of intersection, with Z = 1, a and b can be calculated

d. At high pressure, the slope is positive for all real gases

12. For the reaction: $I^- + ClO_3^- + H_2SO_4 \longrightarrow Cl \quad HSO_4 \quad I_2$

The correct statement(s) in the balanced equation is/are

a. Stoichiometric coefficient of HSO_4^- is 6

b. Iodide is oxidised

c. Sulphur is reduced

d. H_2O is one of the products.

13. The rate of a reaction doubles when its temperature changes from 300 K to 310 K. Activation of such a reaction will be ($R = 8.314\ JK^{-1}$ and $\log 2 = 0.301$)

a. 53.6 kJ mol^{-1} **b.** 48.6 kJ mol^{-1}

c. 58.6 kJ mol^{-1} **d.** 60.5 kJ mol^{-1}

14. The freezing point (in ºC) of a solution containing 0.1 g of $K_3[Fe(CN)_6]$(Mol. Wt. 329) in 100 g of water (K_f = 1.86K kg mol^{-1}) is

a. -2.3×10^{-2} **b.** -5.7×10^{-2}

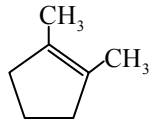
c. -5.7×10^{-3} **d.** -1.2×10^{-2}

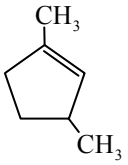
15. The carboxyl functional group (—COOH) is present in

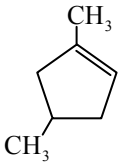
a. picric acid **b.** barbituric acid

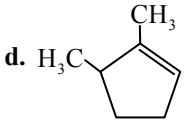
c. ascorbic acid **d.** aspirin

16. Which compound would give 5-keto-2-methyl hexanal upon ozonolysis?

a. 

b. 

c. 

d. 

17. For the process $H_2O(\ell) \longrightarrow H_2O(g)$ at T=100ºC and 1 atmosphere pressure, the correct choice is

a. $\Delta S_{system} > 0$ and $\Delta S_{surrounding} > 0$

b. $\Delta S_{system} > 0$ and $\Delta S_{surrounding} < 0$

c. $\Delta S_{system} < 0$ and $\Delta S_{surrounding} > 0$

d. $\Delta S_{system} < 0$ and $\Delta S_{surrounding} < 0$

18. Which of the following exists as covalent crystals in the solid state?

a. Iodine **b.** Silicon

c. Sulphur **d.** Phosphorus

SECTION 3 Contains 2 Match The Following Type Questions

You will have to match entries in Column I with the entries in Column II.

19. According to Bohr's theory,

E_n = Total energy K_n = Kinetic energy

V_n = Potential energy r_n = Radius of nth orbit

Match the following:

Column I	Column II
(A) V_n/K_n = ?	**1.** 0
(B) If radius of nth orbit $\propto E_n^x$, x = ?	**2.** –1
(C) Angular momentum in lowest orbital	**3.** –2
(D) $\frac{1}{r^n} \propto Z^y$, y = ?	**4.** 1

a. A → 3 ; B → 2 ; C → 1 ; D → 4

b. A → 1 ; B → 2 ; C → 3 ; D → 4

c. A → 4 ; B → 2 ; C → 1 ; D → 3

d. A → 3 ; B → 4 ; C → 1 ; D → 2

20. An aqueous solution of X is added slowly to an aqueous solution of Y as shown in Column - I. The variation in conductivity of these reactions in Column - II.

Column I	Column II
(A) $\underset{X}{(C_2H_5)_3N} + \underset{Y}{CH_3COOH}$	**1.** Conductivity decreases and then increases
(B) $\underset{X}{KI(0.1M)} + \underset{Y}{AgNO_3(0.01\,M)}$	**2.** Conductivity decreases and then does not change much
(C) $\underset{X}{CH_3COOH} + \underset{Y}{KOH}$	**3.** Conductivity increases and then does not change much
(D) $\underset{X}{NaOH} + \underset{Y}{HI}$	**4.** Conductivity does not change much and then increases

a. A → 1 ; B → 2 ; C → 3 ; D → 4

b. A → 2 ; B → 4 ; C → 3 ; D → 1

c. A → 3 ; B → 4 ; C → 2 ; D → 1

d. A → 3 ; B → 4 ; D → 1 ; D → 2

Space for rough work

JEE ADVANCE PAPER-II

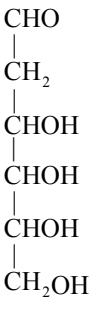
SECTION 1 Contains 8 Questions

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. Amongst the following, the total number of compounds whose aqueous solution turns red litmus paper blue is

KCN	K_2SO_4	$(NH_4)_2C_2O_4$
NaCl	$Zn(NO_3)_2$	$FeCl_3$
K_2CO_3	NH_4NO_3	LiCN

2. When the following aldohexose exists in its D–configuration, the total number of stereoisomers in its pyranose form is



3. The number of resonance structures for N is

4. The concentration of R in the reaction R→P was measured as a function of time and the following data is obtained:

[R] (molar)	1.0	0.75	0.40	0.10
t (min).	0.0	0.05	0.12	0.18

The order of the reaction is

5. The oxidation number of Mn in the product of alkaline oxidative fusion of MnO_2 is

6. The total number of diprotic acids among the following is

H_3PO_4	H_2SO_4	H_3PO_3	H_2CO_3	$H_2S_2O_7$
H_3BO_3	H_3PO_2	H_2CrO_4	H_2SO_3	

7. The number of neutrons emitted when $^{235}_{92}U$ undergoes controlled nuclear fission to $^{142}_{54}Xe$ and $^{90}_{38}Sr$ is

8. In dilute aqueous H_2SO_4, the complex diaquodioxalatoferrate (II) is oxidised by MnO_4^-. For this reaction, the ratio of the rate of change of $[H^+]$ to the rate of change of $[MnO_4^-]$ is

SECTION 2 Contains 8 Multiple Choice Questions

With one or more than one correct option

9. The correct statement(s) for cubic close packed (cep) three dimensional structure is (are)

a. The number of the nearest neighbours of an atom present in the topmost layer is 12

b. The efficiency of atom packing is 74%

c. The number of octahedral and tetrahedral voids per atom are 1 and 2, respectively

d. The unit cell edge length is $2\sqrt{2}$ times the radius of the atom

10. The standard Gibbs energy change at 300 K for the reaction $2A \quad B+C$ is 2494.2 J. At a given time, the composition of the reaction mixture is $[A]=\frac{1}{2}, [B]=2$ and $[C]=\frac{1}{2}$. The reaction proceeds in the:

[$R = 8.314\ J/K/mol, e = 2718$]

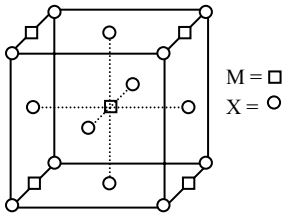
a. forward direction because $Q > K_c$

b. reverse direction because $Q > K_c$

c. forward direction because $Q < K_c$

d. reverse direction because $Q < K_c$

11. A compound M_pX_q has cubic close packing (ccp) arrangement of X. Its unit cell structure is shown below. The empirical formula of the compound is



a. MX **b.** MX_2

c. M_2X **d.** M_5X_{14}

12. Which of the following is the energy of a possible excited state of hydrogen?

a. +13.6 eV **b.** −6.8 eV

c. −3.4 eV **d.** +6.8 eV

13. Assuming 2s-2p mixing is not operative, the paramagnetic species among the following is

a. Be_2 **b.** B_2

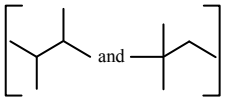
c. C_2 **d.** N_2+

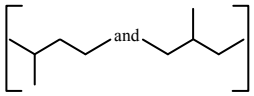
14. In allene(C_3H_4), the type(s) of hybridization of the carbon atoms is (are)

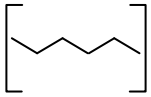
a. sp and sp^3 **b.** sp and sp^2

c. only sp^2 **d.** sp^2 and sp^3

15. Isomers of hexane, based on their branching, can be divided into three distinct classes as shown in the figure

I. 

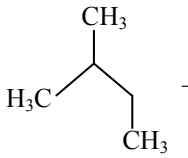
II. 

II. 

The correct order of their boiling point is

a. I > II > III **b.** III > II > I

c. II > III > I **d.** III > I > II

16.  $\xrightarrow{Cl_2, h\nu}$ N (isomeric products)

$C_5H_{11}Cl \xrightarrow{\text{fractional distillation}}$ M(isomeric products)

What are N and M ?

a. 6, 6 **b.** 6, 4

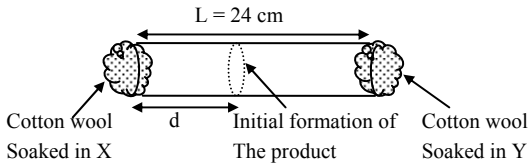
c. 4, 4 **d.** 3, 3

SECTION 3 Contains 2 Paragraph Type Questions

Each paragraph describes an experiment, a situation or a problem. Two multiple choice questions will be asked based on this paragraph. One or more than one option can be correct.

Paragraph for Question Nos. 17 to 18

X and Y are two volatile liquids with molar weights of 10g mol^{-1} and 40g mol^{-1} respectively. Two cotton plugs, one soaked in X and the other soaked in Y, are simultaneously placed at the ends of a tube of length L = 24 cm, as shown in the figure. The tube is filled with an inert gas at 1 atmosphere pressure and a temperature of 300 K. Vapours of X and Y react to form a product which is first observed at a distance d cm from the plug soaked in X. Take X and Y to have equal molecular diameters and assume ideal behaviour for the inert gas and the two vapours.



17. The value of d in cm (shown in the figure), as estimated from Graham's law, is

a. 8 **b.** 12 **c.** 16 **d.** 20

18. The experimental value of d is found to be smaller than the estimate obtained using Graham's law. This is due to

a. larger mean free path for X as compared to that of Y

b. larger mean free path for Y as compared to that of X

c. increased collision frequency of Y with the inert gas as compared to that of X with the inert gas

d. increased collision frequency of X with the inert gas as compared to that of Y with the inert gas

Paragraph for Question Nos. 19 to 20

When 100 mL of 1.0 M HCl was mixed with 100 mL of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increase of 5.7ºC was measured for the beaker and its contents (Expt. 1). Because the enthalpy of neutralization of a strong acid with a strong base is a constant ($-57.0\ kJ\ mol^{-1}$), this experiment could be used to measure the calorimeter constant. In a second experiment (Expt. 2), 100 mL of 2.0 M acetic acid ($K_a = 2.0 \times 10^{-5}$) was mixed with 100 mL of 1.0 M NaOH (under identical conditions to Expt. 1) where a temperature rise of 5.6°C was measured. (Consider heat capacity of all solutions as $4.2\ J\ g^{-1}\ K^{-1}$ and density of all solutions as $1.0\ g\ mL^{-1}$)

19. Enthalpy of dissociation (in $kJ\ mol^{-1}$) of acetic acid obtained from the Expt. 2 is

a. 1.0 **b.** 10.0 **c.** 24.5 **d.** 51.4

20. The pH of the solution after Expt. 2 is

a. 2.8 **b.** 4.7 **c.** 5.0 **d.** 7.0

Space for rough work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	a	c	d	c	c	c	c	c	b
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	c	c	d	b,c	d	c	a	c	a
21.	**22.**	**23.**	**24.**	**25.**	**26.**	**27.**	**28.**	**29.**	**30.**
b	d	b	a	b	a	c	c	c	b

1. **(d)** 2 g of hydrogen occupy volume $= 22.4$ L

1 g of hydrogen occupies volume $= \frac{22.4 \times 1}{2} = 11.2$ L

2. **(a)** ZnS has zinc blende type structure (i.e., ccp structure). The S^{2-} ions are present at the corners of the cube and at the centre of each face. Zinc ions occupy half of the tetrahedral sites. Each zinc ion is surrounded by four sulphide ions which are disposed towards the corner of regular tetrahedral. Similarly, S^{2-} ions surrounded by four Zn^{2+} ions.

3. **(c)** $(\Delta T_f)_{cal} = K_f \times m$

$(\Delta T_f)_{cal} = 1.86 \times 1 = 1.86$

$i = \frac{(\Delta T_f)_{obs}}{(\Delta T_f)_{cal}} = \frac{3.72}{1.86} = 2 = 1 + (n-1)\alpha.$

Hence, $\alpha = 1, \therefore n = 2$

$\therefore$ Two species will be produced from single species which is only possible for $[Pt(H_2O)_3Cl_3], Cl_3H_2O$

4. **(d)** $P_1 = 1$ atm, $V_1 = 12000$ L,

$T_1 = 27 + 273 = 300$ K

$P_2 = 0.5$ atm, $V_2 = ?$

$T_2 = -23 + 273 = 250$ K

$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

or $V_2 = \frac{P_1 V_1 T_2}{P_2 T_1} = \frac{1 \times 12000 \times 250}{0.5 \times 300} = 20,000$ L

5. **(c)** In hydrogen atom, the lowest orbit (n = 1) corresponds to minimum energy (– 13.6 eV).

6. **(c)** $5\begin{matrix}COOH\\|\\COOH\end{matrix} + 2\overset{+7}{K}\overset{+7}{Mn}O_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2\overset{+2}{Mn}SO_4 + 10CO_2 + 8H_2O$

In this reaction oxidation state of Mn change from +7 to +2.

7. **(c)** Molar conductivity $= \frac{1}{\rho M}$

So, its unit will be $\Omega^{-1}cm^2\,mol^{-1}$

8. **(c)** In a double bond connecting two atoms sharing of 4 electrons take place as in $H_2C = CH_2$.

9. **(c)** Decreases the activation energy of both forward and backward reaction.

10. **(b)** $K_h = \frac{K_W}{K_a \times K_b}$

$= \frac{1 \times 10^{-4}}{4.0 \times 10^{-4} \times 1.8 \times 10^{-5}} = 1.4$

11. **(c)** $A + B \rightarrow C$ On doubling the concentration of A rate of reaction increases by four times. Rate $\propto [A]^2$. However on doubling the concentration of B, rate of reaction increases two times. Rate $\propto [B]$

Thus, overall order of reaction = 2 + 1 = 3

12. **(c)** According to Hardy-Schulze rule.

13. **(c)** Heats of combustion are always exothermic except oxidation of N as,

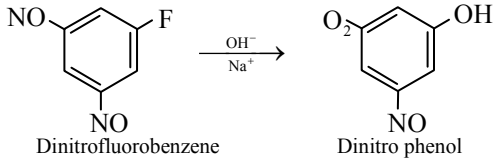
$N_2 + \frac{1}{2}O_2 \longrightarrow N_2O$; $\Delta H = +ve$

$N_2 + O_2 \longrightarrow 2NO$; $\Delta H = +ve$

14. **(d)** $_6X^{14} \xrightarrow{\beta} {}_{6+1}N^{14}$

in $_6X^{14}$ no. of neutrons 14 – 6 = 8.

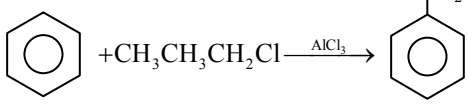
15. **(b, c)**



Dinitrofluorobenzene Dinitro phenol

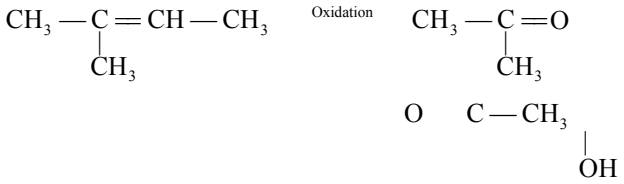
16. **(d)** $I - \overset{H}{\underset{Cl}{\overset{|}{\underset{|}{C^*}}}} - Br$

A carbon atom which is attached to four different atoms or groups is called a chiral or asymmetric carbon atom. such a carbon atom is often marked by an asterisk.

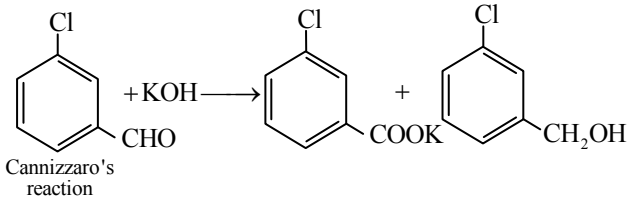
17. **(c)** 

18. **(a)** $CH_3CH_2CH_2CH_2 - Cl + KOH(alc.) \longrightarrow \underset{\text{1-butene}}{CH_3CH_2 - CH = CH_2} + KCl + H_2O$

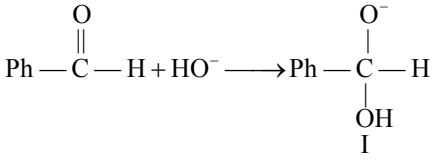
19. **(c)** $CH_3 - \underset{CH_3}{\overset{|}{CH}} - \underset{OH}{\overset{|}{CH}} - CH_3 \xrightarrow{\text{Dehydration}}$



20. (a)



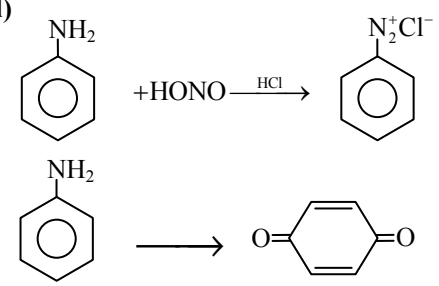
Cannizzaro's reaction



21. (b) $CH_3-\overset{O}{\overset{\|}{C}}-NH_2 + Br_2 + NaOH \longrightarrow CH_3NH_2$

"Hofmann's bromamide reaction"

22. (d)



23. (b) 'Rayon' is man-made fibre which consists of purified cellulose in the form of long threads. Rayon resembles silk in appearance. Hence called as artificial silk.

24. (a) Waxes are esters of higher fatty acids.

25. (b) The anaesthetics produce temporary insensitibility to the vital function of all type of cells, specially of nervous system and are used during surgical operations.

These are classified as

(a) General anasthetic – produces unconsciousness all over the body e.g. N_2O, Cyclopropane, chloroform

(b) Local anasthetic – affect only the part of body e.g. Xylocaine, Procain etc.

26. (a) $_{25}Mn - 3d^5 4s^2$.

27. (c) Hydrometallurgy

$Ag_2S + 4NaCN \longrightarrow 2Na[Ag(CN)_2] + Na_2S$

$2Na[Ag(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4] + 2Ag$

28. (c) $MgCl_2.6H_2O \xrightarrow{heat} MgO + 5H_2O + 2HCl$

29. (c) $Cr_2O_7^{2-} + 8H^+ + 2SO_3^{2-} \longrightarrow 2Cr^{+3} + 3SO_4^{2-} + 4H_2O$

30. (b) $Ag_2S + 4NaCN \qquad \underset{\text{Sodium dicyno argentate}}{2Na\left[Ag(CN)_2\right]} + Na_2S$

$2Na[Ag(CN)_2] + Zn \longrightarrow \underset{\text{Sodium tetracynozincate (ppt)}}{Na_2[Zn(CN)_4]} \quad 2Ag$

JEE Advance Paper-I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
3	8	4	2	7	8	5	6	a	a
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b	a, b, d	a	a	d	b	b	b	a	c

1. (3) $HX \qquad H^+ + X^- \qquad Ka = \dfrac{[H^+][X^-]}{[HX]}$

$HY \qquad H^+ + Y^- \qquad Ka = \dfrac{[H^+][Y^-]}{[HY]}$

$\Lambda_m \text{ for } HX = \Lambda_{m_1} \qquad \Lambda_m \text{ for } HY = \Lambda_{m_2}$

$\Lambda_{m_1} = \dfrac{1}{10}\Lambda_{m_2} \qquad Ka = C\alpha^2$

$Ka_1 = C_1 \times \left(\dfrac{\Lambda_{m_1}}{\Lambda^0_{m_1}}\right)^2 \qquad Ka_2 = C_2 \times \left(\dfrac{\Lambda_{m_2}}{\Lambda^0_{m_2}}\right)^2$

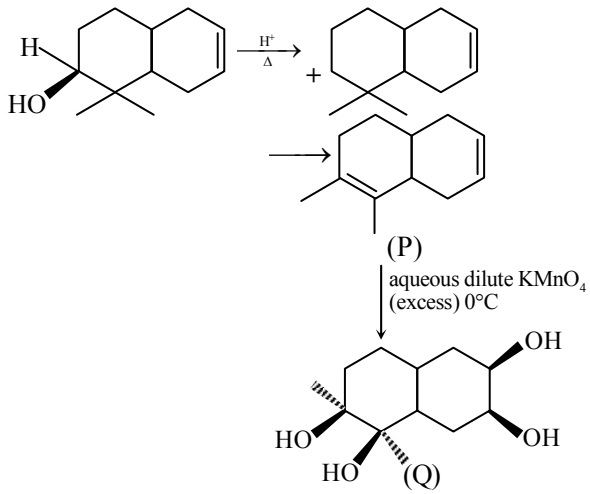
$\dfrac{Ka_1}{Ka_2} = \dfrac{C_1}{C_2} \times \left(\dfrac{\Lambda m_1}{\Lambda m_2}\right)^2$

$= \dfrac{0.01}{0.1} \times \left(\dfrac{1}{10}\right)^2 = 0.001 \Rightarrow pKa_1 - pKa_2 = 3$

2. (8) $_{92}U^{238} \xrightarrow{-6\alpha} {}_{80}X^{214} \xrightarrow{-2\beta} {}_{82}Pb^{214}$

$(6\alpha, 2\beta)$, total 8 particles.

3. (4)



4. **(2)** $-\overset{\overset{O}{\|}}{C}-O^-$ and $-NH_2$ are basic groups in lysine.

5. **(7)** Let the solubility of AgCl is x mol litre^{-1}

AgCl $\quad Ag^+_x + Cl^-_x$ and that of CuCl is y mol litre^{-1}

CuCl $\quad Cu^+_y + Cl^-_y$

$\therefore \quad K_{sp}$ of $AgCl = [Ag^+][Cl^{-1}]$

$1.6\times10^{-10} = x(x+y) \quad \ldots(i)$

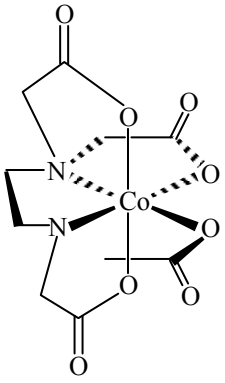
Similarly K_{sp} of $CuCl = [Cu^+][Cl^-]$

$1.6\times10^{-6} = y(x+y) \quad \ldots(ii)$

On solving (i) and (ii) $[Ag^+] = 1.6\times10^{-7}$

$\therefore \quad x = 7$

6. **(8)** Total no. of N – Co – O bond angles is 8.



7. **(5)** $3Br_2 + 3CO_3^{2-} \longrightarrow 5Br^- + BrO_3^- + 3CO_2$

8. **(6)** Let number of glycine units $= n$

Mass of decaeptide = 796

Mass of H_2O needed = 162 g, Total mass = 958 g

$$958\times\frac{47}{100} = 75\times n$$

$$\therefore \quad n = \frac{958\times47}{100\times75} \approx 6$$

9. **(a)** In ccp lattice:

Number of O atoms $\longrightarrow 4$

Number of Octahedral voids $\longrightarrow 4$

Number of tetrahedral voids $\longrightarrow 8$

Number of $Al^{3+} = 4\times m$

Number of $Mg^{2+} = 8\times n$

Due to charge neutrality $4(-2) + 4m(+3) + 8n(+2) = 0$

$$\therefore \quad m = \frac{1}{2} \text{ and } n = \frac{1}{8}$$

10. **(a)**

11. **(b)**

12. **(a, b, d)** $ClO_3^- + 6I^- + 6H_2SO_4 \longrightarrow 3I_2 + Cl^- + 6HSO_4^- + 3H_2O$

13. **(a)** $$0.3010 = \frac{Ea}{2.303R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

$$\Rightarrow \quad 0.3010 = \frac{Ea}{2.303\times8.314\times10^{-3}}\left(\frac{310-300}{310\times300}\right)$$

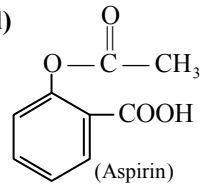
$E_a = 53.6$ kJ mol

14. **(a)** $K_3[Fe(CN)_6] \longrightarrow 3K^+ + [Fe(CN)_6]^{3-}$, $i = 4$

$$\Delta T_f = K_f \times i \times \frac{m}{M}\times\frac{1000}{W} = 1.86\times4\times\frac{0.1}{329}\times\frac{1000}{100}$$

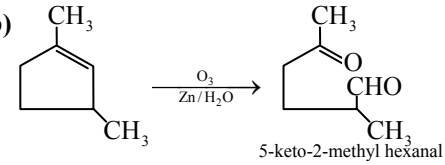
$= 2.3\times10^{-2} \Rightarrow T_f' = -2.3\times10^{-2}$

15. **(d)**



(Aspirin)

16. **(b)**



5-keto-2-methyl hexanal

17. **(b)** At 100°C and 1 atmosphere pressure $H_2O(\ ) \quad H_2O(g)$ is at equilibrium. For equilibrium $\Delta S_{total} = 0$ and $\Delta S_{system} + \Delta S_{surrounding} = 0$

$\therefore \quad \Delta S_{system} > 0$ and $\Delta S_{surrounding} < 0$

18. **(b)** Silicon exists as covalent crystal in the solid state.

19. **(a)** $A \to 3;\ B \to 2;\ C \to 1;\ D \to 4$

20. **(c)** $A \to 3;\ B \to 4;\ C \to 2;\ D \to 1$

(A) $\underset{X}{(C_2H_5)_3} + N + \underset{Y}{CH_3COOH} \longrightarrow (C_2H_5)_3NH^+CH_3COO^-$

Initially conductivity increases due to ion formation after that it becomes practically constant because X alone cannot form ions. Hence (3) is the correct match.

(B) $\underset{X}{KI(0.1\ M)} + \underset{Y}{AgNO_3(0.01\ M)} \longrightarrow AgI\downarrow + KNO_3$

Number of ions in the solution remains constant until all the $AgNO_3$ precipitated as AgI. Thereafter conductance increases due to increase in number of ions. Hence (4) is the correct match.

(C) Initially conductance decreases due to the decrease in the number of $O\bar{H}$ ions thereafter it slowly increases due to the increase in number of H^+ ions. Hence (2) is the correct match.

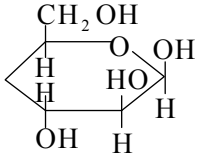
(D) Initially it decreases due to decrease in H^+ ions and then increases due to the increase in $O\bar{H}$ ions. Hence (1) is the correct match.

JEE Advance Paper -II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
3	8	9	5	6	6	3	9	b,c,d	b
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b	c	c	b	b	b	c	d	a	b

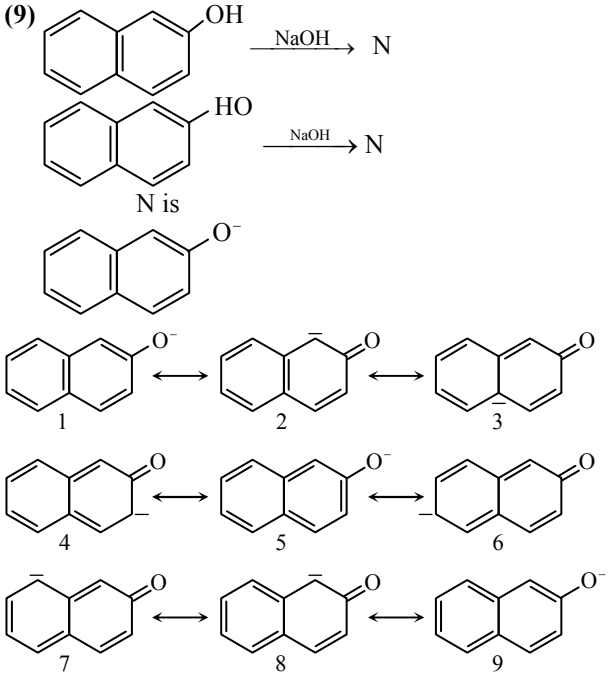
1. **(3)** KCN, K_2CO_3,-LiCN are basic in nature and their aqueous solution turns red litmus paper blue.

2. **(8)**



Total No. of stereoisomers = $2^4 = 16$ which contains 8D – Configuration and 8 – L Configuration.

3. **(9)**



4. **(5)** From two data, (for zero order kinetics)

$$K_I = \frac{x}{t} = \frac{0.25}{0.05} = 5$$

$$\Rightarrow \quad K_{II} = \frac{x}{t} = \frac{0.60}{0.12} = 5$$

5. **(6)** $2MnO_2 + 4KOH + O_2 \rightarrow \underset{\text{(Potassium manganate)}}{2K_2MnO_4} + 2H_2O$

O.S. of Mn = +6 in K_2MnO_4

6. **(6)** $H_2SO_4, H_2CO_3, H_2S_2O_7, H_2CrO_4, H_3PO_3, H_2SO_3$

7. **(3)** $_{92}U^{235} \rightarrow {}_{54}Xe^{142} + {}_{38}Sr^{90} + 3_0n^1$

8. **(9)** $K = \frac{2.303}{t}\log\left(\frac{a_0}{a_0 - x}\right)$, $K = \frac{2.303}{t_{1/8}}\log\left(\frac{a_0}{\frac{1}{8}a_0}\right)$

9. **(b, c, d)** (a) For any atom in topmost layer, coordination number is not 12 since there is no layer above topmost layer, (b) Fact, (c) Fact, (D) $\sqrt{2}\,a = 4R$

So, $a = 2\sqrt{2}\,R$

10. **(b)** $\Delta G = -RT\ln_e K_c$

$2494.2 = 8.314 \times 300 \ln_e K_c$

$\Rightarrow \quad K_c = e^{-1}$

$K_c = e^{-1} = \frac{1}{2.718} = 0.36$

$Q = \frac{(B)(C)}{[A]^2} = \frac{2 \times \frac{1}{2}}{[1/2]^2} = 4$

$Q > K_c$, i.e. backward reaction.

11. **(b)** $X : 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$

$M = \frac{1}{4} \times 4 + 1 = 2; \quad M_2X_4 = MX_2$

12. **(c)** Energy in 1st excited state $= -3.4$ eV

13. **(c)** Assuming that no 2s-2p mixing takes place

(a) $Be_2 \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2$ (diamagnetic)

(b) $B_2 \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \begin{matrix}\pi 2p_x^0 \\ \pi 2p_y^0\end{matrix}$

(diamagnetic)

(c) $C_2 \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \begin{matrix}\pi 2p_x^1 \\ \pi 2p_y^1\end{matrix}$,

$\begin{matrix}\pi 2p_x^1 \\ \pi 2p_y^1\end{matrix}, \begin{matrix}\pi^* 2p_x^0 \\ \pi^* 2p_y^0\end{matrix}, \sigma^* 2p_z^0$(paramagnetic)

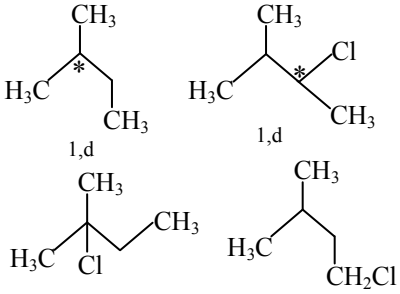
(d) $N_2 \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2,$

$\begin{matrix}\pi 2p_x^2 \\ \pi 2p_y^2\end{matrix}, \begin{matrix}\pi^* 2p_x^0 \\ \pi^* 2p_y^0\end{matrix}, \sigma^* 2p_z^0$(diamagnetic)

14. **(b)** $\overset{H}{\underset{H}{\diagdown}}\!\!>\!C(sp^2) = C(sp) = C(sp^2)\!<\!\overset{H}{\underset{H}{\diagup}}$ (allene)

15. **(b)** III > II > I More the branching in an alkane, lesser will be the surface area, lesser will be the boiling point.

16. **(b)**



1,d

1,d

M→d, 1 cannot be separated by fractional distillation.

For Question Nos. 17 to 18

17. **(c)** $\frac{r_X}{r_Y} = \frac{d}{24-d} = \sqrt{\frac{40}{10}} = 2$

$d = 48 - 2d$

$3d = 48$

$\Rightarrow \quad d = 16$ cm

18. **(d)** As the collision frequency increases then molecular speed decreases than that expected.

For Question Nos. 19 to 20

19. **(a)** $HCl + NaOH \longrightarrow NaCl + H_2O$

$n = 100 \times 1 = 100$ m mole $= 0.1$ mole

Energy evolved due to neutralization of HCl and NaOH $= 0.1 \times 57 = 5.7$ kJ $= 5700$ Joule

Energy used to increase temperature of solution $= 200 \times 4.2 \times 5.7 = 4788$ Joule

Energy used to increase temperature of calorimeter $= 5700 - 4788 = 912$ Joule

ms. $\Delta t = 912$

m.s $\times 5.7 = 912$

$ms = 160$ Joule / °C [Calorimeter constant]

Energy evolved by neutralization of CH_3COOH and NaOH $= 200 \times 4.2 \times 5.6 + 160 \times 5.6 = 5600$ Joule

So energy used in dissociation of 0.1 mole $CH_3COOH = 5700 - 5600 = 100$ Joule

Enthalpy of dissociation $= 1$ kJ / mole

20. **(b)** $CH_3COOH = \frac{1 \times 100}{200} = \frac{1}{2}$

$CH_3CONa = \frac{1 \times 100}{200} = \frac{1}{2}$

$pH = pK_a + \log \frac{[salt]}{[acid]}$

$pH = 5 - \log 2 + \log \frac{1/2}{1/2} \Rightarrow pH = 4.7$

□ □ □

JEE-MAIN: CHEMISTRY MOCK TEST-4

1. The number of moles of SO_2Cl_2 in 13.5 g is
a. 0.1 **b.** 0.2
c. 0.3 **d.** 0.4

2. Arrangement of sulphide ions in zinc blende is
a. simple cubic **b.** hcp
c. bcc **d.** fcc

3. A pressure cooker reduces cooking is increased
a. Heat is more evenly distributed
b. Boiling point of water inside the cooker is increased
c. The high pressure tenderizes the food
d. All of the above

4. Two flasks of equal volume contains SO_2 and CO_2 respectively at $25°C$ and 2 atm pressure. Which of the following is equal in them ?
a. masses of the two gas **b.** number of molecules
c. rates of effusion **d.** molecular structure

5. The ratio of the kinetic energy to the total energy of an electron in a Bohr orbit is
a. –1 **b.** 2
c. 1 : 2 **d.** None of these

6. Oxygen has oxidation states of +2 in the
a. H_2O_2 **b.** CO_2
c. H_2O **d.** OF_2

7. Given $l/a = 0.5\ cm^{-1}$, R = 50 ohm, N = 1.0. The equivalent conductance of the electrolytic cell is
a. 10 $ohm^{-1}\ cm^{-2}\ gm\ eq^{-1}$ **b.** 20 $ohm^{-1}\ cm^{2}\ gm\ eq^{-1}$
c. 300 $ohm^{-1}\ cm^{2}\ gm\ eq^{-1}$ **d.** 100 $ohm^{-1}\ cm^{2}\ gm\ eq^{-1}$

8. In N_2 molecule, the atoms are bonded by
a. One σ, Two π **b.** One σ, One π
c. Two σ, One π **d.** Three σ bonds

9. For the reaction $H_2(g) + I_2(g)$ 2HI(g), the equilibrium constant changes with
a. Total pressure
b. Catalyst
c. The amounts of H_2 and I_2 taken
d. Temperature

10. A certain weak acid has a dissociation constant of 1.0×10^{-4}. The equilibrium constant for its reaction with a strong base is.
a. 1.0×10^{-4} **b.** 1.0×10^{-10}
c. 1.0×10^{10} **d.** 1.0×10^{14}

11. According to Arrhenius theory, the activation energy is
a. The energy it should possess so that it can enter into an effective collision
b. The energy which the molecule should possess in order to undergo reaction
c. The energy it has to acquire further so that it can enter into a effective collison
d. The energy gained by the molecules on colliding with another molecule

12. Which of the following is property of colloid?
a. Scattering of light **b.** They show attraction
c. Dialysis **d.** Emulsion

13. Which of the following is an example of endothermic reaction?
a. $C_2H_2 + 2H_2 \longrightarrow C_2H_6;\ \Delta E = -314.0\,kJ$
b. $C + O_2 \longrightarrow CO_2;\ \Delta E = -393.5\,kJ$
c. $N_2 + O_2 \longrightarrow 2NO;\ \Delta E - 180.5\,kJ$
d. $2H_2 + O_2 \longrightarrow 2H_2O;\ \Delta E + 571.8\,kJ$

14. The nuclear binding energy for *Ar* (39.962384 amu) is: (given mass of proton and neutron are 1.007825 amu and 1.008665 amu respectively)
a. 343.81 MeV **b.** 0.369096 MeV
c. 931 MeV **d.** None of these

15. Which is an electrophile?
a. $AlCl_3$ **b.** CN^- **c.** NH_3 **d.** CH_3OH

16. Cyanide and isocyanide are isomers of type
a. Positional **b.** Functional
c. Tautomer **d.** Structural

17. Acetone will be formed by the ozonolysis of
a. Butene-1 **b.** Butene-2
c. Isobutene **d.** Butyne-2

18. Which of the following reactions gives $H_2C=C=C=CH_2$?
a. $CH_2Br - CBr = CH_2 \xrightarrow{Zn/CH_3OH}$
b. $HC \equiv C - CH_2 - COOH \xrightarrow[40^oC]{Aq.K_2CO_3}$
c. $CH_2Br - C \equiv C - CH_2Br \xrightarrow[Heat]{Zn}$
d. $2CH_2 = CH - CH_2I \longrightarrow$

19. Which of the following does not form phenol or phenoxide?

a. C_6H_5Cl **b.** C_6H_5COOH

c. $C_6H_5N_2Cl$ **d.** $C_6H_5SO_3Na$

20. Identify the final product (Z) in the following sequence of reactions:

$$Me_2C=O+HCN \longrightarrow X \xrightarrow{H_3O^+} Y \xrightarrow[Heat]{H_2SO_4} Z;$$

a. $(CH_3)_2C(OH)COOH$ **b.** $CH_2=C(CH_3)COOH$

c. $HOCH_2CH(CH_3)COOH$ **d.** $CH_3CH=CHCOOH$

21. When propionic acid is treated with aqueous sodium bicarbonate, CO_2 is liberated. The C of CO_2 comes from?

a. methyl group **b.** carboxylic acid group

c. methylene group **d.** bicarbonate group

22. Aniline undergoes condensation to form Schiff base on reacting with

a. Acetyl chloride **b.** Ammonia

c. Acetone **d.** Benzaldehyde

23. The mass average molecular mass and number average molecular mass of a polymer are respectively 40,000 and 30,000. The polydispersity index of polymer will be

a. < 1 **b.** > 1 **c.** 1 **d.** 0

24. Hardening of oils is caused by

a. H_2 **b.** N_2 **c.** O_2 **d.** CO_2

25. Which of the following is molecular disease?

a. Allergy **b.** Cancer

c. German measeles **d.** Sickel-cell-anaemia

26. Hydrogen can be put in halogen group because

a. It has deuterium and tritium as isotopes

b. It forms hydrides like chlorides

c. It contains one electron only

d. It is light

27. Which of the following metal is extracted by amalgamation process?

a. Tin **b.** Silver

c. Copper **d.** Zinc

28. Which of the following hydroxide is insoluble in water?

a. $Be(OH)_2$ **b.** $Mg(OH)_2$

c. $Ca(OH)_2$ **d.** $Ba(OH)_2$

29. The product of oxidation of I^- ion by MnO_4^- in alkaline medium is

a. I_2 **b.** IO_3^-

c. IO_4^- **d.** I_3^-

30. Potassium ferrocyanide is a

a. Normal salt **b.** Mixed salt

c. Double salt **d.** Complex salt

Space for rough work

JEE ADVANCE PAPER-I

SECTION 1 Contains 8 Questions

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. HCl gas is passed into water, yielding a solution of density $1.095\,g\,mL^{-1}$ and containing 30% HCl by weight. Calculate the molarity of the solution.

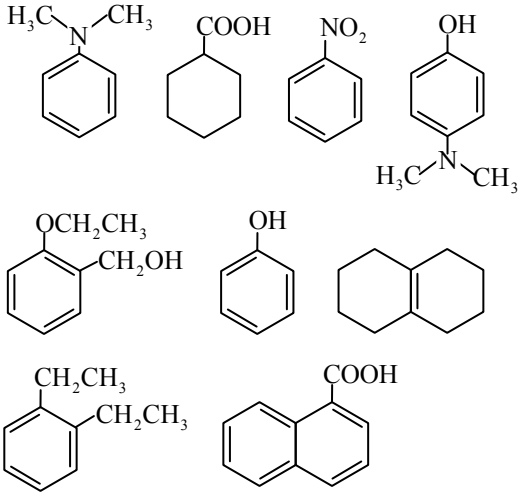
2. A sample contains a mixture of $NaHCO_3$ and Na_2CO_3. HCl is added to 15.0 g of the sample, yielding 11.0 g of NaCl. What percent of the sample is Na_2CO_3 ?

Reactions are:

$$\begin{bmatrix} Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O \\ NaHCO_3 + HCl \longrightarrow NaCl + CO_2 + H_2O \end{bmatrix}$$

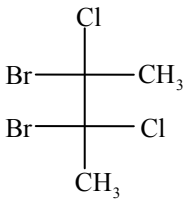
Mw of $NaCl = 58.5$, Mw of $NaHCO_3 = 84$, Mw of $Na_2CO_3 = 106\,g\,mol^{-1}$

3. Amongst the following, the total number of compounds soluble in aqueous NaOH is



4. The total number of contributing structures showing hyperconjugation (involving C—H bonds) for the following carbocation is.

5. The total number(s) of stable conformers with non-zero dipole moment for the following compound is (are)



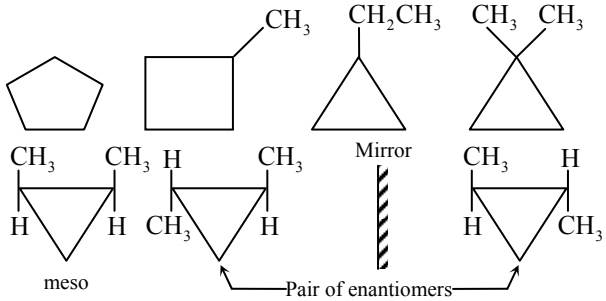
6. $CH_3-CH_2-CH_2-CH_2OH$

$CH_3-CH_2-\underset{\underset{\text{enantiomeric }(\pm)}{|}}{\underset{OH}{CH}}-CH_3$

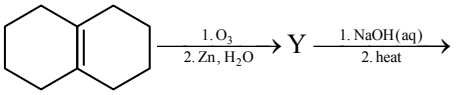
$CH_3-\overset{\overset{CH_3}{|}}{\underset{OH}{C}}-CH_3$; $CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_2OH$

Total number of isomers (including stereoisomers) is

7. Total number of isomers



8. In the scheme given below, the total number of intramolecular aldol condensation products formed from 'Y' is



SECTION 2 Contains 10 Multiple Choice Questions

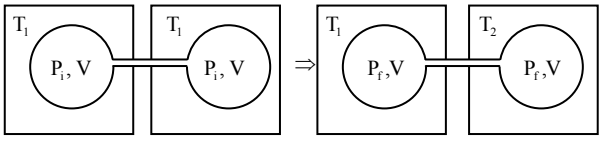
With one or more than one correct option

9. For a dilute solution containing 2.5 g of a non-volatile non-electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is Assuming concentration of solute is 2°C. Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (mm of Hg) of the solution is (take $K_b = 0.76\,K\,kg\,mol^{-1}$)

a. 724 **b.** 740

c. 736 **d.** 718

10. Two closed bulbs of equal volume (V) containing an ideal gas initially at pressure p_i and temperature T_1 are connected through a narrow tube of negligible volume as shown in the figure below. The temperature of one of the bulbs is then I raised to T_2. The final pressure p_f is:



a. $p_i\left(\frac{T_1T_2}{T_1+T_2}\right)$ **b.** $2p_i\left(\frac{T_1}{T_1+T_2}\right)$

c. $2p_i\left(\frac{T_2}{T_1+T_2}\right)$ **d.** $2p_i\left(\frac{T_1T_2}{T_1+T_2}\right)$

11. A stream of electrons from a heated filament was passed between two charged plates kept at a potential difference V esu. If e and m are charge and mass of an electron, respectively, then the value of h/λ (where λ is wavelength associated with electron wave) is given by:

a. meV **b.** 2 meV

c. $\sqrt{meV}$ **d.** $\sqrt{2\ meV}$

12. Reduction of the metal centre in aqueous permanganate ion involves

a. 3 electrons in neutral medium

b. 5 electrons in neutral medium

c. 3 electrons in alkaline medium

d. 5 electrons in acidic medium

13. For the following electrochemical cell at 298 K,

$Pt(s)\,|\,H_2(g, 1\ bar)\,H^+(aq, 1\ M)\,||\,M^{4+}(aq), M^{2+}(aq)\,|\,Pt(s)$

$E_{cell} = 0.092\ V$ when $\frac{[M^{2+}(aq)]}{[M^{4+}(aq)]} = 10^x$.

Given: $E^0_{M^{4+}/M^{2+}} = 0.151\ V; 2.303\frac{RT}{F} = 0.059\ V$

The value of *x* is

a. –2 **b.** –1

c. 1 **d.** 2

14. Which one of the following molecules is expected diamagnetic behaviour?

a. C_2 **b.** N_2

c. O_2 **d.** S_2

15. Aqueous solutions of HNO_3, KOH, CH_3COOH and CH_3COONa of identical concentrations are provided. The pair (s) of solutions which form a buffer upon mixing is (are)

a. HNO_3 and CH_3COOH

b. KOH and CH_3COONa

c. HNO_3 and CH_3COONa

d. CH_3COOH and CH_3COONa

16. The value of $\log_{10}K$ for a reaction A B is

(Given $\Delta_rH^o_{298K} = -54.07\ kJ\ mol^{-1}$, $\Delta_rS^o_{298K} = 10\ JK^{-1}mol^{-1}$ and $R = 8.314\ JK^{-1}mol^{-1}$; $2.303 \times 8.314 \times 298 = 5705$)

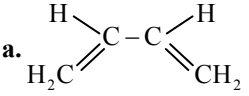
a. 5 **b.** 10 **c.** 95 **d.** 100

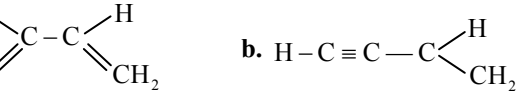
17. The number of structural isomers for C_6H_{14} is

a. 3 **b.** 4

c. 5 **d.** 6

18. Amongst the given options, the compound(s) in which all the atoms are in one plane in all the possible conformations (if any), is (are)

a. 

b. 

c. $H_2C = C = O$ **d.** $H_2C = C = CH_2$

SECTION 3 Contains 2 Matches The Following Type Questions

You will have to match entries in Column I with the entries in Column II.

19. According to Bohr's theory,

E_n = Total energy K_n = Kinetic energy

V_n = Potential energy r_n = Radius of nth orbit

Match the following:

Column I	Column II
(A) $V_n / K_n = ?$	**1.** 0
(B) If radius on nth orbit $\propto E_n^x$, x = ?	**2.** –1
(C) Angular momentum in lowest orbital	**3.** –2
(D) $\frac{1}{r^n} \propto Z^y$, y = ?	**4.** 1

a. A → 3; B → 2; C → 4; D → 1

b. A → 1; B → 2; C → 3; D → 4

c. A → 3; B → 1; C → 2; D → 4

d. A → 3; B → 2; C → 1; D → 4

20. Match the thermodynamic processes given under Column I with the expression given under Column II:

Column I	Column II
(A) Freezing of water at 273 K and 1 atm	**1.** q = 0
(B) Expansion of 1 mol of an ideal gas into a vacuum under isolated conditions	**2.** w = 0
(C) Mixing of equal volumes of two ideal gases at constant	**3.** $\Delta S_{sys} < 0$

temperature and pressure in an isolated container	
(D) Reversible heating of $H_2(g)$ at 1 atm from 300 K to 600 K, followed by reversible cooling to 300 K at 1 atm	**4.** $\Delta U = 0$
	5. $\Delta G = 0$

a. A $\rightarrow$ 3,5; B $\rightarrow$ 1,2,4 ; C $\rightarrow$ 1,2,4; D $\rightarrow$ 1,2,4,5

b. A $\rightarrow$ 1,2,4; B $\rightarrow$ 1,2,4,5; C $\rightarrow$ 3,4; D $\rightarrow$ 1,2,4

c. A $\rightarrow$ 3,5; B $\rightarrow$ 1,2,4; C $\rightarrow$ 1,2,4,5; D $\rightarrow$ 1,2,4

d. A $\rightarrow$ 1,2,4; B $\rightarrow$ 3,5; C $\rightarrow$ 1, 4,5; D $\rightarrow$ 1,2,4

Space for rough work

JEE ADVANCE PAPER-II

SECTION 1 Contains 8 Questions

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. 0.45 g of an acid (mol wt. = 90) required 20 ml of 0.5 N KOH for complete neutralization. Basicity of acid is

2. The co-ordination number of copper in cuprammonium sulphate is

3. The co-ordination number of cobalt in the complex $[Co(en)_2Br_2]Cl_2$ is

4. The primary valence of the metal ion in the co-ordination compoun $K_2[Ni(CN)_4]$ is

5. The oxidation number of Cr in $[Cr(NH_3)_6]Cl_3$ is

6. The number of equivalent $Cr-O$ bonds in CrO_4^{2-} is.

7. The number of the following reagents that produce ppt. with $ZnSO_4$ solution is.

NaOH, N_2CO_3, NaCl, Na_2HPO_4, Na_2S, CH_3CO_3Na

8. The change in the magnetic moment value when $\left[Cu(H_2O)_4\right]^{2+}$ is converted to $\left[Cu(NH_3)_4\right]^{2+}$ is.

SECTION 2 Contains 8 Multiple Choice Questions

With one or more than one correct option

9. A gas described by van der Waal's equation

a. behaves similar to an ideal gas in the limit of large molar volumes

b. behaves similar to an ideal gas in the limit of large pressures

c. is characterized by van der Waal's coefficients that are dependent on the identity of the gas but are independent of the temperature

d. has the pressure that is lower than the pressure exerted by the same gas behaving ideally

10. Assuming that Hund's rule is violated, the bond order and magnetic nature of the diatomic molecule B_2 is

a. 1 and diamagnetic **b.** 0 and diamagnetic

c. 1 and paramagnetic **d.** 0 and paramagnetic

11. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius].

a. $\frac{h^2}{4\pi^2 ma_0^2}$ **b.** $\frac{h^2}{16\pi^2 ma_0^2}$

c. $\frac{h^2}{32\pi^2 ma_0^2}$ **d.** $\frac{h^2}{64\pi^2 ma_0^2}$

12. The equilibrium $2Cu^{I} \quad Cu^{\circ} + Cu^{II}$ in aqueous medium at 25ºC shifts towards the left in the presence of

a. NO_3^- **b.** Cl^- **c.** SCN^- **d.** CN^-

13. The bond energy (in kcal mol^{-1}) of C – C single bond is approximately

a. 1 **b.** 10 **c.** 100 **d.** 1000

14. $N_2 + 3H_2 \quad 2NH_3$ Which is correct statement if N_2 is added at equilibrium condition?

a. The equilibrium will shift to forward direction because according to II[nd] law of thermodynamics the entropy must increases in the direction of spontaneous reaction

b. The condition for equilibrium is $G_{N_2} + 3G_{H_2} = 2G_{NH_3}$ where G is Gibbs free energy per mole of the gaseous species measured at that partial pressure. The condition of equilibrium is unaffected by the use of catalyst, which increases the rate of both the forward and backward reactions to the same extent

c. The catalyst will increase the rate of forward reaction by α and that of backward reaction by β.

d. Catalyst will not alter the rate of either of the reaction

15. For a first order reaction A→P, the temperature (T) dependent rate constant (k) was found to follow the equation $\log k = -(2000)\frac{1}{T} + 6.0$. The pre-exponential factor A and the activation energy Ea, respectively, are

a. $1.0\times10^6 s^{-1}$ and $9.2\,kJmol^{-1}$

b. $6.0 s^{-1}$ and $16.6\,kJmol^{-1}$

c. $1.0\times10^6 s^{-1}$ and $16.6\,kJmol^{-1}$

d. $1.0\times10^6 s^{-1}$ and $38.3\,kJmol^{-1}$

16. The initial rate of hydrolysis of methyl acetate (1 M) by a weak acid (HA, 1M) is 1/100[th] of that of a strong acid (HX, 1M), at 25ºC. The K_a of HA is

a. 1×10^{-4} **b.** 1×10^{-5}

c. 1×10^{-6} **d.** 1×10^{-3}

SECTION 3 Contains 2 Paragraph Type Questions

Each paragraph describes an experiment, a situation or a problem. Two multiple choice questions will be asked based on this paragraph. One or more than one option can be correct.

Paragraph for Question No. 17 to 18

Thermal decomposition of gaseous X_2 to gaseous X at 298 K takes place according to the following equation:

$X_2(g) \rightleftharpoons 2X(g)$

The standard reaction Gibbs energy, $\Delta_r G°$ of this reaction is positive. At the start of the reaction, there is one mole of X_2 and no X. As the reaction proceeds, the number of moles of X formed is given by β. Thus $\beta_{equilibrium}$ is the number of moles of X formed at equilibrium. The reaction is carried out at a constant total pressure of 2 bar. Consider the gases to behave ideally. (Given: R = 0.083 L bar K^{-1} mol^{-1})

17. The equilibrium constant K_p for this reaction at 298 K, in terms of $\beta_{equilibrium}$, is

a. $\frac{8\beta^2_{equilibrium}}{2-\beta_{equilibrium}}$ **b.** $\frac{8\beta^2_{equilibrium}}{4-\beta^2_{equilibrium}}$

c. $\frac{4\beta^2_{equilibrium}}{2-\beta_{equilibrium}}$ **d.** $\frac{4\beta^2_{equilibrium}}{4-\beta^2_{equilibrium}}$

18. The incorrect statement among the following, for this reaction, is

a. Decrease in the total pressure will result in formation of more moles of gaseous X

b. At the start of the reaction, dissociation of gaseous X_2 takes place spontaneously

c. $\beta_{equlibrium} = 0.7$

d. $K_C < 1$

Paragraph for Question No. 19 to 20

Tollen's reagent is used for the detection of aldehyde when a solution of $AgNO_3$ is added to glucose with NH_4OH then gluconic acid is formed

$Ag^+ + e^- \longrightarrow Ag$; $E^o_{red} = 0.8$ V

$C_6H_{12}O_6 + H_2O \longrightarrow$ Gluconic acid

$(C_6H_{12}O_7) + 2H^+ + 2e^-$; $E^o_{oxd} = -0.05$ V

$Ag(NH_3)_2^+ + e^- \longrightarrow Ag(s) + 2NH_3$; $E^o_{oxd} = 0.337$ V

[Use $2.303 \times \frac{RT}{F} = 0.0592$ and $\frac{F}{RT} = 38.92$ at 298 K

19. $2Ag^- + C_6H_{12}O_6 + H_2O \longrightarrow 2Ag(s) + C_6H_{12}O_7 + 2H^+$

Find ln K of this reaction.

a. 66.13 **b.** 58.38

c. 28.30 **d.** 46.29

20. When ammonia is added to the solution, pH is raised to 11. Which half-cell reaction is affected by pH and by how much?

a. E_{oxd} will increase by a factor of 0.65 from E^o_{oxd}

b. E_{oxd} will decrease by a factor of 0.65 from E^o_{oxd}

c. E_{red} will increase by a factor of 0.65 from E^o_{red}

d. E_{red} will decrease by a factor of 0.65 from E^o_{red}

Space for rough work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	d	b	b	a	d	a	a	d	c
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
c	a	a	a	a	b	c	c	b	c
21.	22.	23.	24.	25.	26.	27.	28.	29.	30.
d	d	b	a	b	b	b	a	b	d

1. **(a)** Molecular mass of

$SO_2Cl_2 = 32 + 2\times16 + 35.5\times2 = 135$

$\text{Moles} = \frac{13.5}{135} = 0.1$

2. **(d)** Arrangement of sulphide ions (S^{2-}) in zinc blende (ZnS) is fcc while Zn^{2+} ions occupy alternate tetrahedral voids.

3. **(b)** The temperature at which a liquid boils increases with increase in pressure.

4. **(b)** Equal volumes of all gases under similar conditions of temperature and pressure contain equal number of molecules.

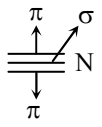
5. **(a)** K.E. = – (T.E.)

6. **(d)** Oxygen have $+2$ oxidation state in OF_2.

7. **(a)** $l/a = 0.5\ cm^{-1}$, R = 50 ohm

$p = \frac{Ra}{l} = \frac{50}{0.5} = 100$

$\Lambda = k\times\frac{1000}{N} = \frac{1}{p}\times\frac{1000}{N} = \frac{1}{100}\times\frac{1000}{1} = 10\ ohm^{-1}\ cm^2\ gm\ eq^{-1}$

8. **(a)** 

9. **(d)** Equilibrium constant changes with temperature, pressure and the concentration of either reactant or product.

10. **(c)** $HA : K_a = 10^{-4}$

$HA + NaOH \rightleftharpoons NaA + H_2O$

Clearly, the reverse reaction is the hydrolysis reaction.

$\Rightarrow \quad K_{Required} = \frac{1}{K_h} = \frac{K_a}{K_w} = \frac{10^{-4}}{10^{-14}} = 10^{10}$

11. **(c)** The definition of activation energy.

12. **(a)** Scattering of light is a property of colloid.

13. **(a)** For exothermic reactions $H_p < H_R$.

For endothermic reactions $H_p > H_R$.

14. **(a)** $_{18}Ar^{40}$ Total no of protons $= 18$

Total no of neutrons $= 22$

Mass defect $= [m\times p + m\times n] - 39.962384$

$= [1.007825\times18 + 1.008665\times22] - 39.962384$

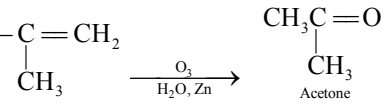
$= [18.14085 + 22.19063] - 39.962384 = 0.369$

Binding energy = mass defect $\times$ 931

$= 0.369\times931 = 343.62\ MeV$

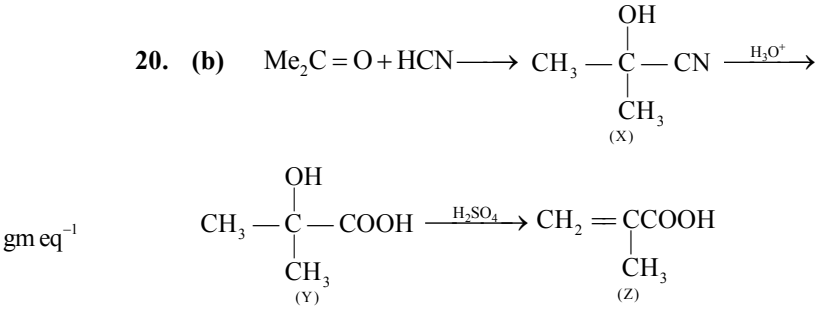
15. **(a)** $AlCl_3$ is lewis acid i.e., electron deficient compound. So it is electrophile.

16. **(b)** $R-\underset{\text{Cyanide}}{C\equiv N}$ and $R-\underset{\text{Isocyanide}}{N\equiv C}$ are functional isomers.

17. **(c)** 

18. **(c)** $CH_2Br - C\equiv C - CH_2Br \xrightarrow[\Delta]{Zn} CH_2 = C = C = CH_2$

19. **(b)** Benzoic acid.

20. **(b)** 

21. **(d)** $CH_3 - CH_2 - COOH + Na\overset{*}{H}CO_3 \longrightarrow$

$CH_3CH_2COONa + H_2O + \overset{*}{C}O_2$

22. **(d)** $C_6H_5NH_2 + O{=}CHC_6H_5 \longrightarrow \underset{\text{Schiff's base}}{C_6H_5 - N{=}CHC_6H_5} + H_2O$

23. **(b)** Average number molecular weight $\overline{M_n} = 30,000$

Average mass molecular weight $\overline{M_w} = 40,000$

Polydispersity index (PDI) $= \frac{\overline{M_w}}{\overline{M_n}} = \frac{40,000}{30,000} = 1.33$

24. **(a)** $\text{Oil (unsaturated)} + H_2 \xrightarrow{Ni} \text{Fat (saturated)}$

25. **(b)** "Cancer" is known as molecular disease.

26. **(b)** Hydrogen, forms hydrides like halides, e.g. HCl.

27. **(b)** $Cu_2Cl_2 + Ag_2S \longrightarrow Cu_2S + 2AgCl$

$2AgCl + Hg \longrightarrow Hg_2Cl_2 + 2Ag$

$AgCl + Hg \longrightarrow Ag + HgCl$

28. **(a)** The solubility of hydroxides of alkaline earth metals in water increases on moving down the group.

29. **(b)** $KI + MnO_4^- \longrightarrow K^+IO_3^- + Mn^{+2}$

30. **(d)** In $K_4Fe(CN)_6$, the species retains its identity in solid as well as in solution state.

JEE Advance Paper -I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
9	9	4	6	3	5	7	1	a	c
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
d	a, d	d	c	c. d	b	c	b, c	d	a

1. **(9)** $M = \dfrac{\%\text{ by weight} \times 10 \times d}{Mw_2} = \dfrac{30 \times 10 \times 1.095}{36.5} = 9\,\text{M}$

2. **(9)** Let x be the percentage of Na_2CO_3. Then, weight of $NaHCO_3 = (15 - x)\text{g}$

Moles of NaCl produced $= \dfrac{11.0\,\text{g}}{58.5\,\text{g}} = 0.18\,\text{mol}$

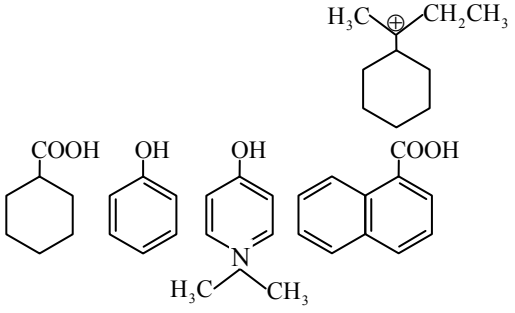
The NaCl is produced by the reaction of $\left(\dfrac{x}{106}\right)$ mol of Na_2CO_3 and $\dfrac{(15-x)}{84}$ mol of $NaHCO_3$. Each mol of Na_2CO_3 produces 2 mol of NaCl.

$\therefore \quad \dfrac{2x}{106} + \dfrac{15-x}{84} = 0.188$ Solve x : $= 13.5\,\text{g}\ Na_2CO_3$,

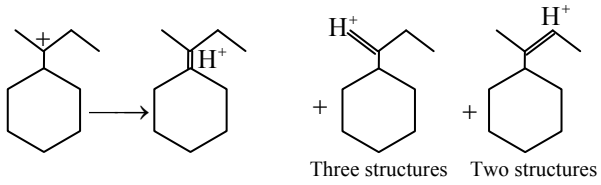
$NaHCO_3 = (15 - 1.35) = 13.6\,\text{g}$

$\%\,Na_2CO_3 = \dfrac{1.35}{15} \times 100 = 9.0\%\,Na_2CO_3$

3. **(4)** Aromatic alcohols and carboxylic acids form salt with NaOH, will dissolve in aqueous NaOH.



4. **(6)** These are total 6α — H to sp^2 carbon and they all can participate in hyperconjugation.



5. **(3)**

6. **(5)**

7. **(7)**

8. **(1)**

9. **(a)** $\Delta T_L = k_b \times m = k_b \times \dfrac{W_s}{M_s \times W_{solution}}$

$2 = 0.76 \times \dfrac{2.5 \times 1000}{M_s \times 100}$ $M_s = \dfrac{0.76 \times 2.5 \times 1000}{100 \times 2} = 9.5$

$\dfrac{760 - x}{760} = X_s = \dfrac{n_s}{n_{solution}} = \dfrac{\frac{W_s}{W_s}}{\frac{W_{solv}}{M_{solv}}} = \dfrac{2.5 \times 18}{9.5 \times 100} = 36.0$

$x = 760 - 36 = 724$

10. **(c)** Initial moles = final moles

$\dfrac{P_i \times V}{RT_1} + \dfrac{P_i \times V}{RT_1} = \dfrac{P_f \times V}{RT_2} + \dfrac{P_f \times V}{RT_1}$

$\dfrac{P_i}{T_1} + \dfrac{P_i}{T_1} = \dfrac{P_f}{T_2} + \dfrac{P_f}{T_1}$ $\dfrac{2P_i}{T_1} = P_f\left[\dfrac{1}{T_2} + \dfrac{1}{T_1}\right]$

$\dfrac{2P_i}{T_1} = P_f\left[\dfrac{T_1 + T_2}{T_1 T_2}\right]$ $P_f = 2P_i \times \left(\dfrac{T_2}{T_1 + T_2}\right)$

11. **(d)** K.E. = eV

$\lambda = \dfrac{h}{\sqrt{2meV}} \Rightarrow \dfrac{h}{\lambda} = \sqrt{2meV}$

12. **(a, d)** In acidic medium

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

In neutral medium,

$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$

Hence, number of electrons loose in acidic and neutral medium are 5 and 3 respectively.

13. **(d)** $E_{cell} = E^o_{cell} - \dfrac{0.059}{2}\log_{10}\dfrac{[M^{2+}][H^+]^2}{[M^{4+}]pH_2}$

$0.092 = 0.151 = \dfrac{0.059}{2}\log_{10} 10^x$

$\therefore \quad x = 2$

14. **(c)** O_2 is expected to be diamagnetic in nature but actually it is paramagnetic.

15. **(c, d)** In option **(c)**, if HNO_3 is present in limiting amount then this mixture will be a buffer and the mixture given in option **(d)**, contains a weak acid (CH_3COOH) and its salt with strong base NaOH, i.e. CH_3COONa.

16. **(b)** $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -54.07\times1000 - 298\times10$

$= -57050 \text{ J mol}^{-1} - 57050 = -5705\log_{10} K$

$\log_{10} K = 10$

Hence (b) is correct.

17. **(c)** C_6H_{14}

$H_3C - CH_2 - CH_2 - CH_2 - CH_2 - CH_3$

$H_3C - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2 - CH_2 - CH_3$

$H_3C - CH - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2 - CH_3$

$H_3C - \underset{\displaystyle CH_3}{\underset{|}{CH}} - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_3$

$H_3C - \underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}} - CH_2 - CH_3$, Hence **(c)** is correct

18. **(b, c)**

19. **(d)** $A \to 3;\ B \to 2;\ C \to 1;\ D \to 4$

20. **(a)** $A \to 3,5;\ B \to 1,2,4;\ C \to 1,2,4;\ D \to 1,2,4,4$

JEE Advance Paper -II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
2	4	6	2	3	4	4	0	a, c, d	a
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	b,c,d	c	b	d	a	b	c	b	a

1. **(2)** $\text{Normality} = N = \dfrac{W_B \times 1000}{\text{Eq. wt} \times V}$

$\therefore \quad \text{Eq. Wt} = \dfrac{0.45\times1000}{0.5\times20} = 45$

$\therefore \quad \text{Basicity} = \dfrac{\text{Molec. Wt}}{\text{Eq. Wt}} = \dfrac{90}{45} = 2$

2. **(4)** In Cuprammonium sulphate $[Cu(NH_3)_4]SO_4$ co-ordination no. of Cu is 4.

3. **(6)** $[Co(en)_2Br_2]Cl_2$

C.N.of Co $= 2\times$ number of bidentate ligand

$+1\times$ number of monodentate ligand $= 2\times2+1\times2 = 6.$

4. **(2)** Primary valencies are also known as oxidation state.

$K_2[Ni(CN)_4],\ 2+x-4=0 \Rightarrow x = +2$

5. **(3)** $x + 6\times(0) + 3\times(-1) = 0$

$x - 3 = 0,\ x + 3$, Oxidation number of Cr is = +3

6. **(4)**

7. **(4)**

8. **(0)**

9. **(a, c, d)** $\left(P + \dfrac{n^2a}{V^2}\right)(V - nb) = nRT$

At low pressure, when the sample occupies a large volume, the molecules are so far apart for most of the time that the intermolecular forces play no significant role, and the gas behaves virtually perfectly.

a and b are characteristic of a gas and are independent of temperature. The term $\left(P + \dfrac{n^2a}{V^2}\right)$ represents the pressure exerted by an ideal gas while P represents the pressure exerted by a real gas.

10. **(a)** $B_2(10) = \sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\pi_{2p_x^2}$

Bond order $= \dfrac{6-4}{2} = 1$ (nature is diamagnetic as no unpaired electron)

11. **(c)** $mvr = \dfrac{nh}{2\pi}$ and $\dfrac{mv^2}{r} = \dfrac{e^2}{r^2} \Rightarrow mv^2r = e^2$

$\Rightarrow \quad v = \dfrac{e^2\times2\pi}{nh} \ \because\ \left(mvr = \dfrac{nh}{2\pi}\right)$

$\Rightarrow \quad \text{K.E.} = \dfrac{1}{2}mv^2 = \dfrac{\frac{1}{2}me^4\times4\pi^2}{n^2h^2} \qquad \ldots(i)$

Expression for $a_0 = \dfrac{h^2}{4\pi^2me^2}$

$\Rightarrow \quad me^2 = \dfrac{h^2}{4\pi^2a_0} \qquad \ldots(ii)$

$\Rightarrow \quad \text{K.E.} = \dfrac{h^2}{8ma_0^2\pi^2}\times\dfrac{1}{n^2} \qquad \ldots(iii)$

For n = 2 K.E. $= \dfrac{h^2}{32\pi^2ma_0^2}$

12. **(c, d)** Cu^{2+} ions will react with CN^- and SCN^- forming $[Cu(CN)_4]^{3-}$ and $[Cu(SCN)_4]^{3-}$ leading the reaction in the backward direction.

$Cu^{2+} + 2CN^- \longrightarrow Cu(CN)_2$

$2Cu(CN)_2 \longrightarrow 2CuCN + (CN)_2$

$CuCN + 3CN^- \longrightarrow [Cu(CN)_4]^{3-}$

$Cu^{2+} + 4SCN^- \longrightarrow [Cu(SCN)_4]^{3-}$

Cu^{2+} also combines with $CuCl_2$ which reacts with Cu to produce CuCl pushing the reaction in the backward direction.

$CuCl_2 + Cu \longrightarrow 2CuCl\downarrow$

13. (c)

14. (b)

15. (d) Given, $\log K = 6 - \frac{2000}{T}$

Since, $\log K = \log A - \frac{Ea}{2.303RT}$

So, $A = 10^6\ sec^{-1}$

and $E_a = 38.3 kJ/mole$

16. (a) Rate in weak acid $= \frac{1}{100}$ (rate in strong acid)

$\therefore \quad [H^+]_{weak\ acid} = \frac{1}{100}[H^+]_{strong\ acid}$

$\therefore \quad [H^+]_{weak\ acid} = \frac{1}{100} M = 10^{-2} M$

$\therefore \quad C\alpha = 10^{-2}$

$\therefore \quad K_a = 10^{-4}$

For Question Nos. 17 to 18

17. (b) Paragraph-1

	$X_2(g)$	$2X(g)$
Initial mole	1	0
$t = t_{eq.}$	$(1-\alpha)$	2α

Given $2\alpha = \beta_{equilibrium}$ So, $\alpha = \frac{\beta_{equilibrium}}{2}$

Total mole at equilibrium $= (1+\alpha) = (1+\left(\frac{\beta_{eq}}{2}\right)$

$$Px_2 = \left[\frac{1-\frac{\beta_{eq.}}{2}}{1+\frac{\beta_{eq.}}{2}} P_{total}\right] = \left[\frac{2-\beta_{eq}}{2+\beta_{eq}} P_{total}\right] = \left[\frac{2-\beta_{eq}}{2+\beta_{eq}} P_{total}\right]$$

$$P_{x(g)} = \left[\frac{\beta_{eq}}{1+\frac{\beta_{eq}}{2}} P_{total}\right] = \left[\frac{2\beta_{eq}}{2+\beta_{eq}} P_{total}\right]$$

So $$K_p = \frac{(Px)^2}{(Px_2)} = \frac{\left[\frac{2\beta_{eq.}}{2+\beta_{eq.}} \times P_{total}\right]^2}{\left[\frac{2-\beta_{eq.}}{(2+\beta_{eq})} \times P_{total}\right]}$$

$$\Rightarrow \quad K_p = \frac{4\beta_{eq.}^2}{4-\beta_{eq.}^2} \times P_{total} = \left(\frac{8\beta_{eq}^2}{4-\beta_{eq}^2}\right)$$

18. (c) (a) Correct statement. As one decrease in pressure reaction will move in the direction where no. of gaseous molecules increases.

(b) Correct statement

At the start of reaction $Q_p < K_p$ so dissociation of X_2 take place spontaneously.

(c) Incorrect statement as

$K_p = \frac{8\beta_{eq}^2}{4-\beta_{eq}^2} = \frac{8\times(0.7)^2}{4-(0.7)^2} > 1$, but

(d) Correct statement.

As $\Delta G° > 0$ & $\Delta G° = -RT\ n K_p$

$\Delta G° > 1$, so K_p should be less than 1.

So $K < 1$

$K_p - K_c(RT)^{\Delta ng.}\ (RT > 1)$

$K_c = \frac{K_p}{RT} \Rightarrow K_c < K_p$

So $K_c < 1$

For Question Nos. 19 to 20

19. (b) $E^o_{cell} = \frac{RT}{nF} \ln K$

$(0.8 - 0.05) = \frac{1}{2} \times \frac{0.0592}{2.303} \ln K$

$\ln k = \frac{(0.8-0.05)\times 22.303}{0.0592} = 58.38$

20. (a) On increasing concentration of NH_3 the concentration of H^+ ion decreases. Therefore, E_{red} increases.

□ □ □

JEE-MAIN: CHEMISTRY MOCK TEST-5

1. One mole of potassium dichromate completely oxidises the following number of moles of ferrous sulphate in acidic medium

a. 1 **b.** 3
c. 5 **d.** 6

2. In a solid lattice, the cation has left a lattice site and is located at an interstitial position, the lattice defect is

a. Frenkel defect
b. Schottky defect
c. F-centre defect
d. Valency defect

3. At 300 K, the vapour pressure of an ideal solution containing 3 mole of A and 2 mole of B is 600 torr. At the same temperature, if 1.5 mole of A and 0.5 mole of C (non-volatile) are added to this solution the vapour pressure of solution increases by 30 torr. What is the value of P_B^o?

a. 940 **b.** 405
c. 90 **d.** None of these

4. If the root mean square speed of helium is 4.75 m s^{-1} at 25°C, then its speed will become 9.50 m s^{-1} at

a. 100°C **b.** 323°C
c. 919°C **d.** 1192°C

5. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon?

a. 3s **b.** 2p
c. 2s **d.** 1s

6. The product of oxidation of I^- with MnO_4^- in alkaline medium is

a. IO_3^- **b.** I_2
c. IO^- **d.** IO_4^-

7. The standard reduction electrode potentials of four elements are $A = -0.250\ V$, $B = -0.136\ V$, $C = -0.126\ V$ and $D = -0.402\ V$. The element that displaces A from its compounds is

a. B **b.** C
c. D **d.** None of these

8. The bond angle in carbon tetrachloride is approximately

a. 90°
b. 109°
c. 120°
d. 180°

9. The formation of nitric oxide by contact process $N_2 + O_2 \rightleftharpoons 2NO$. $\Delta H = 43.200$ kcal is favoured by

a. Low temperature and low pressure
b. Low temperature and high pressure
c. High temperature and high pressure
d. High temperature and excess reactants concentration

10. The pH of an aqueous solution $Mg(OH)_2$ is 9.0. If the solubility product of $Mg(OH)_2$ is 1×10^{-11}, what is $[Mg^{2+}]$?

a. 1×10^{-5} **b.** 1.0×10^{-4}
c. 1×10^{-2} **d.** 0.1

11. If 'I' is the intensity of absorbed light and C is the concentration of AB for the photochemical process $AB + h\nu \rightarrow AB^*$, the rate of formation of AB^* is directly proportional to

a. C **b.** I
c. I^2 **d.** C.I

12. Gold number is maximum for the lyophilic sol is

a. Gelatin **b.** Haemoglobin
c. Sodium oleate **d.** Potato starch

13. For the allotropic change represented by equation C(diamond) $\rightarrow$ C(graphite); the enthalpy change is $\Delta H = -1.89$ kJ. If 6 g of diamond and 6 g of graphite are separately burnt to yield carbon dioxide, the heat liberated in the first case is

a. Less than in the second case by 1.89 kJ
b. More than in the second case by 1.89 kJ
c. Less than in the second case by 11.34 kJ
d. More than in the second case by 0.945 kJ

14. The half-life of $_6C^{14}$ if its K or λ is 2.31×10^{-4} is

a. 2×10^2 yrs **b.** 3×10^3 yrs
c. 3.5×10^4 yrs **d.** 4×10^3 yrs

15. Most stable carbonium ion is

a. $\overset{+}{C}_2H_5$ **b.** $(CH_3)_3\overset{+}{C}$
c. $(C_6H_5)_3\overset{+}{C}$ **d.** $C_6H_5\overset{+}{C}H_2$

16. Which of the following compounds will exhibit *cis-trans* isomerism

a. 2-butene **b.** 2-butyne
c. 2-butanol **d.** Butanone

17. Acetylene gas is obtained by the electrolysis of
a. Sodium fumarate
b. Sodium succinate
c. Sodium maleate
d. Both **(a)** and **(c)**

18. The compound added to prevent chloroform to form phosgene gas is
a. C_2H_5OH
b. CH_3COOH
c. CH_3COCH_3
d. CH_3OH

19. Epoxides are
a. Cyclic ethers
b. Not ethers
c. Aryl-alkyl ethers
d. Ethers with another functional group

20. $NaOH/H^+$ reacts with
a. $C_6H_5OCH_3$ **b.** CH_3OH
c. $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$ **d.** C_2H_5OH

21. In the reaction $C_8H_6O_4 \xrightarrow{\Delta} X \xrightarrow{NH_3}$
The compound *X* is
a. Phthalic anhydride **b.** Phthalic acid
c. *o*-xylene **d.** Benzoic acid

22. In the reaction $CH_3(CH_2)_4CN \xrightarrow{Zn,\ HCl} A \xrightarrow{HONO} B \xrightarrow{O} C$; C is
a. Pentanal **b.** Pentanone
c. 2-Hexanone **d.** Hexanal

23. A polymer containing nitrogen is
a. Bakelite **b.** Dacron
c. Rubber **d.** Nylon-66

24. The base present in DNA, but not in RNA is
a. Guanine **b.** Adenine
c. Uracil **d.** Thymine

25. Which of the following is an antidiabatic drug
a. Insulin **b.** Penicillin
c. Chloroquine **d.** Aspirin

26. The correct sequence of elements in decreasing order of first ionisation energy is
a. Na > Mg > Al **b.** Mg > Na > Al
c. Al > Mg > Na **d.** Mg > Al > Na

27. In the metallurgical extraction of zinc from ZnO the reducing agent used is
a. Carbon monoxide **b.** Sulphur dioxide
c. Carbon dioxide **d.** Nitric oxide

28. In the lime (kiln), the reaction $CaCO_3(s) \longrightarrow CO_2(g)$ goes to completion because
a. Of high temperature
b. CaO is more stable than $CaCO_3$
c. CO_2 escapes simultaneously
d. CaO is not dissociated

29. Amalgams are
a. Highly coloured alloys
b. Always solid
c. Alloys which contain mercury as one of the contents
d. Alloys which have great resistance to abrasion

30. Generally, a group of atoms can function as a ligand if
a. They are positively charged ions
b. They are free radicals
c. They are either neutral molecules or negatively charged ions
d. None of these

Space for rough work

JEE ADVANCE PAPER-I

SECTION 1 Contains 8 Questions

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

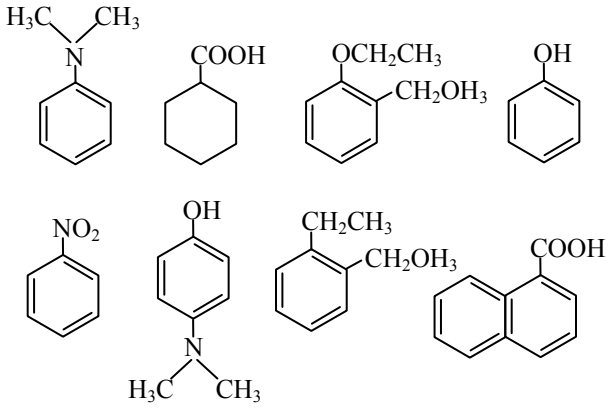
1. The change in the number of unpaired electrons when $\left[Fe(H_2O)_6\right]^{2+}$ is changed into $\left[Fe(CN)_6\right]^{4-}$ is

2. A compound of mol. wt. 180 is acetylated to give a compound of mol. wt. 390. The number of amino groups in the initial compound is?

3. Specific rotations of α-anomer of glucose is $+112°$ and for β-anomer is $+19°$. Specific rotation of equilibrium mixture is 52.6 . Calculate % composition of α-and β-anomers in the equilibrium mixture.

4. Consider all possible isomeric ketones including stereoisomers of MW = 100. All these isomers are independently reacted with $NaBH_4$ (Note: Stereoisomers are also reacted separately). The total number of ketones that give a racemic product(s) is/are

5. Amongst the following, the total number of compounds soluble in aqueous NaOH is



6. The half-life period of a radioactive substance is 2 min. The time taken for 1 g of the substance to reduce to 0.25 g will be

7. $\Delta_f H^-$ of hypothetical MgCl is $-125\,kJ\,mol^{-1}$ and for $MgCl_2$ is $-642\,kJ\,mol^{-1}$. The enthalpy of disproportionation of MgCl is –49x. Find the value of x.

8. Hydrolysis of an alkyl halide (RX) by dilute alkali $[OH^-]$ takes place simultaneously by SN^2 and SN^1 pathways. A plot of $-\frac{1}{[RX]}\frac{d[R-X]}{dt}$ vs. $[OH^-]$ is a straight line of the slope equal to $2\times10^3\ mol^{-1}L\,h^{-1}$ and intercept equal to $1\times10^2\,h^{-1}$. Calculate the initial rate $(mol\,L^{-1}\,min^{-1})$ of consumption of RX when the reaction is carried out taking $1\,mol\,L^{-1}$ of RX and 0.1 mol L^{-1} of $[OH^-]$ ions.

SECTION 2 Contains 10 Multiple Choice Questions

With one or more than one correct option

9. The ionic radii (in Å) of N^{3-}, O^{2-} and F^- are respectively:

a. 1.36, 1.40 and 1.71

b. 1.36, 1.71 and 1.40

c. 1.71, 1.40 and 1.36

d. 1.71, 1.36 and 1.40

10. From the following statements regarding H_2O_2, choose the incorrect statement

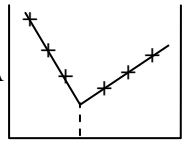
a. It can act only as an oxidizing agent

b. It decomposes on exposure to light

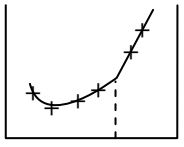
c. It has to be stored in plastic or wax lined glass bottles in dark

d. It has to be kept away from dust

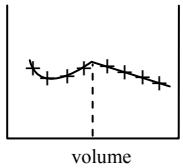
11. $AgNO_3$ (aq.) was added to an aqueous KCl solution gradually and the conductivity of the solution was measured. The plot of conductance (^) versus the volume of $AgNO_3$ is

a.


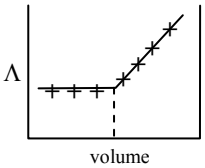
(a)

b.


(b)

c.


(c)

d.


(d)

12. The intermolecular interaction that is dependent on the inverse cube of distance between the molecules is:

a. ion-ion interaction **b.** ion-dipole interaction

c. London force **d.** hydrogen bond

13. According to Molecular Orbital Theory

a. C_2^{2-} is expected to be diamagnetic

b. O_2^{2+} is expected to have a longer bond length than O_2

c. N_2^+ and N_2^- have the same bond order

d. He_2^+ has the same energy as two isolated He atoms

14. For the first order reaction $2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$

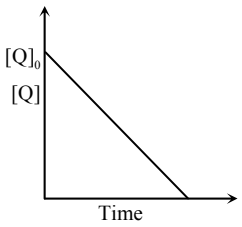
a. the concentration of the reactant decreases exponentially with time

b. the half-life of the reaction decreases with increasing temperature

c. the half-life of the reaction depends on the initial concentration of the reactant

d. the reaction proceeds to 99.6% completion in eight half-life duration

15. In the reaction, $P + Q \longrightarrow R + S$ the time taken for 75% reaction of P is twice the time taken for 50% reaction of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is



a. 2 **b.** 3 **c.** 0 **d.** 1

16. Choose the correct reason(s) for the stability of the lyophobic colloidal particles.

a. Preferential adsorption of ions on their surface from the solution

b. Preferential adsorption of solvent on their surface from the solution

c. Attraction between different particles having opposite charges on their surface

d. Potential difference between the fixed layer and the diffused layer of opposite charges around the colloidal particles

17. The standard enthalpies of formation of $CO_2(g)$, H2O(l) and glucose(s) at 25°C are –400 kJ/mol, –300 kJ/mol and –1300 kJ/mol, respectively. The standard enthalpy of combustion per gram of glucose at 25°C is

a. + 2900 kJ **b.** – 2900 kJ

c. – 16.11 kJ **d.** +16.11 kJ

18. Which of the following compounds is not colored yellow?

a. $Zn_2[Fe(CN)_6]$

b. $K_3[Co(NO_2)_6]$

c. $(NH_4)_3\ [As(Mo_3O_{10})_4]$

d. $BaCrO_4$

SECTION 3 Contains 2 Match The Following Type Questions

You will have to match entries in Column I with the entries in Column II.

19. Match each of the diatomic molecules in Column I with its property/properties in Column II.

Column - I	Column – II
(A) B_2	**1.** Paramagnetic
(B) N_2	**2.** Undergoes oxidation
(C) O_2^-	**3.** Undergoes reduction
(D) O_2	**4.** Bond order ≥ 2
	5. Mixing of 's' and 'p' orbitals

a. A → 5, 3, 1, B → 5, 4, C → 2, 1, D → 4, 1, 2

b. A → 1, 3, 5, B → 1, 2, 4 C → 3, 2, D → 1, 2,

c. A → 1, 3, B → 1, 2, 4 C → 1, 2, 3 D → 4, 5,

d. A→1, 3, 5, B→4, 5, C→1, 2, D→1, 2, 4

[According to MOT]

20. Match the compounds/ions in Column I with their properties/reactions in Column II.

Column I	Column II
(A) C_6H_5CHO	**1.** gives precipitate with 2, 4-dinitrophenyl-hydrazine
(B) $CH_3C \equiv CH$	**2.** gives precipitate with $AgNO_3$
(C) CN^-	**3.** is a nucleophile
(D) I^-	**4.** is involved in cyano-hydrin formation

a. A → 1, 2, 3, B → 2, C → 1, 3, 4, D → 2, 3

b. A → 1, 3, 4, B → 2, 3 C → 1, 2, 3, D → 2

c. A → 1, 3, B → 1, 4, C → 1, 3, D → 2, 3

d. A → 2, 3, B → 1, 2, C → 1, 3, D → 1, 2

JEE ADVANCE PAPER-II

SECTION 1 Contains 8 Questions

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. The number of rings formed in $\left[\mathrm{Ca(EDTA)}\right]^{2-}$ is ……

2. The number of equivalent $\mathrm{Cl-O}$ bonds in $\mathrm{Cl_2O_7}$ is ……

3. In fructose, the possible optical isomers are……

4. Oxidation of glucose is one of the most important reactions in a living cell. What is the number of ATP molecules generated in cells from one molecule of glucose?

5. 10 gm of a mixture of hexane and ethanol reacts with sodium to give 200 ml. of $\mathrm{H_2}$ at $27°\mathrm{C}$ and 760 mm pressure. The percentage of ethanol is……

6. The polymerisation of propene to linear polypene is represented by the reaction

$$n\left(\begin{array}{l}\mathrm{CH_3}\\ |\\ \mathrm{CH=CH_2}\end{array}\right) \longrightarrow \left(\begin{array}{l}\mathrm{CH_3}\\ |\\ \mathrm{CH-CH_2}\end{array}\right)_n$$

Where n has large integral value, the average enthalpies of bond dissociation for $(\mathrm{C=C})$ and $(\mathrm{C-C})$ at 298 K are +590 and $+331\,\mathrm{kJ\,mol^{-1}}$, respectively. The enthalpy of polymerization is $-360\,\mathrm{kJ\,mol^{-1}}$. Find the value of n.

7. In the case of a first order reaction, the time required for 93.75% of reaction to take place is x times that required for half of the reaction. Find the value of x.

8. In 1 L saturated solution of AgCl ($\mathrm{K_{sp}}$ of AgCl $=1.6\times10^{-10}$), 0.1mol of CuCl ($\mathrm{K_{sp}}$ $\mathrm{CuCl}=1.0\times10^{-6}$) is added. The resultant concentration of $\mathrm{Ag^+}$ in the solution is 1.6×10^{-x}. Calculate the value of x.

SECTION 2 Contains 8 Multiple Choice Questions

With one or more than one correct option

9. Which of the following atoms has the highest first ionization energy?

a. Rb **b.** Na

c. K **d.** Sc

10. Among the electrolytes $\mathrm{Na_2SO_4}$, $\mathrm{CaCl_2}$, $\mathrm{Al_2(SO_4)_3}$ and $\mathrm{NH_4Cl}$, the most effective coagulating agent for $\mathrm{Sb_2S_3}$ sol is

a. $\mathrm{Na_2SO_4}$ **b.** $\mathrm{CaCl_2}$ **c.** $\mathrm{Al_2(SO_4)_3}$ **d.** $\mathrm{NH_4Cl}$

11. The compound(s) with TWO lone pairs of electrons on the central atom is(are)

a. $\mathrm{BrF_5}$ **b.** $\mathrm{ClF_3}$

c. $\mathrm{XeF_4}$ **d.** $\mathrm{SF_4}$

12. Solubility product constant ($\mathrm{K_{sp}}$) of salts of types MX, $\mathrm{MX_2}$ and $\mathrm{M_3X}$ at temperature 'T' are 4.0×10^{-8}, 3.2×10^{-14} and 2.7×10^{-15}, respectively. Solubilities (mole $\mathrm{dm^{-3}}$) of the salts at temperature 'T' are in the order

a. $\mathrm{MX > MX_2 > M_3X}$ **b.** $\mathrm{M_3X > MX_2 > MX}$

c. $\mathrm{MX_2 > M_3X > MX}$ **d.** $\mathrm{MX > M_3X > MX_2}$

13. According to the Arrhenius equation

a. a high activation energy usually implies a fast reaction

b. rate constant increases with increase in temperature. This is due to a greater number of collisions whose energy exceeds the activation energy

c. higher the magnitude of activation energy, stronger is the temperature dependence of the rate constant

d. the pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy

14. For a linear plot of log (*x*/m) versus log p in a Freundlich adsorption isotherm, which of the following statements is correct?

a. Both k and 1/n appear in the slope term

b. 1/n appears as the intercept

c. Only 1/n appears as the slope

d. log (1/n) appears as the intercept.

15. When $\mathrm{O_2}$ is adsorbed on a metallic surface, electron transfer occurs from the metal to $\mathrm{O_2}$. The TRUE statement(s) regarding this adsorption is(are)

a. $\mathrm{O_2}$ is physisorbed

b. heat is released

c. occupancy of π^*_{2p} of $\mathrm{O_2}$ is increased

d. bond length of $\mathrm{O_2}$ is increased

16. Methylene blue, from its aqueous solution, is adsorbed on activated charcoal at $25°C$. For this process, the correct statement is

a. The adsorption requires activation at $25°C$

b. The adsorption is accompanied by a decrease in enthalpy

c. The adsorption increases with increase of temperature

d. The adsorption is irreversible

SECTION 3 Contains 2 Paragraph Type Questions

Each paragraph describes an experiment, a situation or a problem. Two multiple choice questions will be asked based on this paragraph. One or more than one option can be correct.

Paragraph for Question Nos. 17 to 18

Carbon–14 is used to determine the age of organic material. The procedure is based on the formation of ^{14}C by neutron capture in the upper atmosphere

$$^{14}_{7}N + {}_{0}n^{1} \longrightarrow {}^{14}_{6}C + {}_{1}n^{1}$$

^{14}C is absorbed by living organisms during photosynthesis. The ^{14}C content is constant in living organism once the plant or animal dies, the uptake of carbon dioxide by it ceases and the level of ^{14}C in the dead being, falls due to the decay which C^{14} undergoes

$$^{14}_{6}C \longrightarrow {}^{14}_{7}N + \beta^{-}$$

The half life period of ^{14}C is 5770 years. The decay constant (λ) can be calculated by using the following formula $\lambda = \frac{0.693}{t_{1/2}}$. The comparison of the β-activity of the dead matter with that of the carbon still in circulation enables measurement of the period of the isolation of the material from the living cycle. The method however, ceases to be accurate over periods longer than 30,000 years. The proportion of ^{14}C to ^{12}C in living matter is $1 : 10^{12}$.

17. Which of the following option is correct?

a. In living organisms, circulation of ^{14}C from atmosphere is high so the carbon content is constant in organism

b. Carbon dating can be used to find out the age of earth crust and rocks

c. Radioactive absorption due to cosmic radiation is equal to the rate of radioactive decay, hence the carbon content remains constant in living organism

d. Carbon dating cannot be used to determine concentration of ^{14}C in dead beings

18. What should be the age of fossil for meaningful determination of its age?

a. 6 years

b. 6000 years

c. 60,000 years

d. It can be used to calculate any age

Paragraph for Question Nos. 19 to 20

Rocket propellants consist of rocket engines powered by propellants. These are used both in space vehicles as well as in offensive weapons such as missiles. The propellants are chemical substances which on ignition provide thrust for the rocket to move forward. These substances are called rocket propellants. A propellant is a combination of an oxidiser and a fuel which when ignited undergoes combustion to release large quantities of hot gases. The passage of hot gases through the nozzle of the rocket motor provides the necessary thrust for the rocket to move forward according to Newton's third law of motion.

19. A biliquid propellant contains

a. Liquid hydrazine

b. A mixture of liquid fuel and a liquid oxidiser

c. A solid rocket fuel

d. A liquid fuel which can also act as an oxidiser

20. A hybrid rocket propellant uses

a. A liquid oxidiser and a solid fuel

b. A composite solid propellant

c. A biliquid propellant

d. A solid, liquid and gas as a propellant

Space for rough work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	a	c	c	d	a	c	b	d	d
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b	d	d	b	c	a	d	a	a	c
21.	**22.**	**23.**	**24.**	**25.**	**26.**	**27.**	**28.**	**29.**	**30.**
a	d	d	d	a	d	a	d	c	c

1. **(d)** $\underset{n=6}{Cr_2O_7^{--}} \rightarrow Cr^{3+}$; $\underset{n=1}{Fe^{++}} \rightarrow Fe^{+++}$

eq. of $K_2Cr_2O_7$ = eq. of $FeSO_4$ $1\times 6 = x\times 1$

2. **(a)** When an ion (generally cation due to its small size) is missing from its normal position and occpy an interstitial site between the lattice points, the lattice defect obtained is known as Frenkel defect.

3. **(c)** $P = P_A^o x_A + P_B^o x_B$

$$600 = P_A^o\left(\frac{3}{3+2}\right) + P_B^o\left(\frac{2}{2+3}\right); 3P_A^o + 2P_B^o = 3000$$

$$630 = P_A^o\left(\frac{4.5}{4.5+2+0.5}\right) + P_B^o\left(\frac{2}{4.5+2+0.5}\right); 4.5P_A^o + 2P_B^o = 4410$$

$1.5P_A^o = 1410; P_A^o = 940$ and $P_B^o = 90$

4. **(c)** r.m.s. speed $\propto \sqrt{T}$

$$\frac{4.75}{9.50} = \sqrt{\frac{298}{T}} \qquad \text{or} \quad \left(\frac{4.75}{9.50}\right)^2 = \frac{298}{T}$$

or $T = 1192\ K$ or $T = 1192 - 273 = 919°C$

5. **(d)** When electron jumps to lower orbit photons are emitted while photons are absorbed when electron jumps to higher orbit, 1s orbital is the lower most, electron in this orbital can absorb photons but cannot emit.

6. **(a)** $6MnO_4^- + I^- + 6OH^- \longrightarrow 6MnO_4^{2-} + IO_3^- + 3H_2O$

7. **(c)** A is displace from D because D have a $E° = -0.402\ V$

8. **(b)** CCl_4 is sp^3 hybridised so bond angle will be approximately $109°$.

9. **(d)** High temperature and excess concentration of the reactant concentration.

10. **(d)** $pH = 9 \ \therefore\ [H^+] = 10^{-9}\ [OH^-] = \dfrac{1\times 10^{-14}}{10^{-9}} = 10^{-5}$

$Mg(OH)_2 \rightleftharpoons Mg^{2+} + 2OH^-$

$K_{sp} = [Mg^{2+}][OH^-]^2 \ \Rightarrow\ 1\times 10^{-11} = [Mg^{2+}][10^{-5}]^2$

$$\Rightarrow\ [Mg^{2+}] = \frac{1\times 10^{-11}}{(10^{-5})^2} = 10^{-1} = 0.1$$

11. **(b)** In photochemical reaction the rate of formation of product is directly proportional to the intensity of absorbed light.

12. **(d)** Gold number shows the protective power of a lyophilic solution. Lesser the gold number, greater will be the protecting power of that colloid. Gelatin is one of the best protective colloid. Among the given colloids, potato starch has maximum gold number.

13. **(d)** $C_{(graphite)} \longrightarrow C_{(diamond)}, \Delta H = 1.9\ kJ$

$C_{(graphite)} + O_2 \longrightarrow CO_2, \Delta H = -\Delta H_1$

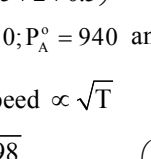
$C_{(diamond)} + O_2 \longrightarrow CO_2, \Delta H = -\Delta H_2$

$(-\Delta H_1) - (-\Delta H_2) = 1.9\ kJ$ or $\Delta H_2 = \Delta H_1 + 1.9$

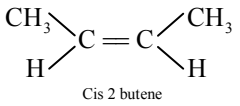
For combustion of $6g, \Delta H_2 > \Delta H_1$ by $1.9/2 = 0.95\ kJ$.

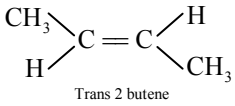
14. **(b)** $t_{1/2} = \dfrac{0.693}{k} = \dfrac{0.693}{2.31\times 10^{-4}} = 0.3\times 10^4\ yrs = 3.0\times 10^3\ yrs.$

15. **(c)** In the triphenyl methyl carbonium ion the π electrons of all the three benzene rings are delocalised with the vacant p-orbital of central carbon atom. So, it is resonance stabilised. It is the most stable of all the carbonium ions given

The ion  is stabilised by hyperconjugation, a second order resonance.

16. **(a)**


Cis 2 butene


Trans 2 butene

Cis-trans isomerism shown by compound which have double or triple bond by which they restrict their rotation, since 2 butyne have no hydrogen on triple bonded carbon.

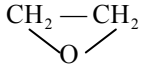
$\underset{\text{2 butyne}}{CH_3 - C \equiv C - CH_3}$ [It does not show cis-trans]

17. **(d)** $\begin{matrix} CH-COONa \\ \| \\ CH-COONa \end{matrix} \longrightarrow \begin{matrix} CHCOO^- \\ \| \\ CHCOO^- \end{matrix} + 2Na^+$

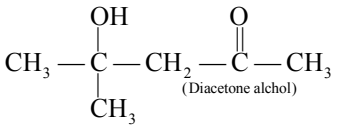
Sodium salt of maleic acid or fumaric acid

$$\begin{matrix} CHCOO^- \\ \| \\ CHCOO^- \end{matrix} \longrightarrow \underset{\text{Acetylene}}{\begin{matrix} CH \\ \| \\ CH \end{matrix}} + 2CO_2 + 2e^-$$

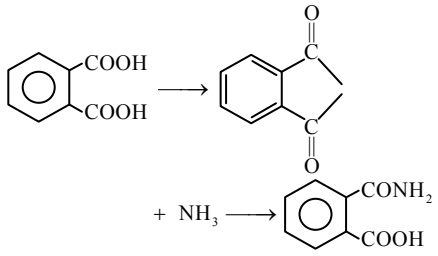
18. **(a)** Because it float over chloroform and prevent its oxidation.

19. **(a)** Cyclic ethers are called epoxies. 

20. **(c)** $2CH_3-CO-CH_3 \xrightarrow{\text{dil NaOH}}$



21. **(a)**

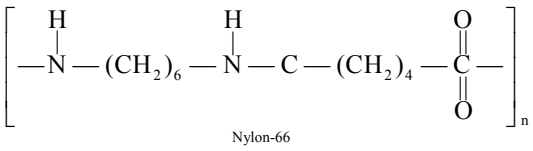


22. **(d)** $CH_3(CH_2)_4CN \xrightarrow{Zn,\ HCl}$

$CH_3(CH_2)_4CH_2NH_2 \xrightarrow{HONO}$

$CH_3(CH_2)_4CH_2OH \xrightarrow{O} \underset{\text{Hexanal}}{CH_3(CH_2)_4CHO}$

23. **(d)** Nylon-66- It is a polymer containing nitrogen



Nylon-66

24. **(d)** Thymine is present in DNA while in RNA there is Uracil.

25. **(a)** Insulin is an antidiabatic drug.

26. **(d)** $Mg > Al > Na$. This is due to the presence of fully filled s-orbital in Mg.

27. **(a)** $ZnO + CO \longrightarrow CO_2 + Zn$

28. **(d)** Lime stone $-CaCO_3$, Clay – silica and alumina
Gypsum $-CaSO_4 .2H_2O$

29. **(c)** Amalgams are alloys which contain mercury as one of the contents.

30. **(c)** Ligand must have capacity to donate lone pair of electrons to form coordinate bond.

JEE Advance Paper -I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
4	5	2	5	4	2	8	5	c	a
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
d	b	a, c	c	b,c,d	b	c	b	b	a

1. **(4)**

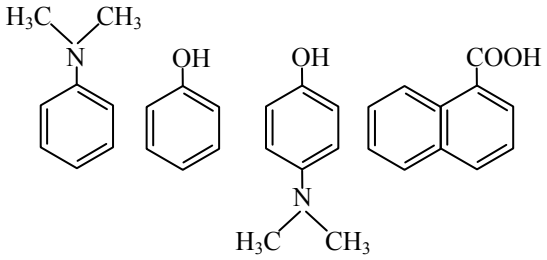
2. **(5)** Difference in mass of compound
$= 390 - 180 = 210$
wt. of CH_3CO – group is $= 43$

Therefore no. of $-NH_2$ group $= \dfrac{210}{43} = 4.88 = 5$.

3. The α-anomer of D-glucose has a specific rotation of +112 degrees in water. The β-anomer of D-glucose has a specific rotation of + 19 degrees (18.7 actually, but rounding up to 19).
If the fraction of glucose present as the α-anomer is *x*and the fraction present as the β-anomer is *y*, and the rotation of the mixture is +52.6 degree, we have
X (+112.2 degree) + y(18.7) = 52.6 degree ……..(*i*)
There is a very little of the open chain form present, so the fraction present as the α-anomer (x) plus the fraction present as the β-anomer (y) should account for all the glucose, i.e., $x + y = 1$ or $y = 1 - x$.
Putting value of y in equation (*i*), we get
x (+112.2 degree) + $(1 - x)$(18.7) = 52.6 degree ……..(*ii*)
Solving equation (*ii*) for (x), we have, $x = 0.36$ or 36%. Thus (y) must be $(1 - 0.36) = 0.64$ or 64%.
So, percentage composition of α- and β-anomers in the equilibrium mixture is 36% and 64% respectively.

4. **(5)**

5. **(4)**



6. **(2)** $1g \xrightarrow{t_{1/2}} 0\cdot5g \xrightarrow{t_{1/2}} 0\cdot25g \quad t_{25\%} = 4\,min$

7. **(8)** $2MgCl \longrightarrow Mg + MgCl_2 \qquad \Delta H = ?$

$Mg(s) + 1/2Cl_2(g) \longrightarrow MgCl \ ; \ \Delta H_1 = -125\,kJ\,mol^{-1}$

$Mg(s) + Cl_2(g) \longrightarrow MgCl_2 \ ; \ \Delta H_2 = -642\,kJ\,mol^{-1}$

$\Delta H = \Delta H_2 - 2\Delta H_1 = -642 - (2 \times -125) = -392\,kJ\,mol^{-1}$

$\therefore \quad -49x = -392 \quad \Rightarrow \quad x = 8$

8. **(5)** $\dfrac{-d[RX]}{dt} = k_2[RX][OH^-]$ (by SN^2 pathway)

$k_2 =$ rate constant of S_N2 reaction

$$\frac{-d[RX]}{dt} = k_2[RX][OH^-] + k_1[RX]$$

$$-\frac{1}{[RX]}\frac{d[RX]}{dt} = k_2[OH^-] + k_1$$

This is the equation of a straight line for $-\dfrac{1}{[RX]}\dfrac{d[RX]}{dt}$ vs $[OH^-]$ plot with slope equal to k_2 and intercept equal to k_1.

From question: $k_2 = 2\times10^3\,mol^{-1}L\,hr^{-1}, k_1 = 1\times10^2\,hr^{-1}$

$[RX] = 1.0\,M$ and $[OH^-] = 0.1\,M$

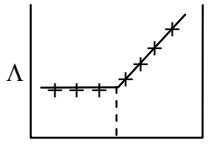
Hence, $\frac{-d[RX]}{dt} = 2\times10^3\times1\times0.1+1\times10^2\times1$

$= 300\,mol\,L^{-1}\,hr^{-1} = 5\,mol\,L^{-1}\,min^{-1}$

9. **(c)** Ionic Radii order: $N^{3-} > O^{2-} > F^-$

10. **(a)** It can act as an oxidising as well as reducing agent.

11. **(d)**



12. **(b)** Ion-dipole interaction

13. **(a,c)** **(a)** C_2^{2-} Total no. of electrons = 14 so it is diamagnetic

(b) O_2^{2+} Bond order = 3; O_2 Bond order = 2

$\therefore$ Bond length in O_2^{2+} is less than bond length in O_2.

(c) Bond order of $N_2^+ = 2.5$; Bond order of $He_2^+ = 1/2$

$\therefore$ Some energy is released during the formation of He_2^+ from two isolated He atoms.

14. **(a, b, d)** For first order reaction $[A] = [A]_0 e^{-kt}$

Hence concentration of $[NO_2]$ decreases exponentially,

Also, $t_{1/2} = \frac{0.693}{K}$, which is independent of concentration and $t_{1/2}$ decreases with the increase of temperature.

$$t_{99.6} = \frac{2.303}{K}\log\left(\frac{100}{0.4}\right)$$

$$t_{99.6} = \frac{2.303}{K}(2.4) = 8\times\frac{0.693}{K} = 8\,t_{1/2}$$

15. **(d)** Overall order of reaction can be decided by the data given $t_{75\%} = 2t_{50\%}$

$\therefore$ It is a first order reaction with respect to P.

From graph [Q] is linearly decreasing with time, i.e., order of reaction with respect to Q is zero and the rate expression is $r = k[P]^1[Q]^0$.

16. **(a, d)** **(a)** Preferential adsorption of ions on surface from the solution

(c) Attraction between particles having same charges on their surface accounts for the Brownian motion.

(d) Definition of Zeta Potential.

17. **(c)** Combustion of glucose

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$$

$\Delta H_{combustion} = (6\times\Delta H_f CO_2 + 6\times\Delta H_f H_2O) - \Delta H_f C_6H_{12}O_6$

$= (6\times-400 + 6\times-300) - (-1300)$

$= -2900\,kJ/mol$

$= -2900/180\,kJ/g$

$= -16.11\,kJ/g$

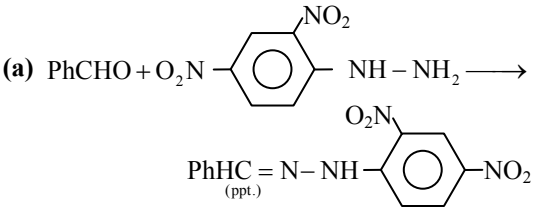
18. **(a)** $Zn_2[Fe(CN)_6]$ is bluish white ppt.

19. **(d)** A → 1, 3, 5, B → 4, 5, C → 1, 2, D → 1, 2, 4 [According to MOT]

20. **(a)** A → 1, 2, 3, B → 2, C → 1, 3, 4, D → 2, 3

(Note: Assuming $AgNO_3$ is ammonical)

(a)



$$PhCHO + Ag_2O \xrightarrow{NH_3} PhCOO^- + \underset{(whiteppt)}{Ag\downarrow}$$

$$PhCHO \xrightarrow{KCN} Ph-\underset{H}{\overset{CN}{\underset{|}{\overset{|}{C}}}}-O^-$$

(b) $CH_3C\equiv CH \xrightarrow[(White\ ppt)]{ammonical\ AgNO_3} CH_3-C\equiv C^-Ag^+\downarrow$

(c) $PhCHO \xrightarrow{KCN} Ph-\underset{H}{\overset{CN}{\underset{|}{\overset{|}{C}}}}-O^-$;

$AgNO_3 + CN^- \longrightarrow AgCN\downarrow$

(d) $AgNO_3 + I^- \longrightarrow AgI\downarrow$

JEE Advance Paper –II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
5	6	8	38	10	5	4	7	d	c
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b, c	d	b,c,d	b,c,d	b	b	c	b	2	1

1. **(5)**

2. **(6)**

3. **(8)** Fructose has three chiral centres and hence $2^3 = 8$ optical isomers are possible.

4. **(38)** $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 38ATP$

5. **(10)** 22400 ml of H_2 is produced by 46 gms of ethanol.

200 ml of H_2 is produced by $\frac{46\times200}{22400} = \frac{23}{56}$

Test Booklet code

A

Mock Test "JEE-Main"

Do not open this Test Booklet until you are asked to do so.
Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions:

1. Immediately fill in the particulars on this page of the Test Booklet with *Blue/Black Ball Point Pen*. *Use of pencil is strictly prohibited.*
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **360.**
5. There are ***three*** parts in the question paper A, B, C consisting of, **Physics, Chemistry** and **Mathematics** having 30 questions in each part of equal weightage. Each question is allotted **4 (four)** marks for each correct response.
6. *Candidates will be awarded marks as stated above in instruction No. 5 for correct response of each question. 1/4 (one fourth) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made if no response is indicated for an item in the answer sheet.*
7. There is only one correct response for each question. Filling up more than one response in each question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instruction 6 above.
8. Use ***Blue/Black Ball Point Pen only*** for writing particulars/marking responses on **Side-1** and **Side-2** of the Answer Sheet. ***Use of pencil is strictly prohibited.***
9. No candidates is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc., except the Admit Card inside the examination hall/room.
10. Rough work is to be done on the space provided for this purpose in the Test Booklet only. This space is given at the bottom of each page and at the end of the booklet.
11. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. ***However, the candidates are allowed to take away this Test Booklet with them.***
12. The CODE for this Booklet is **A**. Make sure that the CODE printed on **Side-2** of the Answer Sheet and also tally the serial number of the Test Booklet and Answer Sheet are the same as that on this booklet. In case of discrepancy, the candidate should immediately report the matter to the invigilator for replacement of both the Test Booklet and the Answer Sheet.
13. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capital letters): _____

Roll Number: in figures ☐☐☐☐☐☐☐☐

in words _____

Examination Centre Number: ☐☐☐☐☐☐☐☐

Name of Examination Centre (in Capital letters): _____

Candidate's Signature: _____ Invigilator's Signature: _____

Sample

***Read the Following Instructions Carefully*:**

1. The candidates should fill in the required particulars on the Test Booklet and Answer Sheet (***Side-1***) with ***Blue/Black Ball Point Pen.***
2. For writing/marking particulars on ***Side-2*** of the Answer Sheet, use ***Blue/Black Ball Point Pen only*.**
3. The candidates should not write their Roll Numbers anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
4. Out of the four options given for each question, only one option is the correct answer.
5. For each ***incorrect response, one-fourth* (*¼*)** of the total marks allotted to the question would be deducted from the total score. ***No deduction*** from the total score, however, will be made ***if no response*** is indicated for an item in the Answer Sheet.
6. Handle the Test Booklet and Answer Sheet with care, *as under no circumstances (except for discrepancy in Test Booklet Code and Answer Sheet Code), another set will be provided.*
7. The candidates are not allowed to do any rough work or writing work on the Answer Sheet. All calculations/writing work are to be done in the space provided for this purpose in the Test Booklet itself, marked 'Space for Rough Work'. This space is given at the bottom of each page and at the end of the booklet.
8. On completion of the test, the candidates must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
9. Each candidate must show on demand his/her Admit Card to the Invigilator.
10. No candidate, without special permission of the Superintendent or Invigilator, should leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign the Attendance Sheet again. Cases where a candidate has not signed the Attendance Sheet a second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case. **The candidates are also required to put their left hand THUMB impression in the space provided in the Attendance Sheet.**
12. Use of Electronic/Manual Calculator and any Electronic Item like mobile phone, pager etc. is prohibited.
13. The candidates are governed by all Rules and Regulations of the JAB/Board with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of the JAB/Board.
14. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
15. **Candidates are not allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, electronic device or any other material except the Admit Card inside the examination hall/room.**

Sample

JEE-MAIN: MATHEMATICS MOCK TEST-1

1. The domain of definition of $f(x)=\frac{\log_2(x+3)}{x^2+3x+2}$ is:

a. $R-\{-1,-2\}$ **b.** $(-2,\infty)$

c. $R-\{-1,-2,-3\}$ **d.** $(-3,\infty)-\{-1,-2\}$

2. How many roots the equation $x-\frac{2}{x-1}=1-\frac{2}{x-1}$ have

a. One **b.** Two

c. Infinite **d.** None

3. If ω is the cube root of unity, then $(3+5\omega+3\omega^2)^2+(3+3\omega+5\omega^2)^2=$

a. 4 **b.** 0

c. – 4 **d.** None of these

4. From the following find the correct relation

a. $(AB)'=A'B'$ **b.** $(AB)'=B'A'$

c. $A^{-1}=\frac{adj\,A}{A}$ **d.** $(AB)^{-1}=A^{-1}B^{-1}$

5. If the roots of the cubic equation $ax^3+bx^2+cx+d=0$ are in *G.P.*, then

a. $c^3a=b^3d$ **b.** $ca^3=bd^3$ **c.** $a^3b=c^3d$ **d.** $ab^3=cd^3$

6. $C_1+2C_2+3C_3+4C_4+....+nC_n=$

a. 2^n **b.** $n.2^n$

c. $n.2^{n-1}$ **d.** $n.2^{n+1}$

7. $1+\frac{(\log_e n)^2}{2!}+\frac{(\log_e n)^4}{4!}+\ldots=$

a. n **b.** $1/n$

c. $\frac{1}{2}(n+n^{-1})$ **d.** $\frac{1}{2}(e^n+e^{-n})$

8. How many words can be made from the letters of the word INSURANCE, if all vowels come together.

a. 18270 **b.** 17280

c. 12780 **d.** None of these

9. There are four machines and it is known that exactly two of them are faulty. They are tested, one by one, in a random order till both the faulty machines are identified. Then, the probability that only two tests are needed, is

a. $\frac{1}{3}$ **b.** $\frac{1}{6}$

c. $\frac{1}{2}$ **d.** $\frac{1}{4}$

10. $\sin\left(\frac{\pi}{10}\right)\sin\left(\frac{3\pi}{10}\right)=$

a. 1/2 **b.** – ½ **c.** 1/4 **d.** 1

11. Find real part of $\cosh^{-1}(1)$

a. – 1 **b.** 1

c. 0 **d.** None of these

12. From a 60 meter high tower angles of depression of the top and bottom of a house are α and β respectively. If the height of the house is $\frac{60\sin(\beta-\alpha)}{x}$, then $x=$

a. sin α sin β **b.** cos α cos β

c. sin α cos β **d.** cos α sin β

13. The function $f(x)=\frac{\log(1+ax)-\log(1-bx)}{x}$ is not defined at $x=0$. The value which should be assigned to f at $x=0$ so that it is continuous at $x=0$, is

a. $a-b$ **b.** $a+b$

c. $\log a+\log b$ **d.** $\log a-\log b$

14. If $x=a\cos^4\theta, y=a\sin^4\theta$, then $\frac{dy}{dx}$, at $\theta=\frac{3\pi}{4}$, is

a. –1 **b.** 1

c. $-a^2$ **d.** a^2

15. The minimum value of $[(5+x)(2+x)]/[1+x]$ for non-negative real x is

a. 12 **b.** 1 **c.** 9 **d.** 8

16. $\int\frac{e^{\tan^{-1}x}}{1+x^2}dx=$

a. $\log(1+x^2)+c$ **b.** $\log e^{\tan^{-1}x}+c$

c. $e^{\tan^{-1}x}+c$ **d.** $\tan^{-1}e^{\tan^{-1}x}+c$

17. The measurement of the area bounded by the co-ordinate axes and the curve $y=\log_e x$ is

a. 1 **b.** 2

c. 3 **d.** ∞

18. The solution of the equation $\frac{dy}{dx}=e^{x-y}+x^2e^{-y}$ is

a. $e^y=e^x+\frac{x^3}{3}+c$ **b.** $e^y=e^x+2x+c$

c. $e^y=e^x+x^3+c$ **d.** $y=e^x+c$

19. The distance between $4x+3y=11$ and $8x+6y=15$, is

a. $\frac{7}{2}$ **b.** 4

c. $\frac{7}{10}$ **d.** None of these

20. The area of a circle whose centre is (*h, k*) and radius *a* is

a. $\pi(h^2+k^2-a^2)$ **b.** πa^2hk

c. πa^2 **d.** None of these

21. The locus of the mid-point of the line segment joining the focus to a moving point on the parabola $y^2=4ax$ is another parabola with directrix

a. $x=-a$ **b.** $x=-\frac{a}{2}$

c. $x=0$ **d.** $x=\frac{a}{2}$

22. If $\vec{a}=(2,5)$ and $\vec{b}=(1,4)$, then the vector parallel to $(\vec{a}+\vec{b})$ is

a. (3, 5) **b.** (1, 1)

c. (1, 3) **d.** (8, 5)

23. The acute angle between the line joining the points (2, 1, –3), (–3,1,7) and a line parallel to $\frac{x-1}{3}=\frac{y}{4}=\frac{z+3}{5}$ through the point (–1, 0, 4) is

a. $\cos^{-1}\left(\frac{7}{5\sqrt{10}}\right)$ **b.** $\cos^{-1}\left(\frac{1}{\sqrt{10}}\right)$

c. $\cos^{-1}\left(\frac{3}{5\sqrt{10}}\right)$ **d.** $\cos^{-1}\left(\frac{1}{5\sqrt{10}}\right)$

24. Which of the following is logically equivalent to $\sim(\sim p\Rightarrow q)$

a. $p\wedge q$ **b.** $p\wedge\sim q$

c. $\sim p\wedge q$ **d.** $\sim p\wedge\sim q$

25. The number of solutions of the system of equations $2x+y-z=7,\ x-3y+2z=1, x+4y-3z=5$ is

a. 3 **b.** 2 **c.** 1 **d.** 0

26. The sum to infinity of the given series $\frac{1}{n}-\frac{1}{2n^2}+\frac{1}{3n^3}-\frac{1}{4n^4}+\ldots$ is

a. $\log_e\left(\frac{n+1}{n}\right)$ **b.** $\log_e\left(\frac{n}{n+1}\right)$

c. $\log_e\left(\frac{n-1}{n}\right)$ **d.** $\log_e\left(\frac{n}{n-1}\right)$

27. The values of *A* and *B* such that the function

$$f(x)=\begin{cases}-2\sin x, & x\le-\frac{\pi}{2}\\ A\sin x+B, & -\frac{\pi}{2}<x<\frac{\pi}{2}\\ \cos x, & x\ge\frac{\pi}{2}\end{cases},$$

is continuous everywhere are

a. $A=0,B=1$ **b.** $A=1,B=1$

c. $A=-1,B=1$ **d.** $A=-1,B=0$

28. $\int x^3\sqrt{3+5x^4}\,dx=$

a. $(3+5x^4)^{3/2}+c$ **b.** $\frac{1}{5}(3+5x^4)^{3/2}+c$

c. $\frac{1}{30}(3+5x^4)^{3/2}+c$ **d.** None of these

29. The equations of tangents to the circle $x^2+y^2-22x-4y+25=0$ which are perpendicular to the line $5x+12y+8=0$ are

a. $12x-5y+8=0, 12x-5y=252$

b. $12x-5y=0, 12x-5y=252$

c. $12x-5y-8=0, 12x-5y+252=0$

d. None of these

30. $\sim p\wedge q$ is logically equivalent to

a. $p\to q$

b. $q\to p$

c. $\sim(p\to q)$

d. $\sim(q\to p)$

Space for rough work

JEE ADVANCE PAPER-I

Time 3 Hours. **Max. Marks 264 (88 for Mathematics)**

Read The Instructions Carefully

Question Paper Format and Marking Scheme:

1. The question paper has three parts: Physics, Chemistry and Mathematics. Each part has three sections.
2. **Section 1** contains 8 questions. The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).
   Marking Scheme: +4 for correct answer and 0 in all other cases.
3. **Section 2** contains 10 multiple choice questions with one or more than one correct option.
   Marking Scheme: +4 for correct answer, 0 if not attempted and –2 in all other cases.
4. **Section 3** contains 2 "match the following" type questions and you will have to match entries in Column I with the entries in Column II.
   Marking Scheme: For each entry in Column I, +2 for correct answer, 0 if not attempted and –1 in all other cases.

NOTE: *It's the mock test as per previous year's papers but sometimes IIT changes the test paper pattern and marking scheme too.*

SECTION 1 (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- Marking scheme:
  +4 If the bubble corresponding to the answer is darkened.
  0 In all other cases.

1. If area enclosed between the curves $y = \ln(x+e)$ and $x = \ln\left(\frac{1}{y}\right)$ and the axis of x is λ sq unit, then the value of 6^{22} must be

2. The number of distinct solutions of the equation $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is

3. Let the curve C be the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If A and B are the points of intersection of C with the line $y = -5$, then the distance between A and B is

4. The minimum number of times a fair coin needs to be tossed, so that the probability of getting at least two heads is at least 0.96 is

5. Let n be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that all the girls stand consecutively in the queue. Let m be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that exactly four girls stand consecutively in the queue. Then the value of $\frac{m}{n}$ is

6. TP and TQ are any two tangents to a parabola and the tangent at a third point R cuts then in P' and Q', then the value of $\frac{TP'}{TP} = \frac{TQ'}{TQ}$ must be

7. Let $f : R \to R$ be a function defined by $f(x) = \begin{cases} [x], & x \le 2 \\ 0, & x > 2 \end{cases}$, where $[x]$ is the greatest integer less than or equal to x. If $I = \int_{-1}^{2} \frac{xf(x^2)}{2 + f(x+1)}\,dx$, then the value of $(4I - 1)$ is

8. A cylindrical container is to be made from certain solid material with the following constraints: It has a fixed inner volume of V mm^3, has a 2 mm thick solid wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container.

If the volume of the material used to make the container is minimum when the inner radius of the container is 10 mm, then the value of $V/250\pi$ is

SECTION 2 (Maximum Marks: 40)

- This section contains **TEN** questions.
- Each question has **FOUR** options (a), (b), (c) and (d). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Marking scheme:
  - +4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened.
  - 0 If none of the bubbles is darkened
  - –2 In all other cases

9. Let ΔPQR be a triangle. Let $\vec{a}=\overrightarrow{QR}, \vec{b}=\overrightarrow{RP},$ and $\vec{c}=\overrightarrow{PQ}.$ If $|\vec{a}|=12, |\vec{b}|=4\sqrt{3}$ and $\vec{b}.\vec{c}=24,$ then which of the following is (are) true?

a. $\frac{|\vec{c}|^2}{2}-|\vec{a}|=12$ **b.** $\frac{|\vec{c}|^2}{2}-|\vec{a}|=30$

c. $|\vec{a}\times\vec{b}+\vec{c}\times\vec{a}|=48\sqrt{3}$ **d.** $\vec{a}.\vec{b}=-72$

10. Let X and Y be two arbitrary, $3\times3,$ non-zero, skew-symmetric matrices and Z be an arbitrary $3\times3,$ non-zero, symmetric matrix. Then which of the following matrices is (are) skew symmetric?

a. $Y^3Z^4-Z^4Y^3$ **b.** $X^{44}+Y^{44}$

c. $X^4Z^3-Z^3X^4$ **d.** $X^{23}+Y^{23}$

11. Which of the following values of α satisfy the equation

$$\begin{vmatrix}(1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2\\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2\\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2\end{vmatrix}=-648\alpha?$$

a. –4 **b.** 9

c. –9 **d.** 4

12. In $R^3,$ consider the planes $P_1: y=0$ and $P_2: x+z=1.$ Let P_3 be a plane, different from P_1 and $P_2,$ which passes through the intersection of P_1 and $P_2.$ If the distance of the point (0, 1, 0) from P_3 is 1 and the distance of a point (α,β,γ) from P_3 is 2, then which of the following relations is (are) true?

a. $2\alpha+\beta+2\gamma+2=0$ **b.** $2\alpha-\beta+2\gamma+4=0$

c. $2\alpha+\beta-2\gamma-10=0$ **d.** $2\alpha-\beta+2\gamma-8=0$

13. In $R^3,$ let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from the two planes $P_1: x+2y-z+1=0$ and $P_2: 2x-y+z-1=0.$ Let M be the locus of the feet of the perpendiculars drawn from the points on L to the plane $P_1.$ Which of the following points lie(s) on M?

a. $\left(0,-\frac{5}{6},-\frac{2}{3}\right)$

b. $\left(-\frac{1}{6},-\frac{1}{3},\frac{1}{6}\right)$

c. $\left(-\frac{5}{6},0,\frac{1}{6}\right)$

d. $\left(-\frac{1}{3},0,\frac{2}{3}\right)$

14. Let P and Q be distinct points on the parabola $y^2=2x$ such that a circle with PQ as diameter passes through the vertex O of the parabola. If P lies in the first quadrant and the area of the triangle ΔOPQ is $3\sqrt{2},$ then which of the following is (are) the coordinates of P?

a. $(4,2\sqrt{2})$ **b.** $(9,3\sqrt{2})$

c. $\left(\frac{1}{4},\frac{1}{\sqrt{2}}\right)$ **d.** $(1,\sqrt{2})$

15. Let $y(x)$ be a solution of the differential equation $(1+e^x)y'+ye^x=1.$ If $y(0)=2,$ then which of the following statements is (are) true ?

a. $y(-4)=0$

b. $y(-2)=0$

c. $y(x)$ has a critical point in the interval $(-1,0)$

d. $y(x)$ has no critical point in the interval $(-1,0)$

16. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py''+Qy'+1=0,$ where P, Q are functions of x, y and y'(here $y'=\frac{dy}{dx}, y''=\frac{d^2y}{dx^2}$), then which of the following statements is (are) true?

a. $P=y+x$

b. $P=y-x$

c. $P+Q=1-x+y+y'+(y')^2$

d. $P-Q=x+y-y'-(y')^2$

17. Let $g: R \to R$ be a differential function with $g(0) = 0, g'(0) = 0$ and $g'(1) \neq 0$.

Let $f(x) = \begin{cases} \frac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases}$ and $h(x) = e^{|x|}$ for all $x \in R$. Let $(f \circ h)(x)$ denote $f(h(x))$ and $(h \circ f)(x)$ denote $h(f(x))$. Then which of the following is (are) true?

a. f is differentiable at $x = 0$

b. h is differentiable at $x = 0$

c. $f \circ h$ is differentiable at $x = 0$

d. $h \circ f$ is differentiable at $x = 0$

18. Let $f(x) = \sin\left(\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)\right)$ for all $x \in R$ and $g(x) = \frac{\pi}{2}\sin x$ for all $x \in R$. Let $(f \text{ o } g)(x)$ denote $f(g(x))$ and $(g \text{ o } f)(x)$ denote $g(f(x))$. Then which of the following is (are) true?

a. Range of f is $\left[-\frac{1}{2}, \frac{1}{2}\right]$

b. Range of f o g is $\left[-\frac{1}{2}, \frac{1}{2}\right]$

c. $\lim_{x \to 0} \frac{f(x)}{g(x)} = \frac{\pi}{6}$

d. There is an $x \in R$ such that $(g \circ f)(x) = 1$

SECTION 3 (Maximum Marks: 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column I** and **Column II**
- **Column I** has **four** entries (A), (B), (C) and (D)
- **Column II** has **five** entries (1), (2), (3), (4) and (5)
- Match the entries in **Column I** with the entries in **Column II**
- One or more entries in **Column I** may match with one or more entries in **Column II**
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below:

  (A) (1) (2) (3) (4) (5)

  (B) (1) (2) (3) (4) (5)

  (C) (1) (2) (3) (4) (5)

  (D) (1) (2) (3) (4) (5)

- For each entry in **Column I,** darken the bubbles of all the matching entries. For example, if entry (A) in **Column I,** matches with entries (2), (3) and (5), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- Marking scheme:

  For each entry in **Column I**

  +2 If only the bubble(s) corresponding to all the correct match(es) is(are) darkened

  0 If none of the bubbles is darkened

  –1 In all other cases

19. Match the Column:

Column I	Column II
(A) In R^2, if the magnitude of the projection vector of the vector $\alpha\hat{i} + \beta\hat{j}$ on $\sqrt{3}\hat{i} + \hat{j}$ is $\sqrt{3}$ and if $\alpha \mid = 2 + \sqrt{3}\beta$, then possible value(s) of $\mid\alpha\mid$ is (are)	**1.** 1
(B) Let a and b be real numbers such that the function $f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ bx + a^2, & x \geq 1 \end{cases}$ is differentiable for all $x \in R$. Then possible value(s) of a is (are)	**2.** 2
(C) Let $\omega \neq 1$ be a complex cube root of unity. If $(3 - 3\omega + 2\omega^2)^{4n+3} + (2 + 3\omega - 3\omega^2)^{4n+3} + (-3 + 2\omega + 3\omega^2)^{4n+3} = 0$, then possible value(s) of n is (are)	**3.** 3
(D) Let the harmonic mean of two positive real numbers a and b be 4. If q is a positive real number such that a, 5, q, b is an arithmetic progression, then the value(s) of $\mid q - a \mid$ is (are)	**4.** 4
	5. 5

20. Match the thermodynamic processes given under Column I with the expression given under Column II:

Column I	Column II
(A) In a triangle ΔXYZ, let a, b and c be the lengths of the sides opposite to the angles X, Y and Z, respectively. If $2(a^2-b^2)=c^2$ and $\lambda=\frac{\sin(X-Y)}{\sin Z}$, then possible values of n for which $\cos(n\pi\lambda)=0$ is (are)	**1.** 1
(B) In a triangle ΔXYZ, let a, b and c be the lengths of the sides opposite to the angles X, Y and Z, respectively. If $1+\cos 2X-2\cos 2Y=2\sin X\sin Y$, then possible value(s) of $\frac{a}{b}$ is (are)	**2.** 2
(C) In R^2, let $\sqrt{3}\hat{i}+\hat{j},\hat{i}+\sqrt{3}\hat{j}$ and $\beta\hat{i}+(1-\beta)\hat{j}$ be the position vectors of X, Y and Z with respect of the origin O, respectively. If the distance of Z from the bisector of the acute angle of $\overrightarrow{OX}$ with $\overrightarrow{OY}$ is $\frac{3}{\sqrt{2}}$, then possible value(s) of $\lvert\beta\rvert$ is (are)	**3.** 3
(D) Suppose that $F(\alpha)$ denotes the area of the region bounded by $x=0$, $x=2$, $y^2=4x$ and $y=\lvert\alpha x-1\rvert+\lvert\alpha x-2\rvert+\alpha x$, where $\alpha\in\{0,1\}$. Then the value(s) of $F(\alpha)+\frac{8}{3}\sqrt{2}$, when $\alpha=0$ and $\alpha=1$, is (are)	**4.** 5
	5. 6

Space for rough work

JEE ADVANCE PAPER-II

Time 3 Hours. **Max. Marks 240 (80 for Mathematics)**

Read The Instructions Carefully

Question Paper Format and Marking Scheme:

1. The question paper has three parts: Physics, Chemistry and Mathematics. Each part has three sections.
2. Section 1 contains 8 questions. The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).
   Marking Scheme: +4 for correct answer and 0 in all other cases.
3. Section 2 contains 8 multiple choice questions with one or more than one correct option.
   Marking Scheme: +4 for correct answer, 0 if not attempted and –2 in all other cases.
4. Section 3 contains 2 "paragraph" type questions. Each paragraph describes an experiment, a situation or a problem. Two multiple choice questions will be asked based on this paragraph. One or more than one option can be correct.
   Marking Scheme: +4 for correct answer, 0 if not attempted and –2 in all other cases.

NOTE: *It's the mock test as per previous year's papers but sometimes IIT changes the test paper pattern and marking scheme too.*

SECTION 1 (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- Marking scheme:
  +4 If the bubble corresponding to the answer is darkened.
  0 In all other cases.

1. Suppose that $\vec{p}, \vec{q}$ and $\vec{r}$ are three non-coplanar vectors in R^3. Let the components of a vector $\vec{s}$ along $\vec{p}, \vec{q}$ and $\vec{r}$ be 4, 3 and 5, respectively. If the components of this vector $\vec{s}$ along $(-\vec{p}+\vec{q}+\vec{r})$, $(\vec{p}-\vec{q}+\vec{r})$ and $(-\vec{p}-\vec{q}+\vec{r})$ are x, y and z, respectively, then the value of $2x+y+z$ is

2. For any integer k, let $\alpha_k = \cos\left(\frac{k\pi}{7}\right) + i\sin\left(\frac{k\pi}{7}\right)$, where $i=\sqrt{-1}$. The value of the expression $\frac{\sum_{k=1}^{12} |\alpha_{k+1}-\alpha_k|}{\sum_{k=1}^{3} |\alpha_{4k-1}-\alpha_{4k-2}|}$ is

3. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is 6 : 11 and the seventh term lies in between 130 and 140, then the common difference of this A.P. is

4. The coefficient of x^9 in the expansion of $(1+x)(1+x^2)(1+x^3)...(1+x^{100})$ is

5. If the normals at the four points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and (x_4, y_4) on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are concurrent, then the value of $(x_1+x_2+x_3+x_4)\times\left(\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}+\frac{1}{x_4}\right)$ must be

6. Let m and n be two positive integers greater than 1. If $\lim_{a\to 0}\left(\frac{e^{\cos(\alpha^n)}-e}{\alpha^m}\right) = -\left(\frac{e}{2}\right)$ then the value of $\frac{m}{n}$ is

7. If $\alpha = \int_0^1 (e^{9x+3\tan^{-1}x})\left(\frac{12+9x^2}{1+x^2}\right)dx$ where $\tan^{-1}x$ takes only principal values, then the value of $\left(\log_e |1+\alpha| \frac{3\pi}{4}\right)$ is

8. Let $f: R \to R$ be a continuous odd function, which vanishes exactly at one point and $f(1)=\frac{1}{2}$. Suppose that $F(x)=\int_{-1}^{x} f(t)\,dt$ for all $x\in[1-1,2]$ and $G(x)=\int_{-1}^{x} t\,|f(f(t))|\,dt$ for all $x\in[-1,2]$. If $\lim_{x\to 1}\frac{F(x)}{G(x)}=\frac{1}{14}$, then the value of $f\left(\frac{1}{2}\right)$ is

SECTION 2 (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (a), (b), (c) and (d). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Marking scheme:
  +4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened.
  0 If none of the bubbles is darkened
  –2 In all other cases

9. Let $f'(x)=\frac{192x^3}{2+\sin^4\pi x}$ for all $x\in R$ with $f\left(\frac{1}{2}\right)=0$. If $m\le\int_{1/2}^{1}f(x)\,dx\le M$, then the possible values of m and M are

a. $m=13, M=24$ **b.** $m=\frac{1}{4}, M=\frac{1}{2}$

c. $M=-11, M=0$ **d.** $M=1, M=12$

10. Let S be the set of all non-zero real numbers α such that the quadratic equation $\alpha x^2-x+\alpha=0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1-x_2|<1$. Which of the following intervals is(are) a subset(s) of S?

a. $\left(-\frac{1}{2},-\frac{1}{\sqrt{5}}\right)$ **b.** $\left(-\frac{1}{\sqrt{5}},0\right)$

c. $\left(0,\frac{1}{\sqrt{5}}\right)$ **d.** $\left(\frac{1}{\sqrt{5}},\frac{1}{2}\right)$

11. If $\alpha=3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta=3\cos^{-1}\left(\frac{4}{9}\right)$, where the inverse trigonometric functions take only the principal values, then the correct option(s) is (are)

a. $\cos\beta>0$ **b.** $\sin\beta<0$

c. $\cos(\alpha+\beta)>0$ **d.** $\cos\alpha<0$

12. Let E_1 and E_2 be two ellipse whose centers are at the origin. The major axes of E_1 and E_2 lie along the x-axis and the y-axis, respectively. Let S be the circle $x^2+(y-1)^2=2$. The straight line $x+y=3$ touches the curves S, E_1 and E_2 at P, Q and R, respectively Suppose that $PQ=PR=\frac{2\sqrt{2}}{3}$. If e_1 and e_2 are the eccentricities of E_1 and E_2, respectively, then the correct expression(s) is(are)

a. $e_1^2+e_2^2=\frac{43}{40}$ **b.** $e_1e_2=\frac{\sqrt{7}}{2\sqrt{10}}$

c. $|e_1^2-e_2^2|=\frac{5}{8}$ **d.** $e_1e_2=\frac{\sqrt{3}}{4}$

13. Consider the hyperbola $H: x^2-y^2=1$ and a circle S with center $N(x_2,0)$. Suppose that H and S touch each other at a point $P(x_1,y_1)$ with $x_1>1$ and $y_1>0$. The common tangent to H and S at P intersects the x-axis at point M. If (l,m) is the centroix of the triangle ΔPMN, then the correct expression(s) is (are)

a. $\frac{dl}{dx_1}=1-\frac{1}{3x_1^2}$ for $x>1$

b. $\frac{dm}{dx_1}=\frac{x_1}{3\left(\sqrt{x_1^2-1}\right)}$ for $x_1>1$

c. $\frac{dl}{dx_1}=1+\frac{1}{3x_1^2}$ for $x_1>1$

d. $\frac{dm}{dy_1}=\frac{1}{3}$ for $y_1>0$

14. The option(s) with the values of a and L that satisfy the following equation is(are)

$$\frac{\int_0^{4\pi}(\sin^6 at+\cos^4 at)\,dt}{\int_0^{\pi}e^t(\sin^6 at+\cos^4 at)\,dt}=L?$$

a. $a=2, L=\frac{e^{4\pi}-1}{e^{\pi}-1}$ **b.** $a=2, L=\frac{e^{4\pi}+1}{e^{\pi}+1}$

c. $a=4, L=\frac{e^{4\pi}-1}{e^{\pi}-1}$ **d.** $a=4, L=\frac{e^{4\pi}+1}{e^{\pi}+1}$

15. Let $f, g:[-1,2]\to R$ be continuous functions which are twice differentiable on the interval (–1, 2). Let the values of f and g at points –1, 0 and 2 be as given in the following table:

	$x=-1$	$x=0$	$x=2$
$f(x)$	3	6	0
$g(x)$	0	1	–1

In each of the intervals (–1, 0) and (0, 2) the function $(f-3g)''$ never vanishes. Then the correct statements(s) is (are)

a. $f'(x)-3g'(x)=0$ has exactly three solutions in $(-1,0)\cup(0,2)$

b. $f'(x)-3g'(x)=0$ has exactly one solution in (–1, 0)

c. $f'(x)-3g'(x)=0$ has exactly one solution in (0, 2)

d. $f'(x)-3g'(x)=0$ has exactly two solutions in (–1, 0) and exactly two solutions in (0, 2)

16. Let $f(x)=7\tan^8 x+7\tan^6 x-3\tan^4 x-3\tan^2 x$ for all $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ Then the correct expression(s) is (are)?

a. $\int_0^{\pi/4} xf(x)\,dx=\frac{1}{12}$ **b.** $\int_0^{\pi/4} f(x)\,dx=0$

c. $\int_0^{\pi/4} xf(x)\,dx=\frac{1}{6}$ **d.** $\int_0^{\pi/4} f(x)\,dx=1$

SECTION 3 (Maximum Marks: 16)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (a), (b), (c) and (d). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Marking scheme:
- +4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened.
- 0 If none of the bubbles is darkened
- –2 In all other cases

Paragraph-I

Let $F:R\to R$ be a thrice differentiable function. Suppose that $F(1)=0, F(3)=-4$ and $F'(x)<0$ for all $x\in\left(\frac{1}{2},3\right)$. Let $f(x)=xF(x)$ for all $x\in R$.

17. The correct statement(s) is (are)

a. $f'(1)<0$

b. $f(2)<0$

c. $f'(x)\neq 0$ for any $x\in(1,3)$

d. $f'(x)=0$ for some $x\in(1,3)$

18. If $\int_1^3 x^2F'(x)\,dx=-12$ and $\int_1^3 F''(x)\,dx=40$, then the correct expression(s) is(are)

a. $9f'(3)+f'(1)-32=0$

b. $\int_1^3 f(x)dx=12$

c. $9f'(3)-f'(1)+32=0$

d. $\int_1^3 f(x)dx=-12$

Paragraph-II

Let n_1 and n_2 be the number of red and black balls, respectively, in box I. Let n_3 and n_4 be the number of red and black balls, respectively, in box II.

19. One of the two boxes, box I and box II, was selected at random and a ball was drawn randomly out of this box. The ball was found to be red. If the probability that this red ball was drawn from box II is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1, n_2, n_3 and n_4 is (are)

a. $n_1=3, n_2=3, n_3=5, n_4=15$

b. $n_1=3, n_2=6, n_3=10, n_4=50$

c. $n_1=8, n_2=6, n_3=5, n_4=20$

d. $n_1=6, n_2=12, n_3=5, n_4=20$

20. A ball is drawn at random from box I and transferred to box II. If the probability of drawing a red ball from box I, after this transfer, is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1 and n_2 is(are)

a. $n_1=4, n_2=6$ **b.** $n_1=2, n_2=3$

c. $n_1=10, n_2=20$ **d.** $n_1=3, n_2=6$

Space for rough work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	d	c	a	a	c	c	d	a	c
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	d	b	a	c	c	d	a	c	c
21.	**22.**	**23.**	**24.**	**25.**	**26.**	**27.**	**28.**	**29.**	**30.**
c	c	a	d	d	a	c	c	a	d

1. **(d)** $x+3>0$ and $x2+3x+2\neq 0$

2. **(d)** If $x\neq 1$, multiplying each term by $(x-1)$, the given equation reduces to $x(x-1)=(x-1)$ or $(x-1)^2=0$ or $x=1$, which is not possible as considering $x\neq 1$. Thus given equation has no roots.

3. **(c)** $(3+5\omega+3\omega^2)^2+(3+3\omega+5\omega^2)^2$

$=(3+3\omega+3\omega^2+2\omega)^2+(3+3\omega+3\omega^2+2\omega^2)^2$

$(1+\omega+\omega^2=0,\omega^3=1)$

$=(2\omega)^2+(2\omega^2)^2=4\omega^2+4\omega^4=4(-1)=-4.$

4. **(b)** It is understandable.

5. **(a)** Let $\frac{A}{R}, A, AR$ be the roots of the equation $ax^3+bx^2+cx+d=0$

then $A^3=$ Product of the roots $=-\frac{d}{a}$

$\Rightarrow A=-\left(\frac{d}{a}\right)^{1/3}$

Since A is a root of the equation.

$\therefore \quad aA^3+bA^2+cA+d=0$

$\Rightarrow \quad a\left(-\frac{d}{a}\right)+b\left(-\frac{d}{a}\right)^{2/3}+c\left(-\frac{d}{a}\right)^{1/3}+d=0$

$\Rightarrow \quad b\left(\frac{d}{a}\right)^{2/3}=c\left(\frac{d}{a}\right)^{1/3} \Rightarrow b^3\frac{d^2}{a^2}=c^3\frac{d}{a} \Rightarrow b^3d=c^3a.$

6. **(c)** Trick: Put $n=1, 2, 3,....$

$S_1=1, S_2=2+2=4$

Now by alternate **(c)**,

put $n=1, 2$

$S_1=1.2^0=1, S_2=2.2^1=4$

7. **(c)** $1+\frac{(\log_e n)^2}{2!}+\frac{(\log_e n)^4}{4!}+\ldots=\frac{e^{\log_e n}+e^{-\log_e n}}{2}=\frac{n+n^{-1}}{2}.$

8. **(d)** IUAENSRNC Obviously required number of words are $\frac{6!}{2!}\times 4!=8640$

9. **(b)** The probability that only two tests are needed = probability that the first machine tested is faulty $\times$ probability that the second machine tested is faulty $=\frac{2}{4}\times\frac{1}{3}=\frac{1}{6}$

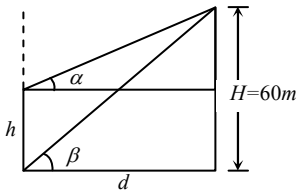
10. **(c)** $\sin\frac{\pi}{10}\sin\frac{3\pi}{10}=\sin 18°.\sin 54°$

$=\sin 18°.\cos 36°=\frac{\sqrt{5}-1}{4}.\frac{\sqrt{5}+1}{4}=\frac{1}{4}.$

11. **(c)** We know that $\cosh^{-1}x=\log\left(x+\sqrt{x^2-1}\right)$

$\therefore \quad \cosh^{-1}(1)=\log\left(1+\sqrt{1^2-1}\right)=\log 1=0.$

12. **(d)**



$H=d\tan\beta$ and $H-h=d\tan\alpha$

$\Rightarrow \quad \frac{60}{60-h}=\frac{\tan\beta}{\tan\alpha}$

$\Rightarrow -h=\frac{60\tan\alpha-60\tan\beta}{\tan\beta} \Rightarrow \quad h=\frac{60\sin(\beta-\alpha)}{\cos\alpha\cos\beta\frac{\sin\beta}{\cos\beta}}$

$\Rightarrow \quad x=\cos\alpha\sin\beta.$

13. **(b)** Since limit of a function is $a+b$ as $x\to 0$, therefore to be continuous at a function, its value must be $a+b$ at $x=0$

$\Rightarrow f(0)=a+b.$

14. **(a)** $y=a\sin^4\theta$

$\Rightarrow \quad \frac{dy}{d\theta}=4a\sin^3\theta\cos\theta$

and $x=a\cos^4\theta$

$\Rightarrow \quad \frac{dx}{d\theta}=-4a\cos^3\theta\sin\theta$

$\therefore \quad \frac{dy}{dx}=\frac{dy/d\theta}{dx/d\theta}=\frac{-\sin^2\theta}{\cos^2\theta}=-\tan^2\theta$

$\therefore \quad \left(\frac{dy}{dx}\right)_{\theta=\frac{3\pi}{4}}=-\tan^2\left(\frac{3\pi}{4}\right)=-1.$

15. **(c)** Given $f(x)=\frac{[(5+x)(2+x)]}{[1+x]}$

$$f(x)=1+\frac{4}{1+x}+(5+x)=(6+x)+\frac{4}{(1+x)}$$

$\Rightarrow \quad f'(x)=1-\frac{4}{(1+x)^2}=0;$

$$x^2+2x-3=0$$

$\Rightarrow \quad x=-3,1$

Now $f''(x)=\frac{8}{(1+x)^3}$,

$f''(-3)=-ve,$

$f''(1)=+ve$

Hence minimum value at $x=1$

$$f(1)=\frac{(5+1)(2+1)}{(1+1)}=\frac{6\times 3}{2}=9.$$

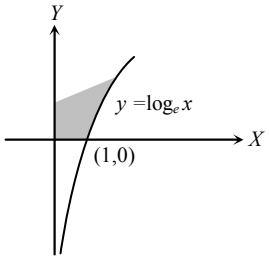
16. **(c)** Putting $t=\tan^{-1}x \Rightarrow dt=\frac{1}{1+x^2}dx,$ we get

$$\int \frac{e^{\tan^{-1}x}}{1+x^2}dx=\int e^t dt = e^t+c=e^{\tan^{-1}x}+c.$$

17. **(d)** Area

$A=\int_0^\infty \log x.dx$

$=(x\log x-x)_0^\infty=\infty$



18. **(a)** $\frac{dy}{dx}=e^{x-y}+x^2e^{-y}=e^{-y}(e^x+x^2)$

$\Rightarrow \quad e^y dy=(x^2+e^x)dx$

Now integrating both sides, we get $e^y=\frac{x^3}{3}+e^x+c.$

19. **(c)** $4x+3y=11$ and $4x+3y=\frac{15}{2}$

Therefore, $D=\left|\frac{11-\frac{15}{2}}{5}\right|=\frac{7}{10}$.

20. **(c)** Since area $=\pi r^2$, where $r=a$

$\Rightarrow$ Area $=\pi a^2.$

21. **(c)** Let $P(h,k)$ be the mid-point of the line segment joining the focus $(a,0)$ and a general point $Q(x,y)$ on the parabola.

Then $h=\frac{x+a}{2}, k=\frac{y}{2}\Rightarrow x=2h-a, y=2k.$

Put these values of x and y in $y^2=4ax$, we get

$4k^2=4a(2h-a)$

$\Rightarrow \quad 4k^2=8ah-4a^2 \Rightarrow k^2=2ah-a^2$

So, locus of $P(h,k)$ is $y^2=2ax-a^2$

$\Rightarrow \quad y^2=2a\left(x-\frac{a}{2}\right)$

Its directrix is

$x-\frac{a}{2}=-\frac{a}{2}\Rightarrow x=0.$

22. **(c)** $\vec{a}+\vec{b}=3\hat{i}+9\hat{j}=3(\hat{i}+3\hat{j})$. Hence it is parallel to $(1,3)$.

23. **(a)** Direction ratio of the line joining the point $(2, 1, -3), (-3, 1, 7)$ are (a_1, b_1, c_1)

$\Rightarrow \quad (-3-2, 1-1, 7-(-3))$

$\Rightarrow \quad (-5, 0, 10)$

Direction ratio of the line parallel to line

$\frac{x-1}{3}=\frac{y}{4}=\frac{z+3}{5}$ are

(a_2, b_2, c_2)

$\Rightarrow \quad (3, 4, 5)$

Angle between two lines,

$$\cos\theta=\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

$$\cos\theta=\frac{(-5\times 3)+(0\times 4)+(10\times 5)}{\sqrt{25+0+100}\sqrt{9+16+25}}$$

$\Rightarrow \quad \cos\theta=\frac{35}{25\sqrt{10}}$

$\Rightarrow \quad \theta=\cos^{-1}\left(\frac{7}{5\sqrt{10}}\right)$

24. **(d)** Since $\sim(p\Rightarrow q)\equiv p\wedge\sim q$

$\sim(\sim p\Rightarrow q)=\sim p\wedge\sim q$

25. **(d)** $\Delta=\begin{vmatrix}2 & 1 & -1\\ 1 & -3 & 2\\ 1 & 4 & -3\end{vmatrix}$

$=2(9-8)-1(-3-2)-1(4+3)=7-7=0$

Hence, number of solutions is zero.

26. **(a)** $\frac{1}{n}-\frac{1}{2n^2}+\frac{1}{3n^3}-\frac{1}{4n^4}+\ldots$

$=\frac{1}{n}-\frac{(1/n)^2}{2}+\frac{(1/n)^3}{3}-\frac{(1/n)^4}{4}+\ldots$

$=\log_e\left(1+\frac{1}{n}\right)=\log_e\left(\frac{n+1}{n}\right).$

27. **(c)** For continuity at all $x \in R$, we must have

$f\left(-\frac{\pi}{2}\right)=\lim_{x\to(-\pi/2)^-}(-2\sin x)$

$=\lim_{x\to(-\pi/2)^+}(A\sin x+B)$

$\Rightarrow \quad 2=-A+B \qquad \ldots(i)$

and $f\left(\frac{\pi}{2}\right)=\lim_{x\to(\pi/2)^-}(A\sin x+B)$

$=\lim_{x\to(\pi/2)^+}(\cos x)$

$\Rightarrow \quad 0=A+B \qquad \ldots(ii)$

From (i) and (ii),

$A=-1$

and $B=1$.

28. **(c)** Put $3+5x^4=t \Rightarrow 20x^3dx=dt$, then

$\int x^3\sqrt{3+5x^4}\,dx=\frac{1}{20}\int t^{1/2}dt$

$=\frac{2}{3}\times\frac{1}{20}.t^{3/2}+c=\frac{1}{30}(3+5x^4)^{3/2}+c.$

29. **(a)** Equation of line perpendicular to $5x+12y+8=0$ is

$12x-5y+k=0.$

Now it is a tangent to the circle, if

Radius of circle

= Distance of line from centre of circle

$\sqrt{121+4-25}=\left|\frac{12(11)-5(2)+k}{\sqrt{144+25}}\right|$

$\Rightarrow \quad k=8$ or $-252.$

Hence equations of tangents are

$12x-5y+8=0$ and

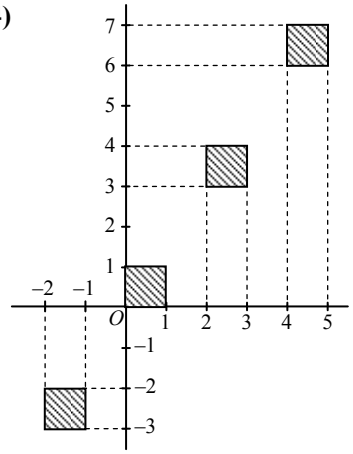
$12x-5y=252$

30. **(d)** $\sim p\wedge q=\sim(q\to p)$.

JEE Advance Paper-I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
4	8	4	8	5	1	0	4	a,c,d	c, d
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b,c	b,d	a,b	a,d	a,c	b,c	a,d	a,b,c	c	c

1. **(4)**



Required area $=4\times(1\times1)=4$ sq unit

$\therefore \quad \lambda=4$

$\sqrt{\lambda\sqrt{\lambda\sqrt{\lambda\sqrt{\lambda\ldots\infty}}}}=\lambda^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\infty}=\lambda^{\frac{1/2}{1-1/2}}=\lambda=4$

2. **(8)** $\frac{5}{4}\cos^2 2x+\cos^4 x+\sin^4 x+\cos^6 x+\sin^6 x=2$

$\Rightarrow \quad \frac{5}{4}\cos^2 2x-5\cos^2 x\sin^2 x=0$

$\Rightarrow \quad \tan^2 2x=1$, where $2x\in[0,4\pi]$

Number of solutions = 8

3. **(4)** Image of $y=-5$ about the line $x+y+4=0$ is $x=1$

$\Rightarrow$ Distance $AB=4$

4. **(8)** Let coin was tossed 'n' times

Probability of getting at least two heads $=1-\left[\frac{1}{2^n}+\frac{n}{2^n}\right]$

$\Rightarrow \quad 1-\left[\frac{n+1}{2^n}\right]\geq 0.96$

$\Rightarrow \quad \frac{2^n}{n+1}\geq 25 \Rightarrow n\geq 8$

5. **(5)** $n=6!.5!$ (5 girls together arranged along with 5 boys) $m={}^5C_4.(7!-2.6!).4!$

(4 out of 5 girls together arranged with others – number of cases all 5 girls are together)

$\frac{m}{n}=\frac{5\cdot5\cdot6!\cdot4!}{6!\cdot5!}$

6. **(1)** Let Parabola be $y^2 = 4ax$ and coordinates of P and Q on this parabola are $P \equiv (at_1^2, 2at_1)$ and $Q \equiv (at_2^2, 2at_2)$; T is the point of intersection of tangents at t_1 and t_2.

$\therefore$ Coordinates of $T \equiv \{at_1, t_2, a(t_1 + t_2)\}$

Similarly, $P' \equiv \{at_3, t_1, a(t_3 + t_1)\}$

$Q' \equiv \{at_2, t_3, a(t_2 + t_3)\}$

Let $TP' : TP = \lambda : 1$

$\therefore \quad \lambda = \dfrac{t_3 - t_2}{t_1 - t_2}$

or $\quad \dfrac{TP'}{TP} = \dfrac{t_3 - t_2}{t_1 - t_2}$

Similarly, $\dfrac{TQ'}{TQ} = \dfrac{t_1 - t_3}{t_1 - t_2}$

or $\quad \dfrac{TP'}{TP} = \dfrac{TQ'}{TQ} = 1$

7. **(0)** $I = \displaystyle\int_{-1}^{0} \frac{x \cdot 0}{2+0}\,dx + \int_{0}^{1} \frac{x \cdot 0}{2+1}\,dx + \int_{1}^{\sqrt{2}} \frac{x \cdot 1}{2+0}\,dx + 0 = \frac{1}{4}$

$\Rightarrow \quad 4I = 1 = 0$

8. **(4)** Let inner radius be r and inner length be ℓ

$\pi r^2 \ell = V$

Volume of material be M,

$M = \pi(r+2)^2(\ell+2) - \pi r^2 \ell$

$\dfrac{dM}{dr} = -\dfrac{4V}{r^2} - \dfrac{8V}{r^3} + 8\pi + 0 + 4\pi r$

$\dfrac{dM}{dr} = 0$ when $r = 10$

$\Rightarrow \quad V = 1000\pi$

$\Rightarrow \quad \dfrac{V}{250\pi} = 4$

9. **(a,c,d)** $|\vec{b} + \vec{c}| = |\vec{a}|$

$\Rightarrow \quad |\vec{b}|^2 + |\vec{c}|^2 = 2\vec{b} \cdot \vec{c} = |\vec{a}|^2$

$\Rightarrow \quad 48 + |\vec{c}|^2 + 48 = 144$

$\Rightarrow \quad |\vec{c}| = 4\sqrt{3} \therefore \dfrac{|\vec{c}|^2}{2} - |\vec{a}| = 12$

Also, $|\vec{a} + \vec{b}| = |\vec{c}|$

$\Rightarrow \quad |\vec{a}|^2 + |\vec{b}|^2 = 2\vec{a} \cdot \vec{b} = |\vec{c}|^2$

$\Rightarrow \quad \vec{a} \cdot \vec{b} = -72 \quad \vec{a} + \vec{b} + \vec{c} = 0$

$\Rightarrow \quad \vec{a} \times \vec{b} = \vec{c} \times \vec{a} \Rightarrow |\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = 2|\vec{a} \times \vec{b}| = 48\sqrt{3}$

10. **(c,d)** $(Y^3Z^4 - Z^4Y^3)^T = (Z^T)^4(Y^T)^3 - (Y^T)^3(Z^T)^4$

$= -Z^4Y^3 + Y^3Z^4$

$\Rightarrow$ symmetric $X^{44} + Y^{44}$ is symmetric

$X^4Z^3 - Z^3X^4$ skew symmetric

$X^{23} + Y^{23}$ skew symmetric.

11. **(b,c)** We get $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 5+2\alpha & 5+4\alpha & 5+6\alpha \end{vmatrix}$

$= -648\alpha \; (R_3 \to R_3 - R_2;\; R_2 \to R_2 - R_1)$

$\begin{vmatrix} \alpha^2 - 2 & 4\alpha^2 - 2 & 9\alpha^2 - 2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 2 & 2 & 2 \end{vmatrix}$

$= -648\alpha \; (R_1 \to R_1 - R_2;\; R_3 \to R_3 - R_2)$

$\Rightarrow \quad \begin{vmatrix} -2\alpha^2 & -5\alpha^2 & -9\alpha^2 - 3 \\ -2\alpha & -2\alpha & 3+6\alpha \\ 0 & 0 & 2 \end{vmatrix} = -648\alpha$

12. **(b,d)** Let the required plane be $x + z + \lambda y - 1 = 0$

$\Rightarrow \quad \dfrac{|\lambda - 1|}{\sqrt{\lambda^2 + 2}} = 1 \Rightarrow \lambda = -\dfrac{1}{2}$

$\Rightarrow \quad P_3 \equiv 2x - y + 2z - 2 = 0$

Distance of P_3 from (α, β, γ) is 2

$\dfrac{|2\alpha - \beta + 2\gamma - 2|}{\sqrt{4 \times 1 + 4}} = 2$

$\Rightarrow \quad 2\alpha - \beta + 2\lambda + 4 = 0$ and $2\alpha - \beta + 2\lambda - 8 = 0$

13. **(a,b)** Line L will be parallel to the line of intersection of P_1 and P_2

Let a, b and c be the direction ratios of line L

$\Rightarrow \quad a + 2b - c = 0$ and $2a - b + c = 0$

$\Rightarrow \quad a : b : c :: 1 : -3 : -5$

Equation of line L is $\dfrac{x-0}{1} = \dfrac{y-0}{-3} = \dfrac{z-0}{-5}$

Again foot of perpendicular from origin to plane P_1 is $\left(-\dfrac{1}{6}, -\dfrac{1}{3}, \dfrac{1}{6}\right)$

$\therefore$ Equation of project of line L on plane P_1 is

$\dfrac{x + \frac{1}{6}}{1} = \dfrac{y + \frac{2}{6}}{-3} = \dfrac{z - \frac{1}{6}}{-5} = k$

Clearly points $\left(0, -\dfrac{5}{6}, -\dfrac{2}{3}\right)$ and $\left(-\dfrac{1}{6}, -\dfrac{1}{3}, \dfrac{1}{6}\right)$ satisfy the line of projection i.e. M

14. **(a,d)** $P(at^2, 2at) \Rightarrow Q\left(\frac{16a}{t^2}, -\frac{8a}{t}\right)$

$\Delta OPQ = \frac{1}{2} OP \cdot OQ$

$\Rightarrow \frac{1}{2}\left|at\sqrt{t^2+4} \cdot \frac{a(4)}{t}\sqrt{\frac{16}{t^2}+4}\right| = 3\sqrt{2}$

$t^2 - 3\sqrt{2}t + 4 = 0$

$\Rightarrow t = \sqrt{2}, 2\sqrt{2}$

Hence, $P(at^2, 2at) = P\left(\frac{t^2}{2}, t\right)$

$t = \sqrt{2} \Rightarrow P(1, \sqrt{2})$

$t = 2\sqrt{2} \Rightarrow P(4, 2\sqrt{2})$

15. **(a,c)** $\frac{dy}{dx} + \frac{ye^x}{1+e^x} = \frac{1}{e^x+1}$

$I.F. = \int e^{\frac{e^x}{1+e^x}dx} = e^{\ln(1+e^x)} = 1+e^x$

$\Rightarrow y(1+e^x) = \int 1dx \Rightarrow y = \frac{x+c}{1+e^x}$

$y(0) = 2$

$\Rightarrow c = 1$

$\Rightarrow y = \frac{x+4}{1+e^x}$

$\Rightarrow y(-4) = 0$

$\Rightarrow y' = \frac{(1+e^x)-(x+4)e^x}{(1+e^x)^2} = 0$

Let $g(x) = \frac{(1+e^x)-(x+4)e^x}{(1+e^x)^2}$

$g(0) = \frac{2-4}{2^2} < 0$

$g(-1) = \frac{\left(1+\frac{1}{e}\right)-\frac{3}{e}}{\left(1+\frac{1}{e}\right)^2} < 0 = \frac{1-\frac{2}{e}}{\left(1+\frac{1}{e}\right)^2} > 0$

$g(0) \cdot g(-1) < 0$. Hence g(x) has a root in between $(-1, 0)$

16. **(b,c)** Let the family of circles be $x^2 + y^2 - \alpha x - \alpha y + c = 0$

On differentiation $2x + 2yy' - \alpha - \alpha y' = 0$

Again on differentiation and substituting 'α' we get

$2x + 2y'^2 + 2yy'' - \left(\frac{2x+2yy'}{1+y'}\right)y'' = 0$

$\Rightarrow (y-x)y'' + y'(1+y'+y'^2) + 1 = 0$

17. **(a,d)** Differentiability of $f(x)$ at $x = 0$

LHD $f'(0^-) = \lim_{\delta\to 0}\left(\frac{f(0)-f(0-\delta)}{\delta}\right) = \lim_{\delta\to 0}\frac{0+g(-\delta)}{\delta} = 0$

RHD $f'(0^+) = \lim_{\delta\to 0}\left(\frac{f(0+\delta)-f(0)}{\delta}\right) = \lim_{\delta\to 0}\frac{g(\delta)}{\delta} = 0$

$\Rightarrow$ $f(x)$ is differentiable at $x = 0$

Differentiabiligy of $h(x)$ at $x = 0$

$h'(0^+) = 1, h(x)$ is an even function

Hence non diff. at $x = 0$

Differentiability of $f(h(x))$ at $x = 0$

$f(h(x)) = g(e^{|x|}) \ \forall \ x \in R$

LHD $f'(h(0^-)) = \lim_{\delta\to 0}\frac{f(h(0))-f(h(0-\delta))}{\delta}$

$= \lim_{\delta\to 0}\frac{g(1)-g(e^\delta)}{\delta} = g'(1)$

RHD $f'(h(0^+)) = \lim_{\delta\to 0}\frac{f(h(0+\delta))-f(h(0))}{\delta}$

$= \lim_{\delta\to 0}\frac{g(e^\delta)-g(1)}{\delta} = g'(1)$ Since $g'(1) \neq 0$

$\Rightarrow$ $f(h(x))$ is non diff. at $x = 0$

Differentiability of $h(f(x))$ at $x = 0$

$h(f(x)) = \begin{cases} e^{(f(x)|}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

LHD. $h'(f(0-\delta)) = \lim_{\delta\to 0}\frac{h(f(0))-h(f(0-\delta))}{\delta}$

$= \lim_{\delta\to 0}\frac{1-e^{|g(-\delta)|}}{|g(-\delta)|} \cdot \frac{|g(-\delta)|}{\delta} = 0$

18. **(a,b,c)** Given $g(x) = \frac{\pi}{2}\sin x \ \ \forall \ x \in R$

$f(x) = \sin\left(\frac{1}{3}g(g(x))\right)$

Also, $g(g(g(x))) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \forall \ x \in R$

Hence, $f(x)$ and $f(g(x)) \in \left[-\frac{1}{2}, \frac{1}{2}\right]$

$\lim_{x\to 0}\frac{f(x)}{g(x)} = \lim_{x\to 0}\frac{\sin\left(\frac{1}{3}g(g(x))\right)}{\frac{1}{3}g(g(x))} \cdot \frac{\frac{1}{3}g(g(x))}{g(x)}$

$\Rightarrow \lim_{x\to 0}\frac{\pi}{6} \cdot \frac{\sin\left(\frac{\pi}{2}\sin x\right)}{\frac{\pi}{2}\sin x} = \frac{\pi}{6}$

Range of $g(f(x)) \in \left[-\frac{\pi}{2}\sin\left(\frac{1}{2}\right), \frac{\pi}{2}\sin\left(\frac{1}{2}\right)\right]$

$\Rightarrow \quad g(f(x)) \neq 1.$

19. (c) (A) $\left|\frac{\sqrt{3}\alpha+\beta}{2}\right| = \sqrt{3}$

$\sqrt{3}\alpha+\beta = \pm 2\sqrt{3}$ . . .(*i*)

Given $\alpha = 2+\sqrt{3}\beta$ . . .(*ii*)

From equation (*i*) and (*ii*), we get $\alpha = 2 \; or \; -1$

So $|\alpha| = 1$ or 2

(B) $f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ bx + a^2, & x \geq 1 \end{cases}$

For continuity $-3a - 2 = b + a^2$

$a^2 + 3a + 2 = -b$ . . .(*i*)

For differentiability $-6a = b$

$6a = -b$

$a^2 - 3a + 2 = 0$

$a = 1, 2$

(C) $(3 - 3\omega + 2\omega^2)^{4n+3} + (2 + 3\omega - 3\omega^2)^{4n+3}$

$+(-3 + 2\omega + 3\omega^2)^{4n+3} = 0$

$(3 - 3\omega + 2\omega^2)^{4n+3} + (\omega(2\omega^2 + 3 - 3\omega))^{4n+3}$

$+(\omega^2(-3\omega + 2\omega^2 + 3)^{4n+3} = 0$

$\Rightarrow \quad (3 - 3\omega + 2\omega^2)^{4n+3} + (1 + \omega^{4n} + \omega^{8n}) = 0$

$\Rightarrow \quad n \neq 3k, k \in N$

(D) Let $a = 5 - d$

$q = 5 + d$

$b = 5 + 2d$

$\Rightarrow \quad |q - a| = |2d|$

Given $\frac{2ab}{a+b} = 4$

$\Rightarrow \quad \frac{ab}{a+b} = 2$

$(5-d)(5+2d) = 2(5-d+5+2d) = 2(10+d)$

$25 + 10d - 5d - 2d^2 = 20 + 2d$

$2d^2 - 3d - 5 = 0$

$d = -1, d = \frac{5}{2} \quad \Rightarrow |2d| = 2, 5$

20. (c) (A) $a^2 - b^2 = \frac{c^2}{2}$ (given)

$4R^2(\sin^2 X - \sin^2 Y) = \frac{4R^2}{2}\sin^2(Z)$

$\Rightarrow \quad 2(\sin(X-Y)\cdot\sin(X+Y) = \sin^2(Z)$

$\Rightarrow \quad 2\cdot\sin(X-Y)\cdot\sin(Z) = \sin^2(Z)$

$\Rightarrow \quad \frac{\sin(X-Y)}{\sin Z} = \frac{1}{2} = \lambda$

$\Rightarrow \quad \cos\left(\frac{n\pi}{2}\right) = 0$ for n = odd integer.

(B) $1 + \cos 2X - 2\cos 2Y = 2\sin X \sin Y$

$\sin^2 X + \sin X \sin Y - 2\sin^2 Y = 0$

$(\sin X - \sin Y)(\sin X + 2\sin Y) = 0$

$\Rightarrow \quad \sin X = \sin Y$

$\Rightarrow \quad \frac{\sin X}{\sin Y} = \frac{a}{b} = 1.$

(C) Here, distance of Z from bisector of

$\overrightarrow{OX}$ and $\overrightarrow{OY} = \frac{3}{\sqrt{2}}$

$\Rightarrow \quad \left(\beta - \frac{1}{2}\right)^2 + \left(\beta - \frac{1}{2}\right)^2 = \frac{9}{2}$

$\Rightarrow \quad \beta = 2, -1$

$\Rightarrow \quad |\beta| = 2, 1$

(D) When $\alpha = 0$

Area $= 6 - \int_0^2 2\sqrt{x}\, dx$

$= 6 - \frac{8\sqrt{2}}{3}$

When $\alpha = 1$

Area $= \int_0^1 (3 - x - 2\sqrt{x})\, dx + \int_1^2 (x + 1 - 2\sqrt{x})\, dx$

$= \left|3x - \frac{x^2}{2} - \frac{4}{3}x^{3/2}\right|_0^1 + \left|\frac{x^2}{2} + x - \frac{4}{3}x^{3/2}\right|_1^2$

$= 5 - \frac{8}{3}\sqrt{2}.$

JEE Advance Paper-II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
9	4	9	8	4	2	9	7	d	a,d
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b,c,d	a,b	a,b,c	a,c	b,c	a,b	a,b,c	c,d	a,b	c,d

1. (9) $\vec{s} = 4\vec{p} + 3\vec{q} + 5\vec{r}$

$\vec{s} = x(-\vec{p} + \vec{q} + \vec{r}) + y(\vec{p} - \vec{q} + \vec{r}) + z(-\vec{p} - \vec{q} + \vec{r})$

$\vec{s} = (-x + y - z)\vec{p} + (x - y - z)\vec{q} + (x + y + z)\vec{r}$

$\Rightarrow \quad -x + y - z = 4$

$\Rightarrow \quad x-y-z=3$

$\Rightarrow \quad x+y+z=5$

On solving we get $x=4,\ y=\frac{9}{2},\ z=-\frac{7}{2}$

$\Rightarrow \quad 2x+y+z=9$

2. **(4)** $\dfrac{\sum_{k=1}^{12}\left|e^{i\frac{k\pi}{7}}\right|\left|e^{i\frac{\pi}{7}}-1\right|}{\sum_{k=1}^{3}\left|e^{i(4k-2)}\right|\left|e^{i\frac{\pi}{7}}-1\right|}=\frac{12}{3}=4$

3. **(9)** Let seventh term be 'a' and common difference be 'd'

Given $\frac{S_7}{S_{11}}=\frac{6}{11} \qquad \Rightarrow a=15d$

Hence, $130<15d<140 \Rightarrow d=9$

4. **(8)** x^9 can be formed in 8 ways

i.e., $x^9, x^{1+8}, x^{2+7}, x^{3+6}, x^{4+5}, x^{1+2+6}, x^{1+3+5}, x^{2+3+4}$ and

coefficient in each case is 1

$\Rightarrow$ Coefficient of $x^9=\underbrace{1+1+1+.....+1}_{8\text{ times}}=8$

5. **(4)** Let point of concurrent is (h,k)

Equation of normal at (x',y') is, $\frac{x-x'}{x'/a^2}=\frac{y_1-y'}{y'/b^2}$

It is passes through (h,k), then

$y'^2\{a^2(h-x')+b^2x'\}^2=b^4k^2x'^2 \qquad ...(i)$

But $\frac{x'^2}{a^2}=\frac{y'^2}{b^2}=1$ or $y'^2=\frac{b^2}{a^2}(a^2-x'^2) \qquad ...(ii)$

Value of y'^2 from equation (*ii*), putting in equation (*i*),

we get $\frac{b^2}{a^2}(a^2-x'^2)\{a^2h+(b^2-a^2)x'\}^2=b^4k^2x'^2$

$\Rightarrow \quad \frac{b^2}{a^2}(a^2-x'^2)\{a^4h^2+(b^2-a^2)^2x'^2+2a^2hx'(b^2-a^2)\}=b^4k^2x'^2$

Arranging above as a fourth degree equation in x', we get

$\Rightarrow \quad -(a^2-b^2)^2x'^4+2ha^2(a^2-b^2)x'^3+x'^2(...)-2a^4h(a^2-b^2)x'+a^6h^2=0$

Above equation being of fourth degree in x', therefore roots of the above equation are x_1,x_2,x_3,x_4 then

$(x_1+x_2+x_3+x_4)=-\frac{2ha^2(a^2-b^2)}{-(a^2-b^2)^2}=\frac{2ha^2}{(a^2-b^2)} \qquad ...(iii)$

$\left(\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}+\frac{1}{x_4}\right)=\frac{\Sigma x_1x_2x_3}{x_1\cdot x_2\cdot x_3\cdot x_4}$

$=\dfrac{\frac{2a^4h(a^2-b^2)}{-(a^2-b^2)^2}}{\frac{a^6h^2}{-(a^2-b^2)^2}}=\frac{2(a^2-b^2)}{a^2h} \qquad ...(iv)$

Multiplying equation (*iii*) and (*iv*), we get

$(x_1+x_2+x_3+x_4)\times\left(\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}+\frac{1}{x_4}\right)=4$

6. **(2)** $\lim_{\alpha\to0}\frac{e^{\cos(\alpha^n)}-e}{\alpha^m}=-\frac{e}{2}$

$\lim_{\alpha\to0}\frac{e\left(e^{(\cos(\alpha)^n-1)}\right)(\cos\alpha^n-1)}{\left(\cos(\alpha^n)-1\right)\alpha^m\alpha^{2n}} \quad \alpha^{2n}=-\frac{e}{2}$

if and only if $2n-m=0$

7. **(9)** $\alpha=\int_0^1 e^{(9x+3\tan^{-1}x)}\left(\frac{12+9x^2}{1+x^2}\right)dx$

Put $9x+3\tan^{-1}x=t$

$\Rightarrow \quad \left(9+\frac{3}{1+x^2}\right)dx=dt$

$\Rightarrow \quad \alpha=\int_0^{9+\frac{3\pi}{4}}e^t\,dt=e^{9+\frac{3\pi}{4}}-1$

$\Rightarrow \quad \left(\log_e|1+\alpha|-\frac{3\pi}{4}\right)=9$

8. **(7)** $G(1)=\int_{-1}^{1}t\,|f(f(t))|\,dt=0 \quad f(-x)=-f(x)$

Given $f(1)=\frac{1}{2}$

$\lim_{x\to1}\frac{F(x)}{G(x)}=\lim_{x\to1}\frac{\frac{F(x)-F(1)}{x-1}}{\frac{G(x)-G(1)}{x-1}}=\frac{f(1)}{|f(f(1))|}=\frac{1}{14}$

$\Rightarrow \quad \frac{1/2}{|f(1/2)|}=\frac{1}{14} \quad \Rightarrow f\left(\frac{1}{2}\right)=7.$

9. **(D)** $\frac{192}{3}\int_{1/2}^{x}t^3dt\le f(x)\le\frac{192}{2}\int_{1/2}^{x}t^3dt$

$16x^4-1\le f(x)\le24x^4-\frac{3}{2}$

$\int_{1/2}^{1}(16x^4-1)dx\le\int_{1/2}^{1}f(x)dx\le\int_{1/2}^{1}\left(24x^2-\frac{3}{2}\right)dx$

$1<\frac{26}{10}\le\int_{1/2}^{1}f(x)dx\le\frac{39}{10}<12$

10. **(a,d)** Here, $0<(x_1-x_2)^2<1$

$\Rightarrow \quad 0<(x_1+x_2)^2-4x_1x_2<1$

$\Rightarrow \quad 0<\frac{1}{\alpha^2}-4<1$

$\Rightarrow \quad \alpha\in\left(-\frac{1}{2},-\frac{1}{\sqrt{5}}\right)\cup\left(\frac{1}{\sqrt{5}},\frac{1}{2}\right)$

11. **(b,c,d)** $\frac{\pi}{2} < \alpha < \pi, \pi < \beta < \frac{3\pi}{2}$

$\Rightarrow \frac{3\pi}{2} < \alpha + \beta < \frac{5\pi}{2}$

$\Rightarrow \sin\beta < 0; \cos\alpha < 0$

$\Rightarrow \cos(\alpha + \beta) > 0.$

12. **(a,b)** For the given line, point of contact for

$E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\left(\frac{a^2}{3}, \frac{b^2}{3}\right)$

and for $E_2 : \frac{x^2}{B^2} + \frac{y^2}{A^2} = 1$ is $\left(\frac{B^2}{3}, \frac{A^2}{3}\right)$

Point of contact $x + y = 3$ of and circle is (1, 2)

Also, general point on $x + y = 3$ can be taken as $\left(1 \mp \frac{r}{\sqrt{2}}, 2 \pm \frac{r}{\sqrt{2}}\right)$ where, $r = \frac{2\sqrt{2}}{3}$

So, required points are $\left(\frac{1}{3}, \frac{8}{3}\right)$ and $\left(\frac{5}{3}, \frac{4}{3}\right)$

Comparing with points of contact of ellipse,

$a^2 = 5, B^2 = 8$

$b^2 = 4, A^2 = 1$

$\therefore \quad e_1 e_2 = \frac{\sqrt{7}}{2\sqrt{10}}$ and $e_1^2 + e_2^2 = \frac{43}{40}$

13. **(a,b,d)** Tanget at P, $xx_1 - yy_1 = 1$ intersects x-axis at $M\left(\frac{1}{x_1}, 0\right)$

Slope of normal $= -\frac{y_1}{x_1} = \frac{y_1 - 0}{x_1 - x_2}$

$\Rightarrow \quad x_2 = 2x_1 \Rightarrow N \equiv (2x_1, 0)$

For centroid $\ell = \frac{3x_1 + \frac{1}{x_1}}{3}, m = \frac{y_1}{3}$

$\frac{d\ell}{dx_1} = 1 - \frac{1}{3x_1^2}$

$\frac{dm}{dy_1} = \frac{1}{3}, \frac{dm}{dx_1} = \frac{1}{3}\frac{dy_1}{dx_1} = \frac{x_1}{3\sqrt{x_1^2 - 1}}$

14. **(a,c)** Let $\int_0^{\pi} e^1(\sin^6 at + \cos^4 at)\, dt = A$

$I = \int_{\pi}^{2\pi} e^1(\sin^6 + at + \cos^4 at)\, dt$

Put $t = \pi + x$

$dt = dx$

for $a = 2$ as well as $a = 4$

Similarly

$\int_{2\pi}^{3\pi} e^1(\sin^6 at + \cos^4 at)\, dt = e^{2\pi} A$

So, $L = \frac{A + e^{\pi} A + e^{2\pi} A + e^{3\pi} A}{A} = \frac{3^{4\pi} - 1}{e^{\pi} - 1}$

For both $a = 2, 4$

15. **(b,c)** Let $H(x) = f(x) - 3g(x)$

$H(-1) = H(0) = H(2) = 3.$

Applying Rolle's Theorem in the interval [–1, 0]

$H'(x) = f'(x) - 3g'(x) = 0$ for atleast one $c \in (-1, 0)$

As $H''(x)$ never vanishes in the interval

$\Rightarrow$ Exactly one $c \in (-1, 0)$ for which $H'(x) = 0$

Similarly, apply Rolle's Theorem in the interval [0, 2]

$\Rightarrow \quad H'(x) = 0$ has exactly one solution in (0, 2)

16. **(a,b)** $f(x) = (7\tan^6 x - 3\tan^2 x)(\tan^2 x + 1)$

$\int_0^{\pi/4} f(x)dx = \int_0^{\pi/4} (7\tan^6 x - 3\tan^2 x)\sec^2 x dx$

$\Rightarrow \quad \int_0^{\pi/4} f(x)\, dx = 0$

$\int_0^{\pi/4} xf(x)dx = \left[x\int f(x)\, dx\right]_0^{\pi/4} - \int_0^{\pi/4}\left[\int f(x)\, dx\right] dx$

$\int_0^{\pi/4} xf(x)\, dx = \frac{1}{12}$

17. **(a,b,c)** (a) $f'(x) = f(x) + xF'(x)$

$f'(1) = F(1) + F'(1)$

$f'(1) = F'(1) < 0$

$f'(1) < 0$

(b) $F(2) = 2F(2)$

$F(x)$ is decreasing and $F(1) = 0$

Hence $F(2) < 0$

$\Rightarrow \quad f(2) < 0$

(c) $f'(x) = F(x) + xF'(x)$

$F(x) < 0 \ \forall \ x \in (1, 3)$

$F'(x) < 0 \ \forall \ x \in (1, 3)$

Hence $f'(x) < 0 \ \forall \ x \in (1, 3)$

JEE-MAIN: MATHEMATICS MOCK TEST-2

1. If $f:[1,\infty)\to[2,\infty)$ is given by $f(x)=x+\frac{1}{x}$ then $f-1(x)$ equals:

a. $\frac{x+\sqrt{x^2-4}}{2}$ **b.** $\frac{x}{1+x^2}$

c. $\frac{x-\sqrt{x^2-4}}{2}$ **d.** $1+\sqrt{x^2-4}$

2. Let α and β be the roots of the equation $x^2+x+1=0$ The equation whose roots are α^{19},β^7 is

a. $x^2-x-1=0$ **b.** $x^2-x+1=0$

c. $x^2+x-1=0$ **d.** $x^2+x+1=0$

3. Let $\omega_n=\cos\left(\frac{2\pi}{n}\right)+i\sin\left(\frac{2\pi}{n}\right), i^2=-1$, then $(x+y\omega_3+z\omega_3^2)\ (x+y\omega_3^2+z\omega_3)$ is equal to

a. 0

b. $x^2+y^2+z^2$

c. $x^2+y^2+z^2-yz-zx-xy$

d. $x^2+y^2+z^2+yz+zx+xy$

4. The value of $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7}-i\cos\frac{2\pi k}{7}\right)$ is

a. –1 **b.** 0

c. $-i$ **d.** i

5. The inverse of the matrix $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$ is

a. $\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$ **b.** $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

c. $\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$ **d.** $\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$

6. The sum of the first five terms of the series $3+4\frac{1}{2}+6\frac{3}{4}+.....$ will be

a. $39\frac{9}{16}$ **b.** $18\frac{3}{16}$

c. $39\frac{7}{16}$ **d.** $13\frac{9}{16}$

7. If $(1+x-2x^2)^6=1+a_1x+a_2x^2+....+a_{12}x^{12}$, then the expression $a_2+a_4+a_6+....+a_{12}$ has the value

a. 32 **b.** 63

c. 64 **d.** None of these

8. $1+\frac{1+2}{2!}+\frac{1+2+3}{3!}+\frac{1+2+3+4}{4!}+\ldots\infty=$

a. e **b.** $3e$ **c.** $e/2$ **d.** $3e/2$

9. $\log_e 2+\log_e\left(1+\frac{1}{2}\right)+\log_e\left(1+\frac{1}{3}\right)+....+\log_e\left(1+\frac{1}{n-1}\right)$ is equal to

a. $\log_e 1$ **b.** $\log_e n$

c. $\log_e(1+n)$ **d.** $\log_e(1-n)$

10. There are 10 lamps in a hall. Each one of them can be switched on independently. The number of ways in which the hall can be illuminated is.

a. 10^2 **b.** 1023

c. 2^{10} **d.** 10 !

11. A fair coin is tossed repeatedly. If tail appears on first four tosses, then the probability of head appearing on fifth toss equals

a. $\frac{1}{2}$ **b.** $\frac{1}{32}$

c. $\frac{31}{32}$ **d.** $\frac{1}{5}$

12. If $x\sin45°\cos^2 60°=\frac{\tan^2 60°\operatorname{cosec}30°}{\sec45°\cot^2 30°}$, then $x=$

a. 2 **b.** 4 **c.** 8 **d.** 16

13. If $\sin\theta+\sin2\theta+\sin3\theta=\sin\alpha$ and $\cos\theta+\cos2\theta+\cos3\theta=\cos\alpha$,then θ is equal to

a. $\alpha/2$ **b.** α **c.** 2α **d.** $\alpha/6$

14. Find imaginary part of $\sin^{-1}\left(\frac{5\sqrt{7}-9i}{16}\right)$

a. $\log 2$ **b.** $-\log 2$

c. 0 **d.** None of these

15. An observer on the top of a tree, finds the angle of depression of a car moving towards the tree to be 30°. After 3 minutes this angle becomes 60°. After how much more time, the car will reach the tree

a. 4 min **b.** 4.5 min **c.** 1.5 min **d.** 2 min

16. Let $f(x)=\begin{cases}\dfrac{x^3+x^2-16x+20}{(x-2)^2}, \text{if } x\neq 2\\ k, \text{if } x=2\end{cases}$. If $f(x)$ be continuous for all x, then $k=$

a. 7 **b.** –7

c. ± 7 **d.** None of these

17. If $y=e^{x+e^{x+e^{x+\ldots\infty}}}$, then $\dfrac{dy}{dx}=$

a. $\dfrac{y}{1-y}$ **b.** $\dfrac{1}{1-y}$

c. $\dfrac{y}{1+y}$ **d.** $\dfrac{y}{y-1}$

18. One maximum point of $\sin^p x\cos^q x$ is

a. $x=\tan^{-1}\sqrt{(p/q)}$ **b.** $x=\tan^{-1}\sqrt{(q/p)}$

c. $x=\tan^{-1}(p/q)$ **d.** $x=\tan^{-1}(q/p)$

19. $\int \dfrac{1}{x(\log x)^2}dx=$

a. $\dfrac{1}{\log x}+c$ **b.** $-\dfrac{1}{\log x}+c$

c. $\log\log x+c$ **d.** $-\log\log x+c$

20. Area bounded by the parabola $y^2=2x$ and the ordinates $x=1, x=4$ is

a. $\dfrac{4\sqrt{2}}{3}sq.unit$ **b.** $\dfrac{28\sqrt{2}}{3}sq.unit$

c. $\dfrac{56}{3}sq.unit$ **d.** None of these

21. The solution of the differential equation $\dfrac{dy}{dx}+\dfrac{1+x^2}{x}=0$ is

a. $y=-\dfrac{1}{2}\tan^{-1}x+c$ **b.** $y+\log x+\dfrac{x^2}{2}+c=0$

c. $y=\dfrac{1}{2}\tan^{-1}x+c$ **d.** $y-\log x-\dfrac{x^2}{2}=c$

22. The solution of the differential equation $\cos y\log(\sec x+\tan x)dx=\cos x\log(\sec y+\tan y)dy$ is

a. $\sec^2 x+\sec^2 y=c$ **b.** $\sec x+\sec y=c$

c. $\sec x-\sec y=c$ **d.** None of these

23. The vertex of an equilateral triangle is (2,–1) and the equation of its base in $x+2y=1$. The length of its sides is

a. $4/\sqrt{15}$ **b.** $2/\sqrt{15}$

c. $4/3\sqrt{3}$ **d.** $1/\sqrt{5}$

24. If the equation $\dfrac{K(x+1)^2}{3}+\dfrac{(y+2)^2}{4}=1$ represents a circle, then $K=$

a. 3/4 **b.** 1 **c.** 4/3 **d.** 12

25. The equation of the common tangent to the curves $y^2=8x$ and $xy=-1$ is

a. $3y=9x+2$ **b.** $y=2x+1$

c. $2y=x+8$ **d.** $y=x+2$

26. The vectors $\vec{a}$ and $\vec{b}$ are non-collinear. The value of x for which the vectors $\vec{c}=(x-2)\vec{a}+\vec{b}$ and $\vec{d}=(2x+1)\vec{a}-\vec{b}$ are collinear, is

a. 1 **b.** $\dfrac{1}{2}$

c. $\dfrac{1}{3}$ **d.** None of these

27. If vectors $\vec{a},\vec{b},\vec{c}$ satisfy the condition $|\vec{a}-\vec{c}|=|\vec{b}-\vec{c}|$, then $(\vec{b}-\vec{a})\cdot\left(\vec{c}-\dfrac{\vec{a}+\vec{b}}{2}\right)$ is equal to

a. 0 **b.** –1 **c.** 1 **d.** 2

28. The angle between the straight lines $\dfrac{x+1}{2}=\dfrac{y-2}{5}=\dfrac{z+3}{4}$ and $\dfrac{x-1}{1}=\dfrac{y+2}{2}=\dfrac{z-3}{-3}$ is

a. 45° **b.** 30° **c.** 60° **d.** 90°

29. A variable plane is at a constant distance p from the origin and meets the axes in A, B and C. The locus of the centroid of the tetrahedron $OABC$ is

a. $x^{-2}+y^{-2}+z^{-2}=16p^{-2}$ **b.** $x^{-2}+y^{-2}+z^{-2}=16p^{-1}$

c. $x^{-2}+y^{-2}+z^{-2}=16$ **d.** None of these

30. $\sim(p\vee q)$ is equal to

a. $\sim p\vee\sim q$ **b.** $\sim p\wedge\sim q$

c. $\sim p\vee q$ **d.** $p\vee\sim q$

Space for rough work

JEE ADVANCE PAPER-I

SECTION 1 Contains 8 Questions.

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. The coefficient of x^9 in the expansion of $(1+x)(1+x^2)(1+x^3).....(1+x^{100})$ is

2. The minimum value of the sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} with $a > 0$ is

3. The value of

$$6+\log_{\frac{3}{2}}\left(\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}}}}\right) \text{is}$$

4. The number of distinct solutions of the equation $\frac{5}{4}\cos^2 2x+\cos^4 x+\sin^4 x+\cos^6 x+\sin^6 x=2$ in the interval $[0, 2\pi]$ is

5. A cylindrical container is to be made from certain solid material with the following constraints: It has a fixed inner volume of V mm^3, has a 2 mm thick solid wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container.
If the volume of the material used to make the container is minimum when the inner radius of the container is 10 mm, then the value of $\frac{V}{250\pi}$ is

6. Let the curve C be the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If A and B are the points of intersection of C with the line $y = -5$, then the distance between A and B is

7. Let $f : R \to R$ be a continuous odd function, which vanishes exactly at one point and $f(1)=\frac{1}{2}$. Suppose that $F(x)=\int_{-1}^{x} f(t)\,dt$ for all $x\in[1-1, 2]$ and $G(x)=\int_{-1}^{x} t\,|f(f(t))|\,dt$ for all $x\in[-1, 2]$. If $\lim_{x\to 1}\frac{F(x)}{G(x)}=\frac{1}{14}$, then the value of $f\left(\frac{1}{2}\right)$ is

8. Consider the set of eight vectors $V=\{a\hat{i}+b\hat{j}+c\hat{k}; a,b,c\in\{-1,1\}\}$. Three non-coplanar vectors can be chosen from V *in* 2^p ways. Then p is.

SECTION 2 Contains 10 Multiple Choice Questions

With one or more than one correct option

9. Let f be a non-negative function defined on the interval $[0, 1]$. If $\int_0^x \sqrt{1-(f'(t))^2}\,dt=\int_0^x f(t)\,dt,\ 0\le x\le 1$, and $f(0) = 0$, then

a. $f\left(\frac{1}{2}\right)<\frac{1}{2}$ and $f\left(\frac{1}{3}\right)>\frac{1}{3}$

b. $f\left(\frac{1}{2}\right)>\frac{1}{2}$ and $f\left(\frac{1}{3}\right)>\frac{1}{3}$

c. $f\left(\frac{1}{2}\right)<\frac{1}{2}$ and $f\left(\frac{1}{3}\right)<\frac{1}{3}$

d. $f\left(\frac{1}{2}\right)>\frac{1}{2}$ and $f\left(\frac{1}{3}\right)<\frac{1}{3}$

10. Let α and β be the roots of $x^2-6x-2=0$, with $\alpha>\beta$. If $a_n=\alpha^n-\beta^n$ for $n\ge 1$, then the value of $\frac{a_{10}-2a_8}{2a_9}$ is

a. 1 **b.** 2
c. 3 **d.** 4

11. Let $z=\cos\theta+i\sin\theta$. Then the value of $\sum_{m=1}^{15}\text{Im}(z^{2m-1})$ at $\theta=2°$ is

a. $\frac{1}{\sin 2°}$ **b.** $\frac{1}{3\sin 2°}$

c. $\frac{1}{2\sin 2°}$ **d.** $\frac{1}{4\sin 2°}$

12. If $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ a & 2 & b\end{bmatrix}$ is a matrix satisfying the equation $AA^T=9I$, where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to:

a. $(2,-1)$ **b.** $(-2,1)$
c. $(2,1)$ **d.** $(-2,-1)$

13. The sum of first 20 terms of the sequence $0.7, 0.77, 0.777, ...$ is

a. $\frac{7}{81}(179-10^{-20})$ **b.** $\frac{7}{9}(99-10^{-20})$

c. $\frac{7}{81}(179+10^{-20})$ **d.** $\frac{7}{9}(99+10^{-20})$

14. Coefficient of x^{11} in the expansion of $(1+x^2)^4(1+^3)^7(1+x^4)^{12}$ is

a. 1051 **b.** 1106
c. 1113 **d.** 1120

15. The number of seven digit integers, with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only, is

a. 55 **b.** 66
c. 77 **d.** 88

16. The mean of the data set comprising of 16 observations is 16. If one of the observation valued 16 is deleted and three new observations values 3, 4 and 5 are added to the data, then the mean of the resultant data, is

a. 16.8 **b.** 16.0
c. 15.8 **d.** 14.0

17. Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b and c denote the lengths of the sides opposite to A, B and C respectively. The value (s) of x for which $a = x^2 + x + 1, b = x^2 - 1$ and $c = 2x + 1$ is (are)

a. $-(2+\sqrt{3})$ **b.** $1+\sqrt{3}$
c. $2+\sqrt{3}$ **d.** $4\sqrt{3}$

18. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$, where $x \in R$ and $k \geq 1$.

Then $f_4(x) - f_6(x)$ equals:

a. $\frac{1}{6}$ **b.** $\frac{1}{3}$
c. $\frac{1}{4}$ **d.** $\frac{1}{12}$

SECTION 3 Contains 2 Match The Following Type Questions

You will have to match entries in Column I with the entries in Column II.

19. Match the Column:

Column I	Column II
(A) In R^2, if the magnitude of the projection vector of the vector $\alpha\hat{i}+\beta\hat{j}$ on $\sqrt{3}\hat{i}+\hat{j}$ is $\sqrt{3}$ and if $\alpha \mid = 2+\sqrt{3}\beta$, then possible value(s) of $\lvert\alpha\rvert$ is (are)	**1.** 1
(B) Let a and b be real numbers such that the function $f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ bx + a^2, & x \geq 1 \end{cases}$ is differentiable for all $x \in R$. Then possible value(s) of a is (are)	**2.** 2
(C) Let $\omega \neq 1$ be a complex cube root of unity. If $(3-3\omega+2\omega^2)^{4n+3} + (2+3\omega-3\omega^2)^{4n+3} + (-3+2\omega+3\omega^2)^{4n+3} = 0$, then possible value(s) of n is (are)	**3.** 3
(D) Let the harmonic mean of two positive real numbers a and b be 4. If q is a positive real number such that a, 5, q, b is an arithmetic progression, then the value(s) of $\lvert q-a \rvert$ is (are)	**4.** 4
	5. 5

20. Consider the following linear equations

$ax + by + cz = 0$
$bx + cy + az = 0$
$cx + ay + bz = 0$

Match the conditions / expressions in Column I with statements in Column II

Column I	Column II
(A) $a+b+c \neq 0$ and $a^2+b^2+c^2 = ab+bc+ca$	**1.** the equations represent planes meeting only at a single point.
(B) $a+b+c = 0$ and $a^2+b^2+c^2 \neq ab+bc+ca$	**2.** the equations represent the line $x = y = z$.
(C) $a+b+c \neq 0$ and $a^2+b^2+c^2 \neq ab+bc+ca$	**3.** the equations represent identical planes.
(D) $a+b+c = 0$ and $a^2+b^2+c^2 = ab+bc+ca$	**4.** the equations represent the whole of the three dimensional space.

Space for rough work

JEE ADVANCE PAPER-II

SECTION 1 Contains 8 Questions.

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is

2. The number of all possible values of θ, where $0 < \theta < \pi$, for which the system of equations $(y+z)\cos 3\theta = (xyz)\sin 3\theta$ $x\sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$ $(xyz)\sin 3\theta = (y+2z)\cos 3\theta + y\sin 3\theta$ have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is

3. Let $\alpha, \beta \in R$ be such that $\lim_{x\to 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals

4. The centres of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be the mid-point of the line segment joining the centres of C_1 and C_2 and C be a circle touching circles C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C, then the radius of the circle C is

5. Suppose that $\vec{p}, \vec{q}$ and $\vec{r}$ are three non-coplanar vectors in R^3. Let the components of a vector $\vec{s}$ along $\vec{p}, \vec{q}$ and $\vec{r}$ be 4, 3 and 5, respectively. If the components of this vector $\vec{s}$ along $(-\vec{p}+\vec{q}+\vec{r})$, $(\vec{p}-\vec{q}+\vec{r})$ and $(-\vec{p}-\vec{q}+\vec{r})$ are x, y and z, respectively, then the value of $2x + y + z$ is

6. Suppose that the foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ are $(f_1, 0)$ and $(f_2, 0)$ where $f_1 > 0$ and $f_2 < 0$. Let P_1 and P_2 be two parabolas with a common vertex at (0, 0) and with foci at $(f_1, 0)$ and $(2f_2, 0)$, respectively. Let T_1 be a tangent to P_1 which passes through $(2f_2, 0)$ and T_2 be a tangent to P_2 which passes through $(f_1, 0)$. The m_1 is the slope of T_1 and m_2 is the slope of T_2, then the value of $\left(\frac{1}{m^2} + m_2^2\right)$ is

7. Let $f : IR \to IR$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is

8. The total number of distinct $x \in R$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 0$ is

SECTION 2 Contains 8 Multiple Choice Questions

With one or more than one correct option

9. If $X = \{4^n - 3n - 1 : n \in N\}$ and $Y = \{9(n-1) : n \in N\}$, where N is the set of natural numbers, then $X \cup Y$ is equal to:

a. N **b.** Y – X **c.** X **d.** Y

10. A value of b for which the equations $x^2 + bx - 1 = 0$, $x^2 + x + b = 0$, Have one root in common is

a. $-\sqrt{2}$ **b.** $-i\sqrt{3}$ **c.** $i\sqrt{5}$ **d.** $\sqrt{2}$

11. The variance of first 50 even natural numbers is:

a. $\frac{833}{4}$ **b.** 833 **c.** 437 **d.** $\frac{437}{4}$

12. The expression $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$ can be written as

a. $\sin A \cos A + 1$ **b.** $\sec A \operatorname{cosec} A + 1$

c. $\tan A + \cot A$ **d.** $\sec A + \operatorname{cosec} A$

13. The value of $\cot\left(\sum_{n=1}^{23} \cot^{-1}\left(1 + \sum_{k=1}^{n} 2k\right)\right)$ is

a. $\frac{23}{25}$ **b.** $\frac{25}{23}$

c. $\frac{23}{24}$ **d.** $\frac{24}{23}$

14. A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 30°. After walking for 10 minutes from A in the same direction, at a point R, he observes that the angle of elevation of the top of the pillar is 60°. Then the time taken (in minutes) by him, from *B* to reach the pillar, is:

a. 6 **b.** 10

c. 20 **d.** 5

15. $\lim_{x\to 0}\frac{(1-\cos 2x)\,(3+\cos x)}{x\tan 4x}$ is equal to

a. $-\frac{1}{4}$ **b.** $\frac{1}{2}$ **c.** 1 **d.** 2

16. Let the function $g:(-\infty,\infty)\to\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ be given by $g(u)=2\tan^{-1}(e^u)-\frac{\pi}{2}$.

Then, g is

a. even and is strictly increasing in $(0,\infty)$

b. odd and is strictly decreasing in $(-\infty,\infty)$

c. odd and is strictly increasing in $(-\infty,\infty)$

d. neither even nor odd, but is strictly increasing in $(-\infty,\infty)$

SECTION 3 Contains 2 Paragraph Type Questions

Each paragraph describes an experiment, a situation or a problem. Two multiple choice questions will be asked based on this paragraph. One or more than one option can be correct.

Paragraph for Question No. 17 to 18

A box B_1 contains 1 white ball, 3 red balls and 2 black balls. Another box B_2 contains 2 white balls, 3 red balls and 4 black balls. A third box B3 contains 3 white balls, 4 red balls and 5 black balls.

17. If 1 ball is drawn from each of the boxes B_1, B_2 and B_3, the probability that all 3 drawn balls are of the same colour is

a. $\frac{82}{648}$ **b.** $\frac{90}{648}$ **c.** $\frac{558}{648}$ **d.** $\frac{566}{648}$

18. If 2 balls are drawn (without replacement) from a randomly selected box and one of the balls is white and the other ball is red, the probability that these 2 balls are drawn from box B_2 is

a. $\frac{116}{181}$ **b.** $\frac{126}{181}$ **c.** $\frac{65}{181}$ **d.** $\frac{55}{181}$

Paragraph for Question No. 19 to 20

Suppose we define the definite integral using the following formula $\int_a^b f(x)dx=\frac{d-a}{2}\quad(f(a)+f(b))$, for more accurate result for $c\in(a,b)F(c)=\frac{c-a}{2}(f(a)+f(c))+\frac{b-c}{2}$ $(f(b)+f(c))$. When $c=\frac{a+b}{2}$,

$\int_a^b f(x)dx=\frac{b-a}{4}(f(a)+f(b)+2f(c))$.

19. $\int_0^{\pi/2}\sin x\,dx$ is equal to

a. $\frac{\pi}{8}(1+\sqrt{2})$ **b.** $\frac{\pi}{4}(1+\sqrt{2})$

c. $\frac{\pi}{8\sqrt{2}}$ **d.** $\frac{\pi}{4\sqrt{2}}$

20. If $f''(x)<0\ \forall x\in(a,b)$ and c is a point such that $a<c<b$, and $(c, f(c))$ is the point lying on the curve for which $F(c)$ is maximum, then $f'(c)$ is equal to

a. $\frac{f(b)-f(a)}{b-a}$ **b.** $\frac{2(f(b)-f(a))}{b-a}$

c. $\frac{2f(b)-f(a)}{2b-a}$ **d.** 0

Space for rough work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	d	c	d	b	a	d	d	b	b
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
a	c	a	b	c	a	a	a	b	b
21.	**22.**	**23.**	**24.**	**25.**	**26.**	**27.**	**28.**	**29.**	**30.**
b	d	b	a	d	c	a	d	a	b

1. **(a)** Use the identity $f(f-1(x))=x$

replace x by $f-1(x)$, in the given function we get

$$f(f-1(x))=f-1(x)+\frac{1}{f^{-1}(x)}$$

$\Rightarrow \quad x=f-1(x)+\frac{1}{f^{-1}(x)}$, solve to find $f-1x.$

2. **(d)** Given $x^2+x+1=0$

$\therefore \quad x=\frac{1}{2}[-1\pm i\sqrt{3}]=\frac{1}{2}(-1+i\sqrt{3}),\frac{1}{2}(-1-i\sqrt{3})=\omega,\omega^2$

But $\alpha^{19}=\omega^{19}=\omega$ and $\beta^7=\omega^{14}=\omega^2.$

Hence the equation will be same.

3. **(c)** $\omega_n=\cos\left(\frac{2\pi}{n}\right)+i\sin\left(\frac{2\pi}{n}\right)$

$\Rightarrow \quad \omega_3=\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}=-\frac{1}{2}+\frac{i\sqrt{3}}{2}=\omega$

and $\omega_3^2=\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)^2=\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}$

$=-\frac{1}{2}-\frac{i\sqrt{3}}{2}=\omega^2.$

$\therefore \quad (x+y\omega_3+z\omega_3^2)(x+y\omega_3^2+z\omega_3)$

$=(x+y\omega+z\omega^2)(x+y\omega^2+z\omega)$

$=x^2+y^2+z^2-xy-yz-zx.$

4. **(d)** $\sum_{k=1}^{6}\left(\sin\frac{2k\pi}{7}-i\cos\frac{2k\pi}{7}\right)$

$=\sum_{k=1}^{6}-i\left(\cos\frac{2k\pi}{7}+i\sin\frac{2k\pi}{7}\right)=-i\left\{\sum_{k=1}^{6}e^{\frac{i2k\pi}{7}}\right\}$

$=-i\{e^{i2\pi/7}+e^{i4\pi/7}+e^{i6\pi/7}+e^{i8\pi/7}+e^{i10\pi/7}+e^{i12\pi/7}\}$

$=-i\left\{e^{i2\pi/7}\frac{(1-e^{i12\pi/7})}{1-e^{i2\pi/7}}\right\}=-i\left\{\frac{e^{i2\pi/7}-e^{i14\pi/7})}{1-e^{i2\pi/7}}\right\}$

$(\because e^{i14\pi/7}=1) \quad =-i\left\{\frac{e^{i2\pi/7}-1}{1-e^{i2\pi/7}}\right\}=i$

5. **(d)** It is understandable.

6. **(a)** Given series is $3+4\frac{1}{2}+6\frac{3}{4}+.....=3+\frac{9}{2}+\frac{27}{4}+.....$

$=3+\frac{3^2}{2}+\frac{3^3}{4}+\frac{3^4}{8}+\frac{3^5}{16}+.....$ (in G.P.)

Here $a=3, r=\frac{3}{2}$, then sum of the five terms

$$S_5=\frac{a(r^n-1)}{r-1}=\frac{3\left[\left(\frac{3}{2}\right)^5-1\right]}{\frac{3}{2}-1}=\frac{1\left[\frac{3^5}{32}-1\right]}{\frac{1}{2}}$$

$$=6\left[\frac{243-32}{32}\right]=\frac{211\times3}{16}=\frac{633}{16}=39\frac{9}{16}.$$

7. **(d)** $(1+x-2x^2)^6=1+a_1x+a_2x^2+....+a_{12}x^{12}$.

Putting $x=1$ and $x=-1$ and adding the results

$64=2(1+a_2+a_4+...)$

$\therefore \quad a_2+a_4+a_6+....+a_{12}=31.$

8. **(d)** $T_n=\frac{\Sigma n}{n!}=\frac{n(n+1)}{2(n)!}$

$=\frac{1}{2}\left[\frac{(n+1)}{(n-1)!}\right]=\frac{1}{2}\left[\frac{n-1}{(n-1)!}+\frac{2}{(n-1)!}\right]$

$=\frac{1}{2}\left[\frac{1}{(n-2)!}+\frac{2}{(n-1)!}\right]=\frac{(e+2e)}{2}=\frac{3e}{2}.$

9. **(b)** The given series reduces to

$\log_e 2+\log_e\left(\frac{3}{2}\right)+\log_e\left(\frac{4}{3}\right)+\ldots+\log_e\left(\frac{n}{n-1}\right)$

$=\log_e 2+\log_e 3-\log_e 2+\log_e 4-\log_e 3+\ldots$

$+\log_e(n)-\log_e(n-1) \ =\log_e n.$

10. **(b)** $2^{10}-1=1023$, corresponds to none of the lamps is being switched on.

11. **(a)** The event that the fifth toss results in a head is independent the event that the first four tosses result in tails.

$\therefore$ Probability of the required event = ½

12. **(c)** $x.\frac{1}{\sqrt{2}}.\frac{1}{4}=\frac{3.2}{\sqrt{2}.3}\Rightarrow\frac{x}{4\sqrt{2}}=\sqrt{2}\Rightarrow x=8$.

13. **(a)** $\sin\theta+\sin3\theta+\sin2\theta=\sin\alpha$

$\Rightarrow \quad 2\sin2\theta\cos\theta+\sin2\theta=\sin\alpha$

$\Rightarrow \quad \sin2\theta(2\cos\theta+1)=\sin\alpha \qquad ...(i)$

Now, $\cos\theta+\cos3\theta+\cos2\theta=\cos\alpha$

$2\cos2\theta\cos\theta+\cos2\theta=\cos\alpha$

$\Rightarrow \quad \cos 2\theta(2\cos\theta+1)=\cos\alpha$ . . .(*ii*)

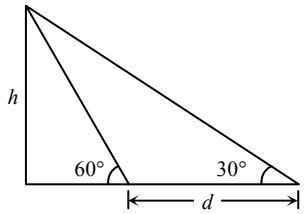
From (*i*) and (*ii*),

$\Rightarrow \quad \tan 2\theta=\tan\alpha \;\Rightarrow\; 2\theta=\alpha \;\Rightarrow\; \theta=\alpha/2$.

14. **(b)** Imaginary part of $\left[\sin^{-1}\left(\frac{5\sqrt{7}}{16}-\frac{9i}{16}\right)\right]$

$=-\log\left[\sqrt{\frac{9}{16}}+\sqrt{1+\frac{9}{16}}\right]=-\log(2).$

15. **(c)**



$d=h\cot 30°-h\cot 60°$ and time = 3 min.

$\therefore \quad$ Speed $=\frac{h(\cot 30^o-\cot 60^o)}{3}$ per minute

It will travel distance $h\cot 60°$ in

$\frac{h\cot 60^o\times 3}{h(\cot 30^o-\cot 60^o)}=1.5$ minute.

16. **(a)** For continuous $\lim\limits_{x\to 2} f(x)=f(2)=k$

$\Rightarrow \quad k=\lim\limits_{x\to 2}\frac{x^3+x^2-16x+20}{(x-2)^2}$

$=\lim\limits_{x\to 2}\frac{(x^2-4x+4)\,(x+5)}{(x-2)^2}=7.$

17. **(a)** $y=e^{x+y} \;\Rightarrow\; \log y=(x+y)\log e$

$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx}=1+\frac{dy}{dx} \;\Rightarrow\; \frac{dy}{dx}=\frac{y}{1-y}.$

18. **(a)** Let $y=\sin^p x.\cos^q x$

$\frac{dy}{dx}=p\sin^{p-1}x.\cos x.\cos^q x+q\cos^{q-1}x.(-\sin x)\sin^p x$

$\frac{dy}{dx}=p\sin^{p-1}x.\cos^{q+1}x-q\cos^{q-1}x.\sin^{p+1}x$

Put $\frac{dy}{dx}=0,$

$\therefore \tan^2 x=\frac{p}{q}$

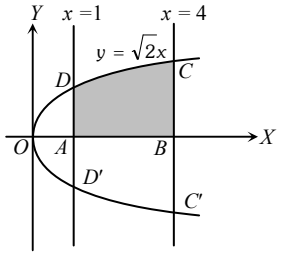
$\Rightarrow \quad \tan x=\pm\sqrt{\frac{p}{q}}$

$\therefore \quad$ Point of maxima $x=\tan^{-1}\sqrt{\frac{p}{q}}.$

19. **(b)** Put $\log x=t\Rightarrow\frac{1}{x}dx=dt,$ then

$\int\frac{1}{x(\log x)^2}dx=\int\frac{1}{t^2}dt=-\frac{1}{t}+c=-\frac{1}{\log x}+c.$

20. **(b)**



Required area $=CDD'C'=2\times ABCD$

$=2\int_1^4\sqrt{2}x^{1/2}\,dx=\frac{28\sqrt{2}}{3}$ sq. unit.

21. **(b)** $\frac{dy}{dx}+\frac{1+x^2}{x}=0 \;\Rightarrow\; dy+\left(\frac{1}{x}+x\right)dx=0$

On integrating, we get $y+\log x+\frac{x^2}{2}+c=0.$

22. **(d)** $\cos y\log(\sec x+\tan x)dx=\cos x\log(\sec y+\tan y)dy$

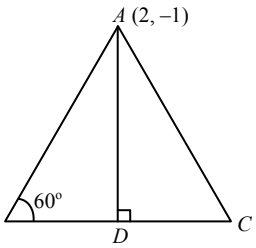
$\Rightarrow \quad \int\sec y\log(\sec y+\tan y)dy$

$=\int\sec x\log(\sec x+\tan x)dx$

Put $\log(\sec x+\tan x)=t$ and $\log(\sec y+\tan y)=z$

$\frac{[\log(\sec x+\tan x)]^2}{2}=\frac{[\log(\sec y+\tan y)]^2}{2}+c.$

23. **(b)**



$|AD|=\left|\frac{2-2-1}{\sqrt{1^2+2^2}}\right|=\frac{1}{\sqrt{5}}$

$\Rightarrow \quad \tan 60°=\frac{AD}{BD}$

$\Rightarrow \quad \sqrt{3}=\frac{1/\sqrt{5}}{BD}$

$\Rightarrow \quad BD=\frac{1}{\sqrt{15}}$

$\Rightarrow \quad BC=2BD=2/\sqrt{15}$.

24. **(a)** It represents a circle, if $a = b$

$\Rightarrow \quad \frac{k}{3} = \frac{1}{4} \Rightarrow k = \frac{3}{4}.$

25. **(d)** Tangent to the curve $y^2 = 8x$ is $y = mx + 2/m.$

So, it must satisfy $xy = -1.$

$\Rightarrow \quad x\left(mx + \frac{2}{m}\right) = -1 \Rightarrow mx^2 + \frac{2}{m}x + 1 = 0$

Since, it has equal roots.

$\therefore \quad D = 0$ z

$\Rightarrow \quad \frac{4}{m^2} - 4m = 0 \Rightarrow m^3 = 1$

$\Rightarrow \quad m = 1$ Hence, equation of common tangent is $y = x + 2.$

26. **(c)** Since $\vec{c} = (x-2)\vec{a} + \vec{b}$ and $\vec{d} = (2x+1)\vec{a} - \vec{b}$ are collinear, therefore $\vec{c} = \lambda\vec{d}$

$\Rightarrow \quad (x-2)\vec{a} + \vec{b} = \lambda(2x+1)\vec{a} - \lambda\vec{b}$

or $\quad [(x-2) - \lambda(2x+1)]\vec{a} + (\lambda+1)\vec{b} = 0$

$(x-2) - \lambda(2x+1) = 0, \lambda + 1 = 0$

($\because$ $\vec{a}, \vec{b}$ are linearly independent)

$\Rightarrow \quad x - 2 + 2x + 1 = 0$

$\Rightarrow \quad x = \frac{1}{3}.$

27. **(a)** $(\vec{b} - \vec{a}) \cdot \left(\vec{c} - \frac{\vec{a} + \vec{b}}{2}\right) = \vec{b} \cdot \vec{c} - \vec{b} \cdot \left(\frac{\vec{a} + \vec{b}}{2}\right) - \vec{a} \cdot \vec{c} + \frac{\vec{a}}{2}(\vec{a} + \vec{b})$

and $|\vec{a} - \vec{c}| = |\vec{b} - \vec{c}|$

$\Rightarrow \quad |\vec{a} - \vec{c}|^2 = |\vec{b} - \vec{c}|^2$

$\therefore \quad \vec{a} + \vec{b} = 2\vec{c}$

Therefore, $(\vec{b} - \vec{a}) \cdot \left(\vec{c} - \frac{\vec{a} + \vec{b}}{2}\right) = 0$

28. **(d)** $\theta = \cos^{-1} \frac{(2 + 10 - 12)}{\sqrt{4 + 25 + 16}\sqrt{1 + 4 + 9}} = \cos^{-1}(0)$

$\Rightarrow \quad \theta = 90°$

29. **(a)** Plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, where $p = \frac{1}{\sqrt{\Sigma(1/a^2)}}$

or $\quad \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{p^2} \qquad \ldots(i)$

Now according to equation, $x = \frac{a}{4}, \; y = \frac{b}{4}, \; z = \frac{c}{4}$

Put the values of x, y, z in (i), we get the locus of the centroid of the tetrahedron.

30. **(a)** $\sim (p \vee q) \equiv \sim p \wedge \sim q$.

JEE Advance Paper-I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
8	8	4	8	4	4	7	5	c	c
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
d	d	c	c	c	d	b	d	c	a

1. **(8)** x^9 can be formed in 8 ways

i.e., $x^9, x^{1+8}, x^{2+7}, x^{3+6}, x^{4+5}, x^{1+2+6}, x^{1+3+5}, x^{2+3+4}$

andcoefficient in each case is 1

$\Rightarrow$ Coefficient of $x^9 = 1 + 1 + 1 + \underbrace{\ldots\ldots}_{8 \text{ times}} + 1 = 8$

2. **(8)** $\frac{a^{-5} + a^{-4} + 3(a^{-3}) + 1 + a^8 + a^{10}}{1+1+3+1+1+1} \geq \left(a^{-5}a^{-4}(a^{-3})^3(1)\,(a^8)(a^{10})\right)^{1/8} = 1$

(using AM $\geq$ GM)

$\therefore \quad a^{-5} + a^{-4} + 3a^{-3} + 1 + a^8 + a^{10} \geq 8$

3. **(4)** $6 + \log_{3/2} \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}}}}} \ldots$

put $x = \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \ldots}$

$x^2 = 4 - \frac{x}{3\sqrt{2}}$

$3\sqrt{2}x^2 = 12\sqrt{2} - x$

$3\sqrt{2}x^2 + x - 12\sqrt{2} = 0$

$\Rightarrow \quad x = \frac{4\sqrt{2}}{3}; \; x = -\sqrt{3/2}$ not possible

$\Rightarrow \quad 6 + \log_{3/2}\left(\frac{1}{3\sqrt{2}} \times \frac{4\sqrt{2}}{3}\right)$

$\Rightarrow \quad 6 + \log_{3/2}\left(\frac{4}{9}\right) \Rightarrow 6 - 2 = 4$

4. **(8)** $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$

$\Rightarrow \quad \frac{5}{4}\cos^2 2x - 5\cos^2 x \sin^2 x = 0$

$\Rightarrow \quad \tan^2 2x = 1$, where $2x \in [0, 4\pi]$

Number of solutions = 8

5. **(4)** Let inner radius be r and inner length be ℓ

$\pi r^2 \ell = V$

Volume of material be M

$M = \pi(r+2)^2(\ell+2) - \pi r^2 \ell$

$\frac{dM}{dr} = -\frac{4V}{r^2} - \frac{8V}{r^3} + 8\pi + 0 + 4\pi r$

$\frac{dM}{dr} = 0$ when $r = 10$

$\Rightarrow \quad V = 1000\pi$

$\Rightarrow \frac{V}{250\pi} = 4$

6. **(4)** Image of $y = -5$

about the line $x + y + 4 = 0$ is $x = 1$

$\Rightarrow$ Distance $AB = 4$

7. **(7)** $G(1) = \int_{-1}^{1} t \,|\, f(f(t))\,|\, dt = 0$

$\Rightarrow \quad f(-x) = -f(x)$

Given $f(1) = \frac{1}{2}$

$$\Rightarrow \quad \lim_{x\to 1}\frac{F(x)}{G(x)} = \lim_{x\to 1}\frac{\frac{F(x)-F(1)}{x-1}}{\frac{G(x)-G(1)}{x-1}} = \frac{f(1)}{|f(f(1))|} = \frac{1}{14}$$

$$\Rightarrow \quad \frac{1/2}{|f(1/2)|} = \frac{1}{14} \quad \Rightarrow f\left(\frac{1}{2}\right) = 7.$$

8. **(5)** Let $(1,1,1), (-1,1,1), (1,-1,1), (-1,-1,-1)$ be vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ rest of the vectors are $-\vec{a}, -\vec{b}, -\vec{c}, -\vec{d}$ and let us find the number of ways of selecting co-planar vectors. Observe that out of any 3 coplanar vectors two will be collinear (anti parallel)

Number of ways of selecting the anti parallel pair = 4

Number of ways of selecting the third vector = 6

Total = 24

Number of non co-planar selections

$= {}^{8}C_3 - 24 = 32 = 2^5$, $p = 5$

Alternate

Required value $= \frac{8\times 6\times 4}{3!}$

$\therefore \quad p = 5$

9. **(c)** $f' = \pm\sqrt{1-f^2}$

$\Rightarrow \quad f(x) = \sin x$ or $f(x) = -\sin x$ (not possible)

$\Rightarrow \quad f(x) = \sin x$

Also, $x > \sin x \;\forall\; x > 0$

10. **(c)** α is a roots of equation

$\alpha^2 - 6\alpha - 2 = 0; \beta^2 - 6B - 2 = 0$

$\alpha^2 - 6\alpha - 2 = 0$

$\Rightarrow \quad \alpha^2 - 2 = 6\alpha$

$$\Rightarrow \quad \frac{a_{10} - 2a_8}{2a_9} = \frac{(\alpha^{10} - \beta^{10}) - 2(\alpha^8 - \beta^8)}{2(\alpha^9 - \beta^9)}$$

$$= \frac{\alpha^8(\alpha^2 - 2) - \beta^8(\beta^2 - 2)}{2(\alpha^9 - \beta^9)}$$

$$= \frac{\alpha^8.(6\alpha) - \beta^8(6\beta)}{2(\alpha^9 - \beta^9)}$$

$$-\frac{6}{2}\frac{(\alpha^9 - \beta^9)}{2(\alpha^9 - \beta^9)} = 3$$

11. **(d)** $X = \sin\theta + \sin 3\theta + ... + \sin 29\theta$

$2(\sin\theta)X = 1 - \cos 2\theta + \cos 2\theta - \cos 4\theta + ... + \cos 28\theta - \cos 30\theta$

$$\Rightarrow \quad X = \frac{1 - \cos 30\theta}{2\sin\theta} = \frac{1}{4\sin 2^\circ}$$

12. **(d)** $AA^T = 9I$

$$\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}\begin{bmatrix} 1 & 2 & a \\ 2 & 1 & 2 \\ 2 & -2 & b \end{bmatrix} = 9I$$

$$\Rightarrow \quad \begin{bmatrix} 9 & 0 & a+4+2b \\ 0 & 9 & 2a+2-2b \\ a+4+2b & 2a+2-2b & a^2+4+b^2 \end{bmatrix} \Rightarrow \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

Equation $a + 4 + 2b = 0$

$\Rightarrow \quad a + 2b = -4$ . . .(*i*)

and $2a + 2 - 2b = 0$

$\Rightarrow \quad 2a - 2b = -2$ . . .(*ii*)

$a^2 + 4 + b^2 = 0$

$\Rightarrow \quad a^2 + b^2 = 5$ . . .(*iii*)

Solving $a = -2, b = -1$

13. **(c)** $0.7 + 0.77 + 0.777 + ... + 0.777...7$

$$= \frac{7}{9}[0.9 + 0.99 + 0.999 + ... + 0.999...9]$$

$$= \frac{7}{9}[(1 - 0.1) + (1 - 0.01) + (1 - 0.001...1) + ... + (1 - 0.000...1)]$$

$$= \frac{7}{9}\left[20 - \left(\frac{1}{10} + \frac{1}{10^2} + \frac{1}{10^3} + ... + \frac{1}{10^{20}}\right)\right]$$

$$= \frac{7}{9}\left[20 - \frac{1}{10}.\frac{1 - \frac{1}{10^{20}}}{1 - \frac{1}{10}}\right] = \frac{7}{9}\left[20 - \frac{1}{9}.\left(\frac{10^{20} - 1}{10^{20}}\right)\right]$$

$$\frac{7}{81}\left[180 - \left(1 - \frac{1}{10^{20}}\right)\right] = \frac{7}{81}[179 + 10^{-20}]$$

14. **(c)** $2x_1 + 3x_2 + 4x_3 = 11$

Possibilities are (0, 1, 2); (1, 3, 0); (2, 1, 1); (4, 1, 0).

$\therefore$ Required coefficients

$= ({}^4C_0 \times {}^7C_1 \times {}^{12}C_2) + ({}^4C_1 \times {}^7C_3 \times {}^{12}C_0) + ({}^4C_2 \times {}^7C_1 \times {}^{12}C_1) + ({}^4C_4 \times {}^7C_1 \times 1)$

$= (1 \times 7 \times 66) + (4 \times 35 \times 1) + (6 \times 7 \times 12) + (1 \times 7)$

$= 462 + 140 + 504 + 4 = 113.$

15. **(c)** Coefficient of x^{10} in $(x + x^2 + x^3)^7$

Coefficient of x^3 in $(1 + x + x^2)^7$

Coefficient of x^3 in

$(1 - x^3)^7 (1 - x)^{-7} = {}^{7+3+7}C_3 - 7 = {}^9C_3 - 7$

$= \dfrac{9 \times 8 \times 7}{6} - 7 = 77$

16. **(d)** $\dfrac{\Sigma x_i}{16} = 16 \Rightarrow \Sigma x_i = 256$

$\dfrac{(\Sigma x_i) - 16 + 3 + 4 + 5}{18} = \dfrac{252}{18} = 14$

17. **(b)** Using cosine rule for $\angle C$

$\dfrac{\sqrt{3}}{2} = \dfrac{(x^2 + x + 1)^2 + (x^2 - 1)^2 - (2x + 1)^2}{2(x^2 + x + 1)(x^2 - 1)}$

$\Rightarrow \sqrt{3} = \dfrac{2x^2 + 2x - 1}{x^2 + x + 1}$

$\Rightarrow (\sqrt{3} - 2)x^2 + (\sqrt{3} - 2)x + (\sqrt{3} + 1) = 0$

$\Rightarrow x = \dfrac{(2 - \sqrt{3}) \pm \sqrt{3}}{2(\sqrt{3} - 2)}$

$\Rightarrow x = -(2 + \sqrt{3}), 1 + \sqrt{3} \Rightarrow x = 1 + \sqrt{3}$ as $(x > 0).$

18. **(d)** $f_k = \dfrac{1}{4}(\sin^k x + \cos^k x)$ $f_6(x) = \dfrac{1}{6}(\sin^6 x + \cos^6 x)$

$f_4(x) = \dfrac{1}{4}(\sin^4 x + \cos^4 x)$

$f_6 K = \dfrac{1}{6}\left[1 - \dfrac{3}{4}\sin^2 2x\right]$

$f_4(x) = \dfrac{1}{4}\left[1 = \dfrac{\sin^2 2x}{2}\right]$

$f_4(x) - f_6(x) = \left[\dfrac{1}{4} - \dfrac{\sin^2 2x}{8}\right] - \left[\dfrac{1}{6} - \dfrac{\sin^2 2x}{8}\right] = \dfrac{1}{4} - \dfrac{1}{6} = \dfrac{1}{12}$

19. **(A)** $\left|\dfrac{\sqrt{3}\alpha + \beta}{2}\right| = \sqrt{3}$

$\sqrt{3}\alpha + \beta = \pm 2\sqrt{3}$ . . .(*i*)

Given $\alpha = 2 + \sqrt{3}\beta$ . . .(*ii*)

From equation (*i*) and (*ii*), we get $\alpha = 2$ *or* -1

So $|\alpha| = 1$ or 2

(B) $f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ bx + a^2, & x \geq 1 \end{cases}$

For continuity $-3a - 2 = b + a^2$

$\Rightarrow a^2 + 3a + 2 = -b$ . . .(*i*)

For differentiability $-6a = b$

$\Rightarrow 6a = -b$

$\Rightarrow a^2 - 3a + 2 = 0$

$\Rightarrow a = 1, 2$

(C) $(3 - 3\omega + 2\omega^2)^{4n+3} + (2 + 3\omega - 3\omega^2)^{4n+3} + (-3 + 2\omega + 3\omega^2)^{4n+3} = 0$

$(3 - 3\omega + 2\omega^2)^{4n+3} + (\omega(2\omega^2 + 3 - 3\omega))^{4n+3} + (\omega^2(-3\omega + 2\omega^2 + 3)^{4n+3} = 0$

$\Rightarrow (3 - 3\omega + 2\omega^2)^{4n+3} + (1 + \omega^{4n} + \omega^{8n}) = 0$

$\Rightarrow n \neq 3k, k \in N$

(D) Let $a = 5 - d$

$q = 5 + d$

$b = 5 + 2d$

$|q - a| = |2d|$

Given $\dfrac{2ab}{a + b} = 4$

$\Rightarrow \dfrac{ab}{a + b} = 2$

$\Rightarrow (5 - d)(5 + 2d) = 2(5 - d + 5 + 2d) = 2(10 + d)$

$\Rightarrow 25 + 10d - 5d - 2d^2 = 20 + 2d$

$\Rightarrow 2d^2 - 3d - 5 = 0$

$\Rightarrow d = -1, d = \dfrac{5}{2}$

$\Rightarrow |2d| = 2, 5$

20. $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -\dfrac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$

(A) If $a + b + c \neq 0$

and $a^2 + b^2 + c^2 = ab + bc + ca$

$\Rightarrow \Delta = 0$ and $a = b = c \neq 0$

$\Rightarrow$ the equations represent identical planes.

(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

$\Rightarrow \Delta = 0$

$\Rightarrow$ the equations have infinitely many solutions.

$ax + by + (a + b)z$

$bx + cy = (b + c)z$

$\Rightarrow (b^2 - ac)y = (b^2 - ac)z \Rightarrow y = z$

$\Rightarrow ax + by + cy = 0 \Rightarrow ax = ay \Rightarrow x = y = z$

(C) $a+b+c\neq 0$

and $a^2+b^2+c^2\neq ab+bc+ca$

$\Rightarrow \quad \Delta\neq 0$

$\Rightarrow$ the equation represent planes meeting at only one point.

(D) $a+b+c=0$

and $a^2+b^2+c^2=ab+bc+ca$

$\Rightarrow \quad a=b=c=0$

$\Rightarrow$ the equation represent whole of the three dimensional space.

JEE Advance Paper -II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
2	3	7	8	9	4	5	2	d	b
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b	b	b	d	d	c	a	d	a	a

1. **(2)** Let $f(x)=x^4-4x^3+12x^2+x-1=0$

$f'(x)=4x^3-12x^2+24x+1=4(x^3-3x^2+6x)+1$

$f''(x)=12x^2-24x+24=12(x^2-2x+2)$

$f''(x)$ has 0 rl roots $f(x)$has maximum 2 distinct real roots as $f(0)=-1$.

2. **(3)** $(y+z)\cos 3\theta-(xyz)\sin 3\theta=0 \qquad \ldots(i)$

$xyz\sin 3\theta=(2\cos 3\theta)z+(2\sin 3\theta)y \qquad \ldots(ii)$

$\therefore \quad (y+z)\cos 3\theta=(2\cos 3\theta)z+(2\sin 3\theta)y=(y+2z)\cos 3\theta+y\sin 3\theta y$

$(\cos 3\theta-2\sin 3\theta)=z\cos 3\theta$ and $y(\sin\theta-\cos 3\theta)=0$

$\Rightarrow \quad \sin 3\theta-\cos 3\theta=0$

$\Rightarrow \quad \sin 3\theta=\cos 3\theta$

$\therefore \quad 3\theta=n\pi+\pi/4$

3. **(7)** $\lim\limits_{x\to 0}\dfrac{x^2\sin\beta x}{\alpha x-\sin x}=\lim\limits_{x\to 0}\dfrac{x^2\left(\beta x-\dfrac{\beta^3x^3}{3!}+\dfrac{\beta^5x^5}{5!}-.....\right)}{\alpha x-\left(x-\dfrac{x^3}{3!}-\dfrac{x^5}{5!}-.....\right)}$

$$=\lim_{x\to 0}\frac{x^3\left(\beta-\dfrac{\beta^3x^2}{3!}+.....\right)}{(\alpha-1)x+\dfrac{x^3}{3!}-\dfrac{x^5}{5!}+.....}=1$$

$\Rightarrow \quad \alpha-1=0$

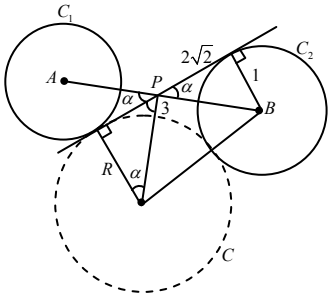
$\Rightarrow \quad \alpha=1,$

Limit $=6\beta=1$

$\Rightarrow \quad \beta=\dfrac{1}{6}$

$\Rightarrow \quad 6(\alpha+\beta)=6\left(1+\dfrac{1}{6}\right)=6\times\dfrac{7}{6}=7$

4. **(8)** $\cos\alpha=\dfrac{2\sqrt{2}}{3}\ \sin\alpha=\dfrac{1}{3}$



$\tan\alpha=\dfrac{2\sqrt{2}}{R}$

$\Rightarrow \quad R=\dfrac{2\sqrt{2}}{\tan\alpha}=8$ units.

5. **(9)** $\vec{s}=4\vec{p}+3\vec{q}+5\vec{r}$

$\vec{s}=x(-\vec{p}+\vec{q}+\vec{r})+y(\vec{p}-\vec{q}+\vec{r})+z(-\vec{p}-\vec{q}+\vec{r})$

$\vec{s}=(-x+y-z)\vec{p}+(x-y-z)\vec{q}+(x+y+z)\vec{r}$

$\Rightarrow \quad -x+y-z=4 \Rightarrow x-y-z=3 \Rightarrow x+y+z=5$

On solving we get $x=4,\ y=\dfrac{9}{2},\ z=-\dfrac{7}{2}$

$\Rightarrow \quad 2x+y+z=9$

6. **(4)** The equation of P_1 is $y^2-8x=0$ and P_2 is $y^2+16x=0$

Tangent to $y^2-5x=0$ passes through (–4, 0)

$\Rightarrow \quad 0=m_1(-4)+\dfrac{2}{m_1}$

$\Rightarrow \quad \dfrac{1}{m_1^2}=2$

Also tangent to $y^2+16x=0$ passes through (2, 0)

$\Rightarrow \quad 0=m_2\times 2-\dfrac{4}{m_2}$

$\Rightarrow \quad m_2^2=2$

$\Rightarrow \quad \dfrac{1}{m_1^2}+m_2^2=4$

7. **(5)** $fx=|x|+|x^2-1|$

$$=\begin{cases}x^2-x-1 & \text{if} \quad x\le -1\\ -x^2-x+1 & \text{if} \quad -1\le x<0\\ -x^2+x+1 & \text{if} \quad 0\le x<1\\ x^2+x-1 & \text{if} \quad x\ge 1\end{cases}$$

So, total no. of local maxima and local minima is = 5

8. **(2)** $x \cdot x^2 \begin{vmatrix} 1 & 1 & 1+x^3 \\ 0 & 2 & 6x^3-1 \\ 0 & 6 & 24x^3-2 \end{vmatrix} = 0$

$\Rightarrow \quad x^3(12x^3+2)=0$

$\Rightarrow \quad 6x^6+x^3-5=0$

$\Rightarrow \quad 6x^6+6x^3-5x^3-5=0$

$\Rightarrow \quad (6x^3-5)(x^3+1)=0$

$\Rightarrow \quad x^3=-1,\ x^3=\frac{5}{6}$

$\Rightarrow \quad x=-1,\ x=\left(\frac{5}{6}\right)^{1/3}$

So, two solution

9. **(d)** $x=4^n-3n-1, n\in N$

$x=(1+3)^n-3n-1, n\in N$

$\Rightarrow \quad X=0,9,54,\ldots\ y=9(n-1), n\in N$

$\Rightarrow \quad y=0,9,18,\ldots \Rightarrow \quad x\cup y=y.$

10. **(b)** $x^2+bx-1=0$

$x^2+x+b=0 \qquad \ldots(i)$

Common root is $(b-1)x-1-b=0$

$\Rightarrow \quad x=\frac{b+1}{b-1}$ This value of x satisfies equation (i)

$\Rightarrow \quad \frac{(b+1)^2}{(b-1)^2}+\frac{b+1}{b-1}+b=0$

$\Rightarrow \quad b=\sqrt{3}i,-\sqrt{3}i,0.$

11. **(b)** Variance $=\frac{\sum x_1^2}{n}-\left(\bar{x}\right)^2$

$\Rightarrow \quad \sigma^2\left(\frac{2^2+4^2+6^2+\ldots+100^2}{50}\right)-\left(\frac{2+4+\ldots+100}{50}\right)^2$

$\Rightarrow \quad \sigma^2=3434-2601=833$

12. **(b)** Exp. $=\frac{\tan^2 A}{\tan A-1}+\frac{1}{\tan A-\tan^2 A}=\frac{1}{\tan A-1}\left[\tan^2 A-\frac{1}{\tan A}\right]$

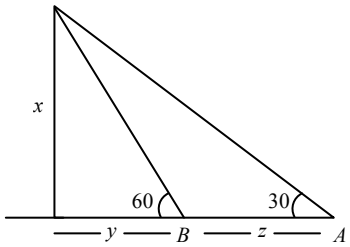
$=\frac{\tan^2 A+\tan A+1}{\tan A}=\tan A+\cot A+1=\sec A.\ cosec\ A+1$

13. **(b)** $\cot\left(\sum_{n=1}^{23}\cot^{-1}(n^2+n+1)\right)$

$\cot\left(\sum_{n=1}^{23}\tan^{-1}\left(\frac{n+1-n}{1+n(n+1)}\right)\right)$

$\Rightarrow \quad \cot\left(\tan^{-1}\left(\frac{23}{325}\right)\right)=\frac{25}{23}.$

14. **(d)** $\tan 30^\circ=\frac{x}{y+z}=\frac{1}{\sqrt{3}}$



$\Rightarrow \quad \sqrt{3}x=y+z$

$\Rightarrow \quad \tan 60^\circ=\frac{x}{y}=\sqrt{3}$

$\Rightarrow \quad x=\sqrt{3}y=y+z$

$3y=y+z$

$\Rightarrow \quad 2y=z$

for 2y distance time

= 10 min.

So for y dist time = 5 min.

15. **(d)** $\lim_{x\to 0}\frac{(1-\cos 2x)(3+\cos x)}{x\tan 4x}$

$=\lim_{x\to 0}\frac{(2\sin^2 x)(3+\cos x)}{x\left(\frac{\tan 4x}{4x}\right)\times 4x}$

$=\lim_{x\to 0}\frac{2\sin^2 x(3+\cos x)}{4x^2}=\frac{2}{4}(3+1)=2$

16. **(c)** $g(u)=2\tan^{-1}(e^u)-\frac{\pi}{2}$

$=2\tan^{-1}e^u-\tan^{-1}e^u-\cot^{-1}e^u=\tan^{-1}e^u-\cot^{-1}e^u$

$g(-x)=-g(x)$

$\Rightarrow \quad g(x)$ is odd and $g'(x)>0$

$\Rightarrow$ increasing.

17. **(a)** P (required) = P (all are white) + P (all are red) + P (all are black)

$=\frac{1}{6}\times\frac{2}{9}\times\frac{3}{12}+\frac{3}{6}\times\frac{3}{9}\times\frac{4}{12}+\frac{2}{6}\times\frac{4}{9}\times\frac{5}{12}\times\frac{5}{12}$

$=\frac{6}{648}+\frac{36}{648}+\frac{40}{648}=\frac{82}{648}.$

18. **(d)** Let A : one ball is white and other is red

E_1 : Both balls are from box B_1

E_2 : Both balls are from box B_2

JEE-MAIN: MATHEMATICS MOCK TEST -3

1. Let $g(x)=1+x-[x]$ and $f(x)=\begin{cases}-1 & x<0\\ 0 & x=0\\ 1 & x>1\end{cases}$. Then for all $x, f\{g(x)\}$ is equal to:

a. x **b.** 1 **c.** $f(x)$ **d.** $g(x)$

2. If $x=\sqrt{6+\sqrt{6+\sqrt{6+....\text{to }\infty}}}$, then

a. x is an irrational number **b.** $2<x<3$

c. $x=3$ **d.** None of these

3. If $z+z^{-1}=1$, then $z^{100}+z^{-100}$ is equal to

a. i **b.** $-I$ **c.** 1 **d.** -1

4. The inverse of $\begin{bmatrix}2 & -3\\ -4 & 2\end{bmatrix}$ is

a. $\frac{-1}{8}\begin{bmatrix}2 & 3\\ 4 & 2\end{bmatrix}$ **b.** $\frac{-1}{8}\begin{bmatrix}3 & 2\\ 2 & 4\end{bmatrix}$

c. $\frac{1}{8}\begin{bmatrix}2 & 3\\ 4 & 2\end{bmatrix}$ **d.** $\frac{1}{8}\begin{bmatrix}3 & 2\\ 2 & 4\end{bmatrix}$

5. The sum of 100 terms of the series $.9+.09+.009.....$ ill be

a. $1-\left(\frac{1}{10}\right)^{100}$ **b.** $1+\left(\frac{1}{10}\right)^{100}$

c. $1-\left(\frac{1}{10}\right)^{106}$ **d.** $1+\left(\frac{1}{10}\right)^{100}$

6. If n is an integer greater than 1, then $a-{}^nC_1(a-1)+{}^nC_2(a-2)+....+(-1)^n(a-n)=$

a. a **b.** 0

c. a^2 **d.** 2^n

7. The sum of $\frac{2}{1!}+\frac{6}{2!}+\frac{12}{3!}+\frac{20}{4!}+\ldots$ is

a. $\frac{3e}{2}$ **b.** e

c. $2e$ **d.** $3e$

8. If the letters of the word KRISNA are arranged in all possible ways and these words are written out as in a dictionary, then the rank of the word KRISNA is

a. 324 **b.** 341

c. 359 **d.** None of these

9. Seven white balls and three black balls are randomly placed in a row. The probability that no two black balls are placed adjacently, equals

a. $\frac{1}{2}$ **b.** $\frac{7}{15}$

c. $\frac{2}{15}$ **d.** $\frac{1}{3}$

10. If $\pi<\alpha<\frac{3\pi}{2}$, then $\sqrt{\frac{1-\cos\alpha}{1+\cos\alpha}}+\sqrt{\frac{1+\cos\alpha}{1-\cos\alpha}}=$

a. $\frac{2}{\sin\alpha}$ **b.** $-\frac{2}{\sin\alpha}$

c. $\frac{1}{\sin\alpha}$ **d.** $-\frac{1}{\sin\alpha}$

11. Find real part of $\cos^{-1}\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)$

a. $\pi/3$ **b.** $\pi/4$

c. $\log\left(\frac{\sqrt{3}-1}{2}\right)$ **d.** None of these

12. A house of height 100 metres subtends a right angle at the window of an opposite house. If the height of the window be 64 metres, then the distance between the two houses is

a. 48 m **b.** 36 m **c.** 54 m **d.** 72 m

13. If $f(x)=\begin{cases}\frac{\sin x}{x}+\cos x, & \text{when } x\neq 0\\ 2, & \text{when } x=0\end{cases}$ then

a. $\lim_{x\to 0+} f(x)\neq 2$ **b.** $\lim_{x\to 0-} f(x)=0$

c. $f(x)$ is continuous at $x=0$ **d.** None of these

14. If $x^y=e^{x-y}$, then $\frac{dy}{dx}=$

a. $\log x.[\log(ex)]^{-2}$ **b.** $\log x.[\log(ex)]^2$

c. $\log x.(\log x)^2$ **d.** None of these

15. 20 is divided into two parts so that product of cube of one quantity and square of the other quantity is maximum. The parts are

a. 10, 10 **b.** 16, 4

c. 8, 12 **d.** 12, 8

16. $\int \frac{1}{\sqrt{x}}\tan^4\sqrt{x}\sec^2\sqrt{x}\,dx=$

a. $2\tan^5\sqrt{x}+c$ **b.** $\frac{1}{5}\tan^5\sqrt{x}+c$

c. $\frac{2}{5}\tan^5\sqrt{x}+c$ **d.** None of these

17. The area bounded by the straight lines $x=0, x=2$ and the curves $y=2^x, y=2x-x^2$ is

a. $\frac{4}{3}-\frac{1}{\log 2}$ **b.** $\frac{3}{\log 2}+\frac{4}{3}$

c. $\frac{4}{\log 2}-1$ **d.** $\frac{3}{\log 2}-\frac{4}{3}$

18. The solution of $\frac{dy}{dx}=e^x(\sin x+\cos x)$ is

a. $y=e^x(\sin x-\cos x)+c$

b. $y=e^x(\cos x-\sin x)+c$

c. $y=e^x\sin x+c$

d. $y=e^x\cos x+c$

19. The product of the perpendiculars drawn from the points $(\pm\sqrt{a^2-b^2},0)$ on the line $\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1$, is

a. a^2 **b.** b^2

c. a^2+b^2 **d.** a^2-b^2

20. Area of the circle in which a chord of length $\sqrt{2}$ makes an angle $\frac{\pi}{2}$ at the centre is

a. $\frac{\pi}{2}$ **b.** 2π

c. π **d.** $\frac{\pi}{4}$

21. The focal chord to $y^2=16x$ is tangent to $(x-6)^2+y^2=2$, then the possible values of the slope of this chord, are

a. $\{-1,1\}$ **b.** $\{-2,2\}$

c. $\{-2,1/2\}$ **d.** $\{2,-1/2\}$

22. If $\vec{a},\vec{b},\vec{c}$ are non-zero vectors such that $\vec{a}\cdot\vec{b}=\vec{a}\cdot\vec{c}$, then which statement is true

a. $\vec{b}=\vec{c}$ **b.** $\vec{a}\perp(\vec{b}-\vec{c})$

c. $\vec{b}=\vec{c}$ or $\vec{a}\perp(\vec{b}-\vec{c})$ **d.** None of these

23. The point of intersection of the lines $\frac{x-5}{3}=\frac{y-7}{-1}=\frac{z+2}{1}, \frac{x+3}{-36}=\frac{y-3}{2}=\frac{z-6}{4}$ is

a. $21,\frac{5}{3},\frac{10}{3}$ **b.** $(2,10,4)$

c. $(-3,3,6)$ **d.** $(5,7,-2)$

24. $\sim(p\wedge q)$ is equal to

a. $\sim p\vee\sim q$ **b.** $\sim p\wedge\sim q$

c. $\sim p\wedge q$ **d.** $p\wedge\sim q$

25. The value of b and c for which the identity $f(x+1)-f(x)=8x+3$ is satisfied, where $f(x)=bx2+cx+d$, are

a. $b=2,c=1$ **b.** $b=4,c=-1$

c. $b=-1,c=4$ **d.** None of these.

26. If H is the harmonic mean between p and q, then the value of $\frac{H}{p}+\frac{H}{q}$ is

a. 2 **b.** $\frac{pq}{p+q}$

c. $\frac{p+q}{pq}$ **d.** None of these

27. Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the enveloped bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then, the number of ways it can be done is

a. 264 **b.** 265

c. 53 **d.** 67

28. *ABCD* is a rectangular field. *A* vertical lamp post of height 12m stands at the corner *A*. If the angle of elevation of its top from *B* is 60° and from C is 45°, then the area of the field is

a. $48\sqrt{2}$ sq. m **b.** $48\sqrt{3}$ sq. m

c. 48 sq. m **d.** $12\sqrt{2}$ sq. m

29. The area bounded by the curves $y=\sqrt{x}, 2y+3=x$ and *x*-axis in the 1st quadrant is:

a. 9 sq unit

b. 27/4 sq unit

c. 36 sq unit

d. 18 sq unit

30. If the line $2x+\sqrt{6}y=2$ touches the hyperbola $x^2-2y^2=4$, then the point of contact is:

a. $(-2,\sqrt{6})$ **b.** $(-5,2\sqrt{6})$

c. $\left(\frac{1}{2},\frac{1}{\sqrt{6}}\right)$ **d.** $(4,-\sqrt{6})$

JEE ADVANCE PAPER-I

SECTION 1 Contains 8 Questions.

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. If z is any complex number satisfying $|z-3-2i| \le 2$, then the minimum value of $|2z-6+5i|$ is

2. Let M be a 3×3 matrix satisfying

$$M\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}-1\\2\\3\end{bmatrix}, M\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}1\\1\\-1\end{bmatrix},$$

$$\text{and } M\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}0\\0\\12\end{bmatrix}.$$

Then the sum of the diagonal entries of M is

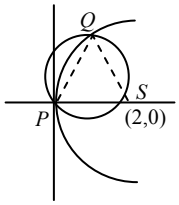
3. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is 6 : 11 and the seventh term lies in between 130 and 140, then the common difference of this A.P. is

4. A pack contains n cards numbered from 1 to *n*. Two consecutive numbered cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller of the numbers on the removed cards is *k*, then $k-20=$ _____

5. Consider a triangle *ABC* and let *a, b* and c denote the lengths of the sides opposite to vertices *A, B* and *C* respectively. Suppose $a=6, b=10$ and the area of the triangle is 15 3. If $\angle ACB$ is obtuse and if r denotes the radius of the in circle of the triangle, then r^2 is equal to

6. Let m and n be two positive integers greater than 1. If $\lim_{\alpha\to 0}\left(\frac{e^{\cos(\alpha^n)}-e}{\alpha^m}\right)=-\left(\frac{e}{2}\right)$ then the value of $\frac{m}{n}$ is

7. Let S be the focus of the parabola $y^2 = 8x$ and let *PQ* be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of the triangle PQS is.



8. Let $n \ge 2$ be an integer. Take n distinct points on a circle and join each pair of points by a line segment. Colour the line segment joining every pair of adjacent points by blue and the rest by red. If the number of red and blue line segments are equal, then the value of *n* is

SECTION 2 Contains 10 Multiple Choice Questions

With one or more than one correct option

9. Let *A* and *B* be two sets containing four and two elements respectively. Then the number of subsets of the set $A\times B$, each having at least three elements is:

a. 219 **b.** 256
c. 275 **d.** 510

10. The real number *k* for which the equation, $2x^3+3x+k=0$ has two distinct real roots in [0, 1]

a. lies between 1 and 2 **b.** lies between 2 and 3
c. lies between –1 and 0 **d.** does not exist

11. If *z* is a complex number of unit modulus and argument θ, then $\arg\left(\frac{1+z}{1+\bar{z}}\right)$ equals

a. $-\theta$ **b.** $\frac{\pi}{2}-\theta$
c. θ **d.** $\pi-\theta$

12. If $A=\begin{bmatrix}5a & -b\\ 3 & 2\end{bmatrix}$ and A adj $A = A\ A^T$, then $5a+b$ is equal to:

a. –1 **b.** 5
c. 4 **d.** 13

13. Three positive numbers form an increasing G.P. If the middle term in this G.P. is doubled, the new numbers are in A.P. Then the common ratio of the G.P. is :

a. $\sqrt{2}+\sqrt{3}$ **b.** $3+\sqrt{2}$
c. $2-\sqrt{3}$ **d.** $2+\sqrt{3}$

14. If the number of terms in the expansion of $\left(1-\frac{2}{x}+\frac{4}{x^2}\right)^n, x\neq 0,$ is 28, then the sum of coefficients of all the terms in this expansion, is:

a. 64 **b.** 2187
c. 243 **d.** 729

15. Let T_n be the number of all possible triangles formed by joining vertices of a n-sides regular polygon. If $T_{n+1} - T_n = 1$ then the value of n is

a. 7 **b.** 5

c. 10 **d.** 8

16. A computer producing factory has only two plants T_1 and T_2. Plant T_1 produces 20% and plant T_2 produces 80% of the total computers produced. 7% of computers produced in the factory turn out to be defective. It is known that P(computer turns out to be defective given that it is produced in plant T_1) $= 10\,P$(computer turns out to be defective given that it is produced in Plant T_2), where $P(E)$ denotes the probability of an event E. A computer produced in the factory is randomly selected and it does not turn out to be defective. Then the probability that it is produced in plant T_2 is

a. $\frac{36}{73}$ **b.** $\frac{47}{79}$

c. $\frac{78}{93}$ **d.** $\frac{75}{83}$

17. A bird is sitting on the top of a vertical pole 20 m high and is elevation from a point O on the ground is $45°$. It flies off horizontally straight away from the point O. After one second, the elevation of the bird from O is reduced to $30°$. Then the speed (in m/s) of the bird is:

a. $40(\sqrt{2}-1)$

b. $40(\sqrt{3}-\sqrt{2})$

c. $20\sqrt{2}$

d. $20(\sqrt{3}-1)$

18. Let $f(x)$ be differentiable on the interval $(0,\infty)$ such that $f(1)=1$, and $\lim_{t\to x}\frac{t^2 f(x)-x^2 f(t)}{t-x}=1$ for each $x>0$. Then $f(x)$ is

a. $\frac{1}{3x}+\frac{2x^2}{3}$

b. $-\frac{1}{3x}+\frac{4x^2}{3}$

c. $-\frac{1}{x}+\frac{2}{x^2}$

d. $\frac{1}{x}$

SECTION 3 Contains 2 Match The Following Type Questions

You will have to match entries in Column I with the entries in Column II.

19. Match the statements given in Column I with the values given in Column II.

Column I	Column II
(A) If $\vec{a}=\hat{j}+\sqrt{3}\hat{k}=-\hat{j}+\sqrt{3}\hat{k}$ and $\vec{c}=2\sqrt{3}\hat{k}$ form a triangle, then the internal angle of the triangle between $\vec{a}$ and $\vec{b}$ is	**1.** $\frac{\pi}{6}$
(B) If $\int_a^b (f(x)-3x)dx = a^2-b^2$, then the value of $f\left(\frac{\pi}{6}\right)$ is	**2.** $\frac{2\pi}{3}$
(C) The value of $\frac{\pi^2}{\ln 3}\int_{7/6}^{5/6}\sec(\pi x)\,dx$ is	**3.** $\frac{\pi}{3}$
(D) The maximum value $\left\lvert \mathrm{Arg}\left(\frac{1}{1-z}\right)\right\rvert$ for $\lvert z\rvert = 1, z\neq 1$ f is given by	**4.** π
	5. $\frac{\pi}{2}$

20. Consider the lines given by $L_1: x+3y-5=0$, $L_2: 3x-ky-1=0$, $L_3: 5x+2y-12=0$ Match the Statements /Expressions in Column I with the Statements/ Expressions in Column II

Column I	Column II
(A) L_1, L_2, L_3 are concurrent, if	**1.** $k=-9$
(B) One of L_1, L_2, L_3 is parallel to at least of the other two, if	**2.** $k=-\frac{6}{5}$
(C) L_1, L_2, L_3 form a triangle, if	**3.** $k=\frac{5}{6}$
(D) L_1, L_2, L_3 do not form a triangle, if	**4.** $k=5$

JEE ADVANCE PAPER-II

SECTION 1 Contains 8 Questions.

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \le d_1(P) + d_2(P) \le 4,$ is _____

2. The slope of the tangent to the curve $(y - x^5)^2 = x(1 + x^2)^2$ at the point (1, 3) is

3. Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations: $3x - y - z = 0$, $-3x + z = 0$ $-3x + 2y + z = 0.$

Then the number of such points for which $x^2 + y^2 + z^2 \le 100$ is

4. Let $\mathrm{f}(\theta) = \sin\left(\tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos 2\theta}}\right)\right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}.$

Then the value of $\frac{\mathrm{d}}{\mathrm{d}(\tan\theta)}(\mathrm{f}(\theta))$ is

5. Two parallel chords of a circle of radius 2 are at a distance $\sqrt{3} + 1$ apart. If the chords subtend at the centre, angles of $\frac{\pi}{k}$ and $\frac{2\pi}{k}$, where $k > 0,$ then the value of $[k]$ is [Note: $[k]$ denotes the largest integer less than or equal to k].

6. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer.

If a, b, c are in geometric progression and the arithmetic mean of a, b, c is b + 2, then the value of $\frac{a^2 + a - 14}{a + 1}$ is

7. Let $\omega = \mathrm{e}^{\mathrm{i}\pi/3}$, and a, b, c, x, y, z be non-zero complex numbers such that

$\mathrm{a} + \mathrm{b} + \mathrm{c} = \mathrm{x}$

$\mathrm{a} + \mathrm{b}\omega + \mathrm{c}\omega^2 = \mathrm{y}$

$\mathrm{a} + \mathrm{b}\omega^2 + \mathrm{c}\omega = \mathrm{z}.$

Then the value of $\frac{|\mathrm{x}|^2 + |\mathrm{y}|^2 + |\mathrm{z}|^2}{|\mathrm{a}|^2 + |\mathrm{b}|^2 + |\mathrm{c}|^2}$ is

8. If the normals of the parabola $y^2 = 4x$ drawn at the end points of its latus rectum are tangents to the circle $(x - 3)^2 + (y + 2)^2 = r^2,$ then the value of r^2 is

SECTION 2 Contains 8 Multiple Choice Questions

With one or more than one correct option

9. Let $p = \lim_{x \to 0+} (1 + \tan^2\sqrt{x})^{\frac{1}{2x}}$ then log p is equal to:

a. 2 **b.** 1 **c.** $\frac{1}{2}$ **d.** $\frac{1}{4}$

10. Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(1 - x)$ and $f'\left(\frac{1}{4}\right) = 0.$ Then

a. $f''(x)$ vanishes at least twice on [0, 1]

b. $f'\left(\frac{1}{2}\right) = 0$

c. $\int_{-1/2}^{1/2} f\left(x + \frac{1}{2}\right)\sin x \, dx = 0$

d. $\int_0^{1/2} f(t)e^{\sin \pi t} dt = \int_{1/2}^{1} f(1-1)e^{\sin \pi t} dt$

11. The following integral $\int_{\pi/4}^{\pi/2} (2\cos ec\, x)^{17} \, dx$ is equal to

a. $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$

b. $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17} du$

c. $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17} du$

d. $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16} du$

12. $\int \frac{x^2 - 1}{x^3\sqrt{2x^4 - 2x^2 + 1}} dx$ is equal to

a. $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^2} + c$ **b.** $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^3} + c$

c. $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + c$ **d.** $\frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + c$

13. If y (x) satisfies the differential equation $y' - y\tan x = 2x\sec x$ and $y(0) = 0$, then.

a. $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$

b. $y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$

c. $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9}$

d. $y'\left(\frac{\pi}{3}\right) = \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}$

14. Given an isosceles triangle, whose one angle is $120°$ and radius of its incircle $= \sqrt{3}$. Then the area of the triangle in sq. units is

a. $7 + 12\sqrt{3}$

b. $12 - 7\sqrt{3}$

c. $12 + 7\sqrt{3}$

d. 4π

15. The circle passing through the point (–1, 0) and touching the y-axis at (0, 2) also passes through the point

a. $\left(-\frac{3}{2}, 0\right)$

b. $\left(-\frac{5}{2}, 2\right)$

c. $\left(-\frac{3}{2}, \frac{5}{2}\right)$

d. (–4, 0)

16 The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the x-axis at Q. If M is the mid-point of the line segment PQ, then the locus of M intersects the latus rectums of the given ellipse at the points

a. $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{2}{7}\right)$

b. $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{\sqrt{19}}{4}\right)$

c. $\left(\pm 2\sqrt{3}, \pm\frac{1}{7}\right)$

d. $\left(\pm 2\sqrt{3}, \pm\frac{4\sqrt{3}}{7}\right)$

SECTION 3 Contains 2 Paragraph Type Questions

Each paragraph describes an experiment, a situation or a problem. Two multiple choice questions will be asked based on this paragraph. One or more than one option can be correct.

Tangents are drawn from the point P(3, 4) to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ touching the ellipse at points A and B.

17. The coordinates of A and B are

a. (3, 0) and (0, 2)

b. $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

c. $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and (0, 2)

d. (3, 0) and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

18. The equation of the locus of the point whose distances from the point P and the line AB are equal, is

a. $9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$

b. $x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$

c. $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$

d. $x^2 + y^2 - 2xy + 27x + 31y - 120 = 0$

Paragraph for Question No. 19 to 20

Let $S = S_1 \cap S_2 \cap S_3$, where $S_1 = \{z \in C : |z| < 4\}$,

$$S_2 = \left\{z \in C : \text{Im}\left[\frac{z - 1 + \sqrt{3}i}{1 - \sqrt{3}i}\right] > 0\right\}$$

and $S_3 = \{z \in C : \text{Re } Z > 0\}$.

19. Area of S =

a. $\frac{10\pi}{3}$ **b.** $\frac{20\pi}{3}$

c. $\frac{16\pi}{3}$ **d.** $\frac{32\pi}{3}$

20. $\min_{z \in S} |1 - 3i - z| =$

a. $\frac{2 - \sqrt{3}}{2}$

b. $\frac{2 + \sqrt{3}}{2}$

c. $\frac{3 - \sqrt{3}}{2}$

d. $\frac{3 + \sqrt{3}}{2}$

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	c	d	a	a	b	d	a	b	b
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b	a	c	a	d	c	d	c	b	c
21.	**22.**	**23.**	**24.**	**25.**	**26.**	**27.**	**28.**	**29.**	**30.**
a	c	a	a	b	a	c	a	a	d

1. **(b)** $f(g(x))=\begin{cases}-1 & g(x)<0\\ 0 & g(x)=0\\ 1 & g(x)>0\end{cases}$

Since $g(x)\geq 1>0$

Hence $g(g(x))=1$

2. **(c)** $x=\sqrt{6+x},\ x>0\Rightarrow x^2=6+x, x>0$

$\Rightarrow \quad x^2-x-6=0, x>0$

$\Rightarrow x=3, x>0.$

3. **(d)** $z+z^{-1}=1 \Rightarrow z^2-z+1=0$

$\Rightarrow z=-\omega$ or $-\omega^2$

For $z=-\omega,\ z^{100}+z^{-100}=(-\omega)^{100}+(-\omega)^{-100}$

$=\omega+\frac{1}{\omega}=\omega+\omega^2=-1$

For $z=-\omega^2,\ z^{100}+z^{-100}=(-\omega^2)^{100}+(-\omega^2)^{-100}$

$=\omega^{200}+\frac{1}{\omega^{200}}=\omega^2+\frac{1}{\omega^2}=\omega^2+\omega$

$=-1.$

4. **(a)** Let The matrix of cofactors of the elements of A viz.

$\begin{bmatrix}c_{11} & c_{12}\\ c_{21} & c_{22}\end{bmatrix}=\begin{bmatrix}2 & -(-4)\\ -(-3) & 2\end{bmatrix}=\begin{bmatrix}2 & 4\\ 3 & 2\end{bmatrix}$

$\therefore$ $adjA=$ transpose of the matrix of cofactors of elements of $A=\begin{bmatrix}2 & 3\\ 4 & 2\end{bmatrix}$

$\therefore$ $A(adj\,A)=|A|I$.

5. **(a)** Series is a G.P. with $a=0.9=\frac{9}{10}$ and $r=\frac{1}{10}=0.1$

$\therefore \quad S_{100}=a\left(\frac{1-r^{100}}{1-r}\right)=\frac{9}{10}\left(\frac{1-\frac{1}{10^{100}}}{1-\frac{1}{10}}\right)=1-\frac{1}{10^{100}}.$

6. **(b)** L.H.S. $=a[C_0-C_1+C_2-C_3+...(-1)^n.C_n]$

$+[C_1-2C_2+3C_3-....+(-1)^{n-1}n.C_n]=a.0+0=0$

7. **(d)** Let $S=\frac{2}{1!}+\frac{6}{2!}+\frac{12}{3!}+\frac{20}{4!}+\ldots$ and let

$S_1=2+6+12+20+\ldots+T_n$

$S_1=\quad 2+6+12+\ldots T_{n-1}+T_n$

$\overline{0=2+4+6+8+\ldots \text{upto } n \text{ terms}-T_n}$

$T_n=2+4+6+8+.....\text{upto } n \text{ terms}$

$\Rightarrow \quad T_n=\frac{n}{2}[2\times 2+(n-1)2]=n(2+n-1)=n(n+1)$

$\therefore$ n^{th} term of given series

$T_n=\frac{n(n+1)}{n!}$ or $T_n=\frac{n(n+1)}{n(n-1)!}$ or $T_n=\frac{1}{(n-2)!}+\frac{2}{(n-1)!}$

Now, sum $=\sum_{n=1}^{\infty}\frac{1}{(n-2)!}+2\sum_{n=1}^{\infty}\frac{1}{(n-1)!}=e+2e=3e.$

8. **(a)** Words starting from A are $5!=120$

Words starting from I are 5 ! = 120
Words starting from KA are 4 ! = 24
Words starting from KI are 4 ! = 24
Words starting from KN are 4 ! = 24
Words starting from KRA are 3 ! = 6
Words starting from KRIA are 2 ! = 2
Words starting from KRIN are 2 ! = 2
Words starting from KRISA are 1 ! = 1
Words starting from KRISNA are 1 ! = 1
Hence rank of the word KRISNA is 324.

9. **(b)** The number of ways of placing 3 black balls without any restriction is ${}^{10}C_3$. Since, we have total 10 places of putting 10 balls in a row. Now the number of ways in which no two black balls put together is equal to the number of ways of choosing 3 places marked '–' out of eight places.

$-W-W-W-W-W-W-W-$

This can be done in ${}^8C_3$ ways.

$\therefore$ Required probability $=\frac{{}^8C_3}{{}^{10}C_3}=\frac{8\times 7\times 6}{10\times 9\times 8}=\frac{7}{15}$

10. **(b)** $\sqrt{\frac{1-\cos\alpha}{1+\cos\alpha}}+\sqrt{\frac{1+\cos\alpha}{1-\cos\alpha}}=\frac{1-\cos\alpha+1+\cos\alpha}{\sqrt{1-\cos^2\alpha}}$

$=\frac{2}{\pm\sin\alpha}=\frac{2}{-\sin\alpha},\left(\text{since } \pi<\alpha<\frac{3\pi}{2}\right).$

11. **(b)** $\because$ Expression $\cos^{-1}(\cos\theta+i\sin\theta)$

$=\sin^{-1}\sqrt{\sin\theta}-i\log(\sqrt{\sin\theta}+\sqrt{1+\sin\theta}),$ where $\theta=\frac{\pi}{6}$

$\therefore$ Real part of $\cos^{-1}\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)=\sin^{-1}\sqrt{\frac{1}{2}}=\frac{\pi}{4}.$

12. **(a)** $64\cot\theta = d$

Also $(100-64)\tan\theta = d$ or $(64)(36)=d^2$,

$\therefore$ $d = 8\times 6 = 48\,m.$

13. **(c)** $f(0+)=f(0-)=2$ and $f(0)=2$

Hence $f(x)$ is continuous at $x=0.$

14. **(a)** $x^y = e^{x-y}$

$\Rightarrow$ $y\log x = x-y$

$\Rightarrow$ $y=\frac{x}{1+\log x}$

$\Rightarrow$ $\frac{dy}{dx}=\log x(1+\log x)^{-2}=\log x[\log ex]^{-2}.$

15. **(d)** Let $x+y=20\Rightarrow y=20-x$

and $x^3.y^2 = z \Rightarrow z = x^3.y^2$

$z = x^3(20-x)^2$

$\Rightarrow$ $z = 400x^3+x^5-40x^4$

$\frac{dz}{dx}=1200x^2+5x^4-160x^3$

Now $\frac{dz}{dx}=0,$ then $x=12, 20$

Now $\frac{d^2z}{dx^2}=2400x+16x^3-480x^2;\ \left(\frac{d^2z}{dx^2}\right)_{x=12}=-ive$

Hence $x=12$ is the point of maxima

$\therefore$ $x=12,\ y=8.$

16. **(c)** $\int \frac{1}{\sqrt{x}}\tan^4\sqrt{x}.\sec^2\sqrt{x}\,dx$

Put $\tan\sqrt{x}=t\Rightarrow\frac{\sec^2\sqrt{x}}{2\sqrt{x}}dx=dt,$

then it reduces to

$2\int t^4dt=\frac{2}{5}(\tan\sqrt{x})^5+c=\frac{2}{5}\tan^5\sqrt{x}+c.$

17. **(d)** Required area $=\int_0^2[2^x-(2x-x^2)]dx$

$=\left[\frac{2^x}{\log 2}-x^2+\frac{x^3}{3}\right]_0^2=\frac{4}{\log 2}-4+\frac{8}{3}-\frac{1}{\log 2}=\frac{3}{\log 2}-\frac{4}{3}.$

18. **(c)** Given equation $\frac{dy}{dx}=e^x(\sin x+\cos x)$

$\Rightarrow$ $dy=e^x(\sin x+\cos x)dx$

On integrating, we get $y=e^x\sin x+c.$

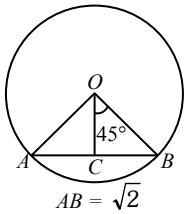
19. **(b)** $\left(\frac{b\sqrt{a^2-b^2}\cos\theta+0-ab}{\sqrt{b^2\cos^2\theta+a^2\sin^2\theta}}\right)\left(\frac{-b\sqrt{a^2-b^2}\cos\theta-ab}{\sqrt{b^2\cos^2\theta+a^2\sin^2\theta}}\right)$

$=\frac{-[b^2(a^2-b^2)\cos^2\theta-a^2b^2]}{(b^2\cos^2\theta+a^2\sin^2\theta)}$

$=\frac{b^2[a^2-a^2\cos^2\theta+b^2\cos^2\theta]}{b^2\cos^2\theta+a^2\sin^2\theta}$

$=\frac{b^2[a^2\sin^2\theta+b^2\cos^2\theta]}{b^2\cos^2\theta+a^2\sin^2\theta}=b^2$

20. **(c)**



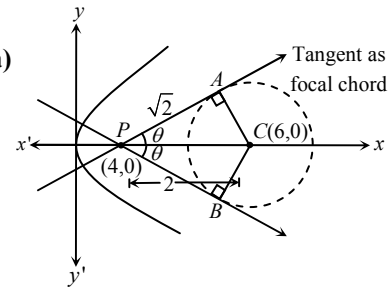
Let AB be the chord of length $\sqrt{2}$, O be centre of the circle and let OC be the perpendicular from O on AB. Then

$AC=BC=\frac{\sqrt{2}}{2}=\frac{1}{\sqrt{2}}$

In $\Delta\,OBC,\ OB=BC\,\text{cosec}\,45°=\frac{1}{\sqrt{2}}.\sqrt{2}=1$

$\therefore$ Area of the circle $=\pi(OB)^2=\pi$

21. **(a)**



Here, the focal chord of $y^2=16x$ is tangent to circle $(x-6)^2+y^2=2.$

$\Rightarrow$ focus of parabola as $(a,0)$ *ie*, $(4,0)$

Now, tangents are drawn from $(4,0)$ to $(x-6)^2+y^2=2.$

Since, PA is tangent to circle.

$\therefore$ $\tan\theta=$ slope of tangent $=\frac{AC}{AP}=\frac{\sqrt{2}}{\sqrt{2}}=1,$ or $\frac{BC}{BP}=-1$

$\therefore$ Slope of focal chord as tangent to circle $=\pm 1.$

22. **(c)** $\vec{a}\cdot\vec{b}=\vec{a}\cdot\vec{c}\Rightarrow\vec{a}\cdot\vec{b}-\vec{a}\cdot\vec{c}=0\Rightarrow\vec{a}\cdot(\vec{b}-\vec{c})=0$

$\Rightarrow$ Either $\vec{b}-\vec{c}=0$ or $\vec{a}=0\Rightarrow\vec{b}=\vec{c}$ or $\vec{a}\perp(\vec{b}-\vec{c}).$

23. **(a)** Given lines are,

$\frac{x-5}{3}=\frac{y-7}{-1}=\frac{z+2}{1}=r_1$, (say)

and $\frac{x+3}{-36}=\frac{y-3}{2}=\frac{z-6}{4}=r_2$, (say)

$\therefore \quad x=3r_1+5=-36r_2-3$,

$y=-r_1+7=3+2r_2$ and $z=r_1-2=4r_2+6$

On solving, we get $x=21, y=\frac{5}{3}, z=\frac{10}{3}$

Trick: Check through options.

24. **(a)** $\sim(p\wedge q)\equiv\sim p\vee\sim q$.

25. **(b)** From the given identity

$b(x+1)^2+c(x+1)+d-(bx^2+cx+d)=8x+3$

$\Rightarrow \quad 2bx+b+c=8x+3$

$\Rightarrow \quad b=4, c=-1$

26. **(a)** As given $H=\frac{2pq}{p+q}$

$\therefore \quad \frac{H}{p}+\frac{H}{q}=\frac{2q}{p+q}+\frac{2p}{p+q}=\frac{2(p+q)}{p+q}=2$.

27. **(c)** Plan: A square matrix M is invertible if f det(M) or $|M|\neq 0$.

Let, $M=\begin{bmatrix} a & b \\ b & c \end{bmatrix}$

(a) Given that $\begin{bmatrix} a \\ b \end{bmatrix}=\begin{bmatrix} b \\ c \end{bmatrix}\Rightarrow a=b=c=\alpha$ (let)

$\Rightarrow \quad M=\begin{bmatrix} \alpha & \alpha \\ \alpha & \alpha \end{bmatrix}\Rightarrow |M|=0$

$\Rightarrow$ M is non-invertible.

(b) Given that $[b\ c]=[a\ b]$

$\Rightarrow \quad a=b=c=\alpha$ (let)

Again $|M|=0$

$\Rightarrow$ M is non-invertible.

(c) As given $M=\begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix}\Rightarrow |M|=ac\neq 0$

($\because$ a and c are non-zero)

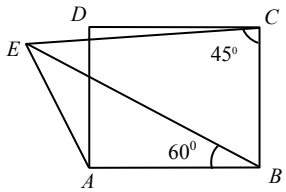
$\Rightarrow$ M is invertible.

(s) $M=\begin{bmatrix} a & b \\ b & c \end{bmatrix}\Rightarrow |M|=ac-b^2\neq 0$

$\because$ ac is not equal to square of an integer.

$\therefore$ M is invertible.

28. **(a)**



Let AE is a vertical lamp-post. Given, AE = 12m

$\tan 45°=\frac{AE}{AC}$

$AC=AE=12$m

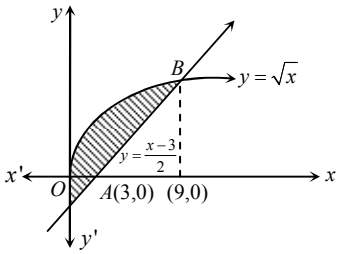
$\tan 60°=\frac{AE}{AB}$

$AB=\frac{AE}{\sqrt{3}}=4\sqrt{3}$

$BC=\sqrt{AC^2-AB^2}=\sqrt{144-48}=\sqrt{96}=4\sqrt{6}$

Area = $= AB\times BC=4\sqrt{3}\times 4\sqrt{6}=48\sqrt{2}$ sq.cm.

29. **(a)**



To find the area between the curves, $y=\sqrt{x}, 2y+3=x$ and x-axis in the Ist quadrant (we can plot the above condition as);

Area of shaded portion $OABO$

$=\int_0^9\sqrt{x}\,dx-\int_3^9\left(\frac{x-3}{2}\right)dx=\left(\frac{x^{3/2}}{3/2}\right)_0^9-\frac{1}{2}\left(\frac{x^2}{2}-3x\right)_3^9$

$=\left(\frac{2}{3}\cdot 27\right)-\frac{1}{2}\left\{\left(\frac{81}{2}-27\right)-\left(\frac{9}{2}-9\right)\right\}$

$=18-\frac{1}{2}(18)=9$ sq unit

30. **(d)** The equation of tangent at (x_1, y_1) is $xx_1-2yy_1=4$, which is same as $2x+\sqrt{6}y=2$

$\therefore \quad \frac{x_1}{2}=-\frac{2y_1}{\sqrt{6}}=\frac{4}{2}$

$\Rightarrow \quad x_1=4$ and $y_1=-\sqrt{6}$

JEE Advance Paper-I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
5	9	9	5	3	2	4	5	a	d
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	b	d	d	b	c	d	a	c	a

1. **(5)** $|2z-6+5i|=2\left|z-\left(3-\frac{5}{2}\right)\right|$

$\geq 2\left|3-\left(3-\frac{5}{2}i\right)\right|$ (corresponding Pt A) $=2\left(\frac{5}{2}\right)=5$

2. **(9)** Let $M=\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

$$M\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}-1\\2\\3\end{bmatrix}\Rightarrow b=-1,\ e=2,\ h=3$$

$$M\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}1\\1\\-1\end{bmatrix}\Rightarrow a=0,\ d=3,\ g=2$$

$$M\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}0\\1\\12\end{bmatrix}\Rightarrow g+h+i=12\Rightarrow i=7$$

$\therefore$ Sum of diagonal elements = 9.

3. **(9)** Let seventh term be 'a' and common difference be'd'

Given $\frac{S_7}{S_{11}}=\frac{6}{11}$

$\Rightarrow a=15d$

Hence, $130<15d<140 \Rightarrow d=9$

4. **(5)** Clearly, $1+2+3+\ldots+n-2\le 1224\le 3+4+\ldots n$

$\Rightarrow \frac{(n-2)(n-1)}{2}\le 1224\le \frac{(n-2)}{2}(3+n)$

$\Rightarrow n^2-3n-2446\le 0$ and $n^2+n-2454\ge 0$

$\Rightarrow 49<n<51$

$\Rightarrow n=50$

$\therefore \frac{n(n+1)}{2}-(2k+1)=1224$

$\Rightarrow k=25\Rightarrow k-20=5$

5. **(3)** $\Delta=\frac{1}{2}ab\sin C$

$\Rightarrow \sin C=\frac{2\Delta}{ab}=\frac{2\times 15\sqrt{3}}{6\times 10}=\frac{\sqrt{3}}{2}\Rightarrow C=120°$

$\Rightarrow c=\sqrt{a^2+b^2-2ab\cos C}$

$=\sqrt{6^2+10^2-2\times 6\times 10\times\cos 120°}=14$

$\therefore r=\frac{\Delta}{s}\Rightarrow r^2=\frac{225\times 3}{\left(\frac{6+10+14}{2}\right)}=3.$

6. **(2)** $\lim_{\alpha\to 0}\frac{e^{\cos(\alpha^n)}-e}{\alpha^m}=-\frac{e}{2}$

$\lim_{\alpha\to 0}\frac{e\left(e^{(\cos(\alpha)^n-1)}\right)(\cos\alpha^n-1)}{\left(\cos(\alpha^n)-1\right)\alpha^m\alpha^{2n}}\quad \alpha^{2n}=-\frac{e}{2}$

if and only if $2n-m=0$

7. **(4)** Solving $y^2=8x$ and $x^2+y^2-2x-4y=0$

Simultaneously, we get (2, 4) and (0, 0)

Focus is (2, 0)

$\therefore$ Are $=\frac{1}{2}\times 2\times 4$ = 4sq. units.

8. **(5)** Number of red lines $={}^nC_2-n$

Number of blue lines = n

Hence, ${}^nC_2-n=n$

$\Rightarrow {}^nC_2=2n$

$\frac{n(n-1)}{2}=2n$

$\Rightarrow n-1=4$

$\Rightarrow n=5.$

9. **(a)** Set A has 4 elements

Set B has 2 elements

$\therefore$ Number of elements in set $(A\times B)=4\times 2=8$

$\therefore$ Total number of subsets of $(A\times B)=2^8=256$

Number of subsets having 0 elements $=8_{C_0}=1$

Number of subsets having 1 element each $=8_{C_1}=8$

$\therefore$ Number of subsets having 2 elements each

$=8_{C_2}=\frac{8!}{2!6!}=\frac{8\times 7}{2}=28$

Number of subsets having at least 3 elements $=256-1-8-28=256-37=219$

10. **(d)** $f(x)=2x^3+3x+k$

$f'(x)=6x^2+3$

$\Rightarrow f'(x)=0\Rightarrow x^2=-\frac{1}{2}$

Not possible. As condition for two distinct real root is $f(\alpha)\,f(\beta)=0$

(where are α,β roots of $f'(x)=0$)

11. **(c)** Let $z=\omega$

Now $\frac{1+z}{1+\bar{z}}=\frac{1+\omega}{1+\omega^2}=\frac{-\omega^2}{-\omega}=\omega$

$\therefore \arg\frac{1+z}{1+z}=\arg\omega=\theta$ (put $z=\cos\theta+i\sin\theta$)

12. **(b)** $|A|I=AA^T$

$\Rightarrow (10a+3b)\begin{bmatrix}1&0\\0&1\end{bmatrix}=\begin{bmatrix}5a&-b\\3&2\end{bmatrix}\begin{bmatrix}5a&3\\-b&2\end{bmatrix}$

$\Rightarrow 25a^2+b^2=10a+3b\ \&\ 15a-2b=0\ \&\ 10a+3b=13$

$\Rightarrow 10a+\frac{3.15a}{2}=13\Rightarrow 65a=2\times 13$

$\Rightarrow \quad a = \frac{2}{5} \Rightarrow \quad 5a = 2$

$\Rightarrow \quad 2a = 6 \Rightarrow \quad b = 3$

$\therefore \quad 5a + b = 5$

13. **(d)** Let the numbers be a, ar, ar^2 is G.P.

Given a, 2ar, ar^2 are in A.P. the $2ar = \frac{a + ar^2}{2} (a \neq 0)$

which gives $r = 2 + \sqrt{3}$, as the G.P. is an increasing G.P.

14. **(d)** Theoretically the number of terms are $2N + 1$ (i.e. odd) But As the number of terms being odd hence considering that number clubbing of terms is done hence the solutions follows:

Number of terms = ${}^{n+2}C_2 = 28$

$\therefore \quad n = 6$

Sum of coefficient = $3^n = 3^6 = 729$

Put $x = 1$

15. **(b)** ${}^{n+1}C_3 - {}^{n}C_3 = 10$

$\Rightarrow$ On solving n = 5

16. **(c)** Let $x = P$ (computer turns out to be defective given that it is produced in plate T_2),

$\Rightarrow \quad \frac{7}{100} = \frac{1}{5}(10x) + \frac{4}{5}x$

$\Rightarrow \quad 7 = 200x + 80x$

$\Rightarrow \quad x = \frac{7}{280}$

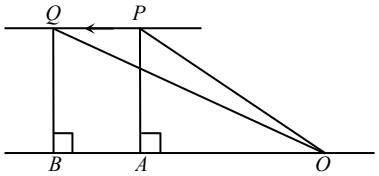
P (produced in T_2 / not defective)

$= \frac{P(A \cap B)}{P(B)}$

$\Rightarrow \quad \frac{\frac{4}{5(1-x)}}{\frac{1}{5}(1-10x) + \frac{4}{5}(1-x)} = \frac{\frac{4}{5}\left(\frac{273}{280}\right)}{\frac{1}{5}\left(\frac{280-70}{280}\right) + \frac{4}{5}\left(\frac{273}{280}\right)}$

$\Rightarrow \quad \frac{4 \times 273}{210 + 4 \times 273} = \frac{2 \times 273}{105 + 2 \times 273} = \frac{546}{651} = \frac{78}{93}$

17. **(d)**



Here, $AP = QB = 20m$

$\angle POA = 45°, \quad \angle QOB = 30°$

$\Rightarrow \quad OA = 20; \quad OB = 20\sqrt{3}$

$\Rightarrow \quad OB - OA = 20(\sqrt{3} - 1)$

Hence distance covered in one second by the bird is $AB = 20(\sqrt{3} - 1)$

Thus speed of bird

$= 20(\sqrt{3} - 1)$ m/s

18. **(a)** $\lim_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$

$\Rightarrow \quad x^2 f'(x) - 2x f(x) + 1 = 0$

$\Rightarrow \quad f(x) = cx^2 + \frac{1}{3x}$

Also $f(1) = 1$

$\Rightarrow \quad c = \frac{2}{3}$

Hence $f(x) = \frac{2}{3}x^2 + \frac{1}{3x}$.

19. **(A)** $\vec{a} - \vec{b} = 1 + 3 = 2$

$|\vec{a}| = b, |\vec{b}| = 2$

$\cos\theta = \frac{2}{2 \times 2} = \frac{1}{2}$

$\theta = \frac{\pi}{3}, \frac{2\pi}{3}$ but its $\frac{2\pi}{3}$ as its opposite to side of maximum length.

(B) $\int_a^b (f(x) - 3x)\, dx = a^2 - b^2$

$\int_a^b f(x) dx = \frac{3}{2}(b^2 - a^2) + a^2 - b^2 = \frac{-a^2 + b^2}{2}$

$\Rightarrow \quad f(x) = x.$

(C) $\frac{\pi^2}{\ln 3}\left(\frac{\ln|(\sec \pi x + \tan \pi x)|_{7/6}^{5/6}}{\pi}\right)$

$= \frac{\pi}{\ln 3}\left(\ln\left|\sec\frac{5\pi}{6} + \tan\frac{5\pi}{6}\right| - \ln\left|\sec\frac{7\pi}{6} + \tan\frac{7\pi}{6}\right|\right) = \pi.$

(D) Let $u = \frac{1}{1-z} \Rightarrow z = 1 - \frac{1}{u}$

$|z| = 1 \Rightarrow \left|1 - \frac{1}{2}\right| = 1$

$\Rightarrow \quad |u - 1| = |u|$

$\therefore$ locus of u is perpendicular bisector of line segment joining 0 and 1

$\Rightarrow$ maximum arg u approaches $\frac{\pi}{2}$ but will not attain.

20. $x+3y-5=0$ and $5x+2y-12=0$ intersect at (2, 1)

Hence $6-k-1=0 \quad k=5$

for L_1, L_2 to be parallel $\frac{1}{3}=\frac{3}{-k} \Rightarrow k=-9$

for L_2, L_3 to be parallel $\frac{3}{5}=\frac{-k}{2} \Rightarrow k=\frac{-6}{5}$.

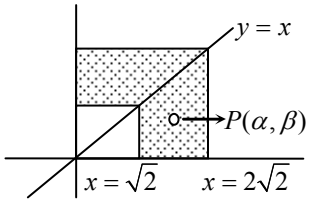
for $k \neq 5, -9, \frac{-6}{5}$ they will form triangle

for $k=5, k=-9, \frac{-6}{5}$ they will not form triangle

JEE Advance Paper-II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
6	8	7	1	5	4	3	2	3	All
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
a	d	a,d	c	d	a	d	a	b	c

1. **(6)**



$2 \le d_1(p)+d_2(p) \le 4$

For $P(\alpha,\beta), \alpha > \beta$

$\Rightarrow \quad 2\sqrt{2} \le 2\alpha \le 4\sqrt{2}$

$\sqrt{2} \le \alpha \le 2\sqrt{2}$

$\Rightarrow$ Area of region $= ((2\sqrt{2})^2 - (\sqrt{2})^2)$

$= 8-2 = 6$ sq. units

2. **(8)** $2(y-x^5)\left(\frac{dy}{dx}-5x^4\right)$

$= 1(1+x^2)^2 + (x)(2(1+x^2)(2x))$

Now put $x=1, y=3$

and $\frac{dy}{dx}=m.\ 2(3-1)(m-5) = 1(4)+(1)(4)(2)$

$\Rightarrow \quad m-5=\frac{12}{4}$

$\Rightarrow \quad m=5+3=8 \ \Rightarrow \ \frac{dy}{dx}=m=8.$

3. **(7)** $3x-y-z=0 \ -3x+z=0$

$\Rightarrow \quad y=0$ and $z=3x$

$\Rightarrow \quad x^2+y^2+z^2 = x^2+z^2 = x^2+9x^2 = 10x^2 \le 100$

$\Rightarrow \quad x^2 \le 10 \Rightarrow \quad x=0, \pm 1, \pm 2, \pm 3$

There are such seven points.

4. **(1)** $f(\theta) = \sin\left(\tan^{-1}\frac{\sin\theta}{\sqrt{2\cos\theta}}\right)$

$= \sin\left(\sin^{-1}\frac{\sin\theta}{\sqrt{\sin^2\theta+\cos 2\theta}}\right)$

$= \frac{\sin\theta}{\sqrt{\sin^2\theta+(\cos^2\theta-\sin^2\theta)}} = \frac{\sin\theta}{|\cos\theta|} = \frac{\sin\theta}{\cos\theta} = \tan\theta$

$\Rightarrow \quad \frac{d}{d(\tan\theta)} f(\theta) = \frac{d(\tan\theta)}{d(\tan\theta)} = 1$

5 **(2)** $2\cos\frac{\pi}{2k}+2\cos\frac{\pi}{k}=\sqrt{3}+1$

$\Rightarrow \quad \cos\frac{\pi}{2k}+\cos\frac{\pi}{k}=\frac{\sqrt{3}+1}{2}$

Let $\frac{\pi}{k}=0, \cos\theta+\cos\frac{\theta}{2}=\frac{\sqrt{3}+1}{2}$

$\Rightarrow \quad 2\cos^2\frac{\theta}{2}-1+\cos\frac{\theta}{2}=\frac{\sqrt{3}+1}{2}$

$\Rightarrow \quad \cos\frac{\theta}{2}=t \ \Rightarrow 2t^2+t-\frac{\sqrt{3}+3}{2}=0$

$\Rightarrow \quad t=\frac{-1\pm\sqrt{1+4(3+\sqrt{3})}}{4}=\frac{-1\pm(2\sqrt{3}+1)}{4}=\frac{-2-2\sqrt{3}}{4}, \frac{\sqrt{3}}{2}$

$\because \quad t\in[-1,1], \cos\frac{\theta}{2}=\frac{\sqrt{3}}{2}$

$\Rightarrow \quad \frac{\theta}{2}=\frac{\pi}{6} \ \Rightarrow \ k=3.$

6. **(4)** $\frac{b}{a}=\frac{c}{b}=$ (integer) $b^2=ac$

$\Rightarrow c=\frac{b^2}{a} \Rightarrow \frac{a+b+c}{3}=b+2$

$a+b+c=3b+6$

$\Rightarrow \quad a-2b+c=6$

$a-2b+\frac{b^2}{a}=6$

$\Rightarrow \quad 1-\frac{2b}{a}+\frac{b^2}{a^2}=\frac{6}{a}$

$\left(\frac{b}{a}-1\right)^2=\frac{6}{a} \quad a=6$ only

7. **(3)** The expression may not attain integral value for all a, b, c

If we consider a = b = c, then x = 3a

$y = a(1+\omega+\omega^2) = a(1+i\sqrt{3})$

$z = a(1+\omega^2+\omega) = a(1+i\sqrt{3})$

$\therefore \quad |x|^2+|y|^2+|z|^2 = 9|a|^2+4|a|^2+4|a|^2 = 17|a|^2$

8. **(2)** Equation of normals are $x+y=3$ and $x-y=3$.

$\Rightarrow$ Distance from $(3,-2)$ on both normals is 'r'

$\Rightarrow \quad \dfrac{|3-2-3|}{\sqrt{2}}=r$

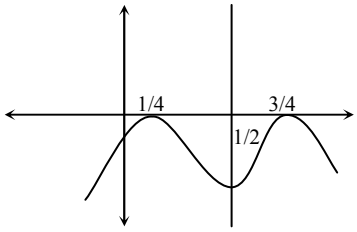
$\Rightarrow \quad r^2=2.$

9 **(3)** $P=\lim\limits_{x\to\infty^+}(1+\tan 2\sqrt{x-1})^{\frac{1}{2x}}$ then $\log p=$

$P=e^{\lim\limits_{x\to 0^+}(1+\tan^2\sqrt{x-1})\frac{1}{2x}}=e^{\lim\limits_{x\to 0^+}\frac{(\tan\sqrt{x})^2}{2(\sqrt{x})^2}}=e^{\frac{1}{2}}$

$\Rightarrow \quad \log P=\log e^{\frac{1}{2}}=\dfrac{1}{2}$

10. **(a,b,c,d)**



$f(x)=f(1-x)$

Put $x=1/2+x$

$\Rightarrow \quad f\left(\dfrac{1}{2}+x\right)=f\left(\dfrac{1}{2}-x\right)$

Hence $f(x+1/2)$ is an even function or $f(x+1/2)\sin x$ an odd function.

Also, $f'(x)=-f'(1-x)$ and for $x=1/2$,

We have $f'(1/2)=0.$

Also, $\int\limits_{1/2}^{1} f(1-t)e^{\sin\pi t}dt=-\int\limits_{1/2}^{0} f(y)e^{\sin\pi t}dy$

(obtained by putting, $1-t=y$).

Since $f'(1/4)=0, f'(3/4)=0.$ Also $f'(1/2)=0$

$\Rightarrow \quad f'(x)=0$ atleast twice in [0, 1] (Rolle's Theorem)

11. **(a)** $\int\limits_{\frac{\pi}{4}}^{\frac{\pi}{2}}(2\operatorname{cosec} x)^{17}dx$

Let $e^u+e^{-u}=2\operatorname{cosec} x, x=\dfrac{\pi}{4}$

$\Rightarrow \quad u=\ln(1+\sqrt{2}), x=\dfrac{\pi}{2}$

$\Rightarrow \quad u=0$

$\Rightarrow \quad \operatorname{cosec} x+\cot x=e^u$ and $\operatorname{cosec} x-\cot x=e^{-u}$

$\Rightarrow \quad \cot x\dfrac{e^u-e^{-u}}{2}\ (e^u-e^{-u})dx=-2\operatorname{cosec} x\cot x dx$

$\Rightarrow \quad -\int(e^u+e^{-u})^{17}\dfrac{(e^u-e^{-u})}{2\operatorname{cosec} x\cot x}du$

$=-2\int\limits_{\ln(1+\sqrt{2})}^{0}(e^u+e^{-u})^{16}du=\int\limits_{0}^{\ln(1+\sqrt{2})}2(e^u+e^{-u})^{16}du$

12. **(d)** $\int\dfrac{\left(\dfrac{1}{x^3}-\dfrac{1}{x^5}\right)dx}{\sqrt{2-\dfrac{2}{x^2}+\dfrac{1}{x^4}}}$

Let $2-\dfrac{2}{x^2}+\dfrac{1}{x^4}=z$

$\Rightarrow \quad \dfrac{1}{4}\int\dfrac{dz}{z}$

$\Rightarrow \quad \dfrac{1}{2}\times\sqrt{z}+c$

$\Rightarrow \quad \dfrac{1}{2}\sqrt{2-\dfrac{2}{x^2}+\dfrac{1}{x^4}}+c$

13. **(a,d)** $y'-y\tan x=2x\sec x$

I.F. $=e^{\int\tan x\,dx}=e^{\log\cos x}=\cos x$

$\therefore \quad y\cos x=\int 2x\sec x.\cos x\,dx$

$\Rightarrow \quad y\cos x=x^2+c$

$\Rightarrow \quad y\cdot\cos x=x^2 \quad (\because y(0)=0)$

$\Rightarrow \quad y=x^2\sec x$

$\therefore \quad y\left(\dfrac{\pi}{4}\right)=\dfrac{\pi^2}{8\sqrt{2}}$ and $y'\left(\dfrac{\pi}{3}\right)=\dfrac{4\pi}{3}+\dfrac{2\pi^2}{3\sqrt{3}}$

14. **(c)** $\Delta=\dfrac{\sqrt{3}}{4}b^2$ . . .(*i*)

Also $\dfrac{\sin 120°}{a}=\dfrac{\sin 30°}{b}$

$\Rightarrow \quad a=\sqrt{3}b$ and $\Delta=\sqrt{3}s$ and $s=\dfrac{1}{2}(a+2b)$

$\Rightarrow \quad \Delta=\dfrac{\sqrt{3}}{2}(a+2b)$ . . .(*ii*)

From equation (*i*) and (*ii*), we get $\Delta=(12+7\sqrt{3})$

15. **(d)** Circle touching y-axis at (0, 2) is

$(x-0)^2+(y-2)^2+\lambda x=0$

passes through (–1, 0)

$\therefore \quad 1+4-\lambda=0\Rightarrow\lambda=5$

$\therefore \quad x^2+y^2+5x-4y+4=0$

Put $y=0 \Rightarrow x=-1,-4$

$\therefore$ Circle passes through (–4, 0)

16. **(a)** Any point on the line can be taken as

$Q \equiv \{(1-3\mu), (\mu-1), (5\mu+2)\}$

$\overline{PQ} = \{-3\mu-2, \mu-3, 5\mu-4\}$

Now $1(-3\mu-2)-4(\mu-3)+3(5\mu-4)=0$

$\Rightarrow \quad -3\mu-2-4\mu+12+15\mu-12=0$

$8\mu = 2 \;\Rightarrow\; \mu = 1/4$

17. **(d)** $y = mx + \sqrt{9m^2+4}$

$4-3m = \sqrt{9m^2+4}$

$16+9m^2-24m = 9m^2+4 \Rightarrow m = \dfrac{12}{24} = \dfrac{1}{2}$

Equation is $y-4 = \dfrac{1}{2}(x-3)$

$2y-8 = x-3$

$\Rightarrow \quad x-2y+5=0$

Let $B = (\alpha, \beta)$

$\Rightarrow \quad \dfrac{x\alpha}{9} + \dfrac{y\beta}{4} - 1 = 0$

$\Rightarrow \quad \dfrac{\alpha/9}{1} = \dfrac{\beta/4}{-2} = \dfrac{-1}{5}$

$\Rightarrow \quad \alpha = -\dfrac{9}{5}, \;\; \beta = \dfrac{8}{5}$

$\Rightarrow \quad B \equiv \left(-\dfrac{9}{5}, \dfrac{8}{5}\right).$

18. **(a)** Locus is parabola

Equation of AB Is $\dfrac{3x}{9} + \dfrac{4y}{4} = 1$

$\Rightarrow \dfrac{x}{3} + y = 1 \Rightarrow x + 3y - 3 = 0$

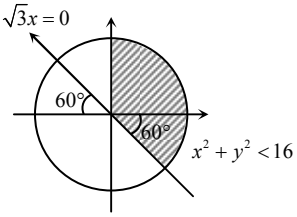
$$(x-3)^2 + (y-4)^2 = \frac{(x+3y-3)^2}{10}$$

$$10x^2 + 90 - 60x + 10y^2 + 160 - 80y$$

$$= x^2 + 9y^2 + 9 + 6xy - 6x - 18y$$

$\Rightarrow \quad 9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0.$

19. **(b)** $y + \sqrt{3}x = 0$



Area of region $S_1 \cap S_2 \cap S_3 =$ shaded area

$$= \frac{\pi \times 4^2}{4} + \frac{4^2 \times \pi}{6}$$

$$= 4^2\pi\left\{\frac{1}{4} + \frac{1}{6}\right\} = \frac{20\pi}{3}$$

20. **(c)** Distance of (1, –3) from $y + \sqrt{3}x = 0$

$$> \left|\frac{-3+\sqrt{3}\times 1}{2}\right| > \frac{3-\sqrt{3}}{2}$$

□ □ □

JEE-MAIN: MATHEMATICS MOCK TEST-4

1. The number of solutions of $\log 4(x-1) = \log 2(x-3)$ is:

a. 3 **b.** 1 **c.** 2 **d.** 0

2. The real roots of the equation $x^2 + 5|x| + 4 = 0$ are

a. $-1, 4$ **b.** $1, 4$

c. $-4, 4$ **d.** None of these

3. If $\frac{1+\sqrt{3}i}{2}$ is a root of equation $x^4 - x^3 + x - 1 = 0$ then its real roots are

a. $1, 1$ **b.** $-1, -1$ **c.** $1, -1$ **d.** $1, 2$

4. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 2 & 0 \\ -1 & 6 & 1 \end{bmatrix}$, then the adjoint of A is

a. $\begin{bmatrix} 2 & -5 & 32 \\ 0 & 1 & -6 \\ 0 & 0 & 2 \end{bmatrix}$ **b.** $\begin{bmatrix} -1 & 0 & 0 \\ -5 & -2 & 0 \\ 1 & -6 & 1 \end{bmatrix}$

c. $\begin{bmatrix} -1 & 0 & 0 \\ -5 & -2 & 0 \\ 1 & -6 & -1 \end{bmatrix}$ **d.** None of these

5. The value of $0.\overline{234}$ is

a. $\frac{232}{990}$ **b.** $\frac{232}{9990}$

c. $\frac{232}{990}$ **d.** $\frac{232}{9909}$

6. The sum of the coefficients of even power of x in the expansion of $(1 + x + x^2 + x^3)^5$ is

a. 256 **b.** 128 **c.** 512 **d.** 64

7. In the expansion of $\frac{1-2x+3x^2}{e^x}$, the coefficient of x^5 will be

a. $\frac{71}{120}$ **b.** $-\frac{71}{120}$

c. $\frac{31}{40}$ **d.** $-\frac{31}{40}$

8. The total number of seven digit numbers the sum of whose digits is even is

a. 9000000 **b.** 4500000

c. 8100000 **d.** None of these

9. If E and F are events with $P(E) \le P(F)$ and $P(E \cap F) > 0$, then

a. occurrence of $E \Rightarrow$ occurrence of F

b. occurrence of $F \Rightarrow$ occurrence of E

c. non-occurrence of $E \Rightarrow$ non-occurrence of F

d. None of the above implication holds

10. $\tan\left(\frac{\pi}{4} + \theta\right) - \tan\left(\frac{\pi}{4} - \theta\right) =$

a. $2\tan 2\theta$ **b.** $2\cot 2\theta$ **c.** $\tan 2\theta$ **d.** $\cot 2\theta$

11. $\tanh(x+y)$ equals

a. $\frac{\tanh x + \tanh y}{1 - \tanh x \tanh y}$ **b.** $\frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}$

c. $\frac{\tanh x - \tanh y}{1 - \tanh x \tanh y}$ **d.** $\frac{\tanh x - \tanh y}{1 + \tanh x \tanh y}$

12. The length of the shadow of a pole inclined at 10° to the vertical towards the sun is 2.05 metres, when the elevation of the sun is 38°. The length of the pole is

a. $\frac{2.05 \sin 38°}{\sin 42°}$ **b.** $\frac{2.05 \sin 42°}{\sin 38°}$

c. $\frac{2.05 \cos 38°}{\cos 42°}$ **d.** None of these

13. If $f(x) = \begin{cases} x^2 \sin\frac{1}{x}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$, then

a. $f(0+0) = 1$ **b.** $f(0-0) = 1$

c. f is continuous at $x = 0$ **d.** None of these

14. If $y = \sqrt{\frac{(x-a)(x-b)}{(x-c)(x-d)}}$, then $\frac{dy}{dx} =$

a. $\frac{y}{2}\left[\frac{1}{x-a} + \frac{1}{x-b} - \frac{1}{x-c} - \frac{1}{x-d}\right]$

b. $y\left[\frac{1}{x-a} + \frac{1}{x-b} - \frac{1}{x-c} - \frac{1}{x-d}\right]$

c. $\frac{1}{2}\left[\frac{1}{x-a} + \frac{1}{x-b} - \frac{1}{x-c} - \frac{1}{x-d}\right]$

d. None of these

15. If $f(x) = \frac{x^2-1}{x^2+1}$, for every real number x, then the minimum value of f

a. Does not exist because f is unbounded

b. Is not attained even though f is bounded

c. Is equal to 1

d. Is equal to –1

16. $\int \frac{a^x}{\sqrt{1-a^{2x}}}dx =$

a. $\frac{1}{\log a}\sin^{-1} a^x + c$ **b.** $\sin^{-1} a^x + c$

c. $\frac{1}{\log a}\cos^{-1} a^x + c$ **d.** $\cos^{-1} a^x + c$

17. The area bounded by the circle $x^2 + y^2 = 4$, line $x = \sqrt{3}y$ and x-axis lying in the first quadrant, is

a. $\frac{\pi}{2}$ **b.** $\frac{\pi}{4}$ **c.** $\frac{\pi}{3}$ **d.** π

18. The general solution of $x^2\frac{dy}{dx} = 2$ is

a. $y = c + \frac{2}{x}$ **b.** $y = c - \frac{2}{x}$ **c.** $y = 2cx$ **d.** $y = c - \frac{3}{x^2}$

19. The length of perpendicular from the point $(a\cos\alpha, a\sin\alpha)$ upon the straight line $y = x\tan\alpha + c, c > 0$ is

a. $c\cos\alpha$ **b.** $c\sin^2\alpha$ **c.** $c\sec^2\alpha$ **d.** $c\cos^2\alpha$

20. The circle passing through point of intersection of the circle $S = 0$ and the line $P = 0$ is

a. $S + \lambda P = 0$

b. $S - \lambda P = 0$ and $\lambda S + P = 0$

c. $P - \lambda S = 0$

d. All of these

21. Axis of a parabola is $y = x$ and vertex and focus are at a distance $\sqrt{2}$ and $2\sqrt{2}$ respectively from the origin. Then equation of the parabola is-

a. $(x-y)^2 = 8(x+y-2)$ **b.** $(x+y)^2 = 2(x+y-2)$

c. $(x-y)^2 = 4(x+y-2)$ **d.** $(x+y)^2 = 2(x-y+2)$

22. If $\vec{a}$ and $\vec{b}$ be unlike vectors, then $\vec{a}\cdot\vec{b} =$

a. $|\vec{a}||\vec{b}|$ **b.** $-|\vec{a}||\vec{b}|$

c. 0 **d.** None of these

23. A line makes the same angle θ, with each of the x and z-axis. If the angle β, which it makes with y-axis is such that $\sin^2\beta = 3\sin^2\theta$, then $\cos^2\theta$ equals

a. $\frac{3}{5}$ **b.** $\frac{2}{3}$

c. $\frac{1}{5}$ **d.** None of these

24. $(\sim(\sim p)) \wedge q$ is equal to

a. $\sim p \wedge q$ **b.** $p \wedge q$ **c.** $p \wedge \sim q$ **d.** $\sim p \wedge \sim q$

25. If α and β are the roots of the equation $x^2 + 6x + \lambda = 0$ and $3\alpha + 2\beta = -20$, then $\lambda =$

a. –8 **b.** –16 **c.** 16 **d.** 8

26. If a, b, c, d are in *H.P.*, then $ab + bc + cd$ is equal to

a. $3ad$ **b.** $(a+b)(c+d)$

c. $3ac$ **d.** None of these

27. If $y = 3x + 6x^2 + 10x^3 + ...$, then the value of x in terms of y is

a. $1-(1-y)^{-1/3}$ **b.** $1-(1+y)^{1/3}$

c. $1+(1+y)^{-1/3}$ **d.** $1-(1+y)^{-1/3}$

28. The number of arrangements of the letters of the word BANANA in which the two *N*'s do not appear adjacently, is

a. 40 **b.** 60

c. 80 **d.** 100

29. Differential coefficient of $\sin^{-1} x$ *w.r.t* $\cos^{-1}\sqrt{1-x^2}$ is

a. 1 **b.** $\frac{1}{1+x^2}$

c. 2 **d.** None of these

30. The interval in which the function x^2e^{-x} is non decreasing, is

a. $(-\infty, 2]$ **b.** $[0, 2]$

c. $[2, \infty)$ **d.** None of these

Space for rough work

JEE ADVANCE PAPER-I

SECTION 1 Contains 8 Questions.

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. For any integer k, let $\alpha_k = \cos\left(\frac{k\pi}{7}\right) + i\sin\left(\frac{k\pi}{7}\right)$, where $i = \sqrt{-1}$. The value of the expression $\frac{\sum_{k=1}^{12} |\alpha_{k+1} - \alpha_k|}{\sum_{k=1}^{3} |\alpha_{4k-1} - \alpha_{4k-2}|}$ is

2. Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1+x)^2 + (1+x)^3 + \ldots + (1+x)^{49} + (1+mx)^{50}$ is $(3n+1)\,{}^{51}C_3$ for some positive integer n. Then the value of n is

3. The minimum number of times a fair coin needs to be tossed, so that the probability of getting at least two heads is at least 0.96 is

4. For any real number x, let $|x|$ denote the largest integer less than or equal to x. Let f be a real valued function defined on the interval $[-10, 10]$ by $f(x) = \begin{cases} x - [x] & \text{if } [x] \text{ is odd} \\ 1 + [x] - x & \text{if } [x] \text{ is even} \end{cases}$

Then the value of $\frac{\pi^2}{10}\int_{-10}^{10} f(x)\cos\pi x\,dx$ is

5. The value of $\int_0^1 4x^3 \left\{\frac{d^2}{dx^2}(1-x^2)^5\right\} dx$ is _____

6. Let f be a real-valued differentiable function on R (the set of all real numbers) such that $f(1) = 1$. If the y-intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P, then the value of $f(-3)$ is equal to

7. The line $2x + y = 1$ is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If this line passes through the point of intersection of the nearest directrix and the x-axis, then the eccentricity of the hyperbola is

8. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors satisfying $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$, then $|2\vec{a} + 5\vec{b} + 5\vec{c}|$ is.

SECTION 2 Contains 10 Multiple Choice Questions

With one or more than one correct option

9. If $y = \sec(\tan^{-1} x)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to

a. $\frac{1}{\sqrt{2}}$ **b.** $\frac{1}{2}$ **c.** 1 **d.** $\sqrt{2}$

10. For function $f(x) = x\cos\frac{1}{x}, x \geq 1$,

a. for atleast one x in interval $[1,\infty), f(x+2) - f(x) < 2$

b. $\lim_{x\to\infty} f'(x) = 1$

c. for all x in the interval $[1,\infty), f(x+2) - f(x) > 2$

d. f' (x) is strcily decreasing in the interval $[1,\infty)$

11. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \ldots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2} g(x)\,dx$ equals

a. $\frac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}} + K$ **b.** $\frac{1}{n-1}(1+nx^n)^{1-\frac{1}{n}} + K$

c. $\frac{1}{n(n+1)}(1+nx^n)^{1+\frac{1}{n}} + K$ **d.** $\frac{1}{n+1}(1+nx^n)^{1+\frac{1}{n}} + K$

12. Let $f : [0,2] \to R$ be a function which is continuous on [0, 2] and is differentiable on (0, 2) with $f(0) = 1$. Let $F(x) = \int_0^{x^2} f(\sqrt{t})\,dt$ for $x \in [0,2]$. If $F'(x) = f'(x)$ for all $x \in (0,2)$, then $F(2)$ equals

a. $e^2 - 1$ **b.** $e^4 - 1$ **c.** $e - 1$ **d.** e^4

13. The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4 + 2x}{\sqrt{1-x^2}}$ in $(-1,1)$ satisfying $f(0) = 0$. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)\,dx$ is

a. $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ **b.** $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ **c.** $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ **d.** $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

14. Let $O(0,0), P(3,4), Q(6,0)$ be the vertices of the triangle OPQ. The point R inside the triangle OPQ is such that the triangles OPR, PQR, OQR are of equal area. The coordinates of R are

a. $\left(\frac{4}{3}, 3\right)$ **b.** $\left(3, \frac{2}{3}\right)$

c. $\left(3, \frac{4}{3}\right)$ **d.** $\left(\frac{4}{3}, \frac{2}{3}\right)$

15. Tangents are drawn to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$, parallel to the straight lien $2x - y = 1$. The points of contact of the tangents on the hyperbola are.

a. $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ **b.** $\left(-\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

c. $3\sqrt{3}, -2\sqrt{2}$ **d.** $-3\sqrt{3}, 2\sqrt{2}$

16. If the vectors $\overline{AB} = 3\hat{i} + 4\hat{k}$ and $\overline{AC} = 5\hat{i} - 2\hat{j} + 4\hat{k}$ are the sides of a triangles ABC, then the length of the median through A is

a. $\sqrt{18}$ **b.** $\sqrt{72}$

c. $\sqrt{33}$ **d.** $\sqrt{45}$

17. In R^3, consider the planes $P_1 : y = 0$ and $P_2 : x + z = 1$. Let P_3 be a plane, different from P_1 and P_2, which passes through the intersection of P_1 and P_2. If the distance of the point (0, 1, 0) from P_3 is 1 and the distance of a point (α, β, γ) from P_3 is 2, then which of the following relations is (are) true?

a. $2\alpha + \beta + 2\gamma + 2 = 0$

b. $2\alpha - \beta + 2\gamma + 4 = 0$

c. $2\alpha + \beta - 2\gamma - 10 = 0$

d. $2\alpha - \beta + 2\gamma - 8 = 0$

18. The negation of $\sim s \vee (\sim r \wedge s)$ is equivalent to

a. $s \wedge \sim r$ **b.** $s \wedge (r \wedge \sim s)$

c. $s \vee (r \vee \sim s)$ **d.** $s \wedge r$

SECTION 3 Contains 2 Match The Following Type Questions

You will have to match entries in Column I with the entries in Column II.

19. Match the statements given in Column I with the interval/union of intervals given in Column II

Column I	Column II
(A) The set $\left\{\text{Re}\left(\frac{2iz}{1-z^2}\right)\right\} : z$ is a complex number, $\lvert z\rvert = 1, z \neq \pm 1$ is	**1.** $(-\infty, -1) \cup (1, \infty)$
(B) The domain of the function $f(x) = \sin^{-1}\left(\frac{8(3)^{x-2}}{1-3^{2(x-1)}}\right)$ is	**2.** $(-\infty, 0) \cup (0, \infty)$
(C) If $f(\theta) = \begin{vmatrix} 1 & \tan\theta & 1 \\ -\tan\theta & 1 & \tan\theta \\ -1 & -\tan\theta & 1 \end{vmatrix}$, then the set $\left\{f(\theta) : 0 \le \theta < \frac{\pi}{2}\right\}$ is	**3.** $[2, \infty)$
(D) If $f(x) = x^{3/2}(3x - 10), x \ge 0$, then f(x) is increasing in	**4.** $(-\infty, -1] \cup [1, \infty)$
	5. $(-\infty, 0] \cup [2, \infty)$

20. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$ Match the conditions/expressions in Column I with statements in Column II

Column I	Column II
(A) If $-1 < x < 1$, then $f(x)$ satisfies	**1.** $0 < f(x) < 1$
(B) If $1 < x < 2$, then $f(x)$ satisfies	**2.** $f(x) < 0$
(C) If $3 < x < 5$, then $f(x)$ satisfies	**3.** $f(x) > 0$
(D) If $x > 5$, then $f(x)$ satisfies	**4.** $f(x) < 1$

Space for rough work

JEE ADVANCE PAPER-II

SECTION 1 Contains 8 Questions.

The answer to each question is a single digit integer ranging from 0 to 9 (both inclusive).

1. The least period of the function $\sin\left(\frac{\pi[x]}{12}\right)+\cos\left(\frac{\pi x}{4}\right)+\tan\left(\frac{\pi[x]}{3}\right)$ is λ, then the value of 201λ must be (where $[\cdot]$ denotes the greatest integer function)

2. If $y=\tan^{-1}\left(\frac{ax-b}{bx+a}\right)$ then the value of $(2008)\left.\frac{dy}{dx}\right|_{x=-1}$ must be

3. The indicated horse power I of an engines is calculated from the formula $I=\frac{PLAN}{33000}$ where $A=\frac{\pi}{4}d^2$. Assuming that error of 10% may have been made in measuring *P, L, N* and *d*. If the greatest possible in *I* is $\lambda\%$ then λ must be

4. If $\int\sin^{-1}\left(\frac{2x+2}{\sqrt{(4x^2+8x+13)}}\right)dx=(x+1)\tan^{-1}\left(\frac{2x+2}{3}\right)$ $+\lambda\ln(4x^2+8x+13)+c$, then the value of -4λ must be

5. *A* particle moves in a straight line with a velocity given by $\frac{dx}{dt}=(x+1)$ (x is the distance travelled). If the time taken by a particle to traverse a distance of 99 m is λ. then the value of $20\lambda\log_{10}e$ must be

6. If a triangle has its orthocenter at (1, 1) and circum centre at $\left(\frac{3}{2},\frac{3}{4}\right)$ and if centroid and nine point centre are (α,β) and (γ,δ) respectively, then the value of $6\alpha+12\beta+4\gamma+8\delta$ must be

7. If the radius of a circle which passes through the point $(2,0)$ and whose centre is the limit of the point of intersection of the lines $3x+5y=1$ and $(2+c)x+5c^2y=1$ as $c\to1$ is $\frac{\sqrt{\lambda}}{25}$, then the value of λ must be

8. *A*, *B*, *C*, *D* are any four points in the space. If $|\overrightarrow{AB}\times\overrightarrow{CD}+\overrightarrow{BC}\times\overrightarrow{AD}+\overrightarrow{CA}\times\overrightarrow{BD}|=\lambda$ (area of ΔABC) then the value of 125λ must be

SECTION 2 Contains 8 Multiple Choice Questions

With one or more than one correct option

9. Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A\times B$ having 3 or more elements is

a. 256 **b.** 220
c. 219 **d.** 211

10. The real number k for which the equation, $2x^3+3x+k=0$ has two distinct real roots in [0, 1]

a. lies between 1 and 2
b. lies between 2 and 3
c. lies between –1 and 0
d. does not exist

11. A complex number *z* is said to be unimodular if $|z|=1$. Suppose z_1 and z_2 are complex numbers such that $\frac{z_1-2z_2}{2-z_1z_2}$ is unimodular and z_2 is not unimodular. Then the point z_1 lies on a

a. straight line parallel to *x*-axis
b. straight line parallel to *y*-axis
c. circle of radius 2
d. circle of radius $\sqrt{2}$

12. If the adjoint of a 3×3 matrix P is $\begin{bmatrix}1&4&4\\2&1&7\\1&1&3\end{bmatrix}$, then the possible value(s) of the determinant of P is (are)

a. –2 **b.** –1
c. 1 **d.** 2

13. If the sum of the first ten term of the series $\left(1\frac{3}{5}\right)^2+\left(2\frac{2}{5}\right)^2+\left(3\frac{1}{5}\right)^2+4^2+\left(4\frac{4}{5}\right)^2+\ldots$, is $\frac{16}{5}m$, then m is equal to:

a. 102 **b.** 101
c. 100 **d.** 99

14. Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is

a. 264 **b.** 265 **c.** 53 **d.** 67

15. Let $\theta, \varphi \in 0, 2\pi$ be such that $2\cos\theta\, 1 - \sin\varphi = \sin^2\theta\left(\tan\frac{\theta}{2} + \cot\frac{\theta}{2}\right)\cos\varphi - 1$ tan $2\pi - \theta > 0$ and $-1 < \sin\theta < -\frac{\sqrt{3}}{2}$. Then φ cannot satisfy:

a. $0 < \varphi < \frac{\pi}{2}$

b. $\frac{\pi}{2} < \varphi < \frac{4\pi}{3}$

c. $\frac{4\pi}{3} < \varphi < \frac{3\pi}{2}$

d. $\frac{3\pi}{2} < \varphi < 2\pi$

16. If $0 < x < 1$, then $\sqrt{1+x^2}\left[\{x\cos(\cot^{-1}x) + \sin(\cot^{-1}x)\}^2 - 1\right]^{1/2}$ is equal to

a. $\frac{x}{\sqrt{1+x^2}}$ **b.** x

c. $x\sqrt{1+x^2}$ **d.** $\sqrt{1+x^2}$

SECTION 3 Contains 2 Paragraph Type Questions

Each paragraph describes an experiment, a situation or a problem. Two multiple choice questions will be asked based on this paragraph. One or more than one option can be correct.

Paragraph for Question No. 17 to 18

Let a_n denote the number of all *n*-digit positive integers formed by the digits 0, 1 or both such that no consecutive digits in them are 0. Let b_n = the number of such *n*-digit integers ending with digit 1 and c_n = the number of such *n*–digit integers ending with digit 0.

17. Which of the following is correct?

a. $a_{17} = a_{16} + a_{15}$ **b.** $c_{17} \neq c_{16} + c_{15}$

c. $b_{17} \neq b_{16} + c_{16}$ **d.** $a_{17} = c_{17} + b_{16}$

18. The value of b_6 is

a. 7 **b.** 8

c. 9 **d.** 11

Paragraph for Question No. 19 to 20

Box 1 contains three cards bearing numbers 1, 2, 3; box 2 contains five cards bearing numbers 1, 2, 3, 4, 5: and box 3 contains seven cards bearing numbers 1, 2, 3, 4, 5, 6, 7. A card is drawn from each of the boxes/ Let x_i be the number on the card drawn from the i^{th} box, $i = 1, 2, 3$.

19. The probability that $x_1 + x_2, x_3$ is odd, is

a. $\frac{29}{105}$ **b.** $\frac{53}{105}$

c. $\frac{57}{105}$ **d.** $\frac{1}{2}$

20. The probability that x_1, x_2, x_3 are in an arithmetic progression, is

a. $\frac{9}{105}$ **b.** $\frac{10}{105}$

c. $\frac{11}{105}$ **d.** $\frac{7}{105}$

Space for rough work

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	d	c	d	a	c	b	b	d	a
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
b	a	c	a	d	a	c	b	a	d
21.	**22.**	**23.**	**24.**	**25.**	**26.**	**27.**	**28.**	**29.**	**30.**
a	b	a	b	b	a	d	a	a	b

1. **(b)** Given $\frac{\log(x-1)}{2\log 2}=\frac{\log(x-3)}{\log 2}$

$\Rightarrow \quad (x-1)=(x-3)^2$

$\Rightarrow \quad x=2,5$

But at $x=2$, given log is not defined.

2. **(d)** $x^2+5|x|+4=0$

$\Rightarrow \quad |x|^2+5|x|+4=0$

$\Rightarrow \quad |x|=-1,-4$, which is not possible. Hence, the given equation has no real root.

3. **(c)** $x^4-x^3+x-1=0$

$\Rightarrow \quad x^3(x-1)+1(x-1)=0$

$x-1=0$ or $x^3+1=0$

$\Rightarrow \quad x=1,-1,\frac{1+\sqrt{3}i}{2},\frac{1-\sqrt{3}i}{2}$

So its real roots are 1 and -1.

4. **(d)** $A=\begin{bmatrix}1&0&0\\5&2&0\\-1&6&1\end{bmatrix}$

$\Rightarrow \quad adj(A)=\begin{bmatrix}2&-5&32\\0&1&-6\\0&0&2\end{bmatrix}^T=\begin{bmatrix}2&0&0\\-5&1&0\\32&-6&2\end{bmatrix}.$

5. **(a)** $0.\overset{\cdot\cdot\cdot\cdot}{234}=0.2343434.....$

$0.2+0.034+0.00034+0.0000034+...$

$=0.2+\frac{34}{1000}+\frac{34}{100000}+\frac{34}{10000000}+.....\infty$

$=\frac{2}{10}+34\left[\frac{1}{10^3}+\frac{1}{10^5}+\frac{1}{10^7}+.....\infty\right]$

$=\frac{2}{10}+34\left[\frac{1/10^3}{1-1/1000}\right]=\frac{2}{10}+34\times\frac{1}{1000}\times\frac{100}{99}$

$=\frac{2}{10}+\frac{34}{990}=\frac{232}{990}.$

6. **(c)** $(1+x+x^2+x^3)^5=(1+x)^5(1+x^2)^5$

$=(1+5x+10x^2+10x^3+5x^4+x^5)$

$\times(1+5x^2+10x^4+10x^6+5x^8+x^{10})$

Therefore the required sum of coefficients

$=(1+10+5).2^5=16\times 32=512$

Note: $2^n=2^5=$ Sum of all the binomial coefficients in the 2nd bracket in which all the powers of x are even.

7. **(b)** $(1-2x+3x^2)e^{-x}=(1-2x+3x^2)\left\{1-\frac{x}{1!}+\frac{x^2}{2!}-\frac{x^3}{3!}+...\right\}$

$\therefore$ The coefficient of $x^5=1\left(-\frac{1}{5!}\right)+(-2)\left(\frac{1}{4!}\right)+3\left(-\frac{1}{3!}\right)$

$=-\frac{1}{120}-\frac{1}{12}-\frac{1}{2}=-\frac{71}{120}.$

8. **(b)** Suppose $x_1x_2x_3x_4x_5x_6x_7$ represents a seven digit number. Then x_1 takes the value $1, 2, 3, \ldots, 9$ and $x_2, x_3, \ldots x_7$ all take values 0, 1, 2, 3, …, 9. If we keep $x_1, x_2, \ldots, x_6$ fixed, then the sum $x_1+x_2+\ldots+x_6$ is either even or odd. Since x_7 takes 10 values 0, 1, 2, …,9, five of the numbers so formed will be even and 5 odd.

Hence the required number of numbers

$=9\,.10\,.10\,.10\,.10\,.10\,.5=4500000.$

9. **(d)** It is given that $P(E)\le P(F)\Rightarrow E\subseteq F$

and $P(E\cap F)>0\ \Rightarrow\ E\subset F$ . . .(*ii*)

(a) Occurrence of $E\ \Rightarrow$ occurrence of F [from Eq.(*i*)]

(b) Occurrence of $F\ \Rightarrow$ occurrence of E [from Eq.(*ii*)]

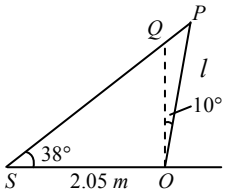
(c) Non-occurrence of $E\ \Rightarrow$ non-occurrence of F [from Eq.(*i*)]

10. **(a)** $\tan\left(\frac{\pi}{4}+\theta\right)-\tan\left(\frac{\pi}{4}-\theta\right)=\frac{1+\tan\theta}{1-\tan\theta}-\frac{1-\tan\theta}{1+\tan\theta}$

$=\frac{4\tan\theta}{1-\tan^2\theta}=2\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)=2\tan 2\theta\,.$

11. **(b)** It is understandable.

12. **(a)**



$\frac{\sin 38^o}{l}=\frac{\sin(SPO)}{2.05}$

$=\frac{\sin(180^o-38^o-90^o-10^o)}{2.05}\Rightarrow\ l=\frac{2.05\sin 38^o}{\sin 42^o}$

13. **(c)** $\lim_{x\to 0^+} f(x) = x^2 \sin\frac{1}{x}$,

but $-1 \le \sin\frac{1}{x} \le 1$ and $x \to 0$

Therefore, $\lim_{x\to 0^+} f(x) = 0 = \lim_{x\to 0^-} f(x) = f(0)$

Hence $f(x)$ is continuous at $x = 0$.

14. **(a)** $y = \sqrt{\left[\frac{(x-a)(x-b)}{(x-c)(x-d)}\right]}$

$\Rightarrow \quad \log y = \frac{1}{2}[\log(x-a) + \log(x-b) - \log(x-c) - \log(x-d)]$

Differentiating *w.r.t.* x we get

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{2}\left[\frac{1}{(x-a)} + \frac{1}{(x-b)} - \frac{1}{(x-c)} - \frac{1}{(x-d)}\right]$$

Thus $\frac{dy}{dx} = \frac{y}{2}\left[\frac{1}{(x-a)} + \frac{1}{(x-b)} - \frac{1}{(x-c)} - \frac{1}{(x-d)}\right]$.

15. **(d)** $f(x) = \frac{x^2-1}{x^2+1} = \frac{x^2+1-2}{x^2+1} = 1 - \frac{2}{x^2+1}$

$\therefore \quad f(x) < 1 \forall x$

and ≥ -1 as $\frac{2}{x^2+1} \le 2$

$\therefore \quad -1 \le f(x) < 1$

Hence $f(x)$ has minimum value –1 and also there is no maximum value.

Alternate: $f'(x) = \frac{(x^2+1)2x - (x^2-1)2x}{(x^2+1)^2} = \frac{4x}{(x^2+1)^2}$

$f'(x) = 0 \Rightarrow x = 0$

$$f''(x) = \frac{(x^2+1)^2 4 - 4x.2(x^2+1)2x}{(x^2+1)^4}$$

$$= \frac{(x^2+1)4 - 16x(x)}{(x^2+1)^3} = \frac{-12x^2+4}{(x^2+1)^3}$$

$\therefore \quad f''(0) > 0$

$\therefore$ There is only one critical point having minima.

Hence $f(x)$ has least value at $x = 0$.

$$f_{\min} = f(0) = \frac{-1}{1} = -1.$$

16. **(a)** Put $a^x = t \Rightarrow a^x \log_e a\, dx = dt$, then

$$\int \frac{a^x}{\sqrt{1-a^{2x}}}dx = \frac{1}{\log_e a}\int \frac{dt}{\sqrt{1-t^2}}$$

$$= \frac{1}{\log_e a}\sin^{-1}(t) + c = \frac{\sin^{-1}(a^x)}{\log_e a} + c.$$

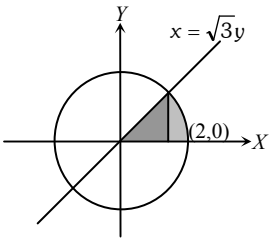
17. **(c)** Required area $= \int_0^{\sqrt{3}} \frac{x}{\sqrt{3}}dx + \int_{\sqrt{3}}^{2}\sqrt{4-x^2}\,dx$

$$= \frac{1}{\sqrt{3}}\left[\frac{x^2}{2}\right]_0^{\sqrt{3}} + \left[\frac{x}{2}\sqrt{4-x^2} + \frac{4}{2}\sin^{-1}\frac{x}{2}\right]_{\sqrt{3}}^{2}$$

$$= \frac{\sqrt{3}}{2} + \left[\pi - \frac{\sqrt{3}}{2} - \frac{2\pi}{3}\right] = \frac{\pi}{3}.$$

Trick: Area of sector made by an arc $= \frac{\theta^c R^2}{2}$

$$= \frac{\pi}{6}.\frac{4}{2} = \frac{\pi}{3}.$$



18. **(b)** $\frac{dy}{dx} = \frac{2}{x^2} \Rightarrow dy = \frac{2}{x^2}dx$, Now integrate it.

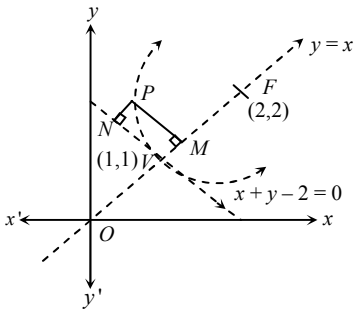
19. **(a)** Here, equation of line is $y = x\tan\alpha + c, c > 0$

Length of the perpendicular drawn on line from point $(a\cos\alpha, a\sin\alpha)$

$$p = \frac{-a\sin\alpha + a\cos\alpha\tan\alpha + c}{\sqrt{1+\tan^2\alpha}};\ p = \frac{c}{\sec\alpha} = c\cos\alpha$$

20. **(d)** It is a fundamental concept.

21. **(a)**



Since, distance of vertex from origin is $\sqrt{2}$ and focus is $2\sqrt{2}$.

$\therefore$ $V(1, 1)$ and $F(2, 2)$ (*ie*, lying on $y = x$)

where, length of latusrectum $= 4a = 4\sqrt{2}$ $(\because a = \sqrt{2})$

$\therefore$ By definition of parabola $PM^2 = (4a)(PN)$

Where, PN is length of perpendicular upon $x + y - 2 = 0$ (*ie*, tangent at vertex).

$$\Rightarrow \quad \frac{(x-y)^2}{2} = 4\sqrt{2}\left(\frac{x+y-2}{\sqrt{2}}\right)$$

$$\Rightarrow \quad (x-y)^2 = 8(x+y-2)$$

22. **(b)** $\vec{a}\cdot\vec{b} = -|\vec{a}||\vec{b}|$, $(\because \cos\theta = -1)$

23. **(a)** Here, $l = \cos\theta, m = \cos\beta, n = \cos\theta$, $(\because l = n)$

Now, $l^2 + m^2 + n^2 = 1$

$\Rightarrow \quad 2\cos^2\theta + \cos^2\beta = 1$

$\Rightarrow \quad 2\cos^2\theta = \sin^2\beta$ Given, $\sin^2\beta = 3\sin^2\theta$

$\Rightarrow \quad 2\cos^2\theta = 3\sin^2\theta \Rightarrow \ 5\cos^2\theta = 3,$

$\therefore \quad \cos^2\theta = \frac{3}{5}.$

24. **(b)** $(\sim(\sim p)) \wedge q = p \wedge q$.

25. **(b)** $\alpha + \beta = -6$ ...(*i*)

$\alpha\beta = \lambda$ ...(*ii*)

and given $3\alpha + 2\beta = -20$ ...(*iii*)

Solving (*i*) and (*iii*), we get $\beta = 2, \alpha = -8$

Substituting these values in (*ii*), we get $\lambda = -16$.

26. **(a)** Since a, b, c, d are in H.P., therefore b is the H.M. of a and c *i.e.* $b = \frac{2ac}{a+c}$ and c is the H.M. of b and d

i.e. $c = \frac{2bd}{b+d}, \ \therefore (a+c)(b+d) = \frac{2ac}{b}.\frac{2bd}{c}$

$\Rightarrow \quad ab + ad + bc + cd = 4ad \Rightarrow ab + bc + cd = 3ad.$

Trick : Check for $a = 1, b = \frac{1}{2}, c = \frac{1}{3}, d = \frac{1}{4}$.

27. **(d)** We have $y = 3x + 6x^2 + 10x^3 + ...$

$\therefore \quad 1 + y = 1 + 3x + 6x^2 + 10x^3 + ...$

$\Rightarrow \quad 1 + y = (1-x)^{-3} \Rightarrow 1 - x = (1+y)^{-1/3} \Rightarrow \ x = 1 - (1+y)^{-1/3}$

28. **(a)** Total number of arrangements of word BANANA

$= \frac{6!}{3!2!} = 60$

The number of arrangements of words BANANA in which two *N*'s appear adjacently $= \frac{5!}{3!} = 20$

Required number of arrangements = 60 – 20 = 40

29. **(a)** Let $y_1 = \sin^{-1}x$ and $y_2 = \cos^{-1}\sqrt{1-x^2}$

Differentiating *w.r.t.* x of y_1 and y_2, we get

$\frac{dy_1}{dx} = \frac{1}{\sqrt{1-x^2}}$

$\frac{dy_2}{dx} = -\frac{1}{\sqrt{1-(1-x^2)}}\frac{1(-2x)}{2\sqrt{1-x}} = \frac{1}{\sqrt{1-x^2}} \Rightarrow \frac{dy_2}{dy_1} = 1.$

30. **(b)** Let $y = f(x) = x^2e^{-x}$

$\Rightarrow \quad \frac{dy}{dx} = 2xe^{-x} - x^2e^{-x} = e^{-x}(2x - x^2)$

Hence $f'(x) \geq 0$ for every $x \in [0, 2]$, therefore it is non-decreasing in [0, 2].

JEE Advance Paper-I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
4	5	3	4	2	9	2	3	a	b,c,d
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
a	b	b	c	a,b	c	b,d	d	b	a

1. **(4)** $\frac{\sum_{k=1}^{12}\left|e^{i\frac{k\pi}{7}}\right|\left|e^{i\frac{\pi}{7}}-1\right|}{\sum_{k=1}^{3}\left|e^{i(4k-2)}\right|\left|e^{i\frac{\pi}{7}}-1\right|} = \frac{12}{3} = 4$

2. **(5)** Coeff. x^2

$\Rightarrow \quad {}^2C_2 + {}^3C_2 + {}^4C_2 + ... + {}^{49}C_2 + {}^{50}C_2m^2 = (3n+1)\,{}^{51}C_3$

$\Rightarrow \quad {}^3C_3 + {}^3C_2 + {}^4C_2 + ... + {}^{49}C_2 + {}^{50}C_2m^2 = (3n+1)\,{}^{51}C_3$

$\Rightarrow \quad {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r \Rightarrow \ {}^{50}C_3 + {}^{50}C_2 \cdot m^2 = (3n+1)\,{}^{51}C_3$

$\Rightarrow \quad {}^{50}C_3 + {}^{50}C_2 + (m^2-1)\,{}^{50}C_2 = 3n\frac{51}{3}\cdot{}^{50}C_2 + {}^{51}C_3$

$\Rightarrow \quad {}^{51}C_3 + (m^2-1)\,{}^{50}C_2 = 51n\cdot{}^{50}C_2 + {}^{51}C_3$

$\Rightarrow \quad m^2 - 1 = 51n \ \ m^2 = 51n + 1$

Min value of m^2 for $51n+1$ is integer for $n = 5$.

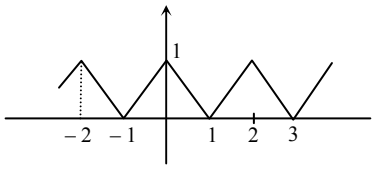
3. **(8)** Let coin was tossed 'n' times

Probability of getting at least two heads $= 1 - \left[\frac{1}{2^n} + \frac{n}{2^n}\right]$

$\Rightarrow \quad 1 - \left[\frac{n+1}{2^n}\right] \geq 0.96$

$\Rightarrow \quad \frac{2^n}{n+1} \geq 25 \ \Rightarrow \ n \geq 8$

4. **(4)**



$f(x) = \begin{cases} x-1, & 1 \leq x < 2 \\ 1-x, & 0 \leq x < 1 \end{cases}$ $f(x)$ is periodic with period 2

$\therefore \quad I = \int_{-10}^{10} f(x)\cos\pi x\,dx$

$= 2\int_0^{10} f(x)\cos\pi x\,dx = 2\times 5\int_0^2 f(x)\cos\pi x\,dx$

$10 = \left[\int_0^1 (1-x)\cos\pi x dx + \int_1^2 (x-1)\cos\pi x\,dx\right] = 10(I_1 + I_2)$

$I_2 = \int_1^2 (x-1)\cos\pi x\,dx$ put $x - 1 = t$

$$I_2 = -\int_0^1 t\cos\pi t\ dt$$

$$I_1 = -\int_0^1 (1-x)\cos\pi x\,dx = -\int_0^1 x\cos(\pi x)dx$$

$$\therefore \quad I = 10\left[-2\int_0^1 x\cos\pi x\,dx\right] = -20\left[x\frac{\sin\pi x}{\pi} + \frac{\cos\pi x}{\pi^2}\right]_0^1$$

$$= -20\left[-\frac{1}{\pi^2} - \frac{1}{\pi^2}\right] = \frac{40}{\pi^2} \therefore \frac{\pi^2}{10} I = 4$$

5. **(2)** $\int_0^1 \underset{\text{I}}{4x^3}\ \frac{d^2}{dx^2}\underset{\text{II}}{(1-x^2)^5}\ dx$

$$= \left[4x^3\frac{d}{dx}(1-x^2)^5\right]_0^1 - \int_0^1 12x^2\frac{d}{dx}(1-x^2)^5 dx$$

$$= \left[4x^3\times 5(1-x^2)^4(-2x)\right]_0^1 - 12\left[\left[x^2(1-x^2)^5\right]_0^1 - \int_0^1 2x\,(1-x^2)^5\ dx\right]$$

$$= 0-0-12[0-0]+12\int_0^1 2x(1-x^2)^5 dx \ = 12\times\left[-\frac{(1-x^2)^6}{6}\right]_0^1$$

$$= 12\left[0+\frac{1}{6}\right] = 2$$

6. **(9)** $y - y_1 = m(x - x_1)$

Put $x = 0$, to get y intercept $y_1 = mx_1 = x_1^3$

$$y_1 - x_1\frac{dy}{dx} = x_1^3$$

$$x\frac{dy}{dx} - y = -x^3$$

$$\frac{dy}{dx} - \frac{y}{x} = -x^2$$

$$e^{\int -\frac{1}{x}dx} = e^{-\ln x} = \frac{1}{x}$$

$$y\times\frac{1}{x} = \int -x^2\times\frac{1}{x}dx$$

$$\frac{y}{x} = -\int x dx \Rightarrow \frac{y}{x} = -\frac{x^2}{2} + c$$

$$\Rightarrow \quad f(x) = -\frac{x^3}{2} + \frac{3}{2}x \therefore f(-3) = 9.$$

7. **(2)** Substituting $\left(\frac{a}{e}, 0\right)$ in $y = -2x + 1$

$$0 = -\frac{2a}{e} + 1$$

$$\frac{2a}{e} = 1 \ \ a = \frac{e}{2}$$

Also, $1 = \sqrt{a^2m^2 - b^2}$

$$1 = a^2m^2 - b^2$$

$$1 = 4a^2 - b^2$$

$$1 = \frac{4e^2}{4} - b^2$$

$$b^2 = e^2 = 1. \text{ Also, } \ b^2 = a^2(e^1 - 1)$$

$$\therefore \quad a = 1, e = 2$$

8. **(3)** $\left|\vec{a} - \vec{b}\right|^2 + \left|\vec{b} - \vec{c}\right|^2 + \left|\vec{c} - \vec{a}\right|^2 = 9$

$$\Rightarrow \quad 2 - 2\,\vec{a}\cdot\vec{b} + 2 - 2\ \vec{b}\cdot\vec{c} + 2 - 2\ \vec{c}\cdot\vec{a} = 9$$

$$\Rightarrow \quad -\frac{3}{2} = \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} \qquad \ldots(i)$$

Now, $|\vec{a} + \vec{b} + \vec{c}|^2 \geq 0$

$$\Rightarrow \quad 1+1+1+2\ \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} \geq 0$$

$$\Rightarrow \quad \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} \geq -\frac{3}{2} \qquad \ldots(ii)$$

Equation (i) and (ii) are simultaneously true

If $\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{c} = \vec{c}\cdot\vec{a} \geq -\frac{1}{2}$

Now, $\left|2\vec{a} + 5\vec{b} + 5\vec{c}\right|^2$

$$= 4 + 25 + 25 + 20\ \vec{a}\cdot\vec{b} + 50\vec{b}\cdot\vec{c} + 20(\vec{a}\cdot\vec{c})$$

$$= 54 - 10 - 25 - 10 = 9 \Rightarrow \ \left|2\vec{a} + 5\vec{b} + 5\vec{c}\right| = 3$$

9 **(a)** $y = \sec(\tan^{-1}x)$

$$\Rightarrow \quad \frac{dy}{dx} = \sec(\tan^{-1}x).\tan(\tan^{-1}x).\frac{1}{1+x^2}$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{x=1} = \frac{\sqrt{2}}{1+1} = \frac{1}{\sqrt{2}}$$

10. **(b,c,d)** For $f(x) = x\cos\left(\frac{1}{x}\right), x \geq 1$

$$f'(x) = \cos\left(\frac{1}{x}\right) + \frac{1}{x}\sin\left(\frac{1}{x}\right) \to 1 \text{ for } x \to \infty$$

$$\text{also } f'(x) = \frac{1}{x^2}\sin\left(\frac{1}{x}\right) - \frac{1}{x^2}\sin\left(\frac{1}{x}\right) - \frac{1}{x^3}\cos\left(\frac{1}{x}\right)$$

$$= \frac{1}{x^3}\cos\left(\frac{1}{x}\right) < 0 \quad \text{for } x \geq 1$$

$$\Rightarrow \quad f'(x) \text{ is decreasing for } [1, \infty)$$

$$\Rightarrow \quad f'(x+2) < f'(x). \text{ Also,}$$

$$\lim_{x\to\infty} f(x+2) - f(x) = \lim_{x\to\infty}\left[(x+2)\cos\frac{1}{x+2} - x\cos\frac{1}{x}\right] = 2$$

$$\therefore \quad f(x+2) - f(x) > 2 \forall x \geq 1$$

11. **(a)** Here $ff(x)=\dfrac{f(x)}{[1+f(x)^n]^{1/n}}=\dfrac{x}{(1+2x^n)^{1/x}}$

$\Rightarrow \quad fff(x)=\dfrac{x}{(1+3x^n)^{1/n}}$

$\Rightarrow \quad g(x)=(f\circ \underset{n \text{ times}}{f\circ\ldots\circ} f)(x)=\dfrac{x}{(1+nx^n)^{1/n}}$

Hence $I=\int x^{n-2}g(x)\,dx=\int\dfrac{x^{n-1}dx}{(1+nx^n)^{1/n}}$

$=\dfrac{1}{n^2}\int\dfrac{n^2x^{n-1}dx}{(1+nx^n)^{1/n}}=\dfrac{1}{n^2}\int\dfrac{\frac{d}{dx}(1+nx^n)}{(1+nx^n)^{1/n}}dx$

$\therefore \quad I=\dfrac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}}+K$

12. **(b)** $F(0)=0$

$F'(x)=2xf(x)=f'(x)$

$f(x)=e^{x^2+c}$

$f(x)=e^{x^2}\ (\because f(0)=1)$

$F(x)=\int\limits_0^{x^2}e^x dx$

$F(x)=e^{x^2}-1 \quad (\because F(0)=0) \Rightarrow F(2)=e^4-1$

13. **(b)** $\dfrac{dy}{dx}+\dfrac{x}{x^2-1}y=\dfrac{x^4+2x}{\sqrt{1-x^2}}$

This is a linear differential equation

I.F. $=e^{\int\frac{x}{x^2-1}dx}=e^{\frac{1}{2}\ln|x^2-1|}=\sqrt{1-x^2}$

$\Rightarrow$ Solution is $y\sqrt{1-x^2}=\int\dfrac{x(x^3+2)}{\sqrt{1-x^2}}\cdot\sqrt{1-x^2}dx$

or $\quad y\sqrt{1-x^2}=\int(x^4+2x)dx=\dfrac{x^5}{5}+x^2+c$

$f(0)=0 \Rightarrow c=0$

$\Rightarrow \quad f(x)\sqrt{1-x^2}=\dfrac{x^5}{5}+x^2$

Now, $\int\limits_{-\sqrt{3}/2}^{\sqrt{3}/2}f(x)dx=\int\limits_{-\sqrt{3}/2}^{\sqrt{3}/2}\dfrac{x^2}{\sqrt{1-x^2}}dx$ (Using property)

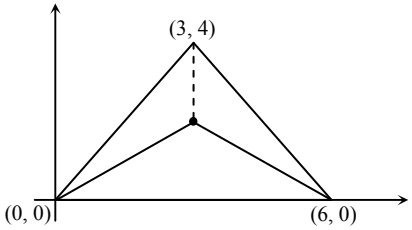
$=2\int\limits_0^{\sqrt{3}/2}\dfrac{x^2}{\sqrt{1-x^2}}dx=2\int\limits_0^{\pi/3}\dfrac{\sin^2\theta}{\cos\theta}\cos\theta d\theta$

(Taking $x=\sin\theta$)

$=2\int\limits_0^{\pi/3}\sin^2\theta d\theta=\left[\dfrac{\theta}{2}-\dfrac{\sin 2\theta}{4}\right]_0^{\pi/3}$

$=2\left(\dfrac{\pi}{6}\right)-2\left(\dfrac{\sqrt{3}}{8}\right)=\dfrac{\pi}{3}-\dfrac{\sqrt{3}}{4}.$

14. **(c)** Since, Δ is isosceles, hence centroid is the desired point.



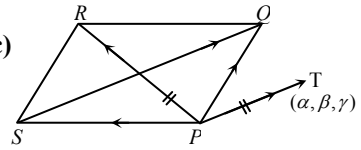
15. **(a,b)** Slope of tangent = m = 2 Equation of tangent in slope form is

$y=mx\pm\sqrt{a^2m^2-b^2}$, $y=2x\pm4\sqrt{2}$

and point of contact is $\left(-\dfrac{ma^2}{c},\dfrac{-b^2}{c}\right)$

$\equiv\left(-\dfrac{2\times9}{\pm4\sqrt{2}},-\dfrac{4}{\pm4\sqrt{2}}\right)\equiv\left(\pm\dfrac{9}{2\sqrt{2}},\pm\dfrac{1}{\sqrt{2}}\right)$

16. **(c)**



Area of base (PQRS) $=\dfrac{1}{2}|\overrightarrow{PR}\times\overrightarrow{SQ}|=\dfrac{1}{2}\begin{vmatrix}\hat i & \hat j & \hat k\\ 3 & 1 & -2\\ 1 & -3 & -4\end{vmatrix}$

$=\dfrac{1}{2}|-10\hat i+10\hat j-10\hat k|=5|\hat i-\hat j+\hat k|=5\sqrt{3}$

Height = proj. of PT on $\hat i-\hat j+\hat k=\left|\dfrac{1-2+3}{\sqrt{3}}\right|=\dfrac{2}{\sqrt{3}}$

Volume $=(5\sqrt{3})\left(\dfrac{2}{\sqrt{3}}\right)=10$ cu. units

17. **(b, d)** Let the required plane be $x+z+\lambda y-1=0$

$\Rightarrow \quad \dfrac{|\lambda-1|}{\sqrt{\lambda^2+2}}=1 \Rightarrow \lambda=-\dfrac{1}{2}$

$\Rightarrow \quad P_3\equiv 2x-y+2z-2=0$

Distance of P_3 from (α,β,γ) is 2 $\dfrac{|2\alpha-\beta+2\gamma-2|}{\sqrt{4\times1+4}}=2$

$\Rightarrow \quad 2\alpha-\beta+2\lambda+4=0$ and $2\alpha-\beta+2\lambda-8=0$

18. **(d)** $\sim[\sim s\vee(\sim r\wedge s)]$

$=\sim(\sim s)\wedge\sim(\sim r\wedge s)$

$=s\wedge(r\vee\sim s)$

$=(s\wedge r)\vee(s\wedge\sim s)$

$=s\wedge r\vee F\ =s\wedge r$

19. **(A)** $z=\dfrac{2i(x+iy)}{1-(x+iy)^2}=\dfrac{2i(x+iy)}{1-(x^2-y^2+2ixy)}$

Using $1-x^2=y^2$

$Z=\dfrac{2ix-2y}{2y^2-2ixy}=-\dfrac{1}{y}.$

$\because \quad -1\le y\le 1\Rightarrow -\dfrac{1}{y}\le -1 \text{ or } -\dfrac{1}{y}\ge 1.$

(B) For domain $-1\le\dfrac{8.3^{x-2}}{1-3^{2(x-1)}}\le 1$

$\Rightarrow \quad -1\le\dfrac{3^x-3^{x-2}}{1-3^{2x-2}}\le 1.$

Case (i): $\dfrac{3^x-3^{x-2}}{1-3^{2x-2}}-1\le 0$

$\Rightarrow \quad \dfrac{(3^x-1)(3^{x-2}-1)}{(3^{2x-2}-1)}\ge 0$

$\Rightarrow \quad x\in(-\infty,0]\cup(1,\infty).$

Case (ii)**:** $\dfrac{3^x-3^{x-2}}{1-3^{2x}-2}+1\ge 0$

$\Rightarrow \quad \dfrac{(3^{x-2}-1)(3^x+1)}{(3^x.3^{x-2}-1)}\ge 0$

$\Rightarrow \quad x\in(-\infty,1)\cup[2,\infty).$

So, $x\in(-\infty,0]\cup[2,\infty)$

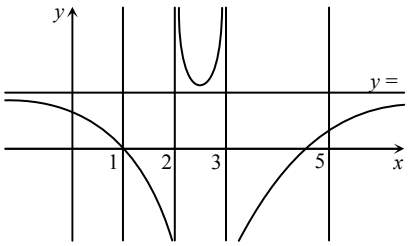
(C) $R_1\to R_1+R_3$

$$f(\theta)=\begin{vmatrix}0 & 0 & 2\\ -\tan\theta & 1 & \tan\theta\\ -1 & -\tan\theta & 1\end{vmatrix}=2(\tan^2\theta+1)=2\sec^2\theta.$$

(D) $f'(x)=\dfrac{3}{2}(x)^{1/2}(3x-10)+(x)^{3/2}\times 3=\dfrac{15}{2}(x)^{1/2}(x-2)$

Increasing, when $x\ge 2$.

20. $f(x)=\dfrac{(x-1)(x-5)}{(x-2)(x-3)}$**.** The graph of $f(x)$ is shown



(A) If $-1<x<1 \Rightarrow 0<f(x)<1$

(B) If $1<x<2 \Rightarrow f(x)<0$

(C) If $3<x<5 \Rightarrow f(x)<0$

(D) If $x>5 \Rightarrow f(x)<1$

JEE Advance Paper-II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
4824	1004	50	3	40	30	1601	500	c	d
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	a,d	b	c	a,c,d	c	a	b	b	c

1. **(4824)** The period of $\sin\left(\dfrac{\pi[x]}{12}\right)$ is $\dfrac{2\pi}{\pi/12}=24$ as

$\sin\left(\dfrac{\pi[x+24]}{12}\right)=\sin\left(\dfrac{\pi([x]+24)}{12}\right)$

$=\sin\left(2\pi+\dfrac{\pi[x]}{12}\right)=\sin\left(\dfrac{\pi[x]}{12}\right)$

Similarly the period of $\tan\left(\dfrac{\pi[x]}{3}\right)$ is 3 and the period of

$\cos\left(\dfrac{\pi x}{4}\right)$ is $\dfrac{2\pi}{\pi/4}=8$

Hence, the period of the given function

$\lambda=$ LCM of 24, 8, 3 = 24

$\therefore \quad 201\,\lambda=201\times 24=4824$

2. **(1004)** $\because\ y=\tan^{-1}\left(\dfrac{ax-b}{bx+a}\right)=\tan^{-1}\left(\dfrac{x-\frac{b}{a}}{1+\frac{b}{a}\cdot x}\right)$

$=\tan^{-1}x-\tan^{-1}\left(\dfrac{b}{a}\right)$

$\therefore \quad \dfrac{dy}{dx}=\dfrac{1}{1+x^2}-0$

$\therefore \quad \left.\dfrac{dy}{dx}\right|_{x=-1}=\dfrac{1}{1+(-1)^2}=\dfrac{1}{1+1}=\dfrac{1}{2}$

$\Rightarrow \quad (2008)\left.\dfrac{dy}{dx}\right|_{x=-1}=2008\times\dfrac{1}{2}=1004$

3. **(50)** $I=\dfrac{PLAN}{33000}=\dfrac{PL\left(\frac{\pi}{4}d^2\right)N}{33000}$

$\therefore \quad \dfrac{\Delta I}{I}=\dfrac{\Delta P}{P}+\dfrac{\Delta L}{L}+\dfrac{2\Delta d}{d}+\dfrac{\Delta N}{N}$

$\Rightarrow \quad \left(\dfrac{\Delta I}{I}\times 100\right)=\left(\dfrac{\Delta P}{P}\times 100\right)+\left(\dfrac{\Delta L}{L}\times 100\right)+2\left(\dfrac{\Delta d}{d}\times 100\right)+\left(\dfrac{\Delta N}{N}\times 100\right)$

$=10\%+10\%+2\times 10\%+10\%=50\%$

$\therefore \quad \lambda=50$

4. **(3)** Let $I=\int\sin^{-1}\left(\dfrac{2x+2}{\sqrt{(4x^2+8x+13)}}\right)dx$

$$I=\int \sin^{-1}\left(\frac{2x+2}{\sqrt{(2x+2)^2+3^2}}\right)dx$$

Put $2x+2=3\tan\theta$

$\therefore \quad 2dx=3\sec^2\theta\, d\theta$

Then $I=\int \theta\cdot\frac{3\sec^2\theta\, d\theta}{2}=\frac{3}{2}\{\theta\cdot\tan\theta-\ln\sec\theta\}+c$

$$=\frac{3}{2}\left\{\left(\frac{2x+2}{3}\right)\cdot\tan^{-1}\left(\frac{2x+2}{3}\right)-\ln\sqrt{1+\left(\frac{2x+2}{3}\right)^2}\right\}+c$$

$$=(x+1)\tan^{-1}\left(\frac{2x+2}{3}\right)-\frac{3}{4}\ln(4x^2+8x+13)+c$$

Hence $\lambda=-\frac{3}{4}$

Then, $-4\lambda=3$

5. **(40)** $\frac{dx}{dt}=x+1$

$\Rightarrow \quad \frac{dx}{x+1}=dt$

$\Rightarrow \quad \ln(x+1)=t+c$

Putting $t=0, x=0$ we get $c=0$

$\Rightarrow \quad t=\ln(x+1)$

For $x=99, t=\ln 100=2\log_e 10$

$\therefore \quad 20\lambda\log_{10}e=20\times 2\log_e 10\times\log_{10}e=40$

6. **(30)** Since, centroid divides the orthocenter and circum center in the ratio $2:1$ (internally) and if centroid $G(x, y)$, then $O(1,1)\overset{2}{\text{———}}G(x,y)\overset{1}{\text{———}}C\left(\frac{3}{2},\frac{2}{4}\right)$

$$x=\frac{2\times\frac{3}{2}+1\times 1}{2+1}=\frac{4}{3} \text{ and } y=\frac{2\times\frac{3}{4}+1\times 1}{2+1}=\frac{5}{6}$$

$\therefore$ Centroid is $\left(\frac{4}{3},\frac{5}{6}\right)$ and nine point centre is the mid point of orthocenter and circumcentre.

$\therefore$ Nine point centre is $\left(\frac{1+(3/2)}{2},\frac{1+(3/4)}{2}\right)$,

i.e., $\left(\frac{5}{4},\frac{7}{8}\right)$

$\Rightarrow \quad a=\frac{4}{3}, \beta=\frac{5}{6}$ and $\gamma=\frac{5}{4}, \delta=\frac{7}{8}$

$\therefore \quad 6\alpha+12\beta+4\gamma+8\delta$

$$=6\times\frac{4}{3}+12\times\frac{5}{6}+4\times\frac{5}{4}+8\times\frac{7}{8}=8+10+5+7=30$$

7. **(1601)** Solving the equation $(2+c)x+5c^2y=1$ and $3x+5y=1$,

$(2+c)x+5c^2\left(\frac{1-3x}{5}\right)=1$, then, $(2+c)x+c^2(1-3x)=1$

$\therefore \quad x=\frac{1-c^2}{2+c-3c^2}$

or $x=\frac{(1+c)(1-c)}{(3c+2)(1-c)}=\frac{1+c}{3c+2}$

$\therefore \quad x=\lim_{c\to 1}\frac{1+c}{3c+2} \Rightarrow \quad x=\frac{2}{5}$

$\therefore \quad y=\frac{1-3x}{5}=\frac{1-(6/5)}{5}=-\frac{1}{25}$

Therefore the centre of the required circle is $\left(\frac{2}{5},-\frac{1}{25}\right)$

but circle passes through (2,0)

$\therefore$ Radius of the required circle $=\sqrt{\left(\frac{2}{5}-2\right)^2+\left(-\frac{1}{25}-0\right)^2}$

$$=\sqrt{\frac{64}{25}+\frac{1}{625}}=\sqrt{\frac{1601}{625}}=\sqrt{\frac{1601}{25}}=\frac{\sqrt{\lambda}}{25}$$

$\therefore \quad \lambda=1601$

8. **(500)** Let PV of *A, B, C* and *D* be $\vec{a}, \vec{b}, \vec{c}$ and $\vec{0}$

$\therefore \quad \overrightarrow{AB}\times\overrightarrow{CD}=(\vec{b}-\vec{a})\times-\vec{c}=-\vec{b}\times\vec{c}+\vec{a}\times\vec{c}$

$\overrightarrow{BC}\times\overrightarrow{AD}=(\vec{c}-\vec{b})\times-\vec{a}=-\vec{c}\times\vec{a}+\vec{b}\times\vec{a}$

and $\overrightarrow{CA}\times\overrightarrow{BD}=(\vec{a}-\vec{c})\times-\vec{b}=-\vec{a}\times\vec{b}+\vec{c}\times\vec{b}$

Adding all we get

$\overrightarrow{AB}\times\overrightarrow{CD}+\overrightarrow{BC}\times\overrightarrow{AD}+\overrightarrow{CA}\times\overrightarrow{BD}=-2(\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a})$

$\therefore \quad |\overrightarrow{AB}\times\overrightarrow{CD}+\overrightarrow{BC}\times\overrightarrow{AD}+\overrightarrow{CA}\times\overrightarrow{BD}|=2|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|$

$=2|(\vec{c}-\vec{a})\times(\vec{b}-\vec{a})|=2|\overrightarrow{AC}\times\overrightarrow{AB}|$

$=4\cdot\frac{1}{2}|\overrightarrow{AC}\times\overrightarrow{AB}|=4$ (Area of ΔABC)

$\therefore \quad \lambda=4$

Then, $125\lambda=500$

9. **(c)** $A=[x, y]$ $B=\{a, b, c, d\}$

$A\times B$ Having $2\times 4=8$ elements

Total subsets of $A\times B$ is $2^8=256$

$\therefore$ Total no. of subsets of $A\times B$ having 3 or more elements

$$=256-\left(\underset{\text{null set}}{1}+\underset{\text{single ton set}}{8}+\underset{\substack{\text{subset having}\\ \text{2 elements}}}{{}^8C_2}\right)$$

$$=256-1-5-28=219$$

10. **(d)** $f(x)=2x^3+3x+k$

$f'(x)=6x^2+3 \;\; f'(x)=0$

$\Rightarrow \quad x^2=-\frac{1}{2}$

Not possible. As condition for two distinct real root is $f(\alpha)\,f(\beta)=0$ (where are α,β roots of $f'(x)=0$)

11. **(c)** $\left|\frac{z_1-2z_2}{2-z_1z_2}\right|=1 \Rightarrow |z_1-2z_2|^2=|2-z_1\bar{z}_2|^2$

$\Rightarrow \quad (z_1-2z_2)(\bar{z}_1-2\bar{z}_2)=(2-z_1\bar{z}_2)(2-\bar{z}_1z_2)$

$\Rightarrow \quad z_1\bar{z}_1+4z_2\bar{z}_2=4+z_1\bar{z}_1\,z_2\bar{z}_2$

$\Rightarrow \quad 4+|z_1|^2|z_2|^2-4|z_2|^2-|z_1|^2=0$

$\Rightarrow \quad (|z_1|^2-1)\cdot(|z_2|^2-4)=0$

But $|z_2|\neq 1, \;\therefore\; |z_2|=2$

Hence, z lies on a circle of radius 2 centered at origin.

12. **(a,d)** $adj\ P=\begin{bmatrix}1 & 4 & 4\\ 2 & 1 & 7\\ 1 & 1 & 3\end{bmatrix}\Rightarrow |adj\ P|=4$

We know, $|adj\ P|=|P|^{n-1}$ (where *n* is order of matrix)

$\Rightarrow \quad |adj\ P|=|P|^2 \Rightarrow 4=|P|^2 \Rightarrow |P|=\pm 2$

13. **(b)** $\left(\frac{8}{5}\right)^2+\left(\frac{12}{5}\right)^2+\left(\frac{16}{5}\right)^2+\left(\frac{20}{5}\right)^2+\left(\frac{24}{5}\right)^2+\ldots$

$=\frac{8^2}{5^2}+\frac{12^2}{5^2}+\frac{16^2}{5^2}+\frac{20^2}{5^2}+\frac{24^2}{5^2}+\ldots. \; T_n=\frac{(4n+4)^2}{5^2}$

$S_n=\frac{1}{5^2}\sum_{n=1}^{10}16(n+1)^2=\frac{16}{25}\sum_{n=1}^{10}(n^2+2n+1)$

$=\frac{16}{25}\left[\frac{10\times11\times21}{6}+\frac{2\times10\times11}{2}+10\right]=\frac{16}{25}\times505=\frac{16}{5}m$

$\Rightarrow \quad m=101$

14. **(c)** Number of required ways

$=5!-\{4\cdot 4\}!-{}^4C_2\cdot 3!+{}^4C_3\cdot 2!-1\}=53.$

15. **(a,c,d,)** Conditions:

$-\tan(\theta)>0\Rightarrow \tan\theta<0$ and $\quad -1<\sin\theta<-\frac{\sqrt{3}}{2}$

$\therefore \quad \theta\in\left(\frac{3\pi}{2},\frac{5\pi}{3}\right)\Rightarrow 0<\cos\theta<\frac{1}{2}$

Also, $2\cos\theta(1-\sin\phi)=\sin^2\theta\left(\frac{1}{\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right)\cos\phi-1$

$\Rightarrow \quad 2\cos\theta-2\cos\theta\sin\phi=2\sin\theta\cos\phi-1$

$\Rightarrow \quad 1+2\cos\theta=2\sin(\theta+\phi)$

$\Rightarrow \quad \sin(\theta+\phi)=\frac{1}{2}+\cos\theta$

$\Rightarrow \quad \frac{1}{2}<\sin(\theta+\phi)<1\Rightarrow\frac{\pi}{2}<\phi<\frac{4\pi}{3}$

16. **(c)** $\sqrt{1+x^2}\left[(x\cos\cot^{-1}x+\sin\cot^{-1}x)^2-1\right]^{1/2}$

$=\sqrt{1+x^2}\left[\left(x\cos\cos^{-1}\frac{x}{\sqrt{1+x^2}}+\sin\sin^{-1}\frac{1}{\sqrt{1+x^2}}\right)^2-1\right]^{1/2}$

$=\sqrt{1+x^2}\left[\left(\frac{x^2}{\sqrt{1+x^2}}+\frac{1}{\sqrt{1+x^2}}\right)^2-1\right]^{1/2}$

$=\sqrt{1+x^2}\,(x^2+1-1)^{1/2}=x\sqrt{1+x^2}.$

17. **(a)** $a_1=1 \quad a_2=2 \quad a_3=3 \quad a_4=5$

a_n is following Fibonacci series. Hence $a_{17}=a_{16}+a_{15}$.

18. **(b)** 1 _ _ _ _ 1

Total no of ways exactly two consecutive two is = 5 ways

Total no of ways exactly three consecutive two is = 2 ways

Total no of ways exactly four consecutive two is = 1 ways

So, $b_6=5+2+1=8$

19. **(b)** Case (i): One odd, 2 even

Total number of ways = 2×2×3+1×3×3+1×2×4 = 29.

Case (ii): AII 3 odd

Number of ways = 2×3×4 = 24

Favourable ways = 53

Required probability = $\frac{53}{3\times5\times7}=\frac{53}{105}$.

20. **(c)** Here $2x_2=x_1+x_3$

$\Rightarrow \quad x_1+x_3=$ even

Hence number of favourable ways

$={}^2C_1\cdot{}^4C_2+{}^1C^1\cdot{}^3C_1=11.$

□ □ □

JEE-MAIN: MATHEMATICS MOCK TEST-5

1. The domain of definition of the function $y4x5$ given by the equation $x+\ \ y=\ \ $ is:

a. $.<x\le 1$ **b.** $x\le x\le 1$

c. $-\infty<x\le .$ **d.** $-\infty<x<12$

2. A real root of the equation $\log_{\cdot\cdot}6\log\ 4\sqrt{x+7}-\sqrt{x}58=.$ is

a. 1 **b.** **c.** ! **d.** "

3. $\cosh 4\alpha+i\beta 5-\cosh 4\alpha-i\beta 5$ is equal to

a. $\sinh\alpha\ \sinh\beta$ **b.** $\cosh\alpha\ \cosh\beta$

c. $i\sinh\alpha\ \sin\beta$ **d.** $\cosh\alpha\ \cos\beta$

4. Matri# $\begin{bmatrix} . & -" & 1 \\ " & . & -5 \\ -1 & 5 & . \end{bmatrix}$ is

a. $rthogonal **b.** %dem&otent

c. 'ke (-symmetric **d.** 'ymmetric

5.)or a sequence $<a_n>/\ a_1=\ \ $ and $\frac{a_{n+1}}{a_n}=\frac{1}{!}2$ Then $\sum_{r=1}^{\cdot} a_r$ is

a. $\frac{\cdot}{}9"+19\times !:$ **b.** $!\left(1-\frac{1}{!\ \cdot}\right)$

c. $41-!\ \cdot 5$ **d.** *one of these

6. +oefficients of $x^r9.\le r\le 4n-15:$ in the e#&ansion of

$4x+!5^{n-1}+4x+!5^{n-}\ 4x+\ \ 5$

$+4x+!5^{n-!}4x+\ \ 5\ \ +222+4x+\ \ 5^{n-1}$

a. ${}^nC_r4!^r-\ {}^n5$ **b.** ${}^nC_r4!^{n-r}-\ {}^{n-r}5$

c. ${}^nC_r4!^r+\ {}^{n-r}5$ **d.** *one of these

7. $1+\frac{}{!;}+\frac{!}{5;}+\frac{"}{7;}+\ldots\infty=$

a. e **b.** e **c.** $e,$ **d.** $e,!$

8. The number of (ays in (hich the follo(ing &ri-es be given to a class of . boys/ first and second Mathematics/ first and second 0hysics/ first +hemistry and first 1nglish is2

a. $.^"\times 19$ **b.** $.^!\times 19^!$

c. $.\ \times 19^"$ **d.** *one of these

9. %f the integers m and n are chosen at random bet(een 1 and 1../ then the &robability that a number of the form 7^m+7^n is divisible by 5/ equals

a. $\frac{1}{"}$ **b.** $\frac{1}{7}$ **c.** $\frac{1}{7}$ **d.** $\frac{1}{"9}$

10. $\sin 4\pi+\theta 5\sin 4\pi-\theta 5\ \text{cosec}\ \theta=$

a. 1 **b.** -1 **c.** $\sin\theta$ **d.** $-\sin\theta$

11. $\sinh\ x$ equals

a. $\cosh\ x-1$ **b.** $\cosh\ x+1$

c. $\frac{1}{}4\cosh\ x-15$ **d.** $\frac{1}{}4\cosh\ x+15$

12. A to(er subtends an angle α at a &oint A in the &lane of its base and the angle of de&ression of the foot of the to(er at a &oint l meters 3ust above A is $\beta 2$ The height of the to(er is

a. $l\ \tan\beta\cot\alpha$ **b.** $l\ \tan\alpha\cot\beta$

c. $l\ \tan\ \alpha\ \tan\beta$ **d.** $l\ \cot\ \alpha\ \cot\beta$

13. The value of k so that the function

$f4x5=\begin{cases} k4\ \ x-x\ \ 5/ & (hen\ \ x<. \\ \cos x/ & (hen\ \ x\ge . \end{cases}$ is continuous at $x=./$ is

a. 1 **b.**

c. " **d.** *one of these

14. %f $y=41+x5^x/$ then $\frac{dy}{dx}=$

a. $41+x5^x\left[\frac{x}{1+x}+\log ex\right]$

b. $\frac{x}{1+x}+\log 41+x5$

c. $41+x5^x\left[\frac{x}{1+x}+\log 41+x5\right]$

d. *one of these

15. $\frac{x}{1+x\tan x}$ is ma#ima at

a. $x=\sin x$ **b.** $x=\cos x$

c. $x=\frac{\pi}{!}$ **d.** $x=\tan x$

16. $\int\frac{1}{x\sqrt{1+\log x}}dx=$

a. $\frac{}{!}41+\log x5^{!,}+c$ **b.** $41+\log x5^{!,}+c$

c. $\sqrt{1+\log x}+c$ **d.** $\sqrt{1+\log x}+c$

17. The area of the triangle formed by the tangent to the hy&erbola $xy=a$ and co-ordinate a#es is

a. a **b.** a **c.** $!a$ **d.** $"a$

18. The solution of $\frac{dy}{dx} = x\log x$ is

a. $y = x^2\log x - \frac{x^2}{2} + c$

b. $y = \frac{x^2}{2}\log x - x^2 + c$

c. $y = \frac{1}{2}x^2 + \frac{1}{2}x^2\log x + c$

d. None of these

19. The distance of point (–2, 3) from the line $x - y = 5$ is

a. $5\sqrt{2}$

b. $2\sqrt{5}$

c. $3\sqrt{5}$

d. $5\sqrt{3}$

20. If the coordinates of one end of the diameter of the circle $x^2 + y^2 - 8x - 4y + c = 0$ are (–3, 2), then the coordinates of other end are

a. (5, 3) **b.** (6, 2)

c. (1, –8) **d.** (11, 2)

21. Let (x, y) by any point on the parabola $y^2 = 4x$. Let P be the point that divides the line segment from $(0,0)$ to (x, y) in the ratio $1:3$. Then, the locus of P is

a. $x^2 = y$

b. $y^2 = 2x$

c. $y^2 = x$

d. $x^2 = 2y$

22. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$, then $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} =$

a. 1

b. 3

c. – 3/2

d. 3/2

23. The direction cosines of three lines passing through the origin are $l_1, m_1, n_1; l_2, m_2, n_2$ and l_3, m_3, n_3. The lines will be coplanar, if

a. $\begin{vmatrix} l_1 & n_1 & m_1 \\ l_2 & n_2 & m_2 \\ l_3 & n_3 & m_3 \end{vmatrix} = 0$ **b.** $\begin{vmatrix} l_1 & m_2 & n_3 \\ l_2 & m_3 & n_1 \\ l_3 & m_1 & n_2 \end{vmatrix} = 0$

c. $l_1l_2l_3 + m_1m_2m_3 + n_1n_2n_3 = 0$

d. None of these

24. $\sim(p \vee (\sim q))$ is equal to

a. $\sim p \vee q$

b. $(\sim p) \wedge q$

c. $\sim p \vee \sim p$

d. $\sim p \wedge \sim q$

25. If the sum of the roots of the equation $\lambda x^2 + 2x + 3\lambda = 0$ be equal to their product, then $\lambda =$

a. 4

b. –4

c. 6

d. None of these

26. If a, b, c and u, v, w are the complex numbers representing the vertices of two triangles such that $c = (1-r)a + rb$ and $w = (1-r)u + rv$, where r is a complex number, than the two triangles

a. have the same area

b. are similar

c. are congruent

d. None of these

27. The rank of the matrix, $A = \begin{bmatrix} 2 & 3 & 1 & 4 \\ 0 & 1 & 2 & -1 \\ 0 & -2 & -4 & 2 \end{bmatrix}$ is

a. 2 **b.** 3

c. 1 **d.** Indeterminate

28. The coefficient of x^n in the expansion of $(1 + x + x^2 + ...)^{-n}$ is

a. 1

b. $(-1)^n$

c. n

d. $n + 1$

29. Three boys and two girls stand in a queue. The probability that the number of boys ahead of every girl is atleast one more than the number of girls ahead of her, is

a. 1/2

b. 1/3

c. 2/3

d. $3/4$

30. The imaginary part of $\tan^{-1}(\cos\theta + i\sin\theta)$ is

a. $\tanh^{-1}(\sin\theta)$

b. $\tanh^{-1}(\infty)$

c. $\frac{1}{2}\tanh^{-1}(\sin\theta)$

d. None of these

JEE ADVANCE PAPER-I

SECTION 1 Contains 8 Questions.

The ans (er to each question is a single digit integer ranging from . to 9 4both inclusive52

1. A variable &lane is at a constant distance p form the origin and meets the a#es in A, B and C2 %f the locus of the centroid of the tetrahedron $OABC$ is $x^{-}+y^{-}+z^{-}=\lambda p^{-}$ then the value of 1=. λ must be

2. The normal to the &arabola $y=7x$ at the &oint 4 / "5 meets it again at 417/ <1 52 %f length of normal chord is λ/ then the value of λ must be

3. An electric com&onent manufactured by BCA'D electronicsE is tested for its defectiveness by a so&histicated testing device2 >et A denote the event Fthe device is defectiveG and B the event Fthe testing device reveals the com&onent to be defectiveG2 'u&&ose $P4A5=\alpha$ and $P4B,A5=P4B\mathrm{L},A\mathrm{I}5=1-\alpha$/ (here . $<\alpha$ H 12 %f the &robability that the com&onent is not defective/ given that the testing device reveals it to be defective/ is λ/ then the value of . .7 λ must be

4. The coefficient of $x^{5\cdot}$ in the &olynomials after &arenthesis have been removed and like terms have been collected in the e#&ansion $41+x5^{1\cdots}+x41+x5^{999}+x41+x5^{997}+222+x^{1\cdots}$ is $\frac{\lambda;}{\mu;\nu;}$/ then the value of $\lambda+\mu+!\nu$ must be 4$\nu>\mu$5

5. The value of

$$\frac{1}{\pi}\left\{1=\sin^{-1}\left(\sin\frac{7\pi}{=}\right)+7\cos^{-1}\left(\cos\frac{\pi}{!}\right)+7\tan^{-1}\left(\tan\frac{5\pi}{"}\right)+..\cot^{-1}\left(\cos\frac{-\pi}{"}\right)\right\}$$

must be

6. >et $\vec{a}/\vec{b}$ and $\vec{c}$ be three non-co&lanar unit vectors such that the angle bet (een every &air of them is $\frac{\pi}{!}$2 %f $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$/ (here p/ q and r are scalars/ then the value of $\frac{p+q+r}{q}$ is IIIIIII

7. %f $f\left(x+\frac{y}{7}/\ x-\frac{y}{7}\right)=xy$/ then the value of $f4=./"75+f47./"75+f41!/55$ must be

8. %f $x=a4\cos t+t\sin t5$@ $y=a4\sin t-t\cos t5$/ then the value of 1 .5$\left.\frac{d\ y}{dx}\right|_{t=\pi,!}$ must be

SECTION 2 Contains 10 Multiple Choice Questions

J ith one or more than one correct o&tion

9. >et A and B be t (o sets containing four and t (o elements res&ectively2 Then the number of subsets of the set $A\times B$/ each having at least three elements is:

a. 19 **b.** 5=

c. 75 **d.** 51.

10. The quadratic equation $p4x5=.$ (ith real coefficients has &urely imaginary roots2 Then the equation $p4p4x55-.$ has

a. only &urely imaginary roots

b. all real roots

c. t (o real and t (o &urely imaginary roots

d. neither real nor &urely imaginary roots

11. A value of θ for (hich $\frac{+!i\sin\theta}{1-i\sin\theta}$ is &urely imaginary/ is:

a. $\frac{\pi}{!}$ **b.** $\frac{\pi}{=}$

c. $\sin^{-1}\left(\frac{\sqrt{!}}{"}\right)$ **d.** $\sin^{-1}\left(\frac{1}{\sqrt{!}}\right)$

12. >et ω be a com&le# cube root of unity (ith $\omega\neq1$ and $P9P_{ij}:$ be a $n\times n$ matri# (ith $p_{ij}=\omega^{i+j}$2 Then (hen $P\neq./\ n=$

a. 57 **b.** 55

c. 57 **d.** 5=

13. %f m is the $A.M.$ of t (o distinct real numbers ℓ and $n4\ell/n>15$ and K1/ K and K! are three geometric means bet (een ℓ and n/ then $G_1^{"}+G^{"}+G_!^{"}$ equals/

a. "ℓ mn **b.** "ℓm n

c. "ℓmn **d.** "ℓ m n

14. >et $S_n = \sum_{k=1}^{"n} 4-15^{\frac{k4k+15}{}} k$ 2 Then S_n can take value4s5

a. 1.5= **b.** 1.77 **c.** 11 . **d.** 1!!

15. %f $!^x = "^{x-1}/$ then $x =$

a. $\frac{\log_!}{\log_! -1}$ **b.** $\frac{}{-\log\ !}$

c. $\frac{1}{1-\log_{"} !}$ **d.** $\frac{\log\ !}{\log\ !-1}$

16. The number of integers greater than =/... that can be formed/ using the digits !/ 5/ =/ 7 and 7/ (ithout re&etition/ is:

a. 1= **b.** 19 **c.** 1 . **d.** 7

17. A shi& is fitted (ith three engines E_1/E and $E_!$2 The engines function inde&endently of each other (ith res&ective &robabilities $\frac{1}{-}/\frac{1}{"}$ and $\frac{1}{"}$2)or the shi& to be o&erational at least t(o of its engines must function2 >et X denote the event that the shi& is o&erational and let X_1/X and $X_!$ denote res&ectively the events that the engines E_1/E And $E_!$ are functioning2 J hich of the follo(ing is 4are5 true M

a. $P9X_1^c \P X := \frac{!}{1=}$

b. 0 91#actly t(o engines of the shi& are functioning $9X := \frac{7}{7}$

c. $P9X \P X\ := \frac{5}{1=}$

d. $P9X \P X_1 := \frac{7}{1=}$

18. >et PQR be a triangle of area Δ (ith $a = /b = \frac{7}{-}$ and $c = \frac{5}{-}/$ (here a/ b and c are the lengths of the sides of the triangle o&&osite to the angles at $P/\ Q$ and R res&ectively2 Then $\frac{\sin P - \sin\ P}{\sin P + \sin\ P}$ equals

a. $\frac{!}{"\Delta}$ **b.** $\frac{"5}{"\Delta}$ **c.** $\left(\frac{!}{"\Delta}\right)$ **d.** $\left(\frac{"5}{"\Delta}\right)$

SECTION 3 Contains 2 Match The Following Type Questions

Nou (ill have to match entries in +olumn % (ith the entries in +olumn %%2

19. Match the statements given in +olumn % (ith the interval,union of intervals given in +olumn %%

Column I	Column II
(A) The set $\left\{Ce\left(\frac{i-}{1--}\right)\right\}$: - is a com&le# number/ $\P - \P = 1/ - \neq \pm 1$ is	**1.** 4–∞/ –15 ∪ 41/ ∞5
(B) The domain of the function $f4\#5 = \sin^{-1}\left(\frac{74!5^{\#-}}{1-!^{\ 4\#-15}}\right)$ is	**2.** 4–∞/ .5 ∪ 4./ ∞5
(C) %f $f4\theta5 = \begin{vmatrix} 1 & \tan\theta & 1 \\ -\tan\theta & 1 & \tan\theta \\ -1 & -\tan\theta & 1 \end{vmatrix}/$ then the set $\left\{f4\theta5 : . \le \theta < \frac{\pi}{-}\right\}$ is	**3.** 9 /∞5
(D) %f $f4\#5 = \#^{!\cdot}\ 4!\# - 1.5/$ $\# \ge ./$ then f4#5 is increasing in	**4.** 4–∞/ –1: ∪91/ ∞5
	5. 4–∞/ .: ∪9 /∞5

20. Match the statements,e#&ressions in +olumn % (ith the values given in +olumn %%2

Column I	Column II
(A) The number of solutions of the equation $xe^{\sin x} - \cos x = .$ in the interval $\left(./\frac{\pi}{-}\right)$	**1.** 1
(B) Oalue 4s5 of k for (hich the &lane $kx + "y + z = ./"x + ky + \ z = .$ and $x + \ y + z = .$ intersect in a straight line	**2.**
(C) Oalue 4s5 of k for (hich $\P x-1\P + \P x - \ \P + \P x+1\P x + \ \P = "k$ has integer solution 4s5	**3.** !
(D) %f $y'y+1$ and $y4.5=1$ then value 4s5 of y 4ln 5	**4.** "
	5. 5

Space for rough work

JEE ADVANCE PAPER-II

SECTION 1 Contains 8 Questions.

The ans (er to each question is a single digit integer ranging from . to 9 4both inclusive52

1. %f the a&&ro#imate value of $\log_{1.}4"2."5$. $abcdef$/ %t is given that $\log_{\cdot\cdot}"=.2=.\ 1$ and $\log_{1.}e=.2"!"!/$ then the value of $abcd$ must be

2. %f $\int 4\sqrt{\tan x}+\sqrt{\cot x}5dx=a\tan^{-1}\left(\frac{\tan x-1}{\sqrt{b\tan x}}\right)+c/$ then the value of $a^{"}+b^5$ must be

3. The solution of the differential equation $\frac{d^!y}{dx^!}-7\frac{d\ y}{dx}=.$ satisfying $y4.5=\frac{1}{7}/y_14.5\,\mathrm{Q}$ .and $y\ 4.5\,\mathrm{Q}$ 1is $y_1=\frac{e^{7x}-7x+7}{\lambda}/$ then the numerical value of λ must be

4. %f m_1 and m are the roots of the equation $x\ +4\sqrt{!}+\ 5x+4\sqrt{!}-15=.$ and if area of the triangle formed by the lines $y=m_1x/\ y=m\ x\ /$ and $y=c$ is $4a+b5c\ /$ then the value of $..7\ \ 4a\ +b\ 5$ must be

5. Tangents are dra (n form $P4=/75$ to the circle $x\ +y\ =r\ /$ then the radius of the circle such that the area of the R formed by tangents and chore of contact is ma#imum must be

6. BPE is any arbitrary &oint on the circum circle of the equilateral triangle of side length = unit/ then the value of $\mathbb{P}\overrightarrow{PA}\mathbb{P}\ +\mathbb{P}\overrightarrow{PB}\mathbb{P}\ +\mathbb{P}\overrightarrow{PC}\mathbb{P}$ must be

7. The lines $\frac{x+"}{!}=\frac{y+=}{5}=\frac{z-1}{-}$ and $!x-\ y+z+5=.$ $=\ x+!y+"z-k$ are co&lanar for k is equal to

8. %f a circle cuts a rectangular hy&erbola $xy=c$ in A, B, C and D are the &arameters of these four &oints be $t_1/t\ /t_!$ and $t_{\cdot\cdot}$ res&ectively/ then the value of $1=t_1t\ t_!t_{\cdot\cdot}$ must be

SECTION 2 Contains 8 Multiple Choice Questions

J ith one or more than one correct o&tion

9. >et $\tan^{-1}y=\tan^{-1}x+\tan^{-1}\left(\frac{x}{1-x}\right)/$ (here $\mathbb{P}x\mathbb{P}<\frac{1}{\sqrt{!}}2$ Then a value of y is

a. $\frac{!x-x^!}{1-!x}$ **b.** $\frac{!x+x^!}{1-!x}$

c. $\frac{!x-x^!}{1+!x}$ **d.** $\frac{!x+x^!}{1+!x}$

10. %f the angles of elevation of the to& of a to (er from three collinear &oints A/ S and +/ on a line leading to the foot of the lo (er/ are !.°/ "5° and =.° res&ectively/ then the ratio/ AS : S+ is

a. $\sqrt{!}:1$ **b.** $\sqrt{!}:\sqrt{\ }$

c. $1:\sqrt{!}$ **d.** $:!$

11. >et $g4x5=\frac{4x-15^n}{\log\cos^m4x-15}\mathbb{Q}$ $.<x<\ /\ \ m$ and n are integers/ $m\neq./n>./$ and let p be the left hand derivative of $\mathbb{P}x-1\mathbb{P}$ at $x=12$ %f $\lim_{x\to1^+}g4x5=p/$ then

a. $n=1/m=1$ **b.** $n=1/m=-1$

c. $n=\ /m=$ **d.** $n>\ /m=n$

12. >et $g4x5=\log4f4x55$ (here $f4x5$ is a t (ice differentiable &ositive function on $4./\infty5$ such that $f4x+15=x\ f4x52$ Then/ for $N=1/\ \ /!/222/$

a. $-"\left\{1+\frac{1}{9}+\frac{1}{5}+222+\frac{1}{4\ \ N-15}\right\}$

b. $"\left\{1+\frac{1}{9}+\frac{1}{5}+222+\frac{1}{4\ \ N-15}\right\}$

c. $-"\left\{1+\frac{1}{9}+\frac{1}{5}+222+\frac{1}{4\ \ N+15}\right\}$

d. $"\left\{1+\frac{1}{9}+\frac{1}{5}+222+\frac{1}{4\ \ N+15}\right\}$

13. >et $I=\int\frac{e^x}{e^{"x}+e^{\ x}+1}dx/\ J=\int\frac{e^{-x}}{e^{-"x}+e^{-\ x}+1}dx2$ Then for an arbitrary constant $C/$ the value of $J-I$ equals

a. $\frac{1}{-}\log\left(\frac{e^{"x}-e^{\ x}+1}{e^{"x}+e^{\ x}+1}\right)+C$

b. $\frac{1}{-}\log\left(\frac{e^{x}+e^{x}+1}{e^{x}-e^{x}+1}\right)+C$

c. $\frac{1}{-}\log\left(\frac{e^{x}-e^{x}+1}{e^{x}+e^{x}+1}\right)+C$

d. $\frac{1}{-}\log\left(\frac{e^{"x}+e^{x}+1}{e^{"x}-e^{x}+1}\right)+C$

14. The integral $\int_{.}^{\pi}\sqrt{1+"\sin\frac{x}{-}-"\sin\frac{x}{-}}\,dx$ equals:

a. $\pi-"$

b. $\frac{\pi}{!}-"-"\sqrt{!}$

c. $"\sqrt{!}-"$

d. $"\sqrt{!}-"-\frac{\pi}{!}$

15. >et the &o&ulation of rabbits surviving at a time t be government by the differential equation $\frac{dp4t5}{dt}=\frac{1}{-}p4t5-\ ..2$ %f $4p5.=1../$ then 4&5t equals:

a. $"..-!..e^{t,}$

b. $!..-\ ..e^{-t,}$

c. $=..-5..e^{t,}$

d. $"..-!..e^{-t,}$

16. A straight line L through the &oint $4!/<5$ is inclined at an angle $=.°$ to the line $\sqrt{!}x+y=12$ %f L also intersects the #-a#is/ then the equation of L is

a. $y+\sqrt{!}x+\ -!\sqrt{!}=.$

b. $y-\sqrt{!}x+\ +!\sqrt{!}=.$

c. $\sqrt{!}y-x+!+\ \sqrt{!}=.$

d. $\sqrt{!}y+x-!+\ \sqrt{!}=.$

SECTION 3 Contains 2 Paragraph Type Questions

1ach &aragra&h describes an e#&eriment/ a situation or a &roblem2 T(o multi&le choice questions (ill be asked based on this &aragra&h2 $ne or more than one o&tion can be correct2

Paragraph for Question No. 17 to 18

+onsider the line

$L_1:\frac{x+1}{!}=\frac{y+}{1}=\frac{z+1}{-}/$

$L\ :\frac{x-}{1}=\frac{y+}{-}=\frac{z-!}{!}$

17. The unit vector &er&endicular to both L_1 and L is

a. $\frac{-\hat{i}+7\hat{j}+7\hat{k}}{\sqrt{99}}$

b. $\frac{-\hat{i}-7\hat{j}+5\hat{k}}{5\sqrt{!}}$

c. $\frac{-\hat{i}+7\hat{j}+5\hat{k}}{5\sqrt{!}}$

d. $\frac{7\hat{i}-7\hat{j}-\hat{k}}{\sqrt{99}}$

18. The distance of the &oint 41/ 1/ 15 from the &lane &assing through the &oint 4T1/ T / T15 and (hose normal is &er&endicular to both the lines L_1 and L is

a. $\frac{}{\sqrt{75}}$

b. $\frac{7}{\sqrt{75}}$

c. $\frac{1!}{\sqrt{75}}$

d. $\frac{!}{\sqrt{75}}$

Paragraph for Question No. 19 to 20

>et $ABCD$ be a square of side length units/ C is the circle through vertices/ A/ B/ C/ D and C_1 is the circle touching all the sides of the square $ABCD$2 L is a line through A2

19. %f P is a &oint on C_1 and U in another &oint on C / then $\frac{PA+PB+PC+PD}{QA+QB+QC+QD}$ is equal to

a. .275

b. 12 5

c. 1

d. .25

20. A line M through A is dra(n &arallel to BD2 0oint S moves such that its distances from the line BD and the verte# A are equal2 %f locus of S cuts M at T and $T_!$ and A+ at T_1/ then area of $\Delta T_1 T\ T_!$ is

a. $\frac{1}{-}$ sq2 unit

b. $\frac{-}{!}$ sq2 unit

c. 1 sq2 unit

d. sq2 unit

ANSWER & SOLUTIONS

JEE-Main

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	a	c	c	b	b	c	a	a	b
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	b	d	c	b	c	b	d	a	d
21.	**22.**	**23.**	**24.**	**25.**	**26.**	**27.**	**28.**	**29.**	**30.**
c	c	a	b	d	b	a	b	a	c

1. **(d)** $2y = 2 - 2x$

$\Rightarrow \quad y = \log_2 2 - 2^x$ for domain $2 - 2x \Rightarrow x < 1$

2. **(a)** $\log_4 \left\{ \log_2 (\sqrt{x+8} - \sqrt{x}) \right\} = 0$

$\Rightarrow \quad 4^0 = \log_2 \left(\sqrt{x+8} - \sqrt{x} \right)$

$\Rightarrow \quad 2^1 = \sqrt{x+8} - \sqrt{x}$

$\Rightarrow \quad 4 = x + 8 + x - 2\sqrt{x^2 + 8x}$

$\Rightarrow \quad 2\sqrt{x^2 + 8x} = 2x + 4$

$\Rightarrow \quad x^2 + 8x = x^2 + 4 + 4x$

$\Rightarrow \quad 4x = 4$

$\Rightarrow \quad x = 1.$

3. **(c)** $\cosh(\alpha + i\beta) - \cosh(\alpha - i\beta)$

$= \cosh\alpha \cosh(i\beta) + \sinh\alpha \sinh(i\beta)$

$- \cosh\alpha \cosh(i\beta) + \sinh\alpha \sinh(i\beta)$

$= 2\sinh\alpha \ \sinh i\beta \ = 2i \sinh\alpha \sin\beta.$

4. **(c)** It is skew-symmetric.

5. **(b)** The sequence is a G.P. with common ratio $\frac{1}{3}$.

Now from $\frac{a(1-r^n)}{1-r}$, $\frac{2[1-(1/3)^{20}]}{1-(1/3)}$

$= 3\left[1 - \frac{1}{3^{20}}\right].$

6. **(b)** We have $(x+3)^{n-1} + (x+3)^{n-2}(x+2) +$

$(x+3)^{n-3}(x+2)^2 + ... + (x+2)^{n-1}$

$= \frac{(x+3)^n - (x+2)^n}{(x+3) - (x+2)} = (x+3)^n - (x+2)^n$

$\left(\because \frac{x^n - a^n}{x-a} = x^{n-1} + x^{n-2}a^1 + x^{n-3}a^2 + ... + a^{n-1}\right)$

Therefore coefficient of x^r in the given expression

$=$ Coefficient of x^r in $[(x+3)^n - (x+2)^n]$

$= {}^nC_r 3^{n-r} - {}^nC_r 2^{n-r} = {}^nC_r (3^{n-r} - 2^{n-r})$

7. **(c)** $S = \frac{1}{1!} + \frac{2}{3!} + \frac{3}{5!} + \frac{4}{7!} + \ldots + \frac{n}{(2n-1)!} + \ldots$

Here $T_n = \frac{1}{2} \cdot \frac{2n}{(2n-1)!} = \frac{1}{2} \frac{(2n-1)+1}{(2n-1)!}$

$= \frac{1}{2} \left\{ \frac{1}{(2n-2)!} + \frac{1}{(2n-1)!} \right\}$

$\Rightarrow \quad S = \sum T_n = \frac{1}{2} \left\{ \frac{e + e^{-1}}{2} + \frac{e - e^{-1}}{2} \right\} = \frac{e}{2}.$

Trick: The sum of this series upto 4 terms is 1.359…and this is value of $e/2$ approximately.

8. **(a)** Four first prizes can be given in 20^4 ways since first prize of Mathematics can be given in 20 ways, first prize of Physics also in 20 ways, similarly first prizes of Chemistry and English can be given in 20 ways each. (Note that a boy can stand first in all the four subjects). Then two second prizes can be given in 19^2 ways since a boy cannot get both the first and second prizes. Hence the required number of ways

$= 20^4 \times 19^2.$

9. **(a)** $7^1 = 7, 7^2 = 49, 7^3 = 34, 7^4 = 2401, ...$

Therefore, for $7^r, r \in N$ the number ends at unit place 7, 9, 3, 1, 7, …..

$\therefore \quad 7^m + 7^n$ will be divisible by 5 if it end at 5 or 0.

But it cannot end at 5. Also for end at 0. For this m and n should be as follows

	m	***n***
1	$4r$	$4r-2$
2	$4r-1$	$4r-3$
3	$4r-2$	$4r$
4	$4r-3$	$4r-1$

For any given value of m, there will be 25 values of n.

Hence, the probability of the required event is $\frac{100 \times 25}{100 \times 100} = \frac{1}{4}$

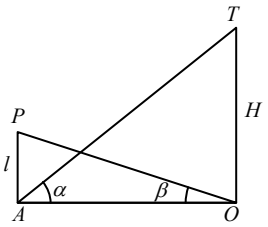
Note: Power of prime numbers have cyclic numbers in their unit place.

10. **(b)** $\sin(\pi + \theta)\sin(\pi - \theta)\text{cosec}^2\theta$

$= -\sin\theta \sin\theta \frac{1}{\sin^2\theta} = -1.$

11. **(c)** $\sinh^2 x = \frac{1}{2}(\cosh 2x - 1)$

12. (b)



)rom figure/ (e can deduce $H = l\tan\alpha\cot\beta$2

13. (d) $f4.–5 = \lim_{x\to.^-} k4\ x-x\ 5 = .$@

$f4.+5 = \lim_{x\to.^+}\cos x = 1$

$\therefore\quad f4.5 = \cos x = 1$

Wence no value of k can make $f4.–5 = 1$2

14. (d) $y = 41+x5^x$

Taking log on both sides/ $\log y = x\log41+x5$

Yifferentiating $w2r2t2\ x$/ (e get

$$\frac{1}{y}\frac{dy}{dx} = \log41+x5 + x\frac{1}{41+x5}$$

Thus $\frac{dy}{dx} = 41+x5^x\left[\frac{x}{1+x} + \log41+x5\right]$

15. (b) %f $\frac{x}{1+x\tan x}$ is ma#ima/ then its reci&rocal $\frac{1+x\tan x}{x}$ (ill be minima2

>et $y = \frac{1+x\tan x}{x} = \frac{1}{x}+\tan x$

$\therefore\quad \frac{dy}{dx} = -\frac{1}{x}+\sec\ x/$

$\frac{d\ y}{dx} = \frac{}{x^!} + \ \sec x\sec x\tan x$

$n &utting $\frac{dy}{dx} = ./\ -\frac{1}{x}+\sec\ x = .$

$\Rightarrow\quad \sec\ x = \frac{1}{x}$

$\Rightarrow\quad x\ = \cos\ x$

$\Rightarrow\quad x = \cos x$

$\therefore\quad \frac{d\ y}{dx} = \frac{}{\cos^!\ x} + \ \sec\ x\tan x$

$= \ \sec\ x4\sec x+\tan x5/$ (hich is &ositive2

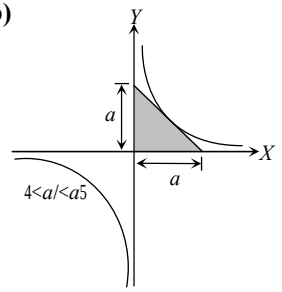
At $x = \cos x/\ \frac{1+x\tan x}{x}$ is minimum2

' o $\frac{x}{1+x\tan x}$ (ill be ma#imum2

16. (c) 0ut $t = 1+\log x \Rightarrow dt = \frac{1}{x}dx/$ then

$$\int\frac{dx}{x\sqrt{1+\log x}} = \int\frac{dt}{t^{1,}} = \ t^{1,}\ + c = \ 41+\log x5^{1,}\ + c2$$

17. (b)



Kiven $xy = a$ or $y = \frac{a}{x}$ 2 2 24i5

There are t (o &oints on the curve $4a/\ a5/4< a/< a5$

The equation of the line at $4a/\ a5$ is/

$$y-a = \left(\frac{dy}{dx}\right)_{4a/a5}4x-a5 = \left(\frac{-a}{x}\right)_{4a/a5}4x-a5$$

$y-a = –4x-a5$ therefore/ equation of the tangent at $4a/a5$ is $x+y = \ a2$ The interce&tion of line $x+y = \ a$ (ith x-a#is is a and (ith y-a#is is a2

$\therefore\quad$ Cequired area $= \frac{1}{}\times\ a\times\ a = \ a\ 2$

18. (d) $\frac{dy}{dx} = x\log x$

$\Rightarrow\quad dy = x\log xdx$

$\Rightarrow\quad \int dy = \int x\log xdx$

$\Rightarrow\quad y = \frac{x}{}\log x - \frac{x}{''}+c2$

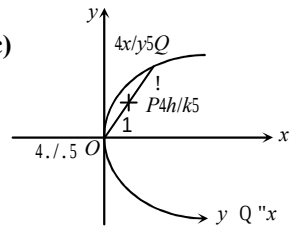
19. (a) Yistance of &oint $4< / !5$ from the line $x-y = 5$ is

$$\left|\frac{-\ -!-5}{\sqrt{\ }}\right| = \left|\frac{-1.}{\sqrt{\ }}\right| = 5\sqrt{\ }2$$

20. (d) $bviously the centre of the circle is $4''/\ 5$ (hich should be the middle &oint of the ends of diameter2

Wence the other end is $411/\ 52$

21. (c)



Sy section formula/ $h=\frac{x+\cdot}{"}/k=\frac{y+\cdot}{"}$

$\therefore \quad x="h/y="k$

'ubstituting in $y \;="x/4"k5 \;="4"h5 \Rightarrow k \;=h$

$r $y \;=x$ is required locus2

22. **(c)** 'quaring $4\vec{a}+\vec{b}+\vec{c}5=./$

(e get $\vec{a}\;+\vec{b}\;+\vec{c}\;+\;\vec{a}\cdot\vec{b}+\;\vec{b}\cdot\vec{c}+\;\vec{c}\cdot\vec{a}=.$

$\Rightarrow \quad P\vec{a}P\;+P\vec{b}P\;+P\vec{c}P\;+\;4\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}5=.$

$\Rightarrow \quad 4\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}5=-!$

$\Rightarrow \quad \vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}=-\frac{!}{}$

23. **(a)** Were/ three given lines are co&lanar if they have common &er&endicular

>et d2c2ls of common &er&endicular be *l/ m/ n*

$\Rightarrow \quad ll_1+mm_1+nn_1=. \qquad 2\,2\,24i5$

$ll\;+mm\;+nn\;=. \qquad 2\,2\,24ii5$

and $ll_!+mm_!+nn_!=. \qquad 2\,2\,24iii5$

'olving 4*ii*5 and 4*iii*5/ (e get

$$\frac{l}{m\,n_!-n\,m_!}=\frac{m}{n\,l_!-n_!l}=\frac{n}{l\,m_!-l_!m}=k$$

$\Rightarrow \quad l=k4m\,n_!-n\,m_!5/m=k4n\,l_!-n_!l\;5/n=k4l\,m_!-l_!m\;5$

'ubstituting in 4*i*5/ (e get

$l_14m\,n_!-n\,m_!5+m_14n\,l_!-n_!l\;5+n_14l\,m_!-l_!m\;5=.$

$$\Rightarrow \quad \begin{vmatrix} l_1 & m_1 & n_1 \\ l & m & n \\ l_! & m_! & n_! \end{vmatrix}=.$$

$$\Rightarrow \quad -\begin{vmatrix} l_1 & n_1 & m_1 \\ l & n & m \\ l_! & n_! & m_! \end{vmatrix}=.$$

24. **(b)** $\sim(p\vee(\sim q))\equiv\sim p\wedge\sim(\sim q)\equiv(\sim p)\wedge q$2

25. **(d)** Dnder condition/

$$--\frac{}{\lambda}=!$$

$\Rightarrow \quad \lambda=--\frac{}{!}$

26. **(b)** 'ince *a, b, c* and *u, v, w* are the vertices of t(o triangles2

Also/ $c=(1-r)a+rb$

and $w=(1-r)u+rv \qquad 2\,2\,24i5$

+onsider $\begin{vmatrix} a & u & 1 \\ b & v & 1 \\ c & w & 1 \end{vmatrix}$

A&&lying $R_!\rightarrow R_!-641-r5R_1+rR\;8$

$$=\begin{vmatrix} a & u & 1 \\ b & v & 1 \\ c-(1-r)a-rb & w-(1-r)u-rv & 1-(1-r)-r \end{vmatrix}$$

$$=\begin{vmatrix} a & u & 1 \\ b & v & 1 \\ . & . & . \end{vmatrix}$$ 9from 1q2 4*i*5:Q .

Wence/ t(o triangle are similar2

27. **(a)** Kiven $A=\begin{bmatrix} & ! & 1 & " \\ . & 1 & & -1 \\ . & - & -" & \end{bmatrix}/\;4R\;\rightarrow\;R\;+R_!5$

$$A=\begin{bmatrix} & ! & 1 & " \\ . & . & . & . \\ . & - & -" & \end{bmatrix}$$

'ince every minor of order ! in A is . and there e#ists a minor order ! *i.e.* $\begin{bmatrix} & ! \\ . & - \end{bmatrix}$ in A (hich is non--ero2 Thus/ rank Q 2

28. **(b)** J e have/ $41+x+x\;+2225^{-n}=941-x5^{-1}:^{-n}=41-x5^n$

$={}^nC_{.}-{}^nC_1x+{}^nC\;x\;+222+4-15^n\;{}^nC_n2x^n$

+oefficient of x^n is $4-15^n\;{}^nC_n=4-15^n2$

29. **(a)** Total number of (ays to arrange ! boys and girls are 5;2 According to given condition/ follo(ing cases may arise2

S K K S S
K K S S S
K S K S S
K S S K S
S K S K S

'o/ number of favourable (ays $=5\times!;\times\;;==.$

$\therefore$ Cequired &robability $=\frac{=.}{1\;.}=\frac{1}{}$

30. **(c)** J e kno((that

$\tan^{-1}4\cos\theta+i\sin\theta5=\frac{\pi}{"}+\frac{i}{"}\log\left(\frac{1+\sin\theta}{1-\sin\theta}\right)/$

$4\cos\theta5>.$

then the %maginary &art of $\tan^{-1}4\cos\theta+i\sin\theta5$ be $\frac{1}{}\tanh^{-1}4\sin\theta5$ / $\left(\because\tanh^{-1}x=\frac{1}{}\log\left(\frac{1+x}{1-x}\right)\right)$

JEE Advance Paper-I

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
5=.	!	1..''	!95''	1!9	''	11	77.	a	d
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
d	bcd	b	ad	abc	b	bd	c	b	a

1. **(2560)** >et the equation of the variable &lane be

$$\left(\frac{x}{a}\right)+\left(\frac{y}{b}\right)+\left(\frac{z}{c}\right)=1 \qquad 2\,2\,24i5$$

Kiven that the &lane is at a distance p form $4./\ ./\ .5$

$$\therefore\quad p=\frac{1}{\sqrt{\left\{\left(\frac{1}{a}\right)+\left(\frac{1}{b}\right)+\left(\frac{1}{c}\right)\right\}}}$$

or $$\frac{1}{p}=\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \qquad 2\,2\,24ii5$$

Also/ the &lane $4i5$ meets the a#es in $A,\ B$ and $C2$ 'o the coordinates of $O,\ A,\ B$ and C are $4./\ ./\ .5/\ 4a/\ ./\ .5/\ 4./\ b/\ .5$ and $4./\ ./\ c5$ res&ectively

>et $4x/\ y/\ z5$ be the centroid of the tetrahedron $OABC/$ then $x=\frac{1}{''}4.+a+.5=\frac{1}{''}a$

'imilarly $y=\frac{1}{''}b$ and $z=\frac{1}{''}c$

or $a=''x/\ b=''y/\ c=''z$

'ubstituting these values in equation $4ii5/$ (e get

$$\frac{1}{p}=\frac{1}{1=x}+\frac{1}{1=y}+\frac{1}{1=z}$$

$\$r \quad x^{-}+y^{-}+z^{-}=1=p^{-}$

$\therefore\quad \lambda=1=$

$\Rightarrow\quad 1=.\lambda=1=.\times 1==\ 5=.$

2. **(3)** +om&aring the given &arabola $4i2e2/\ y\ =\ 7x5$ (ith $y\ =''ax$

$\therefore\quad ''a=7$

$\therefore\quad a=$

'ince/ normal at $4x_1/\ y_1 5$ to the &arabola $y\ =''ax$ is

$$y-y_1=\frac{y_1}{a}4x-x_1 5$$

Were/ $x_1=$ and $y_1=''$

$\therefore\quad$ 1quation of normal is $y-''=-\frac{''}{''}4x-\ 5$

$\Rightarrow\quad y-''=-x+$

$\Rightarrow\quad x+y-==.$ $\qquad 2\,2\,2\,4i5$

'olve equation $4i5$ and $y\ =7x$ then $y\ =74=-y5$

$\Rightarrow\quad y\ +7y-''7=.$

$\Rightarrow\quad 4y+1\ 54y-''5=.$

$\therefore\quad y=-1\quad$ and $\quad y=''$

Then $x=17$ and $x=$

Wence/ &oint of intersection of normal and &arabola are $417/-1\ 5$ and $4\ /\ ''5$ therefore normal meets the &arabola at $417/-1\ 5$ and length of normal chord is distance bet(een their &oints

$$=PQ=\sqrt{417/-1\ 5\ +4-1\ -''5}\ =1=\sqrt{\ }$$

$=\lambda$ 4given5

$\therefore\quad \lambda\ =5,\quad (\ 25\approx !)2$

3. **(1004)** Kiven that $P4A5=\alpha/\ P4B\ ,\ A5=P4B'\ ,\ A'5=1-\alpha$

Thus/ $P4A'5=1-P4A5\ =1-\alpha$

and $P4B\ ,\ A'5=1-P4B'\ ,\ A'5=1-41-\alpha 5=\alpha$ $\qquad 2\,2\,24i5$

$$\therefore\quad P4A'\ ,\ B5=\frac{P4A'\cap B5}{P4B5}=\frac{P4B5-P4A\cap B5}{P4B5}$$

$$=\frac{P4B5-P4A5\ P4B\ ,\ A5}{P4B5}$$

$$\left\{\because\ P4B\ ,\ A5=\frac{P4A\cap B5}{P4A5}\right\}$$

$$=\frac{P4B5-\alpha 41-\alpha 5}{P4B5} \qquad 2\,2\,24ii5$$

Sut $P4B5=P4A5\cdot P4B\ ,\ A5+P4A'5\cdot P4B\ ,\ A'5$

$\alpha\cdot 41-\alpha 5+41-\alpha 5\cdot\alpha$ 6from equation $4i58$

$=\ \alpha 41-\alpha 5$ $\qquad 2\,2\,24iii5$

Outting the value of $P4B5$ from equation $4iii5$ in $4ii5/$ then

$$P4A'\ ,\ B5=\frac{\alpha 41-\alpha 5-\alpha 41-\alpha 5}{\alpha 41-\alpha 5}=\frac{\alpha 41-\alpha 5}{\alpha 41-\alpha 5}=\frac{1}{=}=\lambda$$

4given5

$\therefore\quad ..7\lambda=\ ..7\times\frac{1}{=}=1..''$

4. **(3954)** Dsing the formula for the sum of a geometric &rogression/ (e find

$$41+x5^{1\cdots}+x41+x5^{999}+x\ 41+x5^{997}+\ldots+x^{1\cdots}$$

$$=\frac{41+x5^{1\cdots}\left[1-\left(\frac{x}{1+x}\right)^{1..1}\right]}{\left[1-\left(\frac{x}{1+x}\right)\right]}=\frac{41+x5^{1\cdots}-\frac{x^{1..1}}{41+x5}}{\frac{x+1-x}{41+x5}}=41+x5^{1..1}-x^{1..1}$$

Wence/ the coefficient of $x^{5\cdot}={}^{1..1}C_{5.}=\frac{1..1;}{5.;951;}$

$\therefore\quad \lambda=1..1/\ \mu=5.\ $ and $\ v=951$

$\therefore\quad \lambda+\ \mu+!v=1..1+1..+\ 75!=!95''$

5. **(139)** $\because \sin^{-1}\left(\sin\frac{7\pi}{=}\right)=\pi-\frac{7\pi}{=}=-\frac{\pi}{=}$ /

$\cos^{-1}\left(\cos\frac{\pi}{!}\right)=\frac{\pi}{!}$ /

$\tan^{-1}\left(\tan\frac{5\pi}{"}\right)=\frac{5\pi}{"}-\pi=\frac{\pi}{"}$

And $\cot^{-1}\left\{\cot\left(-\frac{\pi}{"}\right)\right\}=\pi-\frac{\pi}{"}=\frac{!\pi}{"}$

Wence/ required value is

$\frac{1}{\pi}\left\{\ 1=\times-\frac{\pi}{=}+\ \ 7\times\frac{\pi}{!}+\ \ 7\times\frac{\pi}{"}+\ \ \ldots\times\frac{!\pi}{"}\right\}$

$=-!=+17+7+15.=1!9$

6. **(4)** $\mathrm{P}\vec{a}\,\mathrm{P}=\mathrm{P}\vec{b}\,\mathrm{P}=\mathrm{P}\vec{c}\,\mathrm{P}=1$

$\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$

$\vec{a}\cdot 4\vec{b}\times\vec{c}5=p+q4\vec{a}\cdot\vec{b}5+r4\vec{a}\cdot\vec{c}5$ and $9\vec{a}\ \ \vec{b}\ \ \vec{c}:=\frac{1}{\sqrt{\ }}$

$p+\frac{q}{\ }+\frac{r}{\ }=9\vec{a}\ \ \vec{b}\ \ \vec{c}:$ 2 2 24i5

$\frac{p}{\ }+q+\frac{r}{\ }=.$ 2 2 24ii5

$\frac{p}{\ }+\frac{q}{\ }+r=9\vec{a}\ \ \vec{b}\ \ \vec{c}:$ 2 2 24iii5

$\Rightarrow\quad p=r=-q$

$\Rightarrow\quad \frac{p\ +\ q\ +r}{q}="$

7. **(112)** $\therefore\ \ f\left(\ \ x\ +\frac{y}{7}/\ \ x\ -\frac{y}{7}\right)=xy$

$=\sqrt{\left(\ \ x\ +\frac{y}{7}\right)-\left(\ \ x\ -\frac{y}{7}\right)}$

$\therefore\quad f4=./"75+f47./"75+f41!/55$

$=\sqrt{4=.5\ -4"75}\ +\sqrt{47.5\ -4"75}\ +\sqrt{41!5\ -455}\ =!=+="+1$

$=11$

8. **(2880)** $\frac{dy}{dx}=\frac{dy\,,dt}{dx\,,dt}=\frac{a4\cos t-t4-\sin t5-\cos t5}{a4-\sin t+t\cos t+\sin t5}=\tan t$

$\Rightarrow\quad \frac{d\ y}{dx}=\frac{d}{dx}4\tan t5=\frac{d}{dt}4\tan t5\cdot\frac{dt}{dx}=\sec\ \ t\,\frac{1}{at\cos t}=\frac{\sec^{!}t}{at}$

$\left.\frac{d\ y}{dx}\right|_{t=\pi\,,\,!}=\frac{7}{\pi a\,,\,!}=\frac{"}{\pi a}$

$\therefore\quad 1\ .\pi a\left.\frac{d\ y}{dx}\right|_{t=\pi\,,\,!}=1\ .\times\ "=\ \ 77.$

9. **(a)** 'et A has " elements

'et B has elements

$\therefore$ *umber of elements in set $4A\times B5="\times\ \ =7$

$\therefore$ Total number of subsets of $4A\times B5=\ ^{7}=\ 5=$

*umber of subsets having . elements $=7_{C_{.}}=1$

*umber of subsets having 1 element each $=7_{C_1}=7$

$\therefore$ *umber of subsets having elements each

$=7_{C\ }=\frac{7;}{;=;}=\frac{7\times 7}{\ }=\ \ 7$

*umber of subsets having at least ! elements $=\ 5=-1-7-\ \ 7=\ 5=-!7=\ 19$

10. **(d)** $P4x5=ax\ \ +b$ (ith $a/\ b$ of same sign2

$P4P4x55=a4x\ \ +b5\ \ +b$

%f $x\in R$ or $ix\in R$

$\Rightarrow\quad x\ \in R$

$\Rightarrow\quad P4x5\in R$

$\Rightarrow\quad P4P4x55\neq .$

Wence real or &urely imaginary number cannot satisfy $P4P4x55=.2$

11. **(d)** $\frac{+!i\sin\theta}{1-\ i\sin\theta}\times\frac{1+\ i\sin\theta}{1+\ i\sin\theta}$

$-=\sin\ \ \theta=.$ 4)or &urely imaginary5

$\sin\ \ \theta=\frac{1}{!}$

$\sin\theta=\frac{1}{\sqrt{!}}/\ \ \theta=\sin^{-1}\frac{1}{\sqrt{!}}$

12. **(b,c,d)** $P=\begin{bmatrix}\omega & \omega^{!} & \omega^{"} & \ldots & \omega^{n+}\\ \omega^{!} & \omega^{"} & \omega^{5} & \ldots & \omega^{n+!}\\ \vdots & \vdots & \vdots & \vdots & \vdots\\ \omega^{n+} & \omega^{n+!} & \ldots & \ldots & \omega^{\ n+"}\end{bmatrix}$

$P\ =\begin{bmatrix}\omega^{"}+\omega^{=}\ldots & \omega^{5}+\omega^{7}+\omega^{9} & \ldots & \ldots\\ \omega^{5}+\omega^{7}+\omega^{9}\ldots & \ldots & \ldots & \ldots\\ \vdots & \vdots & \vdots & \ldots\\ \omega^{n+"}+\omega^{n+=}\ldots & \ldots & \ldots & \omega^{\ n+"}+\omega^{\ n+=}\ldots\end{bmatrix}$

$P\ =$ *ull matri# if n is a multi&le of !

13. **(b)** Kiven m is A2 M2 bet (een ℓ and n

$\Rightarrow\quad m=\ell+n$

Kiven $\ell/\,G_1/\,G\ /\,G_{!}/\,n$ in K202

$r=\left(\frac{n}{\ell}\right)^{1,"}\Rightarrow r^{"}=\frac{n}{\ell}$

$\therefore \quad G_1 = \ell / G = \ell r \ / G_! = \ell r^!$

'o/ $G_1^{"} + G^{"} + G_!^{"} = \ell^{"} r^{"} 91 + r^{"} + r^7$:

$= \ell^{"} \cdot \left(\frac{n}{\ell}\right)\left[1 + \left(\frac{n}{\ell}\right) + \left(\frac{n}{\ell}\right)\right]$

$= n\ell^! \left[1 + \frac{n}{\ell}\right] = n\ell^! \frac{4n + \ell 5}{\ell} = n\ell 4 \ m5 = "\ell m \ n$

14. **(a,d)** $S_n = \sum_{k=1}^{"n} 4-15^{\frac{k4k+15}{}} k \sum_{r=.}^{4n-15} 44"r + "5 + 4"r + !5 - 4"r + 5 - 4"r + 15\ 5$

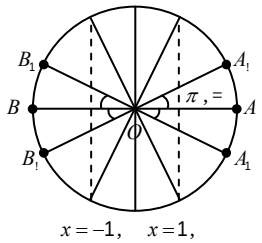
$= \sum_{r=.}^{4n-15} 4 \ 47r + =5 + 47r + "55$

$= \sum_{r=.}^{4n-15} 4! \ r + .5$

$= 1=4n - 15n + .n$

$= "n4"n + 15 = \begin{cases} 1.5= \text{ for n} = 7 \\ 1!! \quad \text{for n} = 9 \end{cases}$

15. **(a,b,c)** log $!^x = 4x - 15$ log $" = 4x - 15$



$x = -1, \quad x = 1,$

$\Rightarrow \quad x \log \ ! = x -$

$\Rightarrow \quad x = \frac{\qquad}{-\log \ !}$

Cearranging/ (e get

$x = \frac{-}{-\frac{1}{\log_!}} = \frac{\log_!}{\log_! \ -1}$

Cearranging again/ $x = \frac{\log_! "}{\log_! "-1} = \frac{\frac{1}{\log_{"} !}}{\frac{1}{\log_{"} !} - 1} = \frac{1}{1 - \log_{"} !}$2

16. **(b) Case-1** Any 5 < digit number Z =. . . is all 5-digits number

Total number Z =. . . using 5-digits Q 5; Q 1 .

Case-2: Dsing "-digits

+an be =27 or 7 " (ays ! (ays (ays

i2e2/ ! (ays Total number $= ! \times " \times ! \times = 7$

Total (ays $= 1 . + 7 = 19$

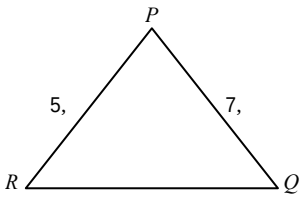
17. **(b,d)** 4a5 $P\left(\frac{X_1^c}{X}\right) \frac{\frac{1}{-} \times \frac{1}{"} \times \frac{1}{"}}{\frac{1}{-} \times \frac{1}{"} \times \frac{!}{"} + \frac{1}{"} \times \frac{1}{"} \times \frac{1}{-} + \frac{1}{-} \times \frac{1}{"} \times \frac{!}{"} + \frac{1}{-} \times \frac{1}{"} \times \frac{1}{"}} = \frac{1}{7}$

4b5 $P\left(\frac{1\#\text{actly t (o enginges are functioning}}{[}\right) = \frac{\frac{7}{!} - \frac{1}{"} \times \frac{1}{"} \times \frac{1}{-}}{\frac{7}{!}} = \frac{7}{7}$

4c5 $P\left(\frac{X}{X}\right) = \frac{\frac{1}{"}\left(\frac{1}{-} \times \frac{!}{"} + \frac{1}{-} \times \frac{1}{"} + \frac{1}{-} \times \frac{1}{"}\right)}{\frac{1}{"}\left(\frac{1}{-} \times \frac{1}{"} + \frac{1}{-} \times \frac{!}{"} + \frac{1}{-} \times \frac{1}{"} + \frac{1}{-} \times \frac{!}{"}\right)} = \frac{5}{7}$

4d5 $P\left(\frac{X}{X_1}\right) = \frac{\frac{1}{-}\left(\frac{1}{"} \times \frac{1}{"} + \frac{1}{"} \times \frac{!}{"} + \frac{1}{"} \times \frac{!}{"}\right)}{\frac{1}{-}\left(\frac{!}{"} \times \frac{!}{"} + \frac{1}{"} \times \frac{1}{"} + \frac{1}{"} \times \frac{!}{"} + \frac{1}{"} \times \frac{!}{"}\right)} = \frac{7}{1=}$

18. **(c)**



$\Rightarrow \quad \frac{\sin P - \ \sin P \cos P}{\sin P + \ \sin P \cos P}$

$\Rightarrow \quad \frac{1 - \cos P}{1 + \cos P} = \frac{!}{!}$

$\cos P = \frac{9}{!5}$

$\Delta = \frac{1}{-} \times \frac{5}{-} \times \frac{7}{-} \times \frac{7\sqrt{=}}{!5}$

$\Rightarrow \quad \Delta = \sqrt{=} \Rightarrow \left(\frac{!}{"\Delta}\right)$

19. **(A)** - $= \frac{\text{i4\#} + \text{iy5}}{1 - 4\# + \text{iy5}} = \frac{\text{i4\#} + \text{iy5}}{1 - 4\# \ - y \ + \ \text{i\#y5}}$

Dsing $1 - \# = y$

$\backslash = \frac{\text{i\#} - \ y}{y \ - \ \text{i\#y}} = -\frac{1}{y}$2

$\because \quad -1 \le y \le 1 \Rightarrow -\frac{1}{y} \le -1 \text{ or } -\frac{1}{y} \ge 12$

(B))or domain

$-1 \le \frac{72!^{\#-}}{1 - !^{\ 4\#-15}} \le 1$

$\Rightarrow \quad -1 \le \frac{!^{\#} - !^{\#-}}{1 - !^{\ \#-}} \le 12$

+ase 4i5: $\frac{!^{\#} - !^{\#-}}{1 - !^{\ \#-}} - 1 \le .$

$\Rightarrow \quad \frac{4!^{\#}-154!^{\#-}-15}{4!^{\#-}-15} \geq .$

$\Rightarrow \quad \# \in 4-\infty/.: \cup 41/\infty 52$

+ase 4ii5: $\frac{!^{\#}-!^{\#-}}{1-!^{\#-}}+1 \geq .$

$\Rightarrow \quad \frac{4!^{\#-}-154!^{\#}+15}{4!^{\#}2!^{\#-}-15} \geq .$

$\Rightarrow \quad \# \in 4-\infty/15 \cup 9 /\infty 52$

'o/ $\# \in 4-\infty/.: \cup 9 /\infty 52$

(C) $C_1 \to C_1 + C_!$

$$f4\theta 5 = \begin{vmatrix} . & . & \\ -\tan\theta & 1 & \tan\theta \\ -1 & -\tan\theta & 1 \end{vmatrix}$$

$= \ 4\tan\ \theta + 15 = \ \sec\ \theta 2$

(D) $f'4\#5 = \frac{!}{}4\#5^{1,}\ 4!\# -1.5 + 4\#5^{!,} \times ! = \frac{15}{}4\#5^{1,}\ 4\# - \ 5$

%ncreasing/ (hen $\# \geq \ 2$

20. **(A)** $f\,l4x5 > ./\forall x \in 4./\pi,\ 5$

$f4.5 < .$ and $f4\pi,\ 5 > .$

'o/ one solution2

(B) >et 4a/ b/ c5 is direction ratio of the intersected line/ then

$ak + "b + c = .$

$"a + kb + \ c = .$

$\frac{a}{7-k} = \frac{b}{"-\ k} = \frac{c}{k\ -1=}$

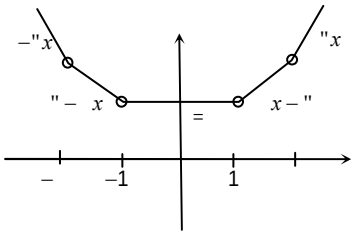
J e must have

$47-k5+\ 4"-\ k5+4k\ -1=5 = .$

$\Rightarrow \quad k = \ /"2$

(C) >et $f4x5 = \text{P}\,x + \ \text{P} + \text{P}\,x + 1\text{P} + \text{P}\,x - 1\text{P} + \text{P}\,x - \ \text{P}$

$\Rightarrow$ k can take value / !/ "/ 52



(D) $\int \frac{dy}{y+1} = \int dx$

$\Rightarrow \quad f4x5 = \ e^{x} - 1$

$\Rightarrow \quad f4\ln\ 5 = !$

JEE Advance Paper-II

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
1	!=	="	55	5	1!5	"	1=	a	a
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**	**18.**	**19.**	**20.**
c	a	c	d	a	b	b	c	a	c

1. **(1)** >et $f4x5 = \log_{1.} x$

$\Rightarrow \quad f'4x5 = \frac{1}{x} \cdot \log_{1.} e$

>et $x = "/\delta x = .2."$

$\therefore \quad f4x + \delta x5 = f4x5 + \delta x \cdot f'4x5$

$\Rightarrow \quad \log_{1.} 4x + \delta x5 = \log_{1.} x + 4.2."5 \times \frac{1}{x}\log_{1.} e$

$\Rightarrow \quad \log_{1.} "2." = \log_{1.} " + 4.2."5 \times \frac{1}{x} \times \log_{1.} e$

$= .2=.\ 1 + 4.2.154.2"!"!5$

$= .2=.\ 1 + .2.."!"! = .2=.=""!\ \approx 1$

2. **(36)** >et $I = \int 4\sqrt{\tan x} + \sqrt{\cot x}5dx$

$= \int \frac{4\sin x + \cos x5}{\sqrt{4\sin x \cos x5}}dx$

Out $\sin x - \cos x = t$

$\Rightarrow \quad 1 - \sin\ x = t$

$\therefore \quad 4\cos x + \sin x5dx = dt$

Then/ $I = \int \frac{dt}{\sqrt{\left(\frac{1-t}{}\right)}} = \sqrt{}\sin^{-1}t = \sqrt{}\tan^{-1}\left(\frac{t}{\sqrt{1-t}}\right) + c$

$= \sqrt{}\tan^{-1}\left(\frac{\sin x - \cos x}{\sqrt{\sin\ x}}\right) + c \ = \sqrt{}\tan^{-1}\left(\frac{\tan x - 1}{\sqrt{\ \tan x}}\right) + c$

J e/ get $a = \sqrt{}/b =$ Then/

$\Rightarrow \quad a^{"} + b^{5} = " + ! \ = !=$

3. **(64)** J e have/ $\frac{d^{!}y\,,dx^{!}}{d\ y\,,dx} = 7$

$\Rightarrow \quad \ln\frac{d\ y}{dx} = 7x + c$

$r \quad \ln y \ = 7x + c$

Outting $x = .$ (e have/

$c = \log y\ 4.5 = \ln 1 = .$

$\therefore \quad \ln y \ = 7x$

$\Rightarrow \quad y \ = e^{7x}$ i2e2/

$y_1 = \frac{e^{7x}}{7} + D$

Again &utting $x = ./$ then $y_1 4.5 = \frac{1}{7} + D$

$\Rightarrow \quad . = \frac{1}{7} + D$

$\therefore \quad D = -\frac{1}{7}$

$\Rightarrow \quad y_1 = \frac{e^{7x}}{7} - \frac{1}{7} \Rightarrow y = \frac{e^{7x}}{="} - \frac{x}{7} + E$

0utting $x = ./$ (e have $y4.5 = \frac{1}{="} - . + E = \frac{1}{7}$

$\therefore \quad E = \frac{1}{7} - \frac{1}{="} = \frac{7}{="}$

Wence/ $y = \frac{e^{7x}}{="} - \frac{x}{7} + \frac{7}{="} = \frac{4e^{7x} - 7x + 75}{="} \Rightarrow \lambda = ="$

4. **(5522)** 'ince m_1 and m are the roots of the equation $x + 4\sqrt{!} + 5x + 4\sqrt{!} - 15 = .$

Then $m_1 + m = -4\sqrt{!} + 5/\ m_1 m = 4\sqrt{!} - 15$

$\therefore \quad m_1 - m = \sqrt{4m_1 + m\ 5 - "m_1 m} = \sqrt{4! + " + "\sqrt{!} - "\sqrt{!} + "5} = \sqrt{11}$

and coordinates of the vertices of the given triangle are $4./\ .5\ 4c, m_1/c5$ and $4c, m\ /c5$

Wence/ the required area of triangle

$$= \frac{1}{-}\left|\frac{c}{m_1} \times c - \frac{c}{m} \times c\right|$$

$$= \frac{1}{-}c\ \left|\left(\frac{1}{m_1} - \frac{1}{m}\right)\right| = \frac{1}{-}c\ \frac{|m - m_1|}{|m_1 m|} = \frac{1}{-}c\ \frac{\sqrt{11}}{4\sqrt{!} - 15}$$

$$= \frac{1}{-}c\ \frac{\sqrt{11}4\sqrt{!} + 15}{4\sqrt{!} - 154\sqrt{!} + 15} = \left(\frac{\sqrt{!!} + \sqrt{11}}{"}\right)c$$

$n com&aring/ $a = \frac{\sqrt{!!}}{"} / b = \frac{\sqrt{11}}{"}$

or $a = \frac{\sqrt{11}}{"} / b = \frac{\sqrt{!!}}{"}$

$\therefore \quad a + b = \frac{!!}{1=} + \frac{11}{1=} = \frac{""}{1=} = \frac{11}{"}$

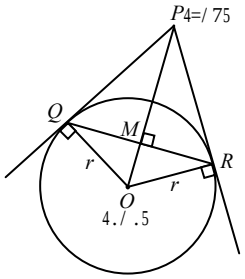
$\Rightarrow \quad ..74a + b\ 5 = \ ..7 \times \frac{11}{"} = 5. \times 11 = 55$

5. **(5)** 1quation of chord of contact $4QR5$ is $=x + 7y - r = .$

$$PM = \frac{\left|=\cdot= + 7 \cdot 7 - r\right|}{\sqrt{4= + 7\ 5}} = \frac{\left|1.. - r\right|}{1.} \text{ and}$$

$$OM = \frac{\left|. + . - r\right|}{\sqrt{4= + 7\ 5}} = \frac{r}{1.}$$

Then $QR = \cdot QM = \sqrt{64OQ5 - 4OM5\ 8} = \sqrt{\left(r - \frac{r"}{1..}\right)}$



Area of $\Delta QPR = \frac{1}{-} \cdot QR \cdot PM$

$\therefore \quad \Delta 4\text{say}5 = \frac{1}{-} \cdot \sqrt{\left(r - \frac{r"}{1..}\right)} \cdot \frac{\left|1.. - r\right|}{1.}$

$\therefore \quad \Delta = \frac{r\ 41.. - r\ 5^!}{1...} = z4\text{say}5$

$\therefore \quad \frac{dz}{dr} = \frac{1}{1...}6r \cdot !41.. - r\ 5 \cdot 4 - r5 + 41.. - r\ 5^! \cdot r8$

$= \frac{r41.. - r\ 5}{1...}61.. - r\ - !r\ 8$

)or ma#imum or minimum $\frac{dz}{dr} = ./$ then (e get

$r = 5/\ 4\ r \neq 1.$ as P is outside the circle5

And $\left.\frac{d\ z}{dr}\right|_{r=5} = -ve$

$\therefore$ Area of triangle is also ma#imum at $r = 52$

6. **(1352)** >et 0O of $P,\ A,\ B$ and C are $\vec{p}/\ \vec{a}/\ \vec{b}$ and $\vec{c}$ res&ectively and $O4\vec{.}.5$ be the circumcircle of the equilateral triangle $ABC2$

Then/ $\frac{\vec{a} + \vec{b} + \vec{c}}{!} = .$ 2 2 24i5

and $\| \vec{p} \| = \| \vec{a} \| = \| \vec{b} \| = \| \vec{c} \| = \frac{=}{\sqrt{!}}$ unit 2 2 24ii5

*o (/ $\| \overrightarrow{PA} \| = \| \vec{a} - \vec{p} \| = a + p - \vec{a} \cdot \vec{p}$

'imilarly $\| \overrightarrow{PB} \| = b + p - \vec{b} \cdot \vec{p}$ and

$\| \overrightarrow{PC} \| = c + p - \vec{c} \cdot \vec{p}$

$\therefore \quad \Sigma \| \overrightarrow{PA} \| = a + b + c + !p - \vec{p}4\vec{a} + \vec{b} + \vec{c}5$

$= =p - .$ 9)rom equation 4i5 and 4ii5:

$= = \frac{4\ =5}{!} = \ 4\ =5\ = 1!5$

7. **(4)** Any &oint on the first line in symmetrical form is $4!r-"/5r--/-\ r+152\%$f the lines are co&lanar/ this &oint must lie on both the &lanes (hich determine the second line2

$\Rightarrow$ $!4!r-"5-\ 45r--5-\ r+1+5=.$ 2 2 24i5

and $4!r-"5+!45r--5+"4-\ r+15-k=.$ 2 2 24ii5

)rom equation 4i5/ (e get $r=$

*o(substituting $r=$ equation 4ii5/ then $k="$

8. **(16)** >et the equation of circle

$x\ +y\ +\ gx+\ fy+k=.$ 2 2 24i5

and the equation of the rectangular hy&erbola is

$xy=c$ 2 2 24ii5

Out $x=ct$ and $y=\dfrac{c}{t}$ in equation 4i5

Then $c\ t\ +\dfrac{c}{t}+\ gct+\dfrac{fc}{t}+k=.$

$\Rightarrow$ $c\ t^{"}+\ gct^{!}+kt\ +\ fct+c\ =.$

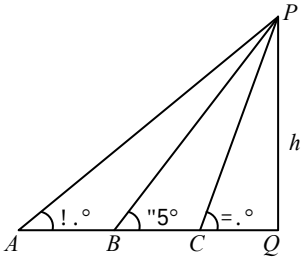
This equation being fourth degree in t2 >et the roots be $t_1/\ t\ /t_!/t_{"}$ then $t_1t\ t_!t_{"}=1$

$\therefore$ $1=t_1t\ t_!t_{"}=1=\times1=1=$

9. **(a)** $\tan^{-1}4y5=\tan^{-1}\left(\dfrac{x+\dfrac{x}{1-x}}{1-\dfrac{x2\ x}{1-x}}\right)$

$\Rightarrow$ $\tan^{-1}4y5=\tan^{-1}\left(\dfrac{!x-x^{!}}{1-!x}\right)\ \Rightarrow\ y=\dfrac{!x-x^{!}}{1-!x}$

10. **(a)** $\dfrac{h}{AQ}=\dfrac{1}{\sqrt{!}}$



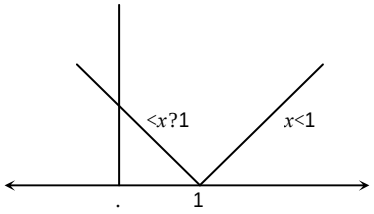
$\Rightarrow$ $AQ=\sqrt{!}h$

'imilarly/ $BQ=h$

$\Rightarrow$ $CQ=\dfrac{h}{\sqrt{!}}$

$\therefore$ $\dfrac{AB}{BC}=\dfrac{AQ-BQ}{BQ-CQ}=\dfrac{4\sqrt{!}-15h}{\left(h-\dfrac{h}{\sqrt{!}}\right)}=\dfrac{\sqrt{!}}{1}$

11. **(c)**)rom gra&h/ $p=-1$



$\Rightarrow$ $\lim\limits_{x\to1^{+}}g4x5=-1\Rightarrow\ \lim\limits_{h\to.}g41+h5=-1$

$\Rightarrow$ $\lim\limits_{h\to.}\left(\dfrac{h^{n}}{\log\cos^{m}h}\right)=-1$

$\Rightarrow$ $\lim\limits_{h\to.}\left(\dfrac{n\cdot h^{n-1}}{m\cdot4-\tan h5}\right)=-\left(\dfrac{n}{m}\right)\lim\limits_{h\to.}\left(\dfrac{h^{n-1}}{\tan h}\right)=-1/$ (hich holds if $n=m=\ 2$

12. **(a)** $g4x+15=\log4f4x+155=\log x+\log4f4x55=\log x+g4x5$

$\Rightarrow$ $g4x+15-g4x5=\log x$

$\Rightarrow$ $g''4x+15-g''4x5=-\dfrac{1}{x}$

$g''\left(1+\dfrac{1}{-}\right)-g''\left(\dfrac{1}{-}\right)=-"$

$g''\left(\ \ +\dfrac{1}{-}\right)-g''\left(1+\dfrac{1}{-}\right)=-\dfrac{"}{9}$ 2222222222

$g''\left(N+\dfrac{1}{-}\right)-g''\left(N-\dfrac{1}{-}\right)=-\dfrac{"}{4\ N-15}$

'umming u& all terms

Wence/ $g''\left(N+\dfrac{1}{-}\right)-g''\left(\dfrac{1}{-}\right)=-"\left(1+\dfrac{1}{9}+222+\dfrac{1}{4\ N-15}\right)2$

13. **(c)** $J-I=\int\dfrac{e^{x}4e^{\ x}-15}{e^{"x}+e^{\ x}+1}dx=\int\dfrac{4Z\ -15}{z^{"}+z\ +1}dz$

(here $z=e^{x}$ $=\int\dfrac{\left(1-\dfrac{1}{z}\right)dz}{\left(z+\dfrac{1}{z}\right)\ -1}=\dfrac{1}{-}\ln\left(\dfrac{e^{x}+e^{-x}-1}{e^{x}+e^{-x}+1}\right)+c$

$\therefore$ $J-I=\dfrac{1}{-}\ln\left(\dfrac{e^{\ x}-e^{x}+1}{e^{\ x}+e^{x}+1}\right)+c2$

14. **(d)** $I=\int_{.}^{\pi}\sqrt{1+"\sin\ \dfrac{x}{-}-"\sin\dfrac{x}{-}}\ \ I=\int_{.}^{\pi}\left|1-\ \sin\dfrac{x}{-}\right|dx$

$I=\int_{.}^{\frac{\pi}{!}}\left(1-\ \sin\dfrac{x}{-}\right)dx+\int_{\frac{\pi}{!}}^{\pi}-\left(1-\ \sin\dfrac{\pi}{-}\right)dx$

$1=\left[x+"\cos\dfrac{x}{-}\right]_{.}^{\frac{\pi}{!}}+\left[-x-"\cos\dfrac{x}{-}\right]_{\frac{\pi}{!}}^{\pi}$

$$1 = \frac{\pi}{!} + "\left(\frac{\sqrt{!}}{} - 1\right) - \left(\pi - \frac{\pi}{!}\right) - "\left(. - \frac{\sqrt{!}}{}\right)$$

$$I = \frac{\pi}{!} + \sqrt{!} - " - \frac{\pi}{!} + \sqrt{!} = "\sqrt{!} - " - \frac{\pi}{!}$$

15. **(a)** Cearranging the equation (e get/

$$\frac{dp4t5}{p4t5 - "..} = \frac{1}{}dt \qquad 2\ 2\ 24i5$$

%ntegrating 415 on both sides (e get

$p4t5 = "... + k\,e^{t,\cdot}$ /

(here k is a constant of integration2

Dsing $p4.5 = 1../$ (e get $k = -!..$

$\therefore$ the relation is $P4t5 = "... - !..e^{1,}$

16. **(b)** %nclination of line $\sqrt{!}x + y = 1$ is 15.°

%nclination of line $L = 15.° \pm =.° = \ 1.°/9.°$

' lo&e of line

$$L = \tan\ 1.° = \tan !.° = \frac{1}{\sqrt{!}}$$

1quation of Q >ine >

$$\Rightarrow \quad y + \ = \frac{1}{\sqrt{!}}4x - !5$$

$$\Rightarrow \quad \sqrt{!}y - x + ! + \sqrt{!} = .$$

17. **(b)** $\begin{vmatrix} i & j & k \\ ! & 1 & \\ 1 & & ! \end{vmatrix} = -\overset{N}{i} - 7\overset{N}{j} + 5\overset{N}{k}$

Wence unit vector (ill be $\frac{-\overset{N}{i} - 7\overset{N}{j} + 5\overset{N}{k}}{5\sqrt{!}}$2

18. **(c)** 0lane is given by $-4x + 15 - 74y + \ 5 + 54z + 15 = .$

$$\Rightarrow \quad x + 7y - 5z + 1. = .$$

$$\Rightarrow \quad \text{distance} \ = \frac{1 + 7 - 5 + 1.}{\sqrt{75}} = \frac{1!}{\sqrt{75}}2$$

19. **(a)** >et A/ S/ + and Y be the com&le# numbers/ $\sqrt{}/-\sqrt{}/\sqrt{}i$

and $-\sqrt{}i$ res&ectively2

$$\Rightarrow \quad \frac{PA\ + PB\ + PC\ + PD}{QA\ + QB\ + QC\ + QD}$$

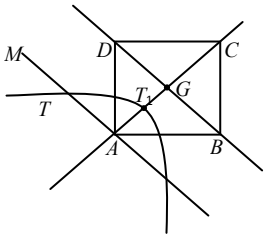
$$= \frac{P z_1 - \sqrt{}P\ + P z_1 - \sqrt{}P\ + P z_1 - \sqrt{}iP\ + P z_1 - \sqrt{}iP}{Pz + \sqrt{}P\ + Pz - \sqrt{}P\ + Pz - \sqrt{}iP\ + Pz - \sqrt{}iP} = \frac{P z_1 P\ +}{P z_1 P\ +} = \frac{!}{"}$$

20. **(c)** $AG = \sqrt{}$

$\therefore \quad AT_1 = T_1G = \frac{1}{\sqrt{}}$ 9as A is the foucs/

T_1 is the verte# and SY is the directri# of &arabola:2

Also $TT_!$ is latus return



$$\therefore \quad TT_! = "\times\frac{1}{\sqrt{}}$$

$\therefore$ Area of $\Delta T_1TT_! = \frac{1}{}\times\frac{1}{\sqrt{}}\times\frac{"}{\sqrt{}} = 1$

□ □ □